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1. SCOPE. This specification establishes the item definition, performance, operating characteristics, reliability, maintainability, subsystems, physical characteristics, general design, installation, and interface requirements for manned aircraft gas turbine engines. This specification also establishes the analysis, inspections, demonstrations, and test procedures required by the Using Service for satisfactory completion and acceptance, of engine qualification Milestones IFR, FFR, ISR, and OCR for aircraft gas turbine engines. This specification also establishes the content and format to be used by the Using Service and/or contractor for the preparation of the acquisition and/or model specifications.

1.1 Structure. The structure of this specification is in accordance with MIL–STD–961 and MIL–STD–490 and complies with the MIL–PRIME format.

1.2 Handbook. This general specification has an accompanying tailoring handbook.

1.3 Deviations. Any projected design for a given application which will result in improvement of system performance, reduced life cycle cost, or reduced development cost through deviation from this specification, or where the requirements of this specification result in compromise in operational capability, will not be considered for approval unless brought to the attention of the Using Service in writing.

1.4 Environmental impact statement. The engine shall be designed so that its operation, maintenance, and repair may be accomplished without violating the most stringent of the environmental regulations applicable to locations where the engine is used or supported. Compliance with environmental regulations shall not prevent the engine from achieving and sustaining mission performance capability. Materials, processes, and environmental control equipment necessary to meet these environmental requirements must currently be available in the Using Service maintenance and supply systems. The design shall not use environmentally unsuitable materials such as ozone depleting fluorocarbons, chlorofluorocarbons and halons, highly volatile organic compounds in solvents and coatings, asbestos containing materials and other hazardous materials or pollutants that will be required to support the engine during development, production, operations, maintenance, and repair.

The contractor shall conduct an environmental analysis of the engine. The analysis shall include environmental life–cycle costs and cost–benefit analyses on alternate procedures and methods for maintenance and repair, material substitution, specialized materials, control technology, treatment processes, process changes, and disposal methods required to support the design.

2. APPLICABLE DOCUMENTS

2.1 General The documents listed in this section are needed to meet the requirements specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

2.2 Government documents

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

TT–S–735	Standard Test Fluids, Hydrocarbon Ref. 4.1.3
VV–F–800	Fuel Oil, Diesel Ref. 3.7.3.1.4

Department of Defense

MIL–F–3056	Gasoline, Automotive, Combat Ref. 3.7.3.1.4
MIL–E–5400	Electronic Equipment, Aerospace, General Specification for Ref. 3.7.4.4
MIL–T–5544	Thread Compound, Antiseize, Graphite–Petrolatum Ref. 3.7.5.1.1
MIL–E–5607	Engine, Gas Turbine, Preparation for Storage and Shipment of, Process of Ref. 5.1
MIL–E–6051	Electromagnetic Compatibility Requirements, System Ref. 3.3.3.2
MIL–C–7024	Calibrating Fluids, Aircraft Fuel System Components Ref. 4.10.3.5.2.1
MIL–S–8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for Ref. 3.1.4.4
MIL–F–16884	Fuel, Naval Distillate Ref. 3.7.3.1.4

MIL–F–17874	Fuel Systems: Aircraft, Installation and Test of Ref. 3.7.3.2.7
MIL–S–19500	Semiconductor Devices, General Specification for Ref. 3.7.4.4
MIL–C–24621	Couplers, Passive, Fiber Optic, General Specification for (Metric) Ref. 3.7.7.3
MIL–H–25579	Hose Assembly, Tetrafluoroethylene, High Temperature, Medium Pressure Ref. 3.1.3.5
MIL–H–27267	Hose, Tetrafluoroethylene, High Temperature, Medium Pressure, Ref. 3.1.3.5
MIL–M–38510	Microcircuits, General Specification for Ref. 3.7.4.4
MIL–C–38999	Connector, Electrical Circular, Miniature, High Density Quick Disconnect (Bayonet, Threaded and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for Ref. 3.7.4.3
MIL–H–46855	Human Engineering Requirements for Military Systems, Equipment and Facilities Ref. 3.5.3
MIL–C–85045	Cables, Fiber Optics, (Metric), General Specification for Ref. 3.7.7.1
MIL–C–85704	Cleaning Compound, Turbine Engine Gas Path Ref. 3.7.14

STANDARDS

Department of Defense

MIL–STD–100	Engineering Drawing Practices Ref. 3.1.5.3
MIL–STD–130	Identification Marking of U.S. Military Property Ref. 3.1.5
MIL–STD–470	Maintainability Program for Systems and Equipment Ref. 3.5.2
MIL–STD–471	Maintainability Verification/Demonstration/Evaluation Ref. 4.5.2
MIL–STD–704	Aircraft Electric Power Characteristics Ref. 3.7.4.1, 3.7.4.2, 4.9.2.1.7

MIL–STD–721	Definitions of Terms for Reliability and Maintainability Ref. 3.5
MIL–STD–785	Reliability Program for Systems and Equipment Development and Production Ref. 3.5.1, 3.7.4.4
MIL–STD–810	Environmental Test Methods and Engineering Guidelines Ref. 4.1.8.3, 3.4.1.10, 3.4.1.11, 4.9.2.1.7, 4.9.2.1.8
MIL–STD–882	System Safety Program Requirements Ref. 3.1.8
MIL–STD–889	Dissimilar Metals Ref. 3.1.3
MIL–STD–1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities Ref. 3.5.3
MIL–STD–1529	Vendor Substantiation for Aerospace Propulsion System Items Ref. 3.1.9.6
MIL–STD–1800	Human Engineering Performance Requirements for Systems Ref. 3.5.3
MIL–STD–1815	Ada Programming Language Ref. 3.8.2
MIL–STD–1818	Electromagnetic Effects, Requirements for Systems Ref. 3.3.3.2, 3.3.3.3
MIL–STD–2084	Maintainability of Avionic and Electronic Systems and Equipment, General Requirements for Ref. 3.8.2.1
MIL–STD–2165	Testability Program for Systems and Equipments Ref. 4.8.2.1
MIL–STD–2173	Reliability–Centered Maintenance Requirements for Naval Aircraft, Weapons Systems and Support Equipment Ref. 4.5.2.3

MS Standards

MS 9006	Recess–Cross, Low Torque Drive, Dimensions and Gage Dimensions for Ref. 3.1.4.3
MS 33537	Insert, Screw Thread, Helical Coil, Inch Series, Coarse and Fine Thread, Standard Assembly Dimensions for Ref. 3.1.4.5

MS 33540	Safety Wiring and Cotter Pinning, General Practices for Ref. 3.1.4.1
MS 33588	Nuts, Self–Locking, Aircraft, Reliability and Maintainability Usage Requirements for Ref. 3.1.4.1
MS 33602	Bolts, Self–Retaining, Aircraft, Reliability and Maintainability, Design and Usage Requirements for Ref. 3.1.4
MS 33750	Recess, High Torque and Conical High Torque, Dimensions of Recess, Gage, and Driver for Ref. 3.1.4.3
MS 33787	Wrenching Element, External Spline, Dimensions for Ref. 3.1.4.6

HANDBOOKS

Department of Defense

MIL–HDBK–235	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems Ref. 3.3.3.3

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094, phone (215) 697–2667.)

2.2.2 Other Government documents, drawings, and publications The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Department of Defense

DODD 5000.2–M	Defense Acquisition Management Documentation and Reports Ref. 4.6.2.1
DODI 5000.2	Defense Acquisition Management Policies and Procedures Ref. 3.6, 4.6.2.1

(Copies of DODDs and DODIs are available from Naval Publications and Forms Directorate, Attn: Code 301, 5801 Tabor Ave, Philadelphia PA 19120–5099.)

National Aeronautics And Space Administration

	U.S. Standard Atmosphere, 1976 Ref. 3.2.1

(Application for copies should be addressed to NASA Center for Aerospace Information (NASA CASI), P O Box 8757, Baltimore MD 21240–0757)

National Bureau Of Standards

Monograph 125	Thermocouples–Calibration Ref. 4.10.3.5.2.1

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office (GPO), Washington DC 20412)

U.S. Code

Title 10 US Code 2366	Major System and Munitions Programs: Survivability Testing and Lethality Testing Required Before Full–Scale Production (Live Fire Test Law) Ref. 4.6.2.1

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, North Capitol and H Streets, N.W., Washington DC 20402)

2.3 Non–Government publications The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONICS INDUSTRIES ASSOCIATION

EIA–455–20	FOTP–20 Measurement of Change in Optical Transmittance Ref. 4.7.7.1
EIA–455–21	Mating Durability of Fiber Optic Interconnecting Devices Ref. 4.7.7.2

(Application for copies should be addressed to Electronics Industries Association, 2001 Eye Street, NW, Washington DC 20006)

SOCIETY OF AUTOMOTIVE ENGINEERS

Aerospace Material Specification

AMS 2432	Short Peening, Computer Monitored Ref. 3.1.3.6

Aerospace Recommended Practice

ARP 755	Gas Turbine Engine Performance Station Identification and Nomenclature Ref. 3.2.1.3
ARP 1179	Aircraft Gas Turbine Engine Exhaust Smoke Measurement Ref. 4.6.1.4

ARP 1256	Continuous Sampling and Measurement of Gaseous Emissions from Aircraft Turbine Engines, Procedures for Ref. 4.6.1.5
ARP 1420	Gas Turbine Engine Inlet Flow Distortion Guidelines Ref. 3.2.2.10

Aerospace Standard

AS 568	Aerospace Size Standard for O–Rings Ref. 3.1.3.3
AS 681	Gas Turbine Engine Steady–State Performance Presentation for Digital Computer Programs Ref. 3.2.1.3
AS 870	Wrenching Configuration, Double Hexagon (12 point), for Threaded Fasteners Ref. 3.1.4.6

(Application for copies should be addressed to the Society of Automotive Engineers Inc., 400 Commonwealth Drive, Warrendale PA 15096; phone (412) 776–4841.)

TECHNICAL REPORTS

AEDC–TR–73–5	Handbook, Uncertainty in Gas Turbine Measurements Ref. 4.10.4.2.1

(Application for copies should be addressed to the National Technical Information Service (NTIS), 5285 Port Royal Road, Springfield VA 22161)

2.4 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. In the event of a conflict within this specification, the contractor shall notify the Using Service in writing for disposition. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2.5 Streamlining This specification has been streamlined. Appendix A lists those documents required for acquisition and is a mandatory part of this specification. Those documents listed in Appendix A have the same status as those referenced directly in section 2 (first tier documents). All other documents referenced through tiering, may be used as guidance and information.

3. REQUIREMENTS. The paragraphs of this section and all references herein comprise the complete set of engine requirements for the gas turbine engine(s) for the (a) airframe. Not withstanding the successful completion of the evaluation requirements of Section 4, the Contractor shall be fully responsible for all features, characteristics, and performance of the engine for all requirements in this section. All requirements that are classified shall be provided in a classified appendix, as part of this specification. Terms, symbols, acronyms, and definitions shall be in accordance with Section 6.

3.1 Design (configuration) and construction.

3.1.1 Item and interface definition. A brief description of the most notable design features of the engine shall be included in the model specification.

3.1.1.1 Item diagrams and schematics. The physical characteristics of the engine shall be described in top–level functional flow diagrams and schematics of the configuration item and include diagrammatic presentations to the level required to identify all essential functions. Copies of the diagrams and schematics shall be provided with the specification.

3.1.1.2 Item drawings Physical features of the engine, in all planes, shall be described in top level detailed dimensional drawings.

3.1.1.3 Interface and installation diagrams and schematics. Engine/airframe interfaces and engine/component accessory interfaces shall be illustrated in top level interface and installation diagrams. Copies of the diagrams and schematics shall be provided with the specification.

3.1.1.4 Interface and installation drawings. The interface and installation, in all planes, shall be shown on top level detailed dimensional drawings. The interface drawings shall include all physical installation requirements necessary for engine installation. The following drawings shall be provided with the specification.

a. Engine installation and interface drawing, figure 1a The drawings shall show mounting details and tolerances for the engine and all installation items, clearances for installation and removal of components and aircraft accessories subject to separate removal, access for adjustments and other maintenance functions and center of gravity of the complete engine, and center of gravity of the engine with fully loaded accessory interfaces. This drawing shall show the engine with fully loaded accessory interfaces and interface dimensions for the attachment of the inlet duct. This drawing shall show the maximum space required by the engine, including tolerances and dimensional changes due to manufacturing, thermal effects, vibration, operating and bending due to externally–applied loads.

b. Electrical installation and interface drawing, figure 1b This drawing shall show and identify external electrical circuit requirements and installation interface connection details for all engine systems.

c. Fiber optic installation and interface drawing, figure 1c This drawing shall show and identify external fiber optic circuit requirements and installation interface connection details for all engine systems.

3.1.1.5 Installation changes Engine features affecting engine installation interfaces shall be specified herein. Changes (additions, deletions, or modifications) shall be submitted to the Using Service for approval.

3.1.1.6 Installation hardware The engine shall be suitable for installation in any engine position in the aircraft without parts replacement except as approved by the Using Service.

3.1.1.7 Connections/interchangeability Connections located in close proximity to each other shall be made physically non–interchangeable.

3.1.1.8 Interface loads The maximum allowable loads and deflections, and their locations, for all controls, externals, and parts that interface with the airframe and are not specified elsewhere in 3.1.1, shall be specified herein.

3.1.1.9 Mounts

3.1.1.9.1 Main mounts The engine mounts shall retain the engine, including retained fluids and externals, at all flight, takeoff and landing, and ground conditions and shall withstand elastic limit loads of (a) without permanent deformation and ultimate tensile strength loads of (b) without complete fracture. A total of (c) mounts shall be provided which have sufficient strength to prevent (d) when subject to a single attachment point failure at any location at the end of the engine mount service life. The locations and descriptions of all engine mounts shall be as specified on figure 1a. The mount system shall accommodate all off–axis loads when a thrust vectoring nozzle is used.

3.1.1.9.2 Ground handling mounts The ground handling mounts shall support the engine, including all engine mounted equipment and externals, components, and operating fluids, under the following maximum inertia load conditions, without deformation to the mounts or damage to the engine: (a) axial, (a) lateral, and (a) vertical acting in combination at the engine center of gravity.

The locations and descriptions for the individual ground handling mounts shall be as specified on figure 1a. The arrangement shall be compatible with ground handling equipment specified herein by the Using Service.

3.1.1.9.3 Engine stiffness The estimated stiffness of the engine in resisting loads and moments applied at the outboard end of the output shaft, relative to the engine mounting points, shall be specified herein. The first "free–free" lateral and vertical engine bending modes shall be specified herein.

3.1.1.10 Inlet connection stresses The maximum allowable static (1g) shear and overhung moment loads at the engine inlet flange shall be specified herein. The maximum allowable loads at the engine inlet flange for the maximum allowable combined maneuver and air loads shall be specified herein. The maximum allowable maneuver loads shall be defined on figure 2. No damage shall occur to the engine inlet interface resulting from the maximum loads specified above, and from airframe deflections due to the externally applied forces of flight maneuver, takeoff, and landing environments.

3.1.1.11 Control system interface and integration The input/output from the airframe/engine shall be defined in table I.

3.1.1.11.1 Thrust/power demand The control system interface shall ensure compatibility between the airframe power demand output signal or load, and the control system.

For mechanical systems, the maximum allowable loads at the thrust/power lever connections shall be specified herein for the static (1g) axial, shear and overhung moment and for the maximum maneuver loads defined on figure 2.

3.1.1.11.1.1 Power lever angle (PLA) The control system interface shall receive and process airframe power lever output angles, movements and signals for fuel control input. Fuel control interpretation of power lever angle signals shall be defined in 3.7.2.1.

3.1.1.11.1.2 Power lever torque The control system interface shall be capable of receiving and processing airframe power lever torque signals for fuel control input.

For systems with a mechanical interface, the power lever torque required to operate the engine power lever through its range of travel shall not exceed 2.8 N–m (25 pound–inches) throughout all flight loads and conditions. The power lever shall not move with the engine operating unless external torque is applied. Movement of the power lever throughout the operating range shall be free of abrupt changes in actuating torque, and the maximum permissible variation shall not exceed 1.13 N–m (10 pound–inches). The maximum torque required to operate any additional levers shall not exceed 2.8 N–m (25 pound–inches). The maximum allowable static (1g) axial, shear and overhung moment loads at the power lever connections shall be specified herein. The maximum allowable loads at these connections shall also be specified for the maximum allowable maneuver loads. The power lever torsional loading limit between incremental stops and the maximum allowable loading when the power lever is against its travel stops shall be specified herein in Newton–meter (pound–inch) units.

3.1.1.11.2 Control signals The engine control system shall receive and transmit signals as required by the airframe, engine, or ground support systems. Control input and output signal requirements, input and output parameters, and related functions shall be specified herein. All control system signals required for the command and automatic control of the engine shall be transmitted and received in accordance with table I.

3.1.1.11.3 Load demand lever For turboshaft and turboprop engines, a means shall be provided at the control interface to adjust for rapid power absorber load changes. The control system interface shall provide for compatibility between the airframe load demand lever output signal and the control system. The load demand lever/signal shall be linear with power absorber load.

3.1.1.11.4 Output speed lever For turboshaft and turboprop engines, a shaft speed lever, if required, shall be provided to adjust and control the delivered shaft speed. The control system interface shall provide for compatibility between the airframe load demand lever output signal and the control system. The lever input shall be linear with governed output shaft speed.

3.1.1.11.5 Fuel shutoff lever For engines with thrust reversers, the fuel control shall process airframe signals, independent of the thrust/power demand signal, for fuel shutoff.

3.1.1.12 Electrical system interface

3.1.1.12.1 Instrumentation system interface The engine shall be instrumented to receive and transmit signals as required by airframe, engine, and ground support systems. The signal characteristics, physical interface, and instrumentation shall be shown in table II.

3.1.1.13 Engine monitoring system (EMS) interface The engine monitoring system shall receive and transmit signals as required by airframe, engine, and ground support systems. A description of the EMS interface shall be specified herein.

3.1.1.14 Fiber optics interface The fiber optics interface if applicable shall be described herein.

3.1.1.15 Bleed air interface The engine shall provide for customer bleed air extraction for aircraft use. The locations, number, and interface dimensions at all customer bleed air ports shall be shown on the engine interface and installation drawings. If required, acceleration bleed air shall be from the (a) stage of the compressor. The maximum allowable bleed airflows, pressures, and temperatures for each port, and the compressor stage(s) from which it is extracted shall be specified herein. The maximum permissible continuous flow capability of each bleed air port in percent of total airflow shall be specified herein. Bleed ducts with external surface temperatures exceeding (b) shall be insulated to prevent hazards from combustible fluid leakage. The bleed port internal pickup points shall be located at positions which have low susceptibility to FOD, and ingestion of sand, dust, ice, moisture, and any other foreign materials contained in the air. Provisions shall be made to prevent high pressure bleed air from entering lower pressure bleed ports. All bleed air ports shall be sized to prevent engine failure in the event of a failure in the aircraft bleed system. The engine provided customer bleed air extraction system shall ensure that no upstream malfunction of the engine will cause specified contamination limits to be exceeded. Where overboard ducting of starting and acceleration bleed air is necessary, the airflow conditions for which provisions must be made shall be specified herein.

3.1.1.15.1 Customer bleed air contamination Under normal operating conditions engine generated substances contained in customer bleed air shall be below the threshold limit of (a). Compressor bleed air extracted from the engine for customer use shall be free of engine generated noxious, toxic, or irritating substances and shall contain no engine generated objectionable odor.

3.1.1.16 Drains and fluid collection devices Fluid drains, drain characteristics, and collection requirements shall be shown on the interface and installation drawing. There shall be no leakage of fluids from any part of the engine except at the drains provided for this purpose. The flow rate into all drains shall not exceed (a) ml ((a) oz) for non–afterburning engine and (b) ml ((b) oz) for afterburning engines. The maximum amount of combustible fluid drainage following shutdown after normal operation shall be (c) ml ((c) oz).

3.1.1.17 Power absorber interface For turboshaft and turboprop engines, the allowable range of characteristics of the shaft power absorber at the power absorber to engine interface shall be completely defined herein. The characteristics shall include but not be limited to: maximum and minimum polar moment of inertia (slug–ft^2), torsional spring constant, torsional damping coefficient as a function of torsional spring constant, the maximum allowable static and dynamic loads on the engine output drive shaft, direction of rotation of the output drive shaft as viewed from the engine inlet, design of the output drive shaft spline and the power absorber shaft maximum misalignment during steady–state and transient operation. Output drive shaft spline bearing air leakage shall be quantified and accommodated in the design of the attached housing so that venting requirements can be determined and accommodated in the design. No resonant frequency shall be transmitted to or from the power absorber through the engine interface. Engine oil shall not be used for propeller lubrication and hydraulic systems.

3.1.1.18 Power take–off (PTO) When PTO pads and drives are provided for driving aircraft accessories, the ratings, clearance envelopes, dimensions, pad and connection details, direction of rotation, accessibility and alignment requirements shall be specified herein and on the engine interface and installation drawings. Customer power extraction limits shall be provided for each pad individually and in combination. The design and configuration of the PTO pads and drives shall be consistent with the requirements of 3.1.1.19.1 and 3.7.18.

3.1.1.19 Starting drivetrain The engine starting torque acceptance capability shall be specified herein and shall be at least (a) times the starting torque required to provide a (b) second acceleration from start initiation to minimum starter cutout speed, under the conditions specified in 3.1.1.19.1. The weakest part of the starting drive system (shear section, clutch, etc.) shall be external to the engine starting drive train. The starter drive pad characteristics shall be included in table III. The direction of rotation when facing the starting pad on the engine shall be specified herein. No resonant frequency shall be transmitted to or from the starting drivetrain through the engine interface.

3.1.1.19.1 Starting torque and speed The required starting torques and drive speeds shall be specified herein and shall include engine drag and engine gearbox drag. Figure 3 shall show the effects, singly and in combination, of ambient temperatures, of altitudes from sea level to the maximum of 3.7.11.3, and for air starts with starter assist, using the worst case fuels of the specification. Figure 3 shall be presented for each of the following:

a. No customer bleed air extraction, no customer power extraction.

b. Maximum allowable customer bleed air extraction, no customer power extraction.

c. No customer bleed air extraction, customer power extraction as specified in the model specification.

d. Maximum allowable customer bleed air extraction, customer power extraction as specified in the model specification.

3.1.1.19.2 Moment of inertia of rotating parts The maximum effective mass moment of inertia (kilogram – meter squared (slug – ft^2)) of engine rotating parts to be rotated by the starter, at the starter drive pad, and the speed ratio between the starter pad and the driven rotor system shall be specified herein.

3.1.1.19.3 Torsional spring constant The torsional spring constant (newton – meter per radian (pound inches per radian)) for the engine starting drive system at the starter drive pad shall be specified herein.

3.1.1.19.4 Starter train backlash The maximum backlash, in radians, of the starting drive system at the starter drive pad shall be specified herein.

3.1.1.20 Thrust reverser interface Interface dimensions required for thrust reverser nacelle installation, clearances, and connections shall be shown on the engine installation and interface drawing. The reverser mechanism shall provide a signal indicating whenever the reverser is in the unlocked condition. Positive mechanical locking, described on the installation drawing, shall prevent reverser deployment in the event of system failure or inadvertent actuation of the control system. Failure of the locking system shall not cause reverser deployment. No resonant frequency shall be transmitted to or from the thrust reverser interface.

3.1.1.21 Exhaust system interface Maximum allowable static (1g) shear, axial and moment loads in all primary axes (x, y, z, as applicable) for any interface connections between the engine and parts of the exhaust system not supplied with the engine shall be specified on figure 5. The maximum allowable loads at any exhaust interface shall also be specified for the maximum allowable maneuver loads as defined in 3.4.1.2.2. No hot exhaust gas leakage is allowed at the interface. Resonant frequencies transmitted to or from the exhaust system through the interface shall not prevent the engine from meeting the requirements of the specification.

3.1.1.22 Compatibility – engine/airframe The interface of the engine with the airframe shall allow the engine to perform as required in the specification.

3.1.1.23 Control and external (C&E) components list The control, external components, and component functional subsystems of the engine, which require qualification testing separate from the engine, shall be listed, see 4.9.2.1.6.

3.1.1.24 Government furnished/loaned property No government furnished/loaned property shall be incorporated in the engine design.

3.1.2 Physical characteristics

3.1.2.1 Dry mass (weight) of complete engine The dry mass (weight) of the complete engine, but without the engine exhaust system, shall not exceed (a) kg ((a) lbm). The engine supplied exhaust system shall not exceed (b) kg ((b) lbm). The weights of engine components which are not mounted on the engine shall be listed and included in the dry weight of the engine. The engine station where the engine center of gravity (cg) occurs shall be specified herein.

3.1.2.2 Mass (weight) of residual fluids The mass (weight) of residual fluids remaining in the engine after operation and drainage, and the drainage attitude of the main rotor axis relative to the level plane, shall be specified herein.

The engine station where the engine center of gravity (cg) occurs with the specified fluid conditions shall be specified herein.

3.1.2.3 Mass (weight) of operating fluid The total mass (weight) of fluids in an operating engine shall be specified herein. The engine station where the engine center of gravity (cg) occurs with the specified fluid conditions shall be specified herein.

3.1.2.4 Mass (weight) of additional equipment The weights of items which are not a part of the engine bill of materials, but which are installed on the engine by the engine manufacturer, shall be listed below:

Item	Weight

The engine station where the engine center of gravity (cg) occurs with the specified equipment shall be specified herein.

3.1.2.5 Mass moment of inertia of complete engine The maximum effective mass moment of inertia (slug – ft^2) of the complete operational engine about three mutually perpendicular axes with the origin at the center of gravity, shall be specified herein. The maximum effective mass moment of inertia (slug – ft^2) about the resultant rotational axis of each engine rotor, together with the direction of rotation when viewed looking forward from the exhaust exit, shall also be specified herein.

3.1.3 Materials, processes, and parts When the engine manufacturer's documents are used for materials and processes, such documents shall be subject to review by the Using Service prior to the start of IFR and, unless specifically disapproved, will be considered released upon approval of FFR, ISR, and OCR. The use of

non–governmental documents shall not constitute waiver of Government inspection. The Using Service reserves the right to inspect any and all processes of manufacture. The use of magnesium and silver shall not be permitted, unless approved by the Using Service. Copper, brass nickel, and cadmium shall not be used in engine parts which are in direct contact with fuel or oil. Dissimilar metal as defined in MIL–STD–889 shall not be used in direct contact with each other. An appropriate surface treatment and finishing system shall be selected from Appendix A of MIL–STD–889 to protect these metals from galvanic corrosion when joined and subjected to the specific environment.

Materials, components and parts shall perform satisfactorily during and after exposure to fuel.

3.1.3.1 Adhesives and sealants The use of adhesive or sealant compounds is permitted only upon specific application approval by the Using Service prior to its incorporation into the design of the engine.

3.1.3.2 Elastomeric materials Elastomeric materials shall have an unlimited shelf life (non–age sensitive) and, on an operational engine, have a minimum of (a) engine life. Materials exposed to fuels and lubricants shall be compatible with such fluids throughout the entire fuel or lubrication system temperature cyclic envelope without experiencing swelling, shrinking or other forms of material deterioration which would impair proper functioning or necessitate replacement to prevent impairment of function. Polychloroprene materials (e.g. neoprene) shall not be used in engine parts which are in direct contact with fuels or lubricants during engine operation.

3.1.3.3 "O" rings, seals, and packings All nonmetallic "O" rings, seals, and packings used in the engine, including vendor supplied components, shall conform to the applicable dimensions and tolerances shown on AS 568. Materials exposed to fuels and lubricants shall be compatible with such fluids throughout the entire fuel or lubrication system temperature cyclic envelope without experiencing swelling, shrinking, or other forms of material deterioration which would impair proper functioning or necessitate replacement to prevent impairment of function. Polychloroprene materials (e.g. neoprene) shall not be used in engine parts which are in direct contact with fuels or lubricants during engine operation.

3.1.3.4 Corrosion protection The materials, coatings, and processes used in the design and manufacture of the complete engine shall be corrosion resistant.

3.1.3.5 Nonmetallic hose Nonmetallic hose shall be in accordance with MIL–H–27267. Hose assemblies shall comply with MIL–H–25579. Hose carrying flammable fluid shall comply with 3.1.8.1.

3.1.3.6 Shot peening For parts dependent upon the residual stresses from shot peening to attain required fatigue life, the following requirements shall apply:

a. Manufacture shall be by computer monitored and controlled shot peening. SAE AMS 2432 shall be the controlling document. Control limits which shutdown the process prior to exceeding critical parameters and documentation of process records (including any 100 percent inspections imposed as a result of process control discrepancies) shall be required.

b. Peening required during repair or rework to attain fatigue life shall be by computer controlled and monitored processes or shall conform to the same specifications and standards used for original manufacture.

3.1.4 Fasteners Fasteners shall conform to the following specifications or standards: (a) . At all joints in control systems, at single attachments, and where the loss of bolts affects safety of flight or ability to control the engine, self–retaining bolts shall be used. Installation and retention of self–retaining bolts (i.e., a slotted or castellated nut) shall be in accordance with MS33602.

3.1.4.1 Securing of fasteners Where cotter pins are used to secure fasteners, the practices in MS33540 shall be followed. The general design and usage limitation of self–locking nuts specified in MS33588 shall be applied. Safety wire is acceptable only on areas of the engine which are not involved in intermediate or lower level of maintenance. Safety wire practice shall conform to MS33540.

3.1.4.2 Clamps Tube support clamps shall be subject to approval by the Using Service.

3.1.4.3 Screw recesses All screw recesses (internal drives) shall conform to MS33750 or MS9006.

3.1.4.4 Screw threads All threads shall conform to the requirements of MIL–S–8879 Classes 3A or 3B. Duplicate parts differing only in thread form are not permitted.

3.1.4.5 Helical coil installation The dimensions and tolerances of the parent material threads intended for use with helical coil inserts shall comply with MS 33537. The installation and use of helical coils shall not cause reduction in life of the affected components.

3.1.4.6 External wrenching elements Wrenching elements for external drives shall conform to AS 870 for twelve point drives, MS 33787 for spline drives.

3.1.5 Nameplate and product marking Equipment, assemblies, modules, and parts shall be marked in accordance with MIL–STD–130.

The engine shall be permanently marked to indicate all connections shown on the engine installation drawing.

3.1.5.1 Engine data plate marking The engine data plate shall include: (a) manufacturer's identification, (b) engine serial number, (c) purchase order or contract number, (d) engine model designation, and (e) warranty notification/information.

3.1.5.2 Warranty marking Each warranted engine, major component, module, and accessory shall have the word "warranted" permanently and clearly imprinted on the nameplate.

3.1.5.3 Part number marking Part number marking shall be in accordance with MIL–STD–100.

3.1.5.4 Critical parts identification and tracking The engine life critical parts/subassemblies shall be identified by serial numbers marked legibly on the part. The parts shall have multiple, non–wearing reference surfaces and have a space designated for marking the number of cycles and time accumulated between each overhaul period.

3.1.6 Transportability The engine shall be suitable for transportation as follows: (a) . Adequate ground handling pads and other features shall be provided to permit installation on and use of appropriate static and mobile ground equipment.

3.1.7 Interchangeability All parts and modules having the same manufacturer's part number shall be functionally and dimensionally interchangeable and shall not degrade engine performance, operability, and durability. Matched parts or selective fits shall be permitted upon Using Service approval. Complete engines shall be interchangeable in multi–engine aircraft (i.e., right side to left side).

3.1.8 Safety System safety program requirements shall be developed and implemented for the engine design, development, tests, production, and operation in accordance with MIL–STD–882.

3.1.8.1 Flammable fluid systems All exterior lines, fittings and components which contain flammable fluids shall be fire resistant (five minutes at 1093°C (2000°F), except that the lubrication oil system and hydraulic system components shall be fire proof (15 minutes at 1093°C (2000°F)). During exposure to the above conditions, the lines and components shall be operating under the worst possible combination of fluid parameters (i.e., flow rate, pressure and temperature) encountered throughout the complete environmental conditions and operating envelope of the engine.

3.1.8.2 Fire shields A fire shield shall be provided to prevent flammable fluid impingement on the engine. The type of attachment, location, dimensions and allowable loads at the interface shall be shown on the engine installation and interface drawing.

3.1.8.3 Explosion–proof All electrical components (except ignitor plug electrodes and devices intended to ignite fuel air mixtures) shall not ignite any explosive mixture surrounding the equipment.

3.1.8.4 Combustible fluid drains Provisions shall be made for automatically clearing the combustion areas of combustible fluids after each false start and for preventing excess combustible fluids from entering the combustion areas after shutdown with the engine in 15 degrees nose up, 20 degrees nose down, and level positions. For engine applications in V/STOL aircraft requiring engine attitude changes, the drain limits of 3.1.1.16 shall not be exceeded for combustible fluids after engine shutdown in a 105 degrees nose up, 20 degrees nose down, and level position. Provisions shall also be made for clearing all vent areas and other pockets or compartments where combustible fluids may collect during or subsequent to operation of the engine. The engine shall be designed so that combustible fluids cannot enter the combustion area when the engine is in, or turned to, a vertical position for maintenance. The maximum allowable quantity of combustible fluid which will discharge from the engine drains after shutdown from normal operation and, if applicable, from augmented operation shall be specified in the model specification.

Overboard fuel leakage, following augmentor operation during flight, shall be controlled at a rate of flow which prevents visual detection of the vapor (see 3.6.1.6). The fluid drain system shall be configured to collect/route the leakage of fuel and oil to a central location, to facilitate detection, and to isolate leakage in excess of the limits of 3.1.1.16.

3.1.8.5 Air and gas leakage The location, amount, temperature and pressure of engine air and gas leakages shall be specified in the model specification. There shall be no leakage of sufficient temperature and concentrated impingement to present a safety hazard or affect installation requirements.

3.1.8.6 Ground safety Warning notices shall be provided, where applicable, for high voltage ignition systems and other high voltage electrical sources, radioactive devices, explosive devices, and high powered optical sources.

3.1.8.7 Jet wake Jet wake temperature and velocity profiles at sea level static standard day conditions when operating at Idle, Intermediate, and Maximum shall be as shown on figure 7. Areas hazardous to personnel and equipment shall be appropriately defined and marked on figure 7.

3.1.9 Design control

3.1.9.1 Standardization Standardization principles, standard parts, materials, processes, tools, subsystems, and components shall be used to the maximum extent possible without compromise in design, performance, operability, or economic life of the engine. All parts, materials, and processes, whether or not identified as a Government, industry, or contractor standard shall be qualified for the intended use as a part of the qualification specified herein. Items already in the Government inventory shall be used to the maximum extent possible where suitable for the intended purpose. Variation in similar components or parts shall be held to the absolute minimum. Proprietary designs shall be kept to a minimum. Under conditions wherein economics of production conflict with standardization objectives, the Using Service shall be requested to select the component desired for use.

3.1.9.2 Design standards Military and industry standards developed specifically for aircraft engines, such as AND, AS, MS, NAS, etc., shall be used unless they are determined by the engine manufacturer to be unsuitable for the intended purpose, subject to Using Service approval.

3.1.9.3 Standard parts The military and industry standard parts developed specifically for use in aircraft engines shall take precedence over any other military standard parts. Where general purpose standards, as defined by envelope dimensions or Qualified Products List (QPL), are used in critical or high strength applications, they shall be identified by the vendor or engine manufacturer's part number. Parts derived from general purpose standards solely on an inspection or selection basis shall be identified by manufacturer parts numbers and all previous identification marks shall be removed.

3.1.9.4 Parts list The parts list for the engine which successfully completes the designated engine milestone evaluations shall constitute the approved parts list for any subsequent engines used for evaluation in that milestone. Changes to the approved parts list shall require approval by the Using Service prior to parts incorporation.

3.1.9.5 Assembly of components and parts Equipment, parts, and components which are not structurally or functionally interchangeable, shall not be physically interchangeable. Parts and components shall be designed such that it is impossible to install them incorrectly, e.g., backwards, upside down, reversed in an assembly, or installed in the wrong location in an assembly. Connections located in close proximity to each other shall be made physically non–interchangeable.

3.1.9.6 Changes in vendors or fabrication process Changes in any vendor, fabrication process, or fabrication source for any component or part shall be in accordance with MIL–STD–1529.

3.2 Performance and operability

3.2.1 Performance characteristics The engine performance characteristics defined in the specification shall be for a minimum engine. Minimum engine performance as defined in 6.4, shall be specified for both new engines and engines deteriorated for the time period specified in 3.2.1.4. Unless otherwise specified, the engine performance characteristics shall be based on:

a. A fuel having a lower heating value of (a) BTU/lb, and otherwise conforming to the fuel specified in 3.7.3.1.1 and oil specified in 3.7.8.1.

b. US Standard Atmosphere (geopotential altitude) (1976).

c. No inlet air distortion.

d. An inlet pressure recovery as defined by the following equations and conditions:
Pt2(ref)/PtO = 1.0, from 0.0 to 1.0 Mach Number.
$Pt2(ref)/PtO = 1.0 – 0.075(M – 1)^{1.35}$ from 1.0 to 5.0 Mach Number.
$Pt2(ref)/PtO = 800/(M^4 + 935)$, above 5.0 Mach Number.
Pt2(ref) = referenced total pressure at the compressor inlet.
PtO = free stream total pressure.
M = flight Mach Number.
For turboshaft applications that incorporate an Inlet Particle Separator (IPS) as part of the engine, all performance shall include the effects of the IPS.

e. The designated exhaust pipe and jet nozzle, fan exit duct and fan nozzle, engine front frame and rear frame features.

f. No customer bleed air extraction.

g. No customer power extraction.

h. Using the engine control system specified in the specification. The specified performance shall be predicated on the tolerance of control system variation which produces poorest performance.

i. Non–standard atmospheres as shown on figure 4.

j. Zero humidity.

k. For turboshaft/turboprop engines, the shaft torque absorber characteristics specified in 3.1.1.17 and 3.7.18.

l. Stall margin/operating line consistent with the requirements of 3.2.2.10 when matched with vehicle flight exhaust. No distortion indicator signal input.

3.2.1.1 Performance and presentation – steady–state The minimum engine steady–state performance requirements shall be defined by the steady–state performance computer program (a) dated (a), for the engine operating envelope of 3.2.2.1, in accordance with the conditions of 3.2.1. Cycle calculations for the minimum engine

computer program shall be based on an average engine with thrust/power and fuel flow related output adjusted to minimum engine performance corresponding to ± (b) . The relationship between minimum and average performance shall be specified throughout the flight envelope. The specification computer program(s) shall calculate minimum performance in accordance with the requirements of 3.2.1. Tables Ia, Ib, Ic, and II shall provide performance points from the performance computer program for new and deteriorated engines. Tables Ia, b and c shall specify performance points at sea level static for the following atmospheric temperatures: (c) . Table II shall specify performance at selected altitude, Mach number and inlet temperature conditions. In the event of a conflict between tables Ia, Ib, Ic, and II and the performance computer program, the computer program shall take precedence. Tables Ia, Ib, Ic, and II shall be presented in both SI (metric) and US Customary (English) units.

Performance shall be specified in the program for the following conditions, singly and any combination: (d) .

Steady–state air flow values for any specific flight condition and principal station properties shall be defined in the computer program. The average steady–state performance shall be specified in the average/status steady–state performance computer program, which shall be updated periodically throughout the development program, as determined by the Using Service.

3.2.1.2 Performance and presentation – transient. The minimum engine transient performance requirements shall be defined by the transient performance computer program (a) dated (a) , for the engine operating envelope of 3.2.2.1, in accordance with the conditions of 3.2.1. Engine thrust transient times from the transient computer program shall be specified in 3.2.2.6, as required therein. Performance and thrust transient times shall be specified in the program for the following conditions, singly and in any combination: (b) . The specification computer program(s) shall calculate minimum performance in accordance with the requirements of 3.2.1.

Transient air flow values for any specific flight condition and bleed/power extraction levels shall be defined in the computer program. The average engine transient performance shall be specified in the average/status transient performance computer program which shall be updated periodically throughout the development program, as determined by the Using Service. The minimum and average/status transient performance shall be specified in (c) cycle deck(s). Engine performance with installed effects shall be included in all transient performance computer programs.

3.2.1.3 Performance computer program Minimum and average/status steady–state and transient performance computer programs shall be constructed in accordance with AS 681. These computer programs shall form a part of the model specification and shall carry a suitable identification and date, including the engine model designation. Compilation and execution of the computer program shall be demonstrated on the (a) computer prior to acceptance by the Using Service. The model specification shall contain a list of symbols in Section 6 and a diagrammatic figure defining station identifications. These symbols and station identifications shall be in accordance with ARP 755. The specification computer program(s) shall calculate minimum performance in accordance with the requirements of 3.2.1.

3.2.1.4 Performance retention Unless otherwise specified, during and at the completion of one cold parts life, the engine thrust/power shall not fall below the minimum engine thrust/power requirements specified in 3.2.1.1 and 3.2.1.2. During and at the completion of any hot section life, the engine thrust/power shall be no less than (a) percent of the actual thrust/power at the beginning of the hot section life at (b) temperature. The specific fuel consumption (SFC) from the lowest part power performance point of table II to sea level static intermediate/maximum continuous thrust/power shall be no more than (c) percent of the actual SFC at the same thrust/power at the beginning of the hot section life.

3.2.1.5 Performance evaluation The performance requirements and presentation of 3.2.1.1 and 3.2.1.2 shall be used as the basis for evaluating engine performance throughout the complete environmental conditions and operating envelope. All steady–state and transient performance data shall be compared to the steady–state and transient minimum performance computer programs of 3.2.1.1 and 3.2.1.2 respectively, at the “as tested” condition. In the

event the engine fails to meet the above success criteria because of catastrophic failures, discretionary engine shutdowns, A/B flameouts, parts replacement, etc., not called out in pre–test data, penalty running requirements shall be established by the Using Service. For test running in contractor test cells, a cell factor(s) shall be applied to test data to account for cell effects. The Using Service shall have approval authority for the cell factor(s).

3.2.2 Operating characteristics and limits The engine operating characteristics and limits shall be as specified in the following paragraphs. All engine steady–state and transient operating limits (maximum, minimum) shall be specified in table VIII and figure 9. The specified limits shall be predicated on the most critical tolerances of the engine.

3.2.2.1 Operating envelope The engine shall meet all the requirements of the specification throughout the complete operating envelope without exceeding any limits. The engine operating limits shall be specified for the environment of figure 4, in accordance with the format shown in figures 9, 10, tables Ia, Ib, Ic, and II. If applicable, the thrust augmentation operating envelope shall be included in figures 9 and 10.

3.2.2.2 Operating attitude and conditions The engine operating attitude limits shall be shown on figure 10. The engine shall meet the requirements of the specification when operating in the unshaded area of figure 11, and operate at least (a) seconds continuously in the shaded areas of figure 10. Operation in the shaded area shall not degrade engine performance or cause any damage. The engine shall start, stop, and be stowed in any of the attitudes shown in the unshaded area figure 10. Engine stowing capability outside of the unshaded area of figure 11 shall be specified herein. The engine shall function satisfactorily for at least (b) seconds in negative g and for at least (c) seconds in zero g conditions.

3.2.2.3 Stopping Upon receiving a signal from the airframe for stopping, the engine shall terminate fuel flow in any operating condition. Stopping of the engine from any power setting (including afterburner) or at any rate shall not a) result in immediate or subsequent exceedance of any engine limits, b) adversely impact engine durability, structural integrity, or operational capability, c) cause rotor bowing or rotor seizure, and d) delay a satisfactory start or afterburner light. The engine shall not experience any post–shutdown fires. No damage to the engine shall result from shutting off the fuel supply by the foregoing means or from shutting off the fuel supply to the engine inlet connection during any engine operating condition. Provisions for stopping the engine by means other than a completely mechanical system shall be subject to approval of the Using Service. All methods for stopping the engine, including emergency provisions, shall be specified herein.

3.2.2.4 Low power conditions

3.2.2.4.1 Idle thrust/power – [ground/flight idle]. Idle shall be the minimum self–sustaining condition of the engine. With the thrust/power demand in the Idle position and with no customer bleed air or power extraction, the thrust/power shall not exceed (a) percent of the Intermediate power available at standard day conditions up to (b) feet. The variation in Idle power, bleed pressure, and HP rotor speed with temperature, altitude, and installation factors shall be as defined in the performance computer program of 3.2.1.1. When the engine incorporates provisions for an additional Idle power setting (i.e., ground Idle), the means of accomplishment, the thrust level, the corresponding rotor speed and the time required to change Idle thrust shall be specified herein.

3.2.2.4.2 No–load condition For turboprop/turboshaft engines, the no–load condition shall be an engine operating condition of zero delivered output shaft torque while maintaining constant steady–state governed output shaft speed with a specified power absorber and operating at altitudes up to (a) km ((a) ft). Engine gas generator speed at no–load shall be specified in the format shown on figure 12.

The engine(s) and torque absorber system shall be capable of transitioning between load and no–load conditions without unacceptable mismatches in rotor speeds or torques defined in 3.2.2.6.2, and without exceeding gas generator operating limits.

3.2.2.5 Stability The engine shall operate satisfactorily during transient operation from one thrust/power setting to another without unstable operation, stall, surge, flameout, A/B instability, or blowouts. Under steady–state operating conditions, engine thrust/power fluctuations shall not exceed (a) percent of the Intermediate rated power (IRP)/Maximum Continuous between Idle and IRP/Maximum Continuous conditions, or (b) percent of the thrust/power available at the power level position and operating condition, whichever is less. During operation above IRP up to Maximum A/B, fluctuations shall not exceed (c) percent of the thrust/power available at that condition. During steady–state operating conditions, the period between major discrete frequency fluctuations, as defined above, shall not be less than (d) seconds. Main combustor and afterburner blowout margins shall be sufficient to protect against blowouts or flameouts during rapid throttle movements under worst case conditions, and engine control and main burner and afterburner stack–ups. Those frequency fluctuations which approach the amplitude or exceed the frequencies specified above shall not be permitted if, in the judgement of the Using Service, they indicate an instability. Stability audits of the engine component system shall be performed to identify critical margin stability areas throughout the complete environmental conditions and operating envelope of the engine. The stability audit methodology shall be specified herein. In addition, the stability audit shall be updated as necessary throughout the development program. Bleed air extraction shall not require thrust/power demand modulation to maintain engine stability and limits. Continuous sinusoidal throttle transients with peak–to–peak thrust/power change demands of (e) % with a frequency up to (f) cycle per second shall not result in a divergent sinusoidal thrust/power response. Thrust/power response phase lag shall be no more than (g) degrees and thrust/power peaks shall be at least (h) % of demand. Symmetrical throttle demands shall not result in divergent symmetrical throttle responses.

3.2.2.6 Transients The engine shall satisfactorily perform any thrust/power transient throughout the operating envelope while meeting the requirements of the specification. Thrust/power requests in any sequence and at any rate shall not result in exceeding any engine operating limit (including overspeed and overtemperature), result in unstable operation of the engine, or cause any mechanical damage. Thrust transient times in all backup control modes shall be the same as those in the primary mode. Thrust transient times for installed engine conditions, including the effects of customer bleed air, customer power extraction, anti–icing bleed, inlet distortion, and the exhaust nozzle, singly or in any combination, for non–standard days throughout the complete environmental conditions and operating envelope, shall not exceed (A) percent of the standard day thrust transient times.

For thrust/power lever movements of 0.5 seconds or less, the transient times below shall be the slowest thrust/power transient times for standard day conditions with: no customer power extraction, no customer bleed air extraction and no engine anti–icing bleed air operation, but with all other engine system bleed air requirements (e.g., acceleration bleed air, cooling bleed air). The time required for the engine to accomplish (B) percent of the thrust/power change shall be the time specified in the transient performance computer program and listed below. The total time required to accomplish the full transient and reach transient stability shall be the time in transient performance computer program and presented below plus (C) seconds. Actual engine acceleration and deceleration time shall be within +0 seconds and (D) seconds of the acceleration and deceleration time specified in the transient performance computer program and below. In the event of a conflict between the transient times specified in the transient performance computer program and those listed below, the times specified in the transient performance computer program shall take precedence.

The standard day engine transient times shall be as follows:

From sea level up to 10,000 feet and 0 to 150 knots (0.0 to 0.25 Mach number) indicated airspeed (IAS):

a. From Idle to Intermediate thrust available (a) seconds. From Intermediate to Idle thrust available, (a) seconds.

b. From Idle to Maximum thrust available, (b) seconds. From Maximum to Idle thrust available, (b) seconds.

c. From Idle to Maximum reverse thrust available, (c) seconds. From Maximum reverse to Idle thrust available, (c) seconds.

d. From 30 percent Intermediate to Intermediate thrust available, (d) seconds. From Intermediate thrust to 30 percent Intermediate thrust available, (d) seconds.

e. From 10 percent Intermediate to 20 percent Intermediate thrust available, (e) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

f. From 20 percent Intermediate to 10 percent Intermediate thrust available, (f) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

g. From 20 percent Intermediate to 30 percent Intermediate thrust available, (g) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

h. From 30 percent Intermediate to 20 percent Intermediate thrust available, (h) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

i. The thrust versus time from Idle and 10, 20, 30, 40, and 50 percent Intermediate to Intermediate thrust and to Maximum thrust shall be presented in the specification as shown in (i).

j. The thrust versus time from Idle and 10, 20, 30, 40, and 50 percent Maximum to Maximum thrust shall be presented in the specification as shown in (j).

From 10,000 feet up to the absolute altitude:

k. From Idle to Intermediate thrust available, (k) seconds. From Intermediate to Idle thrust available, (k) seconds.

l. From Idle to Maximum thrust available, (l) seconds. From Maximum to Idle thrust available, (l) seconds.

For all operating conditions (throughout the complete operating envelope):

m. From Intermediate to Maximum thrust available (m) seconds. From Maximum to Intermediate thrust available (m) seconds.

n. From Intermediate to Maximum reverse thrust available (n) seconds. From Maximum reverse to Intermediate thrust available (n) seconds.

o. From Maximum to Maximum reverse thrust available (o) seconds. From Maximum reverse to Maximum thrust (o) seconds.

TURBOPROP/FIXED WING

The standard day engine power (turboprop) transient times shall be as follows:

Sea level up to 10,000 feet and 0 to 150 knots (0.0 to 0.25 mach number) IAS:

a. No load to Maximum Continuous/Intermediate power available, (aa) seconds. From Maximum Continuous/Intermediate power to no load (aa) seconds.

b. From Idle to Maximum power available (bb) seconds. From Maximum to Idle power available, (bb).

c. From Maximum Continuous/Intermediate power to full reverse power available, (cc) seconds. From full reverse power to Maximum Continuous/Intermediate power available, (cc) seconds.

d. From 30 percent Maximum Continuous/Intermediate to Maximum Continuous/Intermediate power available (dd) seconds. From Maximum Continuous/Intermediate power to 30 percent Maximum Continuous/Intermediate power available, (dd) seconds.

e. From 10 percent Maximum Continuous/Intermediate to 20 percent Maximum Continuous/Intermediate thrust available, (ee) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

f. From 20 percent Maximum Continuous/Intermediate to 10 percent Maximum Continuous/Intermediate thrust available, (ff) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

g. From 20 percent Maximum Continuous/Intermediate to 30 percent Maximum Continuous/Intermediate thrust available, (gg) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

h. From 30 percent Maximum Continuous/Intermediate to 20 percent Maximum Continuous/Intermediate thrust available, (hh) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

i. The power versus time from Idle and 10, 20, 30, 40, and 50 percent Maximum Continuous/Intermediate to Maximum Continuous/Intermediate power and to Maximum power shall be presented in the model specification as shown in (ii) .

From 10,000 feet to absolute altitude:

j. From Idle to Maximum Continuous/Intermediate power available, (jj) seconds. From Maximum Continuous/Intermediate to Idle power available, (jj) seconds.

For all operating conditions:

k. From Maximum power to full reverse power available, (kk) seconds. From full reverse power to Maximum power available, (kk) seconds.

l. From full reverse power to Maximum Continuous/Intermediate power available, (ll) seconds. From Maximum Continuous/Intermediate power to full reverse power available (ll) seconds.

m. From Maximum power to Maximum Continuous/Intermediate power available (mm) seconds. From Maximum Continuous/ Intermediate to Maximum power available (mm) seconds.

TURBOSHAFT/HELICOPTER

For turboshaft engines in helicopters applications, the following requirements apply:

From Sea Level to 3 km (9843 ft), and Mach 0.0 to 0.2:

n. From Idle to Maximum Power, (nn) seconds.

o. From no load to Maximum Power, (oo) seconds.

p. From Maximum Power to no load, (pp) seconds.

From 3 km (9843 ft) to maximum altitude, and Mach 0.0 to 0.2:

q. From no load to Maximum Power, (qq) seconds.

r. From Maximum Power to no load, (rr) seconds.

3.2.2.6.1 Overshoot and undershoot The engine thrust overshoot following any acceleration demand shall not exceed (a) percent of the final stabilized steady–state thrust value. The engine thrust undershoot following any deceleration demand shall not exceed (b) percent of the final stabilized steady–state thrust value.

For turboprop/turboshaft engines, power, output shaft rotor speed and torque shall not exceed (c) percent of the final stabilized steady–state value. During the transient to the stabilized value, output shaft speed and torque shall not exceed the limitations specified in 3.2.2.6.2.

3.2.2.6.2 Droop During any acceleration, no thrust/power droop below (a) percent of the demanded thrust/power value shall occur between the attainment of (b) percent thrust/power change and achievement of transient stability as defined by 3.2.2.6. In addition, there shall be no more than a (c) percent thrust/power droop from the point of transient stability up to attainment of steady–state stability defined in 3.2.2.5.

For turboprop/turboshaft engines, output shaft speed shall not deviate more than ±3 percent from the governed value for any load change not completed within two seconds. During the transient, shaft torque shall not exceed (d) percent of the steady–state operating limit.

3.2.2.7 Windmilling The engine shall be capable of continuous windmilling for (a) hours, without damage to the engine, and without affecting air restart and operational capability. The following information shall be specified (for an uninstalled engine):

a. The limits of windmilling operation, including curves showing windmilling rotor speed and windmilling drag vs. flight Mach number.

b. The oil consumption rate during windmilling operation.

c. In the event of lubrication failure, the time duration and limits of windmilling operation after depletion of oil supply.

d. The power extraction available during windmilling operation as shown on figure 15a.

e. The customer bleed air extraction available during windmilling operation as shown on figure 15b.

3.2.2.8 Reverse thrust For turbofan and turbojet engines, the operating envelope, limitations, and description for thrust reverser operation shall be specified herein. The percentage of total engine thrust used for reverse thrust shall be specified herein.

3.2.2.9 Steady–state and transient airflow The engine maximum allowable steady–state and transient airflow limits shall be shown in table VIII. The transient airflow limits specified shall include augmentor light–off and termination, as applicable, as well as thrust/power demand increase and decrease. The steady–state airflow of any engine shall not vary by more than (a) percent of the steady–state airflow computed by the steady–state computer program of 3.2.1.1 and shown in tables Ia, Ib, Ic, and II. The transient airflow of any engine shall not vary by more than (b) percent of the transient airflow computed by the transient computer program of 3.2.1.2.

Transient airflow shall not deviate from the maximum steady–state airflow by more than (c) percent and shall be controlled to assure compatible engine inlet operation. The engine steady–state and transient airflow limits shall include engine deterioration, control tolerances (including temperature lags and transients), engine–to–engine variations and shall apply to the non–standard atmospheres of this specification as shown on figure 4. The effects of customer air bleed and power extraction shall be identified.

3.2.2.10 Inlet airflow distortion The engine shall not surge, stall, flameout, or incur any damage with the steady–state or time variant inlet distortion (pressure, temperature, or any combination of both) shown on figure 16. The (a) most severe distortion conditions permitted within the operating envelope shall be identified on figure 16.

The aerodynamic interface plane shall be defined as (b) . An engine stability and performance assessment shall be performed using the methodology and inlet distortion descriptors defined in ARP 1420 for total pressure distortion. An engine stability and performance assessment shall also be performed for inlet temperature distortion and for the combination of both pressure and temperature distortion. The temperature distortion methodology and the methodology for combination of both pressure and temperature distortion shall be specified. The effects of inlet airflow distortion on engine performance shall be included in the performance computer programs of 3.2.1.1 and 3.2.1.2.

3.2.2.10.1 Pressure and temperature rate of change The engine shall withstand the following rate–of–change of pressure and temperature, separately or in combination, without surge, stall, flameout, or damage:

Parameters	Rate of Change	Duration
Engine inlet pressure	(a) kPa/sec (psi/sec)	(b) sec
Engine inlet temperature	(c) °C/sec (°F/sec)	(d) sec

3.2.2.11 Gas–path and measurement plane temperature limits

a. The first stage high/low pressure turbine rotor inlet gas temperature limit(s) shall be as follows: (a) .

b. Measurement plane temperature. The maximum allowable steady–state gas temperature averaged over the gas path area at the measuring plane and corresponding to the maximum allowable steady–state gas temperature at the gas generator first stage turbine rotor inlet shall be specified herein, along with the location of the measurement plane (including station identification number). No engine shall operate above the maximum allowable measuring plane turbine temperature. This temperature may be changed prior to the initiation of IFR, FFR, ISR, and OCR based on correlation studies during the development program of the relationship between measured temperature and gas temperature at the measurement plane. The maximum allowable transient measured temperature during starting and acceleration shall also be specified herein. When a direct measuring metal temperature sensing system is provided at the measurement plane, the maximum allowable steady–state metal temperature(s) shall be specified herein. If a direct measuring metal temperature sensing system is not provided at the measuring plane, the maximum allowable measured temperature(s) at the measurement plane (based on the signal indication after the temperature sensor harness plug) shall be specified herein.

3.2.2.12 Surface temperature and heat rejection The maximum operating surface temperatures and heat rejection rates of the engine shall be shown on figure 17a. Accessory pad loadings, compressor bleed air conditions, oil system cooling requirements, and air and gas leakage from engine case flanges and split lines shall be used in establishing engine heat rejection rates. The conditions surrounding the engine, the engine power condition and the oil system temperatures for which the surface temperatures are applicable shall be shown. For components and externals on the surface of the engine, the specified component and accessory surface temperature and heat generation curves may differ from the engine temperature as shown on figure 17a. The engine surface emissivity shall be presented as shown on figure 17b. The heat rejection and surface temperature data presented shall be for the conditions listed below. If heat rejection rates during flight and ground operating conditions other than those listed below are critical or limiting, that data shall be specified:

For turbojet/fan engines

a. Maximum Power, Sea Level static, hot atmosphere of figure 4.

b. Maximum Power, Sea Level, maximum stagnation inlet air temperature for hot atmosphere of figure 4.

c. Maximum Power, 36,089 ft, maximum stagnation inlet air temperature.

d. Maximum Power, highest altitude of figure 4, hot atmosphere of figure 4, subsonic level flight.

e. Low Power, Sea Level, low speed loiter condition, tropical or hot atmosphere of figure 4.

f. Idle Power, Sea Level static, hot atmosphere of figure 4, ground operation.

For turboshaft/turboprop engines

a. Maximum power, sea level static , hot atmosphere of figure 4.

b. Maximum power, 1.2 km (4000 ft), maximum stagnation inlet air temperature for the hot atmosphere of figure 4.

3.2.2.12.1 Component limiting temperatures Engine components mounted on the engine shall not exceed their allowable temperatures when surrounded by still air under the following conditions:

a. Continuous operation (including high and low power settings) with ambient air at the maximum stagnation temperature.

b. Flight shutdown from the most adverse condition and continued soaking with ambient air at maximum stagnation temperature.

c. Ground shutdown with ambient air at the sea level hot resoak temperature conditions of figure 4 with no special cooling.

A tabulation of the maximum ambient air temperature surrounding each component, the limiting surface temperatures for all components, the time period(s) applicable, and the limiting cooling fuel inlet temperature for fuel cooled components, for the most severe engine condition(s), shall be specified in table IX. The specific points of measurement, where the ambient air temperature and surface temperature with respect to the three coordinate axes of the component are maximum, shall be specified. The engine components requiring special cooling and ventilation shall be listed in table IX. Length of operation of these components without special cooling shall be defined in table IX.

Engine gas path flow leakage shall not impact cooling and ventilation. There shall be no backflow of exhaust gases, from any source, through the engine compartment under any operational or shutdown condition.

3.3 Environmental conditions

3.3.1 Atmospheric conditions

3.3.1.1 Humidity The engine shall meet all requirements of this specification when subjected to the following humidity: (a) .

3.3.1.2 Fungus The engine shall meet all specification requirements when subjected to fungus.

3.3.1.3 Corrosive atmosphere The engine shall meet all requirements of this specification during and after exposure to the corrosive atmosphere of table X for (a) corrosion cycles. The engine shall not deteriorate more than (b) percent in thrust/power, or gain (c) percent in specific fuel consumption.

3.3.1.4 Icing conditions The engine shall meet the requirements of this specification while operating in the meteorological conditions shown on figures 19a and 19b and table XI, with not more than (a) percent total loss in thrust/power and (b) percent total increase in specific fuel consumption at all operating conditions. After removal of the icing conditions, the engine shall perform the same as prior to encountering the icing conditions. No damage to the engine shall be allowed as a result of operating in the icing environment.

3.3.2 Ingestion capability (hazard resistance)

3.3.2.1 Bird ingestion The engine shall continue to operate and perform during and after the ingestion of birds as specified in table XII.

3.3.2.2 Foreign object damage (FOD) The engine shall meet the requirements of the specification for (a) inspection intervals of 3.4.1.7.4 without repair after ingestion of foreign objects which produce damage equivalent to a stress concentration factor (K_t) of (b) at the most critical locations of flow path components.

3.3.2.3 Ice ingestion The engine shall operate and perform in accordance with table XIII, during and after ingestion of hailstones and sheet ice at the take–off, cruise, and descent aircraft speeds. The engine shall not be damaged beyond field repair capability after ingesting the hailstones and ice.

3.3.2.4 Sand and dust ingestion The engine shall meet all requirements of the specification during and after the sand and dust ingestion event specified herein. The engine shall ingest air containing sand and dust particles in a concentration of (a) mg sand/m^3. The engine shall ingest the specified course and fine contaminant distribution for (b) and (c) hours, respectively. The engine shall operate at Intermediate/Maximum Continuous thrust/power with the specified concentration of sand and dust particles, with no greater than (d) percent loss in thrust/power, and (e) percent gain in specific fuel consumption.

3.3.2.5 Atmospheric liquid water ingestion The engine shall start and operate satisfactorily with (a) percent of the total airflow weight in the form of water (liquid and vapor), with (b) percent of the liquid water entering the inlet through a segment equivalent to one–third of the inlet area.

3.3.2.6 Armament gas ingestion (AGI) The engine shall meet all the requirements of the specification during and after AGI, except as follows.

If any performance loss occurs during AGI, the engine shall recover within (a) seconds and without pilot intervention, to the thrust/power available immediately prior to the AGI event. In addition, when an AGI signal is present, engine thrust/power shall never fall below (b) percent of the thrust/power available immediately prior to the AGI.

3.3.2.7 Steam ingestion The engine shall meet all specification requirements without stall, surge, flameout, mechanical/structural damage, or permanent performance loss, when the total inlet massflow is at least two percent steam, and has a minimum temperature ramp (increase) of (a) °C/sec ((a) °F/sec) with a (b) °C ((b) °F) temperature rise in a region less than (c) percent of the engine inlet flow area. After a steam ingestion event, the engine shall meet all requirements of the specification.

3.3.3 Electromagnetic environmental effects (E^3) The engine shall meet the electromagnetic environmental effects requirements specified in this section.

3.3.3.1 Electromagnetic interference (EMI) All engine electrical and electronic equipment and subsystems shall meet their performance requirements when subjected to the susceptibility requirements of (a) requirement (b) and shall comply with the emission requirements of (a) requirement (c) . These requirements shall be met for engine operation in all environmental conditions within the operating envelope for all control system operating modes.

3.3.3.2 Intrasystem electromagnetic compatibility EMC The engine shall meet the applicable EMC requirements of MIL–STD–1818, MIL–E–6051, and ADS–37.

3.3.3.3 Intersystem electromagnetic compatibility EMC The engine as installed in the aircraft, shall meet the requirements of the specification when exposed to the applicable external radiated environment of MIL–STD–1818, MIL–HDBK–235, and ADS–37.

3.4 Integrity The engine shall meet the integrity requirements specified in this section.

3.4.1 Structural integrity The engine shall meet the structural integrity requirements specified herein and in the following paragraphs. The integrity requirements contained below shall apply to all parts, components, externals, and expendables as applicable.

3.4.1.1 Design service life The engine shall have a design service life of at least (a) when subject to the design usage of 3.4.1.2.

3.4.1.1.1 Hot parts Hot parts shall have a usable life of (a) times the design service life specified in 3.4.1.1. Hot parts and their lives shall be listed in table XIV.

3.4.1.1.2 Cold parts Cold parts shall have a usable life of (a) times the design service life specified in 3.4.1.1. Cold parts and their lives shall be listed in table XV.

3.4.1.1.3 Expendables The minimum life without replacement of all expendable parts and components shall be equal to the minimum maintenance–free operating period. Expendable parts, components, and their lives shall be listed in table XVI.

3.4.1.1.4 Bearings The mainshaft and gearbox bearings shall have B1.0 lives equal to at least the design service life of the engine. A list of bearings and their lives shall be presented in table XVII.

3.4.1.1.5 Components Engine components shall have a usable life of (a) times the design service life specified in 3.4.1.1. Engine components and their lives shall be listed in table XVIII.

3.4.1.2 Design usage The design usage shall include: (a) .

3.4.1.2.1 Internal environment The internal environment shall be specified herein for the conditions of 3.4.1.2.

3.4.1.2.2 Externally applied forces The engine shall function satisfactorily and no deformation shall occur during or after exposure to the externally applied forces indicated in the Design Load Diagrams of figure 2.

3.4.1.3 Material characterization Material properties shall be based on (a) .

3.4.1.4 Parts classification All engine parts, components, controls and externals and expendables shall be classified for criticality in Table XIX.

3.4.1.5 Durability The engine shall be durable for ___ times the design service life when subjected to the usage of 3.4.1.2.

3.4.1.5.1 High cycle fatigue (HCF) life Engine parts shall not fail when subject to the maximum attainable combined steady–state and vibratory stresses. The vibratory or HCF stress shall be restricted to (a) percent of the material capability. All engine parts shall have a minimum High Cycle Fatigue (HCF) life as follows: (b) .

3.4.1.5.2 Low cycle fatigue (LCF) life Engine parts shall have a minimum LCF life of (a) times the number of (b) cycles that correspond to the parts life of 3.4.1.1.1 for hot parts and 3.4.1.1.2 for cold parts.

3.4.1.5.3 Creep The engine static and rotating parts shall not creep to the extent that it results in unsatisfactory engine operation. Part creep shall not affect disassembly and reassembly of the engine.

3.4.1.6 Strength The engine shall meet all the requirements of the specification during and after exposure to limit loads, singly and in combination. The engine shall not experience catastrophic failure when subjected to ultimate loads, singly and in combination.

3.4.1.6.1 Factors of safety The following factors of safety shall be applied to design usage induced loads to establish limit and ultimate conditions: (a) .

3.4.1.6.2 Blade and disk deflection The blades and disks shall not contact any static parts of the engine other than seals and shrouds. During all phases of engine operation including surge and stall occurrences. Seals and clearances shall remain effective under all internal and external operational loads.

3.4.1.6.3 Containment The engine shall radially contain a fan, compressor, or turbine blade failure at the blade root section in the fillet below the platform which is released within the engine, and shall not result in uncontained engine fire, at any engine operating speed. Engine components with rotating parts shall contain any failed rotating part at any engine operating speed.

3.4.1.6.4 Blade out Subsequent to a single blade failure, with resulting secondary loss of another blade in the same stage at maximum allowable transient speed, the engine shall not experience uncontained fire; catastrophic rotor, bearing, support, or mount failures; overspeed conditions; leakage from flammable fluid lines, or loss of ability to shutdown the engine. The design conditions for a single blade failure with secondary loss as stated above shall be as follows: (a) .

3.4.1.6.5 Overspeed/overtemperature The engine shall meet all the requirements of the specification during and after the following overspeed and overtemperature conditions:

a. Engine rotor speeds of (a) percent of the maximum allowable steady–state speed at the maximum allowable turbine temperature or first stage turbine rotor inlet gas temperature limit of 3.2.2.11 for five minutes.

b. (b) temperature of at least (c) °C ((c) °F) in excess of the maximum allowable temperature or above the first stage turbine rotor inlet gas temperature limit of 3.2.2.11, and at maximum allowable steady–state rotor speed for five minutes.

c. Engine component rotor speeds of (d) percent of the maximum allowable steady–state speed for five minutes.

d. Engine component fuel, lube and hydraulic inlet temperatures of (e) °C ((e) °F) above the maximum allowable operating fluid temperature at the maximum allowable steady–state speed for five minutes.

e. For turboprop/turboshaft engines, the power turbine shaft speed shall not exceed (f) percent of the transient speed limit, or the predicted speed attained following loss of load with the engine at Intermediate power and the power turbine running at the highest rated speed, whichever is greater. The predicted maximum speed shall be specified herein.

3.4.1.6.6 Disk burst speed The minimum loaded disk burst speed of the complete disk assembly shall be (a) percent when the disk is subjected to the maximum temperature gradient and maximum material temperature that will occur for that part.

3.4.1.6.7 Output shaft torque limits For turboprop and turboshaft engines the maximum allowable steady–state delivered shaft torque (mechanical) limit shall be at least (a) percent greater than the rating value.

3.4.1.6.8 Output shaft speed limits For turboprop and turboshaft engines the maximum allowable steady–state delivered shaft speed (mechanical) limit shall be at least (a) percent greater than the value of 3.2.1.1. The shaft shall be able to operate at this speed for at least (b) and function satisfactorily thereafter. Following loss of load, the output shaft speed shall not exceed the maximum shaft speed predicted with the engine at Intermediate power and the output shaft running at the highest rotor speed of 3.2.1.1.

3.4.1.6.9 Pressure vessel/case All engine cases and pressure loaded parts and components shall withstand the ultimate loading conditions defined in 3.4.1.6.1. The cases must remain intact, although permanent deformation and distress, requiring repair or replacement, is permitted. Engine cases shall not fail due to combustion process burning or erosion.

3.4.1.6.10 Pressure balance The engine thrust bearings shall provide sufficient thrust load to ensure satisfactory bearing operation during the design service life.

3.4.1.6.11 Gyroscopic moments The engine shall meet all the requirements of the specification at maximum allowable steady–state engine speeds when subjected to the rotational velocities and accelerations within the flight envelope and the gyroscopic moment conditions specified below:

a. A steady angular velocity of (a) radians per second around any axis in a plane perpendicular to the rotor axis, combined with a ± 1g vertical maneuver load for a total of (a) seconds.

b. A steady angular velocity of (b) radians per second in any axis in a plane perpendicular to the rotor axis for a cyclic life of 10^7 cycles at all load factor conditions within the flight envelope.

3.4.1.7 Damage tolerance Fracture/safety and mission critical engine parts shall be damage tolerant with defects resulting from material quality, manufacturing processing and handling damage during the periods specified in 3.4.1.7.4.

A Damage Tolerance Control Plan shall be prepared to identify and schedule each of the tasks and interfaces in the functional areas of design, material selection, manufacturing control, and inspection of fracture, safety, and mission critical parts. The tasks to be identified in the plan include:

a. Design concepts/material/weight/performance/cost trade studies.

b. Damage tolerance analysis, development testing, and proof of compliance testing tasks.

c. Parts list that identifies fracture, safety and mission critical parts, locations, and special controls required to meet damage tolerance requirements (e.g., material specification controls, quality assurance requirements, etc.).

d. Zoning of drawings for fracture, safety, and mission critical parts to identify critical locations and associated quality control requirements, defect locations, orientation, inspection method, and acceptance standards. The use of alternate procedures for identification of critical locations, etc., on drawings may be proposed.

e. Basic materials fracture data (e.g., K_{IC}, K_C, K_{ISCC}, da/dn).

f. Identification and control of fracture toughness and crack growth rate properties in the material procurement and manufacturing process specifications.

g. Traceability requirements on all tiers of procurement, processing, fabrication, and assembly for fracture critical components. Serialization or time coding requirements for tracking operational exposure of individual components.

h. Quality control requirements during component manufacture. Identification of procedures for certifying and monitoring subcontractor, vendor, and supplier inspection and quality control. Nondestructive inspection requirements for use during depot and base level inspections including supporting manuals (technical orders) and equipment needs.

3.4.1.7.1 Residual strength The residual strength shall be equal to the maximum stress that occurs during design usage conditions. Residual strength requirements shall be established for all damage tolerant designed parts and components. Associated static and dynamic loading conditions for these parts and components shall include: (a) .

3.4.1.7.2 Initial flaw size The assumed minimum initial flaws sizes that exist in a part as a result of material, manufacturing, and processing anomalies shall be as specified below:

Inspection Method	Material	Flaw Type	Flaw Size (Depth X Length)

Assumed initial flaw sizes based on production NDI methods shall have a probability of detection (POD) of (a) at the lower bound (b) percent confidence level (CL). Assumed initial flaw sizes based on the intrinsic material defect distribution shall encompass (c) percent of the defect population when a factor of safety of (d) on residual life is used.

3.4.1.7.3 In–service inspection flaw size The assumed minimum flaw sizes that exist in a part after completion of a depot, intermediate, or base level inspection shall be as specified below:

Inspection Method	Material	Flaw Type	Flaw Size (Depth X Length)

In–service inspection flaw sizes shall have a probability of detection (POD) of (a) at the lower bound (b) percent confidence level (CL).

3.4.1.7.4 Inspection intervals The frequency of inspection shall be as follows:

a. In–service noninspectable: Equal to (a) times the service life specified in 3.4.1.1.

b. Depot, intermediate, or base level inspectable: Equal to (b) times the service life specified in 3.4.1.1.1.

3.4.1.7.5 Flaw growth The initial flaw sizes specified in 3.4.1.7.1 shall not grow to critical size and cause failure of the part due to the application of the required residual strength load within:

a. (a) times the in–service noninspectable interval defined in 3.4.1.7.4, for in–service noninspectable flaws.

b. (b) times the depot, intermediate, or base level inspection intervals specified in 3.4.1.7.4, for depot, intermediate, or base level inspectable flaws.

3.4.1.7.6 Composites Composite parts shall be damage tolerant with defects resulting from material quality, manufacturing processing, and handling damage.

3.4.1.8 Vibration and dynamic response The engine shall be free of vibration and dynamic response that would cause the engine to operate unsatisfactorily and not meet life requirements of 3.4.1.1.

3.4.1.8.1 Vibration limits The location and maximum permissible engine vibration limit of each vibration transducer shall be as specified in table VIII.

3.4.1.8.2 Critical speeds The engine shall be free of resonance conditions at all shaft rotational speeds that would cause the engine not to meet all the requirements of the specification. Critical speeds existing below the engine operating range shall be at least (a) percent below Idle. Critical speeds existing above the maximum operating speed shall be at least (b) percent above the maximum allowable transient shaft rotational speed. The natural frequencies of the mounting system with the engine installed shall be at least (c) percent below Idle shaft rotational speed(s) in all damaging modes of vibration which can be excited by the rotor imbalances.

3.4.1.8.3 Blade, disk, and static structure vibration The engine shall meet all the requirements of the specification when subjected to engine induced natural frequencies and resonances.

3.4.1.9 Catastrophic failure The engine shall incorporate the following fail–safe features or capabilities to eliminate catastrophic failure: (a) .

3.4.1.10 Sustained acceleration Components shall function satisfactorily during and after sustained acceleration, in accordance with MIL–STD–810, and no permanent deformation shall occur.

3.4.1.11 Shock Components shall function satisfactorily after experiencing a shock, as induced in accordance with MIL–STD–810.

3.4.1.12 External surface foreign object damage The engine's external surface, protruding from the airframe, shall meet all requirements of this specification, without repair, after FOD that produces damage equivalent to a stress concentration factor (Kt) of _(a)_.

3.4.2 Mechanical equipment and subsystems integrity The engine's mechanical equipment and subsystems shall meet the integrity requirements specified below:

3.4.3 Avionic/electronic integrity The engine shall meet the electronic system integrity requirements specified below:

3.5 Reliability and maintainability (R&M) The engine shall exhibit characteristics of reliable operation and be designed such that it can be retained in, or restored to a working and usable condition within a specified period of time, when the maintenance is performed in accordance with prescribed procedures and resources. The engine shall be subject to the following:

a. The R&M definitions of Section 6 and MIL–STD–721.

b. The scheduled maintenance intervals of _(a)_.

c. The "Mission and Mission Mix" profile of table VI.

3.5.1 Reliability The engine shall meet the reliability requirements of the following paragraphs in accordance with MIL–STD–785.

3.5.1.1 Reliability quantitative requirements The engine reliability shall be as defined in tables XXIIa and XXIIb.

3.5.2 Maintainability The engine shall meet the maintainability requirements of the following paragraphs in accordance with MIL–STD–470.

3.5.2.1 Maintainability quantitative requirements Maintainability requirements shall be as specified in table XXIV. These values shall be derived from the system maintainability allocations and shall be based on the planned weapons system utilization rate and mission mix.

The engine shall have the following diagnostic capabilities:

a. WRA/LRU fault isolation of installed engine components for at least _(a)_ percent of all failures shall be within _(b)_ minutes.

b. Percent of success which diagnostics tests correctly isolates faulty subassemblies shall be _(c)_ percent.

c. Percent of success which diagnostic tests correctly diagnoses fault by retest shall be _(d)_ percent.

3.5.2.1.1 Excluded maintenance functions All excluded maintenance and repair functions shall be listed in table XXVI.

3.5.2.2 Maintainability qualitative requirements The engine shall meet the qualitative maintainability requirements designated in table XXVII.

3.5.2.3 Maintenance All maintenance events, scheduled and unscheduled, shall be completed at the lowest maintenance level feasible and limited to the least number of actions possible.

3.5.2.3.1 Modules The structural components which define each engine module, and the length of time to remove and replace each module, shall be specified in table XXVIII. All modules shall be capable of removal and replacement at the (a) maintenance level. Disconnects for all external plumbing and wiring associated with a module shall be located as near the front and back planes of the module as possible.

3.5.2.3.2 Maintenance, inspection, and repair cycle The estimated maintenance, inspection and repair periods for the total engine and each module shall be specified in table XXIX, together with the estimated time in manhours required to perform these functions.

3.5.2.3.2.1 Maintenance inspection techniques Engine design shall permit maximum use of non–destructive inspection techniques and multi–purpose test and inspection equipment. Inspection provisions, including access envelopes, shall be shown on the engine configuration and envelope figure.

Provisions for 360 degree inspection of the installed engine shall be made for the fan, compressor, combustor, and turbine sections of the engine. A positive means of slowly rotating the rotor system shall be provided to facilitate inspection. The access port size shall be a minimum of (a) mm ((a) inch). The same tool shall be used for removal and replacement of port covers, plugs, and associated fasteners. The location of access ports shall, as a minimum, permit inspection of the following locations: (b) . Access ports shall be accessible without removing other components.

3.5.2.3.3 Tools Required tooling shall be held to a minimum. The engine shall be capable of being maintained with a set of standard hand tools, as identified by the Using Service. Where provisions for standard hand tools are not feasible, the design shall provide, wherever possible, for special tools and test equipment that are available and in use on other in–service engines, as identified by the Using Service. Any requirement for new special tools, fixtures, and test equipment shall be subject to specific approval of the Using Service. Cadmium–plated tools are permitted with specific approval of the Using Service.

3.5.2.4 Battle damage repair The capability for battle damage repair shall be described. Combat damage repair design techniques shall utilize tools and materials found at organizational and intermediate levels.

3.5.3 Human performance and human engineering The engine shall be designed to be maintained by at least the central ninety percent (90%) of the maintainer population (5th percentile female stature through 95th percentile male stature) wearing a full complement of personal protective equipment, under all environmental conditions (including day/night), defined in the specification and all operational scenarios at operating bases and deployed locations in accordance with MIL–STD–1472, MIL–STD–1800, and MIL–H–46855.

3.6 Combat survivability The engine shall be designed to sustain mission performance in the intended combat threat environment. If the engine is intended for a single aircraft or class of aircraft the engine shall be optimized for the threat environment described in the System Threat Report (STR) defined in DoD instruction 5000.2.

If the engine is partly or wholly intended for a general class of aircraft with combat missions (e.g. patrol/anti–submarine warfare, combat reconnaissance, combat transport, strike fighters/attack, assault helicopter, or other), the engine shall be optimized for the general threat environment described in the threat assessment for those missions or classes of combat.

3.6.1 Susceptibility The engine shall not exceed the detectable signatures and observable characteristics as defined by the Using Service.

3.6.1.1 Noise The near and far field noise levels for an uninstalled engine, vectored or non–vectored, operating at sea level static shall not exceed the levels of figures 21a, 21b, and 21c.

3.6.1.2 IR radiation The propulsion system contribution to the total platform signature shall be specified herein. The engine shall not exceed the maximum IR signature levels of (a) , while meeting the requirements of 3.3, after one hot section life.

3.6.1.2.1 IR suppression system An infrared suppression system shall be required. The absolute (non–contrast) intensity limits and IR detection limits against specific threats shall be as defined by the Using Service for the uninstalled engine with and without suppression.

a. A description of the system shall be provided, including the method of actuation if ON–OFF capable, operating limitations in the suppression mode, and fail–safe provisions.

b. The detailed effects of IR suppression system operation on thrust, SFC, and other performance parameters shall be included in the engine performance computer programs of 3.2.1.1 and 3.2.1.2.

3.6.1.3 UV radiation The following shall be the absolute (non–contrast) UV signatures for the uninstalled engine: (a) .

3.6.1.4 Smoke The engine shall not emit detectable smoke contrails or smoke puffs during any transient or steady state operation when using any primary or alternate fuel as specified by the Using Service. The maximum smoke number shall not exceed (a) .

3.6.1.5 Gaseous emissions The engine shall not produce by–product exhaust emissions greater than the levels specified on figures 20a and 20b when using any primary or alternate fuel as specified by the Using Service.

3.6.1.6 Fuel streaming/vapor puffing The engine shall not emit visible unburned fuel (fuel streams or vapor puffs) at any power setting at or above Flight Idle or during any transient or steady state operation when using any primary or alternate fuel as specified by the Using Service.

3.6.1.7 Water vapor contrails The engine shall not emit water vapor contrails during any transient or steady state operation when using any primary or alternate fuel as specified by the Using Service.

3.6.1.8 Radar cross section (RCS) With all RCS reducing features, materials, and devices installed, the engine static RCS signature levels shall not exceed the weapons system interface control document budget allocation for the propulsion system contribution to the aircraft inlet and exhaust nozzle returns. Budget levels shall not exceed the values defined in table XXX given in sector averages in median and mean (dBsm units) and linear mean (meters squared units). Sector average shall be computed by adding all the 0.05 degree data (using 512 integrations per data point). For each sector, the linear mean in meters squared shall be computed by taking the anti log of each of the 0.05 degree dBsm points and dividing the sum by the number of data points. Engine dynamic RCS signature return shall be presented in the array format shown in table XXX. The array shows the RCS magnitude of the rotating components as a function of frequency and engine rotation and represents the mean dynamic return computed over a full 360–degree rotation of the first stage fan blades with scattering from all stationary elements eliminated. Dynamic returns due to rotating fan module components shall not exceed the weapons system interface control document budget allocation for this contributor to the total aircraft inlet return.

3.6.1.9 Radar absorbent materials (RAMs) and coatings The ingestion requirements of 3.3.1.3 and 3.3.2.4 combined shall not deteriorate the RCS signature level for any radar absorbent material (RAM) and coating by more than (a) dBsm.

3.6.2 Vulnerability The engine shall be designed to control, reduce, tolerate, and resist finite degradation caused by interaction with weapon effects (e.g., enemy or friendly nuclear effects, chemical and biological weapons, ballistic threats, or other area weapons) as specified below.

3.6.2.1 Ballistic weapons The engine, including externally mounted components, shall tolerate ballistic weapon threats as follows: (a) .

3.6.2.1.1 Vulnerability – components and externals Components and externals shall confer vulnerability reduction to the engine by means of:

a. The size, location, and packaging of engine components and externals including the interconnecting elements (e.g., piping, wires, and cables) shall minimize the vulnerable area to the maximum extent allowed by other requirements of this specification.

b. Non–flight–critical parts and externals, including the interconnecting elements (e.g., piping, wires, and cables), shall be positioned to protect flight–critical engine parts from ballistic damage to the maximum extent allowed by other requirements of this specification.

c. Components and externals shall exhibit minimum vulnerability by means of redundancy, separation, shielding, fault tolerance, damage recognition, analysis, and automatic control algorithm reconfiguration.

3.6.2.1.2 Static structure No case or gas pressure loaded non–rotating component, except airfoils, shall collapse or rupture when subject to its maximum operating pressure and a round hole of (a) diameter placed anywhere on the structure.

3.6.2.2 Directed energy weapons (DEW) effects The engine shall meet the specific directed energy weapons vulnerability requirements of (a).

3.6.2.3 Nuclear weapons effects The engine shall have the following operational capabilities and balanced hardness in nuclear environment that includes blast/gust, dust, thermal, transient radiation effects, and electromagnetic pulse: (a).

3.6.2.4 Chemical/biological agent effects The engine shall survive contamination, from chemical and biological agents, and decontamination without losing mission capability as follows: (a). The engine shall be capable of being decontaminated within 30 minutes using the procedures and decontamination agents defined by (b). Engine materials shall be compatible with the decontamination agents. The engine shall be capable of being operated and maintained by personnel in nuclear, biological and chemical (NBC) protective ensembles. The engine shall withstand the chemical/biological environments defined by the applicable System Threat Report (STR).

3.6.2.5 Fuel ingestion effects The engine shall be capable of ingesting fuel via the inlet duct under the conditions specified in the following subparagraphs.

3.6.2.5.1 Fuel ingestion – steady–flow The engine shall be capable of detecting and mitigating, by engine fuel control action, the effects of steady fuel flow via the inlet duct with the fuel and plume characteristics and flow rates as follows: (a). The swirl displacement angles for core and fan flow paths shall be specified by the contractor.

3.6.2.5.1.1 Fuel ingestion – cooling air contamination Contamination of engine internal cooling air from fuel ingestion shall not degrade engine life nor engine parts life below that specified in 3.4.1.1.1 and 3.4.1.1.2.

3.6.2.5.2 Fuel ingestion – transient (“quick dump”) The engine shall ingest transient fuel without permanent damage or performance degradation under operating conditions as follows: (a). Engine instability is permitted during the event, but normal engine function shall be reestablished within (b) seconds of the ingestion event with no pilot input.

3.7 Subsystems

3.7.1 Anti–icing/de–icing system The anti–icing/de–icing system, the system actuating method and system limitations shall be described. Operation of the anti–icing/de–icing system shall not result in a loss of performance greater than that specified in 3.3.1.4. Within (a) seconds following termination of the anti–icing/de–icing system, engine performance shall return to the level prior to system actuation.

The anti–icing/de–icing system shall allow (no/no detrimental) accumulation of ice on any engine part while operating under the icing condition specified in 3.3.1.4. Operation of the anti–icing/de–icing system throughout the operating envelope shall not damage the engine. The performance and operability effects of anti–icing/de–icing system

operation in a non–icing environment shall be specified. The anti–icing/de–icing system shall provide a signal for indicating that the anti–icing/de–icing system is operating. The fail–safe mode for the ice protection system shall allow safe return to base in icing conditions and non–icing conditions without damage to the engine. If an inlet particle separator (IPS) is used, ice formation on the IPS shall not be permitted. Limited ice accumulation shall be permitted in the IPS scavenge system if it does not affect IPS operation. The service life of the anti–icing/de–icing system shall be specified if it is other than one cold part life.

3.7.2 Control system The control system shall be redundant, fault tolerant, and automatically control the engine to meet all specification requirements. The engine control system, airframe/engine control system integration, all subcomponents, interfaces, and related functional operability and performance shall be described. The control system shall provide (a) percent fault coverage for any single control system malfunction or failure. The control system malfunctions or failures that cannot be fault isolated shall be specified.

For engines with afterburner, augmentor ignition shall be automatically activated when the control system receives the appropriate airframe throttle position signal.

Control system interpretation for all airframe demanded engine thrust/power requirements shall be specified herein.

Regions of control limiting functions as a function of altitude, Mach number, and thrust/power setting for standard, hot, and cold day atmospheres of figure 4, shall be shown in the format of figure 24. The limiting values used to establish the various regions shall be specified on the figure.

Reprogramming capability shall be provided at the LRU level installed in the aircraft or as required by the Using Service.

For turboshaft/prop applications, the engine control system shall provide for load sharing and torque matching for multiple turboshaft engines.

3.7.2.1 Control system performance The control system fault accommodation shall detect failures that shall result in fail–operational, fail–degraded, and fail–safe engine operation, during and after control system malfunction(s) or failure(s). The engine control system shall provide fully modulated thrust/power response, and prevent exceedance of aerothermodynamic and mechanical limits when subjected to unrestricted thrust/power demand inputs. Thrust/power signal input versus engine thrust/power output shall be linear between Idle and Intermediate/Maximum Continuous. The relationship between control input and power output shall be shown on figure 25. For thrust reversers, the relationship between reverse thrust and thrust demand input signals shall be linear and free of abrupt changes.

For turboshaft/turboprop engines, the system shall allow the selection of any output shaft speed between (a) and (a) percent, inclusive, at all powers equal to and greater than no load. The system shall have the capability to automatically match the output shaft torque of both engines of a twin engine installation within (b) percent of the torque available from a single engine at Maximum Continuous Power.

3.7.2.1.1 Backup control (BUC) A BUC shall be considered part of the control system. A BUC, in conjunction with a redundant control system shall be required for single engine aircraft. During transfer from the primary mode to the backup mode and during operation, the BUC shall provide, as a minimum, a safe return to base. The description of the BUC, and the capabilities and limitations of the engine when operating with the BUC shall be specified. Transfer from the primary system to the BUC and back to the primary system shall be possible at any power setting without engine instability or change to the control system input. Transfer from primary control to BUC shall be (a) after system malfunction or failure. Thrust/power demand input signal relationships shall be identical to those of the primary control system. The BUC shall use a separate metered fuel flow path in single engine applications. The BUC shall prevent the engine from exceeding any limits.

3.7.2.2 Control system adjustments Control designs requiring mechanical adjustment of the engine control system shall not be permitted except upon approval by the Using Service.

3.7.2.3 Overspeed protection system The overspeed protection system shall prevent all rotors from exceeding their overspeed limits. The rotor speeds at which the overspeed protection occurs shall be specified.

For turboshaft and turboprop applications, the measurement of the power rotor system speeds shall be sensed directly with no intermediate mechanical devices such as gears, flexible shafts, or clutches.

3.7.3 Fuel system The engine fuel system shall be described herein. Provisions shall be made for installing separate flowmeter(s) for combustor and augmentor fuel flows for test and diagnostic purposes.

3.7.3.1 Fuels The engine shall meet the requirements of the specification while using the fuels as required in the following paragraphs.

3.7.3.1.1 Primary fuel The primary fuels shall be as specified in table XXXIa. The engine shall meet the requirements of the specification when using any of the fuels conforming to and having any of the variations in characteristics permitted by the specifications listed in table XXXIa. The engine shall meet all requirement of the specification after transition between fuels.

3.7.3.1.2 Alternate fuel The alternate fuel(s) the engine shall be capable of using shall be specified in table XXXIb. Any limitations on the engine when using alternate fuels shall be specified herein.

3.7.3.1.3 Restricted fuel The restricted fuel(s) the engine shall be capable of using shall be specified in table XXXIc. Any limitations on the engine when using restricted fuels shall be specified herein.

3.7.3.1.4 Emergency fuel The engine shall be capable of operating for a time period of at least (a) hours from sea level to (b) (km) ((b) ft) altitude, throughout a range from (c) to (c) percent of IRP/Maximum continuous thrust/power, and at no greater than (d) percent of the specific fuel consumption of 3.2.1.1 when using automotive or marine diesel fuels (VV–F–800, MIL–F–16884, and MIL–G–3056). Any other limitations on the engine when using emergency fuels shall be specified herein.

3.7.3.2 Fuel system performance

3.7.3.2.1 Fuel contamination The engine shall meet specification requirements when supplied with fuel contaminated in an amount up to the extent specified in table XXXIIa, and when supplied with (a) liters ((a) gallons) of fuel, contaminated per table XXXIIb, with the filter in full bypass. Engines for land based rotor craft and engines with unassisted (suction) fuel systems shall meet specification requirements with fuel contaminated in amounts specified in table XXXIIc, with fuel filter not in bypass mode.

3.7.3.2.2 Fuel system performance with external assistance The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

Normal Operation:

a. Fuel temperature: From a minimum of (a) °C ((a) °F) when using worst case primary fuel or (b) centistoke, whichever is worse, to a maximum of no less than (c) °C ((c) °F).

b. Fuel pressure: From a minimum of (d) kPa, ((d) psi) above the true vapor pressure of the fuel to a maximum of (e) kPa, ((e) psig) with a vapor/liquid volume ratio of zero (0).

c. Fuel pressure surges: The fuel system shall accept inlet pressure surges up to (f) kPa, ((f) psig) occurring at a maximum rate of change of up to (g) kPa, ((g) psi) per second. Engine induced pressure surges at the fuel inlet flange shall not exceed (h) kPa, ((h) psi) above the fuel inlet pressure during augmentor initiation or termination.

3.7.3.2.3 Fuel system performance with no external assistance The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

a. Fuel Temperature: From a minimum of (a) °C, ((a) °F) when using worst case primary fuel or (b) centistoke, whichever is worse, to a maximum of no less than (c) °C ((c) °F).

b. Fuel Pressures: From a minimum of (d) above the true vapor pressure of the fuel to a maximum of (e) .

c. Vapor/liquid volume ratio from zero (0) to (f) .

3.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor When the vapor/liquid ratio is greater than (a) at the fuel inlet, it shall not be necessary for the fuel system to supply fuel for engine operation. However, after the conditions of 3.7.3.2.2 or 3.7.3.2.3 have been re–established, it shall be possible to accomplish normal starts and operate satisfactorily thereafter. Vapor relief provisions shall be incorporated to vent excess vapor during pump priming.

3.7.3.2.5 Fuel pump priming The fuel pump shall be self priming and be capable of a dry lift of (a) meter, ((a) feet) at an absolute pump inlet pressure of (b) kPa, ((b) psia), using the fuels of 3.7.3.1 with the Reid Vapor Pressure not to exceed (c) kPa, ((c) psi), and a fuel temperature not to exceed (d) °C, ((d) °F).

3.7.3.2.6 Fuel lubricity The engine shall operate satisfactorily when using fuel with a lubricity value equivalent to a 0.75 – 0.85 mm (0.030 – 0.033 inch) Wear Scar Diameter (WSD) as measured by the Ball–On–Cylinder Lubricity Evaluator (BOCLE).

3.7.3.2.7 Fuel system performance with water saturated fuel The engine shall meet the requirements of this specification when supplied with fuel saturated with water in accordance with MIL–F–17874 at worst case fuel pressure and temperature conditions.

3.7.3.2.8 Fuel filter Fuel filtration on the engine shall be provided, as needed, for the engine to meet the requirements of the specification. The fuel filter shall consist of cleanable/non–cleanable filter(s), and shall be made of a material that does not cause fuel contamination due to media migration. The filtration ratings of the filter elements, contaminated per table XXXIIa, shall be as follows:

Capacity		Filtration Microns or Mesh	Capacity Liter (gal) to Impending Bypass	Liter (gal) Impending Bypass to Bypass
Type	Location			

The filter assembly shall incorporate a pressure relief bypass and shall prevent the discharge of filter contaminant through the bypass. The filter assembly shall emit a signal for (a) indication of impending and actual filter bypass. The filter assembly shall have (b) indications of both impending and actual filter bypass. The indications shall remain visible until manually reset. A drain valve shall be provided at the low point of the filter assembly to drain the filter element cavity and to purge the fuel system of air following filter element replacement. The filter shall be removable without spills or draining of the fuel lines. Fuel pressure surges during operation within the limits specified in 3.7.3.2.2 shall not cause false activation of the impending or actual bypass indicators.

3.7.3.2.9 Fuel flow limit The maximum and minimum fuel flow shall be specified herein and in table VIII.

3.7.4 Electrical system

3.7.4.1 Electrical power The primary electrical power for the engine shall be redundant and supplied by the engine and shall be described herein. Electrical equipment shall accept externally supplied power complying with MIL–STD–704 (a) .

3.7.4.1.1 Generator/alternator The engine generator/alternator shall provide the required electrical power from the specified minimum ignition exciter speed to the maximum engine rotor speed. The generator/alternator performance shall not degrade during and after continuous operation at a speed equivalent to (a) percent of the maximum allowable steady–state engine speed under full electrical load. The generator/alternator performance shall not degrade following the overspeed of 3.4.1.6.5, with no electrical load. The generator/alternator housing shall completely contain all damage if a mechanical failure should occur when operating at or below maximum transient rotor speed.

3.7.4.2 Alternate/external electrical power The engine shall automatically accept airframe provided electrical power in accordance with MIL–STD–704 (a) for both backup use and for ground power during engine–off checkouts. Power transients allowed by MIL–STD–704 (a) shall not result in engine performance and operability degradation. The alternate/external power supply shall be described.

3.7.4.3 Electrical connectors and cables Electrical connectors shall comply with MIL–C–38999, Series III. Electrical connectors and cables shall have sufficient spare pins in the connectors and wires in the cables to permit future growth and shall be repairable. The quantity of pins and wires is subject to Using Service approval. Electrical connections located in close proximity to each other shall be made physically non–interchangeable. At the sea level cold atmospheric temperature of figure 4 it shall be possible to flex electrical cable and conductors during routine maintenance without damage to these items and to connect or disconnect electrical connectors using normal maintenance procedures. Electrical connectors shall perform satisfactorily in the engine bay environment.

For turboshaft applications, electrical connectors shall be MIL–C–38999, series (a) type (or equivalent metric) and shall also comply with the series III, paragraphs 1.2.1.2 and 4.7.1.5 of MIL–C–38999. The engine shall have not more than three electrical connectors for connection to the airframe.

3.7.4.4 Electronic components Electronic components shall meet the requirements of MIL–E–5400, Class 1 equipment and the reliability program requirements of MIL–STD–785. Electrical parts shall be derated in accordance with (a).

All integrated circuits shall be qualified in accordance with MIL–M–38510. Semiconductor devices shall be qualified in accordance with MIL–S–19500.

3.7.4.5 Electrical grounding The grounding system for all electrical and electronic components of the engine shall be compatible and consistent with the aircraft grounding scheme. All components shall be grounded to the engine system. Power returns shall be isolated internally from the chassis of each component. A separate pin shall be provided on each power connector as a current return for each source of power supplied by the alternate/external electrical power system.

a. Safety grounding. A pin on each power connector which is connected internally to the component chassis shall be provided to permit internal grounding of the component through wiring. No circuit shall utilize ground wire as a power return.

b. Shield grounding. Cable shields shall be terminated at each end to the back shell of connectors that are conductive and provide bonding of shields 360° peripherally around the connector.

3.7.4.6 Electrical bonding The internal and external bonding requirements shall be in accordance with (a). For composite materials, bonding shall be accomplished at impedance levels consistent with the materials in use. Bonding provisions shall be compatible with other requirements imposed on the system for corrosion control.

3.7.4.7 Ground isolation The engine/aircraft interface shall not use aircraft structure for power or signal return. The isolation between power returns/signal returns and structure shall be at least 1 megohm at DC.

3.7.4.8 Potting compounds The use of potting compounds shall not be permitted except where specifically approved by the Using Service. When approved, the potting compounds and their application(s) shall be described.

3.7.5 Ignition system(s) The engine ignition system(s) shall be electrically self–sufficient and shall not require power external to the engine. A means shall be provided to deactivate the engine ignition system(s), and it shall be described herein.

The engine ignition system(s) shall light the combustor(s) under carbon and water fouling conditions.

a. Main Ignition System. The main engine ignition system shall be described herein to include power source, ratings in terms of stored energy level and delivered energy level to each ignitor, in joules per spark and frequency of spark, and type (continuous, manual, automatic, etc). For single engine applications the ignition system shall be redundant, shall provide a minimum of two separate ignitors and two independent exciter output circuits. Each separate ignitor and circuit shall release sufficient energy for all ground and air starting requirements.

b. Augmentor Ignition System. If applicable, the augmentor ignition system shall be described herein.

3.7.5.1 Ignition system fouling

3.7.5.1.1 Carbon fouling The spark ignitors of the ignition system shall provide sparking performance with spark gaps completely covered, filled, or bridged with an application of graphite petrolatum compound, MIL–T–5544. With the minimum power input of 4.7.5, and these carbon fouling conditions, the sparking rate shall not be less than the minimum design value.

3.7.5.1.2 Water fouling The spark ignitors of the ignition system shall provide sparking performance when thoroughly drenched with water or frosted over. With the minimum power input of 4.7.5, and the water fouling conditions, the sparking rate shall not be less than the minimum design value.

3.7.6 Engine monitoring system (EMS) The EMS and the equipment which make up the system shall be described herein. If a ground support system is required, it shall include all the (a) support equipment needed to achieve the overall EMS requirements. Hardware and software shall be provided to monitor and record the aircraft and engine parameters defined in tables VIIa and VIIb during flight and ground operation, and to provide data in an organized format for ground analytical condition checkout capability, parts life tracking, warranties, and scheduling maintenance actions. The malfunction of any EMS hardware or software shall not affect engine performance or operability. The EMS shall be compatible with existing engine maintenance systems. The engine monitoring system shall be completely functional after failure of any other subsystem of the engine, including engine control and electrical systems.

3.7.6.1 EMS fault detection/isolation The EMS shall provide (a) percent fault detection and isolation of all faults. The system's functions required to achieve this level of detection/isolation faults shall be described. The on–board engine diagnostics shall provide fault detection/isolation to the faulty LRU/WRA to meet the reliability and maintainability requirements as specified in 3.5. On–board engine diagnostics system false alarms shall be considered as failures and shall be used in the reliability and maintainability calculations per 3.5. The mission critical equipment and functions of the on–board diagnostic system which monitor mission and safety critical parameters shall be defined in the applicable contract. The on–board engine diagnostics system shall not cause a failure of any other mission or safety critical system.

3.7.6.2 On–board engine diagnostic functions The on–board engine diagnostics system shall perform the following functions:

a. Acquire data to determine engine performance status and performance trends, and mission profile data (altitude, Mach number, alpha, beta, etc.).

b. Acquire data for parts life tracking.

c. Acquire limit exceedance data (a) seconds prior to the event until (b) seconds after.

d. Acquire pilot initiated data records.

For carrier based aircraft, the on–board engine diagnostics system shall perform:

e. Takeoff thrust/power check and provide a signal to the airframe. The thrust/power shall be within (c) percent of the actual net thrust/power. The EMS shall be able to determine engine mission capability and provide a signal to the airframe.

3.7.7 Optical systems Engine subsystems that use optical systems, including fiber optic system(s), shall be identified herein. The fiber optic system shall be described herein. Optical system hardware shall operate satisfactorily in the engine bay environment.

The optical system shall be powered by the engine's primary electrical power system. Switching from the primary to the alternate/external electrical power system shall not affect the operation of the optical system. Optical sensors (e.g., A/B flame detectors, optical pyrometers) shall maintain accuracy and not lose calibration during the life of the engine module in which it is installed.

The optical system components shall be in accordance with the following component specifications: (a) .

3.7.7.1 Fiber optic cables Fiber optic cables shall comply with MIL–C–85045, and shall have sufficient spare fibers to permit future growth. The quantity of fiber is subject to Using Service approval. At the sea level cold atmospheric temperature of figure 4, it shall be possible to flex fiber optic cables during installation and routine maintenance without damage. Installation of a fiber optic cable shall not induce a change in optical transmittance that exceeds (a) dB. Repairs to fiber optic cables shall not degrade performance beyond the limits of this paragraph.

3.7.7.2 Fiber optic connectors Fiber optic connectors shall be in accordance with (a) . Multiple terminus optical connectors shall have sufficient spare terminals to permit future growth. Optical connections located in close proximity to each other shall be made physically noninterchangeable. At the sea level cold temperature of figure 4, it shall be possible to connect or disconnect optical connectors using normal maintenance procedures, without damage to the fiber cables or connectors. Connectors shall be resistant to engine bay contamination and to the bleed air contamination of 3.1.1.15.1.

It shall be possible to remove these contaminants from the ends of the transmissive elements without causing damage to them. During maintenance operation in which a connector is disconnected, the procedure for removing contaminants from the ends of the transmissive elements before reconnection shall be described.

3.7.7.3 Fiber optic passive couplers Fiber optic passive couplers shall comply with MIL–C–24621.

3.7.7.4 Fiber optic high altitude radiation resistance All fiber optic systems shall not have degradation in performance due to radiation effects at altitudes above (a) km ((a) ft).

3.7.8 Lubrication system The engine lubrication system physical and functional characteristics shall be described herein.

Any coke and sludge build–up shall not prevent the engine from meeting the specification requirements. A signal for continuous remote indication of the oil pressure shall be provided. If the lubrication and hydraulic systems share a common reservoir, the lubrication system shall be protected from total depletion in the event of a hydraulic failure.

3.7.8.1 Lubrication oil The engine shall meet the requirements of the specification when using the following lubricating oil(s): (a) .

3.7.8.1.1 Oil pressure and temperature limits The maximum and minimum operating oil pressure limits shall be (a) and (b) , respectively. The maximum transient and maximum allowable steady–state oil temperature limits shall be (c) and (d) , respectively. The above limits shall also be listed in table VIII. The maximum and minimum oil

pressures during starting and initial operation at the sea level cold atmospheric temperature of figure 4 or (e) centistoke oil viscosity, whichever is worse, shall be (f) and (g) respectively. Minimum and maximum oil pressures during starting and initial operation shall not persist for more than (h) minutes.

3.7.8.1.2 Oil consumption limits The maximum allowable oil consumption rate shall not exceed (a) liters/hour ((a) gallons/hour). Oil consumption limits shall also be specified in table VIII.

3.7.8.1.3 Oil flow interruption/depletion The engine shall meet the requirements of the specification during and after operation at (a) thrust/power for a period of at least (b) seconds with no oil (only air) supplied to the oil pump inlet. The engine shall be capable of operating without oil for an additional (c) minutes at (d) thrust/power without rotor seizure.

3.7.8.2 Lubrication system components and features

3.7.8.2.1 Oil reservoir Unvented oil reservoirs shall meet the pressure vessel requirements of 3.4. Deaeration features shall be specified herein.

The capacity of the oil reservoir shall be:

Usable Oil	_____ liters	(_____ Gallons)
Gulping Volume Oil	_____ liters	(_____ Gallons)
Unusable Oil	_____ liters	(_____ Gallons)
Expansion Space (equivalent oil)	_____ liters	(_____ Gallons)
Total Enclosed Reservoir Volume (equivalent oil)	_____ liters	(_____ Gallons)

3.7.8.2.1.1 Oil reservoir external features The oil reservoir shall contain features to determine the oil level and to service and drain the reservoir for attitudes of 3.2.2.2. An oil quantity signal (a) be provided to the airframe. The reservoir shall have a (b) system and shall have features for overfill protection.

3.7.8.2.2 Oil drains Drain ports shall be provided at appropriate low points in the oil system for draining the oil and taking oil samples while the engine is installed in the aircraft.

3.7.8.2.3 Oil filters Oil filter type(s), micron size, and capacity shall be specified herein. Primary filter assemblies shall incorporate a pressure relief bypass, an impending bypass indicator, and a bypass indicator. The indications shall remain visible until manually reset. The time from impending bypass to bypass shall be specified herein.

3.7.8.2.4 Oil debris monitor Oil debris monitors shall be installed in locations where wear particles in the oil would most likely be detected.

3.7.8.2.5 Oil coolers The type and number of oil coolers used in the oil system shall be specified herein.

3.7.9 Hydraulic system The hydraulic system shall be described. The hydraulic system fluid(s) shall be (a) . Nominal and maximum working pressure and maximum operating fluid temperature shall be (b) . Peak pressure (ripple or surge) shall not exceed (c) percent of maximum system working pressure. The allowable pressure ripple shall be defined as less than or equal to (d) Pa abs ((d) psia) at (e) hertz. The hydraulic system shall withstand a proof pressure of (f) times the maximum working pressure. The total volume and the usable quantity of hydraulic fluid in the reservoir shall be (g) . Hydraulic systems which are integral to the fuel system shall meet the requirements of 3.7.3.

3.7.9.1 Hydraulic ground test provisions Ground test provisions shall be provided. Fluid leakage shall not occur during ground test. Self–sealing ground test couplings shall be provided with the hydraulic system. These couplings shall be secured with sealing caps when not used.

3.7.9.2 Hydraulic system fire and safety hazards The hydraulic system design shall minimize fire hazards caused by proximity of system plumbing and components to heat sources and electrical equipment. All hydraulic operated services which are essential to safe engine operation shall have redundancy or shall have provisions for emergency activation.

3.7.9.3 Hydraulic system air removal Entrapped air shall not cause sustained loss of system pressure or degradation of system operational performance. There shall be provisions for bleeding air from the system at critical points for maintenance purpose.

3.7.9.4 Hydraulic fluid filters Filter(s), type(s), location(s), and filtration rating(s) in microns (nominal and absolute particle sizes) shall be specified herein. Each filter element shall be accessible and easily removed. Each filter housing shall incorporate an automatic shutoff device to prevent fluid drainage during filter element removal. Each filter assembly shall supply indication of impending bypass and bypass. Each indication shall remain visible until manually reset. Whenever activation of the indicator is a result of the exceedance of the differential pressure limit, the indicator shall remain in the fault mode until removal, cleaning, or replacement of the filter element is accomplished.

Filters for the hydraulic system shall meet the following requirements:

Supply line – Filters shall be located such that all fluid from the engine pumps and Ground Test Equipment (GTE) pressure connections will be filtered prior to entering components of the system.

Return line – Filters shall be located such that all fluid will be circulated through the filter prior to entering the return line relief valve, pumps or reservoir.

Other – Additional filters at critical component locations shall be provided where filter protection is necessary.

3.7.10 Pneumatic system The functional and physical characteristics of the pneumatic system shall be described.

3.7.10.1 Pneumatic air contamination Pneumatic system components requiring fan or compressor air shall operate satisfactorily with the inlet air contaminated as defined in 3.3.2.4.

3.7.11 Starting The engine shall satisfactorily make ground and air starts, and shall restart under the conditions, limitations, and procedures, specified in this section. The engine shall start with inlet air at any temperature between and including the extreme hot and cold atmospheres of figure 4, with fuel inlet temperatures between and including the maximum fuel temperature specified in 3.7.3.1 and the minimum inlet fuel temperature either equal to the cold atmospheric temperature of figure 4, or that temperature which corresponds to a fuel viscosity of 12 centistoke, whichever is warmer. Starts shall be made with the primary fuel of 3.7.3.1.1 and oil(s) of 3.7.8.1, unless specified otherwise. A start shall be considered satisfactory when the engine rotor is accelerated from either rest or windmilling speed to Idle, when the minimum assist torque is provided by a starter or ram air, as specified in 3.1.1.19.1, and the engine stays within the engine starting and operating limits of figure 9.

3.7.11.1 Ground starts The engine shall start at the specified ambient temperatures and altitudes, in times equal to or less than those shown in figure 26a. The engine shall be capable of an unlimited number of consecutive start cycles, with a maximum time interval of (a) seconds between the completion of one cycle and the initiation of the next cycle. The wind condition envelopes for ground starting shall be shown on figure 26b. Additionally, special starting requirements apply as follows:

a. Hot temperature soak start. The engine shall start, after a (b) hour hot soak at the temperatures specified on figure 4, using (c) fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperatures shown on figure 4.

Following shutdown, the engine shall start, after a resoak period of (d) minutes at the temperatures specified on figure 4, using (c) fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperature shown on figure 4.

b. Cold temperature soak start. The engine shall start, using (e) fuel and (f) oil, after a cold soak for a period of (g) hours with inlet air, fuel, and oil at the minimum cold ambient temperature shown on figure 4.

3.7.11.2 Air starts Windmilling, spooldown, and starter assisted airstart envelopes and maximum start times, with customer extractions shall be defined on figure(s) 27. Air starts within figure(s) 27 shall be considered acceptable when the engine achieves Idle speed within the air start times specified and no starting or operating limits are exceeded. The engine shall be capable of consecutive start cycles, with a maximum time interval of (a) seconds between the completion of one cycle and the initiation of the next cycle. Starting time shall be measured from the initiation of the starting sequence to the attainment of (b) percent of stabilized uncorrected engine Idle rotor speed. During air starts the engine rotor load shall be at least (c) percent of the total maximum accessory drive load specified for the drives shown in table V.

For turboprop engines, the engine shall start at the post loiter altitude start envelope of figure 10 after being inoperative for at least (d) hours.

3.7.11.3 Starting limits The starting altitude for ground starts and air starts shall be up to (a) km, ((a) ft) and (b) km, ((b) ft), respectively. The maximum turbine temperature for ground and air starts shall be specified. Differences in the altitude starting limits for “hot” and “cold” engines shall be specified. Limits with maximum customer bleed air and maximum customer power extraction shall be specified. The conditions where engine starting is the most difficult shall be specified. For turboshaft and turboprop applications, the post loiter altitude start limit and the associated loiter period (i.e., time the engine is inoperative) shall be specified.

3.7.11.4 Starting procedure The starting procedure shall be simple and shall not require critical timing. The engine shall be capable of starting with the thrust/power demand at Idle or above and, after initiation of the starting sequence, being accelerated immediately to any power request. This shall be accomplished within specified engine starting limits and without stall. A description of the starting procedure shall be specified herein.

3.7.11.5 Automatic relight The engine shall incorporate an automatic rapid relight system which shall immediately re–establish engine operation (without any pilot action) after a main combustor flameout or augmentor blowout has occurred. The automatic relight system shall detect any flameout that may have occurred and initiate a sequence for automatic recovery anywhere in the automatic relight operating envelope shown on figure 10. The automatic relight system shall be described herein.

3.7.12 Exhaust nozzle system The exhaust nozzle physical and operational characteristics, including emergency or manual provisions, shall be described.

Any exhaust nozzle system failure shall allow safe return to base. For variable exhaust nozzles, failure of one segment of a multi–segmented nozzle shall not prevent operation of the nozzle.

Failure of any single part of the actuating mechanism on a controllable nozzle shall not cause asymmetric thrust relative to the nozzle. The operational mode, and limits resulting from any single part failure in the actuating mechanism of a variable nozzle, shall be specified herein. Provisions shall be made to actuate the nozzle for ground maintenance checkouts, rigging, and adjustment without the engine running. Life of the engine exhaust system, special components, features, materials, and applied coatings shall be specified in table XIV.

3.7.12.1 Exhaust nozzle external asymmetrical air pressure loads The engine shall meet the requirements of the specification when subject to the asymmetrical and all other air pressure loads resulting from the external flow field pressure distribution on the nozzle and exhaust systems. The limits shall be specified herein.

3.7.12.2 Vectoring nozzle The thrust vectoring nozzle system shall be described. The engine thrust vectoring system shall revert to the fail–safe mode after receiving a fail–safe/standby signal from the airframe. Nozzle feedback signals, if required, shall be provided to the airframe and shall include: (a) .

3.7.12.2.1 Vectoring nozzle angle and rate The vectoring nozzle shall have a maximum slew rate of at least _(a)_ degree/sec at _(b)_ flight condition. The nozzle shall provide a maximum vectored deflection angle of _(c)_ degrees from the symmetric axial thrust position with an accuracy of _(d)_ degrees over the entire deflection range. The nozzle shall vector without interruption, in a constant, steady movement.

3.7.12.2.2 Vectoring nozzle failure accommodation Failure accommodation for the nozzle system shall be described. Any system failure shall not preclude the nozzle from reverting to a fail–safe position.

3.7.13 Augmentation system If thrust/power augmentation system(s) are used, the system(s) shall be described in the subparagraphs below. This system shall be free of destructive operating modes and shall meet all requirements of the specification. The engine shall provide thrust/power augmentor operation, as required via the airframe signal, including initiation and termination, throughout the augmentor operating envelope. The amount of augmentation shall be controllable, and transition from non–augmented to augmented operation shall be free of large abrupt thrust/power changes. The augmentor shall not induce an unacceptable loss of engine operability, either in operation or during ignition or cancellation over the required envelope.

3.7.13.1 Afterburner (fuel injection) system If utilized, the afterburner system shall be described. Initiation of the afterburner system shall be accomplished by placing the thrust/power demand into any position in the afterburner range. Termination of the afterburner system shall be accomplished by placing the thrust/power demand in any position other than in the afterburner range.

3.7.13.2 Water injection system The water injection system, if utilized, shall be described.

3.7.13.2.1 Water injection system fluid If a water injection system is utilized, the water solution used shall be specified herein. The water solution shall use demineralized or distilled water, shall be visually clear, and shall conform to the following purity standards:

Impurities	Quality
Total Solids (max) ppm	35
Dissolved Solids (max) ppm	25
pH Factor	6.5 – 7.5

3.7.14 Wash system The wash system shall be described. An integral wash system shall be provided with the engine. The wash system shall not allow standing water to accumulate in the engine. The composition of the washing medium shall be in accordance with MIL–C–85704. Wash system interface connections shall be compatible with all existing Using Service support equipment.

3.7.15 Brake system Turboshaft or turboprop engines shall incorporate sufficient braking to prevent rotation of the propeller when exposed up to _(a)_ km/hr (_(a)_ mph) wind from any direction while grounded with the lower half of its projected area blanked off (and the engine shutdown), or while in flight with the propeller feathered at all airspeeds up to 0.5 Mach. The torque capacity of the output shaft brake shall be specified herein.

3.7.16 Negative torque limiter For turboshaft or turboprop engines, following an engine failure or malfunction a means to automatically prevent excessive negative torque shall be provided. The engine shall return to normal operation when the malfunction is corrected. The negative torque limiter shall be described.

3.7.16.1 Multiple power section engines For multiple power section turboshaft engines, a means shall be provided to automatically prevent an inoperative power section from absorbing power from the remaining operating power section(s) and the shaft power absorber at any flight condition or power setting. Inoperative engine(s) in the power section shall not result in the operating engine(s) exceeding design or operating limits. Similarly, failure(s) in the power absorber propulsor, gearbox/transmission, or drive system shall not result in the operating engine(s) or

power absorber system exceeding design and operating limits. A manual override shall be provided. A signal shall be provided to identify the section that has been separated. Separation and reconnection of a power section shall not cause any adverse affects on the performance and durability of the engine(s) and power train, and shall not affect safety of flight.

Multiple power section engines shall provide for load sharing and torque matching required by the engine control system in 3.7.2.

3.7.17 Power absorber (engine supplied) The engine supplied power absorber shall be described herein. The power absorber shall include any remote engine supplied mechanism which absorbs power from the engine cycle.

3.7.17.1 Feathering Engines using propellers shall withstand all loads and temperatures imposed by feathering of the propeller from any operating condition. All multi–engine applications shall incorporate auto–feather systems. If fuel shutoff is required, it shall be accomplished automatically by the control system. If engine/gearbox lubrication oil shut–off is required, it shall not occur until rotation has ceased.

3.7.17.2 Reverse pitch operation For engines using propellers, the operational capability and limits with the propeller in reverse pitch shall be described.

3.7.18 Gearbox The engine shall be equipped with a gearbox to provide mechanical power to components required for engine and aircraft operation and as a means to transmit torque to the engine for starting. The mechanical power characteristics at each pad shall be specified in Table V. The gearbox lubricant shall be the same type as the engine lubricating oil.

The accessory gearbox shall be capable of simultaneous operation of all the drives when each drive is subjected to (a) times the maximum permissible torque or power rating specified for the individual drive. The steady state and transient peak load capabilities shall be specified herein.

3.7.19 Tubing/plumbing system The tubing and plumbing system shall function satisfactorily while meeting the requirements of this specification.

3.7.19.1 Tubing/plumbing identification Method(s) for identifying tubing and plumbing lines shall be described. All methods shall be visible in red light conditions.

3.7.19.2 Tubing/plumbing clearances Clearance and supports shall prevent physical contact of tubing with other tubing, components, or structures. Clearance and support shall be adequate for wrenching, installation and removal.

3.7.20 Inlet particle separator (IPS) For turboshaft engines equipped with an engine supplied inlet particle separator, the IPS shall be described. The separator scavenge exhaust (exit) duct shall be located to prevent re–ingestion of separated particles and prevent interference or contamination with other systems. The engine shall meet all requirements of the specification while utilizing the IPS.

3.8 Software resources

3.8.1 Software integrity The software integrity shall be in accordance with (a).

3.8.2 Software performance and design The Contractor shall design, develop, test, document, and deliver operational, test, and support software programs to meet and maintain equipment performance specified herein. The detailed software requirements and computational components shall be described. All programming language shall be in accordance with MIL–STD–1815. If required by the weapon system contractor, the equipment shall provide an output over the multiplex busses which uniquely identifies the operational configuration of the equipment and software combination. The operational software shall perform all computations, data processing, and interfacing functions necessary for the engine to meet the performance and operability requirements of the specification.

3.8.2.1 Built–in test and inspectability Built–in–Test (BIT) shall be implemented to accomplish failure detection and failure isolation to the WRA and LRU without the assistance of support equipment at the organizational level. The testability of the engine system electronics shall be in accordance with MIL–STD–2084. The failure status provided by the BIT to the airframe, in accordance with the Interface Control Document (ICD), for determining mission capability, shall be specified herein. A nonvolatile record of BIT failure data shall be retained in memory. The equipment shall include provisions via data bus to allow both ROM and RAM contents of the subject computer to be inspected (monitored).

The BIT shall monitor both the thrust vectoring nozzle deflection rate and position and shall be capable of detecting failure of the thrust vectoring nozzle.

3.8.2.2 Computer reprogramming The engine system shall have the computer capability for making software changes without requiring removal of memory hardware, and without altering the aircraft installed configuration. Safety design provisions shall be provided to prevent unauthorized reprogramming. A description of the physical reprogramming procedure shall be provided. Each software version shall be identified by a unique software version number. This number shall be verified during the loading process.

3.8.3 Spare resources Prior to the completion of the _(a)_ engine Milestone, the electronic fuel control and engine monitoring system shall utilize no greater than _(b)_ and _(c)_ percent, respectively, of the available throughput during the worst case (most heavily loaded) execution of the operational software for each minor frame. The memory reserve shall be at least _(d)_ percent for control systems, and at least _(e)_ percent for engine monitoring systems.

3.9 Engine system controls and externals (C&E) evaluation

Note: Section 3.9 shall not contain requirements. Section 4.9 shall contain evaluation requirements for engine systems controls and externals that have not been specified in other sections of this specification.

3.10 Engine qualification For the purposes of qualifying the engine during Engineering and Manufacturing Development (E&MD), the engine shall be evaluated in accordance with Section 4.10.

Note: Section 3.10 shall not contain requirements. Section 4.10 shall contain evaluation requirements for engine tests that have not been specified in other sections of this specification.

4. EVALUATION AND QUALIFICATION Evaluation of the engine for acceptance for Initial Flight Release (IFR), Full Flight Release (FFR), Initial Service Release (ISR), Operational Capability Release (OCR), and for delivery shall be accomplished by inspection, analysis, demonstration and/or test as specified in this section. The tests required in this section shall be conducted to evaluate the engine design during each qualification milestone. Successful completion of individual evaluation criteria does not necessarily demonstrate satisfactory compliance of any requirement of Section 3. Detected failures or deficiencies during testing constitutes non–compliance with the evaluation paragraph and re–evaluation shall be performed to the satisfaction of the Using Service. Evaluation procedures and pass/fail criteria not specified in this section, shall be established by the Contractor and be subject to review and approval/disapproval by the Using Service, prior to the initiation of qualification milestones.

4.1 Design (configuration) and construction

4.1.1 Item and interface definition The requirements of 3.1.1 shall be evaluated by inspection.

4.1.1.1 Item diagrams and schematics The requirements of 3.1.1.1 shall be evaluated by inspection.

4.1.1.2 Item drawings The requirements of 3.1.1.2 shall be evaluated by inspection.

4.1.1.3 Interface and installation diagrams and schematics The requirements of 3.1.1.3 shall be evaluated by demonstration and inspection.

4.1.1.4 Interface and installation drawings The requirements of 3.1.1.4 shall be evaluated by inspection, demonstration, and test.

4.1.1.5 Installation changes The requirements of 3.1.1.5 shall be evaluated by inspection of interface documents and installation demonstration.

4.1.1.6 Installation hardware The requirements of 3.1.1.6 shall be evaluated by inspection and demonstration.

4.1.1.7 Connections/interchangeability The requirements of 3.1.1.7 shall be evaluated by inspection and demonstration.

4.1.1.8 Interface loads The requirements of 3.1.1.8 shall be evaluated by test.

4.1.1.9 Mounts

4.1.1.9.1 Main mounts The requirements of 3.1.1.9.1 shall be evaluated by analysis and test.

4.1.1.9.2 Ground handling mounts The requirements of 3.1.1.9.2 shall be evaluated by analysis, demonstration, and test.

4.1.1.9.3 Engine stiffness The requirements of 3.1.1.9.3 shall be evaluated by analysis, demonstration, and test.

4.1.1.10 Inlet connection stresses The requirements of 3.1.1.10 shall be evaluated by inspection and analysis.

4.1.1.11 Control system interface and integration The requirements of 3.1.1.11 shall be evaluated by inspection, demonstration, and test.

4.1.1.11.1 Thrust/power demands The requirements of 3.1.1.11.1 shall be evaluated by inspection, demonstration, and test.

4.1.1.11.1.1 Power lever angle (PLA) The requirements of 3.1.1.11.1.1 shall be evaluated by inspection and demonstration.

4.1.1.11.1.2 Power lever torque The requirements of 3.1.1.11.1.2 shall be evaluated by demonstration and test.

4.1.1.11.2 Control signals The requirements of 3.1.1.11.2 shall be evaluated by inspection, demonstration, and test.

4.1.1.11.3 Load demand lever The requirements of 3.1.1.11.3 shall be evaluated by demonstration and test.

4.1.1.11.4 Output speed lever The requirements of 3.1.1.11.4 shall be evaluated by demonstration and test.

4.1.1.11.5 Fuel shutoff lever The requirements of 3.1.1.11.5 shall be evaluated by inspection, demonstration, and test.

4.1.1.12 Electrical system interface

4.1.1.12.1 Instrumentation system interface The requirements of 3.1.1.12.1 shall be evaluated by inspection, demonstration, and test.

4.1.1.13 Engine monitoring system (EMS) interface The requirements of 3.1.1.13 shall be evaluated by inspection, demonstration, and test.

4.1.1.14 Fiber optics interface The requirements of 3.1.1.14 shall be evaluated by analysis, demonstration, and test.

4.1.1.15 Bleed air interface The requirements of 3.1.1.15 shall be evaluated by demonstration and test.

4.1.1.15.1 Customer bleed air contamination The requirement of 3.1.1.15.1 shall be evaluated by analysis and test.

4.1.1.16 Drains and fluid collection devices The requirements of 3.1.1.16 shall be evaluated by analysis, inspection, and test.

4.1.1.17 Power absorber interface The requirements of 3.1.1.17 shall be evaluated by analysis, inspection, and test. The maximum allowable misalignment of the power absorber shall be tested.

4.1.1.18 Power take–off (PTO) The requirements of 3.1.1.18 shall be evaluated by demonstration and test.

4.1.1.19 Starting drivetrain The requirements of 3.1.1.19 shall be evaluated by analysis and tests.

4.1.1.19.1 Starting torque and speed The requirements of 3.1.1.19.1 shall be evaluated by test and analysis.

4.1.1.19.2 Moment of inertia of rotating parts The requirements of 3.1.1.19.2 shall be evaluated by analysis, demonstration, and test.

4.1.1.19.3 Torsional spring constant The requirements of 3.1.1.19.3 shall be evaluated by test.

4.1.1.19.4 Starter train backlash The requirements of 3.1.1.19.4 shall be evaluated by inspection and test.

4.1.1.20 Thrust reverser interface The requirements of 3.1.1.20 shall be evaluated by inspection and demonstration.

4.1.1.21 Exhaust system interface The requirements of 3.1.1.21 shall be evaluated by analysis and demonstration.

4.1.1.22 Compatibility – engine/airframe The requirement of 3.1.1.22 shall be evaluated by test and demonstration.

4.1.1.23 Control and external (C&E) components list The requirement of 3.1.1.23 shall be evaluated by inspection and analysis.

4.1.1.24 Government furnished/loaned property The requirement of 3.1.1.24 shall be evaluated by inspection.

4.1.2 Physical characteristics

4.1.2.1 Dry mass (weight) of complete engine The requirement of 3.1.2.1 shall be evaluated by test.

4.1.2.2 Mass (weight) of residual fluids The requirement of 3.1.2.2 shall be evaluated by test.

4.1.2.3 Mass (weight) of operating fluid The requirements of 3.1.2.3 shall be evaluated by test.

4.1.2.4 Mass (weight) of additional equipment The weight of the equipment listed in 3.1.2.4 shall be evaluated by test.

4.1.2.5 Mass moment of inertia of complete engine The requirement of 3.1.2.5 shall be evaluated by analysis and test.

4.1.3 Materials, processes, and parts The requirements of 3.1.3 shall be evaluated by analysis and inspection. When applicable, engine materials, components, and parts shall be tested for fuel resistance using TT–S–735, type I and III test fluids.

4.1.3.1 Adhesives and sealants The use of adhesives and sealants shall be evaluated by demonstration and test.

4.1.3.2 Elastomeric materials The requirements of 3.1.3.2 shall be evaluated by inspections and tests.

4.1.3.3 "O" rings, seals, and packings The requirements of 3.1.3.3 shall be evaluated by inspection and test.

4.1.3.4 Corrosion protection The requirements of 3.1.3.4 shall be evaluated by demonstration and test.

4.1.3.5 Nonmetallic hose The requirement of 3.1.3.5 shall be evaluated by inspection, demonstration, and test.

4.1.3.6 Shot peening The requirement of 3.1.3.6 shall be evaluated by analysis, test, and demonstration via the structural integrity evaluations of 3.4.1, and as follows:

a. Prior to or concurrent with the preliminary design review, the contractor shall conduct the following analyses:

(1) A strength and life analysis that identifies the magnitude of life enhancement required from peening, assesses the potential for degradation in life enhancements during operation and support, and identifies appropriate safeguards against degradation including the establishment of a proposed inspection interval that accounts for likely degradation.

(2) A report describing planned manufacturing process controls and nondestructive inspection.

b. Life certification and qualification testing (along with the strength and life analysis) shall include evaluation of the proposed inspection interval.

c. All Engineering & Manufacturing Development and life certification or qualification testing shall be conducted on parts peened as per production requirements. The hardware configuration for testing necessary to evaluate a proposed inspection interval shall be approved by the Using Service prior to the start of testing.

4.1.4 Fasteners The requirements of 3.1.4 shall be evaluated by test.

4.1.4.1 Securing of fasteners The requirements of 3.1.4.1 shall be evaluated by inspection.

4.1.4.2 Clamps The requirements of 3.1.4.2 shall be evaluated by inspection and tests.

4.1.4.3 Screw recesses The requirements of 3.1.4.3 shall be evaluated by inspection and tests.

4.1.4.4 Screw threads The requirements of 3.1.4.4 shall be evaluated by inspection and demonstration.

4.1.4.5 Helical coil installation The requirements of 3.1.4.5 shall be evaluated by inspection and demonstration.

4.1.4.6 External wrenching elements The requirements of 3.1.4.6 shall be evaluated by inspection and demonstration.

4.1.5 Nameplate and product marking This requirement of 3.1.5 shall be evaluated by inspection.

4.1.5.1 Engine data plate marking The requirement of 3.1.5.1 shall be evaluated by inspection.

4.1.5.2 Warranty marking The requirements of 3.1.5.2 shall be evaluated by inspection.

4.1.5.3 Part number marking The requirement of 3.1.5.3 shall be evaluated by inspection.

4.1.5.4 Critical parts identification and tracking The requirement of 3.1.5.4 shall be evaluated by inspection.

4.1.6 Transportability The requirements of 3.1.6 shall be evaluated by analysis and test.

4.1.7 Interchangeability The requirements of 3.1.7 shall be evaluated by demonstration and test.

4.1.8 Safety The requirements of 3.1.8 shall be evaluated by analysis, inspection, demonstration, and test.

4.1.8.1 Flammable fluid systems The requirements of 3.1.8.1 shall be evaluated as follows: (a) .

4.1.8.2 Fire shield The requirements of 3.1.8.2 shall be evaluated by test.

4.1.8.3 Explosion–proof The requirements of 3.1.8.3 shall be evaluated by component test in accordance with MIL–STD–810.

4.1.8.4 Combustible fluid drains The requirements of 3.1.8.4 shall be evaluated by inspection and tests.

4.1.8.5 Air and gas leakage The air and gas leakage requirements of 3.1.8.5 shall be evaluated by test.

4.1.8.6 Ground safety The requirements 3.1.8.6 shall be evaluated by inspection.

4.1.8.7 Jet wake The requirements of 3.1.8.7 shall be evaluated by analysis and test. For vectoring nozzles, the full range of thrust vectoring angles in 5º increments shall be tested.

4.1.9 Design control

4.1.9.1 Standardization The requirements of 3.1.9.1 shall be evaluated by inspection and demonstration.

4.1.9.2 Design standards The requirements of 3.1.9.2 shall be evaluated by inspection.

4.1.9.3 Standard parts The requirement of 3.1.9.3 shall be evaluated by inspection.

4.1.9.4 Parts list The requirements of 3.1.9.4 shall be evaluated by inspection prior to the initiation of each milestone.

4.1.9.5 Assembly of components and parts The requirement of 3.1.9.5 shall be evaluated by demonstration.

4.1.9.6 Changes in vendors or fabrication process The requirements of 3.1.9.6 shall be evaluated by inspection.

4.2 Performance and operability

4.2.1 Performance characteristics The requirements of 3.2.1 shall be evaluated by inspection, analysis, demonstration, and test.

4.2.1.1 Performance and presentation – steady–state The requirements of 3.2.1.1 shall be evaluated by analysis, demonstration, and test.

4.2.1.2 Performance and presentation – transient The requirements of 3.2.1.2 shall be evaluated by analysis, demonstrations, and tests.

4.2.1.3 Performance computer program The requirements of 3.2.1.3 shall be evaluated by inspection, demonstration, analysis, and test.

4.2.1.4 Performance retention The requirements of 3.2.1.4 shall be evaluated by test.

4.2.1.5 Performance evaluation The requirements of 3.2.1.5 shall be evaluated by analysis, demonstrations, and test.

4.2.2 Operating characteristics and limits The requirements of 3.2.2 shall be evaluated by analysis, demonstrations, and tests.

4.2.2.1 Operating envelope The requirements of 3.2.2.1 shall be evaluated by analysis, demonstration, and test.

4.2.2.2 Operating attitude and conditions The requirements of 3.2.2.2 shall be evaluated by analysis, demonstration, and test.

4.2.2.3 Stopping The requirements of 3.2.2.3 shall be evaluated by demonstration, analysis, and test.

4.2.2.4 Low power conditions

4.2.2.4.1 Idle thrust/power – [ground/flight idle] The requirements of 3.2.2.4.1 shall be evaluated analysis, demonstration, and test.

4.2.2.4.2 No–load condition The requirements of 3.2.2.4.2 shall be evaluated by analysis, demonstration and test.

4.2.2.5 Stability The requirements of 3.2.2.5 shall be evaluated by analysis, demonstrations and tests. Testing shall be conducted under the following conditions: (a) . Tests shall be performed in each control mode and with control mode transfer. Stability audits shall be performed to identify marginal stability areas in the engine envelope.

4.2.2.6 Transients The requirement of 3.2.2.6 shall be evaluated by analysis, demonstration, and test.

4.2.2.6.1 Overshoot and undershoot The requirement of 3.2.2.6.1 shall be evaluated by analysis, demonstration, and test.

4.2.2.6.2 Droop The requirements of 3.2.2.6.2 shall be evaluated by analysis, demonstration, and test.

4.2.2.7 Windmilling The requirements of 3.2.2.7 shall be evaluated by analysis, demonstration, and test.

4.2.2.8 Reverse thrust The requirements of 3.2.2.8 shall be evaluated by test.

4.2.2.9 Steady–state and transient airflow The requirements of 3.2.2.9 shall be evaluated by analysis, demonstration, and test.

4.2.2.10 Inlet airflow distortion The requirements of 3.2.2.10 shall be evaluated by analysis, demonstration, and test throughout the flight envelope.

4.2.2.10.1 Pressure and temperature rate of change The requirement of 3.2.2.10.1 shall be evaluated by analysis, demonstration, and test.

4.2.2.11 Gas–path and measurement plane temperature limits The requirements of 3.2.2.11 shall be evaluated by analysis, demonstration, and test.

4.2.2.12 Surface temperature and heat rejection The requirements of 3.2.2.12 shall be evaluated by analysis, demonstration, and test.

4.2.2.12.1 Component limiting temperatures The requirement of 3.2.2.12.1 shall be evaluated by analysis, demonstration, and test.

A test shall be conducted on an instrumented engine in a simulated or actual engine bay.

4.3 Environmental conditions

4.3.1 Atmospheric conditions

4.3.1.1 Humidity The requirements of 3.3.1.1 shall be evaluated by analysis, demonstration, and test.

4.3.1.2 Fungus The requirements of 3.3.1.2 shall be evaluated by analysis, demonstration, and test.

4.3.1.3 Corrosive atmosphere The requirements of 3.3.1.3 shall be evaluated by engine test, on a new or newly overhauled engine, in accordance with table X and figure 18. Prior to starting the corrosion susceptibility test, the engine shall be disassembled sufficiently to inspect the surface condition of all parts normally exposed to atmospheric conditions. Detailed photographic coverage of these parts shall be taken. Upon reassembly and after an initial calibration, including data required to establish compliance with transients of 3.2.1.2, the engine shall be subjected to the cycles of 3.3.1.3 for 24 hours each in accordance with table X. Should engine performance during the test deteriorate more than the amount specified in 3.3.1.3 from that determined during the initial calibration, water washing shall be accomplished in accordance with 3.7.14. Water wash shall be used only to correct for performance loss. External water wash is not permitted. If performance cannot be recovered after water washing, the engine shall be disassembled and inspected to determine the effect of the corrosion testing on performance loss. If the unrecoverable performance loss is determined to be caused by a problem not related to corrosion, the engine shall be repaired and reassembled and the test continued. During the test, the engine shall be subjected to internal inspections after every tenth cycle to detect any evidence of corrosion or progression of corrosion of internal parts. Additional inspection may be conducted with approval of the Using Service.

After completion of the corrosion susceptibility test, the cleaning procedure of 3.7.14, Wash system, shall be accomplished prior to recalibration of the engine. During recalibration the applicable transient performance specified in 3.2.1.2 shall be demonstrated. Following recalibration, the engine shall be disassembled and inspected for evidence of corrosion. Detailed photographs shall be taken of all parts which show evidence of corrosion. Metallurgical analyses that completely characterize the types of corrosion found and test specimen evidence shall be available for review.

The corrosion susceptibility test will be considered to be satisfactorily completed when upon recalibration: (1) the engine non–recoverable performance deterioration at steady–state measured temperatures or engine pressure ratio (EPR) at Maximum Continuous and above does not exceed the deterioration specified in 3.3.1.3, for the same value of measured temperature or EPR obtained during the initial calibration, (2) the engine exhibits not more than the increase in specific fuel consumption specified in 3.3.1.3, for the same values of measured temperature or EPR obtained during the initial calibration at Idle and above, (3) the transients conducted are in accordance with the specification requirements, and (4) the extent of corrosion evident at test completion satisfies the following criteria for all engine parts:

a. All internal parts exposed to gas path air, upon cleaning, shall show no impairment of their function due to corrosion. Minor corrosive attack is acceptable only when the part design criteria (e.g., fatigue resistance) are not affected.

b. All corrosion–protected parts shall indicate no effects of corrosion upon cleaning and stripping of the protective schemes. Recoating of parts designed for recoating shall restore such parts to as–new condition.

c. All other parts shall show no corrosion that affects component integrity or prescribed maintenance procedures.

4.3.1.4 Icing conditions The requirements of 3.3.1.4 shall be evaluated by test as follows:

a. This part shall consist of two runs at each of several engine thrust/power settings under each of the conditions in Part 1 of table XI. The engine thrust settings shall be: Idle, 25, 50, and 75 and 100 percent IRP (Maximum Continuous). At each icing condition and at each thrust/power setting, the engine shall be operated for a period of not less than (a) minutes. During each period, at intervals after ice buildup, the engine shall be rapidly accelerated to Intermediate (Maximum Continuous) thrust/power to demonstrate acceleration response.

b. This part shall consist of a (b) minute run at Idle with no throttle movement, followed by an acceleration to Maximum thrust/power at the end of the period. During this run the engine shall be operated under the conditions listed in Part 2 of table XI. At each of the test thrust/power demand settings above, thrust/power shall be measured to evaluate performance loss, without any power lever movement.

If the engine incorporates an anti–icing system, the above tests shall be performed using the anti–icing system to demonstrate the requirements of 3.3.1.4.

4.3.2 Ingestion capability (hazard resistance)

4.3.2.1 Bird ingestion The requirements of 3.3.2.1 shall be evaluated by analysis, demonstration, and test. The contractor shall specify in the pretest data the critical target area for bird ingestion.

4.3.2.2 Foreign object damage (FOD) The requirements of 3.3.2.2 shall be evaluated by analysis, demonstration, and test.

Simulated foreign object damage shall be applied to the (a) critical stage blades at one or more sections of the (b) of the airfoil. The damage applied shall produce at least the stress concentration factor (K_t) of 3.3.2.2. Following the foreign object damage application, the damaged blades shall be tested to the life required in 3.4.1.5.1 and 3.4.1.5.2. At the completion of the test there shall be no evidence of blade failure or flaw sizes beyond values allowed by the in–service inspection flaw size of 3.4.1.7.3 as the result of the foreign object damage. Sufficient instrumentation for monitoring the structure of the engine shall be included in the test engine.

4.3.2.3 Ice ingestion The requirements of 3.3.2.3 shall be evaluated by analysis, demonstration, and test. The test procedure shall require the engine to run for (a) minutes following ice ingestion, before it is shut down for inspection. During the ingestion test, high speed photographic coverage shall be taken. Sufficient instrumentation for monitoring the structure of the engine shall be included in the test engine.

4.3.2.4 Sand and dust ingestion The requirements of 3.3.2.4 shall be evaluated by analysis, demonstration, and test. During the engine test, the coarse sand and dust shall be ingested first, with the fine particle sand and dust ingested afterward. An engine disassembly and inspection shall be conducted between the coarse and fine sand tests as specified by the Using Service.

For turbofan and turbojet engines, the engine shall be tested at Intermediate thrust, with sand and dust ingested at the concentration levels and for the length of time specified in 3.3.2.4. During each hour of operation, at least one deceleration to Idle and acceleration to Maximum augmentation shall be made, with power lever movements of 0.5

seconds or less. If an anti–icing system is provided, ten periods of one minute operation of the anti–icing system shall be performed during the first test hour. During the entire test, maximum customer bleed air shall be extracted from the engine. The customer bleed air shall be continually filtered and the total deposits measured and recorded. Following the post–test performance check, the engine shall be disassembled to determine the extent of erosion, and the degree to which the contaminant may have entered critical areas in the engine. The test will be considered satisfactorily completed when the criteria of 3.3.2.4 have been met and the teardown inspection reveals no failure or evidence of impending failure.

For turboshaft and turboprop engines, the engine shall be tested at Maximum Continuous with sand and dust ingested at the concentration levels and the length of time specified in 3.3.2.4. With the coarser sand, the engine shall be tested for 50 hours at Maximum Continuous power. With the finer sand, the engine shall be tested for 9 hours at Maximum Continuous, 27 hours at Intermediate, and 18 hours at Maximum, for a total of 54 hours. The engine shall be tested with the IPS (Inlet Particle Separator) if it is an inherent part of the engine design. Engines with an IPS should perform additional testing at reduced power settings to determine the effectiveness of the separator and engine vulnerability to sand and dust. During each hour of operation, at least one deceleration to Idle and acceleration to Maximum Continuous/Intermediate/Maximum shall be made, with power lever movements of 0.5 seconds or less. If an anti–icing system is provided, ten periods of one minute operation of the anti–icing system shall be performed during the first hour. During the entire test, maximum customer bleed air shall be extracted from the engine. The customer bleed air shall be continually filtered, and the total deposits measured and recorded. If an engine internal washing system is provided, it shall be demonstrated during the test. Following the post–test performance check, the engine shall be disassembled to determine the extent of sand erosion, and the degree to which sand may have entered critical areas in the engine. The test will be considered satisfactorily completed when the criteria of 3.3.2.4 have been met and the teardown inspection reveals no failure or evidence of impending failure.

4.3.2.5 Atmospheric liquid water ingestion The requirements of 3.3.2.5 shall be evaluated by analysis, inspection, and test. The engine shall start and operate satisfactorily from ground Idle to Maximum power with water ingested at varying rates of (a) to (b) percent with no airbleed, (c) to (d) percent with air bleed, and with a minimum of (e) percent (up to the limit of the engine) of total inlet mass flow in the form of water entering the engine inlet through a segment equivalent to one third the inlet area. Idle–max–Idle throttle transients shall be conducted at each test condition. The engine shall be operated at each condition for five minutes. During the test, the effects of the water ingestion on engine performance shall be noted and recorded.

Following the performance check, the engine shall be disassembled sufficiently for inspection. This test shall be considered to be satisfactorily completed when adequate clearances were maintained, no damaging or detrimental rubbing occurred during the test, the performance has not deteriorated, and the gas–flow path parts show no damage.

4.3.2.6 Armament gas ingestion (AGI) The requirements of 3.3.2.6 shall be evaluated by test.

4.3.2.7 Steam ingestion The requirements of 3.3.2.7 shall be evaluated by test. Steam shall be introduced into the engine inlet as 0.5, 1.0, 1.5 and 2.0% of the total mass flow. Initial temperature of the steam shall be 245°C (475°F) and gradually raised to obtain a total temperature rise of 100°C (212°F) in the 25% region of the inlet that received the steam.

4.3.3 Electromagnetic environmental effects (E^3) The requirements of 3.3.3 shall be evaluated by analysis, demonstration and test.

4.3.3.1 Electromagnetic interference (EMI) The requirements of 3.3.3.1 shall be evaluated by test in accordance with the following procedures of (a).

4.3.3.2 Intrasystem electromagnetic compatibility EMC The requirement of 3.3.3.4 shall be evaluated by test on the engine as installed on the aircraft

4.3.3.3 Intersystem electromagnetic compatibility EMC The requirement of 3.3.3.5 shall be evaluated by test on the engine as installed on the aircraft.

4.4 Integrity The requirements of 3.4 shall be evaluated by analysis, inspection, demonstration, and test.

4.4.1 Structural integrity The requirements of 3.4.1 shall be evaluated by analysis, inspection, demonstration, and test.

4.4.1.1 Design service life The requirements of 3.4.1.1 shall be evaluated by analysis, inspection, demonstration, and test.

4.4.1.1.1 Hot parts The requirement of 3.4.1.1.1 shall be evaluated by analyses and tests.

4.4.1.1.2 Cold parts The requirement of 3.4.1.1.2 shall be evaluated by analyses and tests.

4.4.1.1.3 Expendables The requirement of 3.4.1.1.3 shall be evaluated by analyses and tests.

4.4.1.1.4 Bearings The requirement of 3.4.1.1.4 shall be evaluated by analyses and tests.

4.4.1.1.5 Components The requirement of 3.4.1.1.5 shall be evaluated by analyses and tests.

4.4.1.2 Design usage The requirements of 3.4.1.2 shall be evaluated by analysis, inspection, demonstration, and test.

4.4.1.2.1 Internal environment The requirement of 3.4.1.2.1 shall be evaluated by analysis and test.

4.4.1.2.2 Externally applied forces The requirement of 3.4.1.2.2 shall be evaluated by analysis and test.

4.4.1.3 Material characterization The requirement of 3.4.1.3 shall be evaluated by analysis, test, and inspection.

4.4.1.4 Parts classification The requirement of 3.4.1.4 shall be evaluated by analysis and inspection.

4.4.1.5 Durability The requirements of 3.4.1.5 shall be evaluated by a strength and life analysis, inspection, demonstration, and part, component, and full–scale engine tests.

4.4.1.5.1 High cycle fatigue (HCF) life The requirements of 3.4.1.5.1 shall be evaluated by analysis and test. A stair–step test shall be conducted before and after the specified engine test(s). The engine shall undergo an increasing speed stair–step run consisting of (a) periods of (b) duration each, at equal increasing rotational speed increments beginning at Idle and continuing up to and including the maximum rotor speed.

4.4.1.5.2 Low cycle fatigue (LCF) life The requirements of 3.4.1.5.2 shall be evaluated by analyses and tests.

4.4.1.5.3 Creep The requirements of 3.4.1.5.3 shall be evaluated by analysis and test.

4.4.1.6 Strength The requirements of 3.4.1.6. shall be evaluated by structural analysis and part, component, and full–scale engine tests.

4.4.1.6.1 Factors of safety The requirements of 3.4.1.6.1 shall be evaluated by the analyses and tests.

4.4.1.6.2 Blade and disk deflection The requirements of 3.4.1.6.2 shall be evaluated by the analyses and tests.

4.4.1.6.3 Containment The requirements of 3.4.1.6.3 shall be evaluated by analysis and test.

4.4.1.6.4 Blade out The requirements of 3.4.1.6.4 shall be evaluated by analysis and test.

4.4.1.6.5 Overspeed/overtemperature The requirements of 3.4.1.6.5 shall be evaluated by analysis and test.

4.4.1.6.6 Disk burst speed The requirements of 3.4.1.6.6 shall be evaluated by analysis and test.

4.4.1.6.7 Output shaft torque limits The requirements of 3.4.1.6.7 shall be evaluated by analysis and test.

4.4.1.6.8 Output shaft speed limits The requirements of 3.4.1.6.8 shall be evaluated by analysis and test.

4.4.1.6.9 Pressure vessel/case The requirements of 3.4.1.6.9 shall be evaluated by analyses and tests.

4.4.1.6.10 Pressure balance The requirement of 3.4.1.6.10 shall be evaluated by analysis and test.

4.4.1.6.11 Gyroscopic moments The requirements of 3.4.1.6.11 shall be evaluated by analysis and test.

4.4.1.7 Damage tolerance The requirements of 3.4.1.7 shall be evaluated by analysis and test.

4.4.1.7.1 Residual strength The requirements of 3.4.1.7.1 shall be evaluated by the analyses and tests.

4.4.1.7.2 Initial flaw size The requirements of 3.4.1.7.2 shall be evaluated by analysis, inspection, demonstration, and test.

4.4.1.7.3 In–service inspection flaw size The requirements of 3.4.1.7.3 shall be evaluated by analysis, inspection, and test.

4.4.1.7.4 Inspection intervals The requirements of 3.4.1.7.4 shall be evaluated by analyses and tests.

4.4.1.7.5 Flaw growth The requirements of 3.4.1.7.5 shall be evaluated by analyses and tests.

4.4.1.7.6 Composites The requirements of 3.4.1.7.6 shall be evaluated by analyses and tests.

4.4.1.8 Vibration and dynamic response The requirements of 3.4.1.8 shall be evaluated by analyses and tests.

4.4.1.8.1 Vibration limits The requirements of 3.4.1.8.1 shall be evaluated by analysis and test.

4.4.1.8.2 Critical speeds The requirements of 3.4.1.8.2 shall be evaluated by analysis and test.

4.4.1.8.3 Blade, disk, and static structure vibration The requirements of 3.4.1.8.3 shall be evaluated by analysis and test.

4.4.1.9 Catastrophic failure The requirements of 3.4.1.9 shall be evaluated by analysis and test.

4.4.1.10 Sustained acceleration The requirement of 3.4.1.10 shall be evaluated by test.

4.4.1.11 Shock The requirement of 3.4.1.11 shall be evaluated by test.

4.4.1.12 External surface foreign object damage The requirement of 3.4.1.12 shall be evaluated by analysis, demonstration, and test.

4.4.2 Mechanical equipment and subsystem integrity The requirements of 3.4.2 shall be evaluated by analysis, inspection, demonstration, and test.

4.4.3 Avionic/electronic integrity The requirements of 3.4.3 shall be evaluated by analysis, inspection, demonstration, and test.

4.5 Reliability and maintainability (R&M) The requirements of 3.5 shall be evaluated by analysis, inspection, demonstration, and test.

4.5.1 Reliability The requirements of 3.5.1 shall be evaluated by inspection, analysis, demonstration and test.

4.5.1.1 Reliability quantitative requirements The requirements of 3.5.1.1 and tables XXIIa and XXIIb shall be evaluated by test and analyses in table XXIIIa. Progress toward meeting any operational reliability requirements shall be assessed with data from the contractor's Failure Reporting and Corrective Action System (FRACAS) during the development phase, but from the Using Service data collection system during the operational phase, when service data becomes available.

4.5.2 Maintainability The requirements of 3.5.2 shall be evaluated by analysis, demonstration, and test. Evaluation of qualitative and quantitative maintainability requirements shall be performed in accordance with MIL–STD–471, and the Maintainability Test Program shall be prepared by the Contractor and approved by the Using Service. Evaluations shall be conducted using production hardware, field tools and support equipment, protective clothing (e.g., chemical gear, cold weather gear, etc.) and by the Using Service's fifth (5th) percentile female stature through ninety–fifth (95th) percentile male stature.

4.5.2.1 Maintainability quantitative requirements The requirements of 3.5.2.1 shall be evaluated as follows:

a. Total maintenance man–hour/EFH shall be determined via demonstration and analysis, and shall be based on the following:

(1) Frequency of failure based on the most up–to–date reliability predictions, results of development testing and like–item field data where available.

(2) Maintainability demonstration task times as determined utilizing Using Service personnel and procedures for task times that are demonstrated. The number and skill level of personnel shall be consistent with current Using Service maintenance policy.

(3) Maintainability predictions for task times that are not demonstrated.

b. Mean time of repair shall be measured via demonstration.

c. Engine replacement time shall be measured via demonstration.

d. WRA/LRU replacement times shall be measured using established Using Service procedures.

e. Diagnostics:

(1) WRA/LRU fault isolation of installed engine components shall be evaluated by demonstration. This demonstration shall be based on failure modes which can be simulated. At least one failure mode shall be selected from each WRA/LRU which comprises at least the top 80 percent of the total WRA/LRU removal rate. Also, the top 30 failure mode drivers shall be included in the demonstration.

(2) Percent of success which diagnostic's tests correctly isolates faulty subassemblies shall be evaluated by demonstration. Data called for in table XXIV shall be documented during the demonstration.

(3) Percent of success in which diagnostic tests correctly diagnose fault by retest shall be evaluated by analysis of data gathered during development, documenting the number of times that the diagnostic testing properly isolated problems or gave false readings.

4.5.2.1.1 Excluded maintenance functions The requirements of 3.5.2.1.1 shall be evaluated by inspection and analysis.

4.5.2.2 Maintainability qualitative requirements The requirements of 3.5.2.2 shall be evaluated by demonstration and inspection.

4.5.2.3 Maintenance The requirements of 3.5.2.3 shall be evaluated by demonstration, analysis, and test. A reliability centered maintenance analysis in accordance with MIL–STD–2173 shall be the basis for establishing preventative maintenance requirements (scheduled maintenance). All maintenance actions (scheduled and unscheduled) shall be evaluated through demonstrations conducted as specified in 4.5.2.

4.5.2.3.1 Modules The requirements of 3.5.2.3.1 shall be evaluated by inspection and demonstration.

4.5.2.3.2 Maintenance, inspection, and repair cycle. The requirements of 3.5.2.3.2 shall be evaluated by demonstration.

4.5.2.3.2.1 Maintenance inspection techniques The requirements of 3.5.2.3.2.1 shall be evaluated by inspection and demonstration on an installed engine.

4.5.2.3.3 Tools The requirements of 3.5.2.3.3 shall be evaluated by inspection and demonstration.

4.5.2.4 Battle damage repair The requirements of 3.5.2.4 shall be evaluated by analysis and demonstration.

4.5.3 Human performance and human engineering The requirements of 3.5.3 shall be evaluated by demonstration and analysis.

4.6 Combat survivability Survivability development test and evaluation (DT&E) shall be conducted on the engine to evaluate critical survivability characteristics at specified development milestones.

4.6.1 Susceptibility The requirements of 3.6.1 shall be evaluated by analyses and tests.

4.6.1.1 Noise The requirements of 3.6.1.1 shall be evaluated by analysis and test.

4.6.1.2 IR radiation The requirements of 3.6.1.2 shall be evaluated by analysis, demonstration, and test. Contrast spectral radiant intensity measurements shall be acquired over the 1.5 to 14 micron region of the IR spectrum. The propulsion system shall be integrated into the airframe structure so that measured signature data accounts for IR radiation emitted and reflected from internal flow path components, exhaust nozzle convergent and divergent flaps, and external expansion ramp surfaces. Plume radiation shall be measured.

4.6.1.2.1 IR suppression system The requirements of 3.6.1.2.1 shall be evaluated by analysis and test.

4.6.1.3 UV radiation The requirements of 3.6.1.3 shall be evaluated by analysis and test.

4.6.1.4 Smoke The requirements of 3.6.1.4 shall be evaluated by analysis, demonstration and test in accordance with ARP 1179.

4.6.1.5 Gaseous emissions The requirements of 3.6.1.5 shall be evaluated by analysis, demonstration, and test in accordance with ARP 1256.

4.6.1.6 Fuel streaming/vapor puffing The requirements of 3.6.1.6 shall be evaluated by analysis and test.

4.6.1.7 Water vapor contrails The requirements of 3.6.1.7 shall be evaluated by analysis and test as follows: (a) .

4.6.1.8 Radar cross section (RCS) The requirements of 3.6.1.8 shall be evaluated by analysis, demonstration, and test. Static and dynamic measurements shall be evaluated by test on an uninstalled engine including engine inlet device (if required), and any engine components which may contribute to the RCS signature. RCS measurement shall be acquired either on a full scale inlet fixture or a “zero” duct length fixture with traceable correlation to the full scale inlet. RCS measurement shall be made on full scale engine hardware. Conic cut data shall be taken using either swept or stepped frequency techniques. The RCS measurement for engine nose and tail shall be 0, plus and minus 60 degrees

in azimuth as measured from the engine inlet or aft centerline. Data shall be measured 0, plus and minus 10, 20, and 30 degrees in elevation. In addition, median values for each frequency band, for 0, plus and minus 10, 30, and 60 degrees about the nose and tail shall be measured. RCS degradation shall be determined after each individual ingestion or durability test and shall not be cumulative.

4.6.1.9 Radar absorbent materials (RAMs) and coatings The requirements of 3.6.1.9 shall be evaluated with engines fitted with all the radar absorbent devices, RAMs, and RAM coatings specified for production, and that have satisfactorily completed the sand ingestion and corrosion tests of 4.3.2.4 and 4.3.1.3, respectively.

4.6.2 Vulnerability The requirements of 3.6.2 shall be evaluated by analysis and test.

4.6.2.1 Ballistic weapons The requirements of 3.6.2.1 shall be evaluated by a combination of analyses, transfer of data from other programs, and ballistic testing of replicas, components, systems, major assemblies and running engines under conditions that simulate realistic combat. The evaluation shall be structured to provide design information at times supportive of program design milestones, evaluate achievement of specification requirements, and evaluate the actual engine ballistic hardness achieved at a measure of expected combat performance. The test evaluation shall be tailored to support intended present, or future use of the engine in systems covered under the Live Fire Test Law (Title 10 USC 2366) and as discussed by DODI 5000.2, and DODD 5000.2M.

4.6.2.1.1 Vulnerability – components and externals The requirements of 3.6.2.1.1 shall be evaluated by analysis and Live Fire Test and Evaluation (LFT&E).

4.6.2.1.2 Static structure The requirements of 3.6.2.1.2 shall be evaluated by analyses, demonstration, and test.

4.6.2.2 Directed energy weapons (DEW) effects The requirements of 3.6.2.2 shall be evaluated by analysis and test.

4.6.2.3 Nuclear weapons effects The requirement of 3.6.2.3 shall be evaluated by the analyses, demonstrations, and tests.

4.6.2.4 Chemical/biological agent effects The requirements of 3.6.2.4 shall be evaluated by analysis, demonstration, and test. Material compatibility shall be shown by coupon tests or existing data, as applicable. Prior to any exposure to agents or decontaminants, the critical functions/properties of the engine shall be demonstrated to measure engine performance and material properties after the decontamination.

4.6.2.5 Fuel ingestion effects The requirements of the subparagraphs of 3.6.2.5 shall be evaluated by analysis and test.

4.6.2.5.1 Fuel ingestion – steady–flow The requirements of 3.6.2.5.1 shall be evaluated by test. Fuel shall be ingested at flow rates in steps of (a) with cooling periods of (b) between injections. Each fuel ingestion condition shall be conducted with the engine operating at (c). The engine shall be inspected for damage at periodic intervals. The fuel injection mechanism shall simulate airframe fuel system behavior and characteristics.

4.6.2.5.1.1 Fuel ingestion – cooling air contamination The requirements of 3.6.2.5.1.1 shall be evaluated by inspection and analysis.

4.6.2.5.2 Fuel ingestion – transient (“quick dump”) The requirement of 3.6.2.5.2 shall be evaluated by analysis and test. The engine shall ingest transient fuel, without permanent damage or performance degradation, of a stoichiometrically carbureted column of inlet duct air at a stagnation temperature of (a) °F and stagnation pressure of (b) psia at (c) thrust/power. The test shall subject the engine to an injection of fuel which effectively carburetes the complete inlet air column for a length corresponding to the amount of air contained between engine face and exhaust nozzle exit.

4.7 Subsystems

4.7.1 Anti–icing/de–icing system The requirement of 3.7.1 shall be evaluated by analysis, demonstration, and test.

4.7.2 Control system The requirements of 3.7.2 shall be evaluated by analysis, demonstration, and test.

4.7.2.1 Control system performance The requirements of 3.7.2.1 shall be evaluated by analysis, demonstration, and test.

4.7.2.1.1 Backup control (BUC) The requirements of 3.7.2.1.1 shall be evaluated by engine and component analysis, demonstration, and test.

4.7.2.2 Control system adjustments The requirements of 3.7.2.2 shall be evaluated by engine and component test and demonstration.

4.7.2.3 Overspeed protection system The requirements of 3.7.2.3 shall be evaluated by analysis, demonstration, and test.

For turboshaft/turboprop applications the engine shall be set up in a test cell with a shaft power absorber that can be instantaneously decoupled from the engine. The power turbine output shaft speed shall be adjusted to the highest speed with the engine operating at Maximum Continuous power setting. After reaching a stabilized condition, the shaft power absorber shall be instantaneously disconnected from the engine. This test shall be considered satisfactorily completed if the power turbine overspeed logic prevents destructive overspeed of the power turbine and no destructive overspeed of engine rotors.

4.7.3 Fuel system The requirements of 3.7.3 shall be evaluated by inspection.

4.7.3.1 Fuels The requirements of 3.7.3.1 shall be evaluated by test as required in the evaluation paragraphs below.

4.7.3.1.1 Primary fuel Unless noted otherwise within the specification, engines shall be evaluated using the worst case fuel(s) below: (a) .

4.7.3.1.2 Alternate fuel The requirements of 3.7.3.1.2 shall be evaluated by test.

4.7.3.1.3 Restricted fuel The requirements of 3.7.3.1.3 shall be evaluated by test.

4.7.3.1.4 Emergency fuel The requirements of 3.7.3.1.4 shall be evaluated by test.

4.7.3.2 Fuel system performance

4.7.3.2.1 Fuel contamination The requirements of 3.7.3.2.1 shall be evaluated by test.

4.7.3.2.2 Fuel system performance with external assistance The requirements of 3.7.3.2.2 shall be evaluated by demonstration and test.

4.7.3.2.3 Fuel system performance with no external assistance The requirements of 3.7.3.2.3 shall be evaluated by analysis, demonstration, and test.

4.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor The requirements of 3.7.3.2.4 shall be evaluated by test.

4.7.3.2.5 Fuel pump priming The requirements of 3.7.3.2.5 shall be evaluated by test.

4.7.3.2.6 Fuel lubricity The requirement of 3.7.3.2.6 shall be evaluated by analysis, demonstration, and test.

4.7.3.2.7 Fuel system performance with water saturated fuel The requirements of 3.7.3.2.7 shall be evaluated by analysis, demonstration, and test.

4.7.3.2.8 Fuel filter The requirements of 3.7.3.2.8 shall be evaluated by analysis, demonstration, and test.

4.7.3.2.9 Fuel flow limits The requirements of 3.7.3.2.9 shall be evaluated by analysis, demonstration, and test.

4.7.4 Electrical system

4.7.4.1 Electrical power The requirements of 3.7.4.1 shall be evaluated by demonstration and test.

4.7.4.1.1 Generator/alternator The requirements of 3.7.4.1.1 shall be evaluated by analysis, demonstration, and test.

4.7.4.2 Alternate/external electrical power The requirements of 3.7.4.2 shall be evaluated by demonstration and test.

4.7.4.3 Electrical connectors and cables The requirements of 3.7.4.3 shall be evaluated by analysis, demonstration, and test.

4.7.4.4 Electronic components The requirements of 3.7.4.4 shall be evaluated by analysis, demonstration, and test. The following test are required: (a) .

4.7.4.5 Electrical grounding The requirements of 3.7.4.5 shall be evaluated by analysis and test.

4.7.4.6 Electrical bonding The requirements of 3.7.4.6 shall be evaluated by inspection and test.

4.7.4.7 Ground isolation The requirements of 3.7.4.7 shall be evaluated by inspection and test.

4.7.4.8 Potting compounds The requirements of 3.7.4.8 shall be evaluated by demonstration, inspection, and test. Removable potting compounds shall be removed and replaced following the engine and component endurance test. The components shall then be functionally tested to ensure that no damage exists as a result of potting compound removal and replacement.

4.7.5 Ignition system(s) The requirement of 3.7.5 shall be evaluated by tests including fouling tests. The test procedure and schedule shall be as follows: (a) .

4.7.5.1 Ignition system fouling

4.7.5.1.1 Carbon fouling The requirement of 3.7.5.1.1 shall be evaluated by test.

4.7.5.1.2 Water fouling The requirement of 3.7.5.1.2 shall be evaluated by test.

4.7.6 Engine monitoring system (EMS) The requirement of 3.7.6 shall be evaluated by inspection, analysis, and test.

4.7.6.1 EMS fault detection/isolation The requirement of 3.7.6.1 shall be evaluated by inspection, demonstration, and test.

4.7.6.2 On–board engine diagnostic functions The requirement of 3.7.6.2 shall be evaluated by inspection, demonstration, and test.

4.7.7 Optical systems The requirements of 3.7.7 shall be evaluated by analysis, demonstration, and test.

4.7.7.1 Fiber optic cables The requirements of 3.7.7.1 shall be evaluated by analysis, demonstration, and test. All cables shall be tested for change in optical transmittance in accordance with EIA–455–20 following installation on the engine.

4.7.7.2 Fiber optic connectors The requirements of 3.7.7.2 shall be evaluated by analysis, demonstration, and test. In addition, the mating durability test of EIA–455–21 shall be performed on connectors that have been contaminated with the contaminants specified in 3.7.7.2 and subsequently cleaned.

4.7.7.3 Fiber optic passive couplers The requirements of 3.7.7.3 shall be evaluated by analysis, demonstration, and test.

4.7.7.4 Fiber optic high altitude radiation resistance The requirement of 3.7.7.4 shall be evaluated by analysis and test.

4.7.8 Lubrication system The requirements of 3.7.8 shall be evaluated by analysis, demonstration, and test.

4.7.8.1 Lubrication oil The requirement of 3.7.8.1 shall be evaluated by analysis, demonstration, and test.

4.7.8.1.1 Oil pressure and temperature limits The requirement of 3.7.8.1.1 shall be evaluated by analysis, demonstration, and test.

4.7.8.1.2 Oil consumption limits The requirement of 3.7.8.1.2 shall be evaluated by analysis, demonstration, and test.

4.7.8.1.3 Oil flow interruption/depletion The requirement of 3.7.8.1.3 shall be evaluated by test.

4.7.8.2 Lubrication system components and features

4.7.8.2.1 Oil reservoir The requirements of 3.7.8.2.1 shall be evaluated by analysis, inspection, and test.

4.7.8.2.1.1 Oil reservoir external features The requirements of 3.7.8.2.1.1 shall be evaluated by inspection, demonstration, and test.

4.7.8.2.2 Oil drains The requirements of 3.7.8.2.2 shall be evaluated by demonstration and inspection.

4.7.8.2.3 Oil filters The requirements of 3.7.8.2.3 shall be evaluated by demonstration and inspection.

4.7.8.2.4 Oil debris monitor The requirements of 3.7.8.2.4 shall be evaluated by analysis, demonstration, and test.

4.7.8.2.5 Oil coolers The requirements of 3.7.8.2.5 shall be evaluated by analysis, demonstration, and inspection.

4.7.9 Hydraulic system The requirements of 3.7.9 shall be evaluated by analysis, demonstration, and test.

4.7.9.1 Hydraulic ground test provisions The requirements of 3.7.9.1 shall be evaluated by analysis, demonstration, inspection, and test.

4.7.9.2 Hydraulic system fire and safety hazards The requirements of 3.7.9.2 shall be evaluated by analysis, demonstration, inspection, and test.

4.7.9.3 Hydraulic system air removal The requirements of 3.7.9.3 shall be evaluated by analysis, demonstration, and test.

4.7.9.4 Hydraulic fluid filters The requirements of 3.7.9.4 shall be evaluated by analysis, demonstration, inspection, and test.

4.7.10 Pneumatic system The requirements of 3.7.10 shall be evaluated by analysis, demonstration, inspection, and test.

4.7.10.1 Pneumatic air contamination The requirement of 3.7.10.1 shall be evaluated by analysis and test.

4.7.11 Starting The requirements of 3.7.11 shall be evaluated by analysis, demonstration, and test.

4.7.11.1 Ground starts The requirements of 3.7.11.1 shall be evaluated by analysis and test.

4.7.11.2 Air starts The requirements of 3.7.11.2 shall be evaluated by analysis and test.

4.7.11.3 Starting limits The requirements of 3.7.11.3 shall be evaluated by analysis and test.

4.7.11.4 Starting procedures The requirements of 3.7.11.4 shall be evaluated by demonstration and test.

4.7.11.5 Automatic relight The requirements of 3.7.11.5 shall be evaluated by demonstration and test.

4.7.12 Exhaust nozzle system The requirements of 3.7.12 shall be evaluated by analysis, inspection, and test. Nozzle structural integrity shall be tested during operation at maximum ram pressure conditions. Actuator scraper and seal design shall be evaluated for effectiveness and wear from engine generated debris in a heated environment.

4.7.12.1 Exhaust nozzle external asymmetrical air pressure loads The requirements of 3.7.12.1 shall be evaluated by analysis and demonstration for each engine and airframe combination.

4.7.12.2 Vectoring nozzle i i

4.7.12.2.1 Vectoring nozzle angle and rate The requirements of 3.7.12.2.1 shall be evaluated by demonstration and test.

4.7.12.2.2 Vectoring nozzle failure accommodation The requirement of 3.7.12.2.2 shall be evaluated by analysis, demonstration, and test.

4.7.13 Augmentation system The requirement of 3.7.13 shall be evaluated by analysis, demonstration, and test.

4.7.13.1 Afterburner (fuel injection) system The requirement of 3.7.13.1 shall be evaluated by analysis, demonstration, and test.

4.7.13.2 Water injection system The requirement of 3.7.13.2 shall be evaluated by analysis, demonstration, and test.

4.7.13.2.1 Water injection system fluid The requirements of 3.7.13.2.1 shall be evaluated by inspection and test.

4.7.14 Wash system The requirement of 3.7.14 shall be evaluated by demonstration and test.

4.7.15 Brake system The requirement of 3.7.15 shall be evaluated by demonstration and test.

4.7.16 Negative torque limiter The requirement of 3.7.16 shall be evaluated by analysis and test.

4.7.16.1 Multiple power section engines The requirement of 3.7.16.1 shall be evaluated by analysis, demonstration, and test.

4.7.17 Power absorber (engine supplied) The requirement of 3.7.17 shall be evaluated by analysis, demonstration, and test.

4.7.17.1 Feathering The requirement of 3.7.17.1 shall be evaluated by analysis, demonstration, and test.

4.7.17.2 Reverse pitch operation The requirement of 3.7.17.2 shall be evaluated by analysis, demonstration, and test.

4.7.18 Gearbox The requirement of 3.7.18 shall be evaluated by analysis, demonstration, and test.

4.7.19 Tubing/plumbing system The requirement of 3.7.19 shall be evaluated by analysis, inspection, and test. Natural frequencies of all tubing lines shall be examined and compared to dominant engine and installation operating speeds (gear pump speeds, blade passing frequencies, etc.).

4.7.19.1 Tubing/plumbing identification The requirement of 3.7.19.1 shall be evaluated by inspection and demonstration.

4.7.19.2 Tubing/plumbing clearances The requirement of 3.7.19.2 shall be evaluated by inspection, demonstration, and test.

4.7.20 Inlet particle separator (IPS) The requirements of 3.7.20 shall be evaluated by engine test.

4.8 Software resources

4.8.1 Software integrity The requirements of 3.8.1 shall be evaluated in accordance with (a).

4.8.2 Software performance and design The requirements of 3.8.2 shall be evaluated by inspection, analysis, demonstration, and test. The functional performance of the complete computer system shall be demonstrated after integration into the computer system hardware during all engine testing.

4.8.2.1 Built–in–test and inspectability The requirements of 3.8.2.1 shall be evaluated by demonstration and test. The equipment fault detection and isolation capabilities shall be evaluated per MIL–STD–2165.

4.8.2.2 Computer reprogramming The requirements of 3.8.2.2 shall be evaluated by demonstration and test.

4.8.3 Spare resources The requirements of 3.8.3 shall be evaluated by demonstration and test.

4.9 Engine system controls and externals (C&E) evaluation The engine system and C&E's shall be evaluated by tests in accordance with the following paragraphs.

4.9.1 Engine systems tests

4.9.1.1 Endurance tests The endurance tests in the subparagraphs below shall be required for the class of engines specified herein.

4.9.1.1.1 Accelerated mission test (AMT)

For turbofan and turbojet engines the following tests shall apply:

For IFR:

An engine conforming to the IFR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the engine accelerated mission duty cycles defined in figure 28, table XX, and tables XXXIIIa and XXXIIIb.

A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service.

The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

For FFR:

An engine conforming to the FFR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined in figure 28, table XX, and tables XXXIIIa and XXXIIIb. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after engine operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service. The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

For ISR:

An engine conforming to the ISR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined in figure 28, table XX, and tables XXXIIIa and XXXIIIb. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after engine operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service. The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

For OCR:

An engine conforming to the OCR configuration, but containing the same cold parts used in the ISR test, shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined in figure 28, table XX, and tables XXXIIIa and XXXIIIb. A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after engine operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service. The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

The test operation for all engine milestones shall be conducted as follows:

(1) A fixed level of engine inlet air temperature, fuel inlet temperature, and pressure shall be set for each mission cycle as defined by the Strength and Life Analysis Report and defined in figure 28, table XX, and tables XXXIIIa and XXXIIIb.

(2) Air passing over the engine shall correspond to the engine inlet air temperature for that mission condition, within facility constraints.

(3) Upon completion of each mission cycle, the engine shall be shutdown prior to the start of a subsequent cycle. During the shutdown period the engine shall be motored or windmilled until the _(h)_ measured temperature stabilizes below _(i)_ °C (_(i)_ °F), when the engine is not rotating.

(4) Condition monitoring and diagnostic system installed hardware and software shall be in the production configuration The condition monitoring and diagnostic system shall be operated in a manner analogous to its proposed service operation throughout the test and the output data shall be correlated to observations made during the test.

(5) The customer power takeoff drives and customer bleed air extraction level shall be representative of the maximum required at that condition.

(6) The engine anti–icing/de–icing system shall be actuated at periods during the test to simulate mission operation.

(7) The exhaust duct, inlet duct and bleed air connections shall be loaded to the maximum allowable loads specified in 3.1.1.8 and in the pretest data of 4.10.4.1.1 respectively, at the discretion of the Using Service.

(8) Fuel inlet temperature shall be maintained at the maximum fuel inlet temperature specified in 3.7.3.2.2 and 3.7.3.2.3 for at least _(j)_ percent of the duration of the mission oriented schedule.

(9) The thrust/power demand shall be advanced or retarded in not more than 0.50 seconds unless otherwise specified. The time for changing thrust shall be charged to the duration of the lower thrust setting.

(10) The oil pressure, if adjustable, shall be adjusted at the beginning of the test to the minimum steady–state value specified in 3.7.8.1.1. No further adjustments shall be permitted during the test except when authorized by the Using Service.

(11) The total quantity of fluid leakage from the engine drains shall be recorded and shall not exceed values specified in 3.1.1.16.

(12) Special engine control system features (e.g., overspeed, overtemperature) shall be cycled through their functions periodically as specified in the pre–test data of 4.3.5.1.

(13) At the completion of the test, the engine oil drain provisions specified in 3.7.8.2.2 shall be demonstrated.

(14) If the engine provides special features (i.e., IR suppressors, fuel heaters etc.), these items shall be activated during the test to simulate mission operation.

(15) This test is to be run using the distortion pattern(s) of 3.2.2.10.

(16) The test exhaust system (if different from vehicle flight exhaust) shall produce the same aerodynamics/thermodynamic cycle match as the flight exhaust of the primary using application.

4.9.1.1.2a Missionized durability test

For turbofan, turbojet, turboprop and turboshaft engines, the following tests apply:

For IFR:

An engine conforming to the IFR configuration shall be tested to the missionized cycles of figure 28, table XX, and table XXXIV for _(a)_ hours/cycles. The test shall be conducted so that the _(b)_ turbine temperature shall be equal to _(c)_ °C (_(c)_ °F) at thrust/power settings of IRP/Maximum Continuous and above.

For FFR:

An engine conforming to the FFR configuration shall be tested to the missionized cycles of figure 28, table XX, and table XXXIV for (d) hours/cycles. The test shall be conducted so that the (e) turbine temperature shall be equal to (f) °C ((f) °F) at thrust/power settings of IRP/Maximum Continuous and above.

For ISR:

An engine conforming to the ISR configuration shall be tested to the missionized cycles of figure 28, table XX, and table XXXIV for (g) hours/cycles. The test shall be conducted so that the (h) turbine temperature shall be equal to (i) °C ((i) °F) at thrust/power settings of IRP/Maximum Continuous and above.

For OCR:

An engine conforming to the OCR configuration shall be tested to the missionized cycles of figure 28, table XX, and table XXXIV for (j) hours/cycles. The test shall be conducted so that the (k) turbine temperature shall be equal to (l) °C ((l) °F) at thrust/power settings of IRP/Maximum Continuous and above.

4.9.1.1.2b Cyclic durability test

For turboshaft and turboprop engines, the following tests shall apply:

For IFR:

An engine conforming to the IFR configuration shall be tested for (a) hours of the durability cycles defined in table XXXV. The engine shall be tested such that the (b) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuel/oil of (c) and the shaft torque absorber characteristics of 3.7.17.

For FFR:

An engine conforming to the FFR configuration shall be tested for (d) hours of the durability cycles defined in table XXXV. The engine shall be tested such that the (e) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuel/oil of (f) and the shaft torque absorber characteristics of 3.7.17.

For ISR:

An engine conforming to the ISR configuration shall be tested for (g) hours of the durability cycles defined in table XXXV. The engine shall be tested such that the (h) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on two engines, using the fuels/oils of (i), respectively, and the shaft torque absorber characteristics representative of different vehicle applications, as applicable.

For OSR:

An engine conforming to the OSR configuration shall be tested for (j) hours of the durability cycles defined in table XXXV. The engine shall be tested such that the (k) turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on two engines using the fuels/oils of (l), respectively, and the shaft torque absorber characteristics representative of different vehicle applications, as applicable.

4.9.1.2 Altitude test The performance requirements of 3.2 shall be evaluated by test. An engine conforming to the same parts list and configuration as the endurance test engine of 4.9.1.1, in accordance with 4.10.3.2, shall be subjected to altitude tests at several selected conditions within the operating limits envelope specified in Figure 9 (P_2/T_2). Pre–test data shall be in accordance with 4.10.4.1.1. The test points shall include the effects of power extraction, bleed air extraction, inlet recovery, inlet distortion, anti–icing/de–icing, and windmilling on engine performance and stability. No adjustments shall be made to the engine without approval of the Using Service. The altitude tests shall be accomplished using the fuels and oils of 3.7.3.1.1 and 3.7.8.1, respectively. Fuel temperature shall be varied over a range sufficient to encompass all anticipated engine operating environments. If the engine has a continuous ignition system, it shall be in operation at all times after the start sequence has been completed. Data to be taken and recorded during the test shall be as specified in 4.10.4. Overall true RMS velocity measurements and acceleration spectrograms shall be obtained for each transducer mounted on the engine case and engine gearbox case at the engine speeds and thrusts selected for the test. The points selected shall include at least the altitude representative points and the point in the complete environmental conditions and operating envelope where the highest engine vibrational levels are generated. Critical components of the engine shall be identified on each spectrogram.

The following altitude test procedures apply to each of the following engine qualification milestones.

For IFR:

a. Altitude Performance Points. The tests shall consist of no fewer than (a) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than (b) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of (c) operating conditions and (d) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (e) core turnaround speeds as defined by the Using Service. A success rate of (f) shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at (g) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at (h) flight test condition(s) selected by the Using Service.

For FFR

a. Altitude Performance Points. The tests shall consist of no fewer than (aa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than (bb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of (cc) operating conditions and (dd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (ee) core turnaround speeds as defined by the Using Service. A success rate of (ff) shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at (gg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at (hh) flight test condition(s) selected by the Using Service.

For ISR

a. Altitude Performance Points. The tests shall consist of no fewer than (aaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than (bbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power, and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of (ccc) operating conditions and (ddd) distortion

patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eee) core turnaround speeds as defined by the Using Service. A success rate of (fff) shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at (ggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at (hhh) flight test condition(s) selected by the Using Service.

For OCR

a. Altitude Performance Points. The tests shall consist of no fewer than (aaaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than (bbbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power, and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of (cccc) operating conditions and (dddd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eeee) core turnaround speeds as defined by the Using Service. A success rate of (ffff) shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at (gggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at (hhhh) flight test condition(s) selected by the Using Service.

4.9.1.3 Engine systems tests pass/fail criteria The test shall be satisfactorily completed when the following conditions are met:

a. The engine shall meet the pass/fail criteria of the test plan.

b. The engine shall meet the inspection criteria of 4.10.3.5.5.

c. The engine and engine components meet the service limits of 4.10.3.4.1.

d. There are no part failures or any parts failing integrity requirements of 3.4.

e. All failures and problems encountered during the test, and any additional testing that may be required has been analyzed and resolved to the satisfaction of the Using Service.(i.e., penalty runs)

f. The test engine and all initial (start of test) components are operated satisfactorily to the end of the test and an analysis utilizing engine calibration results establishes that the performance characteristics of 3.2 and the starting requirements of 3.7.11 have been met.

g. Inspection of engine parts and engine components throughout the test, including the final inspection, shows that the parts life requirements of the Milestone have been met.

h. Test results indicate compliance with the reliability and maintainability requirements specified in 3.5.

i. The engine weight meets the requirements of 3.1.2.1.

j. The engine has not exceeded any limits of table VIII.

4.9.2 Controls and externals (C&E) component tests

4.9.2.1 C&E component operational tests Simulated mission operational tests to evaluate the life specified in 3.4.1.1 shall be conducted on all C&E systems and components. The ignition system shall be tested in accordance with 3.7.5. All C&E tests shall be conducted on test assemblies that have the same part number, consisting of groups of related components arranged and interconnected to simulate their normal relationship and function on the engine. However, subassemblies or components of a system may be tested separately if such separation does not prevent simulation of the complete function of the components or subassemblies. If practical, components shall be mounted in their normal position as mounted on the engine. No adjustments shall be made subsequent to the component calibration.

4.9.2.1.1 C&E component calibration Prior to the initiation of the C&E tests, each component which has input–output relationships shall be subjected to a calibration. The calibration shall be extensive enough to cover the engine steady–state and dynamic ranges of operation of the component and shall indicate conformance with the design tolerance range of the component. The engine control components shall be shown to conform to accuracy, stability and response requirement(s) stated in the component test procedure. Each calibration shall be recorded. Prior to initiation of the calibration, the contractor shall develop a list of components that do not require calibration. This list shall be subject to approval by the Using Service. Components not requiring calibration shall be operated under normal operating conditions to demonstrate satisfactory functioning.

4.9.2.1.2 C&E component test procedures Prior to the start of any testing, all components shall be cleaned of any corrosion–preventive compounds used for storage protection to simulate the condition of the part as installed on the engine. Test assemblies or components shall be subjected to operating loads simulating those encountered on the engine. Sufficient instrumentation shall be provided to indicate the performance of each component and to indicate that the functional relationships of components are maintained as required by the applicable test schedule. Functional checks shall be performed at the end of each test or group of tests and at other times, at the option of the Contractor, to indicate that no calibrated component has changed its calibration beyond allowable service limits and that the function of uncalibrated components is unimpaired. All components shall be tested with the primary operating fluids of 3.7.3.1.1, 3.7.8.1, and 3.7.9. All items driven by the engine gearbox shall be operated at maximum allowable axial and angular misalignment conditions at the drive pad. All components shall be subjected to the vibration test of 4.9.2.1.8.

4.9.2.1.3 C&E component missionized tests All engine components, excluding the ignition system, shall be subjected to simulated mission operational tests (or cycles) intended to produce the same equivalent component damage as would be achieved by real time tests based on the weapon system mission and mission mix of table VI. The simulated mission operational test cycle shall be defined by the contractor and the method of development shall be specified herein. The test shall include maximum and minimum fluid temperatures expected during engine operation and shall include maximum hot and cold soak temperatures of figure 4.

Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure and temperature values corresponding to those occurring throughout the range of engine operation. A sufficient number of test cycles, interspersed throughout the test, shall include air contaminated as follows:

a. The amount of engine lubricating oil shall be consistent with what has leaked into the flow path in previous engine testing.

b. A salt concentration of 0.2 parts salt (NaCl) per million parts of air by weight (salt shall be introduced using 4.0 percent water solution).

c. Distilled water to saturate the air at 52°C (125°F) at an ambient pressure of 14.7 psia.

d. Sand and dust concentration and particle size distribution of 3.3.2.4.

4.9.2.1.4 C&E component accelerated aging Upon completion of the C&E calibrations, all components containing age–sensitive non–metallic parts shall be placed in an air oven and maintained in an ambient temperature not less than the sea level hot ground soak temperature of Figure 4 for (a) hours. Components may be aged individually or in test assemblies.

4.9.2.1.5 C&E component recalibration, disassembly, and inspection Upon completion of the preceding tests, component calibrations of 4.9.2.1.1 shall be repeated. Components not subjected to calibration shall be operated under normal operating conditions to demonstrate satisfactory functioning. During recalibration, the same fluids, inputs, and operating cycles shall be used as in the calibration. Each recalibration shall be recorded. All components shall then be completely disassembled and inspected for indications of failure or excessive wear.

4.9.2.1.6 C&E component environmental test All engine electrical components or sub–components, including electrical connectors shall be subjected to environmental component testing in accordance with table XXXVI and as noted herein. These tests may be conducted on test assemblies or individual components, either of which may be new or not previously subjected to any testing. When a single hermetically sealed component is used in a series of the tests below, the component need not be disassembled for inspection until the last test of such series has been completed. In addition, hermetically sealed components need not be subjected to the explosion–proof, sand and dust, and fungus tests. Prior to disassembly, a test to determine hermetic seal integrity shall be performed. Failure of the hermetic seals during any test shall disqualify that component. The components shall be inspected for defects or damage which may have been incurred during any of the tests performed.

4.9.2.1.7 C&E component sand and dust All electronic components shall be subjected to a sand and dust test in accordance with MIL–STD–810. At the conclusion of the test, and prior to any cleaning of the components, four power supply transients in accordance with MIL–STD–704 shall be applied to the component. Criteria for passing the test shall be as defined in MIL–STD–810.

4.9.2.1.8 C&E component vibration All components shall be subjected to a vibration test in accordance with MIL–STD–810. Components may be tested in test assemblies or as individual units. During the test the component shall be subjected to its maximum limiting component temperature specified herein.

4.9.2.1.9 C&E component test pass/fail criteria The component tests shall be considered to be satisfactorily completed when, in the judgment of the Using Service:

a. During the tests, component performance and function were within established limits.

b. During the tests, there was no fluid leakage from any component other than that of a nature and rate specified in the specification.

c. During the tests, there was no hang–up or hesitation of any component.

d. Recalibrations indicate that no component has changed its calibration beyond allowable service limits.

e. The component teardown inspection shows no indication of failed, excessively worn, and distorted parts. Measurements shall be taken and compared with the contractor's drawing dimensions and tolerances or with similar measurements made prior to the test.

4.10 Engine qualification The engine shall be evaluated during each engine milestone by analysis, inspection, demonstration, and test in accordance with this section, and table XXXVII. Each evaluation in each milestone must pass its respective criteria and gain approval of the Using Service, prior to proceeding to the next milestone. All evaluations shall have quantified pass–fail criteria based on expected failure modes and this criteria shall be approved by the Using Service prior to initiation of each milestone.

4.10.1 Qualification milestones The engine shall be evaluated in accordance with table XXXVII for satisfactory completion of each of the engine qualification milestones defined below.

4.10.1.1 Initial flight release (IFR) The IFR milestone shall evaluate safe flight capability in a specific portion of the flight envelope. The acceptability of the engine design for IFR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVII and the approval of the Using Service.

4.10.1.2 Full flight release (FFR) The FFR milestone shall evaluate the engine's operational and performance capability throughout the entire flight envelope. The acceptability of the engine design for FFR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVI, and the approval of the Using Service.

4.10.1.3 Initial service release (ISR) The ISR milestone shall evaluate the initial operational capability and production configuration of the engine. The acceptability of the engine design for ISR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVII and the approval of the Using Service.

4.10.1.4 Operational capability release (OCR) The OCR milestone shall establish the production engine configuration for unlimited production release. The acceptability of the engine design for OCR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVII and the approval of the Using Service.

4.10.2 Evaluation responsibility The contractor is responsible for the performance of all evaluations (inspections, analyses, demonstrations, and tests) as specified herein. The contractor may use any facility suitable for the performance of the evaluation specified herein, unless disapproved by the Using Service. The Using Service reserves the right to perform any of the evaluations set forth in this specification when necessary to ensure supplies and services conform to prescribed provisions. Specific evaluations required may be waived by the Using Service provided the contractor submits documentation with appropriate justification for approval.

4.10.2.1 Responsibility for compliance All articles shall meet the requirements of sections 3, 4, and 5. The evaluations set forth in this specification shall become a part of the contractor's overall design and product evaluation program. The absence of any evaluation requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Using Service for acceptance comply with all requirements of the contract. Sampling inspections, as part of manufacturing operations, is an acceptable practice to ascertain conformance to the requirements, however, this does not authorize submission of defective parts, either indicated or actual, nor does it commit the Using Service to accept defective material.

4.10.2.2 Quality conformance inspection Test articles and test apparatus shall be subject to inspection by authorized Government officials who will be given the necessary information to determine conformance with this specification.

4.10.2.2.1 Analysis inspection Computer programs and calculations used in all analyses shall be subject to review and approval by the Using Service.

4.10.3 Manner of testing Unless otherwise noted, the following testing methods, conditions and procedures shall be required for evaluations for full engine system qualification in accordance with table XXXVII.

4.10.3.1 Test surveillance Each test and demonstration herein shall be subject to witnessing by authorized Using Service representatives. At convenient times prior to the tests and during teardown inspections, the engine and components shall be examined to determine if they conform to all requirements of the contract and specifications under which they were built. At no time shall any part of the engine or component be disassembled, adjusted, cleaned, replaced, or removed without prior approval of the Using Service representative.

4.10.3.2 Test article configuration and parts list Prior to the initiation of IFR, FFR, ISR, and OCR, the contractor shall submit a proposed parts list, which identifies the specific configuration, including any software logic version, for each test article. This parts list shall define the proposed engine configuration for each respective milestone. The parts list and configuration of each milestone test article and the differences from the proposed milestone parts list and configuration shall be identified and justified in the pretest data of 4.10.4.1.1 and shall be included in the detailed test procedure. The test article configuration for each evaluation conducted in accordance with table XXXVII shall be the approved engine qualification milestone parts list and configuration. The evaluations shall be modified, as appropriate, due to changes incorporated in the approved parts list and configuration, when required by the Using Service. The parts list for the engine configuration which successfully completes the IFR, FFR, ISR, and OCR evaluations shall be the approved parts list for the respective engine model. The mixing of parts of the same or different design such as different vendors components and parts in a multiple assembly like a segmented stator assembly or different designed blades in a disc is not allowed without Using Service approval. All parts shall be considered as having zero time at the start of a test, as appropriate, unless otherwise specified by the Using Service. While all parts do not have to be new, any part which fails the pass–fail criteria during the test shall be cause for rejection of that part or test article as determined by the Using Service. Both hardware and software are considered to be test articles. Hardware configuration shall be identified by part number and associated drawings. Software configuration shall be identified by a specific source code listing and associated logic diagrams.

4.10.3.2.1 Test article The particular article intended for a specific milestone test or demonstration shall be officially designated by the contractor prior to the start of the test.

4.10.3.2.2 Installation interfaces Inlet, exhaust, and air bleed connections shall be representative of airframe/engine installed environments. The engine power takeoff (PTO) shall duplicate the load distribution defined in 3.7.19

4.10.3.3 Test equipment

4.10.3.3.1 Automatic recording equipment Automatic steady–state and transient recording equipment shall be used to record data during the execution of those parts of the tests required to meet the objectives of each evaluation. Automatic continuous recording equipment shall be used for tests requiring the evaluation of test article variables versus time.

4.10.3.3.2 Vibration measuring equipment and response characteristics The vibration measurement and analysis equipment shall operate over a frequency band of at least 3 Hz to 10 kHz and produce acceleration spectrograms with a confidence level of 95 percent. For turbojet and turbofan engines the maximum allowable effective filter bandwidth of the spectrum analysis equipment shall be 30 Hz. For turboshaft and turboprop engines, the

maximum allowable effective filter bandwidth of the spectrum analysis equipment shall be 3 Hz up to 1,000 Hz and 6 Hz above 1,000 Hz. The equipment shall be calibrated as a complete system. The system frequency response, when calibrated by applying a known sinusoidal motion to the pickup, shall not deviate by more than ±3 dB from the known sinusoidal input at frequencies from 3 Hz to 10 kHz. If high bypass filters are required when measuring overall velocity levels, the filters shall not exceed 3 dB down at frequencies of 30, 70, or 110 Hz, as appropriate, with a roll–off of at least 18 dB per octave. High pass filters shall not be used to produce velocity/acceleration spectrograms.

4.10.3.3.3 Test stand dynamic characteristics The test stand shall not induce damaging or detrimental resonances into the engine at any test or operating condition.

4.10.3.3.4 Starter test Starting shall be performed with a starter that has torque characteristics within (a) percent of the minimum required torque shown on figure 3.

4.10.3.4 Test conditions

4.10.3.4.1 Oil servicing and consumption The oil system shall be serviced at the start of each test or demonstration. Any and all additions of oil added after initial servicing shall be recorded. Oil shall be drained only when authorized by the Using Service. The use of external oil filters shall not be permitted. Oil consumption shall be measured and the average consumption rate determined.

4.10.3.4.2 Fluid properties for test

4.10.3.4.2.1 Fuel properties Each test and demonstration shall be conducted using the primary and alternate fuels specified in 4.7.3.1.1 and 4.7.3.1.2, unless otherwise specified. If more than one primary fuel is specified, the fuel used shall be the fuel which makes the test the most difficult as determined by the Using Service.

4.10.3.4.2.2 Oil properties Each test and demonstration shall be conducted using the oils specified in 3.7.8.1. If more than one oil is specified, the oil used will be the oil that makes the test most difficult as determined by the Using Service.

4.10.3.4.2.3 Hydraulic fluid properties Each test and demonstration shall be conducted using the hydraulic fluids specified in 3.7.9.

4.10.3.5 Test, calibrations and procedures

4.10.3.5.1 Pretest evaluation

4.10.3.5.1.1 Pretest dry mass (weight) of engine Prior to initiation of the calibration, the dry weight of the engine, as specified in 3.1.2.1, shall be evaluated in accordance with 4.10.4.1.2. The dry mass (weight) shall be measured before the engine has been serviced with fuel and oil.

4.10.3.5.1.2 Pretest power lever torque For hydromechanical controls, prior to the test, power lever torque loads shall be measured for evaluation of 3.1.1.11.1.2.

4.10.3.5.2 Calibration

4.10.3.5.2.1 Control and engine monitoring system calibration Prior to the initiation of the engine calibration all components necessary for proper control and engine monitoring systems operation and scheduling shall be checked on the bench and on the engine to establish proper functioning and calibration over the applicable ranges of engine operating conditions. If thermocouples are used, the relationship between temperatures and output signal shall be in accordance with the National Bureau of Standards Monograph 125. The accuracy of the signals in relation to the actual measured temperature and transient time response characteristics shall be specified. All fuel nozzles and fuel carrying components of the engine control system shall undergo bench calibrations using fluid in accordance with MIL–C–7024.

4.10.3.5.2.2 Engine calibration The engine calibration shall be accomplished to establish the performance characteristics of the complete engine. Prior to the beginning of the calibration the engine may be cleaned using the wash procedure specified in 3.7.14 and engine controls may be adjusted but shall not be readjusted throughout the calibration. Calibrations shall be made initially with no customer power extraction and no bleed air extraction other than that required for continuous engine operation. Data required in 4.10.4.1.3 and 4.10.4.1.4 shall be recorded. During calibration, conformance with the leakage and shutdown drainage requirement of 3.1.1.16 shall be demonstrated. The engine shall be calibrated using fuel per 3.7.3.1.1 and oil per 3.7.8.1. The following shall be obtained:

a. Data to establish compliance with 3.2.1 and sea level performance of tables Ia, b and c. The use of outside ambient temperature (OAT), to calibrate the engine in lieu of tables Ia, Ib, and Ic conditions is acceptable with Using Service approval. (a) power settings shall be included in the performance calibration. For turboprop/ turboshaft engines, sufficient data to establish the accuracy of the torque sensor signals shall be recorded.

b. Data to establish compliance with 3.2.2.5 Stability, 3.2.2.6 Transients, and 3.7.11 Starting requirements at sea level static conditions. All transient performance data shall be used to update the transient status performance computer program of 3.2.1.2 and 3.2.1.3.

c. Repeat items “a” and “b” conditions with maximum permissible bleed air extraction for that condition. The maximum permissible bleed air extraction includes customer and anti–icing bleed air flow, if applicable.

d. Repeat item “c” with customer/accessory power extraction as specified by tables Ia, Ib, and Ic, except for starting.

e. Data required to characterize engine performance as a function of engine flight hours and cycles.

f. Data showing adequate distorted stall/surge margin remaining with (b) distortion pattern (screen) selected by the Using Service. In addition, data establishing the distorted stall/surge margin shall be determined at (c) engine thrust setting selected by the Using Service.

Additionally, for turboprop engines, the following shall be obtained with the engine fitted with the power absorbing device specified in the specification:

g. Data to establish compliance with 3.7.17.1 Feathering and 3.7.17.2 Reverse pitch operation and load/no load operation.

4.10.3.5.2.2.1 Customer bleed air analysis calibration. Prior to and during the (a) (select either altitude or endurance) test, a customer bleed air analysis shall be performed. The customer bleed air shall be sampled from each bleed air outlet during an Intermediate/Maximum Continuous power run. A sample of air entering the engine inlet shall be taken at the same time the bleed air samples are obtained. The samples shall be properly identified and processed through laboratory analysis to determine whether the contaminant levels are within the limits specified in 3.1.1.15.1.

4.10.3.5.3 Inspections, maintenance, and calibration The following inspections, maintenance and calibrations shall be performed:

a. Using established procedures, inspections shall be performed at intervals equivalent to the inspection periods specified in 3.4.1.7.4. Inspections and partial disassembles at other times during the test are permitted when warranted by performance deterioration and after approval from the Using Service is obtained.

b. Prior to and following each inspection interval, an abbreviated engine calibration may be performed. If performance deterioration exceeds 3.2.1.4, the engine may be cleaned. If cleaning is necessary, the washing system and procedure specified in 3.7.14 shall be used prior to initiation of the next test segment.

c. Maintenance shall be performed on a basis equivalent to the maintenance procedures specified in 3.5.2.3. Parts replacements during the test, except for consumables defined by maintenance instructions, shall not be permitted without approval of the Using Service. A record of all consumable and nonconsumable parts replaced during the test shall be maintained by the contractor. A failure analysis report shall be prepared for any failures resulting in the replacement or repair of non–consumable parts. Non–consumable parts replaced during the test shall require retest equivalent to the severity and test hours of the full engine test.

4.10.3.5.4 Recalibration After completion of the tests specified in 4.9.1, a recalibration in accordance with the requirements of 4.10.3.5.2.2 and 4.10.3.5.2.2.1 shall be made on the engine. The recalibration may be preceded by a run during which the cleaning procedure of 3.7.14 may be applied. The fuel and oil used shall be the same as those used during the initial calibration.

After completion of the engine recalibration, all components necessary for proper control and engine monitoring systems operation and scheduling shall be rechecked to establish proper functioning and calibration in accordance with 4.10.3.5.2.1.

The performance of these components shall meet their design service limits.

4.10.3.5.5 Engine disassembly and inspection For tests requiring engine disassembly the Using Service shall be notified of the inspection commencement date prior to each inspection. Prior to cleaning, the engine parts shall be given a “dirty” inspection for abnormal conditions. The dirty inspection shall be completed before any parts are cleaned. Parts shall then be cleaned and a “clean” inspection shall be performed. Part measurements shall be taken, as necessary, to determine excessive wear and distortion. Those measurements shall be compared with the manufacturer’s drawing dimensions and tolerances and with similar measurements made prior to the test. Non–Destructive Inspection (NDI) techniques shall be in accordance with 3.4.1.7. During the clean inspection a visual examination and condition assessment shall be conducted. The contractor shall maintain all data and results of non–destructive inspection tests. All NDI inspection data and findings shall be made available for Using Service review during the inspections. All data of 4.10.4 shall be made available for Using Service review upon request.

As a result of the inspection, parts shall be categorized as follows:

Category A. No repair required.

Category B. Repair required before further use.

Category C. Outside repair limits.

4.10.4 Data and reporting Data and reporting for tests conducted in this specification shall be as required below.

4.10.4.1 Types of data Data shall be recorded during tests in accordance with the following subparagraphs.

4.10.4.1.1 Pretest data Pretest data shall be made available to the Using Service for approval prior to the initiation of each test or demonstration. The data shall include:

a. Detailed test and maintenance plans. The detailed test and maintenance plans to be used by test and maintenance personnel during the conduct of the test. Test procedures shall be in accordance with the test plan.

b. Parts list and configuration of test article. Parts list and configuration of the test article and justification for any differences shall be in accordance with 4.10.3.2.

c. Service/Wear Limits. Service limits to be used by maintenance personnel in the field and repair facilities.

4.10.4.1.2 Preliminary data Photographs and other pertinent data shall be obtained and recorded at the time the test article is prepared for the test. For engines, dry mass (weight) and center of gravity shall be pertinent data. If the mass (weight) is measured after the engine has been serviced with fuel and oil and subsequently drained, the dry mass (weight) may be calculated by subtracting the mass (weight) of the residual fluids specified in 3.1.2.2 from the measured engine mass (weight).

4.10.4.1.3 Data – steady–state During operation at each specified steady–state condition and after performance stabilization,as a minimum data as specified in table XXXVIII shall be recorded.

4.10.4.1.4 Data – transient As a minimum, data specified in table XXXVIII shall be recorded.

4.10.4.1.5 Starting data During each start, data shall be recorded to evaluate requirements of 3.1.1.19.1, 3.7.11.3, and 3.7.11, as a minimum.

4.10.4.1.6 Accreditable test time Total test time for a period less than (a) minutes shall not be credited to the official run time, except when shorter periods are a test requirement.

4.10.4.1.7 Miscellaneous data The date, test title, test article designation, and serial number shall be recorded on each page of the log sheet.

4.10.4.1.8 Test notes Notes shall be added to the log sheets to record all incidents and irregular functioning of the test article and equipment and the corrective measures taken. All test article software changes shall be recorded on the log sheets.

4.10.4.2 Accuracy of data For all test article calibrations and tests or demonstrations, the reported data shall have a steady–state accuracy within the tolerances specified herein. The accuracy of transient data and the corresponding instrument calibration methods shall be described in the test plans. All instruments and equipment shall be calibrated as necessary to ensure the required degree of accuracy is maintained. No credit or debit shall be permitted for measurement uncertainty to establish compliance with the requirements of this specification. The contractor shall minimize measurement uncertainty errors. The instrument calibration methods shall be submitted to the Using Service for approval prior to the initiation of the test. The “as tested” accuracies as defined by the test facility shall be included in the test report.

4.10.4.2.1 Measurement uncertainty analysis A detailed measurement uncertainty analysis addressing all elemental precision and bias errors shall be performed in accordance with AEDC–TR–73–5 or equivalent approved by the Using Service. As a minimum, the analysis shall cover the three categories of errors: calibration errors, data acquisition errors, and data reduction errors. Analysis will be performed for the following measured parameters: airflow, fuel flow, torque, temperature, pressure, rotor speeds, (Ng & Np), SFC, output power, engine mass, and vibration.

4.10.4.3 Atmospheric data

4.10.4.3.1 Barometer reading The barometric reading shall be corrected for temperature and shall be recorded at intervals of (a) hours, but not exceeding three hours.

4.10.4.3.2 Relative humidity Relative humidity data readings shall be recorded at intervals of (a), but not exceeding three hours if required for performance calculations. The relative humidity shall be measured in the vicinity of the test article. A wet and dry thermometer or a sling psychrometer shall be used for these measurements.

4.10.4.4 Fluid data Samples of fuel, oil, and hydraulic fluid, shall be taken at the start and completion of each test or demonstration. The sample shall be analyzed for physical and chemical properties to determine conformance with applicable fluid specifications. The results of these analyses shall be included as part of the test report.

5. PACKAGING

5.1 Preparation for storage and shipment The engine shall be prepared for storage and shipment in accordance with MIL–E–5607. The level of preservation, the type of shipping container, and the furnishing of engine historical records shall be in accordance with contract requirements. A packing list shall be furnished with each engine and shall include all components and tools which are not installed, but are shipped with the engine.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use Engines covered by this specification are intended for air vehicle propulsion.

6.2 Acquisition requirements Acquisition documents will be tailored based on the information in the tailoring handbook and peculiar program requirements, they must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

6.3 Consideration of data requirements The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475–1 exempts the requirement for a DD Form 1423. DIDs that are required should be included in the contract.

6.4 Definitions. Definitions shall be used as specified below:

a. General definitions.

Accessories. Items of engine–mounted equipment not furnished by the engine contractor or airframe mounted equipment, which are required for aircraft operation or as auxiliaries for engine operation.

Alternate Fuel. Fuel on which the aircraft can be flown without operational restrictions but which can have long term durability or maintainability impact if used for continuous operation (multiple flights). Alternate fuels are used only on an occasional or intermittent basis.

Article. An individual item or particular unit which may be a component (e.g., fuel pump) or a system when considered as a whole (e.g., engine).

Automatic Ignition. Similar to manual ignition except the engine control activates the system. No pilot action is needed. In addition to the benefit and usage shown for the manual ignition system, the automatic system may be activated during decelerations at high altitudes, especially in the upper left corner of the flight envelope, or during rapid maneuvers of the aircraft.

Capture Area. Capture area is the projected physical area of the aircraft inlet.

Cold Engine. (for starting only) A “cold” engine is defined as one which has been allowed to windmill until the low pressure turbine exit gas temperature is within 55°C (100°F) of the engine compressor inlet temperature before a start is attempted.

Cold Parts. Those parts not listed as hot parts.

Control Limiting Temperature. The maximum value built into the engine control to limit the operating value at a commanded power setting and flight condition.

Continuous Ignition. A low energy level, low spark rate (relative to the system used to ignite the combustor during an engine start) ignition system that is constantly fired regardless of operating condition or power setting. No pilot action is needed to activate the system.

Critical Size. The crack size where unstable growth occurs.

Damage Tolerance. The ability of the engine to resist failure due to the presence of flaws, cracks, or other damage for a specified period of unrepaired usage.

Damage Tolerant. Resistant to failure due to the presence of flaws, cracks or other damage for a specified period of unrepaired usage.

Depot, Intermediate, or Base Level Inspectable Structure. Structure is depot, intermediate, or base level inspectable if the nature and extent of damage will be detected utilizing one or more selected NDI procedures.

Derating. Using an item in such a manner that the stresses applied during operation are lower than the stresses the item was designed to withstand.

Design Service Life. The life defined by the Using Service (used for design purposes) that the engine is expected to attain during operational service.

Design Usage. The engine usage conditions (used for design purposes) specified in 3.4.3.

Deteriorated Minimum Engine. Deteriorated minimum engine is the lowest thrust/power output and highest SFC after operation for the time period and environment specified in 3.2.1.4.

Diagram. A diagram is a sketch or outline giving only interface or mounting information for a part or part of an assembly without all the details of the item shown(i.e., diagrams show bolt hole patterns, flanges, electrical plug sockets, etc.). A diagram shows the relationship between the parts of a whole assembly. It does not show many details.

Dissimilar Control Technology. Other than digital electronic technology, e.g. hydromechanical, fluidic, analog.

Domestic Object Damage. Domestic object damage (DOD) is caused by engine parts coming loose and striking the engine. See foreign object damage (FOD).

Drawing. A drawing is a sketch showing major dimensions, clearances, and details of a part or assembly. It can show every detail you need to manufacture the part, assembly, or engine. Materials of composition are shown on the sketch.

Durable. Resistant to cracking, corrosion, deterioration, thermal degradation, delamination, wear, etc., for a specified period of time.

Durability. The ability of the engine to resist cracking, corrosion, deterioration, thermal degradation, delamination, wear, etc., for a specified period of time.

Durability Critical. Failure results in a significant economic impact to the system but will not necessarily impair flight safety or mission capability.

Durability Non–Critical. Failure results in a minor economic impact to the system but will not impair flight safety or mission capability.

Duty Cycle. A composite derived from the mission profiles and mission mix.

Dynamic Response. Response due to forced functions. Forced functions include: surge, stall, flutter, and resonance.

Emergency Fuel. Fuel which significantly limits the overhaul life of the engine and imposes operational restrictions on the aircraft as well.

Engine. The complete propulsion system (including all parts, components, and expendables) delivered by the engine contractor.

Engine Component. Items of equipment, furnished as part of and qualified with the engine, whose size, conformation, and dynamic and static characteristics are essential to attain the engine performance specified in the engine specification. Fuel pumps, engine controls, jet nozzle and actuators, anti–icing valves, and the temperature sensing system or devices are included in this category. Components may require separate qualification, calibration, and adjustment.

Engine Part. A piece, or two or more pieces joined together, that are not normally subject to disassembly without destruction of the designed use.

Fail–To–Trail. A failure mode where the thrust vectoring nozzle is automatically positioned into the zero vectoring angle. This is also known as the axisymmetrical position.

Fail Operational. The ability to detect any single or multiple failure(s) and to continue to operate the engine after detection and still be capable of meeting all defined mission and performance requirements.

Fail Degraded. The ability to continue to operate the engine after any single or multiple failure(s) and retain a defined minimum operational capability as defined in the ICD and the definition of levels of Flying Qualities.

Fail Safe. The ability to continue to operate the engine safely or safely shut–down following the failure or partial failure of one or more elements.

Flaw Growth Interval. The predicted typical propagation life required to grow an assumed initial flaw to its critical size.

Flight Hours. The time duration beginning at takeoff and ending at landing.

Foreign Object Damage. Foreign object damage (FOD) is caused by non–engine parts striking the engine. See domestic object damage (DOD).

Fracture Critical. Failure will result in probable loss of the aircraft or degradation in mission capability. There are two categories under fracture critical; safety critical and mission critical.

Frequency Response. The steady–state output of the system to input sinusoids of varying frequency. The output for a linear system can be completely described in terms of the amplitude ratio of the output sinusoid to the input sinusoid. The amplitude ratio, gain, and phase are functions of the frequency of the input sinusoid.

Global Plot. A global plot (polar plot) provides azimuth display on the angular scale, frequency displayed on the radial scale and RCS displayed as color levels.

Hot Engine. (for starting only) A “hot” engine is defined as one where a start is attempted within 10 seconds after a flameout or shutdown.

Hot Parts. Parts exposed to the hot gas stream.

Impending Failure. The condition of a part where loss of performance or function would take place if the part was continued to be used for its part life required by the specification.

In–Service Noninspectable Structure. Structure is in–service noninspectable if either damage size, accessibility, or maintenance requirements preclude inspection.

Inspection. The physical examination of an article, item, drawing, or part.

Installed Configuration. When the engine is tested in the airframe of the aircraft, this is known as the installed configuration.

Latent Failure. Any undetected condition which prevents the part or component from performing its intended function. This includes undetected conditions between missions and during missions.

Limit Load. The maximum load expected to be encountered when operated for the design service life at design usage conditions. The factor of safety associated with this load is defined as the limit load.

Loaded Disk. The condition where all blades are attached to the disk.

Maximum Allowable Temperature. The maximum allowable gas temperature, as specified for either steady–state or transient condition, is the limit beyond which operation of the engine is not allowed.

Maximum Operating Temperature. The highest temperature the engine can obtain in normal operation at a commanded power setting considering all effects due to engine–to–engine variation, deterioration, installation factors, control limits/tolerances, environment, and operating point.

Manual Ignition. Similar to continuous ignition except the ignitor is fired only when the system is activated by the pilot via switch. This system also uses a low energy level, low spark rate ignition system. The benefit of this system is the ignitor lasts longer than a continuous ignition ignitor since it's not used as often. This system would be used in instances where a flameout risk is high (i.e., rain or icing conditions and/or during takeoff or descent) or during a carrier approach under any condition.

Minimum Engine. "Minimum engine" is a performance level having the lowest thrust/power output and highest SFC at or over a time period and environment specified by the Using Service and considering all effects due to control system, engine–to–engine variation, operating time/deterioration, flight point, and operating condition.

Mission. Period beginning with the start of engine prior to flight and ending with the engine shutdown at the completion of the flight. Specific missions include: familiarization, air combat maneuvers, navigation, and air–to–ground weapons.

Mission Critical. Failure will generate a significant operational impact by degrading mission capabilities to the extent of creating an indirect safety impact on the weapon system or results in less than Level II handling qualities.

Mission Mix. The frequency each mission is flown in a finite period of time (e.g., 5% for familiarization and 25% for air combat maneuvers).

Mission Profile. A representation of a specific mission in terms of flight conditions and usage parameters.

New Minimum Engine. New minimum engine is the lowest thrust/power output and highest SFC when the engine is new.

Non–Operating Environment. The environment to which the engine is exposed during logistics/storage/maintenance/transportation activities.

Operational Life. The life of the engine when exposed to the operational usage.

Operational Usage. The usage that the engine is exposed to during actual service operation.

Primary Flight Control Surfaces. Movable aircraft surfaces which control the direction of flight. In most cases, they are airfoils that control the pitch, yaw, and lateral stability of the aircraft in flight. Rudder, aileron, elevator, or stabilator are considered primary flight control surfaces.

Primary Fuels. Fuels on which the engine is designed to operate continuously without restrictions.

Residual Strength. The load carrying capability of a part at any time during the design service life. Residual strength accounts for the presence of damage and the growth of damage as a function of design usage and must maintain limit load capability.

Restricted Fuel. Fuel which imposes operational restrictions on the aircraft when used.

Safety Critical. Failure will result in the probable loss of the aircraft or hazard to personnel due to direct part failure or by causing other progressive part failures.

Safety Limit. The predicted typical propagation life required to grow an assumed initial flaw to its critical size.

Satisfactorily. The words “satisfactorily” or “satisfactory” as used in this specification in conjunction with words or terms relating to operation or performance of the engine described in this model specification shall mean: Under the condition specified, throughout the entire operating envelope, the engine operating characteristics and performance are not affected, and the operating and physical limits shown in the specification are not exceeded and no permanent deformation or other damage to the engine occurs.

Satisfactory. The words “satisfactorily” or “satisfactory,” as used in this specification in conjunction with words or terms relating to operation or performance of the engine described in the model specification, shall mean: Under the condition specified, throughout the complete environmental conditions and operating envelope, the engine operating characteristics and performance are not affected, and the operating and physical limits shown in the specification are not exceeded and no permanent deformation or other damage to the engine occurs.

Small Engines. A small engine is an engine with fan inlet corrected flow requirements of 100 pps or less or compressor inlet corrected flow of 20 pps or less.

Smoke Puff. A smoke puff is visually detectable unburned fuel droplets that are produced when a combustor and/or afterburner is initiated (light–off) or terminated (shutdown).

Stall. Stall is an engine compression system flow instability which does not result in the loss of engine control. A condition in the engine compression system where one or more blades or stages is operating with separated flow.

Steady–state Stability. Steady–state stability is the condition at a constant power setting in which thrust fluctuations do not exceed a stated margin.

Strength. The ability of the engine to withstand limit loading conditions while maintaining operational capability.

Surge. Surge is the loss of engine control resulting from an engine compression system flow instability.

Surge Margin. Surge margin is the calculated and/or demonstrated tolerance of the engine to adverse operating conditions while maintaining the required steady–state and transient performance capability.

Throughput. The time required for a computer circuit to process the input signal.

Transient. Transient include augmentor light–off and termination, as applicable, as well as thrust/power demand increase and decrease.

Transient Stability. Transient stability is the condition following a transient in which thrust fluctuations do not exceed a stated margin.

Transparent to the Pilot. Any event the pilot can not detect. This includes any significant difference of aircraft handling ability or aircraft stability that may or may not occur by having the thrust vectoring nozzle assembly attached or not having a thrust vectoring assembly on the aircraft.

Ultimate Load. The maximum load that a material, part, component, or engine must withstand without catastrophic failure.

Unweathered Fuel. Fuel with chemical and physical properties that have not been degraded by environmental effects such as temperature and pressure cycles.

Usable Life. Life without repair or replacement.

Usage Parameters. Those parameters defining engine operation (e.g., ambient temperature, rotor speed, power/temperature dwell times, A/B lights, time at A/B, etc.).

b. Reliability definitions.

Reliability definitions shall be listed herein and consistent with MIL–STD–721 when possible.

6.5 Symbols, subscripts, abbreviations and acronyms. The symbols, subscripts, abbreviations and acronyms used in this specification are listed below and shall be in accordance with AS 681 and ARP 755.

A/B	Afterburner
A/C	Aircraft
ACGIH	American Conference of Government Industrial Hygienists
ADS	Automated Data System
AEDC	Arnold Engineering and Development Center
AEMS	Aircraft Engine Management System
AFB	Air Force Base
AFR	Air Force Regulation
AFSC	Air Force System Command
AGI	Armament Gas Ingestion
AIA	Aerospace Industries Association
AMS	Aerospace Material Specification
AMT	Accelerated Mission Test
AOA	Angle of Attack
APC	Aircraft Power Control
API	Armor Piecing Incendiary
APU	Auxiliary Power Unit
ARD	Aerospace Resource Document
ARP	Aerospace Recommended Practices
AS	Aerospace Standards
ASC	Aeronautical Systems Center
ASCII	American Standard Code for Information Interchange
ASIP	Aircraft Structural Integrity Program

ASME	American Society of Mechanical Engineers
ASMET	Accelerated Simulated Mission Engine Test
ASTM	American Society for Testing and Material
ASW	Anti Submarine Warfare
ATE	Automatic Test Equipment
ATF	Advanced Tactical Fighter
Aviation 3M	Aviation Maintenance–Material–Management System
AVIP	Avionics Integrity Program
BIT	Built in Test
BOCLE	Ball–on–Cylinder Lubricity Evaluator
BPI	Bytes Per Second
BUC	Backup Control
C&E	Controls and Externals
CAMS	Core Automated Maintenance System
CDRL	Contractor Data Requirements List
CEMS	Comprehensive Engine Management System
CERT	Combined Environmental Reliability Test
CFE	Contractor Furnished Equipment
CG	Center of Gravity
CL	Confidence Level
COMNAVAIRLANT	Commander Naval Air Atlantic Fleet
COMNAVAIRPAC	Commander Naval Air Pacific Fleet
CONUS	Continental United States
CR	Component Removal
CSCIs	Computer Software Configuration Items
dBsm	Decibels per sq meter
DDR&E	Deputy Director of Research and Engineering
DECU	Digital Electronic Control Unit
DEM/EVAL	Demonstration/Evaluation

DEW	Directed Energy Weapons
DID	Data Item Descriptor
DOD	Domestic Object Damage
DT&E	Development Tests and Evaluation
DTD	Damage Tolerant Design
E&MD	Engineering and Manufacturing Development
ECC	Engine Cycle Counter
ECOMTRAK	Engine Component Tracking System
ECP	Engineering Change Proposal
ECS	Environmental Control System
EDSA	European Distribution System Aircraft
EEPROM	Erasable Electronically Programmable Read Only Memory
EFH	Engine Flight Hours
EGT	Exhaust Gas Temperature
EIA	Electronic Industries Association
EM	Electromagnetic
EMC	Electromagnetic Compatibility
EMD	Engineering and Manufacturing Development
EME	Electromagnetic Environment
EMI	Electromagnetic Interference
EMP	Electromagnetic Pulse
EMS	Engine Monitoring System
EMT	Elapsed Maintenance Time
EMV	Electromagnetic Vulnerability
ENSIP	Engine Structural Integrity Program
EOH	Engine Operating Hours
EPR	Engine Pressure Ratio
ER	Engine Removal
ESS	Environmental Stress Screening

FADEC	Full Authority Digital Engine Control
FDDM	Frequency Division Demultiplexer
FDM	Frequency Division Multiplexer
FER	Failure Engine Removals
FETT	First Engine to Test
FFR	Full Flight Release
FITS	Fuel Ingestion Tolerance Specification
FMECA	Failure Modes, Effects and Criticality Analysis
FOD	Foreign Object Damage
FRACAS	Failure Reporting and Corrective Action System
FSD	Full Scale Development
FSII	Fuel System Icing Inhibitor
GdB	Giga decibel
GFE	Government Furnished Equipment
GGT	Gas Generator Temperature
HC	Hydrocarbon
HCF	High Cycle Fatigue
HEMP	High Energy Electromagnetic Pulse
HOL	Higher Order Language
HPT	High Pressure Turbine
IAS	Indicated Airspeed
ICD	Interface Control Document
IEC	International Electrotechnical Commission
IECMS	In–Flight Engine Condition Monitoring System
IFR	Initial Flight Release
IFSD	In–Flight Shut Down
IGV	Inlet Guide Vanes
ILS	Integrated Logistics Support
IPS	Inlet Particle Separator

IR	Infrared
IRP	Intermediate Rated Power
IRTP	Integrated Reliability Test Program
ISA	International Standard Atmosphere
ISR	Initial Service Release
JBD	Jet Blast Deflector
JLFT	Joint Live Fire Test
JOAP	Joint Oil Analysis Program
JTCG/AS	Joint Tactical Coordination Group on Aircraft Survivability
LCC	Life Cycle Costs
LCF	Low Cycle Fatigue
LD	Laser Diode
LE	Leading Edge
LED	Light Emitting Diode
LFT&E	Live Fire Test and Evaluation
LHV	Lower Heating Value
LI	Lubricity Improver
LO	Low Observable
LPT	Low Pressure Turbine
LRU	Line Replaceable Unit
LWC	Liquid Water Content
MA	Maintenance Action
MATV	Multi Axis Thrust Vectoring
MAU	Maintenance Actions Unscheduled
MECSIP	Mechanical Equipment Structural Integrity Program
MEFH	Mean Engine Flight Hours
MMH/EFH	Maintenance Man Hours/Engine Flight Hours
MNS	Mission Need Statement
MOPP	Mission Oriented Protective Posture

MTBF	Mean Time Between Failures
MTBMA	Mean Time Between Maintenance Action
MTTR	Mean Time to Repair
MULDEM	Multiplexer/Demultiplexer
NACA	National Advisory Committee for Aeronautics
NADEP	Naval Air Depot
NALDA	Naval Engine Logistics Data Analysis
NAPTC	Naval Air Propulsion Test Center
NATF	Navy Advanced Tactical Fighter
NAVAIR	Naval Air Systems Command
NAVSEA	Naval Sea Systems Command
NAWC	Naval Air Warfare Center
NAWCWD	Naval Air Warfare Center Weapons Division
NBC	Nuclear, Biological and Chemical
NBCCS	NBC Contamination Survivability
NDI	Non–destructive Inspection
NMC	Not Mission Capable
NOX	Nitrogen Oxide
OCR	Operational Capability Release
OEI	One engine inoperative
OMC	Organic Matrix Composites
OPNAVINST	Naval Operations Instruction
PFR	Preliminary Flight Release
PLA	Power Lever Angle
PNdB	Perceived Noise Level
POD	Probability of Detection
PPM	Parts Per Million
PTO	Power Takeoff
QAP	Quality Assurance Provision

QEC	Quick Engine Change
QPL	Qualified Products List
R&M	Reliability and Maintainability
RAM	Random Access Memory
RAM	Radar Absorbent Material
RCI	Reliability Critical Items
RCS	Radar Cross Section
RFP	Request for Proposal
RMS	Root Mean Square
RVP	Reid Vapor Pressure
SAE	Society of Automotive Engineers
SCN	Specification Change Notice
SDIP	Software Development Integrity Program
SFC	Specific Fuel Consumption
SHP	Shaft Horsepower
SI	International System of Units
SLS	Sea Level, Static
SNR	Signal to Noise Ratio
SRA	Service Replaceable Assembly
SRU	Service Replaceable Unit
STAR	System Threat Assessment Report
STR	System Threat Report
SURVIAC	Survivability and Vulnerability Information Analysis Center
TAC	Total Accumulated Cycles
TBO	Time Between Overhauls
TDDM	Time Division Demultiplexer
TDM	Time Division Multiplexer
TEMS	Turbine Engine Monitoring System
TJ/TF	Turbojet and Turbofan

TLV	Threshold Limit Value
TP/TS	Turboprop and Turboshaft
TSFC	Thrust Specific Fuel Consumption
USA	United States Army
USAF	United States Air Force
USC	United States Code
USN	United States Navy
UV	Ultraviolet
V/L	Vapor–Liquid Ratio
V/STOL	Vertical/Short Takeoff & Landing
VLF	Very Low Frequency
VPC	Vertical Parity Check
WOW	Weight of Wheels
WRA	Weapon Replacement Assembly
WRA/LRU	Weapons Replacement Assembly/Line Replaceable Unit
WSD	Wear Scar Diameter

6.6 Material safety data sheets Contracting officers shall identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED–STD–313. The pertinent Government mailing addresses for submission of data are listed in FED–STD–313.

6.7 Subject term (key word) listing

Engine, aircraft
Gas turbine engine, aircraft
Propulsion
Turbofan
Turbojet
Turboprop
Turboshaft

6.8 Metrication. The engine shall be of metric design in accordance with Public Laws 94–168 and 100–418. The engine shall conform to the SI Standard described in American Society for Testing and Material (ASTM) E380, Standard for Metric Practice (also numbered American National Standards Institute Z210.1) and Institute of Electrical & Electronics Engineer (IEEE) Standard 268, as listed in the DOD Index of Specifications and Standards. All new metric items (requiring a new part identification) or any new design practice created/designed to replace any customary system of measurement shall be integrated into the SI Metric System of Measurement in accordance with Public Laws 94–168 and 100–418. The specification shall contain both metric and English units. The metric units shall be primary. English units shall be provided within parenthesis.

6.9 Responsible engineering office. The office responsible for development and technical maintenance of this standard is ASC/ENFP, Bldg 125, 2335 Seventh St Ste 6, Wright-Patterson AFB OH 45433–7809. Requests for additional information or assistance on this standard can be obtained from ASC/ENFP, Wright–Patterson AFB OH 45433–7809; DSN 785–9590, Commercial (513) 255–9590. Any information obtained relating to Government contracts must be obtained through contracting officers. The responsible Navy engineering office is the Naval Air Systems Command, AIR–536, Arlington, VA 22243–5360, DSN 664–3290 Commercial (703) 604–3290. The responsible Army engineering office is the Aviation Troop Command, ATCOM (AMSAT–R–EPT), 4300 Goodfellow Blvd, St. Louis, MO 63120–1798, DSN 693–1644, Commercial (314) 263–1644.

6.10 Streamlining. For acquisitions, the required portions of this specification's tier reference documents, will be limited to the portion(s) referenced in the applicable paragraphs.

6.11 Spec tree. Due to the extensive number of references, a specification tree is not provided for this document. Only first tier documents are contractually binding.

6.12 Tables and figures Due to the extensive number of tables and figures, they are numbered according to their reference within Appendix A (in some cases they are not referenced in numeric order). All tables and figures follow Appendix A.

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

ENGINES, AIRCRAFT, TURBINE, HANDBOOK FOR

10. SCOPE. This handbook provides tailoring guidance and background information for individual paragraphs of the General Engine Specification (JSGS–87231A). The handbook has been developed and coordinated by technical personnel from the Air Force, Navy, and Army and has been approved as an official guide for use by Government personnel for assistance in tailoring the general engine specification for acquisition and model specifications.

RATIONALE

The scope of the specification must state the intended use of the document. The scope of the handbook must indicate that the handbook is to be used as a guide for Using Service personnel for tailoring the General Engine Specification for specific applications.

GUIDANCE

The following should be used to tailor the specification paragraph:

The Using Service should be designated and the type of document should be specified: i.e. acquisition or model specification. The Using Service should use “acquisition” in a request for proposal and “model” when the specification is ready to be signed with the contractor. The contractor should specify the engine model designation.

LESSONS LEARNED

None.

10.1 Structure The handbook structure replicates the structure of the specification except that it places each corresponding Section 3 requirement and Section 4 evaluation together.

RATIONALE

The General Engine Specification must be structured in accordance with MIL–STD–961 and MIL–PRIME format. For easier reference, each Section 3 and corresponding Section 4 paragraph is combined in the handbook text to show how individual requirements are to be evaluated.

GUIDANCE

Background:

Blanks in the General Engine Specification may be left open in the acquisition specification to allow flexibility for the contractors to meet or exceed certain requirements. This complies with acquisition streamlining initiatives by not over–specifying how to design and build an engine. The model specification that the contractor provides to the Using Service should have all blanks filled.

LESSONS LEARNED

None.

10.2 Handbook The handbook provides assistance for tailoring the General Engine Specification. Rationale is provided to justify the requirement in the specification. Guidance gives recommendations on how to tailor the specification paragraph. In cases where blanks appear in the specification paragraph, the handbook guidance paragraph provides recommended values or text that the Using Service may use to insert in the blank. When

contractors are expected to fill blanks, the Guidance paragraph will state that. The Using Service makes the final decision on whom fills in the blanks in the specification. Background is included in some Guidance paragraphs to provide additional information on the specification paragraph or on the recommendations in Guidance. Finally, Lessons Learned is provided to give valuable insight to events in the past that may impact the tailoring of the specification.

RATIONALE

This paragraph provides a summary of the handbook.

GUIDANCE

This paragraph should not be included in the acquisition or model specification. This paragraph should be labeled, “Not Used”, since the handbook is only applicable to the general specification.

Background:

The handbook is a dynamic document that should evolve with engine technology and mission requirements. Government and Industry personnel are encouraged to contact the custodians of the specification to provide new input for this document.

LESSONS LEARNED

Handbooks have been used to link past events and experiences with current design and performance requirements.

10.3 Deviations Any projected design for a given application which will result in improvement of system performance, reduced life cycle cost, or reduced development cost through deviation from this specification, or where the requirements of this specification result in compromise in operational capability, will not be considered for approval unless brought to the attention of the Using Service in writing.

RATIONALE

The contractor must have the means to specify deviations. It also allows for new concepts and technology transfer. This requires the contractor to submit, in writing to the Using Service, specification paragraphs they are unable to meet.

GUIDANCE

Background:

This deviation applies during the acquisition phase. After the acquisition phase, the deviations are handled by the ECP/SCN process.

LESSONS LEARNED

In the past, contractors have not notified the Using Service of known deviations before or during development and waited until the production phase to notify the Using Service.

10.4 Environmental impact statement The engine shall be designed so that its operation, maintenance, and repair may be accomplished without violating the most stringent of the environmental regulations applicable to locations where the engine is used or supported. Compliance with environmental regulations shall not prevent the engine from achieving and sustaining mission performance capability. Materials, processes, and environmental control equipment necessary to meet these environmental requirements must currently be available in the Using Service maintenance and supply systems. The design shall not use environmentally unsuitable materials such as ozone

depleting fluorocarbons, chlorofluorocarbons and halons, highly volatile organic compounds in solvents and coatings, asbestos containing materials, and other hazardous materials or pollutants that will be required to support the engine during development, production, operations, maintenance, and repair.

The contractor shall conduct an environmental analysis of the engine. The analysis shall include environmental life–cycle costs and cost–benefit analyses on alternate procedures and methods for maintenance and repair, material substitution, specialized materials, control technology, treatment processes, process changes, and disposal methods required to support the design.

RATIONALE

This paragraph requires that the engine and its support will comply with all environmental regulations and will not adversely affect the environment.

GUIDANCE

Background:

This requirement includes all federal, state, and local environmental regulations.

LESSONS LEARNED

NADEPs in California had problems with state laws in the use and disposal of certain fluids used in the repair of engines.

10.5 Responsible engineering office The responsible engineering office (REO) for this appendix is ASC/ENFP, Wright-Patterson AFB, OH 45433–7809, DSN 785–9590, commercial 513–255–9590.

20. APPLICABLE DOCUMENTS

20.1 General The documents listed in this section are needed to meet the requirements specified in sections 30, 40, and 50 of this appendix. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 30, 40, and 50 of this appendix, whether or not they are listed. Some of the documents listed in section 20 may be documents which have been cancelled. These documents are listed for informational purposes only; copies of these documents are not available from the preparing activity.

20.2 Government documents

20.2.1 Specifications, standards, and handbooks The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Note: R – Rationale, G – Guidance, B – Background, and L – Lessons Learned.

SPECIFICATIONS

Federal

TT–S–735	Standard Test Fluids, Hydrocarbon Ref. 4.1.3
VV–F–800	Fuel Oil, Diesel Ref. 3.7.3.1.4, B4.7.3.1.4

Department of Defense

MIL–G–3056	Gasoline, Automotive, Combat Ref. 3.7.3.1.4, B4.7.3.1.4, L3.7.3.1
MIL–E–5007	Engines, Aircraft, Turbojet and Turbofan, General Specification for Ref. L3.1.1.11.1.2, L3.2.1.3
MIL–B–5087	Bonding, Electrical and Lightning Protection, for Aerospace Systems Ref. G3.7.4.6, B3.7.4.6
MIL–E–5400	Electronic Equipment, Aerospace, General Specification for Ref. 3.7.4.4
MIL–T–5544	Thread Compound, Antiseize, Graphite–Petrolatum Ref. 3.7.5.1.1
MIL–G–5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145 Ref. L3.7.3.1, B4.7.3.1.4
MIL–E–5607	Engine, Gas Turbine, Preparation for Storage and Shipment of, Process of Ref. 5.1

MIL–T–5624	Turbine Fuel, Aviation, Grades JP–4, JP–5, and JP–5/JP–8ST Ref. G4.7.3.1.1, G4.7.3.2.4, G4.7.3.2.6, G4.7.3.2.7, G3.7.11.1
MIL–E–6051	Electromagnetic Compatibility Requirements, System Ref. 3.3.3.2, B4.3.3.2
MIL–C–7024	Calibrating Fluids, Aircraft Fuel System Components Ref. 4.10.3.5.2.1
MIL–F–7179	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapon Systems Ref. B3.1.3.1
MIL–S–7742	Screw Threads, Standard, Optimum Selected Series: General Specification for Ref. B.3.1.4.4
MIL–L–7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0–148 Ref. G3.7.8.1, G3.7.11.1, B3.7.8.1, B4.7.11.3, L4.10.3.4.2.2,
MIL–I–8500	Interchangeability and Replaceability of Component Parts for Aerospace Vehicles Ref. B4.1.9.5
MIL–S–8802	Sealing Compound, Temperature–Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion Ref. B3.1.3.1
MIL–S–8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for Ref. 3.1.4.4, R3.1.4.4, B4.1.4.4
MIL–D–12468	Decontaminating Agent, STB Ref. B3.6.2.4
MIL–F–16884	Fuel, Naval Distillate Ref. 3.7.3.1.4
MIL–F–17874	Fuel Systems: Aircraft, Installation and Test of Ref. 3.7.3.2.7, G4.7.3.2.7
MIL–S–19500	Semiconductor Devices, General Specification for Ref. 3.7.4.4
MIL–O–19838	Oil Systems, Aircraft, Installation and Test of Ref. B3.7.8.2.2
MIL–S–22473	Sealing, Locking and Retaining Compounds: (Single Component) Ref. B3.1.3.1

MIL–L–23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0–156 Ref. G3.7.8.1, G3.7.1.1, B3.7.8.1, L4.10.3.4.2.2
MIL–B–23964	Bolt, Self–Retaining, Positive Locking Ref. R3.1.4, G3.1.4
MIL–C–24621	Couplers, Passive, Fiber Optic, General Specification for Ref. 3.7.7.3, G3.7.7, G4.7.7.3, B3.7.7.3, B4.7.7.3
MIL–R–24720	Receivers, Digital, Fiber Optic, Shipboard (Metric), General Specification for Ref. G3.7.7
MIL–T–24721	Transmitters, Light Signal, Digital, Fiber Optic, Shipboard (Metric), General Specification for Ref. G3.7.7, B3.7.7
MIL–A–24726	Attenuators, Fiber Optic, Shipboard, General Specification for Ref. G3.7.7, B3.7.7
MIL–M–24731	Multiplexers, Demultiplexers, Multiplexers/Demultiplexers (MULDEMS), Frequency–Division, Fiber Optic Interfaceable, Shipboard (Metric), General Specification for Ref. G3.7.7, B3.7.7
MIL–T–24735	Transmitters, Light Signal, Analog, Fiber Optic, Shipboard (Metric), General Specification for Ref. G3.7.7, B3.7.7
MIL–M–24736	Multiplexers, Demultiplexers, Multiplexers/Demultiplexers (MULDEMS), Time–Division, Fiber Optic Interfaceable, Shipboard (Metric), General Specification for Ref. G3.7.7, B3.7.7
MIL–R–24737	Receivers, Light Signal Analog, Fiber Optic, Shipboard (Metric), General Specification for Ref. G3.7.7
MIL–W–25140	Weight and Balance Control System (for Aircraft and Rotorcraft) Ref. R3.1.2.1, R3.1.2.2
MIL–H–25579	Hose Assembly, Tetrafluoroethylene, High Temperature, Medium Pressure Ref. 3.1.3.5, B4.1.3.5
MIL–H–27267	Hose, Tetrafluoroethylene, High Temperature, Medium Pressure Ref. 3.1.3.5, B4.1.3.5
MIL–I–27686	Inhibitor, Icing, Fuel System Ref. B3.7.3.2.7

MIL–C–28876	Connectors, Fiber Optic, Circular, Plug and Receptacle Style, Multiple Removable Termini, General Specification for Ref. B3.7.7.2
MIL–C–38373	Cap, Fluid Tank Filler Ref. B3.7.8.2.1.1
MIL–M–38510	Microcircuits, General Specification for Ref. 3.7.4.4
MIL–C–38999	Connector, Electrical Circular, Miniature, High Density Quick Disconnect (Bayonet, Threaded and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for Ref. 3.7.4.3, B3.7.7.2
MIL–R–46082	Retaining Compounds, Single Component, Anaerobic Ref. B3.1.3.1
MIL–S–46163	Sealing, Lubricating and Wicking Compounds: Thread–Locking, Anaerobic, Single–Component Ref. B3.1.3.1
MIL–H–46855	Human Engineering Requirements for Military Systems, Equipment and Facilities Ref. 3.5.3
MIL–D–50030	Decontaminating Agent, DS2 Ref. B3.6.2.4
MIL–T–83133	Turbine Fuels, Aviation, Kerosene Types, NATO F–34 (JP–8) and NATO F–35 Ref. G4.7.3.1.1, G4.7.3.2.4
MIL–H–83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H–537 Ref. B3.7.9
MIL–C–83522	Connectors, Fiber Optic, Single Terminus, General Specification for Ref. B3.7.7.2
MIL–C–83526	Connectors, Fiber Optic, Circular, Environmental Resistant, Hermaphroditic, General Specification for Ref. B3.7.7.2
MIL–C–85045	Cables, Fiber Optics, (Metric), General Specification for Ref. 3.7.7.1, R3.7.7.1, B3.7.7.1
MIL–I–85470	Inhibitor, Icing, Fuel System, High Flash NATO Code Number S–1745 Ref. B3.7.3.2.7

MIL–C–85704	Cleaning Compound, Turbine Engine Gas Path Ref. 3.7.14
MIL–L–87100	Lubricating Oil, Aircraft Turbine Engine, Polyphenyl Ether Base Ref. L3.7.8.1
AFGS–87219	Electrical Power System, Aircraft Ref. B3.7.4.8
MIL–H–87227	Hydraulic Power Systems Ref. G4.7.9
AFGS–87233	Support Equipment System Ref. B3.1.6, B3.1.1.9.2, G3.1.6
MIL–A–87244	Avionic/Electronic Integrity Program Requirements (AVIP) Ref. G3.4.3, B3.7.4.4
AFGS–87249	Requirements for the Integrity of Mechanical Equipment and Subsystems (MECSIP) Ref. G3.4.2

STANDARDS

Department of Defense

MIL–STD–100	Engineering Drawing Practices Ref. 3.1.5.3
MIL–STD–130	Identification Marking of U.S. Military Property Ref. 3.1.5
MIL–STD–210	Climatic Information to Determine Design and Test Requirements for Military Systems and Equipment Ref. G3.2.1.1, B3.2.1, B3.2.1.1, B3.3.2.5
MIL–STD–454	Electronic Equipment, Standard General Requirements for Ref. B4.3.1.2
MIL–STD–461	Control of Electromagnetic Interference Emissions and Susceptibility, Requirements for the Ref. B3.3.3, B4.3.3, G3.3.3.1, B3.3.3.1, B4.3.3.1, B4.7.4.6
MIL–STD–462	Electromagnetic Interference Characteristics, Measurement of Ref. B4.3.3, G4.3.3.1, B4.3.3.1, B4.7.4.6
MIL–STD–470	Maintainability Program for Systems and Equipment Ref. 3.5.2, G3.5.2.2, B3.5.2
MIL–STD–471	Maintainability Verification/Demonstration/Evaluation Ref. 4.5.2, G3.5.2.2, B4.5

MIL–STD–704	Aircraft Electric Power Characteristics Ref. 3.7.4.1, 3.7.4.2, 4.9.2.1.7, R3.7.4.2, G3.7.4.1, G3.7.4.2, B4.1.8.3
MIL–STD–721	Definitions of Terms for Reliability and Maintainability Ref. 3.5, G3.5.2.1, B3.5.2.1
MIL–STD–756	Reliability Modeling and Prediction Ref. B3.5.1.1
MIL–STD–781	Reliability Testing and Engineering Development, Qualification, and Production Ref. G4.7.4.4, B4.5, B4.5.1, B3.5.1.1, B4.5.1.1, G4.5.1
MIL–STD–785	Reliability Program for Systems and Equipment Development and Production Ref. 3.5.1, 3.7.4.4, G3.5.1, G4.7.4.4, B3.5.1, B3.5.1.1
MIL–STD–810	Environmental Test Methods and Engineering Guidelines Ref. 4.1.8.3, 3.4.1.10, 3.4.1.11, 4.9.2.1.7, 4.9.2.1.8, G4.3.1.1, G4.3.1.2, G4.4.1.10, G4.4.1.11, B4.1.8.3, B3.3.1.1, B4.3.1.1, B4.3.1.2, B3.4.1.11
MIL–STD–838	Lubrication of Military Equipment Ref. B3.7.8.1
MIL–STD–841	Marking of Aircraft and Missile Propulsion System Parts Fabricated from Critical High Temperature Alloys Ref. B3.1.5.4
MIL–STD–882	System Safety Program Requirements Ref. 3.1.8, G3.5.1, B3.1.8
MIL–STD–889	Dissimilar Metals Ref. 3.1.3, L3.1.3
MIL–STD–961	Military Specifications and Associated Documents, Preparation of Ref. R4.10.2.1
MIL–STD–973	Configuration Management Ref. B3.4.1.7
MIL–STD–965	Parts Control Program Ref. G3.5.2.2, B3.1.5.4
MIL–STD–1388–1	Logistic Support Analysis Ref. G3.5.2.2
MIL–STD–1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities Ref. 3.5.3, G3.5.2.2, G3.5.3

MIL–STD–1515	Fastener Systems for Aerospace Applications Ref. B3.1.4
MIL–STD–1521	Technical Reviews and Audits for Systems, Equipments, and Computer Software Ref. G3.8.1, B3.8.1
MIL–STD–1529	Vendor Substantiation for Aerospace Propulsion System Items Ref. 3.1.9.6
MIL–STD–1553	Digital Time Division Command/Response Multiplex Data Bus Ref. B3.1.1.11, B3.1.1.13, B3.7.2, L3.7.6
MIL–STD–1559	Numbers, Serial, Aircraft Gas Turbine Engine and Engine Module, Assignment of Ref. B3.1.5.1
MIL–STD–1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems Ref. B3.3.1.3
MIL–STD–1629	Procedures for Performing a Failure Mode Effects and Criticality Analysis Ref. B3.5.1
MIL–STD–1773	Fiber Optics Mechanization of an Aircraft Internal Time Division Command/Response Multiplex Data Bus Ref. B3.7.2, G3.1.1.14
MIL–STD–1783	Engine Structural Integrity Program (ENSIP) Ref. B3.4.1
MIL–STD–1795	Lightning Protection of Aerospace Vehicles and Hardware Ref. B3.3.3.1
MIL–STD–1798	Mechanical Equipment and Subsystems Integrity Program Ref. B3.4.2
MIL–STD–1799	Survivability, Aeronautical Systems (For Combat Mission Effectiveness) Ref. G4.6.2.1, B4.6
MIL–STD–1800	Human Engineering Performance Requirements for Systems Ref. 3.5.3, G3.5.2.2, G3.5.3
MIL–STD–1803	Software Development Integrity Program (SDIP) Ref. G3.8.1, G4.8.1, B3.8.1
MIL–STD–1812	Type Designation, Assignment and Method for Obtaining Ref. B3.1.5.1

MIL–STD–1815	Ada Programming Language Ref. 3.8.2
MIL–STD–1818	Electromagnetic Effects Requirements for Systems Ref. 3.3.3.2, 3.3.3.3, G3.7.4.6, B3.3.3.1, B4.3.3.2
MIL–STD–2069	Requirements for Aircraft Nonnuclear Survivability Program Ref. G4.6.2.1, B4.6
MIL–STD–2084	Maintainability of Avionic and Electronic Systems and Equipment, General Requirements For Ref. 3.8.2.1, G3.8.2.1
MIL–STD–2165	Testability Program for Systems and Equipments Ref. 4.8.2.1, G4.8.2.1
DOD–STD–2167	Defense System Software Development Ref. G3.8.1, B3.8.1
DOD–STD–2168	Defense System Software Quality Program Ref. G4.8.1, B4.8.2
MIL–STD–2173	Reliability–Centered Maintenance Requirements for Naval Aircraft, Weapons Systems and Support Equipment Ref. 4.5.2.3

MS Standards

MS 9006	Recess–Cross, Low Torque Drive, Dimensions and Gage Dimensions for Ref. 3.1.4.3
MS 9825	Clamp, Loop–Cushioned, Joggled, Corrosion Resisting Steel, .204 Hole Ref. L3.1.4.2
MS 9826	Clamp, Loop–Cushioned, Joggled, Corrosion Resisting Steel, .265 Hole Ref. L3.1.4.2
MS 14169	Circular Spline and Adapter Details Engine Driven Accessories Ref. B3.7.18
MS 14184	Nonmetallic Shaft–Coupling Details Engine Driven Accessories Ref. B3.7.18
MS 18069	Compounds, Adhesive, Aircraft Design and Use Limitations for Threaded Components Retained, Sealed or Locked by Ref. R3.1.3.1
MS 24476	Adapter, Pressure Lubricating Oil Servicing, Aircraft Ref. B3.7.8.2.1.1

MS 33537	Insert, Screw Thread, Helical Coil, Inch Series, Coarse and Fine Thread, Standard Assembly Dimensions for Ref. 3.1.4.5, B4.1.4.5
MS 33540	Safety Wiring and Cotter Pinning, General Practices for Ref. 3.1.4.1, B3.1.4.1
MS 33588	Nuts, Self–Locking, Aircraft, Reliability and Maintainability Usage Requirements for Ref. 3.1.4.1, B3.1.4.1
MS 33602	Bolts, Self–Retaining, Aircraft, Reliability and Maintainability, Design and Usage Requirements for Ref. 3.1.4
MS 33750	Recess, High Torque and Conical High Torque, Dimensions of Recess, Gage, and Driver for Ref. 3.1.4.3
MS 33781	Recess – Torque–Set, Dimensions of Recess, Gage, and Driver for Ref. L3.1.4.3
MS 33787	Wrenching Element, External Spline, Dimensions for Ref. 3.1.4.6

HANDBOOKS

Department of Defense

MIL–HDBK–5	Metallic Materials and Elements for Aerospace Vehicle Structures Ref. B4.1.3
MIL–HDBK–189	Reliability Growth Management Ref. B4.5.1.1
MIL–HDBK–217	Reliability Prediction of Electronic Equipment Ref. G3.7.4.4
MIL–HDBK–235	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems Ref. 3.3.3.3
MIL–HDBK–781	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production Ref. B4.5.1
MIL–HDBK–784	Guidelines–Design to Minimize Contamination and to Facilitate Decontamination of Military Vehicles and Other Equipment: Interiors and Exteriors Ref. B3.6.2.4

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094, phone (215) 697–2667.)

RATIONALE

This paragraph is needed for a convenient list and location of all the documents referenced in the specification.

GUIDANCE

The following should be used to tailor the specification paragraph:

For the engine model specification this paragraph may read: "The following documents of the exact issue shown form a part of this specification to the extent specified herein."

Background:

The paragraph words are a MIL–STD–961 requirement.

LESSONS LEARNED

None.

20.2.2 Other Government documents, drawings, and publications Other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Department of Defense

DODD 5000.2–M	Defense Acquisition Management Documentation and Reports Ref. 4.6.2.1
DODI 5000.2	Defense Acquisition Management Policies and Procedures Ref. 3.6, 4.6.2.1, R3.6.1

(Application for copies should be addressed to Naval Publications & Forms Directorate, Attn: Code 301, 5801 Tabor Ave., Philadelphia PA 19120–5099)

Federal Aviation Regulation

FAR Part 25	Airworthiness Standards: Transport Category Airplanes Ref. B3.3.1.4
FAR Part 33	Airworthiness Standards: Aircraft Engines Ref. B3.3.2.5, B3.1.1.9.1
FAR part 36	Noise Standards: Aircraft Type and Airworthiness Certification Ref. B3.6.1.1

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office (GPO), Washington DC 20412)

National Bureau of Standards

Monograph 125	Thermocouples–Calibration Ref.–4.10.3.5.2.1

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office (GPO), Washington DC 20412)

National Oceanic and Atmospheric Administration

U.S. STANDARD ATMOSPHERE, 1976
Ref. 3.2.1, G3.6.1.2, G3.6.1.3, B3.2.1

U.S. STANDARD ATMOSPHERE, 1962
Ref. B3.2.1

(Application for copies should be addressed to NOAA Distribution Branch, N/CG33, National Ocean Service, Riverdale MD 20737)

Naval Air Instructions

NAVAIRINST 4870.2	Shot Peening of Aircraft Components Ref.–B3.1.3.6

(Application for copies should be addressed to Naval Publications & Forms Directorate, Attn: Code 301, 5801 Tabor Ave., Philadelphia PA 19120–5099)

Naval Air System Command

NAVSEA TE–000–AB–GTP–010	Parts Derating Requirements and Application Manual for Navy Electronic Equipment Ref. G3.7.4.4

(Application for copies should be addressed to Naval Publications & Forms Directorate, Attn: Code 301, 5801 Tabor Ave., Philadelphia PA 19120–5099)

Naval Operation Instruction

OPNAVINST 4790.2	The Naval Aviation Maintenance Program Ref.–B3.5.2.1, B3.5.2.3

(Application for copies should be addressed to Naval Publications & Forms Directorate, Attn: Code 301, 5801 Tabor Ave., Philadelphia PA 19120–5099)

Naval Sea Systems

NAVSEA 0967–LP–597–1011	Parts Application and Reliability Information Manual for Navy Electronic Equipment Ref. G3.7.4.4

(Application for copies should be addressed to Naval Publications & Forms Directorate, Attn: Code 301, 5801 Tabor Ave., Philadelphia PA 19120–5099)

RATIONALE

This paragraph is needed for a convenient list and location of the documents referenced in the specification.

GUIDANCE

None.

LESSONS LEARNED

None.

20.3 Non–Government publications The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation

American Society for Testing And Materials (ASTM)

ASTM D 439	Standard Specification for Automotive Gasoline Ref. B4.7.3.1.4
ASTM D 1655	Aviation Turbine Fuels Ref. G4.7.3.1.1, B4.7.3.1.1
ASTM E 380	International System of Units (SI) (the Modernized Metric System), Use of the (DoD adopted) Ref. G3.2.1.3

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103–1187.)

American Society of Mechanical Engineers (ASME)

ASME 68–GT–37	Problems and Solutions for Sand environment Operation of Helicopters Gas Turbines Ref. L3.3.2.4
ASME 70–GT–96	Environment Problems Affecting Helicopter Engine Operation Ref. L3.3.1.3, L3.3.2.4
ASME 84–GT–262	Aircraft Engine Control Mode Analysis Ref. B3.2.1.1

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York NY 10017.)

Electronics Industries Association (EIA)

EIA–455–20	FOTP–20 Measurement of Change in Optical Transmittance Ref. 4.7.7.1,
EIA–455–21	Mating Durability of Fiber Optic Interconnecting Devices Ref. 4.7.7.2, B4.7.7.2

(Application for copies should be addressed to Electronics Industries Association, 2001 Pennsylvania Ave Street, NW, Washington DC 20006.)

International Electrotechnical Commission (IEC)

IEC 179	Precision Sound Level Meters Ref. G4.6.1.1
IEC 225	Octave, Half–Octave, and Third–Octave Band Filters Intended for the Analysis of Sounds and Vibration Ref. G4.6.1.1

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York NY 10036.)

National Advisory Committee for Aeronautics (NACA)

NACA TN 1855	Recommended Values of Meteorological Factors to be Considered in the Design of Aircraft Ice–Prevention Equipment Ref. B3.3.1.4
NACA TN 2569	A Summary of Meteorological Conditions Associated with Aircraft Icing and a Proposed Method of Selecting Design Criterions for Ice–Protection Equipment Ref. B3.3.1.4

(Application for copies should be addressed to National Technical Information Service (NTIS), 5285 Port Royal Rd, Springfield VA 22161)

Society of Automotive Engineers (SAE)

Aerospace Information Report

AIR 713	Determining, Presenting, and Substantiating Turbine Engine Starting and Motoring Characteristics, Guide for Ref. B3.7.11.2
AIR 781	Determining Engine Starter Drive Torque Requirements, Guide for Ref. B3.7.11.2
AIR 1087	Aircraft Accessory Drag Torque During Engine Starts Ref. B3.7.11.2
AIR 1419	Inlet Total–Pressure–Distortion Considerations for Gas–Turbine Engines Ref. B3.2.2.10

AIR 1794	Ball–on–Cylinder (BOC) Aircraft Turbine Fuel Lubricity Tester Ref. G4.7.3.2.6

Aerospace Material Specification

AMS 2432	Shot Peening, Computer Monitored Ref. 3.1.3.6

Aerospace Resource Document

ARD 50015	A Current Assessment of the Inlet/Engine Inlet Temperature Distortion Problem Ref. B3.2.2.5, B3.2.2.10

Aerospace Recommended Practice

ARP 492	Aircraft Engine Fuel Pump Cavitation Endurance Test Ref. G4.7.3.2.4
ARP 755	Gas Turbine Engine Performance Station Identification and Nomenclature Ref. 3.2.1.3
ARP 949	Turbine Engine Starting System Design Requirements Ref. B3.7.11.2
ARP 1179	Aircraft Gas Turbine Engine Exhaust Smoke Measurement Ref. 4.6.1.4, L4.6.1.4
ARP 1256	Continuous Sampling and Measurement of Gaseous Emissions from Aircraft Turbine Engines, Procedures for Ref. 4.6.1.5, B4.6.1.5
ARP 1420	Gas Turbine Engine Inlet Flow Distortion Guidelines Ref. 3.2.2.10, G3.2.2.10, B3.2.2.5, B4.2.2.5, B3.2.2.10, L4.2.2.10
ARP 1587	Aircraft Gas Turbine Engine Monitoring System Guide Ref. B3.7.6
ARP 1797	Aircraft and Aircraft Engine Fuel Pump Low Lubricity Fluid Endurance Test Ref. G4.7.3.2.6
ARP 1870	Aerospace Systems Electrical Bonding and Grounding for Electromagnetic Compatibility and Safety Ref. B3.7.4.6

Aerospace Standard

AS 568	Aerospace Size Standard for O–Ring Sizes Ref. 3.1.3.3, G4.1.3.3
AS 681	Gas Turbine Engine Steady–State Performance Presentation for Digital Computer Programs Ref. 3.2.1.3, 60.5, G3.2.1.3
AS 848	Attachment Points, Engine, Remote Trimming, Main Fuel Control Ref. G3.7.2.2
AS 870	Wrenching Configuration, Double Hexagon (12 point), for Threaded Fasteners Ref. 3.1.4.6
AS 1055	Fire Testing of flexible Hose, Tube Assemblies, Coils, Fittings and Similar System Components Ref. G4.1.8.1, B4.7.9.2
AS 1606	Gas Turbine Starter (Jet Fuel Starter) Specification Ref. B3.7.11.1
AS 4273	In Draft Form, Not Released Ref. G4.1.8

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale PA 15096.)

Technical Reports

AEDC–TR–73–5	Handbook, Uncertainty in Gas Turbine Measurements Ref. 4.10.4.2.1
AEDC–TR–83–42	Guidelines for Validation of Turbine Engine Status Decks Ref. B4.2.1.1, B4.2.1.2
AFAPL–TR–74–64	Aircraft Exhaust Pollution and it Effect on the US Air Force Ref. B3.6.1.4
AFWAL–TR–83–2079	Weibull Analysis Handbook Ref. B3.5.1.1, B4.5.1.1, B4.4.1.5.2
FAA REPORT JAN 1971	Crash Fire Hazard Evaluation of Jet Fuels Ref. B3.1.1.15
NAVY Rsch Lab Rpt 9025	Aircraft Icing Mishaps on U.S. Navy Aircraft Ref. L3.3.1.4, L3.3.2.3
Kaman Report No. R–169	Amount of Dust Recirculated by a Hovering Helicopter Ref. L3.3.2.4

(Application for copies of test reports should be addressed to National Technical Information Service (NTIS), 5285 Port Royal Rd, Springfield VA 22161)

Legal Documents

Title 10 US Code 2366	Major System and Munitions Programs: Survivability Testing and Lethality Testing Required Before Full–scale Production (Live Fire Test Law) Ref. 4.6.2.1
	Federal Clean Air Act Ref. R4.6.1.5, B3.6.1.5

Nuclear, Biological and Chemical Documents

AR 70–71	Nuclear Biological and Chemical Contamination Survivability of Army Materiel Ref. G3.6.2.4, B4.6.2.4

(Application for copies should be addressed to the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111–5094, phone (215) 697–2667.)

	US Army Nuclear, Chemical Agency Letter DAMO–SW Ref. B3.6.2.4
FM 3–5	NBC Decontamination Ref. B3.6.2.4

(Application for copies should be addressed to the Commander, Army AG Publications Center, 2800 Eastern Blvd., Baltimore MD 21220)

Miscellaneous

AFR 80–36	Civil Airworthiness Standards for U.S. Air Force Transport Aircraft Ref. B3.6.1.1
AFR 161–35	Hazardous Noise Exposure Ref. B3.6.1.1

(Application for copies of regulations should be addressed to National Technical Information Service (NTIS), 5285 Port Royal Rd, Springfield VA 22161)

AFSC DH 1–4	Electromagnetic Compatibility Ref. B3.7.4.6

(Application for copies of AFSC DHs should be addressed to ASC/ENOSD, Bldg 125, 2335 Seventh St. Ste 6, Wright–Patterson AFB OH 45433–7809)

	Aviation Week Sept 14 1992 pg 76 Ref. B3.3.2.1
NATO STANAG 3824 F&L	Guide Specification (Minimum Quality Standard) for Aviation Gasoline Ref. B3.7.3.1.2

(Application for copies of NATO STANAG should be addressed to the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111–5094, phone (215) 697–2667.)

ADS–25	Engine Performance Data Ref. B3.2.1.3
ADS–37	Electromagnetic Environmental Effects (E^3), Management, Design, and Test Requirements Ref. 3.3.3.2, 3.3.3.3, B3.3.3.1, B4.3.3.2

(Application for copies of ADSs should be addressed to USA ATCOM, 4300 Goodfellow Blvd, St Louis MO 63120–1798.)

DATA LIST 4920–EG–081	Maintenance Trailer Ref. B3.1.6
SEA AIR 947	Engine Erosion Protection Ref. L3.3.2.4

(Application for copies of SEA AIRs should be addressed to Commander, NAVSEA, SEA 05Q42, Naval Sea Systems Command, 2531 National Center, Bldg 3, Washington DC 20362 .)

Joint NAVAIR 16–1–540/ Air Force TO 1–1–689	Avionic Cleaning and Corrosion Prevention/Control Technical Manual Ref. B3.1.3.4

(Application for copies of NAVAIRs should be addressed to Commanding Officer, NAVAIR Attn: AIR–51122, Naval Air Systems Command, 1421 Jefferson Davis Hwy, Arlington VA 22243)

Development of the Lycoming Inertial Particle Separator, Gas Turbine Conference & Product Show
Ref. L3.3.2.4

Evaluation of the Dust Cloud Generated by Helicopter Blade Downwash, Proceedings of the 7th Annual National Conference of Environmental Effects on Aircraft and Propulsion Systems
Ref. L3.3.2.4

Engine/Airframe Interface Considerations for Future Open–Ocean ASW Aircraft, Proceedings of the 7th Annual National Conference of Environmental Effects on Aircraft and Propulsion Systems
Ref. L3.3.2.5

Catapult Steam Ingestion Tests of Three Turbofan Engines in the A–7 Aircraft
Ref. L3.3.2.7

Steam Ingestion by Aircraft Gas Turbine Engines, Proceedings of the 7th Annual National Conference of Environmental Effects on Aircraft and Propulsion Systems
Ref. L4.3.2.7

RATIONALE

This paragraph is needed for a convenient list and location of all the documents referenced in the specification.

GUIDANCE

None.

LESSONS LEARNED

None.

20.4 Order of Precedence In the event of a conflict between the documents referenced herein and the contents of this specification, the content of this specification shall be considered a superseding requirement. In the event of a conflict within this specification, the contractor shall notify the Using Service in writing for disposition. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

RATIONALE

This paragraph is needed to create a hierarchy to prevent confusion and avoid conflicts between referenced documents.

GUIDANCE

The following should be used to tailor the specification paragraph:

The words of paragraph 20.4 should appear in the specification.

Background:

None.

LESSONS LEARNED

In previous programs there have been cases where acquisition specifications contained conflicting requirements and the engine contractor has chosen to design the least stringent requirement rather than consult the government position. As a result, the government was made aware of the conflict late in the program, but the contractor design was already finalized.

20.5 Streamlining This specification has been streamlined. Those documents listed in Appendix A which are required for acquisition have the same status as those referenced directly in section 2 (first tier documents). All other documents referenced through tiering, may be used for guidance and information.

RATIONALE

Acquisition streamlining is needed to promote innovative and cost–effective acquisition requirements and acquisition strategies that will result in the most efficient utilization of resources to produce quality weapons systems and products.

GUIDANCE

The following should be used to tailor the specification paragraph:

The words of paragraph 20.5 should appear in the specification used in requests for proposals (RFP). An RFP specification is a military specification.

Background:

Acquisition streamlining is based on the concept that by applying pertinent contract requirements and allowing early industry involvement in recommending the most cost–effective solutions, the Department of Defense can reduce the cost and time of system acquisition and life cycle cost without degrading system effectiveness.

This requirement comes from MIL–HDBK–800 paragraph on “Streamlining amendment/revision to a procurement specification”.

LESSONS LEARNED

None.

30. REQUIREMENTS The paragraphs of this section and all references herein comprise the complete set of engine requirements for the gas turbine engine(s) for the (a) airframe. Not withstanding the successful completion of the evaluation requirements of Section 4, the Contractor shall be fully responsible for all features, characteristics, and performance of the engine for all requirements in this section. All requirements that are classified shall be provided in a classified appendix, as part of this specification. Terms, symbols, acronyms, and definitions shall be in accordance with Section 6.

REQUIREMENT RATIONALE (30.)

The Using Service must have requirements for the engine to serve its purpose in the airframe.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should insert the airframe designation or delete any reference to the airframe.

REQUIREMENT LESSONS LEARNED

None.

40. EVALUATION AND QUALIFICATION Evaluation of the engine for acceptance for Initial Flight Release (IFR), Full Flight Release (FFR), Initial Service Release (ISR), Operational Capability Release (OCR), and for delivery shall be accomplished by inspection, analysis, demonstration and/or test as specified in this section. The tests required in this section shall be conducted to evaluate the engine design during each qualification milestone. Successful completion of individual evaluation criteria does not necessarily demonstrate satisfactory compliance of any requirement of Section 3. Detected failures or deficiencies during testing constitutes non–compliance with the evaluation paragraph and re–evaluation shall be performed to the satisfaction of the Using Service. Evaluation procedures and pass/fail criteria not specified in this section, shall be established by the Contractor and be subject to review and approval/disapproval by the Using Service, prior to the initiation of qualification milestones.

EVALUATION RATIONALE (40.)

All requirements must be evaluated and results judged by the Using Service.

EVALUATION GUIDANCE

Background:

The intention of this header paragraph is to specify that the evaluations are conducted for the purpose of progressing through a qualification program, giving the Using Service an indicator of engine capability and maturity at any given milestone. The qualification program (see 4.10) requires the Contractor to successfully conduct a set of singular or multiple engine and component tests, analyses, demonstrations, and inspections, as specified, at each milestone. The Using Service should determine the success criteria of each evaluation. Upon the development engine's successful completion of the evaluation requirements in each milestone, the Using Service should be reasonably assured of the engine's capabilities, and assess continued advancement of the development program into the following milestone. The evaluations specified have been thoroughly researched and developed to provide the Using Service with the best indicator of probable engine mission success while keeping qualification costs at a reasonable level. Individual tests or combination of tests in this section are normally conducted at specific conditions. They should be considered to be good demonstrators of the engine for qualification evaluation purposes only, and should not be considered as demonstrations for full compliance with Section 3 requirements.

EVALUATION LESSONS LEARNED

None.

3.1 Design (configuration) and construction

4.1 Design (configuration) and construction

3.1.1 Item and interface definition A brief description of the most notable design features of the engine shall be included in the model specification.

REQUIREMENT RATIONALE (3.1.1)

A description of the engine design is necessary as part of the total definition of the engine.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The item definition should include, but not be limited to, a description of such components as integral particulate separator; compressor including number of stages, variable geometry provisions, starting and acceleration bleeds, airflow bypass ratio; type of combustor, including methods of fuel injection; turbine components, including number of stages, cooling provisions; augmentation provisions, exhaust nozzle, including variable geometry features; thrust reversers; speed decreaser engine control features; gearbox provisions; type of lubrication and scavenge system; fuel and electrical systems; starting and ignition system; number and location of main rotor bearings; instrumentation and performance indicating provisions, and location of borescope ports and any other feature deemed essential .

REQUIREMENT LESSONS LEARNED

Past engine specification have almost always included an item and interface description.

4.1.1 Item and interface definition The requirements of 3.1.1 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.1)

The engine must be inspected to ensure that the written description adequately describes the engine.

EVALUATION GUIDANCE

Background:

Each detail of the engine described in the model specification should be compared with the corresponding feature of the engine to evaluate whether the description accurately and completely describes the engine. The Using Service should compare the specification and ICD Item Definition requirements for consistency.

EVALUATION LESSONS LEARNED

Engine descriptions in the past have not always been complete. This has led to disagreements between the Using Service and the contractor.

3.1.1.1 Item diagrams and schematics The physical characteristics of the engine shall be described in top–level functional flow diagrams and schematics of the configuration item and include diagrammatic presentations to the level required to identify all essential functions. Copies of the diagrams and schematics shall be provided with the specification.

REQUIREMENT RATIONALE (3.1.1.1)

Diagrams and schematics are needed to identify physical characteristics.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: The requirements in 3.1.1.1 through and including 3.1.1.4 can be deleted from the specification and placed in the Interface Control Document (ICD). The ICD can then be made part of the contract.

Background:

Item diagrams for lubrication system, fuel system, electrical system, and engine control system should be included in the model specification. Second level and lower diagrams and schematics should be required per CDRL item. Computer Aided Design (CAD) engine definitions may be used to meet this requirement. These CAD drawings may be provided in Initial Graphics Exchange System (IGES) format.

REQUIREMENT LESSONS LEARNED

Past engine model specifications have included diagrams and schematics of the engine. The use of CAD has been used on commercial engine applications with reductions in overall cost and time.

4.1.1.1 Item diagrams and schematics The requirements of 3.1.1.1 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.1.1)

The diagrams and schematics must be evaluated to ensure that they are functionally and physically correct.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.1.2 Item drawing Physical features of the engine, in all planes, shall be described in top level detailed dimensional drawings.

REQUIREMENT RATIONALE (3.1.1.2)

Drawings defining the physical characteristics of the engine are essential to ensure compatibility with the engine bay and airframe.

REQUIREMENT GUIDANCE

Background:

The Using Service should request lower tier drawings in the CDRL.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.2 Item drawings The requirements of 3.1.1.2 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.1.2)

The drawings must be evaluated to ensure that they are functionally and physically correct.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.1.3 Interface and installation diagrams and schematics Engine/airframe interfaces and engine/component accessory interfaces shall be illustrated in top level interface and installation diagrams. Copies of the diagrams and schematics shall be provided with the specification.

REQUIREMENT RATIONALE (3.1.1.3)

The interfaces between the airframe and the engine must be established and controlled to ensure compatibility. The installation interface diagrams and schematics are required to ensure physical compatibility between engine and airframe.

REQUIREMENT GUIDANCE

Background:

Diagrams and schematics for lubrication system, fuel system, electrical system, EMS, and engine control system should be included in the model specification. Top level and lower diagrams and schematics should be required per CDRL item.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.3 Interface and installation diagrams and schematics The requirements of 3.1.1.3 shall be evaluated by demonstration and inspection.

EVALUATION RATIONALE (4.1.1.3)

The diagrams and schematics must be evaluated to ensure that they are functionally and physically correct.

EVALUATION GUIDANCE

Background:

All engine interfaces with the airframe should be evaluated using data obtained from demonstrations and inspections conducted during the development program. The engine/airframe integration plan and interface control document should be reviewed to ensure that no incompatibilities exist.

EVALUATION LESSONS LEARNED

None.

3.1.1.4 Interface and installation drawings The interface and installation, in all planes, shall be shown on top level detailed dimensional drawings. The interface drawings shall include all physical installation requirements necessary for engine installation. The following drawings shall be provided with the specification.

a. Engine installation and interface drawing, figure 1a. The drawings shall show mounting details and tolerances for the engine and all installation items, clearances for installation and removal of components and aircraft

accessories subject to separate removal, access for adjustments and other maintenance functions and center of gravity of the complete engine, and center of gravity of the engine with fully loaded accessory interfaces. This drawing shall show the engine with fully loaded accessory interfaces and interface dimensions for the attachment of the inlet duct. This drawing shall show the maximum space required by the engine, including tolerances and dimensional changes due to manufacturing, thermal effects, vibration, operating and bending due to externally–applied loads.

b. Electrical installation and interface drawing, figure 1b. This drawing shall show and identify external electrical circuit requirements and installation interface connection details for all engine systems.

c. Fiber optic installation and interface drawing, figure 1c. This drawing shall show and identify external fiber optic circuit requirements and installation interface connection details for all engine systems.

REQUIREMENT RATIONALE (3.1.1.4)

Drawings defining the physical characteristics and installation of the engine are essential to ensure compatibility with the engine bay and airframe.

REQUIREMENT GUIDANCE

Background:

The engine/airframe interfaces should be described in sufficient detail to ensure compatibility throughout the ground and flight envelope. Details defining the interfaces should be shown on the engine installation and interface drawings and any supplemental tables and figures required. The interfaces should include, but not be limited to, engine mounts, gearbox pads and drives, engine inlet, electrical connections, bleed and horsepower extraction, exhaust nozzle/reverser, control connections, starting, instrumentation, condition monitoring, cooling, anti–icing, fuel system, ignition system, power shaft absorber, water and fuel injection, ground test equipment, and data bus requirements.

Interface defined elsewhere in the specification should also be listed on the installation tables, figures, and drawings. An airframe/engine integration plan and an interface control document defining the working relationship and responsibilities of the engine and airframe contractors should be established.

REQUIREMENT LESSONS LEARNED

Early definition of the engine/airframe interface has been vital to a successful system development program.

4.1.1.4 Interface and installation drawings The requirements of 3.1.1.4 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.1.4)

The engine/airframe interface must be checked to ensure compatibility.

EVALUATION GUIDANCE

Background:

All engine interfaces with the airframe should be evaluated using data obtained from measurements, demonstrations, and tests conducted during the development program. The engine/airframe integration plan and interface control document should be reviewed to ensure that no incompatibilities exist.

EVALUATION LESSONS LEARNED

None.

3.1.1.5 Installation changes Engine features affecting engine installation interfaces shall be specified herein. Changes (additions, deletions, or modifications) shall be submitted to the Using Service for approval.

REQUIREMENT RATIONALE (3.1.1.53.1.1.5)

Interface control must be maintained to assure installation compatibility.

REQUIREMENT GUIDANCE

Background:

Coordination should be made with the airframe contractor in order to determine the feasibility and impact of interface changes. Using Service contractual procedures should be followed for coordination and approval.

EVALUATION LESSONS LEARNED

A requirement concerning installation changes has existed since 1946.

4.1.1.5 Installation changes The requirements of 3.1.1.5 shall be evaluated by inspection of interface documents and installation demonstration.

EVALUATION RATIONALE (4.1.1.5)

Interface information can only be evaluated by inspection and demonstration.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.1.6 Installation hardware The engine shall be suitable for installation in any engine position in the aircraft without parts replacement except as approved by the Using Service.

REQUIREMENT RATIONALE (3.1.1.6)

There must be no installation hardware necessary to install the engine.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.6 Installation hardware The requirements of 3.1.1.6 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.1.6)

Evaluation is necessary to ensure no hardware is required for engine installation.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.1.7 Connections/interchangeability Connections located in close proximity to each other shall be made physically non–interchangeable.

REQUIREMENT RATIONALE (3.1.1.7)

There is a need to ease installation and maintenance. Also, an assurance is needed to prevent incorrect connections.

REQUIREMENT GUIDANCE

Background:

The engine should be permanently marked to indicate all instrumentation, fluid, air, electrical, and fiber optic connections shown on the engine installation drawing.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.7 Connections/interchangeability The requirements of 3.1.1.7 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.1.7)

There is a need to demonstrate all connections on the engine.

EVALUATION GUIDANCE

Background:

The inspection and demonstration should be conducted with the maintenance demonstration of 3.5.2.3.

EVALUATION LESSONS LEARNED

None.

3.1.1.8 Interface loads The maximum allowable loads and deflections, and their locations, for all controls, externals, and parts that interface with the airframe and are not specified elsewhere in 3.1.1, shall be specified herein.

REQUIREMENT RATIONALE (3.1.1.8)

This information is required in order to ensure engine compatibility with the airframe.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: The Contractor should specify the required information.

The requirements in 3.1.1.8 can be deleted from the specification and placed in the Interface Control Document (ICD). The ICD can then be made part of the contract.

Background:

This paragraph should include all interface loads and deflections not specified in subsequent paragraphs in Section 3.1.1 (e.g., anti–icing/de–icing, fuel, electrical, hydraulic, pneumatic, water injection, wash, and any other items that interfaces with the airframe).

REQUIREMENT LESSONS LEARNED

The USN had an engine mount wear problem on an aircraft model in the late 1980's. The contributing cause was unknown stress levels the airframe side of the mount system imparted on the engine mounts. Higher than expected levels were discovered in flight test when both airframe and engine sides of the mount system were instrumented. The calculated stresses, from a previously used model simulation, were erroneous and didn't predict this wear problem. This shows that measured values are needed.

4.1.1.8 Interface loads The requirements of 3.1.1.8 shall be evaluated by test.

EVALUATION RATIONALE (4.1.1.8)

This requirement is essential for the airframe contractor and can only be evaluated by inspection and tests.

EVALUATION GUIDANCE

Background:

The airframe contractor and the Using Service should inspect the information for completeness and compatibility with the airframe.

EVALUATION LESSONS LEARNED

None.

3.1.1.9 Mounts

4.1.1.9 Mounts

3.1.1.9.1 Main mounts The engine mounts shall retain the engine, including retained fluids and externals, at all flight, takeoff and landing, and ground conditions and shall withstand elastic limit loads of (a) without permanent deformation and ultimate tensile strength loads of (b) without complete fracture. A total of (c) mounts shall be provided which have sufficient strength to prevent (d) when subject to a single attachment point failure at any location at the end of the engine mount service life. The locations and descriptions of all engine mounts shall be as specified in figure 1a. The mount system shall accommodate all off–axis loads when a thrust vectoring nozzle is used.

REQUIREMENT RATIONALE (3.1.1.9.1)

Engine mounts are necessary to attach the engine to the aircraft. Elastic and ultimate tensile strength load limits must be specified in order to ensure that both engine and aircraft designs are sufficiently strong for normal operations and safe for a limited range of crash landings throughout the specified service life. Mounts are required to have sufficient strength to protect the engine against a worst case single attachment point failure in order to ensure safety of flight (i.e., fuel fire and single engine loss of power) and ground safety (i.e., crashed engines or aircraft) (see LESSONS LEARNED). The number, locations, and descriptions of all engine mounts must be specified in order to ensure proper engine installation into the aircraft.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph except when the airframe's engine mounting requirements have been previously established:

(a), (b): The contractor should specify the maximum system limits in units of force and in reference to the engine. The specified values should include, but not be limited to, the effects of the following requirements and specific design characteristics: externally applied forces (i.e., accelerations) of 3.4.1.2.2, gyroscopic moments of 3.4.1.6.11, all airframe loads which are supported through the engine structure (if such loads exist) and safety factors of 3.4.1.6.1, cyclic fatigue, engine mass, material strength/mechanics and service life. The contractor should specify the bending moment limits in the axial, vertical, and lateral directions. (NOTE: For competitive engine development programs the Using Service should ensure that the engine and aircraft specifications are compatible with the maximum loads of the worst case engine and airframe combination.)

(c): The Contractor should specify the number of mounts.

(d): The Using Service should specify that engine mounts have sufficient strength to prevent "a reduction in engine power and change in engine position" for single engine aircraft or "loss of engine retention" for multi–engine aircraft.

Figure 1a is where the contractor should specify the locations and description of engine mounts to ensure that both engine and aircraft are designed to a common goal. Redundant mounts should also be indicated on figure 1a. NOTE: The Using Service should ensure that limited crash loading accelerations are specified and identified on the figures, so the contractor can specify ultimate load limits.

Background:

For engines which utilize thrust vectoring, the vector forces should be determined by the engine contractor. These forces will be based upon a particular engine/airframe combination. Because of the higher vertical loads imposed during thrust vectoring, it is essential that the engine mount system sustain the worst case load predictions that the aircraft would obtain. The engine mounting system must be suitable for the thrust vectoring nozzle arrangement and the mount system must be durable enough to withstand the various loads that will be imposed when the nozzle is vectored at selected angles.

Part 33 of the FAR requires that engine mounts have both elastic and ultimate strength integrity.

REQUIREMENT LESSONS LEARNED

Engine mounts have failed in flight. The right engine (JT8–PW) on a Boeing 737–200 jetliner fell off shortly after takeoff from O'Hare International Airport on 20 January 1989 (Piedmont flight 1480). According to an eyewitness report, "There was a lot of smoke coming out of one engine, and we saw it leaning, almost falling off, and then it fell off..." (ref: Washington Post, Jan 21, 1989). Similar incidents involving 737s were a Southwest Airlines flight leaving Dallas on January 3, 1986 and USAir flight leaving Philadelphia on December 5, 1987 (ref: Washington Post, January 22, 1989).

The single attachment point guidance for multiple engine aircraft will likely result in two forward mounts and one aft mount, with an aft mount failure possibly resulting in partial engine separation (e.g., inelastic mount elongation) and engine shutdown. Guidance for single engine aircraft will likely result in a design with four mounts arranged in a rectangular or diamond shaped pattern, without loss of engine power or change in engine position (33 percent redundancy factor and stresses in the elastic range) after single attachment point failure.

4.1.1.9.1 Main mounts The requirements of 3.1.1.9.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.1.1.9.1)

Elastic and ultimate tensile strength load limits must be evaluated in order to ensure that the engine is sufficiently strong for normal operations and safe for limited crash landings or single point attachment failures throughout the service life.

EVALUATION GUIDANCE

Background:

To minimize cost prior to preflight qualification, engine mount requirements should be evaluated by analysis of the worst case engine mount failures and their consequences. Engine mount testing should be conducted after the completion of endurance test cycling and prior to production qualification. See 3.1.1.9.1 REQUIREMENT GUIDANCE for load criteria which should be included in the evaluation.

Thrust vectoring nozzles impose new structural loads upon the engine and the airframe. These loads must be examined carefully by analysis of test data. Determination as to whether the mount system will withstand these new forces is a factor of mount system strength, durability, and mount system life considerations.

The engine mounting system must be suitable to the thrust vectoring nozzle arrangement and the mount system should be durable enough to withstand the various loads that will be imposed by directing the engine thrust at selected angles.

The load calculations done by the engine contractor during the design phase should be evaluated during sea level demonstrations and tests.

EVALUATION LESSONS LEARNED

The most commonly known failure mode for the engine mounting system was metal fatigue of the mounting bolts.

3.1.1.9.2 Ground handling mounts The ground handling mounts shall support the engine, including all engine mounted equipment and externals, components, and operating fluids, under the following maximum inertia load conditions, without deformation to the mounts or damage to the engine: (a) axial, (a) lateral, and (a) vertical acting in combination at the engine center of gravity.

The locations and descriptions for the individual ground handling mounts shall be as specified on figure 1a. The arrangement shall be compatible with ground handling equipment specified herein by the Using Service. .

REQUIREMENT RATIONALE (3.1.1.9.2)

Ground handling mounts are required to provide a means to lift the engine during the installation/removal from the aircraft and for ground transportation and maintenance.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should specify at least 4g axial, 2g lateral, and 3g vertical.

Figure 1a should be formatted and provided in the specification by the Using Service for the contractor to provide the required information.

Background:

For more information on handling mount requirements, see MIL–S–87233. The mounts should be designed to be compatible with existing engine transportation and maintenance equipment. Life cycle cost studies, in conjunction with operational requirement studies, should be conducted to determine if particular or existing engine handling equipment should be used.

If adapters are necessary, they should be designed and provided by the engine manufacturer as specified in the contract.

REQUIREMENT LESSONS LEARNED

Ground handling mounts and related support equipment have not always been adequately designed and compatible. Engine damage has occurred due to engine support equipment not containing sufficient shock absorbing capability, thereby transmitting high “g” forces into the engine. Also, engine ground handling mounts have been damaged or have failed because the mounts are not adequately designed for the loads imposed during engine transportation, maintenance, and installation/removal from the air vehicle.

4.1.1.9.2 Ground handling mounts The requirements of 3.1.1.9.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.1.1.9.2)

The intent of this paragraph is to evaluate that ground handling mounts provided on the engine are adequate for ground handling, transportation, and maintenance of the engine.

EVALUATION GUIDANCE

The following should be transferred verbatim to the specification paragraph:

The following procedures shall be demonstrated:

a. On–base ground transportation

b. Engine installation/removal from the air vehicle

c. Engine maintenance tasks

d. Other ground handling tasks peculiar to the engine.

Background:

Tests should be conducted to load levels sufficient to evaluate limit load and ultimate load operational requirements and to evaluate that minimum strength components can meet the load requirements assuming that the test components have average strength capability.

EVALUATION LESSONS LEARNED

None.

3.1.1.9.3 Engine stiffness The estimated stiffness of the engine in resisting loads and moments applied at the outboard end of the output shaft, relative to the engine mounting points, shall be specified herein. The first “free–free” lateral and vertical engine bending modes shall be specified herein.

REQUIREMENT RATIONALE (3.1.1.9.3)

Aircraft maneuvers with turboprop/turboshaft engines cause large moments about the propeller shaft. This may increase propeller shaft and engine/gearbox case deflection thereby requiring better structural rigidity.

REQUIREMENT GUIDANCE

Background:

Engine stiffness is the total deflection of the engine at the output shaft/propeller relative to airframe mounting points.

The loads should include, but not be limited to, the effects of externally applied forces, gyroscopic moments, safety factors, cyclic fatigue, material strength, and service life.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.9.3 Engine stiffness The requirements of 3.1.1.9.3 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.1.1.9.3)

The test of the engine stiffness is required to ensure compatibility with the air vehicle.

EVALUATION GUIDANCE

Background:

Engine stiffness should be evaluated prior to preflight qualification since the design may be impacted. See externally applied forces of 4.4.1.2.2.

EVALUATION LESSONS LEARNED

None.

3.1.1.10 Inlet connection stresses The maximum allowable static (1g) shear and overhung moment loads at the engine inlet flange shall be specified herein. The maximum allowable loads at the engine inlet flange for the maximum allowable combined maneuver and air loads shall be specified herein. The maximum allowable maneuver loads shall be defined on figure 2. No damage shall occur to the engine inlet interface resulting from the maximum loads specified above, and from airframe deflections due to the externally applied forces of flight maneuver, takeoff, and landing environments.

REQUIREMENT RATIONALE (3.1.1.10)

This paragraph provides interface information to assure that the engine is compatible with the airframe.

REQUIREMENT GUIDANCE

Background:

The wording of the requirement eliminates any misunderstanding between static and dynamic loading.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.10 Inlet connection stresses The requirements of 3.1.1.10 shall be evaluated by inspection and analysis.

EVALUATION RATIONALE (4.1.1.10)

The requirements of 3.1.1.10 need to be evaluated by analysis and inspection.

EVALUATION GUIDANCE

Background:

The analysis and inspection should insure that not only the maximum allowable static loads are considered, but also that maximum allowable combined maneuver and air loads are not exceeded.

EVALUATION LESSONS LEARNED

None.

3.1.1.11 Control system interface and integration The input/output from the airframe/engine shall be defined in the following paragraphs.

REQUIREMENT RATIONALE (3.1.1.11)

All signals (mechanical, electrical, pneumatic, fluid, etc.), to and from the airframe, required to properly control the engine must be defined in the following paragraphs.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The contractor should define the inputs/outputs in the following paragraphs.

The requirements in 3.1.1.11 can be deleted from the specification and placed in the Interface Control Document (ICD). The ICD can then be made part of the contract.

Background:

The input and feedback signals required for engine control and operation should include but not be limited to throttle control signals to aircraft instrumentation and condition monitoring systems, control of power and bleed extraction systems, and switching signals for backup control and electrical power features.

For new aircraft programs that use electronic engine control systems, the interface may be routed through a MIL–STD–1553 multiplex data bus. However, some update rates can be too slow, some signals must be processed realtime. Therefore, the MIL–STD–1553 multiplex bus is not necessarily called out as a requirement.

The description of the data bus or communication network should consist of but not be limited to: protocol, architecture, data rate, word capacity, bus controller, remote terminals, and shared and/or dedicated communication and electronic control interfaces of the engine, flight air inlet control, and the EMS, as applicable. As a minimum, the data bus should be redundantly interfaced to each channel to ensure fail–operational capability.

For multi–engine aircraft with electrical control signals going to data buses and flight controls, lines should be used to separate identical signals/wires from each engine control system.

The Using Service should double check with the airframe/interface specification to ensure that the control system/interface requirements are consistent with cockpit input/output signal requirements.

REQUIREMENT LESSONS LEARNED

Past multi–engine aircraft with engine FADECs had one data bus terminal or line tied into the same signal for all engine controls. When one terminal broke, all FADECs were unable to send that signal to or through the data bus.

4.1.1.11 Control system interface and integration The requirements of 3.1.1.11 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.1.11)

The control system interface must be evaluated to ensure that no incompatibilities exist.

EVALUATION GUIDANCE

Background:

Each input and feedback signal between the control and the airframe should be measured and documented and compared to the values listed in the specification.

Component and full engine tests will serve as further evaluation that the proper signals are present.

EVALUATION LESSONS LEARNED

None.

3.1.1.11.1 Thrust/power demand The control system interface shall ensure compatibility between the airframe power demand output signal or load, and the control system.

For mechanical systems, the maximum allowable loads at the thrust/power lever connections shall be specified herein for the static (1g) axial, shear and overhung moment and for the maximum maneuver loads defined on figure 2.

REQUIREMENT RATIONALE (3.1.1.11.1)

The control system interface must interpret cockpit thrust/power demand electrical or mechanical signals.

For electrical systems, the linkage from the pilot's lever to the control may be by a direct reading potentiometer signal, or a multiplex bus signal via the aircraft computer, to control a torque motor or other drive unit on the engine throttle linkage.

For mechanical systems, the maximum static and maneuver loads are required to provide mechanical systems interface information, to prevent cable breakage, and to ensure the manufacturer takes into consideration the maximum maneuver loads.

REQUIREMENT GUIDANCE

Background:

The Using Service should confirm that the airframe or interface specification requires an index plate for mechanic's reference. Positive prevention of incorrect rigging should be required so that errors are not made.

Thrust/power demand controls, besides the 'old' mechanical cable linkage from pilots lever to the engine throttle linkage, can also be an "electrical linkage". This electrical linkage is from a potentiometer signal input from the pilot's lever to control a torque motor or other drive unit on the engine throttle linkage. The maximum allowable loads on lever connection are for the mechanical cable linkage systems to prevent cable breakage. This maximum load requirement is not needed for electrical linkages.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.11.1 Thrust/power demand The requirements of 3.1.1.11.1 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.1.11.1)

The thrust/power demand interface must be evaluated to ensure that no incompatibilities exist between the airframe output and the control system.

EVALUATION GUIDANCE

Background:

Engine control response to the thrust/power demand is evaluated by various tests and demonstrations conducted throughout engine development.

EVALUATION LESSONS LEARNED

None.

3.1.1.11.1.1 Power lever angle (PLA) The control system interface shall receive and process airframe power lever output angles, movements and signals for fuel control input. Fuel control interpretation of power lever angle signals shall be defined in 3.7.2.1.

REQUIREMENT RATIONALE (3.1.1.11.1.1)

The control system interface must relay the proper electrical power lever angle signal to the control unit.

The interpretation of PLA signals must be specified early in development for mechanical systems because of the long lead design decisions that need to be made for mechanical control design.

REQUIREMENT GUIDANCE

Background:

For electronic interfaces, this requirement should require proper cockpit signal processing by the interface, and the cockpit power lever angle requirement should be part of the airframe/interface specification.

For mechanical systems, the Using Service should make sure the airframe/interface specification requirements are consistent. The following PLAs are normally used in today's aircraft:

Power Lever				
Augmented Engines	Augmented Engines with Thrust Reverser	Non–augmented Engines	Non–augmented Engines with Thrust Reverser	Power Lever Angle
Maximum	Maximum	Maximum	Maximum	127° to 130°
Intermediate	Intermediate Idle		Idle	97° to 100° 35° to 38°
Idle	Maximum	Idle	Maximum	15° to 18° 0° to 3°
Off	Reverser	Off	Reverser	

A single power lever should be provided in the cockpit to modulate thrust. The power lever should have a total travel of 129 $\pm$1° and should provide the fuel shutoff, except for thrust reversing engines. The length of the dwell bands should be 3.0 $\pm$0.25°. Positive stops should be located at zero and 130 degrees positions. If a Flight Idle setting is required, it should be activated by an electrical signal provided by the aircraft (e.g., weight on aircraft wheels) without requiring the power lever to be moved from the Idle detent. The power lever position tolerance for any given value of a control variable should be within $\pm$1.5 degrees. For engines with thrust reverser, partial reverse thrust should be attained with the power lever angle setting between 3° and 35°. In this mode stopping the engine can be accomplished only by using the fuel shutoff lever. Rigging pin locations should be provided at Idle and Intermediate thrust for turbojet and turbofan engines, and Maximum power for turboshaft and turboprop engines. If additional rigging pin locations are required, the setting should be specified.

For engines with thrust reversers, the control system interface should process a fuel shutoff signal from the airframe that is independent of the power lever signal.

REQUIREMENT LESSONS LEARNED

The range of motion for the thrust levels indicated are based on previous problems with controlling thrust during carrier landings. Major efforts and funds had to be expended on two recent Navy aircraft programs to correct a problem with thrust vs. throttle angle. The power range for afterburner had to be reduced and Idle changed to a reduced angle to provide more travel from Idle to Intermediate for improved approach power control.

4.1.1.11.1.1 Power lever angle (PLA) The requirements of 3.1.1.11.1.1 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.1.11.1.1)

The engine control interface must be evaluated to ensure that no incompatibilities and inconsistencies exist between the cockpit power lever angle and the response by the engine control.

EVALUATION GUIDANCE

Background:

Engine control interface response is evaluated by various tests and demonstrations conducted throughout the development program.

EVALUATION LESSONS LEARNED

In early F–18 flight testing the pilots discovered that the PLA dead band ranges were too small to assure good aircraft control during carrier landings.

3.1.1.11.1.2 Power lever torque The control system interface shall be capable of receiving and processing airframe power lever torque signals for fuel control input.

For systems with a mechanical interface, the power lever torque required to operate the engine power lever through its range of travel shall not exceed 2.8 N–m (25 pound–inches) throughout all flight loads and conditions. The power lever shall not move with the engine operating unless external torque is applied. Movement of the power lever throughout the operating range shall be free of abrupt changes in actuating torque, and the maximum permissible variation shall not exceed 1.13 N–m (10 pound–inches). The maximum torque required to operate any additional levers shall not exceed 2.8 N–m (25 pound–inches). The maximum allowable static (1g) axial, shear and overhung moment loads at the power lever connections shall be specified herein. The maximum allowable loads at these connections shall also be specified for the maximum allowable maneuver loads. The power lever torsional loading limit between incremental stops and the maximum allowable loading when the power lever is against its travel stops shall be specified herein in Newton–meter (pound–inch) units.

REQUIREMENT RATIONALE (3.1.1.11.1.2)

The intent of this paragraph is to provide information for electronic and mechanical linkages at the engine interface to establish the maximum lever torques and variation. The requirement to specify torque differences (engine on/off), the relationship between the torque and lever angle, and the torsional loading limits between stops, is to define interface requirements. The lever movement is required to be free of abrupt changes and maximum variances to assure that the pilot’s strength and endurance are not overly taxed.

REQUIREMENT GUIDANCE

Background:

In the case of mechanical control interfaces, the torque should be specified in the airframe/interface specification if it is different when the engine is not running. Also, the Using Service should ensure the airframe or ICD specifies the relationship between power lever torque and power lever angle. The Using Service should make sure the airframe/interface specification requirements are consistent with the mechanical interface requirements of 3.1.1.11.1.2. The cockpit requirement for power lever torque for mechanical systems is provided for information because of the long lead design decisions that need to be made for mechanical control design.

In the case of electrical control interfaces, the Using Service should ensure that cockpit torque requirements are defined in the airframe/interface specification. In addition, the Using Service should make sure that the airframe requirements precludes inadvertent power lever movement, eases operation for all pilots, and enhances operational safety. The lever is required to remain fixed while the engine is running because there have been instances where the lever moved because of vibration, "g" load, etc. The lever movement is required to be free of abrupt changes, and a maximum variable of 1.13 N–m (10 pound–inches) ensures that the pilot's strength and endurance are not overly taxed.

REQUIREMENT LESSONS LEARNED

The maximum allowable torque, between Idle and Maximum, historically has been 15 pound–inches since AN–E–30 was developed. A review of the J79, J85, T56, T58 and TF34 model specifications revealed that the maximum allowable torque was greater than 15 pound–inches. During the Tri–Service coordination meeting for MIL–E–5007D the limiting value was increased to 25 pound–inches by the USAF.

4.1.1.11.1.2 Power lever torque The requirements of 3.1.1.11.1.2 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.1.11.1.2)

The power lever torque must be evaluated to ensure compatibility with the engine control interface.

EVALUATION GUIDANCE

Background:

Engine control interface response is evaluated by various tests and demonstrations conducted throughout the development program.

EVALUATION LESSONS LEARNED

None.

3.1.1.11.2 Control signals The engine control system shall receive and transmit signals as required by the airframe, engine, or ground support systems. Control input and output signal requirements, input and output parameters, and related functions shall be specified herein. All control system signals required for the command and automatic control of the engine shall be transmitted and received in accordance with table III.

REQUIREMENT RATIONALE (3.1.1.11.2)

The intent of this paragraph is to provide interface information pertaining to the external engine control system signals and functions.

REQUIREMENT GUIDANCE

Background:

The control system input/output signals can be electrical, mechanical, hydraulic, or pneumatic. The input signals may include aircraft Mach number, altitude, armament reset, Idle exhaust nozzle reset, etc. The output signals may include engine parameters for cockpit display and/or engine condition monitoring equipment (for fault detection and isolation to the control system component weapon replacement assembly (WRA) level) and other interrogation systems external to the engine.

For complex data systems it may be desirable to separate this section into sub–sections to clarify what interfaces are required. Possible headings are: Discrete Inputs, Scaled Inputs, Digital Inputs, Discrete Outputs, Scaled Outputs, Digital Outputs.

Electronic fuel controls have greater signal capacities than hydro–mechanical controls. The electronic controls also possess a much greater processing capability, and there have been tendencies to integrate the control computer to the aircraft computers. This may result in the degradation of control system performance and should be avoided.

This requirement contains interface information that must be defined early in the development of the weapon system. There should be documented coordination between the engine contractor and the airframe contractor. Each must be aware of the equipment's capability and input and output signals that will be used.

REQUIREMENT LESSONS LEARNED

In a recent Navy engine program, the following separately wired input/output signals to the engine control system were required:

Input Signals	Output Signals
Thrust/power demand	Turbine exit temperature
Speed lockup discrete	Core speed
APC throttle command	Fuel flow
1553 data bus (redundant)	1553 data bus (redundant)
	PLA position

4.1.1.11.2 Control signals The requirements of 3.1.1.11.2 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.1.11.2)

The control signals must be evaluated to ensure that no incompatibility exists.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.1.11.3 Load demand lever For turboshaft and turboprop engines, a means shall be provided at the control interface to adjust for rapid power absorber load changes. The control system interface shall provide for compatibility between the airframe load demand lever output signal and the control system. The load demand lever/signal shall be linear with power absorber load.

REQUIREMENT RATIONALE (3.1.1.11.3)

In some applications the load demand lever is used to control the power extracted from the engine, in conjunction with the speed lever.

REQUIREMENT GUIDANCE

For turbofan and turbojet engines, replace the text with "Not Applicable".

Background:

When required, a load demand lever/signal is utilized on turboshaft and turboprop engines only. The Using Service should make sure the airframe/interface specification requirements are consistent with the mechanical interface requirements of 3.1.1.11.3. The cockpit requirements for load demand lever for mechanical systems ARE necessary in the engine specification because of the long lead design decisions that need to be made for mechanical control design. The interface should properly interpret the load demand lever from the cockpit which normally has a total travel of 89–91 degrees, with positive stops at the zero (0) and 90 degree positions. The minimum load position usually is at the zero (0) degrees and the maximum at 90 degrees. For electronic interfaces, the cockpit requirements in 3.1.1.11.3 should be only in the airframe/interface specification.

REQUIREMENT LESSONS LEARNED

This has been a necessary requirement for all helicopters in the past. A load demand lever with mechanical cable linkage has been used on hydromechanical systems.

4.1.1.11.3 Load demand lever The requirements of 3.1.1.11.3 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.1.11.3)

The load demand lever must be evaluated to ensure that no incompatibility exists.

EVALUATION GUIDANCE

For turbofan and turbojet engines, replace the text with “Not Applicable”.

Background:

Engine control interface response is evaluated by various tests and demonstrations conducted throughout the development program.

EVALUATION LESSONS LEARNED

None.

3.1.1.11.4 Output speed lever For turboshaft and turboprop engines, a shaft speed lever, if required, shall be provided to adjust and control the delivered shaft speed. The control system interface shall provide for compatibility between the airframe load demand lever output signal and the control system. The lever input shall be linear with governed output shaft speed.

REQUIREMENT RATIONALE (3.1.1.11.4)

In some applications, the output speed lever controls the speed of the output shaft, in conjunction with the load demand lever.

REQUIREMENT GUIDANCE

For turbofan and turbojet engines, replace the text with “Not Applicable”.

Background:

When required, an output speed lever is utilized on turboshaft and turboprop engines only. For mechanical systems, the Using Service should make sure the airframe/interface specification requirements are consistent with the mechanical interface requirements of 3.1.1.11.4. The cockpit requirements for output speed lever for mechanical systems ARE necessary in the engine specification because of the long lead design decisions that need to be made for mechanical control design.

An output speed lever normally has a total travel of 89–91 degrees, with positive stops at the zero (0) and 90 degree positions. The minimum governed output shaft speed should be at the zero position and the maximum at 90 degrees. For electronic interfaces, the cockpit requirements in 3.1.1.11.4 should be only in the airframe/interface specification.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.11.4 Output speed lever The requirements of 3.1.1.11.4 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.1.11.4)

The output speed lever must be evaluated to ensure that no incompatibility exists.

EVALUATION GUIDANCE

For turbofan and turbojet engines, replace the text with “Not Applicable”.

Background:

Engine control interface response is evaluated by various tests and demonstrations conducted throughout the development program.

EVALUATION LESSONS LEARNED

None.

3.1.1.11.5 Fuel shutoff lever For engines with thrust reversers, the fuel control shall process airframe signals, independent of the thrust/power demand signal, for fuel shutoff.

REQUIREMENT RATIONALE (3.1.1.11.5)

Engine with thrust reversers are not capable of shutdown by cockpit thrust/power demand mechanism, therefore, a signal separate from the thrust/power demand mechanism must be provided to shutoff fuel to the engine.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

Normally, a separate lever in the cockpit with a total travel of 89° $\pm$1° provides the fuel shutoff signal. Usually, lever angles in the cockpit are as shown below.

Shutoff Lever	
Shutoff	0° to 40°
Open	55° to 90°

4.1.1.11.5 Fuel shutoff lever The requirements of 3.1.1.11.5 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.1.11.5)

The fuel shutoff signal must be evaluated for proper processing by the fuel control.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.1.12 Electrical system interface

4.1.1.12 Electrical system interface

3.1.1.12.1 Instrumentation system interface The engine shall be instrumented to receive and transmit signals as required by airframe, engine, and ground support systems. The signal characteristics, physical interface, and instrumentation shall be shown in table IV.

REQUIREMENT RATIONALE (3.1.1.12.1)

Instrumentation required by the aircrew for monitoring the condition of the engine should be defined. The physical and functional interface with the airframe must be specified to enable the installation to be accomplished.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

Table IV should be filled in by the contractor.

The requirements in 3.1.1.12.1 can be deleted from the specification and placed in the Interface Control Document (ICD). The ICD can then be made part of the contract.

Background:

Instrumentation should be provided to display the power setting and information to determine normal and abnormal engine operation. Engine power settings may be determined by engine speed, exhaust nozzle position, engine pressure ratio, exhaust gas or turbine temperature, or fuel flow. Engine condition parameters may include oil pressure, temperature, and quantity; engine speeds; and various caution and warning lights. The instrumentation requirements may be more stringent for single engine aircraft.

REQUIREMENT LESSONS LEARNED

A turbine overtemperature warning system was retrofitted into the pilot headset in one fighter aircraft because the pilot’s attention to the environment outside the cockpit prevented them from noticing an over limit condition.

4.1.1.12.1 Instrumentation system interface The requirements of 3.1.1.12.1 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.1.12.1)

Instrumentation must be evaluated to ensure that the engine performance parameters are adequate and accurate.

EVALUATION GUIDANCE

Background:

An analysis should be accomplished to evaluate the accuracy of instrumentation. During development testing, the instrumentation should be installed and tested to the extremes of engine operation.

EVALUATION LESSONS LEARNED

Pilot comments have been valuable for information and types of engine data that may be required.

3.1.1.13 Engine monitoring system (EMS) interface The engine monitoring system shall receive and transmit signals as required by airframe, engine, and ground support systems. A description of the EMS interface shall be specified herein.

REQUIREMENT RATIONALE (3.1.1.13)

The physical and functional interface between the engine, airframe, and support equipment must be defined to ensure compatibility. Caution and warning data from an EMS may be sent to another device for display (an electronic pilot's aid for example). However, this device may include logic to prioritize the order/time for display of this information to the pilot. If this logic is not understood by both the engine and airframe, a priority message requiring pilot action may inadvertently not be displayed.

REQUIREMENT GUIDANCE

Background:

The EMS should consider all aspects of engine maintenance from visual inspections to the most sophisticated life usage counters, exceedance recorders, and performance measurement equipment. The interface should comply with the requirements of the Standard Interface for Data Transfer Equipment.

The EMS should also interface and be compatible with existing and current development maintenance and support systems (e.g., CAMS, CEMS, IMIS, and GIMADS) and interface with the airframe via a MIL–STD–1553 Vehicle Maintenance Mux Bus.

REQUIREMENT LESSONS LEARNED

Engine condition monitoring systems have been successfully used in fleet engines to schedule maintenance and diagnose engine problems.

4.1.1.13 Engine monitoring system (EMS) interface The requirements of 3.1.1.13 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.1.13)

The suitability of physical and functional interface between the EMS and the airframe or support equipment must be evaluated.

EVALUATION GUIDANCE

Background:

During the engine development program, the engine condition monitoring equipment, both engine and off–engine mounted systems, should be operated throughout the testing to ensure that it performs as intended. The accompanying support equipment should be used during the testing to demonstrate the adequacy of this equipment as well as the interfaces. Full compliance for off–engine mounted systems may not be evaluated until aircraft flight testing is completed.

EVALUATION LESSONS LEARNED

None.

3.1.1.14 Fiber optics interface The fiber optics interface if applicable shall be described herein.

REQUIREMENT RATIONALE (3.1.1.14)

If fiber optics are used as a part of the engine control or monitoring systems, then the interface requirements for them must be specified.

REQUIREMENT GUIDANCE

MIL–STD–1773 defines the interface electronics/requirements for fiber optic mechanization of an aircraft internal time division command/response multiplex data bus. This standard should be used to tailor the specification paragraph.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.14 Fiber optics interface The requirements of 3.1.1.14 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.1.1.14)

The fiber optic interface must be evaluated.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.1.15 Bleed air interface The engine shall provide for customer bleed air extraction for aircraft use. The locations, number, and interface dimensions at all customer bleed air ports shall be shown on the engine interface and installation drawings. If required, acceleration bleed air shall be from the (a) stage of the compressor. The maximum allowable bleed airflows, pressures, and temperatures for each port, and the compressor stage(s) from which it is extracted shall be specified herein. The maximum permissible continuous flow capability of each bleed air port in percent of total airflow shall be specified herein. Bleed ducts with external surface temperatures exceeding (b) shall be insulated to prevent hazards from combustible fluid leakage. The bleed port internal pickup points shall be located at positions which have low susceptibility to FOD, and ingestion of sand, dust, ice, moisture, and any other foreign materials contained in the air. Provisions shall be made to prevent high pressure bleed air from entering lower pressure bleed ports. All bleed air ports shall be sized to prevent engine failure in the event of a failure in the aircraft bleed system. The engine provided customer bleed air extraction system shall ensure that no upstream malfunction of the engine will cause specified contamination limits to be exceeded. Where overboard ducting of starting and acceleration bleed air is necessary, the airflow conditions for which provisions must be made shall be specified herein.

REQUIREMENT RATIONALE (3.1.1.15)

Bleed air may be required for various purposes including environmental control, anti–icing, component/accessory cooling, pneumatic operation of components, aircraft boundary layer control, attitude control, and high lift. Bleed air characteristics must be defined so that the systems using engine air bleed provisions can be properly designed.

The starting and accelerating capability of an engine may depend on the use of compressor bleed air during the starting sequence and during the acceleration process. The need for bleed air during starting and/or acceleration should be known by the aircraft designer so a compatible system can be developed for the airframe and engine interface and integration.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The contractor should identify the stage.

(b): A value of 390°C (734°F).

Background:

Bleed ducts should be analyzed to determine their potential as ignition sources for flammable fluids to which they may be exposed. Ducts assessed as potential ignition sources should be insulated. 390°C has been used in the past, however the engine bay installation (i.e., cooling mass flow, cooling air temperatures) and placement of bleed pipes and flammable fluid lines (location of flammable fluid sources) may require a different temperature limit.

The temperature for autogenous ignition is a function of numerous variables and relationships. At near static conditions the surface ignition temperature is approximately 340°C (650°F) and rises to 621°C (1150°F) at a wind velocity of 0.6 m/sec (2 ft/sec). FAA report January 1971, “Crash fire hazard evaluation of jet fuels” indicates that for kerosene a range of surface ignition temperature from 215°C (417°F) with a confined system to 650°C (1200°F) with an open steel plate can occur. Based on all of the above the existing 370°C (700°F) requirement is considered to be a reasonable limiting temperature for uninsulated ducts.

A potential fire hazard exists where bleed air is taken from the engine at high pressure/temperature locations. Consideration should be given to using an engine–furnished shutoff device at the bleed source where such hazard conditions can exist. Alternate solutions are the use of insulation on all high temperature bleed lines and adequate ventilation of the engine compartment.

Some aircraft/engine bleed systems require a minimum bleed pressure or energy level for satisfactory operation. Examples are aircraft ECS and engine bleed air start systems. Many helicopters use bleed air for “cross–bleed starting” for two engine aircraft and “buddy starting” for single engine aircraft. As an example with the T800 engine, bleed air energy available should be 15 to 30 watts per kilowatt of delivered shaft power at Maximum Continuous rated power. The pressure of the bleed air should be at least twice atmospheric pressure when the bleed air energy equals 15 watts at Idle power and the temperature of the bleed air should not exceed 400°C (750°F) at Maximum power.

Bleed flow is required for airframe uses such as cabin pressurization, air conditioning and anti–icing. These are airframe systems and the bleed flow will vary according to airframe requirements during flight. It is necessary that no thrust/power demand adjustment be required for the bleed flow modulation. This would impose an additional work load on the pilot when they least need it. Allowing high pressure air to flow into the low pressure stages would result in surge, stall, or performance deterioration. Sonic flow will limit the air flow through a given port size. By selecting the area of the port, the maximum air flow can be controlled. This will act as a safety valve to protect the engine against any failure modes down stream of the aircraft bleed system. Some aircraft concepts, such as aircraft with wing blowing or V/STOL attitude control, will need bleed in excess of the normal bleed limit. This is accommodated by increasing the size of the bleed orifice to either increase flow or lower pressure losses.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.15 Bleed air interface The requirements of 3.1.1.15 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.1.15)

The bleed air interface information needs to be confirmed by demonstration and tests.

EVALUATION GUIDANCE

Background:

Bleed air tests are usually conducted in conjunction with engine performance tests at sea level and altitude. The tests should include the airframe requirements and the maximum bleed flow specified by the engine contractor. This will eliminate any operability problems if the bleed flow is increased due to airframe changes.

Measurement inspection of ports and locations, surface temperature measurements, and engine tests at various bleed rates up to maximum specified flow rates should be performed.

Low power settings may require a large amount of anti–ice airflow; therefore, the pilot may be instructed to use high power in icing conditions.

EVALUATION LESSONS LEARNED

None.

3.1.1.15.1 Customer bleed air contamination Under normal operating conditions engine generated substances contained in customer bleed air shall be below the threshold limit of (a). Compressor bleed air extracted from the engine for customer use shall be free of engine generated noxious, toxic, or irritating substances and shall contain no engine generated objectionable odor.

REQUIREMENT RATIONALE (3.1.1.15.1)

Since aircraft gas turbine engine compressor bleed air is normally used for cabin air conditioning, bleed air must be supplied with minimal contaminants to assure no possibility of a health hazard to the crew members, or to interfere with their ability to perform the mission.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Engine generated substances contained in the bleed air should be within the threshold limit values specified below.

Substance	Parts per million
Carbon dioxide	5000.0
Carbon monoxide	50.0
Ethanol	1000.0
Fluorine (as HF)	0.1
Hydrogen peroxide	1.0
Aviation fuels	250.0
Methyl alcohol	200.0
Methyl bromide	20.0
Nitrogen oxides	5.0
Acrolein	0.1
Oil breakdown products (e.g., aldehydes)	1.0
Ozone	0.1

The bleed air should not contain a total of more than 5.0 mg (0.3 lbm x 10^{-6}) of engine generated particles per m^3 (ft^3) of bleed air.

Background:

Where substances other than those listed are contributed to the extracted air by engine operation, the engine manufacturer should report the substances and contamination in parts per million to the Using Service to determine maximum limits. When two or more engine generated substances are present their combined effect should be determined and reported. In the absence of information to the contrary, the combined effects of the different substances should be considered additive. If cleaning fluids are specified for use during normal engine maintenance, consideration should be given to their effect on bleed air contamination.

The Using Service should check with the aircraft system specification to make sure all engine generated particles/contaminates in the bleed air have requirements that agree with the system specification.

The source of bleed air contamination may be composed primarily of engine lubricating oil and its decomposition products. Contamination becomes a particularly difficult problem on high pressure ratio engines in which air temperatures of 370 to 425°C (700–800°F) are sufficiently high to cause decomposition of the oil. When decomposition products (organic carbonyl and organic hydroperoxide compounds) are present in concentrations as high as 0.5 to 1.0 parts per million (ppm), anyone breathing this air may experience severe irritation, nausea, and vomiting.

REQUIREMENT LESSONS LEARNED

The Threshold Limit Values (TLVs) for the substances shown in the table were set by the American Conference of Governmental Industrial Hygienists (ACGIH). These values are considered safe levels for continuous exposure of up to eight hours.

Problems have also occurred on engines with air operated components that are contaminated by foreign particles such as blade abradable material. These parts need to be designed to tolerate the contaminants or a filtering system needs to be included.

4.1.1.15.1 Customer bleed air contamination The requirement of 3.1.1.15.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.1.1.15.1)

Bleed air contamination, because of its importance to crew safety, must be measured.

EVALUATION GUIDANCE

Background:

Bleed air contamination levels should be evaluated on a test engine at Idle, Maximum Continuous, and Maximum power settings. Bleed air samples should be collected from all customer bleed ports and from the engine inlet air. Samples should be submitted to a qualified chemistry laboratory for analysis. Bleed air samples should be taken periodically during the course of the endurance testing. Considerations should be made for evaluating the bleed air contamination levels of the altitude qualification engine.

EVALUATION LESSONS LEARNED

One oil breakdown product, acrolein, is a highly active product which has shown to be quite difficult to measure accurately because it cannot be separated easily from the other oil breakdown products.

3.1.1.16 Drains and fluid collection devices Fluid drains, drain characteristics, and collection requirements shall be shown on the interface and installation drawing. There shall be no leakage of fluids from any part of the engine except at the drains provided for this purpose. The flow rate into all drains shall not exceed _(a)_ ml (_(a)_ oz) for non–afterburning engine and _(b)_ ml (_(b)_ oz) for afterburning engines. The maximum amount of combustible fluid drainage following shutdown after normal operation shall be _(c)_ ml (_(c)_ oz).

REQUIREMENT RATIONALE (3.1.1.16)

Drains and fluid collection devices are necessary to remove flammable fluids from areas conducive to combustion. Drain location is subject to interface definition. A fluid collection device/system is necessary to prevent fluid seepage/leakage into potentially hazardous areas of the engine. Most Army aircraft now require an environmental collection bottle.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): A value of five (5) ml (.169 oz) per minute

(b): A value of ten (10) ml (.338 oz) per minute

(c): A value up to 100 ml (3.38 oz) for helicopter engines, a value up to 400 ml (13.52 oz) for fixed wing aircraft.

Background:

Typically the interface for the engine fluid drain should define the characteristics of the fluid, including the maximum flow rate from failed components. Drains that can be separated into individual lines are useful when it is necessary to isolate leaking components during troubleshooting. Maximum fluid rates should be specified for each individual system (i.e., fuel, oil, hydraulic) for when the engine is and is not running.

Drain lines that cannot be separated for troubleshooting can mask problems with individual component leakage. The total fluid leakage rate may fall within limits, but a single component may be exceeding its own leakage limits.

The survivability and vulnerability community should be consulted to determine restrictions on drain hole sizes.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.16 Drains and fluid collection devices The requirements of 3.1.1.16 shall be evaluated by analysis, inspection, and test.

EVALUATION RATIONALE (4.1.1.16)

This evaluates the adequacy of the routing and drain system size.

EVALUATION GUIDANCE

Background:

During engine endurance and altitude testing, the source and amount of fluid drainage should be evaluated.

EVALUATION LESSONS LEARNED

None.

3.1.1.17 Power absorber interface For turboshaft and turboprop engines, the allowable range of characteristics of the shaft power absorber at the power absorber to engine interface shall be completely defined herein. The characteristics shall include but not be limited to: maximum and minimum polar moment of inertia (slug–ft^2), torsional spring constant, torsional damping coefficient as a function of torsional spring constant, the maximum allowable static and dynamic loads on the engine output drive shaft, direction of rotation of the output drive shaft as viewed from the engine inlet, design of the output drive shaft spline and the power absorber shaft maximum misalignment during steady–state and transient operation. Output drive shaft spline bearing air leakage shall be quantified and accommodated in the design of the attached housing so that venting requirements can be determined and accommodated in the design. No resonant frequency shall be transmitted to or from the power absorber through the engine interface. Engine oil shall not be used for propeller lubrication and hydraulic systems.

REQUIREMENT RATIONALE (3.1.1.17)

Power absorber to engine interface characteristics must be defined for compatibility.

REQUIREMENT GUIDANCE

This paragraph applies only to turboshaft and turboprop applications. For turbojet/fan engines, delete the text and insert “Not Applicable.”

The following should be used for tailoring the specification paragraph: For engines using engine oil to actuate the propeller, delete the last sentence and replace with the following:

If engine oil is necessary for actuation of the propeller, the propeller lubrication/hydraulic system should ensure debris from the propeller mechanism does not contaminate the engine oil or cause engine oil starvation due to failure of the propeller.

REQUIREMENT LESSONS LEARNED

In a past engine program a propeller failure caused the failure of an engine/propulsion system. Engine oil was used to lubricate and provide hydraulic pressure to the propeller, and debris from the failed propeller mechanism contaminated the engine oil and induced engine failure.

4.1.1.17 Power absorber interface The requirements of 3.1.1.17 shall be evaluated by analysis, inspection, and test. The maximum allowable misalignment of the power absorber shall be tested.

EVALUATION RATIONALE (4.1.1.17)

The characteristics of the power absorber to engine interface can only be evaluated by inspection and test.

EVALUATION GUIDANCE

This paragraph applies only to turboshaft and turboprop applications. For turbojet/fan engines, delete the text and insert "Not Applicable."

Background:

If possible, the actual aircraft propeller should be used as the power absorber. Helicopter systems may be difficult to simulate accurately at Idle and/or no load conditions.

EVALUATION LESSONS LEARNED

Along with spline fatigue life demonstration, misalignment has indicated problems with turboshaft engine output torque measurement accuracy, especially with phase shift systems.

3.1.1.18 Power take–off (PTO) When PTO pads and drives are provided for driving aircraft accessories, the ratings, clearance envelopes, dimensions, pad and connection details, direction of rotation, accessibility and alignment requirements shall be specified herein and on the engine interface and installation drawings. Customer power extraction limits shall be provided for each pad individually and in combination. The design and configuration of the PTO pads and drives shall be consistent with the requirements of 3.1.1.19.1 and 3.7.18.

REQUIREMENT RATIONALE (3.1.1.18)

The PTO is a drive pad therefore its characteristics must be furnished for interface information, similar to 3.7.18.

REQUIREMENT GUIDANCE

Background:

PTOs are being used more often on modern engine designs in lieu of integral engine gearboxes. The single PTO provides for simpler engine removal and installation.

REQUIREMENT LESSONS LEARNED

The extended PTO shaft is affected by more than maneuver loads. In particular, hard carrier landings are detrimental to the shaft.

4.1.1.18 Power take–off (PTO) The requirements of 3.1.1.18 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.1.18)

There is a need to demonstrate the clearance envelopes, dimensions, and accessibility. A test is required to evaluate the capability of the PTO.

EVALUATION GUIDANCE

Background:

Since the PTO is a driving pad, it must conform to 3.7.18. Also, the starting drive torque is usually applied through the PTO and must be consistent with 3.1.1.19.1. Final evaluation of the PTO should occur during flight test when it is subjected to maneuver loads.

EVALUATION LESSONS LEARNED

The F110 engine had a PTO that developed problems during carrier landings.

3.1.1.19 Starting drivetrain The engine starting torque acceptance capability shall be specified herein and shall be at least (a) times the starting torque required to provide a (b) second acceleration from start initiation to minimum starter cutout speed, under the conditions specified in 3.1.1.19.1. The weakest part of the starting drive system (shear section, clutch, etc.) shall be external to the engine starting drive train. The starter drive pad characteristics shall be included in table V. The direction of rotation when facing the starting pad on the engine shall be specified herein. No resonant frequency shall be transmitted to or from the starting drivetrain through the engine interface.

REQUIREMENT RATIONALE (3.1.1.19)

The engine starter drivetrain must be capable of accepting a load above that required for a normal start in order to provide protection from impact loads.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): A value of a minimum of 3.33 times.

(b): A value of 15 seconds.

The requirements in 3.1.1.19 through and including 3.1.1.19.4 can be deleted from the specification and placed in the Interface Control Document (ICD). The ICD can then be made part of the contract.

Background:

An engine drivetrain should be designed to accept the impact loads that can occur during start–up (initial valve opening in hydraulic or pneumatic, or engagement of a mechanical starter jaw clutch on electrical or other types). The factor of 3.33 is specified to provide this safety margin. A 15–second start initiation time to minimum starter cutout speed is an adequate minimum since most engines require 30 to 40 seconds to start (which includes the 15–second time period from start initiation to starter cutout).

It is not practical or desirable to put a shear section within the engine. All starting systems should have a shear section incorporated at the interface between the engine and the starter system. The shear section is a coupling which can be replaced without requiring return of either the starter or engine to the overhaul facility. The general specification existing for air, hydraulic, and electrical starters lists basic design requirements to be used as a guide in developing a starter system. These specifications require a shear section but allow the designer to select the value to be compatible with the engine torque requirements.

There is no requirement defining starter direction of rotation. This would be restrictive because of the extra gears, envelope space, etc. that might be needed to comply. Current day starters turn in either direction.

REQUIREMENTS LESSONS LEARNED

The original safety factor was 2.0 with a 15–second start. This has resulted in a number of engine removals because of major damage to the engine starter drivetrain. The factor was increased to 2.5 then to 3.33. The fact that there have been no reports of drive train failures on recently developed engines indicates that the factor is adequate.

4.1.1.19 Starting drivetrain. The requirements of 3.1.1.19 shall be evaluated by analysis and tests.

EVALUATION RATIONALE (4.1.1.19)

The starting drivetrain must be evaluated by analysis and test.

EVALUATION GUIDANCE

Background:

Tests should be conducted at extreme cold conditions since this imposes the greatest load on the starter.

EVALUATION LESSONS LEARNED

None.

3.1.1.19.1 Starting torque and speed. The required starting torques and drive speeds shall be specified herein and shall include engine drag and engine gearbox drag. Figure 3 shall show the effects, singly and in combination, of ambient temperatures, of altitudes from sea level to the maximum of 3.7.11.3, and for air starts with starter assist, using the worst case fuels of the specification. Figure 3 shall be presented for each of the following:

a. No customer bleed air extraction, no customer power extraction.

b. Maximum allowable customer bleed air extraction, no customer power extraction.

c. No customer bleed air extraction, customer power extraction as specified in the model specification.

d. Maximum allowable customer bleed air extraction, customer power extraction as specified in the model specification.

REQUIREMENT RATIONALE (3.1.1.19.1)

Since the engine starter and starting system are not supplied by the engine manufacturer, it is necessary that sufficient information be supplied to permit procurement of a starting system.

REQUIREMENT GUIDANCE

Background:

This is interface information to reduce logistic load. There should be every effort to use existing starter systems for engines of comparable size.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.19.1 Starting torque and speed. The requirements of 3.1.1.19.1 shall be evaluated by test and analysis.

EVALUATION RATIONALE (4.1.1.19.1)

Starting torque and speed requirements must be evaluated to procure the proper starting system.

EVALUATION GUIDANCE

Background:

The procedure for accomplishing the test should be specified in the pre–test data. The test should be conducted at the temperature extremes of figure 4 and using the fuels and oils of 3.7.3.1 and 3.7.8.1, respectively.

EVALUATION LESSONS LEARNED

Extreme cold conditions are the most difficult and taxing for starter systems.

3.1.1.19.2 Moment of inertia of rotating parts. The maximum effective mass moment of inertia (kilogram – meter squared (slug – ft^2)) of engine rotating parts to be rotated by the starter, at the starter drive pad, and the speed ratio between the starter pad and the driven rotor system shall be specified herein.

REQUIREMENT RATIONALE (3.1.1.19.2)

Since the engine starter and starting system are not supplied by the engine manufacturer, it is necessary that sufficient information be supplied to permit procurement of a starting system. This provides the moment of inertia of rotating parts for system dynamic information. This requirement is necessary so that Using Service can thoroughly evaluate engine systems (i.e., transmission, starter, rotor, assembly) and diagnose various problems throughout the life of the engine. It should be noted that redesigns are generally paid for by the Government. If improper designs could be caught early, tremendous cost savings would be achieved. Additionally, if the Government decided to second source a particular component (starter), this requirement is of great significance.

REQUIREMENT GUIDANCE

Background:

This is interface information to reduce logistic load. There should be every effort to use existing starter systems for engines of comparable size.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.19.2 Moment of inertia of rotating parts. The requirements of 3.1.1.19.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.1.1.19.2)

The moment of inertia of rotating parts is interface information that is needed in the early part of the development.

EVALUATION GUIDANCE

Background:

The procedure for accomplishing this test should be specified in the pre–test data. The test should be conducted at the temperature extremes specified and using the fuels and oils as required.

EVALUATION LESSONS LEARNED

None.

3.1.1.19.3 Torsional spring constant. The torsional spring constant (newton – meter per radian (pound inches per radian)) for the engine starting drive system at the starter drive pad shall be specified herein.

REQUIREMENT RATIONALE (3.1.1.19.3)

Since the engine starter and starting system may not be supplied by the engine manufacturer, it is necessary that sufficient information be supplied to permit procurement of a starting system. This provides the torsional spring constant.

This requirement is necessary so that Using Service can thoroughly evaluate engine systems (i.e. transmission, starter, rotor, assembly) and diagnose various problems throughout the life of the engine. It should be noted that redesigns are generally paid for by the Government. If improper designs could be caught early, tremendous cost savings would be achieved. Additionally, if the Government decided to second source a particular component (starter), this requirement would be of great significance.

REQUIREMENT GUIDANCE

Background:

This is interface information to reduce logistic load. There should be every effort to use existing starter systems for engines of comparable size.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.19.3 Torsional spring constant. The requirements of 3.1.1.19.3 shall be evaluated by test.

EVALUATION RATIONALE (4.1.1.19.3)

The torsional spring constant must be evaluated to procure the proper starting system.

EVALUATION GUIDANCE

Background:

The procedure for accomplishing this test should be specified in the pre–test data. The test should be conducted at the temperature extremes specified and using the fuels and oils as required.

EVALUATION LESSONS LEARNED

None.

3.1.1.19.4 Starter train backlash. The maximum backlash, in radians, of the starting drive system at the starter drive pad shall be specified herein.

REQUIREMENT RATIONALE (3.1.1.19.4)

This requirement is necessary so that the Using Service can thoroughly evaluate engine systems (i.e., transmission, starter, rotor, assembly) and diagnose various problems throughout the life of the engine. It should be noted that redesigns are generally paid for by the Government. If improper designs could be caught early, tremendous cost savings would be achieved. Additionally, if the Government decided to second source a particular component (starter), this requirement would be of great significance.

Since the engine starter and starting system may not be supplied by the engine manufacturer, it is necessary that sufficient information be supplied to permit procurement of a starting system. This provides the starter train backlash information.

REQUIREMENT GUIDANCE

Background:

Backlash is the amount by which the width of a tooth space exceeds the thickness of the engaging tooth measured on the pitch circle. Backlash does not adversely affect proper gear function except for lost motion upon reversal of gear rotation. Backlash inevitably occurs because of necessary fabrication tolerances on tooth thickness and center distance plus need for clearance to accommodate lubricant and thermal expansion. This is interface information to reduce logistic load. There should be every effort to use existing starter systems for engines of comparable size.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.19.4 Starter train backlash The requirements of 3.1.1.19.4 shall be evaluated by inspection and test.

EVALUATION RATIONALE (4.1.1.19.4)

Starter train backlash must be evaluated to procure the proper starting system.

EVALUATION GUIDANCE

Background:

The procedure for accomplishing this test should be specified in the pre–test data. Testing should be conducted during IFR at the temperature extremes specified and using the fuels and oils as required.

EVALUATION LESSONS LEARNED

None.

3.1.1.20 Thrust reverser interface Interface dimensions required for thrust reverser nacelle installation, clearances, and connections shall be shown on the engine installation and interface drawing. The reverser mechanism shall provide a signal indicating whenever the reverser is in the unlocked condition. Positive mechanical locking, described on the installation drawing, shall prevent reverser deployment in the event of system failure or inadvertent actuation of the control system. Failure of the locking system shall not cause reverser deployment. No resonant frequency shall be transmitted to or from the thrust reverser interface.

REQUIREMENT RATIONALE (3.1.1.20)

This requirement provides interface information and requires a signal for cockpit indication for operation. Thrust reversers that inadvertently deploy, affect aircraft control and safety; and therefore should operate only upon command.

An inflight failure of the thrust reverser locking system should not cause reverser deployment. Such deployment can cause uncontrollable forces on the aircraft and result in a crash.

REQUIREMENT GUIDANCE

Background:

The reverser should revert to a stowed position during failure modes. The thrust reverser system should be designed so that no single failure or malfunction under any anticipated flight or ground condition results in reverse thrust.

Thrust reversers are used for braking commercial jet transports. Military aircraft generally use other devices. The thrust reverser system should be properly documented for airframe information.

REQUIREMENT LESSONS LEARNED

Thrust reverser actuation systems that have a single motor connected to the actuators via flex shafts have encountered binding. Actuator systems that use multiple hydraulic cylinders have shown better reliability and are more safe than single motor systems.

4.1.1.20 Thrust reverser interface The requirements of 3.1.1.20 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.1.20)

The thrust reverser interface/compatibility information needs to be evaluated by inspection and demonstration.

EVALUATION GUIDANCE

Background:

The dimension and envelope should be checked using a full scale mockup. Mechanical locking and unlocking of the actuation system should be demonstrated during engine testing.

EVALUATION LESSONS LEARNED

None.

3.1.1.21 Exhaust system interface Maximum allowable static (1g) shear, axial and moment loads in all primary axes (x, y, z, as applicable) for any interface connections between the engine and parts of the exhaust system not supplied with the engine shall be specified on figure 5. The maximum allowable loads at any exhaust interface shall also be specified for the maximum allowable maneuver loads as defined in 3.4.1.2.2. No hot exhaust gas leakage is allowed at the interface. Resonant frequencies transmitted to or from the exhaust system through the interface shall not prevent the engine from meeting the requirements of the specification.

REQUIREMENT RATIONALE (3.1.1.21)

Load and moment information for exhaust interface connection is necessary for the airframe contractor to design parts which may interface with the exhaust system.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The shear, axial, and moment loads shown on figure 5 should include loads due to asymmetric aerodynamic pressure and aircraft seal resistance as well as the forces shown on figure 2. For vectoring nozzles, all loads due to vectord forces should also be included on figure 5.

Background:

Exhaust attachments should be of the bolted type in order to assure adequate sealing of the hot gases and yet provide for ease of assembly and disassembly. The other commonly used means of attachment, the V–Band Clamp type requires accurate alignment on assembly to prevent hot gas leakage. This is not practical on large engines.

Aircraft with highly integrated/unique exhaust systems may require separate installation/removal of the engine and exhaust system components, due to configuration layout or limited physical access. These designs may need a more “rapid–disconnect” system of installation and mounting.

REQUIREMENT LESSONS LEARNED

Exhaust nozzle loads, installation, mounting, and removal have been significant issues for certain aircraft arrangements. Thrust vectoring nozzles increase interface loads. Vibratory and acoustical effects have been significant on nozzle design and life.

4.1.1.21 Exhaust system interface The requirements of 3.1.1.21 shall be evaluated by analysis and demonstration.

EVALUATION RATIONALE (4.1.1.21)

The exhaust system interface information needs to be confirmed by analysis and demonstration.

EVALUATION GUIDANCE

Background:

The exhaust nozzle is not only subjected to flight maneuver forces but also aerodynamic and vibratory loads.

EVALUATION LESSONS LEARNED

The Army was concerned on the T700 engine that attachments could possibly be statically loaded to the maximum allowable limit, in which case the connection would be overloaded in a dynamic situation.

3.1.1.22 Compatibility – engine/airframe The interface of the engine with the airframe shall allow the engine to perform as required in the specification.

REQUIREMENT RATIONALE (3.1.1.22)

This is a system level requirement that ties all the installation factors together to ensure that all the engine/airframe interfaces will work correctly.

REQUIREMENT GUIDANCE

Background:

The Using Service should ensure that there is a one inch clearance between the airframe engine bay/nacelle and the engine per SD–24.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.22 Compatibility – engine/airframe The requirement of 3.1.1.22 shall be evaluated by test and demonstration.

EVALUATION RATIONALE (4.1.1.22)

The requirement needs to be evaluated by test and demonstration since an analysis may not reveal all possible problems.

EVALUATION GUIDANCE

Background:

This requirement should be evaluated by test and demonstration throughout the engine development program.

EVALUATION LESSONS LEARNED

None.

3.1.1.23 Control and external (C&E) components list The control, external components, and component functional subsystems of the engine, which require qualification testing separate from the engine, shall be listed, see 4.9.2.1.6.

EVALUATION RATIONALE (3.1.1.23)

Listings of all controls and externals in the model specification is required to enable the contractor and subcontractors to develop the appropriate critical item development specifications required for adequate design, development, evaluation, and testing of engine components.

REQUIREMENT GUIDANCE

Background:

All control and external components should be listed in the specification; examples include fuel pumps, controls, actuators, valves, electrical component, EMS and/or BIT hardware.

This list is used in table VI which requires qualification tests for each control and external.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.23 Control and external (C&E) components list The requirement of 3.1.1.23 shall be evaluated by inspection and analysis.

EVALUATION RATIONALE (4.1.1.23)

Control and external components require qualification testing to establish their integrity prior to commencement of engine qualification testing.

EVALUATION GUIDANCE

Background:

The actual test requirements for the controls and externals listed in 3.1.1.23 are specified in 4.9.2.

EVALUATION LESSONS LEARNED

None.

3.1.1.24 Government furnished/loaned property No government furnished/loaned property shall be incorporated in the engine design.

REQUIREMENT RATIONALE (3.1.1.24)

Government property must not be used in engine designs so that the contractor is solely responsible for the engine's performance, operability, and durability.

REQUIREMENTS GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.1.1.24 Government furnished/loaned property The requirement of 3.1.1.24 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.1.24)

The use of Government property is prohibited.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.2 Physical characteristics

4.1.2 Physical characteristics

3.1.2.1 Dry mass (weight) of complete engine The dry mass (weight) of the complete engine, but without the engine exhaust system, shall not exceed (a) kg ((a) lbm). The engine supplied exhaust system shall not exceed (b) kg ((b) lbm). The weights of engine components which are not mounted on the engine shall be listed and included in the dry weight of the engine. The engine station where the engine center of gravity (cg) occurs shall be specified herein.

REQUIREMENT RATIONALE (3.1.2.1)

The dry mass (weight) of the engine must be established for weapon system use to establish the empty weight of a system in accordance with MIL–W–25140. It is also an inventory item in the DD 365–1 of the weight and balance handbook.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The contractor should specify the dry mass (weight). The dry mass (weight) of any armor used to protect the engine should also be included in the dry mass weight of the engine.

(b): The contractor should provide dry mass (weight) of the exhaust system

Background:

The “complete engine” should include all engine contractor supplied components including technology features, coatings, and exhaust system components attached to the engine while in the airframe installation. Weight should be specified separately for engine and exhaust if the exhaust system is heavy or otherwise unconventional. The associated weights should be identified individually to separate basic turbomachinery from other components.

Examples of engine components not mounted on the engine include: life recorders, electronic control units, and vectorable nozzles.

Items which are not included in the aircraft installed weight (i.e., cover plates, dust covers, etc.) should not be included in the dry weight.

REQUIREMENT LESSONS LEARNED

None.

4.1.2.1 Dry mass (weight) of complete engine The requirement of 3.1.2.1 shall be evaluated by test.

EVALUATION RATIONALE (4.1.2.1)

The dry mass (weight) of the complete engine must be established to be consistent with the aircraft requirements.

EVALUATION GUIDANCE

Background:

Each qualification and production engine should be weighed, using the same procedure, to ensure consistency. The Using Service should make sure that weighing equipment has been calibrated within the last six months and weighing accuracy is within 0.1%.

EVALUATION LESSONS LEARNED

Weighing each qualification and production engine will provide data to determine effects of engine–to–engine variation.

3.1.2.2 Mass (weight) of residual fluids The mass (weight) of residual fluids remaining in the engine after operation and drainage, and the drainage attitude of the main rotor axis relative to the level plane, shall be specified herein.

The engine station where the engine center of gravity (cg) occurs with the specified fluid conditions shall be specified herein.

REQUIREMENT RATIONALE (3.1.2.2)

The mass (weight) of residual fluids remaining in the engine after operation and drainage should be specified in order to accurately account for the engine mass (weight). The weight of residual fluids is needed as part of MIL–W–25140 to establish basic weight and is part of the running log weight maintained in DD 365–3 of the weight and balance handbook.

REQUIREMENT GUIDANCE

Background:

Residual fluids include fuel, oil, hydraulic fluid, etc.

REQUIREMENTS LESSONS LEARNED

None.

4.1.2.2 Mass (weight) of residual fluids The requirement of 3.1.2.2 shall be evaluated by test.

EVALUATION RATIONALE (4.1.2.2)

Accurate determination of the mass (weight) of residual fluids in the engine is required for aircraft mass (weight) and balance calculations.

EVALUATION GUIDANCE

Background:

The mass (weight) of all residual fluids (fuel, oil, hydraulic, etc.) remaining in the engine after operation and drainage should be determined while the engine is in a single specified attitude of the main rotor axis relative to the level plane. The residual fluid mass (weight) should be determined by calculating the difference between the mass (weight) of an engine which has been operated and drained and the dry mass (weight) of the same engine before being serviced with fluids. Fluids should only be drained through lines or other connections specified for normal maintenance practices.

EVALUATION LESSONS LEARNED

None.

3.1.2.3 Mass (weight) of operating fluid The total mass (weight) of fluids in an operating engine shall be specified herein. The engine station where the engine center of gravity (cg) occurs with the specified fluid conditions shall be specified herein.

REQUIREMENT RATIONALE (3.1.2.3)

The mass (weight) of fluids within the engine during operation should be specified in order to accurately account for the engine mass (weight) in the aircraft's mass (weight) and balance computations.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.1.2.3 Mass (weight) of operating fluid. The requirements of 3.1.2.3 shall be evaluated by test.

EVALUATION RATIONALE (4.1.2.3)

Accurate determination of the total mass (weight) of fluids in the operating engine (i.e., with a full oil tank, prime oil, and fuel systems, etc) is required for aircraft mass (weight) and balance computations.

EVALUATION GUIDANCE

Background:

The mass (weight) of fluids in the operating engine should be determined by calculating the difference between the mass (weight) of an operational engine and the dry mass (weight) of the same engine before servicing with fluids.

EVALUATION LESSONS LEARNED

None.

3.1.2.4 Mass (weight) of additional equipment. The weights of items which are not a part of the engine bill of materials, but which are installed on the engine by the engine manufacturer, shall be listed below:

Item	Weight

The engine station where the engine center of gravity (cg) occurs with the specified equipment shall be specified herein.

REQUIREMENT RATIONALE (3.1.2.4)

The weights of non–engine furnished equipment mounted on, and delivered with, the engine should be specified in order to accurately account for the engine weight in the aircraft's weight and balance handbook (DD 365–1).

REQUIREMENT GUIDANCE

Background:

The weights of items of equipment (e.g., airframe brackets, attaching clips, wiring harnesses, shields, cover plates, etc.) which are not part of the engine bill of material, but which are installed on the engine by the engine contractor, should be listed and totaled in the engine specification.

REQUIREMENT LESSONS LEARNED

Some aircraft parts are best installed during engine assembly, especially those parts which require opening the engine fluid lines for installation. Also, some engines with multiple applications may require cover plates for features not used. These part weights must be included in the total aircraft weight and balance computations.

4.1.2.4 Mass (weight) of additional equipment The weight of the equipment listed in 3.1.2.4 shall be evaluated by test.

EVALUATION RATIONALE (4.1.2.4)

Determination of the total weight of the engine, including parts furnished by the airframe manufacturer, must be accomplished for accurate aircraft weight and balance calculations.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.2.5 Mass moment of inertia of complete engine The maximum effective mass moment of inertia (slug – ft^2) of the complete operational engine about three mutually perpendicular axes with the origin at the center of gravity, shall be specified herein. The maximum effective mass moment of inertia (slug – ft^2) about the resultant rotational axis of each engine rotor, together with the direction of rotation when viewed looking forward from the exhaust exit, shall also be specified herein.

REQUIREMENT RATIONALE (3.1.2.5)

The maximum mass moment of inertia of the complete engine about three mutually perpendicular axes is required for aircraft structural design.

REQUIREMENT GUIDANCE

Background:

The mass moment of inertia of the engine about its' center of gravity should be based on the summation of the dry mass (weight) of the complete engine, and the mass (weight) of operating fluids, and should be specified about the engine's pitch, roll, and yaw axes. For turbofan and turbojet applications, the maximum effective mass moment of inertia about the rotational axis of each rotor, together with the direction of rotation when viewed from the exhaust end looking forward, should be specified.

For turboshaft applications, the maximum effective mass moment of inertia of each engine rotor system about the resultant rotational axis, together with the effective direction of the inertia and the direction and location of the resultant axis, should be specified. For engines with geared rotor systems, the shaft to which all inertias of each rotor system have been algebraically referred, should be specified. The maximum effective mass moment of inertia of the complete power output system, including the reduction gear train (referred to the output shaft speed), should be specified.

REQUIREMENT LESSONS LEARNED

None.

4.1.2.5 Mass moment of inertia of complete engine The requirement of 3.1.2.5 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.1.2.5)

The mass moments of inertia must be accurately determined so that it is available for use in aircraft design considerations.

EVALUATION GUIDANCE

Background:

The mass moments of inertia should be determined at each of the milestones. The method used should be the most accurate means available, consistent with the size and weight of the engine. The accuracy of the method used should be determined and specified.

The mass moments of inertia for the complete operational engine should be used for determining forces required as a result of rotational inertia and the moments of inertia of the rotors should be used to determine forces required as a result of gyroscopic moments.

EVALUATION LESSONS LEARNED

None.

3.1.3 Materials, processes and parts When the engine manufacturer's documents are used for materials and processes, such documents shall be subject to review by the Using Service prior to the start of IFR and, unless specifically disapproved, will be considered released upon approval of FFR, ISR, and OCR. The use of non–governmental documents shall not constitute waiver of Government inspection. The Using Service reserves the right to inspect any and all processes of manufacture. The use of magnesium and silver shall not be permitted, unless approved by the Using Service. Copper, brass, nickel, and cadmium shall not be used in engine parts which are in direct contact with fuel or oil. Dissimilar metal as defined in MIL–STD–889 shall not be used in direct contact with each other. An appropriate surface treatment and finishing system shall be selected from Appendix A of MIL–STD–889 to protect these metals from galvanic corrosion when joined and subjected to the specific environment.

Materials, components, and parts shall perform satisfactorily during and after exposure to fuel.

REQUIREMENT RATIONALE (3.1.3)

All aspects of materials and associated processes must be subject to approval of the Using Service, with special emphasis given to avoid the use of magnesium and dissimilar metals to prevent galvanic corrosion failures. For example, copper should be prevented from coming in contact with fuel or oil during engine operation. Copper acts as a catalyst, adversely affecting the thermal stability and increases the rate of deterioration in these fluids.

REQUIREMENT GUIDANCE

Background: In a specific turbojet engine model there was a compressor case and compressor blades made of titanium. When the blades failed during excitation, they produced pieces that rubbed against the case and caused titanium fires. If material selection for these blades and/or case excluded titanium, then the fires could have possibly been avoided. The use of copper, brass, and cadmium is not allowed in the fuel systems due to possible corrosion of aircraft parts which are in contact with fuel.

REQUIREMENT LESSONS LEARNED

Magnesium is restricted because it is extremely susceptible to corrosion, especially in a marine environment. Even a small “pin hole” break in the protective coating will allow corrosion to take place under the remainder of the protective coating. On visits (1978–1979) to the NADEPS, it was observed that J79 engine magnesium accessory gearboxes have been scrapped due to extensive corrosion. An Air Force engine with a magnesium gearbox experienced galvanic corrosion due to dissimilar metals being used at the front frame mount hole/towershaft interface.

The T76 engine was originally developed with a magnesium propeller gearbox. Major corrosion problems were incurred with the gearbox (after the coating was scratched during normal maintenance), and the magnesium was later replaced by aluminum.

Copper and cadmium can be eliminated from the fuel and oil system design without undue hardship. MIL–STD–889, Appendix A contains some methods of combining dissimilar methods.

4.1.3 Materials, processes, and parts The requirements of 3.1.3 shall be evaluated by analysis and inspection. When applicable, engine materials, components, and parts shall be tested for fuel resistance using TT–S–735, type I and III test fluids.

EVALUATION RATIONALE (4.1.3)

Material properties must be established by test using standardized, accepted practices.

EVALUATION GUIDANCE

Background:

Specific allowances used for design shall be statistically derived in accordance with MIL–HDBK–5 or other applicable documents. Special “allowances” which represent candidate or final solutions should be developed by test in accordance with the test plan established under the overall integrity program for the specific application. The material property characterization plan should be referenced. The contractor, in developing this plan, should review existing data on candidate materials and processes and develop a test plan to generate the data to fully characterize the material for the intended application. The test plan should contain a materials listing and identification of properties that should be characterized for each material. Material variability should be examined and means of insuring minimum properties in critical parts should be developed in accordance with integrity program guidance. The Using Service should require this plan on the CDRL.

EVALUATION LESSONS LEARNED

None.

3.1.3.1 Adhesives and sealants The use of adhesive or sealant compounds is permitted only upon specific application approval by the Using Service prior to its incorporation into the design of the engine.

REQUIREMENT RATIONALE (3.1.3.1)

This assures that only those adhesives and sealants acceptable to the Using Service will be used and are governed as a minimum by MS 18069.

REQUIREMENT GUIDANCE

Background:

The engine contractor should specify how, when, and where adhesives and sealants will be applied to prevent sealing and adhesive failures. Adhesives and sealants should not be used as a locking feature.

Adhesives and sealants should be in compliance with current federal environmental hazardous materials regulations. Provided there is no current industry or government standard, the Using Service may utilize the following requirements in the main paragraph if sealing or locking compounds are used on threads. The use of retaining, sealing, and locking compounds applied to the threads or bearing surfaces of aircraft threaded components should:

a. not be used on threaded components except those that are used in a permanent or semi–permanent installation.

b. be applied to the threads and bearing surfaces before assembly of components.

c. not be used on threaded fasteners unless the externally threaded fastener is preloaded at installation. The preload should be greater than 20% of the ultimate strength of the fastener.

d. not be used as a primary locking or retaining method in those applications where the failure of the compound to seal, retain, or lock would endanger the safety of personnel or would render the equipment inoperative or cause its destruction.

e. not be used in structural and control system applications except as an auxiliary locking or retaining method.

f. not be used on threaded components for which a torque or preload value is specified for assembly, unless it is feasible to discard and replace these by new components. Threaded fasteners previously treated with such compounds should not be reused. These fasteners should be replaced by new fasteners newly coated.

g. be in accordance with MIL–S–22473, MIL–S–46163, MIL–R–46082, and the following requirements:

(1) The compound should not accelerate galvanic or stress corrosion of cadmium plating, corrosion resistance steels, aluminum alloys, or titanium alloys in naval aircraft environments.

(2) The compound should not cause degradation of the finishes prescribed in MIL–F–7179 or other approved finish materials.

(3) The compound should not cause degradation or reduce adhesion of tank sealant materials.

(4) The compound should not accelerate or cause degradation of aircraft fuels, lubricants, and hydraulic fluids, which may be in contact with compound.

(5) The compound should be inert in the present aircraft fuels, lubricant, and hydraulic fluids.

h. should be used only where no other satisfactory means exist.

REQUIREMENT LESSONS LEARNED

Adhesives and sealants have complicated maintenance and repair (i.e., having proper type available and using correct age control). They should be minimized. An afterburner liner delta P problem was caused by excessive use of room temperature curing silicone based sealant. In the past silicone was used in the fan section of the engine to seal gaps at the blade platforms. This silicone broke off and plugged cooling holes in the afterburner liner.

4.1.3.1 Adhesives and sealants The use of adhesives and sealants shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.3.1)

The use of adhesives and sealants can only be evaluated by demonstration and test.

EVALUATION GUIDANCE

Background:

The adhesives and sealants should be evaluated for corrosion, engine fluid loss, loosening of fasteners, and integrity of joined structures. The list of adhesives and sealants should be provided for approval 60 days after contract award.

EVALUATION LESSONS LEARNED

None.

3.1.3.2 Elastomeric materials Elastomeric materials shall have an unlimited shelf life (non–age sensitive) and, on an operational engine, have a minimum of (a) engine life. Materials exposed to fuels and lubricants shall be compatible with such fluids throughout the entire fuel or lubrication system temperature cyclic envelope without experiencing swelling, shrinking or other forms of material deterioration which would impair proper functioning or necessitate replacement to prevent impairment of function. Polychloroprene materials (e.g., neoprene) shall not be used in engine parts which are in direct contact with fuels or lubricants during engine operation.

REQUIREMENT RATIONALE (3.1.3.2)

This requirement assures that the quality of elastomeric material used in the manufacture of elastomeric parts will not be lessened as a result of time spent in storage and during engine operation.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): up to one design service life

Background:

The elastomeric material must not be allowed to deteriorate, to permit voids to form in the seals and, in turn allow the fluid to leak past the seal. “Unlimited shelf life” for elastomers is equivalent to the term “non–age sensitive” elastomers. This requirement may be covered in the R&M section.

REQUIREMENT LESSONS LEARNED

None.

4.1.3.2 Elastomeric materials The requirements of 3.1.3.2 shall be evaluated by inspection and tests.

EVALUATION RATIONALE (4.1.3.2)

The evaluation of elastomeric materials needs to be accomplished by inspection and tests.

EVALUATION GUIDANCE

Background:

The cause of all fluid leaks should be determined and logged.

EVALUATION LESSONS LEARNED

None.

3.1.3.3 "O" rings, seals, and packings All nonmetallic "O" rings, seals, and packings used in the engine, including vendor supplied components, shall conform to the applicable dimensions and tolerances shown on AS 568. Materials exposed to fuels and lubricants shall be compatible with such fluids throughout the entire fuel or lubrication system temperature cyclic envelope without experiencing swelling, shrinking, or other forms of material deterioration which would impair proper functioning or necessitate replacement to prevent impairment of function. Polychloroprene materials (e.g., neoprene) shall not be used in engine parts which are in direct contact with fuels or lubricants during engine operation.

REQUIREMENT RATIONALE (3.1.3.3)

It is necessary to limit the sizes of non–metallic "O" rings, seals, and packings used in engines and reduce the proliferation of unnecessary sizes. Not all "O" rings are made of elastomeric materials.

REQUIREMENT GUIDANCE

Background:

"O" rings, seals, and packings should be compatible with fuels and lubricants and neither the seal/packing or the fuel/lubricant should be affected. Particular attention should be given to "O" rings used for high temperature applications since they may be susceptible to excessive leakage when exposed to low temperatures. An example of use of a non elastomeric material would be polytetraflouoroethylene "O" ring in transmissions, variable geometry actuators, gearboxes, etc.

REQUIREMENT LESSONS LEARNED

None.

4.1.3.3 "O" rings, seals, and packings The requirements of 3.1.3.3 shall be evaluated by inspection and test.

EVALUATION RATIONALE (4.1.3.3)

Approved selection of "O" rings, seals, and packings can only be evaluated by inspection and tests.

EVALUATION GUIDANCE

The Using Service should confirm that the "O" rings, seals, and packings conform to AS 568. Extreme temperature test should evaluate the seals under extreme environmental conditions.

EVALUATION LESSONS LEARNED

"O" rings have failed at extreme cold temperatures.

3.1.3.4 Corrosion protection The materials, coatings, and processes used in the design and manufacture of the complete engine shall be corrosion resistant.

REQUIREMENT RATIONALE (3.1.3.4)

This requirement assures proper corrosion protection is applied to the materials, coatings, and processes by the contractor or vendor to reduce chance of failure of the engine hardware due to corrosion.

REQUIREMENT GUIDANCE

Background:

The engine contractor should submit a list of all protective treatments. Past experience with coatings and processes should be used when existing materials are proposed. Metallurgist are continually making small changes to materials, coatings, and processes.

Where a metal and non–metal are coupled, in presence of 3.5% salt water, the galvanic voltage formed should not exceed 0.15 volts without the interface couple being protected from corrosion. The cathodic material should be topcoated except where it needs to be bare to perform an engine function. Acid rain equivalent to Ph 3.5 using sulfuric acid solutions should be used to qualify the engine components for resistance to rain and condensed humidity (diurnal cycling through the dew point). Processes and materials used to maintain corrosion prevention should be used and verified for their technical capability. The design shall be maintained using the Joint NAVAIR 16–1–540/Air Force TO 1–1–689, except where the design includes unique materials or design detail that cannot be serviced by the manual. In this instance, a system should be maintained with a unique corrosion prevention technical order with the procedures and materials validated and verified to the system requirement prior to approval for use on the system production design hardware.

REQUIREMENT LESSONS LEARNED

Messages from COMNAVAIRLANT and COMNAVAIRPAC have addressed the general problem of corrosion on numerous occasions in the past.

4.1.3.4 Corrosion protection The requirements of 3.1.3.4 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.3.4)

Corrosion protection must be evaluated by demonstration and test.

EVALUATION GUIDANCE

Background:

The engine contractor usually evaluates all coatings and processes by strip samples or component tests. This requirement should be evaluated in 4.3.1.3 (Corrosion atmosphere test).

EVALUATION LESSONS LEARNED

Corrosion has been a major problem and has resulted in huge man–hour expenditures for repair. Corrosion can be a slow and subtle process that may take years to appear during actual field use. It may appear only in certain localities or result from a particular mission. In the past, contractors have attempted to satisfy this requirement by material coupon test (e.g., a sample material is immersed in a corrosive solution and a salt spray environment, however the material is not exposed to the thermal environment normally seen in an operating engine). Coupon tests have not been considered acceptable substitutes for the corrosive atmosphere test.

3.1.3.5 Nonmetallic hose Nonmetallic hose shall be in accordance with MIL–H–27267. Hose assemblies shall comply with MIL–H–25579. Hose carrying flammable fluid shall comply with 3.1.8.1.

REQUIREMENT RATIONALE (3.1.3.5)

This requirement assures standard nonmetallic hose and hose assemblies will be used to reduce the potential for failure or leaking. This also limits the sizes for hose and hose assemblies in order to reduce the proliferation of unnecessary sizes.

REQUIREMENT GUIDANCE

Background:

The nonmetallic hose and hose assemblies should be capable of resisting failure due to such items as corrosion, stress, overtightening torque, vacuum, temperature, workmanship, and tolerances. This requirement may be covered in the R&M section.

REQUIREMENT LESSONS LEARNED

Some engine contractors prefer not to use nonmetallic hose.

4.1.3.5 Nonmetallic hose The requirement of 3.1.3.5 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.3.5)

Nonmetallic hose must be evaluated by inspection, demonstration, and test.

EVALUATION GUIDANCE

Background:

The nonmetallic hose must be inspected to insure it conforms to MIL–H–27267 and MIL–H–25579. Demonstration will show if it can withstand stress and overtightening torque. Engine tests will determine its suitability to the engine environment.

EVALUATION LESSONS LEARNED

Hard tubing can usually be substituted for nonmetallic hose when any problems occur.

3.1.3.6 Shot peening For parts dependent upon the residual stresses from shot peening to attain required fatigue life, the following requirements shall apply:

a. Manufacture shall be by computer monitored and controlled shot peening. SAE AMS 2432 shall be the controlling document. Control limits which shutdown the process prior to exceeding critical parameters and documentation of process records (including any 100 percent inspections imposed as a result of process control discrepancies) shall be required.

b. Peening required during repair or rework to attain fatigue life shall be by computer controlled and monitored processes or shall conform to the same specifications and standards used for original manufacture.

REQUIREMENT RATIONALE (3.1.3.6)

Shot peening is a recognized method of improving the fatigue characteristics and damage tolerance in manufactured metallic aircraft and engine components. The large magnitude of improvement possible, the difficulty of evaluating the improvement through inspection, and the potential for in–service degradation of peening benefits require establishing a uniform policy for shot peening.

REQUIREMENT GUIDANCE

Background:

This requirement originated from NAVAIR INSTRUCTION 4870.2 dated 4 June 90. The Using Service should make sure that strict control of shot peening and good understanding of its benefits are documented or shown.

REQUIREMENT LESSONS LEARNED

The US Navy experienced fleet problems from shot peened material left in production engines during build–up on the assembly line. This caused significant logistical problems for that engine model in determining which production engines were affected and consequently required disassembly/inspection for shot peened debris. This raised concerns on having strict quality control of shot peening because of its major contribution to life enhancement in some parts.

4.1.3.6 Shot peening The requirement of 3.1.3.6 shall be evaluated by analysis, test, and demonstration via the structural integrity evaluations of 3.4.1, and as follows:

a. Prior to or concurrent with the preliminary design review, the contractor shall conduct the following analyses:

(1) A strength and life analysis that identifies the magnitude of life enhancement required from peening, assesses the potential for degradation in life enhancements during operation and support, and identifies appropriate safeguards against degradation including the establishment of a proposed inspection interval that accounts for likely degradation.

(2) A report describing planned manufacturing process controls and nondestructive inspection.

b. Life certification and qualification testing (along with the strength and life analysis) shall include evaluation of the proposed inspection interval.

c. All Engineering & Manufacturing Development and life certification or qualification testing shall be conducted on parts peened as per production requirements. The hardware configuration for testing necessary to evaluate a proposed inspection interval shall be approved by the Using Service prior to the start of testing.

EVALUATION RATIONALE (4.1.3.6)

The shot peening process must be evaluated for effectiveness and accuracy.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.4 Fasteners Fasteners shall conform to the following specifications or standards: (a) . At all joints in control systems, at single attachments, and where the loss of bolts affects safety of flight or ability to control the engine, self–retaining bolts shall be used. Installation and retention of self–retaining bolts (i.e., a slotted or castellated nut) shall be in accordance with MS33602.

REQUIREMENT RATIONALE (3.1.4)

Threaded fasteners and other connections must be selected to meet strength and standardization requirements for the application.

This requirement assures that self–retaining positive locking bolts, in accordance with MIL–B–23964, by virtue of their positive locking elements, are used to reduce the chance of bolt loss (slipping out) at joints in control systems, at single attachments, and where loss of the bolt would affect safety of flight or ability to control the engine. Specifically, bolt loss would be prevented by means of the positive locking elements even if the nut and cotter pin were lost.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): Either the Using Service or the Contractor should fill in the blank. The Using Service may fill the blank with MIL–B–23964.

Background:

MIL–STD–1515, Section 400, should be researched for propulsion system fastener requirements.

REQUIREMENT LESSONS LEARNED

Silver plating used as an anti–seize agent on bolts in the turbine section reacted with the disk material at elevated temperatures resulting in the initiation of cracks in the disks. Tack welded captive nuts, if not properly aligned, have resulted in failures due to backing out of the fasteners. The use of silver plate on inner threads of bolts in order to retain torque of a nut at high temperatures encountered in component operation can lead to the brazement of nut to bolt. Consequently, in the 1970's some exhaust nozzle controls had to have their compressor bleed–air motor bolts sawed off during overhaul.

4.1.4 Fasteners The requirements of 3.1.4 shall be evaluated by test.

EVALUATION RATIONALE (4.1.4)

Engine testing is required to evaluate the adequacy and characteristics of fasteners.

EVALUATION GUIDANCE

Background:

Fastener torque retention, strength, and removal characteristics should be evaluated in the engine endurance test. An inspection of each engine submitted for qualification testing should be accomplished to evaluate that all fasteners conform to the parts list. The condition of the fasteners at the completion of tests should be used to determine the acceptability of each for the application.

EVALUATION LESSONS LEARNED

None.

3.1.4.1 Securing of fasteners Where cotter pins are used to secure fasteners, the practices in MS 33540 shall be followed. The general design and usage limitation of self–locking nuts specified in MS 33588 shall be applied. Safety wire is acceptable only on areas of the engine which are not involved in intermediate or lower level of maintenance. Safety wire practice shall conform to MS 33540.

REQUIREMENT RATIONALE (3.1.4.1)

This requirement assures the proper methods of securing fasteners that are employed to reduce the chance of fastener loss.

REQUIREMENT GUIDANCE

Background:

MS 33540 and MS 33588 prescribe methods to restrain the rotation and loosening of assembled fasteners. To reduce the chance of FOD and to save labor, safety wire is only permitted where it is not involved in low level or intermediate maintenance.

REQUIREMENT LESSONS LEARNED

Safety wire was once extensively used throughout the engine to secure fasteners. Poor housekeeping resulted in frequent FOD by stray pieces of safety wire.

4.1.4.1 Securing of fasteners The requirements of 3.1.4.1 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.4.1)

Securing of fasteners must be evaluated by inspection.

EVALUATION GUIDANCE

Background:

An inspection of each engine submitted for qualification testing should be accomplished to evaluate whether all fasteners conform to the parts list. The condition of the fasteners at the completion of tests should be used to determine the acceptability of each for the application.

EVALUATION LESSONS LEARNED

None.

3.1.4.2 Clamps Tube support clamps shall be subject to approval by the Using Service.

REQUIREMENT RATIONALE (3.1.4.2)

This requirement assures only proper tube support clamps will be used so that the potential for tube failures will be minimized.

REQUIREMENT GUIDANCE

Background:

Tube support clamps restrict tube motion. Too much restriction can damage the tube. There is no standard for clamps.

REQUIREMENT LESSONS LEARNED

MS 9825 and MS 9826 were used for clamp standards. Failure of clamps was observed and these standards were made inactive for design after September 1972. No superseding standards have been developed. Past clamp designs have used a vibration dampening material to directly contact the tube surface.

4.1.4.2 Clamps The requirements of 3.1.4.2 shall be evaluated by inspection and tests.

EVALUATION RATIONALE (4.1.4.2)

Clamps for tubes need to be evaluated by inspection and tests.

EVALUATION GUIDANCE

Background:

An inspection of each engine submitted for qualification testing should be accomplished to evaluate whether all clamps conform to the parts list. The condition of the clamps at the completion of tests should be used to determine the acceptability of each for the application.

EVALUATION LESSONS LEARNED

The engine contractor usually uses his own specifications for clamps.

3.1.4.3 Screw recesses All screw recesses (internal drives) shall conform to MS 33750 or MS 9006.

REQUIREMENT RATIONALE (3.1.4.3)

This requirement assures screw recesses conform to specific proven MS designs to permit proper assembly and removal torques.

REQUIREMENT GUIDANCE

Background:

The design of screw recesses attempts to maximize the contact area between the installation tool and the recess. The head of the fastener should have the maximum strength to resist tension and shear forces.

REQUIREMENT LESSONS LEARNED

MS 33781 has been deleted because of problems with recesses of this type in service.

4.1.4.3 Screw recesses The requirements of 3.1.4.3 shall be evaluated by inspection and tests.

EVALUATION RATIONALE (4.1.4.3)

Screw recesses need to be evaluated by inspection and tests.

EVALUATION GUIDANCE

Background:

An inspection of each engine submitted for qualification testing should be accomplished to evaluate whether all recesses conform to the MS drawings. The condition of the screw recesses at the completion of tests should be used to determine the acceptability of each for the application.

EVALUATION LESSONS LEARNED

The T700 engine specification added the following: “Or an internal hex may be used for control adjustments and in special applications such as for holding parts while tightening nuts”.

3.1.4.4 Screw threads All threads shall conform to the requirements of MIL–S–8879 Classes 3A or 3B. Duplicate parts differing only in thread form are not permitted.

REQUIREMENT RATIONALE (3.1.4.4)

This assures the thread form for screw threads complies with MIL–S–8879 Class 3A or 3B.

REQUIREMENT GUIDANCE

Background:

The Class 3A and 3B screw thread forms are “rolled screw threads”. This results in good grain flow of material and a very strong thread. There is also a minimum screw thread root radius which reduces stress concentration. The two screw thread forms help to insure the integrity of fastener systems and reduce the possibility of engine failures. MIL–S–7742 is not applicable for new designs as of 31 December 1991.

REQUIREMENT LESSONS LEARNED

None.

4.1.4.4 Screw threads The requirements of 3.1.4.4 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.4.4)

Screw threads need to be evaluated by inspection and demonstration.

EVALUATION GUIDANCE

Background:

An inspection of each engine submitted for qualification testing should be accomplished to evaluate whether all screw threads conform to the MIL–S–8879. The condition of the screw threads at the completion of tests should be used to determine the acceptability of each for the application.

EVALUATION LESSONS LEARNED

None.

3.1.4.5 Helical coil installation The dimensions and tolerances of the parent material threads intended for use with helical coil inserts shall comply with MS 33537. The installation and use of helical coils shall not cause reduction in life of the affected components.

REQUIREMENT RATIONALE (3.1.4.5)

This requirement reduces the possibility of losing the bolt or stud and the helical coil insert.

REQUIREMENT GUIDANCE

Background:

Experience–proven standardized dimensions and tolerances help to insure a proper fit of the insert into the parent material and the installed bolt or stud into the insert.

REQUIREMENT LESSONS LEARNED

None.

4.1.4.5 Helical coil installation The requirements of 3.1.4.5 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.4.5)

Helical coil installations need to be evaluated by inspection and demonstration.

EVALUATION GUIDANCE

Background:

An inspection of each engine submitted for qualification testing should be accomplished to evaluate whether all helical coil installation conform to MS 33537. The condition of the fasteners at the completion of tests should be used to determine the acceptability of each for the application.

EVALUATION LESSONS LEARNED

None.

3.1.4.6 External wrenching elements Wrenching elements for external drives shall conform to AS 870 for twelve point drives, MS 33787 for spline drives.

REQUIREMENT RATIONALE (3.1.4.6)

This assures the wrenching elements for external drives conform to performance–proven standards. Conformation to the standards reduces proliferation of unnecessary sizes which increases the logistics system.

REQUIREMENT GUIDANCE

Background:

The standards cover external fastener requirements of most engines in operation and presently planned for the future.

REQUIREMENT LESSONS LEARNED

None.

4.1.4.6 External wrenching elements The requirements of 3.1.4.6 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.4.6)

External wrenching elements need to be evaluated by inspection and demonstration.

EVALUATION GUIDANCE

Background:

An inspection of each engine submitted for qualification testing should be accomplished to evaluate that all external wrenching elements conform to the AS and MS documents of 3.1.4.6. The condition of the external wrenching elements at the completion of tests should be used to determine the acceptability of each for the application.

EVALUATION LESSONS LEARNED

None.

3.1.5 Nameplate and product marking Equipment, assemblies, modules, and parts shall be marked in accordance with MIL–STD–130.

The engine shall be permanently marked to indicate all connections shown on the engine installation drawing.

REQUIREMENT RATIONALE (3.1.5)

All engine components require marking or identification.

REQUIREMENT GUIDANCE

Background:

The markings should be visible through normal access openings on the engine and/or aircraft. The Using Services only allows color red marking when adjacent non–red color marking is easily readable in red light conditions.

REQUIREMENT LESSONS LEARNED

Components have had to be removed to examine nameplates and markings during inspection and maintenance due to the poor placement of those markings.

4.1.5 Nameplate and product marking This requirement of 3.1.5 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.5)

Evaluation of property marking or identification can only be accomplished by inspection.

EVALUATION GUIDANCE

Background:

Evaluation should be made on completeness, accuracy, and visibility of the information supplied.

EVALUATION LESSONS LEARNED

None.

3.1.5.1 Engine data plate marking The engine data plate shall include: (a) manufacturer's identification, (b) engine serial number, (c) purchase order or contract number, (d) engine model designation, and (e) warranty notification/information.

REQUIREMENT RATIONALE (3.1.5.1)

All U.S. military property requires proper identification.

REQUIREMENT GUIDANCE

Background:

Engine serial numbers should be in accordance with MIL–STD–1559 and engine model designations in accordance with MIL–STD–1812.

REQUIREMENT LESSONS LEARNED

None.

4.1.5.1 Engine data plate marking The requirement of 3.1.5.1 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.5.1)

The requirement can only be evaluated by inspection.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.5.2 Warranty marking Each warranted engine, major component, module, and accessory shall have the word "warranted" permanently and clearly imprinted on the nameplate.

REQUIREMENT RATIONALE (3.1.5.2)

Marking warranted parts is needed for maintenance and logistics.

REQUIREMENT GUIDANCE

Background:

Unique warranty conditions should be specified in the contract.

REQUIREMENT LESSONS LEARNED

None.

4.1.5.2 Warranty marking The requirements of 3.1.5.2 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.5.2)

Evaluation of warranty marking can only be accomplished by inspection.

EVALUATION GUIDANCE

Background:

The warranted parts list should be used during inspection to check that all warranted parts are marked per 3.1.5.2.

EVALUATION LESSONS LEARNED

None.

3.1.5.3 Part number marking Part number marking shall be in accordance with MIL–STD–100.

REQUIREMENT RATIONALE (3.1.5.3)

The maximum number of digits or characters must be limited due to computer limitations used for inventory/logistical control.

REQUIREMENT GUIDANCE

Background:

Engine part numbers should be limited to a total of 15 digits and/or characters to be consistent with the capability of the Using Service parts life tracking system and the asset management system.

REQUIREMENT LESSONS LEARNED

This requirement was first imposed by NAVAIR during the LAMPS MK–3 program.

4.1.5.3 Part number marking The requirement of 3.1.5.3 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.5.3)

The number of digits or characters used for part number can only be evaluated by inspection.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.5.4 Critical parts identification and tracking The engine life critical parts/subassemblies shall be identified by serial numbers marked legibly on the part. The parts shall have multiple, non–wearing reference surfaces and have a space designated for marking the number of cycles and time accumulated between each overhaul period.

REQUIREMENT RATIONALE (3.1.5.4)

Identification of life critical parts is necessary for adequate tracking during the maintenance and overhaul process.

REQUIREMENT GUIDANCE

Background:

The critical part or assembly serial number should be legibly and visibly marked on their surfaces as prescribed in the parts control program. MIL–STD–965 should be adhered to in the parts control program. Provisions should be made during the design process for adequate identification of life limited fracture and fatigue critical parts, and for components that have established time between overhauls (TBOs). The requirements of MIL–STD–841 for marking of parts should be followed.

REQUIREMENT LESSONS LEARNED

None.

4.1.5.4 Critical parts identification and tracking The requirement of 3.1.5.4 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.5.4)

The requirement must be evaluated by inspection.

EVALUATION GUIDANCE

Background:

Testing of every part of the engine attempts to simulate the real world environment of thermal cycling, mechanical wear, chemical cleaning, and other deteriorative effects that the engine will encounter. Parts should be marked prior to testing, and evaluation of durability of identification should be conducted upon completion of component and engine tests.

EVALUATION LESSONS LEARNED

In the past parts identification has worn off in use.

3.1.6 Transportability The engine shall be suitable for transportation as follows: (a) . Adequate ground handling pads and other features shall be provided to permit installation on and use of appropriate static and mobile ground equipment.

REQUIREMENT RATIONALE (3.1.6)

The engine must be compatible with the logistic system equipment for ease of transportation.

REQUIREMENT GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph:

(a): The main engine mounts shall be used to support the engine during transportation. The engine shall be restrained by the handling or flight mounts for engine shipment. The engine shall be suitable for transportation from factory to aircraft, delivery to transport aircraft, and delivery to overhaul facilities. The engine shall be capable of withstanding

the externally applied forces and vibrational loads induced during transportation without impact to structural life or engine performance as defined by figure 6. Land shipment of the engine shall be by spring or air ride trucks, or by railroad, as determined by the Using Service. The method of protecting the engine shall be specified.

Background:

Engine transportability should give consideration to total logistic mobility. Turbine engine bearings, seals, mounts, and joint hard coats are subject to damage during transportation and can adversely affect both performance and durability. Damage during engine transportation has been a concern in the propulsion community for many years. The newer turbine engines, primarily dual rotor systems (some have three rotors), have compounded this concern—more bearings, seals, etc. Adding to the problem, the newer engines have tighter clearances designed to lower margins, and they possess more sophisticated coatings (some brittle) and material systems than in the past. Small engines have been shipped in reusable or throwaway foam, metal, or wood containers. A life cycle cost analysis should be conducted to determine the most economical method of shipping engines and whether peculiar shipping stands/isolators should be developed to ship engines between the engine and the air vehicle manufacturer and/or between the field and overhaul facility. For more information see AFGS–87233. The Army standard aircraft maintenance (engine transportation) trailer (NSN 1750–01–086–1653) conforms to data list 4920–EG–081.

The use of air ride semi–trailers has been recommended and used for highway shipment to minimize shipping concerns. However, there are major drawbacks to restricting movement of military engines by air ride transport. Some of these drawbacks are discussed below:

a. Air ride is approximately 30 percent more costly than spring ride truck transport.

b. Air ride is not always available, particularly outside of the CONUS; this results in an adverse effect on readiness when delays are caused by air ride truck availability.

c. Inspection of air ride equipment to assure it is in working order usually is not conducted.

d. Air ride characteristics may be worse than other suspension types depending on the position of the engine on the trailer bed and level of load applied.

The development and use of shock isolating adapters are preferred and recommended to alleviate the above concerns.

REQUIREMENT LESSONS LEARNED

Use of flight mounts during shipment has reduced engine weight since the ground handling mounts are not required to withstand the shipping loads.

4.1.6 Transportability The requirements of 3.1.6 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.1.6)

Transportability needs to be evaluated by analysis and test to ensure the engine will not be damaged during shipment.

EVALUATION GUIDANCE

In addition to the required testing, consideration should be given to:

a. Methods of engine protection

b. A package drop test

c. Air transportation. For example, the European distribution system aircraft (EDSA) C–23A provides no active damping between the engine and the aircraft and therefore, the inherent damping characteristics of the aircraft and the anticipated landing, takeoff, and load spectrums should be evaluated.

EVALUATION LESSONS LEARNED

Evaluation of engine highway shipment capabilities has been attempted in the past by placing an instrumented engine upon a semi–trailer, both air and spring ride, and driving at various speeds over a predetermined course. The results of this type of testing have tended to be inconclusive due to the numerous variables (driver, trail material, bogie spread, etc.) involved. In addition, the impact to engine performance or structural life (bearings, seals, coating, etc.) has not been fully investigated.

3.1.7 Interchangeability All parts and modules having the same manufacturer's part number shall be functionally and dimensionally interchangeable and shall not degrade engine performance, operability, and durability. Matched parts or selective fits shall be permitted upon Using Service approval. Complete engines shall be interchangeable in multi–engine aircraft (i.e., right side to left side).

REQUIREMENT RATIONALE (3.1.7)

It is essential that parts with the same part number be interchangeable to reduce logistic support requirements, minimize maintenance/repair problems, and assure that engine performance and operability are not compromised.

REQUIREMENT GUIDANCE

Background:

All parts having the same manufacturer's part number should be functionally and physically interchangeable and replaceable with each other with no effect on installation and performance, except that matched parts or selective fits may be permitted where necessary to meet other design requirements. The use of matched parts and selective fits should be held to a minimum. The replaceable units should be functionally and physically interchangeable without selective fittings. A list of non–interchangeable, matched, and selected fit parts should be identified and submitted to the Using Service during the engine development program to focus attention on these exceptions and force early, improved design changes. Due to logistical concerns, the Using Service needs to be aware of configurations (i.e., bleed/access port, etc.) peculiar to left/right side engine configurations which would preclude interchangeability of right/left engines/modules.

REQUIREMENT LESSONS LEARNED

Functional and physical interchangeability of components and parts is a desired design goal. There have been many instances where engines designed with matched split cases had high scrapping costs. When unusual damage occurs to one–half of the case and the other half is not damaged, both halves must be condemned. Maintenance of matched sets throughout distribution and overhaul cycles is extremely costly.

4.1.7 Interchangeability The requirements of 3.1.7 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.1.7)

Testing and demonstration is required to evaluate the functional and physical interchangeability of engine parts.

EVALUATION GUIDANCE

Background:

Interchangeability of engine parts can be evaluated during the course of routine assembly, maintenance, and testing. Documentation of this evaluation is recommended to insure complete coverage of all parts. Consideration should be given to parts that have been slightly modified to ensure that parts which are functionally not suitable cannot be installed on the engine.

EVALUATION LESSONS LEARNED

None.

3.1.8 Safety System safety program requirements shall be developed and implemented for the engine design, development, tests, production, and operation in accordance with MIL–STD–882.

REQUIREMENT RATIONALE (3.1.8)

Standard and uniform safety considerations must be used for engine safety.

REQUIREMENT GUIDANCE

Background:

MIL–STD–882 outlines a safety program for the complete life cycle of systems. Using Service specialists involved in safety relative to aircraft equipment and systems generally provide engine safety program requirements in the contract. They would be expected to tailor the requirement in the standard to the particular application. Since the requirements are extensive, they are not usually incorporated into the model specification. Warning notices should be provided, where applicable, for high voltage ignition systems and other high voltage electrical sources, radioactive devices, and explosive devices.

REQUIREMENT LESSONS LEARNED

None.

4.1.8 Safety The requirements of 3.1.8 shall be evaluated by analysis, inspection, demonstration, and test.

EVALUATION RATIONALE (4.1.8)

The safety program should be monitored and evaluated throughout the engine's development.

EVALUATION GUIDANCE

Background:

The Using Service should evaluate the safety program throughout the development program via normal tests, inspection, etc. required by this specification.

EVALUATION LESSONS LEARNED

None.

3.1.8.1 Flammable fluid systems All exterior lines, fittings and components which contain flammable fluids shall be fire resistant (five minutes at 1093°C (2000°F), except that the lubrication oil system and hydraulic system components shall be fire proof (15 minutes at 1093°C (2000°F)). During exposure to the above conditions, the lines and components shall be operating under the worst possible combination of fluid parameters (i.e., flow rate, pressure, and temperature) encountered throughout the complete environmental conditions and operating envelope of the engine.

REQUIREMENT RATIONALE (3.1.8.1)

Exterior lines, fittings, and components carrying flammable fluids must operate within a fire environment for a specified period of time. This requirement is to ensure that these parts will not contribute to the fire until corrective action can be taken or the crew can escape.

REQUIREMENT GUIDANCE

Background:

The Using Service should make sure the aircraft flammable fluid systems have airframe supplied shut–off valves at points prior to delivery to the engine. If the airframe does not, then the engine fluid system should be fireproofed.

REQUIREMENT LESSONS LEARNED

The most adverse conditions have occurred at Idle when flow rates are low and heat transfer capability is minimized.

4.1.8.1 Flammable fluid systems The requirements of 3.1.8.1 shall be evaluated as follows: (a) .

EVALUATION RATIONALE (4.1.8.1)

Evaluation of the fire resistance and fireproof requirements is essential for safety.

EVALUATION GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph:

(a): Testing of the exterior lines, fittings, and components containing flammable fluids shall be accomplished by feeding the fluids at the minimum flow rate, highest fluid temperature, and highest system operating pressure encountered during engine Idle power or windmilling conditions. The unit being tested shall be directly exposed to the flame. Component orientation and direction and location of flame impingement during testing shall be subject to Using Service approval. A visual inspection for leakage shall be made during and after the test. Sufficient instrumentation shall be included to continuously record critical parameters. Remotely controlled color video and stop–action photographic test facility equipment shall be provided to record and monitor all fire tests. The entire test shall be taped for test substantiation. Voice input on video tape shall be provided for clarification of test times and conditions. Thermocouple(s) shall be provided at the interface between flame pattern and surface to monitor and record flame temperature during testing to verify that flame temperature does not fall below pre–test burner calibration limits. To provide for accurate readings, thermocouple(s) shall be mounted directly within the flame pattern. AS 1055 shall be used for conducting the fire test. AS 4273 shall be used for fire testing components.

The test shall be considered successful, if there is no leakage during and five minutes after removal of the flame. Analysis may be conducted via computer modeling to evaluate the requirement with approval from the Using Service.

EVALUATION LESSONS LEARNED

Component seals, covers, and fittings are primary areas of concern for leakage. Component orientation with respect to the flame pattern during test should simulate realistic engine compartment fire conditions.

3.1.8.2 Fire shield A fire shield shall be provided to prevent flammable fluid impingement on the engine. The type of attachment, location, dimensions, and allowable loads at the interface shall be shown on the engine installation and interface drawing.

REQUIREMENT RATIONALE (3.1.8.2)

Standard and uniform safety considerations must be used for engine safety. The purpose of a fire shield is to (1) prevent external leakage of combustion liquids from reaching the parts of the engine hot enough to ignite them and (2) confines exhaust system fires to the exhaust area.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.1.8.2 Fire shield The requirements of 3.1.8.2 shall be evaluated by test.

EVALUATION RATIONALE (4.1.8.2)

This requirement can only be evaluated by test.

EVALUATION GUIDANCE

Background:

This test may be done concurrently with the flammable fluid systems test.

EVALUATION LESSONS LEARNED

None.

3.1.8.3 Explosion–proof All electrical components (except ignitor plug electrodes and devices intended to ignite fuel air mixtures) shall not ignite any explosive mixture surrounding the equipment.

REQUIREMENT RATIONALE (3.1.8.3)

All electrical components must be explosion–proof, since an explosive atmosphere may exist near the component.

REQUIREMENT GUIDANCE

Background:

Electrical system components should be designed to be non–spark producing or should be sealed sufficiently to prevent entrance of combustible materials.

REQUIREMENT LESSONS LEARNED

During the development of the USAF Turbine Engine Monitoring System (TEMS), it was determined some of the connectors would be mated/unmated in a hazardous environment with power applied between the mating halves. Any spark of sufficient energy from this connector could create an explosion. It was determined that electrical connectors should not allow a spark to propagate outside of its shell of enclosure into the hazardous environment.

4.1.8.3 Explosion–proof The requirements of 3.1.8.3 shall be evaluated by component test in accordance with MIL–STD–810.

EVALUATION RATIONALE (4.1.8.3)

The electrical components must be tested to evaluate their explosion–proof properties.

EVALUATION GUIDANCE

Background:

All electrical components, including electrical connectors, should be subjected to explosion–proof testing in accordance with MIL–STD–810. Hermetically sealed components, and passive devices such as thermocouples and chip detectors with no make and break contacts do not need to be tested.

As another test option, the following could be implemented:

During the test, components shall have maximum input voltage applied to them and shall be operated continuously at their maximum loads. During each altitude condition, all make and break contacts shall be operated at least ten times. An overvoltage power supply transient shall be applied to the components at least four times during each altitude condition. At least four of these power supply transients shall be applied during operation of make and break contacts. Power supply transients shall consist of the application of an overvoltage for the times specified in MIL–STD–704. The four points selected for performance of an overvoltage shall include the following 180, 160, 140, and 125 volts ac RMS line to neutral or 50, 43, 37, and 31.5 volts dc. Ignition components or systems shall be operated continuously. Electrodes of spark ignitors shall be mounted in such a manner that the explosive vapor in the test chamber shall not be contacted. Electrically self–sufficient ignition systems shall be exempted from the application of power supply transients. Failure criteria shall be defined in MIL–STD–810, Method 511. Test altitudes shall be every 10,000 above 20,000 ft up to the absolute altitude and every 5000 ft below 20,000 ft. The mixture shall be verified to be an explosive mixture at each altitude during the test procedure.

EVALUATION LESSONS LEARNED

None.

3.1.8.4 Combustible fluid drains Provisions shall be made for automatically clearing the combustion areas of combustible fluids after each false start and for preventing excess combustible fluids from entering the combustion areas after shutdown with the engine in 15 degrees nose up, 20 degrees nose down, and level positions. For engine applications in V/STOL aircraft requiring engine attitude changes, the drain limits of 3.1.1.16 shall not be exceeded for combustible fluids after engine shutdown in a 105 degrees nose up, 20 degrees nose down, and level position. Provisions shall also be made for clearing all vent areas and other pockets or compartments where combustible fluids may collect during or subsequent to operation of the engine. The engine shall be designed so that combustible fluids cannot enter the combustion area when the engine is in, or turned to, a vertical position for maintenance. The maximum allowable quantity of combustible fluid which will discharge from the engine drains after shutdown from normal operation and, if applicable, from augmented operation shall be specified in the model specification.

Overboard fuel leakage, following augmentor operation during flight, shall be controlled at a rate of flow which prevents visual detection of the vapor (see 3.6.1.6). The fluid drain system shall be configured to collect/route the leakage of fuel and oil to a central location, to facilitate detection, and to isolate leakage in excess of the limits of 3.1.1.16.

REQUIREMENT RATIONALE (3.1.8.4)

This requirement assures that combustible fluids do not remain in the combustion area after a false start. This reduces the possibility of a “hot” start on the next attempt. The various engine positions are specified to allow for different engine mounting positions for different engine applications. The no fuel leakage requirements from the engine in the vertical position is to eliminate the possible spillage of fluid on maintenance personnel or in the working area.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

Engine contractors generally have had no problem complying with these requirements.

4.1.8.4 Combustible fluid drains The requirements of 3.1.8.4 shall be evaluated by inspection and tests.

EVALUATION RATIONALE (4.1.8.4)

Combustible fluid drains can only be evaluated by inspection and engine tests.

EVALUATION GUIDANCE

Background:

The requirement should be evaluated by tests throughout the engine development program.

EVALUATION LESSONS LEARNED

Engine testing has identified leakage rates, from drains, in the test cell.

3.1.8.5 Air and gas leakage The location, amount, temperature, and pressure of engine air and gas leakages shall be specified in the model specification. There shall be no leakage of sufficient temperature and concentrated impingement to present a safety hazard or affect installation requirements.

REQUIREMENT RATIONALE (3.1.8.5)

The leakage requirement provides interface information and limits leakage to a tolerable level.

REQUIREMENT GUIDANCE

Leakage of air and gas can interfere with engine bay cooling and cause safety problems.

REQUIREMENT LESSONS LEARNED

Hot gas leakage has caused engine control failures in the engine tests. Reverse flow of hot gases into helicopter engine bays/nacelles has caused failure of electrical and fluid lines and components, precipitating fires.

4.1.8.5 Air and gas leakage The air and gas leakage requirements of 3.1.8.5 shall be evaluated by test.

EVALUATION RATIONALE (4.1.8.5)

Air and gas leakage can only be evaluated by test.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.8.6 Ground safety Warning notices shall be provided, where applicable, for high voltage ignition systems and other high voltage electrical sources, radioactive devices, explosive devices, and high powered optical sources.

REQUIREMENT RATIONALE (3.1.8.6)

Standard and uniform safety considerations must be used for engine safety.

REQUIREMENT GUIDANCE

Ground maintenance personnel use warning notices to avoid safety problems during engine bay area maintenance procedures.

REQUIREMENT LESSONS LEARNED

Warning notices have given maintenance personnel needed warnings that they need to increase safety precautions.

4.1.8.6 Ground safety The requirements 3.1.8.6 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.8.6)

Ground safety warning notices can only be evaluated by inspection.

EVALUATION GUIDANCE

Background:

For engines that may require service while under red–light conditions, the Using Service should ensure that all warning notices can be visibly read under red–light conditions.

EVALUATION LESSONS LEARNED

None.

3.1.8.7 Jet wake Jet wake temperature and velocity profiles at sea level static standard day conditions when operating at Idle, Intermediate, and Maximum shall be as shown on figure 7. Areas hazardous to personnel and equipment shall be appropriately defined and marked on figure 7.

REQUIREMENT RATIONALE (3.1.8.7)

It is important to ensure that the engine/airframe is compatible with ground or shipboard equipment such as the Jet Blast Deflector (JBD).

Jet wake diagrams aid the aircraft manufacturer in initial aircraft design and in development of safe ground and shipboard operating procedures.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

The contractor should present the requirement in accordance with the format of figure 7.

Background:

The jet wake should not become a hazard to equipment and personnel. On shipboard operations where space is at a premium, the hazardous zones must be minimized, especially during Idle and Taxi. The heat load on equipment and areas where the jet wake will impinge should be within the limits of the impinged equipment whenever practical.

REQUIREMENT LESSONS LEARNED

One recent aircraft/engine combination had a compatibility problem with the shipboard Jet Blast Deflectors. The JBDs were barely adequate in dissipating the heat imposed on it from the wake, and the deflected gas was hazardously hot. It is important that the engine/airframe combination is compatible with existing equipment, and that the safety of ground personnel is not compromised. Some aircraft systems experienced jet wake impingement on portions of the airframe at certain flight conditions which affected airframe structural integrity. Jet wakes have caused personnel injury and weapons/ordnance and equipment damage.

4.1.8.7 Jet wake The requirements of 3.1.8.7 shall be evaluated by analysis and test. For vectoring nozzles, the full range of thrust vectoring angles in 5 degree increments shall be tested.

EVALUATION RATIONALE (4.1.8.7)

Temperature and velocity characteristics must be evaluated for aircraft, ground equipment designers, and for development of ground operating procedures.

EVALUATION GUIDANCE

Background:

A sea level static test is satisfactory for ground operating data. Engineering judgments can be made about flight conditions from computer models.

EVALUATION LESSONS LEARNED

None.

3.1.9 Design control

4.1.9 Design control

3.1.9.1 Standardization Standardization principles, standard parts, materials, processes, tools, subsystems, and components shall be used to the maximum extent possible without compromise in design, performance, operability, or economic life of the engine. All parts, materials, and processes, whether or not identified as a Government, industry, or contractor standard shall be qualified for the intended use as a part of the qualification specified herein. Items already in the Government inventory shall be used to the maximum extent possible where suitable for the intended purpose. Variation in similar components or parts shall be held to the absolute minimum. Proprietary designs shall be kept to a minimum. Under conditions wherein economics of production conflict with standardization objectives, the Using Service shall be requested to select the component desired for use.

REQUIREMENT RATIONALE (3.1.9.1)

Standardization is needed to reduce the number of parts in the logistic system, the number of manhours required to maintain the system, and avoid significant increase in manhours required to track assets in the system.

REQUIREMENT GUIDANCE

Background:

This requirement helps to assure that, to the maximum extent possible, standardization will be incorporated in the design of gas turbine engines even when economics of production conflict with standardization objectives but not to compromise engine design or performance.

REQUIREMENT LESSONS LEARNED

In cast parts, contractors have requested changes in design to lower rejection rates. This has caused reduction in standardization and an increase in ILS tracking efforts but has decreased the cost of parts.

4.1.9.1 Standardization The requirements of 3.1.9.1 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.1.9.1)

Standardization principles, standard parts, materials, processes, tools, subsystem, and components can only be evaluated by inspection and demonstration.

EVALUATION GUIDANCE

Background:

Evaluation of standardization is an ongoing process during development and into production.

EVALUATION LESSONS LEARNED

None.

3.1.9.2 Design standards Military and industry standards developed specifically for aircraft engines, such as AND, AS, MS, NAS, etc., shall be used unless they are determined by the engine manufacturer to be unsuitable for the intended purpose, subject to Using Service approval.

REQUIREMENT RATIONALE (3.1.9.2)

MS and AND design standards should be incorporated in engine designs to assure the use of accepted and proven methods and procedures.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.1.9.2 Design standards The requirements of 3.1.9.2 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.9.2)

Inspection of the manufacturer's documents and parts is the best manner of determining the acceptability of the design standards.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.9.3 Standard parts The military and industry standard parts developed specifically for use in aircraft engines shall take precedence over any other military standard parts. Where general purpose standards, as defined by envelope dimensions or Qualified Products List (QPL), are used in critical or high strength applications, they shall be identified by the vendor or engine manufacturer's part number. Parts derived from general purpose standards solely on an inspection or selection basis shall be identified by manufacturer parts numbers and all previous identification marks shall be removed.

REQUIREMENT RATIONALE (3.1.9.3)

Parts which have already proven successful in previous aircraft engine operating environments stand a greater chance of success than parts of a general nature, due to the harsh operating environment of aircraft engines.

REQUIREMENT GUIDANCE

Background:

Military standard parts help to assure that the best available parts are used in engines. This is accomplished by giving precedence to those military standard parts developed specifically for use in aircraft engines and by requiring that parts selected from general purpose standards solely on an inspection or selection basis be identified only by manufacturer parts numbers (no other markings permissible) to avoid confusing these parts with the unselected standard parts.

REQUIREMENT LESSONS LEARNED

None.

4.1.9.3 Standard parts The requirement of 3.1.9.3 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.9.3)

The use of various standard parts can only be evaluated by inspection.

EVALUATION GUIDANCE

Background:

The Using Service inspectors at the contractor plant must check parts and the parts list.

EVALUATION LESSONS LEARNED

None.

3.1.9.4 Parts list The parts list for the engine which successfully completes the designated engine milestone evaluations shall constitute the approved parts list for any subsequent engines used for evaluation in that milestone. Changes to the approved parts list shall require approval by the Using Service prior to parts incorporation.

REQUIREMENT RATIONALE (3.1.9.4)

A parts list from engines that successfully completed qualification tests assures the best possible parts for the production engines.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.1.9.4 Parts list The requirements of 3.1.9.4 shall be evaluated by inspection prior to the initiation of each milestone.

EVALUATION RATIONALE (4.1.9.4)

The parts list must be evaluated to ensure an accurate description of the engine.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.1.9.5 Assembly of components and parts Equipment, parts, and components which are not structurally or functionally interchangeable, shall not be physically interchangeable. Parts and components shall be designed such that it is impossible to install them incorrectly, e.g., backwards, upside down, reversed in an assembly, or installed in the wrong location in an assembly. Connections located in close proximity to each other shall be made physically non–interchangeable.

REQUIREMENT RATIONALE (3.1.9.5)

The frequency of equipment, parts, and component assembly removal is such that “Murphy proofing” is required to ensure that system/flight safety of the engine is not compromised.

REQUIREMENT GUIDANCE

Background:

Non–metallic “O” ring seals and packings should remain on the part onto which they are installed when the associated mating part is not installed, and the associated mating part should be designed to prevent cutting or other damage to the “O” ring seals and packings.

REQUIREMENT LESSONS LEARNED

Past experience has shown that catastrophic damage could result when the design of a hydraulic filter allows easy installation onto a connection intended for a lube filter. Lube or hydraulic system failure may result which in turn could cause engine failure. Another typical situation would be if a plug were installed in reverse causing reversed diode polarity, resulting in failed diodes or other polarity sensitive devices.

4.1.9.5 Assembly of components and parts The requirement of 3.1.9.5 shall be evaluated by demonstration.

EVALUATION RATIONALE (4.1.9.5)

Evaluation of the non–interchangeability (“Murphy proofing”) of parts must be by demonstration.

EVALUATION GUIDANCE

Background:

The non–interchangeability demonstrations should be performed after the design is baselined. MIL–I–8500 should be used as a guide.

EVALUATION LESSONS LEARNED

None.

3.1.9.6 Changes in vendors or fabrication process Changes in any vendor, fabrication process, or fabrication source for any component or part shall be in accordance with MIL–STD–1529.

REQUIREMENT RATIONALE (3.1.9.6)

The Using Service must have control of vendors or fabrication processes to prevent the use of inferior parts.

REQUIREMENT GUIDANCE

Background:

The contractor should prepare and submit a list of those parts, components, and assemblies which require substantiation tests to qualify an alternative vendor source or process. The specific test(s) required to qualify parts as engine parts should be defined and submitted with the list. The fabrication source and process of selected vendor components will be included in this list. The contractor should be responsible for insuring that all parts, components, and assemblies on the substantiation list comply with the qualified fabrication source and process, and that any changes to those sources or processes are effectively controlled.

The contractor should be responsible for performance of the substantiation test to establish satisfactory alternate vendors or fabrication sources or processes. A fabrication source is defined as the prime physical source producing the part, component, or assembly. Changes of fabrication location, such as to another plant of an individual vendor, should be construed as a change of fabrication source.

REQUIREMENT LESSONS LEARNED

Change in vendors or processes was a source of inferior parts being introduced into fleet equipment in the past.

4.1.9.6 Changes in vendors or fabrication process The requirements of 3.1.9.6 shall be evaluated by inspection.

EVALUATION RATIONALE (4.1.9.6)

The Using Service must monitor all changes in vendors or fabrication processes to ensure acceptable parts for the engine.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.2 Performance and operability

4.2 Performance and operability

3.2.1 Performance characteristics The engine performance characteristics defined in the specification shall be for a minimum engine. Minimum engine performance as defined in 6.4, shall be specified for both new engines and engines deteriorated for the time period specified in 3.2.1.4. Unless otherwise specified, the engine performance characteristics shall be based on:

a. A fuel having a lower heating value of (a) BTU/lb, and otherwise conforming to the fuel specified in 3.7.3.1.1 and oil specified in 3.7.8.1.

b. US Standard Atmosphere (geopotential altitude) (1976).

c. No inlet air distortion.

d. An inlet pressure recovery as defined by the following equations and conditions:

Pt2(ref)/PtO = 1.0, from 0.0 to 1.0 Mach Number.

$Pt2(ref)/PtO = 1.0 – 0.075(M – 1)^{1.35}$ from 1.0 to 5.0 Mach Number.

$Pt2(ref)/PtO = 800/(M^4 + 935)$, above 5.0 Mach Number.

Pt2(ref) = referenced total pressure at the compressor inlet.

PtO = free stream total pressure.

M = flight Mach Number.

For turboshaft applications that incorporate an Inlet Particle Separator (IPS) as part of the engine, all performance shall include the effects of the IPS.

e. The designated exhaust pipe and jet nozzle, fan exit duct and fan nozzle, engine front frame and rear frame features.

f. No customer bleed air extraction.

g. No customer power extraction.

h. Using the engine control system specified in the specification. The specified performance shall be predicated on the tolerance of control system variation which produces poorest performance.

i. Non–standard atmospheres as shown on figure 4.

j. Zero humidity.

k. For turboshaft/turboprop engines, the shaft torque absorber characteristics specified in 3.1.1.17 and 3.7.18.

l. Stall margin/operating line consistent with the requirements of 3.2.2.10 when matched with vehicle flight exhaust. No distortion indicator signal input.

REQUIREMENT RATIONALE (3.2.1)

Engine performance characteristics must be standardized to allow easy comparison of engine models and for assessing an engine's capability. Without a common base line it is difficult to conduct a one–to–one comparison of proposals from different engine manufacturers. Performance must be stated as the minimum engine in order to assure that the resultant aircraft performance estimates will be achievable.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 18,300 (if JP–5 is the only primary fuel); 18,400 (if JP–4 and/or JP–8 is a primary fuel).

Background:

The following is background for corresponding subparagraphs in the specification paragraph:

a. The Lower Heating Value (LHV) of primary fuels is 18,300 and 18,400 BTU for JP–5 and JP–4/8, respectively. These LHVs should be used to ensure that accepted engines will produce better than specification SFCs, since these represent the worst LHVs for the particular fuels. For alternate fuels, the SFC should be corrected for the worst LHV specified by the Using Service.

b. U.S. Standard Atmosphere 1976 is a universally agreed upon standard of temperatures and pressures for a range of altitudes sufficient for aircraft operations. Geopotential altitude is designated since it produces a more conservative definition of engine altitude capability. At a given atmospheric static pressure geopotential altitude is lower than geometric altitude. Current recommended text reflects the 1976 Standard Atmosphere edition which is the same as the 1962 edition for altitudes below 50km (164,050 ft), but has new values for higher altitudes.

c. Zero inlet air distortion is chosen as the standard, since air distortion has an effect on performance. Zero distortion is a baseline from which performance effects can be specified with various degrees of inlet air distortion. The inlet venturi or bellmouth should be defined by the engine contractor at the start of the development program. Performance credit or debit should not be allowed for FOD screens and bellmouths. The Air Force ATF program based the engine performance on a specific level of inlet distortion and stall margin.

d. Inlet pressure recovery is dependent on the inlet design and methods of shock recovery. Every inlet design has a different characteristic pressure recovery relationship. Some relationship must be selected as a standard baseline.

The relationship selected in the current recommended text has been used in turbojet/fan specifications as far back as historical records exist. For turboshaft/prop engines the pressure recovery is assumed to be 1.0 (since they do not fly supersonic). An integrated Inlet Particle Separator (IPS) will have a detrimental impact on engine performance that must be reflected in the performance presentation. Neglecting the IPS effects would result in incorrect performance estimates and airframe performance shortfalls.

e. An exhaust system should conform aerodynamically with the exhaust nozzle shown on the engine installation and interface drawing and have the thermal and pressure growth properties of the material(s) as specified thereon. The exhaust system characteristics should be based on steady–state operating conditions of the primary nozzle, primary plug, fan exit duct, and fan nozzle (including the pylon shape and other surface areas affecting performance), and should include all other details affecting performance. These exhaust system performance characteristics should be presented in the steady–state and transient computer programs of 3.2.1.1 and 3.2.1.2. If referee nozzles are used, then the referee nozzle performance should be substantiated by test and included in the steady–state and transient performance computer programs when generating performance. The preferred referee nozzle for turboshaft engines is a straight circular duct for consistency of performance quotes. Substantiating performance of referee nozzles is necessary for engine performance bookkeeping when changing from the referee to flight nozzle. Thrust (and SFC) with vectoring nozzles should be based on axial thrust (i.e., zero vector angle). Thrust and (SFC) for fixed geometry non–axially aligned nozzles should be based on resultant gross thrust minus ram drag. Thrust (and SFC) of multiple nozzle (e.g., V/STOL) arrangements should specify the thrust or force components of each nozzle.

Engine cycle performance should include the effects of specified front and rear face devices, frames, coatings, and features (if applicable).

f. Zero customer bleed air extraction should be specified since it is a baseline from which performance effects can be obtained. Customer bleed air other than that required for continuous engine operation should be as cited in table II.

g. Zero customer power extraction, other than that required for continuous operation of the engine should be cited in table II.

h. Control system tolerance on such items as variable geometry schedules, stability bleed schedules, etc. can affect performance. In order to assure that no engine will perform worse than specified, it is desirable to fix the control system characteristics at those tolerances which will produce the poorest engine performance.

i. Figure 4 is based on MIL–STD–210B except for the tropical atmosphere which is based on MIL–STD–210A. Non–standard atmospheres should reference either MIL–STD–210 or a special table/figure.

j. A dry atmosphere should be assumed unless otherwise specified elsewhere in the specification.

k. For turboprop/turboshaft engines, it is necessary to know the mechanical and performance characteristics and limitations of the output shaft torque absorber. This is particularly needed for matching rotational speeds, observing torque/power/speed limitations, and designing the engine control for transient operation.

l. Engine performance should be determined using the operating line/stall margin needed for the operational engine. This includes matching with the vehicle inlet and exhaust systems and engine surge control bleed ducting when present.

Minimum performance should be obtained without exceeding specific values for certain parameters (i.e., N1, N2, EGT, T4.1, TSFC, etc). These parameters are dependent on the control system.

Minimum performance should be defined as the lowest thrust and highest fuel flow or SFC which may occur at the specified measured turbine temperatures or commanded power setting (as limited by fan speed, core speed, turbine temperature, etc.). SFC (fuel flow) ratings should be at the specified measured thrust. The minimum performance

quotations of thrust/power, specific fuel consumption, and limiting temperature should be a reasonable representation of a $\pm$2 or $\pm$3 sigma (based on Using Service approval) production engine based upon the component performance levels and control tolerances known or assumed at the time the specification is developed. Stall margin should be consistent with the stability requirements of 3.2.2.5.

Installed performance is required by the airframe contractor to predict or determine aircraft performance. As a result, the performance computer program of 3.2.1.3 should be capable of including engine installation effects (e.g., inlet distortion, humidity, inlet pressure recovery, customer bleed and power extraction, alternate exhaust characteristics, etc.) and be capable of integration with an airframe installation computer code.

The usual definition of minimum engine (for temperature limited or scheduled engines) is –3 sigma. For engines with fan speed control, minimum thrust occurs on either + or – 3 sigma depending upon flight point, operating time, and control logic. While some minimum engines use characteristics for $\pm$2 sigma, $\pm$3 sigma is recommended for additional margin for the Using Service and higher acceptance rate for the engine manufacturer. The "+/–" is used because –3 sigma is not always the poorest engine for both thrust and SFC. The engine with lowest thrust is either –3 sigma or +3 sigma depending upon engine control scheme and the flight condition. The engine with lowest component efficiencies and highest SFC is –3 sigma. A minimum engine, having both low thrust and high SFC at the same time, only exists at flight points where the engine is on a temperature limit, not fan speed. Aircraft flight performance, based on factor definitions of minimum engine, are not as accurate as cycle simulations using modified component performance.

Aircraft installed performance normally includes more items and losses than found in typical engine company customer programs. In other words, aircraft performance cannot be determined by engine company programs except for relatively simple installations. Recent CFE aircraft/engine programs have required the airframe contractor to deliver an integrated airframe/engine installed performance code as part of the proposal

REQUIREMENT LESSONS LEARNED

None.

4.2.1 Performance characteristics The requirements of 3.2.1 shall be evaluated by inspection, analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.1)

The engine's performance must be evaluated by inspection, analysis, demonstration, and test. The engine submitted for testing would be assumed to have average engine performance characteristics, and analysis would be accomplished to calculate minimum performance.

EVALUATION GUIDANCE

Background:

For turbofan/jet engines, some parameters used to evaluate performance include net thrust, fuel flow, rotor speeds, inlet airflow, turbine gas path, or blade metal temperature. For turboshaft/prop engines the parameters should include shaft horsepower, torque, airflow, fuel flow, and measured turbine gas path or blade metal temperature. This evaluation should be satisfied by the utilization of a combination of tests listed in the engine milestones (i.e., altitude, sea level, steady–state, transient, AMT, corrosion, etc.). The installation bleed air and horsepower extraction values should be provided by the airframe contractor.

EVALUATION LESSONS LEARNED

None.

3.2.1.1 Performance and presentation – steady–state The minimum engine steady–state performance requirements shall be defined by the steady–state performance computer program (a) dated (a) , for the engine operating envelope of 3.2.2.1, in accordance with the conditions of 3.2.1. Cycle calculations for the minimum engine computer program shall be based on an average engine with thrust/power and fuel flow related output adjusted to minimum engine performance corresponding to + (b) . The relationship between minimum and average performance shall be specified throughout the flight envelope. The specification computer program(s) shall calculate minimum performance in accordance with the requirements of 3.2.1. Tables Ia, Ib, Ic and table II shall provide performance points from the performance computer program for new and deteriorated engines. Tables Ia, Ib, and Ic shall specify performance points at sea level static for the following atmospheric temperatures: (c) . Table II shall specify performance at selected altitude, Mach number and inlet temperature conditions. In the event of a conflict between tables Ia, Ib, Ic, and table II and the performance computer program, the computer program shall take precedence. Tables Ia, Ib, Ic and table II shall be presented in both SI (metric) and US Customary (English) units.

Performance shall be specified in the program for the following conditions, singly and any combination: (d) .

Steady–state air flow values for any specific flight condition and principal station properties shall be defined in the computer program. The average steady–state performance shall be specified in the average/status steady–state performance computer program, which shall be updated periodically throughout the development program, as determined by the Using Service.

REQUIREMENT RATIONALE (3.2.1.1)

All engines must meet or exceed minimum performance to ensure the capability of the weapon system to accomplish established missions. Performance computer programs provide a means of assessing performance without actual testing of engine hardware. Engine performance information is of critical importance in the development of the aircraft mission capability, and influences airframe inlet design, takeoff, landing, and various other operational characteristics to be generated and assessed by the airframe contractor. Engine performance under various conditions/loads (i.e., altitude, sea level, varying temperatures, bleed air extraction, power extraction, attitudes, power settings, etc.) also needs to be specified to assess aircraft performance under these conditions.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph.

(a): Insert the computer program number and date of its release.

(b): The engine contractor should provide the average to minimum calculations which may be a + sigma variation, factors applied to the performance parameters or cycle calculations based on modified component performance.

(c): Insert either a reference to a MIL–STD–210 percent risk day (i.e., 5% risk cold day) or a specific temperature (for example, –40 degree F cold day).

(d): Inlet air distortion, relative humidity, ram recovery, customer air bleed, customer power extraction, nozzle effects, parasitic flows, engine anti–icing, and variable geometry, etc., as applicable, for non–standard days throughout the complete environmental conditions and operating envelope.

The following is recommended to be transferred verbatim into the specification paragraph: The steady–state computer program shall present pressure, temperature, and mass flow rates of bleed air from Idle to the highest power setting. These presentations shall include the following conditions: operating altitudes, air inlet temperatures, flight speeds, and the effects of bleed air extraction on engine performance. Compressor bleed air required for compressor surge protection, which operates continuously during steady–state engine operation in a surge–sensitive regime, shall be specified as to the operating envelope involved.

The Navy recommends the following to be transferred verbatim into the specification paragraph: The margin or relationship between average and minimum performance levels for thrust and fuel flow (specific fuel consumption) throughout the flight envelope shall be specified.

Background:

Minimum steady–state engine performance should be established early in the development program. Minimum steady–state performance should be based on average engine components and average control system schedules. The overall engine performance parameters (e.g., airflow, fuel flow, thrust/power, shaft power) should be adjusted for engine–to–engine variations to produce minimum engine performance. Alternately, minimum performance can be based on cycle calculations using modified component performance. The latter approach with modified component performance is preferred for accuracy of cycle parameters and performance, for engines with moderate to high sigma and deterioration effects, and for aircraft needing more accurate definition of exhaust stream properties. The performance requirements should be integrated with other engine design requirements such as operability, durability, and reliability to produce a satisfactory flight system.

The atmospheric conditions for tables Ia, Ib, Ic, and II have historically required cold, standard, and hot and/or tropical days. MIL–STD–210 should be consulted to choose an appropriate percent risk temperature day, or for special requirements, the Using Service can choose a specific temperature. The USN traditionally uses cold day –51°C (–60°F), standard day 15°C (59°F), tropical day 32.2C (89.9°F), and hot day 49°C (120°F) for tables Ia, Ib, and Ic temperatures, respectively. The USAF uses MIL–STD–210 percent risk days temperatures. The Army in tables Ia, Ib, and Ic uses Sea Level static standard day and Sea Level static hot day 50°C (122°F). An important Army table II point for helicopters is 1.2 km (3937 ft) 35°C (95°F). In addition, to define sea level and altitude temperatures for various type days, the USN uses figure 4, which is based on MIL–STD–210 one percent risk day temperatures, except for SLS cold day (20 percent risk day). Figure 4 also includes tropical day temperatures which are not included in the latest version of MIL–STD–210.

Performance presentation via computer programs, plots, data and reports should be consistent. The computer program performance should take precedence over tables Ia, Ib, Ic, and II and curves for establishing minimum steady–state engine performance (including thrust/power, fuel flow, SFC). Tables Ia, Ib, Ic, and II reflect the performance requirements that the engine contractor must meet. Seemingly small performance table changes can greatly impact the engine or control design. Selection of performance points which cover the operating extremes and critical mission points should be done with care because they are very powerful drivers on the engine design. Minimum engine performance should be defined in terms of minimum thrust/power and maximum specific fuel consumption and should represent the poorest performing engine $\pm$ 2 to 3 sigma that will pass the engine acceptance criteria. Thrust/power and SFC characteristics due to altitude pressure lapse and ambient temperature lapse rates should agree with actual engine performance data. When defining tables Ia, Ib, Ic and table II it may be desirable to have part power points above Idle of tables Ia, Ib, and Ic, and include the lowest corrected airflow cruise and loiter points in table II. This would provide a sea level performance point that is equivalent in corrected airflow (engine cycle match point) to an altitude cruise point. That is, the lowest corrected airflow in table II should also be included in the power hook in tables Ia, Ib, and Ic. The performance tables specified should define the steady–state values for thrust/power, airflow, fuel flow, rotor speeds, and temperatures critical to overall weapon system performance. One performance table should present sea level static uninstalled performance, which should form the basis for the sea level acceptance tests for production engines. These tables should also be used in calculations for performance retention evaluation. The performance parameters should be presented in tables Ia, Ib, Ic, and II format for sea level and altitude conditions and should be chosen to:

a. Assure sufficient engine thrust/power at crucial mission points such as takeoff, climb, or transonic acceleration.

b. Assure that fuel usage during climb, loiter and cruise is consistent with the overall weapon system design and mission.

There should be no derates applied between these tables, i.e., a minimum engine at SLS condition should also represent a minimum engine at the altitude performance points. Thrust droop should be defined for a cold engine (at Idle power) which makes a burst to takeoff power settings and included in aircraft performance. Consideration of minimum thrust during thermal stabilization should be included in the assessment of minimum takeoff thrust requirements.

The aircraft performance specification requirements are reflected in the engine specification via the table II characteristics. Table II engine performance is required to allow the aircraft to meet its system performance levels. Table II should include combinations of flight points, installation factors, alternate front frames, and exhausts, if applicable.

Engines which require airflow and/or exhaust flow for supplemental lift systems will need special consideration in the performance program and tables. If the ducting and nozzles are to be engine contractor furnished equipment, thrust, duct pressure losses and nozzle characteristics should be defined in the specification and computer program. If the supplemental lift system is furnished by the aircraft contractor, the mass flow, pressure and temperature characteristics should be specified for the engine/airframe interface and in the engine performance computer program. When the performance is obtained by means of augmentation, it should be so designated. If liquid injection is provided, the liquid composition, flows, and limitations on use should be specified.

Any detrimental performance effects caused by an integral IPS (i.e.., the IPS is provided by the engine manufacturer) should be modeled in the performance computer program and should be included in tables Ia, Ib, Ic, and II. If the IPS is not part of the engine, the computer model should be able to accept inlet losses associated with an airframe IPS to predict installed performance.

Performance curves may be required by the Using Service and should be delivered via CDRL. The performance curves are used to obtain quick overview information. The curves should present engine performance under standard conditions, including altitude effects (pressure level and Reynolds number), at sea level, and at each 10,000 foot increment up to, and including the absolute altitude of the engine and also at altitudes of 15,000 and 36,089 feet. The method for interpolation should be specified, should include any discontinuities in the curves, and be consistent with the computer program. Each performance curve should show the variation of net thrust with Mach numbers for the following power settings: maximum, minimum augmented, two other augmented conditions, Intermediate, Maximum Continuous, 90, 75, and 60 percent of IRP/Maximum Continuous, and Idle. Curves should show the airflow rate (in lbs/sec) (for turbofan engines, show fan, core, and total airflow), specific fuel consumption in (lbs/hr/lb), engine RPM, and other engine thrust variables (such as engine pressure ratio). These curves should be drawn in accordance with the format of figure 8a. Curves should be provided to indicate maximum reverse thrust obtainable versus ambient temperature at sea level, 2,500 and 5,000 feet altitudes for Mach numbers of 0, 0.1, and 0.2. Additional curves may be required by the Using Service for clarity or to describe the special characteristics of an engine. The family of curves should be presented to ease visual interpolation of engine data; i.e., the lines could extend past operating boundaries or limits. For engines incorporating ejector nozzles, the ejector total pressure, and associated secondary airflow characteristics curves should be presented as follows:

a. Secondary airflow variation with Mach number (format of figure 8b for the altitudes and thrust conditions specified above.

b. Secondary air total pressure recovery variation with Mach number (format of figure 8c for the altitudes and thrust conditions specified above.

c. The minimum secondary airflow required for nozzle cooling or to make performance for the altitude and thrust condition specified above.

Some aircraft have very involved engine and airframe ejector nozzle configurations which cannot be described as above or independent of the aircraft flow system

The steady–state status/average performance computer program should be defined in the statement of work and required for submittal in the CDRL. The status performance computer program should be used to define the margin between the status model and the minimum engine performance requirements. The margin or relationship between average/status and minimum performance levels for thrust/power and fuel flow (specific fuel consumption) throughout the flight envelope should be approximately 2–3% for turbojet/turbofan engines and 5–6% for smaller turboshaft/turboprop engines. The margins used should consider engine class, maturity, production history, and data base. For identified deficiencies in status–to–minimum margin, a plan should be presented that defines the engine design/configuration changes necessary to achieve adequate production margin. The average engine should reflect the predicted nominal engine characteristics, taken from contractor data base and rig tests, until actual engine performance data can be used from the first engine to test (FETT). After the FETT the average/status performance computer programs should reflect actual engine test data. Status "average engine" should reflect a true arithmetic average of sufficient accepted production engine samples. The specification average/minimum engine program remains the same until re–negotiated with the Using Service.

The capability for determining installation effects should be included in the computer program. Installed engine performance should include effects of distortion, relative humidity, test specific instrumentation, ram pressure recovery, customer air bleed, customer power extraction, nozzle effects, parasitic flows, engine anti–icing, windmilling, and variable geometry, etc. as applicable.

For more information concerning methodology for generation of performance margin consult ASME Paper 84–GT–262 by H. Brown and J.A. Elgin.

REQUIREMENT LESSONS LEARNED

Performance curves were used in the past to clarify performance requirements to Generals and Admirals (G and A curves), and for operators in the test cells before the advent of computer programs. Older computer programs produced varied output depending on the sequence of the input parameters. Some of these programs had to be reprogrammed to adapt to Using Service computers. Secondary and tertiary airflow was only estimated in the past, since airflow instability had a pronounced affect.

4.2.1.1 Performance and presentation – steady–state The requirements of 3.2.1.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.1.1)

Evaluation of engine performance capability must be accomplished by analysis, demonstration, and test. The only reliable, yet safe, way of determining an engine's ability to meet specified requirements is to subject the engine to simulated inlet and exhaust conditions under which it will be used in flight.

EVALUATION GUIDANCE

Background:

The steady–state performance and computer program should be evaluated initially by analysis and component rig test, and subsequently by full–up engine testing. Testing should be conducted at extreme conditions/loads (i.e., altitude, sea level, varying temperatures and pressures, bleed air extraction, power extraction, attitudes, power settings, etc.).

A sea level static acceptance test of the altitude performance qualification engine should be performed in the contractor's production type test cell and repeated in a government test cell. Altitude testing is usually performed in a government test facility. The engine should be the production configuration with nominal production control schedules. The USN prefers conducting performance tests with a minimum performance engine to ensure that all production engines will exceed minimum performance. The test condition should result in the measurement of

steady–state performance at stabilized engine conditions. Engine bleed air and power extraction should be taken from the engine if called for by tables Ia, Ib, Ic, and II. Instrumentation should be capable of measuring thrust/power, fuel flow, exhaust gas temperature, and airflow, as well as internal pressure and temperatures. The specific instrumentation requirements and accuracies should be defined in the test plans. Control parameters such as rotor speeds and variable geometry position should also be measured.

Customer bleed air evaluations should be accomplished at sea level and at altitude cruise condition, from Idle to Maximum power, for inlet air temperatures over the range of sea level hot and cold atmospheric temperatures of figure 4 (a minimum of five test temperatures). The test should be conducted in such a manner as to demonstrate the amount of customer bleed air available over and above that required for engine systems, such as for engine anti–icing systems, and to establish the relationship between bleed air extraction and engine performance loss.

Measurement of scale force should be the primary method of determining thrust for tables Ia, Ib, Ic, and II evaluation. Other methods of determining thrust, such as the use of nozzle coefficients, should be used to confirm the thrust as measured by the scale force method. The data reduction program should be approved by the Using Service and should be fully approved prior to the start of testing. Test cell performance correction factors should be evaluated by back–to–back indoor/outdoor testing or a calculation procedure approved by the Using Service. If installation of test instrumentation results in significant performance losses, the engine performance results may be adjusted with the concurrence of the Using Service. Once the official tables Ia, Ib, Ic, and II evaluation begins, no further adjustments of the engine, its components, or fuel control should be allowed unless specifically authorized by the Using Service. Throughout tables Ia, Ib, Ic, and II evaluation testing, the measured performance should not be adjusted for engine degradation.

When conducting altitude performance tests, consideration should be given to collection of data for status deck development and evaluation, including temperature and pressure lapse rates and installation effects (e.g.. air bleed and horsepower extraction, inlet distortion). The objective is to characterize the engine components across the full range of their maps, including Reynolds number, clearance, Mach number, and other effects. This may require steady–state power hooks at several flight conditions other than those listed in tables Ia, Ib, Ic and table II. Additional guidance may be found in AEDC–TR–83–42.

The points in tables Ia, Ib, Ic, and II are performance points of the steady–state performance computer program and may not represent all the points required for demonstration or testing.

EVALUATION LESSONS LEARNED

Experience has shown that in an attempt to achieve a high level of performance, engines have been built up so tight that heavy rubs and excessive deterioration resulted. In the past, some contractors delivered high performing engines for performance tests and low performing (cold running) engines for durability tests. It has proven difficult to measure secondary and tertiary air for exhaust nozzles.

3.2.1.2 Performance and presentation – transient The minimum engine transient performance requirements shall be defined by the transient performance computer program (a) dated (a), for the engine operating envelope of 3.2.2.1, in accordance with the conditions of 3.2.1. Engine thrust transient times from the transient computer program shall be specified in 3.2.2.6, as required therein. Performance and thrust transient times shall be specified in the program for the following conditions, singly and in any combination: (b). The specification computer program(s) shall calculate minimum performance in accordance with the requirements of 3.2.1.

Transient air flow values for any specific flight condition and bleed/power extraction levels shall be defined in the computer program. The average engine transient performance shall be specified in the average/status transient performance computer program which shall be updated periodically throughout the development program, as determined by the Using Service. The minimum and average/status transient performance shall be specified in (c) cycle deck(s). Engine performance with installed effects shall be included in all transient performance computer programs.

REQUIREMENT RATIONALE (3.2.1.2)

All engines must meet or exceed minimum performance to ensure that the weapon system can accomplish established missions. Performance computer programs provide a means of assessing that performance without actual testing of engine hardware. Engine performance information is of critical importance in the development of the aircraft mission capability, and influences airframe inlet design, takeoff, landing, and various other operational characteristics to be generated and assessed by the airframe contractor. Engine performance under various conditions/loads (i.e., altitude, sea level, varying temperatures, bleed air extraction, power extraction, attitudes, power settings etc.) also needs to be specified to assess aircraft performance under these conditions.

The transient computer program provides a means to evaluate engine or fuel control design changes. This computer model of the engine is needed in solving transient engine problems involving engine stall margin, transient time, and combustion operation.

Transient engine performance is needed to calculate engine and vehicle flight behavior such as takeoff, landing, glidepath control, speed brake sizing, engine thrust steering, augmentor lightoff, transient spooldowns, airstarting, etc.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Insert the computer program number and date of its release.

(b): Inlet air distortion, relative humidity, ram recovery, customer air bleed, customer power extraction, nozzle effects, parasitic flows, engine anti–icing, and variable geometry etc., as applicable, for non–standard days throughout the complete environmental conditions and operating envelope.

(c): The Using Service should specify one or two separate cycle decks

The following is recommended to be transferred verbatim into the specification paragraph: The transient performance computer program shall present pressure, temperature, and mass flow rates of bleed air from Idle to Maximum power. These presentations shall include the following conditions: operating altitudes, air inlet temperatures, flight speeds, and the effects of bleed air extraction on engine performance. The effects of air bleed for compressor surge protection during starting and acceleration shall be modeled in the program.

Background:

The minimum specification transient computer program should reflect transient behavior using minimum specification engine hardware and control tolerances. Status transient programs can model specific test engines. The status performance computer program should be used to define the margin between the status model and the minimum engine performance requirements. For identified deficiencies in status–to–minimum margin, a plan should be presented defining the engine design/configuration changes necessary to achieve adequate production margin. The average engine should reflect the predicted nominal engine characteristics, taken from contractor data base and rig tests, until actual engine performance data can be used from the first engine to test (FETT). After the FETT the average/status performance computer programs should reflect actual engine test data. Minimum and average transient performance may, at the option of the Using Service, be specified in two separate cycle decks or may be incorporated in one. If separate from the minimum engine transient performance computer program, the status/average transient performance computer program should be defined in the statement of work and required for submittal in the CDRL. The advantage of having two separate cycle decks is that the minimum deck will remain constant, and only the average/status deck will need to be updated.

Starting, acceleration, and deceleration operation should be closely checked since these represent conditions during which spikes (temperature, rpm) may occur.

REQUIREMENT LESSONS LEARNED

Adequate planning and design attention during initial development is required to understand compressor stall characteristics, including those in the starting and non idle regions. In one engine development program, the initial design did not have a start/acceleration bleed system or the associated control functions/logic. During the course of development, it was discovered that such a system was required to meet the engine specification. This resulted in in–flight restrictions during initial testing, schedule slippage, and engine design modification.

4.2.1.2 Performance and presentation – transient The requirements of 3.2.1.2 shall be evaluated by analysis, demonstrations, and tests.

EVALUATION RATIONALE (4.2.1.2)

The test engine's transient performance must be compared to the transient computer program to ensure the model accurately predicts the engine's transient capability.

EVALUATION GUIDANCE

Background:

The transient performance and computer program should be evaluated initially by analysis and component rig test and subsequently by full–up engine testing. Testing should be conducted at extreme conditions/loads (i.e., altitude, mach number, sea level, varying temperatures, bleed air extraction, power extraction, attitudes, power settings, etc.).

A sea level static acceptance test of the altitude performance qualification engine should be performed in the contractor's production type test cell and repeated in a government test cell. Altitude testing is usually performed in a government test facility. The engine should be the production configuration with nominal production control schedules. The USN prefers conducting performance tests with a minimum engine to ensure that all production engines will exceed minimum performance. Engine bleed air and power extraction should be taken from the engine if called for by tables Ia, Ib, Ic and table II. Instrumentation should be capable of measuring thrust, fuel flow, exhaust gas temperature, and airflow, as well as internal pressure and temperatures. The specific instrumentation requirements and accuracies should be defined in the test plans. Control parameters such as rotor speeds and variable geometry position should also be measured.

Bleed air evaluations should be accomplished at sea level and at altitude cruise condition, from Idle to Maximum power, for inlet air temperatures over the range of sea level hot and cold atmospheric temperatures of figure 4 (a minimum of five test temperatures). The test should be conducted in such a manner as to demonstrate the amount of customer bleed air available over and above that required for engine systems, such as starting and acceleration, and to establish the relationship between bleed air extraction and engine performance loss.

Measurement of scale force should be the primary method of determining thrust for transient evaluation. Other methods of determining thrust, such as the use of nozzle coefficients, should be used to confirm the thrust as measured by the scale force method. The data reduction program should be approved by the Using Service and should be fully approved prior to the start of testing. Test cell performance correction factors should be verified by back–to–back indoor/outdoor testing or a calculation procedure approved by the Using Service. If installation of test instrumentation results in significant performance losses, the engine performance results may be adjusted with the concurrence of the Using Service. Once the official tables Ia, Ib, Ic, and II evaluation begins, no further adjustments of the engine, its components, or fuel control will be allowed unless specifically authorized by the Using Service. Throughout tables Ia, Ib, Ic, and II evaluation testing, the measured performance should not be adjusted for engine degradation. When conducting altitude performance tests, consideration should be given to collection of data for the purpose of status deck development and evaluation, including temperature and pressure lapse rates. The objective is to characterize the engine components across the full range of their maps, including Reynolds number, clearance, Mach number, and other effects. This may require steady–state power hooks at several flight conditions other than those listed in tables Ia, Ib, Ic, and II. Additional guidance may be found in AEDC–TR–83–42.

EVALUATION LESSONS LEARNED

In the past, transient computer programs have not accurately simulated engine operability to be able to predict surge or stall problems.

3.2.1.3 Performance computer program Minimum and average/status steady–state and transient performance computer programs shall be constructed in accordance with AS 681. These computer programs shall form a part of the model specification and shall carry a suitable identification and date, including the engine model designation. Compilation and execution of the computer program shall be demonstrated on the (a) computer prior to acceptance by the Using Service. The model specification shall contain a list of symbols in Section 6 and a diagrammatic figure defining station identifications. These symbols and station identifications shall be in accordance with ARP 755. The specification computer program(s) shall calculate minimum performance in accordance with the requirements of 3.2.1.

REQUIREMENT RATIONALE (3.2.1.3)

The computer programs define engine performance throughout the complete operating envelope. The format of the programs must be specified to ensure software compatibility.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should fill in name/type of computer.

The following should be transferred verbatim into the specification paragraph: The computer program shall be prepared in accordance with the practices of AS 681, except as modified herein.

a. Program Requirements. The performance program shall be a thermodynamic cycle simulation in which component identity is maintained; e.g., the compressor, turbine and combustor must each be identifiable as entities in the model logic as required to obtain and maintain an accurate simulation. The computer program shall be submitted in source language compatible with the computer specified.

b. Program Capabilities. The program shall be capable of operating throughout the complete environmental conditions and operating envelope. The program shall also be capable of operating at ambient static pressures up to 32 in. Hg abs. Compilation shall not be necessary for each different run. Capability for determination of installation effects shall be included in the computer program. Effects of distortion, relative humidity, test specific instrumentation, ram recovery, customer air bleed, customer power extraction, nozzle effects, parasitic flows, engine anti–icing, windmilling, surge bleed ducting, coating installation, and variable geometry, etc, shall be included as applicable. Installation effects which are interactive with engine characteristics shall be modeled as dummy subroutines for Using Service inclusions and shall be called within the appropriate parameter convergence logic of the engine model. The ability to thermodynamically match with complex exhaust/suppressor system models shall also be included. The distortion limits of stall–free and surge–free operation shall be included in the performance computer program. The program shall provide the ability to input component performance modifiers. The computer program shall include the engine inlet and exhaust systems as discrete modules. The performance computer program shall be capable of calculating windmilling performance and being integrated into airframe installation programs to determine installed windmilling performance.

c. Documentation Requirements. A user's manual and source program card image tape shall be provided.

(l) User's Manual. In addition to those items specified in AS 681, the user's manual shall contain a general description of the simulation techniques, general overall model flow chart and a clear explanation of the calculation

process and related assumptions for all engine components. The user's manual shall include a tabulation of all parameter limits and reference to all engine limits described in the model specification; e.g., measured temperature, speed. Additionally, the user's manual shall include graphs of all empirical functions used, reference values for normalized parameters, an index of subroutines including their inputs, outputs and functions, and a listing of all test case program inputs and the corresponding required outputs. All input parameters, output parameters, and parameters used in calculations shall be identified along with corresponding units for each parameter. The manual shall describe the relationship between average and minimum engine performance and how it is calculated in the computer program. The user's manual for Government users should include component maps, principal factors, nomenclature, and basis for internal performance.

(2) Source Program. The Using Service computer program shall be in source form, and shall include all program subroutines with comment card images to identify subroutines and their functions. Sufficient comment card images shall appear in the program code to elucidate the calculation process. The first card images on the tape shall be comment cards indicating the contractor, engine type designation, program number, and date. The format of the card image tape shall be as specified by the Using Service.

d. Inputs/Outputs. The program output listing shall be organized so that the input data shall print separately from the output. The program output listing shall show the engine designation, computer program number, and date. On classified computer programs, provision shall be made in the program for a line on the output sheet which can print declassification and authority. The program shall have optional features to print internal station properties such as physical flow, total pressure, and total temperature. Output is to be labeled as to performance basis; i.e., average, minimum, new, deteriorated, spec, status, etc.

e. All program inputs must be independently variable and the programs shall be capable of sequentially accepting multiple numerical changes to the computer inputs. In addition to the inputs listed in AS 681, the program shall provide inputs of fuel heating value and measured temperature. The program shall be capable of calculating all the required output parameters of AS 681, with input options of net thrust, measured temperature and engine pressure ratio, in addition to the options of rating code and power lever angle or power code, as specified in AS 681. In addition to the required output parameters of AS 681, the program shall output the gas temperature at the first stage turbine rotor inlet of each turbine and the measured gas or metal temperatures at the measurement plane (contractor shall parenthetically insert the measurement plane station identification here). The program shall have the capability of accepting inputs and providing outputs of all temperatures in the unit of degree Celsius (C). The program shall have the capability of accepting inputs and providing outputs in the U.S. Customary System of Units and, additionally, in the International System of Units (SI) in accordance with ASTM E 380. SI units shall be made selectively available.

In the event of conflict between SI and US Customary Values (due to improper conversion), the US Customary Value shall prevail.

Background:

The engine steady–state and transient performance computer programs should be cycle balancing thermodynamic models of the engine and controls. The transient performance model should be capable of generating the transient performance points within one percent (1%) of the steady–state performance computer program for thrust, airflow, and fuel flow. All other parameters (i.e., rotor speed, pressure, and temperature) should be within the engine–to–engine variation of production acceptance engines. Subsequent changes to the engine specification performance levels require that tables Ia, Ib, Ic, and II be updated to agree with the performance computer program.

The specification computer program(s) need to be able to calculate minimum engine performance for both new engines and engines deteriorated for the time period and environment specified in 3.2.1.4. Both are needed in order to determine the true worst or minimum performance over the time period. This is because the lowest thrust/power output and highest SFC often occurs at different operating times and with different sigma components. For many engines, thrust increases as the engine deteriorates and the lowest thrust occurs with a new engine. For all engines, the highest

SFC occurs with a deteriorated engine. Aircraft performance needs to be analyzed on both a new and deteriorated basis in order to determine the true worst or minimum aircraft performance. Two approaches have been used in the past to calculate minimum engine and deteriorated engine performance. One approach uses new average engine cycle calculations with factors applied to thrust/power, fuel flow, and SFC output to represent the effects of deterioration and engine–to engine variation. The other approach uses cycle calculations with component performance modified to represent different sigma engines at different points in time. The first approach is acceptable if the sigma variation and deterioration effects are small. The second approach is preferred if the sigma and deterioration effects are larger or if additional output is needed, such as more accurate exhaust properties for special application aircraft. The second approach also allows direct output of all table I/II parameters by surveying different sigma engines at different points in time. Several models or versions may be required since a single set of component modifiers will not produce worst case thrust, fuel flow, and exhaust gas temperature, simultaneously. The first approach requires additional factors for other table I/II parameters such as maximum rotor speeds and gas temperatures.

All aerothermodynamic cycle parameters used in component performance determination and cycle calculations should be selectively available by means of a user input. All program output characteristics should present status levels of performance with the exception of the guaranteed performance parameters which will be characterized as both status and minimum performance.

Expanded input and output capability should be required to allow the performance prediction to be tailored for special analysis. For example, the model may be used to support field service evaluations (including deterioration effects) or to evaluate the effect of proposed design changes on status and/or minimum engine performance.

Expanded inputs should include adders and scalers used to modify the performance of each of the engine components, such as: component efficiencies, burner pressure loss, duct pressure losses, intermediate case pressure loss, nozzle discharge coefficient, leakages and cooling flows, and turbine flow parameters. Similarly the ability to adjust engine control schedules should be provided through expanded input.

Expanded output should include all control and aerothermodynamic cycle parameters used in component performance evaluation and engine power management, such as: component efficiencies, fan and compressor surge margins, turbine flow parameters, and nozzle areas. For the US Army the computer programs should also comply with ADS–25.

March 1992: The Navy computer is a “VAX 11/780” and is located at Naval Air Systems Command. The Navy card image tape format is 1600 BPI, 9 track, non–labeled, ASCII, and compatible with the VAX VMS operation environment.

REQUIREMENT LESSONS LEARNED

A requirement for a computer program first appeared in MIL–E–5007B in January 1959. It has since been continually improved, based upon experience, and use of the programs. Since 1960, aircraft manufacturers have provided computer programs along with engine model specification performance graphs. The last vehicle competition based on specification performance curves and correction factors, rather than the customer computer program, was the F–14 competition circa 1968. After that, the customer computer program became the official basis of aircraft performance.

4.2.1.3 Performance computer program The requirements of 3.2.1.3 shall be evaluated by inspection, demonstration, analysis, and test.

EVALUATION RATIONALE (4.2.1.3)

The use of actual engine and components data is the only acceptable method of evaluating the computer programs.

EVALUATION GUIDANCE

Background:

Substantiating data analysis for performance and computer program evaluation should be submitted to the Using Service. The engine transient computer program output should be within one percent (1%) of the steady–state performance computer program output at the performance points. The contractor should demonstrate the successful operation of the computer program(s) using all input mode options at all performance points on the specified computer. Failure of the computer program to produce valid output within the engine operating envelope should be cause for the computer program(s) to be judged as unacceptable. The test cases for each input mode option should cover the operating envelope and power range of the engine. The contractor should arrange the test case inputs in such a manner as to demonstrate the capability of the computer program(s) to accurately calculate engine data regardless of the order of the inputs or the option modes being exercised. Each test case should be evaluated to ensure that the computer program(s) is operating properly and the data produced for each test case represents the performance of the engine including program options, limits, cutbacks, and error messages. More than one item can be evaluated with each point.

The contractor's evaluation of the computer program(s) should include a comparison of the engine performance steady–state data with the figures and curves in the model specification, characteristics summary, and user's manuals for consistency and accuracy. Engine test data should be used to evaluate the computer program predictions of performance. If there are significant discrepancies between the engine test data and the computer program, a status computer program should be generated, upon Using Service approval, to represent current engine performance levels. A sufficient number of test cases, as determined by the Using Service, should be used for evaluation to test/analyze all areas of the operating envelope.

Evaluation of the minimum engine model is important in that the minimum performance quotations of thrust/power, specific fuel consumption, and limiting temperature should be a reasonable representation of a $\pm$ 2 or 3 sigma production engine. It is based upon the component performance levels known or assumed at the time the specification is developed. This evaluation requires the consideration of several factors: (1) that the quoted component performance characteristics are realistic and attainable, (2) that the computer representations of the component aerothermodynamic characteristics are of adequate fidelity, and (3) that the laws of continuity, momentum, and energy conservation are applied in a valid manner.

EVALUATION LESSONS LEARNED

In the past, some computer programs had problems predicting engine performance at the extremes of the operating envelope. Performance programs have not always reflected all the hardware changes that could affect performance.

Unrealistic component performance predictions, errors in bookkeeping of loss terms, and programming mistakes have often been discovered in the computer models, and in some cases have resulted in unattainable performance quotes.

On a past vehicle competition (F–18, 1975), several candidate engines had very detailed transient computer programs available. When exercised, these engines had significantly varying transient performance. When the Government RFP was issued, all engine vendors changed their transient performance to agree with the Government specification (some up and some down).

3.2.1.4 Performance retention Unless otherwise specified, during and at the completion of one cold parts life, the engine thrust/power shall not fall below the minimum engine thrust/power requirements specified in 3.2.1.1 and 3.2.1.2. During and at the completion of any hot section life, the engine thrust/power shall be no less than (a) percent of the actual thrust/power at the beginning of the hot section life at (b) temperature. The specific fuel consumption (SFC) from the lowest part power performance point of table II to sea level static intermediate/maximum continuous thrust/power shall be no more than (c) percent of the actual SFC at the same thrust/power at the beginning of the hot section life.

REQUIREMENT RATIONALE (3.2.1.4)

This requirement sets limits on power and specific fuel consumption (SFC) degradation which is needed to ensure an adequate performance life of the engine for the Using Service.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value from 95% to 100%.

(b): For constant temperature machines insert “constant turbine (station no. ___)” with the engine station number in parentheses.

For machines whose turbine temperatures vary (i.e., control modes based on rotor speed or pressure ratio), the words “at ___ temperature” should be deleted unless the engine performance is based on a specific turbine temperature. In this case, the EPR or rotor speed should be included in the blank.

(c): A value from 100% to 105%.

Background:

The engine should retain at least 95 percent of the initial power, and SFC should not increase to more than 105 percent of the initial SFC when operated for any hot parts life specified and in accordance with the mission and mission mix of table XIV and the ambient temperatures of figure 4. The amount of performance degradation during each hot section life should be specified to preclude large performance drops. It is assumed an average engine would be used for the accelerated life test and an average performance degradation of 3 to 5 percent would be expected (3–5 percent depends upon what flight point, control scheme, and sigma engine). This requirement limits the performance drop during a hot section life test so that a poor performing engine (e.g., high turbine temperature and high thrust) would not have a large performance drop of 7 to 10 percent. The engine should complete the hot parts life interval without replacement of parts specified in table XIV. Turbine temperature should not exceed a limit consistent with hot section life limits. The contractor should specify the station number of the turbine rotor inlet.

Performance retention is characterized by power, turbine, or exhaust gas temperature, and SFC retention at all SLS specification performance points. Performance retention requirements should be established in such a way that any engine should be able to deteriorate (at the established rate) to the maximum allowable turbine temperature (SFC) for operational use and always provide the specified percent of rated power for the required number of engine operating hours between scheduled refurbishments.

T5 control scheduled and power managed engines should not exceed maximum allowable temperatures during or at the completion of any hot section life. “Power managed” engines schedule desired thrust/power to engine pressure ratio (EPR).

Deterioration characteristics based on engine manufacturer field data will include the environmental and ingestion effects experienced by individual post production engines. This level of deterioration should be higher than that obtained in qualification testing in “clean air”. Conversely, deterioration in non–production test engines may be high because of tight clearances and thermal mismatch of materials. Field deterioration for land based helicopters is significantly higher than on qualification tests due to actual environmental and ingestion effects. Comparison and development of field deterioration data, performance retention requirements, and performance test re–calibration requirements needs to consider the environmental and ingestion effects of each engine class or application.

REQUIREMENT LESSONS LEARNED

Hot day takeoff, hot day low altitude penetration, and supersonic operation have been areas where past weapon systems have failed to maintain adequate system performance, and therefore, have driven engines back to overhaul prematurely. Thrust available from deteriorated engines at key operating points, such as catapult launch and waveoff on a tropical/hot day, can affect flight safety. Thrust available for vertical takeoff, hover, and landing for helicopters and V/STOL aircraft is important for flight safety and mission accomplishment. Performance deterioration is a principle cause of engines being pulled for maintenance on helicopters. Water washing was sometimes used in the field for performance recovery.

Some recent CFE engine/aircraft programs have had performance retention warranties with vehicle performance guarantees based on performance of deteriorated or “warranty” engines. This was because of concern for deterioration rates and engine–to–engine variation expected in service.

4.2.1.4 Performance retention The requirements of 3.2.1.4 shall be evaluated by test.

EVALUATION RATIONALE (4.2.1.4)

Demonstration of performance retention is needed to ensure compliance with the specification and provides information on the expected performance life of the engine for the intended application. Computer simulations and predictions of performance retention are not realistic or accurate to evaluate performance retention.

EVALUATION GUIDANCE

Background:

The Using Service will determine the cycles of testing and the equivalent engine flight hours. Test cycles and time can be determined from table VI and the hot section life.

For turboshaft/turboprop applications, power is delivered through a rotating shaft. A dynamometer that uses shaft rpm and torque is used to measure power in terms of horsepower.

Scale force measurement is the preferred method for determining power for turbofan/turbojet applications. The test setup, instrumentation, and data reduction procedures should be acceptable to the Using Service and should be approved prior to the test start. Test calibrations should be held to a minimum but will not count toward endurance time.

EVALUATION LESSONS LEARNED

Past SLS endurance testing (AMT/ASMET) on high performance turbofan engines revealed that SFC and operating line deterioration cannot be determined by SLS endurance test alone. High ram temperature and pressure gradients and upper left hand corner cold temperatures appear to impose significant additional engine deterioration on modern tight clearance engines. Therefore, an AMT which includes sea level, high Mach, and high altitude conditions should be accomplished since it is more representative of expected operational usage.

Deterioration in the field is also affected by environmental, design, and operating factors such as sand, rain, ice, thermal mismatch of materials, design clearances, and hard landings on carrier decks. New engines in test development can have problems with initial design clearances. One engine on its initial runs lost 5% performance in the first 40 hours because of clearances that were too tight. Engines in helicopter service can also lose 15–20% power in 500–1000 hours due to engine design and environmental factors.

3.2.1.5 Performance evaluation The performance requirements and presentation of 3.2.1.1 and 3.2.1.2 shall be used as the basis for evaluating engine performance throughout the complete environmental conditions and operating envelope. All steady–state and transient performance data shall be compared to the steady–state and transient minimum performance computer programs of 3.2.1.1 and 3.2.1.2 respectively, at the “as tested” condition. In the event the engine fails to meet the above success criteria because of catastrophic failures, discretionary engine shutdowns, A/B flameouts, parts replacement, etc., not called out in pre–test data, penalty running requirements shall be established by the Using Service. For test running in contractor test cells, a cell factor(s) shall be applied to test data to account for cell effects. The Using Service shall have approval authority for the cell factor(s).

REQUIREMENT RATIONALE (3.2.1.5)

The methods used to evaluate performance must be established to confirm performance assessments during qualification testing.

REQUIREMENT GUIDANCE

Background:

Navy practice is to not allow test data to be corrected to a desired flight condition. The performance computer deck should be operated at the as tested condition and the results compared to those obtained from actual testing. Performance correction programs should not be used. Before running the deck to “as tested” conditions, the appropriate input parameters must be carefully determined. As a minimum these should include engine inlet total pressure and temperature, ambient (exhaust nozzle discharge static) pressure and fuel heating value and fuel type. Depending on the engine control logic, additional inputs may be required, such as control Mach number signal from the aircraft and fuel specific gravity. Other control features must also be considered. Some engines reset variable geometry as a function of whether the aircraft is climbing or descending. Additionally, all parties must agree whether credit/debit should be allowed for observed variable geometry or control schedule/sensor biases, or whether the deck is to be run with “nominal” values for these parameters. These and other factors complicate the task of running the deck at “at tested” conditions.

Air Force practice is to allow adjustment factors derived from engine test data. The adjustment factors and methodology are subject to approval by the Air Force. Data adjustment or correction to standard conditions allows convenient tracking of progress during the development phase, as well as trending of engine performance for statistical process control during the production phase.

While the specification performance requirements define the minimum acceptable altitude performance levels for the in–service population, during the development phase evaluation testing, each test engine is not likely to be minimum but more likely to be “about average”. Thus a strict comparison of the performance of these few engines against minimum requirements provides little confidence on how the production population will perform. There are two ways to approach this problem. One approach is to attempt to guarantee that the test article is a minimum engine, such as by select fitting engine components and/or trimming engine controls. Another approach is to require that the evaluation engine demonstrate some margin above the minimum specification requirements as a success criteria for the performance qualification test. The amount of margin required is a function of expected engine–to–engine variability and an assessment of the “degree of averageness” of the test article, and should be negotiated prior to the test (each qualification engine is first subjected to a sea level acceptance test, and this data can be used for the assessment). In the past, the Air Force has accomplished this goal by using average rather than minimum engine performance as the specification requirements.

REQUIREMENT LESSONS LEARNED

Correction factors provided by contractors for theta and delta have historically contained many inaccuracies, especially for large temperature variations from standard day conditions.

4.2.1.5 Performance evaluation The requirements of 3.2.1.5 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.1.5)

Various analyses, demonstrations, and tests required in this specification are used to evaluate engine performance.

EVALUATION GUIDANCE

Background:

A battery of tests including altitude and sea level testing throughout the complete environmental conditions and operating envelope should be conducted and the results checked against the results from the computer program.

Engine performance defined by the computer program of 3.2.1.1 and 3.2.1.2 should be substantiated by sufficient test data to assure that theoretical assumptions used to develop the computer program performance presentation are correct. The definition of ambient temperature and pressure lapse rates and the sensitivity to engine installation effects should be a part of the substantiation test data. All major engine components and model predictions should be evaluated by analysis and comparison to actual engine performance.

The substantiation test data and analysis should be submitted by CDRL to the Using Service prior to completion of IFR, FFR, ISR, and OCR. The report should include uncertainty analysis for the performance measurands and methods for calculating thrust/power and SFC and airflow, etc..

EVALUATION LESSONS LEARNED

Past engines had performance calculation programs and performance computer programs which incorporated numerous subroutines for corrections to the data. These included differences in flight conditions (temperature, pressure) between desired and actual conditions tested. Corrections were also made for performance loss from test cell effects, internal engine station rakes, and engine control schedules. Questionable corrections changed an engine's performance from a fail to a pass.

3.2.2 Operating characteristics and limits The engine operating characteristics and limits shall be as specified in the following paragraphs. All engine steady–state and transient operating limits (maximum, minimum) shall be specified in table VIII and figure 9. The specified limits shall be predicated on the most critical tolerances of the engine.

REQUIREMENT RATIONALE (3.2.2)

The engine's operating characteristics and limits must be specified in order for the Using Service to know the full range of the engine's capabilities.

The limits are the limiting conditions within which the engine must meet all specified steady–state and transient performance characteristics.

REQUIREMENT GUIDANCE

Background:

The engine and control system have a tolerance band, for each parameter with a limit, which should be taken into account when the limits are specified.

REQUIREMENT LESSONS LEARNED

None.

4.2.2 Operating characteristics and limits The requirements of 3.2.2 shall be evaluated by analysis, demonstrations, and tests.

EVALUATION RATIONALE (4.2.2)

All engine operating characteristics and limits must be evaluated by analysis, demonstrations, and tests.

EVALUATION GUIDANCE

Background:

Test results should be analyzed for worst case tolerance in the engine and the control system.

EVALUATION LESSONS LEARNED

The full impact of some operating limits is not discovered until the engine has been in service for a number of years.

3.2.2.1 Operating envelope The engine shall meet all the requirements of the specification throughout the complete operating envelope without exceeding any limits. The engine operating limits shall be specified for the environment of figure 4, in accordance with the format shown in figures 9 and 10 and tables 1a, Ib, Ic, and II. If applicable, the thrust augmentation operating envelope shall be included in figures 9 and 10.

REQUIREMENT RATIONALE (3.2.2.1)

This requirement defines the operating envelope within which the engine must meet its functional, performance, and durability requirements (aerothermodynamic and mechanical limitations).

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph: If the referenced figures are insufficient to describe the operating envelope, either the Using Service or contractor can add a table.

Background:

The specified limits should be predicated on the most critical parameters and characteristics of the engine. The absolute altitude of the engine and the range of Mach numbers applicable at standard day, cold, tropical, and hot atmospheric conditions should be specified on figure 10. Figure 9 should represent the operating envelope for pressure/temperature. Both transient and steady–state operations should be specified. The engine operating envelopes should meet or exceed the envelope requirements of all current and anticipated aircraft applications for the engine.

Design requirements are set by envelope extremes, such as the maximum inlet pressure, which helps define the combustor case maximum pressure requirements and blade aero–elastic requirements for blade vibration. Low inlet pressure and temperatures impose design requirements on the combustor and augmentor performance parameters. Altitude and Mach number extremes will impact the cooling requirements of the engine lubrication system. Bearing loads need to be quantified throughout the engine envelope. In summary, most parts of the engine are, in some way, impacted by the engine envelope requirements.

REQUIREMENT LESSONS LEARNED

In–service engines have operated outside of the operating envelope. The aircraft envelope is normally within the engine envelope.

4.2.2.1 Operating envelope The requirements of 3.2.2.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.1)

Engine testing in ground test facilities is required to demonstrate satisfactory engine operation and performance throughout the operating envelope of the engine.

EVALUATION GUIDANCE

Background:

Engine testing should be conducted to demonstrate that the engine can meet the functional, performance, and durability requirements of this specification throughout the operating envelope of figure 10. All the sea level and altitude tests specified should be used to demonstrate satisfactory operation throughout the operating envelope. Test demonstration figures should show both the operating envelope and demonstration points. The Using Service and contractor should negotiate the test points, and the testing should be accomplished in an altitude test facility.

Characteristics of the engine which should be evaluated around the envelope include: steady–state and transient performance, engine stability, starting, internal stresses and temperatures, augmentor operation, lubrication system operation, control system operation, and inlet distortion tolerance.

EVALUATION LESSONS LEARNED

Thorough testing of the engine around the envelope extremes in an altitude test facility has been a cost effective method of reducing the possibility of encountering engine problems during flight testing and deployment of the weapon system. Early identification of engine problems and demonstration of corrective changes around the engine envelope has been accomplished with this testing. Experience has shown that augmentors on afterburning engines should be tested for stable combustion in the upper left hand corner of the envelope as well as at maximum Mach number conditions. Maximum inlet pressure conditions on the exhaust nozzles have caused buckling of nozzle components. This test is, therefore, of particular importance to engines with variable geometry exhaust nozzles. There have been cases where oil hiding has been observed in the engine gearbox at high inlet pressure. Therefore, the behavior of lubrication and fuel systems should be carefully monitored at flight envelope extremes and during altitude starts.

3.2.2.2 Operating attitude and conditions The engine operating attitude limits shall be shown on figure 11. The engine shall meet the requirements of the specification when operating in the unshaded area of figure 11, and operate at least (a) seconds continuously in the shaded areas of Figure 11. Operation in the shaded area shall not degrade engine performance or cause any damage. The engine shall start, stop, and be stowed in any of the attitudes shown in the unshaded area figure 11. Engine stowing capability outside of the unshaded area of figure 11 shall be specified herein. The engine shall function satisfactorily for at least (b) seconds in negative g and for at least (c) seconds in zero g conditions.

REQUIREMENT RATIONALE (3.2.2.2)

The engine is required to operate throughout the attitudes imposed by the aircraft, such as takeoff, climb, inverted flight, air combat maneuvers, stowage, and terrain following. Time duration of these factors is a significant consideration to the engine design.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of at least 30 seconds.

(b): A value of at least 60 seconds.

(c): A value of at least 30 seconds.

Background:

For Army V/STOL aircraft requiring engine attitude changes, the starting and stopping attitude limits should be not less than 105 degrees nose up, 20 degrees nose down, and 30 degrees roll to each side. Above 6 km, continuous operation is not required at nose up attitudes greater than 45 degrees.

REQUIREMENT LESSONS LEARNED

Extended flight operation in zero or negative “g” conditions resulted in oil system malfunctions, such as oil starvation, bearing sump flooding, gearbox flooding, oil foaming problems, and seal leaks. Special provisions, such as an auxiliary lube system may be necessary in the lubrication system for extended inverted flight operation.

4.2.2.2 Operating attitude and conditions The requirements of 3.2.2.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.2)

Testing is required to demonstrate satisfactory engine functional capability under a variety of attitude and maneuvering conditions.

EVALUATION GUIDANCE

Background:

The engine should be subjected to an attitude test to meet the requirements of 3.2.2.2. Engine capability to operate for 60 seconds at negative "g" and 30 seconds at zero "g" conditions should be evaluated by analysis or by a rig test of the lubrication oil system. The engine should be started and operated at Intermediate thrust/power demand position, for at least 30 minutes, at each of the test points shown in the unshaded area of figure 11. The engine should also be operated at Intermediate thrust/power demand position for at least 30 seconds at each of the test points shown in the shaded area of figure 11. This test should be considered completed when the engine starts satisfactorily, remains within all operating limits, and there is no evidence of mechanical damage.

The qualification test program should have a strong foundation of component tests, especially in the oil system component area. Oil tanks, oil pumps, scavenge pumps, gearboxes, deareators, and bearing compartments should all be component tested in various simulated flight conditions. It may be possible to test complete small engines at various attitudes.

Figure 11 defines engine attitude limits (roll and pitch) under static ground test stand conditions and should not be construed as necessarily defining engine attitude limits during flight (engine attitude capability is usually limited by lubrication system design). The forces generated during maneuvering flight may have a combined or resultant effect that permits engine pitch and roll attitudes that exceed the limits of figure 11 without encountering an engine limiting condition.

EVALUATION LESSONS LEARNED

None.

3.2.2.3 Stopping Upon receiving a signal from the airframe for stopping, the engine shall terminate fuel flow in any operating condition. Stopping of the engine from any power setting (including afterburner) or at any rate shall not a) result in immediate or subsequent exceedance of any engine limits, b) adversely impact engine durability, structural integrity, or operational capability, c) cause rotor bowing or rotor seizure, and d) delay a satisfactory start or afterburner light. The engine shall not experience any post–shutdown fires. No damage to the engine shall result from shutting off the fuel supply by the foregoing means or from shutting off the fuel supply to the engine inlet connection during any engine operating condition. Provisions for stopping the engine by means other than a completely mechanical system shall be subject to approval of the Using Service. All methods for stopping the engine, including emergency provisions, shall be specified herein.

REQUIREMENT RATIONALE (3.2.2.3)

A simple method must be provided to stop the engine for operational and safety considerations.

REQUIREMENT GUIDANCE

Background:

Shutdown procedures should avoid rotor seizure or excessive oil leakage on subsequent starts. If the provisions for normal stopping include systems other than mechanical, an additional completely mechanical emergency system for shutting off all fuel flow to the engine should be supplied. For turboprop or turboshaft engines, adjusting the load of the shaft power absorber should not be required when shutting down the engine.

Cool down periods at Idle before shutdown have been beneficial in preventing undesirable effects of a high power shutdown.

REQUIREMENT LESSONS LEARNED

Hot shutdowns (termination of fuel when the engine is above a stabilized Idle condition) has caused rotor seizure and bowed rotors. This resulted in delaying the subsequent start.

4.2.2.3 Stopping The requirements of 3.2.2.3 shall be evaluated by demonstration, analysis, and test.

EVALUATION RATIONALE (4.2.2.3)

Engine stopping or shutdown should be evaluated by test. Analysis and demonstrations are needed for evaluating stopping and subsequent restarting from high power settings.

EVALUATION GUIDANCE

Background:

Engine shutdowns during ground testing should be accomplished in accordance with established procedures. Engine shutdowns should also be accomplished during the altitude testing to evaluate the adequacy of altitude shutdown procedures at a wide range of temperature conditions.

EVALUATION LESSONS LEARNED

Multi–engine patrol aircraft have routinely shut down engines in flight to conserve fuel.

3.2.2.4 Low power conditions

4.2.2.4 Low power conditions

3.2.2.4.1 Idle thrust/power – [ground/flight idle] Idle shall be the minimum self–sustaining condition of the engine. With the thrust/power demand in the Idle position and with no customer bleed air or power extraction, the thrust/power shall not exceed (a) percent of the Intermediate power available at standard day conditions up to (b) feet. The variation in Idle power, bleed pressure, and HP rotor speed with temperature, altitude, and installation factors shall be as defined in the performance computer program of 3.2.1.1. When the engine incorporates provisions for an additional Idle power setting (i.e., ground Idle), the means of accomplishment, the thrust level, the corresponding rotor speed and the time required to change Idle thrust shall be specified herein.

REQUIREMENT RATIONALE (3.2.2.4.1)

Idle power levels must be consistent with aircraft limitations/requirements and allow the engine to maintain steady–state, stall–free operation. Keeping the power as low as possible also lowers the wear on the aircraft brakes and reduces the possibility of aircraft skids on wet or icy pavements.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of 5 percent.

(b): 15,000 feet or the ground start upper limit.

Background:

In determining Idle power, consideration must be made of the following: (1) landing and taxi speed requirements, (2) bleed pressure to assure minimum aircraft environmental control system performance, (3) engine stability and acceleration times when doing engine transients, and (4) minimum gearbox speeds. Maximum Idle power levels and minimum gearbox speeds and bleed pressures should be defined in this specification. The steady–state computer program should be used to define actual thrust, bleed pressure, and rotor speeds.

It is imperative that once the engine cycle has been selected, the relationship between combustor burner pressure and power in the region near Idle should be accurately characterized. Combustion blowout limits should be defined in the Idle region to prevent flameout during decelerations.

Ground and flight Idle speeds must be compatible with power requirements and still provide adequate stall margin for off–idle and bodie accelerations. Attention must be given to the trade–off between Idle thrust, ECS bleed pressure, and compressor off–idle stability margin. Idle speed should be limited to minimize power while taxiing.

REQUIREMENT LESSONS LEARNED

The specific value of 5 percent is approximately the capability of state–of–the–art engines.

Some engines reduced ground Idle power by opening the variable exhaust nozzles while taxiing. Opening the nozzles inflight had the favorable effect of increasing rotor speeds and transient responses at low power settings. In the open position, the nozzles did not interfere with the ground plane during field or carrier landings.

Computer simulations have been in significant error during some engine early development phases because of the inaccuracies of component maps at near Idle operation and inaccurate definitions of Idle power. This has led to costly system redesigns in aircraft ECS and braking systems or in the engine control system. Minimum bleed pressure at altitude can affect the bleed takeoff locations(s), heat exchanger sizing, and overall ECS design. Arbitrary idle control schedules in early program phases have resulted in errors in calculation for mission fuel, payload range, airplane sizing, and engine or cycle comparisons. This results from credit being given for fuel burned and distance gained on Idle descent mission segments when Idle is not well defined at altitude. Definition of Idle power can also affect operation and stability of secondary flow systems and flows used for exhaust or aircraft cooling. Off–idle and core stalls have plagued engines where the inherent compressor stall margin at the Idle power selected was not sufficient to meet acceleration time requirements. In this case, it would be prudent to select a higher power setting for Idle and identify this change to the airframe contractor as soon as possible to minimize the overall system impact.

4.2.2.4.1 Idle thrust/power – [ground/flight idle] The requirements of 3.2.2.4.1 shall be evaluated analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.4.1)

The Idle power must be evaluated to ensure that it does not exceed the specified values and that it will be compatible with aircraft requirements.

EVALUATION GUIDANCE

Background:

Ground and flight Idle power as a function of inlet temperature, altitude, and flight Mach should be evaluated in sea level and altitude facilities. Use of a measured power from a scale force load cell for thrust and from a torque meter with waterbrakes for shaft horsepower are the preferred methods. Calculation of power is less accurate in most cases.

Measurement of stall margin in the Idle region is highly desirable to ensure engine acceleration capability.

EVALUATION LESSONS LEARNED

For turbofan/turbojet engines, experience has shown that stability problems have been encountered on numerous engines in the Idle speed range.

3.2.2.4.2 No–load condition For turboprop/turboshaft engines, the no–load condition shall be an engine operating condition of zero delivered output shaft torque while maintaining constant steady–state governed output shaft speed with a specified power absorber and operating at altitudes up to _(a)_ km (_(a)_ ft). Engine gas generator speed at no–load shall be specified in the format shown on figure 12.

The engine(s) and torque absorber system shall be capable of transitioning between load and no–load conditions without unacceptable mismatches in rotor speeds or torques defined in 3.2.2.6.2, and without exceeding gas generator operating limits.

REQUIREMENT RATIONALE (3.2.2.4.2)

No–load conditions must be consistent with aircraft limitations and prevent output shaft overspeed when power absorber torque is zero. Transitions to and from no–load are important in control system design and flight dynamics.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Typical altitudes are from sea level to 4.5 km (15,000 ft) for helicopters and sea level to 9.0 km (30,000 ft.) for fixed wing aircraft.

REQUIREMENT LESSONS LEARNED

Helicopters have often encountered no–load conditions during combat and terrain following/masking maneuvers when the engine(s) de–coupled from the main rotor system. Transitions to and from no–load have resulted in helicopter rotor speed and rotor thrust droop, shaft overspeed, overtorque, exceedance of engine operating limits, and unacceptable flight dynamics.

4.2.2.4.2 No–load condition The requirements of 3.2.2.4.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.4.2)

The no–load condition must be evaluated to ensure that the engine does not exceed the specified values of operating limits and that the engine is compatible with the aircraft requirements. Transitions to and from no–load must be evaluated and tested on a transient basis for performance and control characteristics. Operation at no–load is a power setting lower than idling with a coupled rotor system.

EVALUATION GUIDANCE

Background:

Transitions to and from no–load must be evaluated and tested on a transient basis for performance and control characteristics. The power absorber used in the test should replicate key characteristics of the principal application.

EVALUATION LESSONS LEARNED

None.

3.2.2.5 Stability The engine shall operate satisfactorily during transient operation from one thrust/power setting to another without unstable operation, stall, surge, flameout, A/B instability, or blowouts. Under steady–state operating conditions, engine thrust/power fluctuations shall not exceed (a) percent of the Intermediate rated power (IRP)/Maximum Continuous between Idle and IRP/Maximum Continuous conditions, or (b) percent of the thrust/power available at the power level position and operating condition, whichever is less. During operation above IRP up to Maximum A/B, fluctuations shall not exceed (c) percent of the thrust/power available at that condition. During steady–state operating conditions, the period between major discrete frequency fluctuations, as defined above, shall not be less than (d) seconds. Main combustor and afterburner blowout margins shall be sufficient to protect against blowouts or flameouts during rapid throttle movements under worst case conditions, and engine control and main burner and afterburner stack–ups. Those frequency fluctuations which approach the amplitude or exceed the frequencies specified above shall not be permitted if, in the judgement of the Using Service, they indicate an instability. Stability audits of the engine component system shall be performed to identify critical margin stability areas throughout the complete environmental conditions and operating envelope of the engine. The stability audit methodology shall be specified herein. In addition, the stability audit shall be updated as necessary throughout the development program. Bleed air extraction shall not require thrust/power demand modulation to maintain engine stability and limits. Continuous sinusoidal throttle transients with peak–to–peak thrust/power change demands of (e) % with a frequency up to (f) cycle per second shall not result in a divergent sinusoidal thrust/power response. Thrust/power response phase lag shall be no more than (g) degrees and thrust/power peaks shall be at least (h) % of demand. Symmetrical throttle demands shall not result in divergent symmetrical throttle responses.

REQUIREMENT RATIONALE (3.2.2.5)

A stable operational and stall/surge–free engine is important to the operation of the aircraft. Stall or surge can result in loss of aircraft and personnel. Stable operation within all temperature and rotor speed limits during power transients is necessary to ensure mechanical integrity and parts durability and allow pilots to keep attention to other aspects of flight.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of 1%.

(b): A value of 5%.

(c): A value of 1%.

(d): A value of 5 seconds.

(e): A value of +5%

(f): A value of 1 cycle/second.

(g): A value of 135.

(h): A value of 50%.

Figure 13 should be used as a guide for determining the requirements of “e” through “h”. Figure 13 may also be included in the model specification to clarify the requirement.

Background:

SAE ARD 50015, Section 3, identifies a methodology for temperature stability assessments. ARP 1420 identifies the methodology for pressure stability assessments. A stability audit CDRL should be prepared requiring the engine contractor to submit updated stability audits to the Using Service at each milestone, or more often as necessary.

REQUIREMENT LESSONS LEARNED

Nonrecoverable stalls have been encountered in some engines. Investigations involving engine flow paths, control systems, and afterburner operating conditions have been required to establish redesign and operating limitations necessary for satisfactory operation. The problems have reduced the operational utility of the aircraft. Engine stalls have been encountered due to external influences such as inlet distortion during aircraft maneuvers, ingestion of aircraft gun gas, and improper design of aircraft bleed systems that allowed high pressure bleed air to flow into the compressor. It has been found that determining Idle and bodie stall margins and burner blowout margins early in the development process is necessary for a successful and timely program. Burner blowout margin problems and off idle/bodie stall problems have been encountered late in the development cycle; these have impacted early production and operational engines. Problems were encountered because of a lack of understanding (data base) of the stackup of engine–to–engine and control system variations.

4.2.2.5 Stability The requirements of 3.2.2.5 shall be evaluated by analysis, demonstrations, and tests. Testing shall be conducted under the following conditions: (a) . Tests shall be performed in each control mode and with control mode transfer. Stability audits shall be performed to identify marginal stability areas in the engine envelope.

EVALUATION RATIONALE (4.2.2.5)

Stability audits and tests need to be conducted during the engine development program to provide assurance that there is adequate stall margin available for the intended aircraft application.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service and contractor, with airframe contractor input, should determine the most critical flight conditions, and these should become the conditions for the stability tests and audits.

Background:

Stability audits and tests should be performed to identify marginal stability areas within the engine and flight envelope, and the results should delineate the steady–state and transient stall–free envelopes. The stability audit methodology should be presented in appropriate documentation such as the Interface Control Document and should be in accordance with ARP 1420. The stability audit should identify the critical conditions for stability tests. Stability tests should include evaluation of the effects of inlet airflow distortion, customer air bleed, power extraction, exhaust nozzle backpressure, and thrust reverser installation factors. Engine and component tests should be conducted to evaluate engine stability and ensure stall–free operation prior to aircraft operational tests, to ensure stable operation and to evaluate margins. Stability tests should be conducted at:

a. critical stability margin and flight conditions

b. high altitude, low mach conditions

c. sea level takeoff – static operation

d. cruise operation

e. envelope extremes

f. conditions examining extreme engine or aircraft effects on engine stability

g. control tolerances and engine–to–engine variations.

EVALUATION LESSONS LEARNED

Improper test planning and execution has led to improper conclusions regarding engine stability margins. Raised operating line tests afford some measure of assuring adequate surge margin to account for engine–to–engine variations and control tolerances.

3.2.2.6 Transients The engine shall satisfactorily perform any thrust/power transient throughout the operating envelope while meeting the requirements of the specification. Thrust/power requests in any sequence and at any rate shall not result in exceeding any engine operating limit (including overspeed and overtemperature), result in unstable operation of the engine, or cause any mechanical damage. Thrust transient times in all backup control modes shall be the same as those in the primary mode. Thrust transient times for installed engine conditions, including the effects of customer bleed air, customer power extraction, anti–icing bleed, inlet distortion, and the exhaust nozzle, singly or in any combination, for non–standard days throughout the complete environmental conditions and operating envelope, shall not exceed (A) percent of the standard day thrust transient times.

For thrust/power lever movements of 0.5–seconds or less, the transient times below shall be the slowest thrust/power transient times for standard day conditions with: no customer power extraction, no customer bleed air extraction and no engine anti–icing bleed air operation, but with all other engine system bleed air requirements (e.g., acceleration bleed air, cooling bleed air). The time required for the engine to accomplish (B) percent of the thrust/power change shall be the time specified in the transient performance computer program and listed below. The total time required to accomplish the full transient and reach transient stability shall be the time in transient performance computer program and presented below plus (C) seconds. Actual engine acceleration and deceleration time shall be within +0 seconds and (D) seconds of the acceleration and deceleration time specified in the transient performance computer program and below. In the event of a conflict between the transient times specified in the transient performance computer program and those listed below, the times specified in the transient performance computer program shall take precedence.

The standard day engine transient times shall be as follows:

From sea level up to 10,000 feet and 0 to 150 knots (0.0 to 0.25 Mach number) indicated airspeed (IAS):

a. From Idle to Intermediate thrust available (a) seconds. From Intermediate to Idle thrust available, (a) seconds.

b. From Idle to Maximum thrust available, (b) seconds. From Maximum to Idle thrust available, (b) seconds.

c. From Idle to Maximum reverse thrust available, (c) seconds. From Maximum reverse to Idle thrust available, (c) seconds.

d. From 30 percent Intermediate to Intermediate thrust available, (d) seconds. From Intermediate thrust to 30 percent Intermediate thrust available, (d) seconds.

e. From 10 percent Intermediate to 20 percent Intermediate thrust available, (e) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

f. From 20 percent Intermediate to 10 percent Intermediate thrust available, (f) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

g. From 20 percent Intermediate to 30 percent Intermediate thrust available, (g) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

h. From 30 percent Intermediate to 20 percent Intermediate thrust available, (h) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

i. The thrust versus time from Idle and 10, 20, 30, 40 and 50 percent Intermediate to Intermediate thrust and to Maximum thrust shall be presented in the specification as shown in (i).

j. The thrust versus time from Idle and 10, 20, 30, 40 and 50 percent Maximum to Maximum thrust shall be presented in the specification as shown in (j).

From 10,000 feet up to the absolute altitude:

k. From Idle to Intermediate thrust available, (k) seconds. From Intermediate to Idle thrust available, (k) seconds.

l. From Idle to Maximum thrust available, (l) seconds. From Maximum to Idle thrust available, (l) seconds.

For all operating conditions (throughout the complete operating envelope):

m. From Intermediate to Maximum thrust available (m) seconds. From Maximum to Intermediate thrust available (m) seconds.

n. From Intermediate to Maximum reverse thrust available (n) seconds. From Maximum reverse to Intermediate thrust available (n) seconds.

o. From Maximum to Maximum reverse thrust available (o) seconds. From Maximum reverse to Maximum thrust (o) seconds.

TURBOPROP/FIXED WING

The standard day engine power (turboprop) transient times shall be as follows:

Sea level up to 10,000 feet and 0 to 150 knots (0.0 to 0.25 mach number) IAS:

a. No load to Maximum Continuous/Intermediate power available, (aa) seconds. From Maximum Continuous/Intermediate power to no load (aa) seconds.

b. From Idle to Maximum power available (bb) seconds. From Maximum to Idle power available, (bb).

c. From Maximum Continuous/Intermediate power to full reverse power available, (cc) seconds. From full reverse power to Maximum Continuous/Intermediate power available, (cc) seconds.

d. From 30 percent Maximum Continuous/Intermediate to Maximum Continuous/Intermediate power available (dd) seconds. From Maximum Continuous/Intermediate power to 30 percent Maximum Continuous/Intermediate power available, (dd) seconds.

e. From 10 percent Maximum Continuous/Intermediate to 20 percent Maximum Continuous/Intermediate thrust available, (ee) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

f. From 20 percent Maximum Continuous/Intermediate to 10 percent Maximum Continuous/Intermediate thrust available, (ff) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

g. From 20 percent Maximum Continuous/Intermediate to 30 percent Maximum Continuous/Intermediate thrust available, (gg) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

h. From 30 percent Maximum Continuous/Intermediate to 20 percent Maximum Continuous/Intermediate thrust available, (hh) seconds. Power lever movement shall not exceed 125 percent of the power lever movement required to obtain a corresponding steady–state change.

i. The power versus time from Idle and 10, 20, 30, 40 and 50 percent Maximum Continuous/Intermediate to Maximum Continuous/Intermediate power and to Maximum power shall be presented in the model specification as shown in (ii) .

From 10,000 feet to absolute altitude:

j. From Idle to Maximum Continuous/Intermediate power available, (jj) seconds. From Maximum Continuous/Intermediate to Idle power available, (jj) seconds.

For all operating conditions:

k. From Maximum power to full reverse power available, (kk) seconds. From full reverse power to Maximum power available, (kk) seconds.

l. From full reverse power to Maximum Continuous/Intermediate power available, (ll) seconds. From Maximum Continuous/Intermediate power to full reverse power available (ll) seconds.

m. From Maximum power to Maximum Continuous/Intermediate power available (mm) seconds. From Maximum Continuous/ Intermediate to Maximum power available (mm) seconds.

TURBOSHAFT/HELICOPTER

For turboshaft engines in helicopters applications, the following requirements apply:

From Sea Level to 3 km (9843 ft), and Mach 0.0 to 0.2:

n. From Idle to Maximum Power, (nn) seconds.

o. From no load to Maximum Power, (oo) seconds.

p. From Maximum Power to no load, (pp) seconds.

From 3 km (9843 ft) to maximum altitude, and Mach 0.0 to 0.2:

q. From no load to Maximum Power, (qq) seconds.

r. From Maximum Power to no load, (rr) seconds.

REQUIREMENT RATIONALE (3.2.2.6)

Thrust/power transient characteristics must be defined to meet aircraft mission performance requirements during critical phases of aircraft operation (i.e., takeoff, go–around/wave–off, and combat aircraft acceleration). Power transient acceleration and deceleration times are critical because of their impact on engine life and operability (i.e., compressor stall, combustor stability).

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(A): 125%

(B): 95% – 98%

(C): 5 seconds (small engines) to 10 seconds (large engines)

(D): a value between 0.2 – 0.5 seconds

The following transient criteria is recommended for turbojet and turbofan (TJ/TF) fighter engines. The times should be negotiated among the engine contractor, airframe contractor, and the Using Service for each specific program. Transient times may be expected to vary depending on the class of engine (i.e., bypass ratio, augmentation), and its mission (fighter, bomber, transport, trainer). For turbofan and turbojet engines, the flight envelope is relatively large with a wide range in transient times. The transient times should be specified graphically on a flight envelope to clearly define the capability. The Using Service may opt to leave the blanks as is, and then the contractor should complete the blanks.

(a): 5, 5

(b): 7, 7

(c): 5, 5

(d): 4, 4

(e): 0.5 to 1.0

(f): 0.5 to 1.0

(g): 0.5 to 1.0

(h): 0.5 to 1.0

(i): Contractor should provide in a table or figure

(j): Contractor should provide in a table or figure

(k): 12, 12

(l): 14, 14

(m): 2, 2

(n): 2, 2

(o): 2, 2

The following power transient times are recommended for turboprop (TP) engines. The times should be negotiated among the engine contractor, airframe contractor, and the Using Service for each specific program. The Using Service may opt to leave blanks as is, and then the contractor should complete the blanks. For this application the transient responses should not be compromised by application or removal of load to the output shaft in any sequence and at any rate.

(aa): 5, 5

(bb): 7, 7

(cc): 3, 3

(dd): 4, 4

(ee): 0.5

(ff): 0.5

(gg): 0.5

(hh): 0.5

(ii): Contractor should provide in a table or figure

(jj): 12, 12

(kk): 2, 2

(ll): 2, 2

(mm): 2, 2

The following times are recommended for turboshaft engines in helicopter applications.

(nn): 7

(oo): 3

(pp): 3 for attack/scout, 5 for cargo/utility

(qq): 8

(rr): 8

Note that helicopter transient to and from no–load are at constant commanded output shaft speed. Transients to and from Idle use the minimum self–sustaining engine and output shaft speeds for Idle.

Background:

The tolerance band on thrust transients for all engines delivered to the Using Service should be +0 and –___ seconds. The minus time (in seconds) should be negotiated between the contractors and Using Service. Quicker acceleration times are desirable as long as they do not excessively use up available surge margin, cause engine compatibility problems with the airframe, and reduce engine life. All production engines should have faster transient times than those specified herein, because the specification transient times are based on the poorest performing engine. A very rapid acceleration time is not desirable since this could cause additional problems.

Transient characteristics from initial PLA movement up to 95% or 98%, depending on engine size, thrust change should be shown on figure 14. The transient characteristics between 95% or 98%, and 100% of the final transient stabilized thrust (transient stability) should be shown on figure 14. Transient characteristics from transient stability up to steady–state stability should be shown on figure 14. Transient figures permit a better pictorial presentation of the critical transient area from initiation of the transient to final stabilization.

Component maps should be sufficiently accurate in the Idle and sub–idle range to characterize ground and airstart transient times to within 10 percent. Actual engine Idle to Intermediate and maximum transient times should be accurate to within 5 percent of the transient performance computer program.

Power loss should be defined for a cold engine transient from Idle to Intermediate and Idle to Maximum power. The cold engine transient time should be consistent with SLS Idle to Intermediate and Idle to Maximum power times. This requirement represents the USAF “tiger start” and is the first start of the day when the Idle dwell time does not exceed 90 seconds. This scenario represents alert status/base escape requirements.

Special system needs should be considered when transient requirements are established. For example, some applications have a need for expanded inlet airflow distortion limits in order to provide for the desired maneuvering capability.

Engine stall margin available for distortion may be increased at the expense of transient acceleration time if system requirements allow. System requirements for customer power or bleed air capability should also be considered and balanced with distortion and acceleration capabilities. Some systems may require increased Idle speeds (ground or flight) which will alter transient time requirements.

REQUIREMENT LESSONS LEARNED

Some operational engines have experienced hung accelerations in the upper left hand corner of the operating envelope.

In the process of reducing transient times considerable internal stall margin is consumed. This resulted in operability problems in many fighter aircraft.

Crisp and repeatable engine response is needed during aerial refueling and aircraft carrier approaches.

4.2.2.6 Transients The requirement of 3.2.2.6 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.6)

Transient characteristics must be evaluated to ensure that the mission requirements of the aircraft can be met with the engine.

EVALUATION GUIDANCE

Background:

Transient evaluation should be performed on all qualification engines, especially during the altitude test. In addition to thrust/power, the engine measured turbine temperature, rotor speeds, fuel flow, and compressor discharge pressure, as well as other pertinent parameters, should be recorded during each transient. Acceleration and deceleration times should be evaluated during the acceptance tests on all engines delivered to the Using Service.

EVALUATION LESSONS LEARNED

On a past USAF engine, when a modified turbine temperature probe design was tested, a power overshoot characteristic occurred. This allowed the engine to meet the specification requirement in terms of time from Idle to 95 percent Maximum Continuous/Intermediate, but the thrust subsequently dipped below 95 percent before finally trimming to a stable value.

3.2.2.6.1 Overshoot and undershoot The engine thrust overshoot following any acceleration demand shall not exceed (a) percent of the final stabilized steady–state thrust value. The engine thrust undershoot following any deceleration demand shall not exceed (b) percent of the final stabilized steady–state thrust value.

For turboprop/turboshaft engines, power, output shaft rotor speed and torque shall not exceed (c) percent of the final stabilized steady–state value. During the transient to the stabilized value, output shaft speed and torque shall not exceed the limitations specified in 3.2.2.6.2.

REQUIREMENT RATIONALE (3.2.2.6.1)

Overshoot and undershoot need to be specified to assess control limiting functions (overspeed and overtemperature), surge margin loss and aircraft system performance. Undershoot is especially important during landing or aerial refueling.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 2%–3%

(b): 2%–3%

(c): 2%–3%

Background:

This requirement is most critical between 95% and 100% power. The overshoot or undershoot should not exceed the recommended values in order to prevent using up surge margin and excessive engine life. The amount of temperature overshoot or undershoot generally increases with engine bypass ratio, and a direct correlation may be established for a given engine between the amount of overshoot or undershoot and the time at Idle prior to acceleration or the time at Intermediate/Maximum prior to deceleration. This paragraph should be tailored to address EGT overshoot or undershoot depending on the engine control mode. Note that the engine should be capable of limiting the overshoot or undershoot.

Helicopter control systems typically try to govern output shaft speed during power transients. Observing rotor system torque limits is typically left to pilot action. The engine control has anticipators to increase transient response, lessen overshoots, and lessen speed/performance droop.

REQUIREMENT LESSONS LEARNED

Several Air Force engines have been found to experience turbine temperature overshoot after a burst from Idle power to Maximum speed. The overshoot was found to be caused primarily by a large loss in high pressure turbine efficiency, due to open blade tip clearances caused by the differing thermal responses of the turbine disk, blades, and case. The typical amount of temperature overshoot was approximately 50 degrees.

In one engine, the turbine temperature overshoot resulted in a thrust droop whenever the engine's turbine temperature limit was encountered (e.g., a new engine on a hot day or a deteriorated engine on any day.) The maximum temperature overshoot/thrust droop occurred 45 to 90 seconds after the throttle movement. Engine rotor temperatures were found to be a big driver on the amount of turbine temperature overshoot.

A cold rotor resulted in a large overshoot – running time at Idle from a "cold" shutdown condition warmed the rotor and decreased the overshoot. Conversely, running time at Idle on a "hot" rotor cooled the rotor and increased the overshoot.

The "worst case" overshoot condition occurred on a burst to Maximum power with less than 5 minutes at Idle after a shutdown period of 4 hours or more. Aircraft on alert can suffer overtemperatures or low take–off thrust since no stabilization time is possible.

Naval aircraft can also experience overtemperatures and low thrust on catapult launch on a tropical/hot day. Thrust undershoots/overshoots and lengthy stabilization times have resulted from hysteresis in the control system. In attempts to compensate for low thrust and overtemperatures on a hot day takeoff/launch the Using Service has had to implement turbine "fix kits".

4.2.2.6.1 Overshoot and undershoot The requirement of 3.2.2.6.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.6.1)

Power/thrust overshoot must be evaluated to prevent potential performance losses at critical points, minimize engine stalls, and preserve engine life.

EVALUATION GUIDANCE

Background:

Power/thrust overshoot problems should be exposed during various sea level and altitude qualification performance tests. Testing with and without customer bleed air and power extraction should be mandatory.

EVALUATION LESSONS LEARNED

Power/thrust overshoot or undershoot has not occurred on all engines or particular models. It can be very elusive to find and may occur only on a deteriorated engine.

3.2.2.6.2 Droop During any acceleration, no thrust/power droop below (a) percent of the demanded thrust/power value shall occur between the attainment of (b) percent thrust/power change and achievement of transient stability as defined by 3.2.2.6. In addition, there shall be no more than a (c) percent thrust/power droop from the point of transient stability up to attainment of steady–state stability defined in 3.2.2.5.

For turboprop/turboshaft engines, output shaft speed shall not deviate more than ±3 percent from the governed value for any load change not completed within two seconds. During the transient, shaft torque shall not exceed (d) percent of the steady–state operating limit.

REQUIREMENT RATIONALE (3.2.2.6.2)

During a change in power level there is a rapid change in gas temperature. All the parts of the engine do not respond thermally in the same manner to the temperature change. These differential rates of thermal expansion between cases, disks, and other parts can result in a power/thrust droop. This can be a critical condition during and immediately after catapult launch or hot day takeoff from high altitude airfields. It can also be noteworthy for V/STOL aircraft.

For helicopters, rotor speed falloff or “rotor droop” is an important consideration for low level flight dynamics, designing the engine control, and aircraft interface. Thrust loss occurs when the pilot commands a sudden increase in blade pitch (rotor power required) and the rotor slows down while waiting for engine power to increase. When engine power increases, it may exceed the rotor system torque limit (at the lower speed) and thereby lengthen the time to accelerate and achieve thrust. If the rotor speed or thrust droop becomes too large, generators can come offline, sink rates can become excessive, and the pilot’s ability to recover from extreme droop is reduced.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value between 95% and 98%.

(b): A value between 95% and 98%.

(c): A value between 0% and 2%.

(d): The Using Service should provide a value.

Background:

The engine thrust droop should not exceed 2% in order to maintain aircraft performance demands. The so–called “thrust droop” phenomenon that has been encountered on many engines is due to transient thermal response mismatch between the massive turbine rotor disk and the turbine case. A cold rotor (i.e., one at or near Idle for less than about 15 minutes) expands more slowly than the case during an acceleration, resulting in transiently open clearances and high

turbine temperatures. On engines with automatic temperature limiting controls, the control will cut back fuel flow to hold the temperature limit, resulting in the "thrust droop". On engines with no automatic temperature limiting, thrust is maintained but exhaust gas temperature (EGT) may exceed limits, resulting in aborted take–offs or require maintenance actions. Thus this paragraph should be tailored to address thrust droop or EGT overshoot depending on the engine control mode. Note also that automatic turbine clearance control devices may be utilized to counteract the problem in severe situations; however, they have an associated weight penalty on the engine.

REQUIREMENT LESSONS LEARNED

In one instance of thrust droop, the problem was reduced or eliminated by the addition of an active turbine clearance control system. In another instance, the engine added "droop compensation" in the control followed by "extended life" turbine kits.

Transient thrust droop has often been overlooked by airframe and engine contractors in the vehicle proposal phase. This was from the intent to obtain "good performance" at key operating points and to avoid unfavorable performance evaluation comparisons. The assumption was often made that the thrust droop could be compensated for and thereby neglected in the proposal phase. The reality, though, was that many cold engines experienced some amount of thrust droop when operating on a temperature limit.

An exception to the cold engine droop phenomena was the small engine class used in helicopters and light transport. Since the engines were physically small, parts and components heated up rapidly and cold engine thrust droop was less obvious. Thrust droop on helicopters resulted from decreasing rotor speed with sudden changes in blade pitch.

4.2.2.6.2 Droop The requirements of 3.2.2.6.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.6.2)

Power/thrust droop must be evaluated to prevent potential performance losses at critical points, minimize engine stalls, and preserve engine life.

EVALUATION GUIDANCE

Background:

Power/thrust droop problems should be exposed during various sea level and altitude qualification performance tests. Testing with and without customer bleed air and power extraction should be mandatory. In theory, the thrust droop may occur whenever a cold engine of sufficient size is operating on a temperature limit. Since a deteriorated engine is more often on a temperature limit, the thrust loss may be more apparent. The droop can also vary with engine–to–engine variation.

EVALUATION LESSONS LEARNED

Power/thrust droop has not occurred on all engines or particular models. It can be very elusive to find and may occur only on a deteriorated engine.

3.2.2.7 Windmilling The engine shall be capable of continuous windmilling for _(a)_ hours, without damage to the engine, and without affecting air restart and operational capability. The following information shall be specified (for an uninstalled engine):

a. The limits of windmilling operation, including curves showing windmilling rotor speed and windmilling drag vs. flight Mach number.

b. The oil consumption rate during windmilling operation.

c. In the event of lubrication failure, the time duration and limits of windmilling operation after depletion of oil supply.

d. The power extraction available during windmilling operation as shown on figure 15a.

e. The customer bleed air extraction available during windmilling operation as shown on figure 15b.

REQUIREMENT RATIONALE (3.2.2.7)

The windmilling capability of the engine should be compatible with the intended weapon system's usage and with all airframe systems which use bleed air and/or power extraction.

Installed windmilling performance is needed to determine aircraft get–home capability and engine–out trim for cruise, takeoff, or landing. Installed windmilling performance/capability can vary significantly from traditional specification curves with high loss or buried engine installations.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A period equal to the longest mission as determined by the Using Service

REQUIREMENT LESSONS LEARNED

Multi–engine aircraft have used windmilling techniques to conserve fuel during cruising and to conserve brakes while taxiing.

In past programs, acceptable minimum times for subparagraph c. have been one hour at full speed (USAF fighter program), and one hour at 30% speed (USN fighter program). Rotor axial loading during windmilling is a force opposite to normal loading which has caused oil seals to leak.

4.2.2.7 Windmilling The requirements of 3.2.2.7 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.7)

Windmilling should be evaluated to ensure that oil leakage and other related problems will not occur in flight.

EVALUATION GUIDANCE

Background:

During altitude testing, a functional test run of windmilling drag vs. flight Mach number and windmilling oil consumption should be accomplished. Estimates of windmilling time after oil supply depletion and the power extraction capability should be determined from these tests. Evaluation of windmilling should be done in the air starts evaluation test.

Windmilling data (oil consumption, rotor speeds, etc.) should be analyzed to evaluate windmilling capability.

An assessment should be made of windmilling performance installed in the aircraft using the performance computer program(s) of 3.2.1.3. This is necessary for high loss or buried type installations which affect system flow balance and pressure ratio across the engine.

EVALUATION LESSONS LEARNED

None.

3.2.2.8 Reverse thrust For turbofan and turbojet engines, the operating envelope, limitations, and description for thrust reverser operation shall be specified herein. The percentage of total engine thrust used for reverse thrust shall be specified herein.

REQUIREMENT RATIONALE (3.2.2.8)

The reverser may be required to slow the aircraft after touch down, aircraft maneuvering, and/or aircraft braking and backing.

REQUIREMENT GUIDANCE

For turboprop and turboshaft engines replace the specification paragraph text with “Not applicable”.

The following is recommended to be transferred verbatim into the specification paragraph:

The reverser system shall be capable of operating from the fully stowed position to the full deployed position in no more than two seconds and from the fully deployed position to the fully stowed position in no more than five seconds at all power settings and throughout the reverser operating envelope.

Background:

The mechanism and thrust reverser operating envelope should be defined by the contractor together with the reverser’s level of performance of reverse thrust, deployment time, and stowing time. Reverser performance should be included in the steady–state and transient performance computer programs. Thrust reversers should be designed to avoid re–ingestion of exhaust products and foreign object damage. Aircraft backing requirements, if any, should be included in the thrust reverser design.

Reverse thrust systems are mainly used to brake large turbofan/jet powered aircraft after landing. Military fighter and attack aircraft tend to use wheel brakes, arresting gear systems, or drag parachute system for braking. Thrust reversers have the disadvantage of providing a weight penalty to the aircraft.

REQUIREMENT LESSONS LEARNED

Past problems have included re–ingestion of hot exhaust products, foreign object damage due to exhaust gas impingement on runways, and engine stalls due to transient operation.

4.2.2.8 Reverse thrust The requirements of 3.2.2.8 shall be evaluated by test.

EVALUATION RATIONALE (4.2.2.8)

The performance and operating characteristics of the thrust reverser must be evaluated.

EVALUATION GUIDANCE

For turboprop and turboshaft engines, replace the specification paragraph text with “Not applicable”.

Background:

Tests should include component and engine tests. Component tests may include model tests to establish the aerodynamic characteristics, and mechanical tests should demonstrate functional characteristics of the thrust reverser structure.

Engine reverser tests should include operation throughout the full range of engine power settings and thrust reverser positions. Full range transients from full forward thrust to full reverse and full reverse to full forward thrust should be accomplished to demonstrate transient requirements and satisfactory operation.

EVALUATION LESSONS LEARNED

Experience has shown that strain gages on fan and turbine blades are very useful in defining blade stress levels during reverser operation. This type of testing is useful to establish operating limitations relative to re–ingestion of reverser exhaust gas into the engine inlet.

3.2.2.9 Steady–state and transient airflow The engine maximum allowable steady–state and transient airflow limits shall be shown in table VIII. The transient airflow limits specified shall include augmentor light–off and termination, as applicable, as well as thrust/power demand increase and decrease. The steady–state airflow of any engine shall not vary by more than (a) percent of the steady–state airflow computed by the steady–state computer program of 3.2.1.1 and shown in tables 1a, Ib, Ic, and II. The transient airflow of any engine shall not vary by more than (b) percent of the transient airflow computed by the transient computer program of 3.2.1.2.

Transient airflow shall not deviate from the maximum steady–state airflow by more than (c) percent and shall be controlled to assure compatible engine inlet operation. The engine steady–state and transient airflow limits shall include engine deterioration, control tolerances (including temperature lags and transients), engine–to–engine variations and shall apply to the non–standard atmospheres of this specification as shown on figure 4. The effects of customer air bleed and power extraction shall be identified.

REQUIREMENT RATIONALE (3.2.2.9)

Air flow values and tolerances must be established to meet power requirements and to ensure inlet compatibility. Steady–state operational tolerances must be determined and controlled so that the lower and upper tolerance air flow limit still allow stable engine operation. Transient air flow deviation from maximum steady–state air flow must be established to ensure stall–free operation.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 3%

(b): 3%

(c): 3%

The maximum and minimum airflow limits, including the deviation during transient operation to be specified in table VIII are normally provided by the contractor.

Background:

Air flow limits are a fallout of trade studies that optimize installed performance based on mission requirements while maintaining compatible inlet/engine operation during steady–state and transient operation. Both maximum and minimum air flow limits must be considered to assure stall–free operation. The engine control system must maintain upper and lower airflow limits both during steady–state and transient operation. Engine–to–engine air flow variations must be considered within the desired limits. Steady–state and transient air flow values for any specific flight condition and bleed/power extraction levels are required in the computer programs of 3.2.1.1 and 3.2.1.2.

REQUIREMENT LESSONS LEARNED

Gusts, maneuvers, and abrupt altitude changes can cause transient airflow to exceed maximum steady–state limits and lead to engine stall. Low inlet mass flow (near Idle/minimum air flow limits) has caused planar pulse distortion generated in the inlet resulting in engine stall. This has classically been associated with supersonic operation where it

is known as buzz; however, it has also existed in the subsonic region of the envelope at levels severe enough to cause engine stall. If the inlet has been properly characterized at low mass flows, the lower air flow limit can be properly scheduled in the engine control to maintain compatible inlet/engine operation. If the engine control system is limited in its capability, inlet redesign may be necessary.

4.2.2.9 Steady–state and transient airflow The requirements of 3.2.2.9 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.9)

Evaluation of air flow is required to ensure compatibility with the aircraft inlet.

EVALUATION GUIDANCE

Background:

Accurate evaluation of air flow limits and requirements can be obtained only through controlled test conditions. Altitude wind tunnel testing with adequate instrumentation and instrumented flight testing are methods that may be used for evaluation. There are several methods of measuring air flow, such as venturis, bellmouths, and through fan/compressor pumping characteristics. It is desirable to confirm the air flow values by measurement with at least two different techniques.

EVALUATION LESSONS LEARNED

Experience has shown that air flow limits must be maintained throughout the entire engine population. One engine program experienced difficulty because control system stack–up and engine–to–engine variation caused some engines to exceed the required air flow limit; for stable operation, an air flow topping device was added to the control system to maintain stable engine operation throughout the engine envelope during cold weather operation.

3.2.2.10 Inlet airflow distortion The engine shall not surge, stall, flameout, or incur any damage with the steady–state or time variant inlet distortion (pressure, temperature, or any combination of both) shown on figure 16. The (a) most severe distortion conditions permitted within the operating envelope shall be identified on figure 16. The aerodynamic interface plane shall be defined as (b). An engine stability and performance assessment shall be performed using the methodology and inlet distortion descriptors defined in ARP 1420 for total pressure distortion. An engine stability and performance assessment shall also be performed for inlet temperature distortion and for the combination of both pressure and temperature distortion. The temperature distortion methodology and the methodology for combination of both pressure and temperature distortion shall be specified. The effects of inlet airflow distortion on engine performance shall be included in the performance computer programs of 3.2.1.1 and 3.2.1.2.

REQUIREMENT RATIONALE (3.2.2.10)

Aircraft inlet distortion is an important consideration in the design of gas turbine engines. The distortion limits provide interface information for making engines and aircraft compatible.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The number of distortion conditions (i.e., patterns, screens) should be specified by the Using Service in conjunction with the airframe and engine contractors.

(b): The aerodynamic interface plane should be defined by the airframe and engine contractors. If no agreement can be made between the contractors, the Using Service should make the decision.

As an option, the following may be transferred verbatim into the specification paragraph to supplement the requirement:

An engine stability and performance assessment for total pressure and total temperature distortion shall be performed using the methodology and inlet distortion descriptors defined below:

a. A distortion descriptor methodology indicating the maximum distortion limits at which the engine shall operate without stall, rotating stall, and surge, throughout the complete environmental conditions and operating envelope, due to steady–state and time–variant inlet air total pressure and temperature variation, shall be defined in the specification. Where exhaust nozzle back pressure effects on the fan and/or gas generator affects the tolerance of the engine to inlet air pressure variation, the effect shall be specified in the specification. Engine stability and performance assessments shall be performed using the methodology and inlet distortion descriptor elements defined in ARP 1420. This information shall be provided with the specification. The effects of inlet airflow distortion on engine performance shall be incorporated into the performance computer programs of 3.2.1.1 and 3.2.1.2.

b. The engine contractor, using the distortion descriptor methodology shall provide a minimum of five sets of pressure distortion patterns (screens), that, with Using Service approval, reflect the most severe conditions permitted within the limits of the distortion descriptor specified for the engine. Three of these distortion patterns (screens), exceeding the most severe inlet spatial and planar pressure conditions expected in the weapon system installation, shall be used for IFR testing. All distortion patterns (screens) shall be used during ISR and OCR engine testing. One of the patterns (screens), as required by the Using Service, shall be used for the Acceptance Test surge/stall margin test. For each set of inlet flow distortion data specified, the engine interface operating conditions shall be defined, as applicable, in terms of Mach number, altitude, power setting, customer bleed air, customer power extraction, etc. Measurements of the engine inlet total pressure, temperature, and flow variation shall be made at the engine/inlet aerodynamic interface, which shall be defined in the specification. All inlet instrumentation utilized in measuring airflow, pressures and temperatures, the arrangement, location, response, and instrumentation accuracies shall be consistent with the guidelines of ARP 1420 and defined in the specification. For each set of specified inlet flow distortion data, the total airflow, average total pressure recovery and pressure and temperature for each individual probe shall be specified. The available surge margin, maximum distortion limits, and the differences between the contractor's methodology and ARP 1420 shall be presented in the specification.

Background:

An in–depth discussion of inlet total pressure distortion for gas turbine engines is contained in SAE AIR 1419 and provides the technical background and development of the guidelines of ARP 1420. A current assessment of the inlet/engine temperature distortion problem is contained in SAE ARD 50015. The aircraft engine industry is most knowledgeable about steady–state total pressure distortion. Pressure distortion is the most prevalent disturbance, easiest to measure, and can be duplicated by screens. Time variant pressure distortion requires more complex measurement system and elaborate equipment to duplicate in the laboratory and test cell. Engine performance degradation, including power loss due to combustor blowout and/or compressor instability, has been attributed to engine inlet total temperature distortion. It can be a time–variant spatial temperature distortion or a temperature ramp. The armament gas ingestion requirement (3.3.2.6) is a form of time variant distortion. Steam ingestion (3.3.2.7) is a combined form of pressure and temperature distortion.

ARP 1420 provides generalized parameters for inlet airflow total pressure distortion. The distortion limits should be defined in terms of both spatial and planar content. When the inlet flow descriptions and key inlet operating conditions are established, a table should be constructed by the contractor to specify the engine requirements.

Consideration should be given to:

a. mach number and altitude

b. inlet angle of attack and yaw

c. inlet flow margins

d. inlet flow descriptions (steady–state and dynamic pressure and temperatures)

e. planar wave description (frequency and amplitude)

f. steady–state and dynamic flow disturbances

g. planar pulse distortion under low inlet mass flow conditions

h. cross wind effect during ground operation

i. control tolerances and engine–to–engine variations

j. compression system bleed and vane schedule

k. shift in compressor operating line

REQUIREMENT LESSONS LEARNED

Early definition of the inlet distortion limits has proven to be necessary for successful integration of the engine into the aircraft and should be provided by the engine manufacturer as part of the trade–offs that must be made during the design and development process. Early model tests of the inlet provide assurance in the validity of the distortion limits. Inlet distortion has resulted in premature fan blade failure and has also been shown to have an impact on steady–state and transient augmentor operations. Reported inlet distortion causes include aircraft maneuvers, afterburner operation on fan engines, ingestion of gases from armament firing, ingestion of steam from catapult launch systems, water ingestion, bird ingestion, ingestion of engine exhaust gases from thrust reverser systems, and ingestion of engine exhaust during helicopter and V/STOL aircraft operations. Fighter aircraft were reported to be most susceptible to inlet distortion in the upper left hand corner of the aircraft flight envelope. Aircraft that fly with speeds in excess of Mach 1 have experienced engine surge caused by aircraft inlet shock waves.

4.2.2.10 Inlet airflow distortion The requirements of 3.2.2.10 shall be evaluated by analysis, demonstration, and test throughout the flight envelope.

EVALUATION RATIONALE (4.2.2.10)

Inlet distortion tests are required to demonstrate the engine's tolerance to inlet distortion. Analysis is needed for points in the flight envelope where distortion screens or other distortion generating devices are not tested.

EVALUATION GUIDANCE

Background:

Inlet pressure distortion tolerance can be evaluated by several techniques: distortion screens or distortion simulators in engine test facilities, and actual aircraft inlets in wind tunnel facilities. If screens are used, the distortion patterns should be selected on the basis of peak instantaneous dynamic distortion level from inlet tests. A minimum of five screens should be tested. One of the screens should be capable of surging the engine. Distortion sensitivity is determined by throttling the engine to surge at constant rotor speed behind each screen. The process is repeated at several levels of rotor speeds for each screen. Consideration should be given to testing classical patterns (e.g.. pure circumferential and pure radial content) as well as flight patterns for the purposes of methodology development and evaluation. Fan distortion transfer characteristics are determined by measuring distortion in front of the fan and the

compressor. Accurate measurement of distortion transfer requires adequate instrumentation and screen rotation. By rotating the screen to (usually 3) different orientations, the distorted flow field is more accurately mapped. Care should be taken that the engine match point does not shift during the test. This may be accomplished by use of a remotely–controlled screen rotator (so the engine does not have to be shut down to rotate the screen) and by locking the engine control settings. Also, the effect of fan operating line on distortion transfer may be significant, so it is recommended that distortion transfer data be collected at a high (near surge) operating line as well as the normal operating line.

Using a screen capable of surging the engine is needed to evaluate the engine surge margin during the Acceptance Test and to evaluate the contractor's distortion descriptor methodology. These tests should include rapid throttle transients and bodies.

Inlet temperature distortion tolerance can be evaluated by one of the following techniques in a test chamber: a controlled rocket charge in front of the engine, a hydrogen–fueled distortion generator, or a hot–air injection system. Flight tests are also used to evaluate engine distortion by firing rockets in flight. The effects of customer bleed air, power extraction, and exhaust nozzle back pressure on engine tolerance to inlet airflow distortion should be determined. Tests should be accomplished to evaluate the engine's capability to accept distortion generated by airframe inlets.

A complete aircraft inlet can be used during the engine development for distortion evaluation, but it does not ensure total engine/airframe integration. The test must simulate the most severe cases of distortion at the flight condition and/or maneuver at which they occur. The attachment of the aircraft inlet to an engine in a test chamber only simulates straight and level flight at a very low flight speed.

The results of the evaluation should be incorporated into the transient performance computer program for a better understanding of the effects of distortion.

EVALUATION LESSONS LEARNED

Aircraft inlet tests have not always simulated the worse case pressure distortion for the installed engine. The tests used scale models that did not have engines installed. These tests did not always produce the correct Strouhal Number or the Reynolds Number that are experienced in the full scale aircraft with the engine installed. In spite of these drawbacks, distortion screens, calibrated for the proper distortion level, have proven to be effective, economical tools for engine distortion testing. Other tools for generating inlet distortion were a planar wave generator or a distortion valve. Distortion transfer coefficient is extremely sensitive to the accurate mapping of the distorted flow field. Experience has shown that the distortion transfer coefficient may be in error by as much as 50% if screen rotations are not used, even with the full complement of 40 probes per ARP 1420. In some cases instrumentation is less than the recommended levels due to inoperative probes or missing rakes. In one test program, the engine manufacturer elected not to remove the engine gearbox to install rakes on the lower half of the compressor face, leaving a gap of greater than 100 degrees unmapped. The performance computer program was also used for analytical distortion evaluation.

The earliest cases where armament gas ingestion caused significant engine surge problems occurred during the 1950s on the F–86 and F–94 aircraft. During the late 1960's, the A–7 aircraft encountered steam ingestion induced surges during catapult launch on aircraft carrier. A 1979 study documented the effects of gas ingestion from a nose–mounted cannon on the A–10 aircraft.

3.2.2.10.1 Pressure and temperature rate of change The engine shall withstand the following rate–of–change of pressure and temperature, separately or in combination, without surge, stall, flameout, or damage:

Parameters	Rate–of–change	Duration
Engine inlet pressure	(a) kPa/sec (psi/sec)	(b) sec
Engine inlet temperature	(c) °C/sec (°F/sec)	(d) sec

REQUIREMENT RATIONALE (3.2.2.10.1)

Rate of pressure and temperature changes must be specified since large changes in pressure and temperature may occur at the engine face due to ingestion of turbine engine exhaust, gun gas, or nuclear and non–nuclear explosions.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a) – (d): The Using Service should provide values.

Background:

The maximum rates of pressure and temperature change required for the engine should be determined from examination of the intended mission of the aircraft. Rate of change of pressure may be set by nuclear or conventional weapons effects or climb/dive requirements on the aircraft. Temperature rate of change may be determined by the effects of rocket/gun gas ingestion.

Temporary losses in stall margin result when pressure and temperature changes cause substantial variations in density of the air entering the engine. Engine exhaust from minimum interval takeoff operation of tactical and strategic aircraft can reduce the surge margin of engines.

REQUIREMENT LESSONS LEARNED

Fighter/attack aircraft in the past have experienced compressor stall when exposed to rapid temperature changes caused by rocket/gun gas ingestion.

4.2.2.10.1 Pressure and temperature rate of change The requirement of 3.2.2.10.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.10.1)

Analysis and/or test must be conducted to predict the engine stability characteristics during maximum rates of pressure and temperature change.

EVALUATION GUIDANCE

Background:

Analytical predictions should be evaluated by testing if technically and economically feasible using instrumented component rigs, core engines, and/or full engines, as appropriate. Pressure distortion testing should be accomplished using an air jet distortion generator or distortion screens in front of the engine face. Use of a shock tunnel to generate appropriate full face overpressure transients should be considered. High response pressure probes mounted in an approved array should be placed directly behind the distortion device to verify the pressure values presented to the inlet of the compression component being tested. Temperature changes due to ingestion of rocket exhaust or gun gas can be simulated by discharging the device mounted in an appropriate location ahead of the engine. Temperature measurement sensors should be mounted in an array of rakes to validate the temperature changes and rates being experienced by the engine. The inlet temperature distortion tests can be done in flight test aircraft.

EVALUATION LESSONS LEARNED

The Defense Nuclear Agency has developed a test technique to satisfactorily demonstrate the adequacy of turbomachinery against the stated overpressure requirement. Tests on the J85 engine have shown that turbomachinery can be aerodynamically and/or mechanically sensitive to relatively small increases in full face overpressure that results from nuclear and non–nuclear explosions. These tests showed that the engine failed before any part of the airframe failed. Therefore, engine survivability from a nuclear or non–nuclear explosion is a serious concern.

3.2.2.11 Gas–path and measurement plane temperature limits

a. The first stage high/low pressure turbine rotor inlet gas temperature limit(s) shall be as follows: (a) .

b. Measurement plane temperature. The maximum allowable steady–state gas temperature averaged over the gas path area at the measuring plane and corresponding to the maximum allowable steady–state gas temperature at the gas generator first stage turbine rotor inlet shall be specified herein, along with the location of the measurement plane (including station identification number). No engine shall operate above the maximum allowable measuring plane turbine temperature. This temperature may be changed prior to the initiation of IFR, FFR, ISR, and OCR based on correlation studies during the development program of the relationship between measured temperature and gas temperature at the measurement plane. The maximum allowable transient measured temperature during starting and acceleration shall also be specified herein. When a direct measuring metal temperature sensing system is provided at the measurement plane, the maximum allowable steady–state metal temperature(s) shall be specified herein. If a direct measuring metal temperature sensing system is not provided at the measuring plane, the maximum allowable measured temperature(s) at the measurement plane (based on the signal indication after the temperature sensor harness plug) shall be specified herein.

REQUIREMENT RATIONALE (3.2.2.11)

The temperature of the hot gas in a gas turbine is important to reliability, endurance, and performance and limits must be established.

REQUIREMENT GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph:

(a): Maximum allowable operating gas temperature(s): The maximum allowable operating steady–state and transient gas temperature(s) at the first stage high/low pressure turbine rotor inlet(s) shall be specified for the range of engine inlet air temperatures defined by figure 9. No engine shall operate above the maximum allowable first stage turbine rotor inlet temperature. Additionally, the maximum allowable operating transient gas temperature(s) during starting and acceleration shall be specified.

Background:

If short period ratings (less than 30 minutes) are used, the maximum allowable temperature will exceed the highest value for continuous operation. The Army requires the engine to be capable of operating at the maximum allowable temperature for 12 seconds during accelerations above Idle. A maximum allowable starting temperature that can be tolerated by the engine for 12 seconds should also be specified.

Some recent Navy engines have had short period or emergency thrust ratings (“APR”) above Intermediate power. These engines exceeded the normal maximum operating temperatures at Intermediate. The maximum allowable temperature limit(s) should allow for droop compensation of engines operating at the highest steady–state temperature. This condition may last for one to three minutes per application. When feasible, the maximum allowable temperature limit(s) and material selection should allow for near–term thrust/power growth or “throttle push”.

REQUIREMENT LESSONS LEARNED

None.

4.2.2.11 Gas–path and measurement plane temperature limits The requirements of 3.2.2.11 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.11)

All engine limits must be evaluated.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

The full impact of operating limits is not discovered until the engine has been in service for a number of years.

3.2.2.12 Surface temperature and heat rejection The maximum operating surface temperatures and heat rejection rates of the engine shall be shown on figure 17a. Accessory pad loadings, compressor bleed air conditions, oil system cooling requirements, and air and gas leakage from engine case flanges and split lines shall be used in establishing engine heat rejection rates. The conditions surrounding the engine, the engine power condition, and the oil system temperatures for which the surface temperatures are applicable shall be shown. For components and externals on the surface of the engine, the specified component and accessory surface temperature and heat generation curves may differ from the engine temperature as shown on figure 17a. The engine surface emissivity shall be presented as shown on figure 17b. The heat rejection and surface temperature data presented shall be for the conditions listed below. If heat rejection rates during flight and ground operating conditions other than those listed below are critical or limiting, that data shall be specified:

For turbojet/fan engines

a. Maximum Power, Sea Level static, hot atmosphere of figure 4.

b. Maximum Power, Sea Level, maximum stagnation inlet air temperature for hot atmosphere of figure 4.

c. Maximum Power, 36,089 ft, maximum stagnation inlet air temperature.

d. Maximum Power, highest altitude of figure 4, hot atmosphere of figure 4, subsonic level flight.

e. Low Power, Sea Level, low speed loiter condition, tropical or hot atmosphere of figure 4.

f. Idle Power, Sea Level static, hot atmosphere of figure 4, ground operation.

For turboshaft/turboprop engines

a. Maximum power, sea level static , hot atmosphere of figure 4.

b. Maximum power, 1.2 km (4000 ft), maximum stagnation inlet air temperature for the hot atmosphere of figure 4.

REQUIREMENT RATIONALE (3.2.2.12)

The surface temperature and heat rejection must be controlled to ensure compatibility between the engine and the aircraft for safe and durable operation. This paragraph provides interface data necessary for the airframe contractor to determine the amount of heat which must be dissipated by the airframe from the engine skin. The surface temperatures and emissivity are necessary to determine the radiant heat transfer to the inner skin of the airframe engine compartment.

Multiple flight points are required to survey bay cooling flows and thermal soakback. Aircraft with largely composite structure are more difficult to cool and more vulnerable to engine surface heat and temperature.

REQUIREMENT GUIDANCE

Background:

For this requirement, the engine surface includes all external components and associated tubing and plumbing and is not limited to the outer casing.

The conditions to be specified in subparagraphs a – c are generally considered the most severe heat rejection conditions.

As a minimum, the following factors should be considered in the analysis: horsepower extraction, compressor bleed air conditions, anti–ice operation, oil system cooling requirements, engine skin temperature and radiation properties, and air/gas leakage from flange joints and split lines.

Skin temperature has an influence on the adjacent airframe structure and significantly affects the weight and cost of that structure and the engine bay cooling requirements.

REQUIREMENT LESSONS LEARNED

In the past, there were errors made by the requirement to specify component and accessory surface temperatures. Components are items furnished by the engine manufacturer and component surface temperatures are the engine manufacturer's responsibility. Externals are items mounted on the engine and not furnished by the engine manufacturer. Accessory surface temperatures are not addressed in this specification.

4.2.2.12 Surface temperature and heat rejection The requirements of 3.2.2.12 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.2.2.12)

Surface temperatures and heat rejection rates must be evaluated by analysis, demonstration, or test.

EVALUATION GUIDANCE

Background:

The required evaluation should include, but not be limited to the considerations listed in 3.2.2.12. The determination of whether to use analysis or test should be based on the accuracy and relative simplicity of each method.

The engine should be tested at the highest inlet temperature it expects to encounter in the flight envelope. All other temperatures, including surface temperatures, are proportional to the inlet air temperature. Heat rejection also is a function of temperature and pressure.

EVALUATION LESSONS LEARNED

None.

3.2.2.12.1 Component limiting temperatures Engine components mounted on the engine shall not exceed their allowable temperatures when surrounded by still air under the following conditions:

a. Continuous operation (including high and low power settings) with ambient air at the maximum stagnation temperature.

b. Flight shutdown from the most adverse condition and continued soaking with ambient air at maximum stagnation temperature.

c. Ground shutdown with ambient air at the sea level hot resoak temperature conditions of figure 4 with no special cooling.

A tabulation of the maximum ambient air temperature surrounding each component, the limiting surface temperatures for all components, the time period(s) applicable, and the limiting cooling fuel inlet temperature for fuel cooled components, for the most severe engine condition(s), shall be specified in table IX. The specific points of measurement, where the ambient air temperature and surface temperature with respect to the three coordinate axes of the component are maximum, shall be specified. The engine components requiring special cooling and ventilation shall be listed in table IX. Length of operation of these components without special cooling shall be defined in table IX.

Engine gas path flow leakage shall not impact cooling and ventilation. There shall be no backflow of exhaust gases, from any source, through the engine compartment under any operational or shutdown condition.

REQUIREMENT RATIONALE (3.2.2.12.1)

The Using Service needs to ensure that the engine components will not exceed their limiting temperatures under the most adverse conditions of heat generation and/or rejection.

REQUIREMENT GUIDANCE

Background:

The contractor should establish the maximum allowable temperature that the components can withstand and list these components and corresponding limits in the engine model specification so that adequate cooling schemes can be provided.

REQUIREMENT LESSONS LEARNED

Non–unidirectional flow in the engine compartments has resulted in overheating of components. Overheated fuel, which was intended for cooling electronic components, damaged electronic controls, resulting in an engine overspeed and turbine failure.

4.2.2.12.1 Component limiting temperatures The requirement of 3.2.2.12.1 shall be evaluated by analysis, demonstration, and test.

A test shall be conducted on an instrumented engine in a simulated or actual engine bay.

EVALUATION RATIONALE (4.2.2.12.1)

Component limiting temperatures must be evaluated to ensure that they are not exceeded.

EVALUATION GUIDANCE

Background:

Applicable cooling data should be obtained for the conditions tested for all individually cooled components/externals and for each section of the engine case (compressor case, combustor case, turbine case, and exhaust duct). Sufficient data should be obtained to ensure that the engine and all critical components are within the specified temperature limits.

An instrumented engine in a simulated or actual engine bay should be tested with maximum allowable fuel and ambient air temperatures. Data should be taken at critical component locations at the power and airflow conditions that produce the highest component temperatures. Cooling data should be obtained for the conditions tested for all individually cooled components/externals and for each section of the engine case (compressor case, combustor case, turbine case, and exhaust duct). Sufficient data should be obtained to ensure that all critical components are within specified temperature limits.

EVALUATION LESSONS LEARNED

None.

3.3 Environmental conditions

4.3 Environmental conditions

3.3.1 Atmospheric conditions

4.3.1 Atmospheric conditions

3.3.1.1 Humidity The engine shall meet all requirements of this specification when subjected to the following humidity: (a) .

REQUIREMENT RATIONALE (3.3.1.1)

Humidity must be required because it may physically and chemically deteriorate engine material. Humidity can lead to: (1) swelling of materials due to moisture absorption, (2) loss of physical strength, (3) changes in mechanical properties, (4) degradation of electrical and thermal properties in insulation materials, (5) electrical shorts due to condensation, (6) binding of moving parts due to corrosion or fouling of lubricants, and (7) deterioration of electrical components.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): A value of 95 percent humidity on a 90–degree day.

Background:

This requirement is intended to assure that certain components will not be affected by exposure to high humidity conditions, such as in tropical areas.

The 95% humidity requirement is based on MIL–STD–810 requirements.

REQUIREMENT LESSONS LEARNED

Chief design problems have resulted because of the long term effects of humidity related corrosion of metals due to oxidation and of effects on electrical and electronic components.

4.3.1.1 Humidity The requirements of 3.3.1.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.3.1.1)

The deteriorating effects of humidity must be determined. Components must be tested and/or the materials of construction analyzed to evaluate their ability to withstand high humidity conditions.

EVALUATION GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph: The requirements of 3.3.1.1 shall be evaluated by test as follows: Components shall be subjected to a humidity test in accordance with MIL–STD–810, METHOD 507.3 (Humidity), Procedure III (Aggravated) for a test duration of 15 cycles (360 hours). The components shall be subjected to pre– and post–test calibrations and shall not be operated during the test. At the

end of the tenth cycle (not less than 240 hours), electrical/electronic components shall be dried without forced convection for 12 hours in air at a temperature not greater than 21°C (70°F) and a relative humidity not less than 50 percent. At the end of this period, the component shall be subjected to four power supply overvoltage transients and transient spike voltages. The component shall then complete the humidity test. After the post–test calibration, the component shall be disassembled and inspected. There shall be no corrosion or other defects present that will affect the function or structural strength of the component. Additional pass/fail criteria shall be in accordance with MIL–STD–810. MIL–STD–810 cannot be invoked on a blanket basis but must be tailored to suit a particular operating environment.

Background:

A full–up engine test to satisfy the humidity test requirement cannot be accomplished in the engine corrosion test. The engine corrosion test does not contain the humidity requirements of 3.3.1.1. Operation of the component during the corrosion test deters humidity buildup within the component, and would diminish the ability of the facility to maintain a specified humidity level. Therefore, engine components should be subjected to a humidity test in accordance with MIL–STD–810.

EVALUATION LESSONS LEARNED

Past engines have had flow path coatings on compressor cases that swelled when exposed to humidity. When the engine was started there was an interference in the blade tip clearances for the compressor blades. This caused some blade tip corner losses or extreme rubbing in the flow path coating. This would not have been discovered until production, if the humidity test had not been run during development.

3.3.1.2 Fungus The engine shall meet all specification requirements when subjected to fungus.

REQUIREMENT RATIONALE (3.3.1.2)

Fungus requirements must be specified because it can create problems such as metal corrosion, grease hardening, reduction of insulation effectiveness, and bridging of conductors causing electrical shorts and arcing. These problems are particularly severe in tropical and Naval environments.

REQUIREMENT GUIDANCE

Background:

New engine designs are now including porous ceramic material layers on hot section or afterburner parts. These materials are cooled with compressor/fan bleed or discharge air. The air flows through the “pores” in this material. Any entrained particles larger than the pore sizes, will be deposited in this material. This can degrade durability and operability.

Fungus may be resisted by the selection of non–nutrient materials and the use of special fungus resistant coating on nutrient materials, as well as other methods, as approved by the Using Service.

Adding a requirement that selection of all non–nutrient materials be mandatory is a very difficult requirement because this would preclude the use of polymer composite materials. Some polymer composite materials in gas turbine engines contain fillers that are fungus nutrient materials.

REQUIREMENT LESSONS LEARNED

Anti–icing additives in fuel have reduced micro–biological contamination in fuel tanks.

Some of the typical fungus–related problems experienced in Southeast Asia:

a. Micro–organisms digest organic materials as a normal metabolic process thus degrading the substrate, reducing the surface tension, and increasing moisture penetration.

b. Enzymes and organic acids produced during metabolism diffuse out of the cells and onto the materials and cause metal corrosion, glass etching, hardening of grease, and other physical and chemical changes to the materials, such as insulation resistance and arcing.

c. The physical presence of micro–organisms produces living bridges across components which have resulted in electrical failures.

4.3.1.2 Fungus The requirements of 3.3.1.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.3.1.2)

The engine must be resistant to fungus growth and fungus related problems. Fungus must be evaluated by satisfactory completion of a component fungus test.

EVALUATION GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph: The requirements of 3.3.1.2 shall be evaluated by test as follows: Components shall be subjected to a fungus test in accordance with MIL–STD–810, METHOD 508.4.

Background:

A full–up engine test to satisfy the fungus test requirement cannot be easily accomplished, therefore engine components should be subjected to a fungus test in accordance with MIL–STD–810, Method 508, which is the recognized standard military fungus test. The contractor should review each engine component for the materials used that comprise all parts and determine those materials that are fungus resistant. For electrical components, MIL–STD–454, Requirement 4 should be used.

EVALUATION LESSONS LEARNED

Past engine materials have contained fungus food (e.g., carbon particles in flow path coatings) that was found to promote fungus growth during testing for fungus.

3.3.1.3 Corrosive atmosphere The engine shall meet all requirements of this specification during and after exposure to the corrosive atmosphere of table X for (a) corrosion cycles. The engine shall not deteriorate more than (b) percent in thrust/power, or gain (c) percent in specific fuel consumption.

REQUIREMENT RATIONALE (3.3.1.3)

This requirement is specified because during engine operation in the proximity of the sea/salt air environment, engine materials can be susceptible to corrosive attack. The air and gas path areas of the engine can be particularly susceptible to corrosive attack. Also, if external engine hardware, such as fasteners and clamps, corrode in service, drilling and cutting bolts to maintain or repair engines result in high maintenance costs.

REQUIREMENT GUIDANCE

This requirement is applicable for aircraft deployed and operating in a corrosive environment, such as on aircraft carriers, at or around seacoast bases, and in some desert environments such as in the Middle East, where sand composition contains a relatively large concentration of salt. If the weapon system will not operate in such environments, then this paragraph and 4.3.1.3 should be labeled “Not Applicable.”

The following should be used for tailoring the specification paragraph:

(a): A value of at least 100 cycles.

(b): The number should not exceed 5 percent.

(c): The number should not exceed 5 percent.

Background:

Engine components should not exhibit evidence of failure or impending failure.

Engine corrosion will vary with environment and exposure time. Critical corrosion parameters are time at temperature and salt concentration. Materials may continue to corrode after exposure to salt laden air. There are many environments that are conducive to corrosion. Operation in proximity to salt water or salt laden air is one of the biggest causes of corrosion. The burning of fossil fuels causes air pollution and an acidic environment. Exposure time will depend on aircraft installation, mission, and physical location of the aircraft. Naval aircraft, particularly helicopters, are very susceptible to corrosion.

The corrosion of parts in the engine is a large contributor to increased maintenance and aircraft downtime. Some parts require special protective coatings (see 3.1.3.4, Corrosion protection) to make them corrosion resistant. For example, no corrosion resistant material could match the high strength–to–weight ratio of certain steel alloys. Therefore, a corrosion resistant coating needs to be used. These coatings must withstand extreme temperatures and thermal stresses as well as environmental effects. Refer to MIL–STD–1568 for selection of materials and processes for corrosion prevention and control.

The state–of–the–art of combustors is increasing the required hot section inspection to 1000 hours and longer. As a result, engine inspection periods are longer with less possibility of uncovering a corrosion problem in the engine. Therefore, there is a need for ensuring engine parts do not corrode for longer periods of time.

Corrosion may take place while the engine is operating or when the aircraft is parked on the ground in a static condition. Cold corrosion continues long after the engine is removed from the corrosive environment. Hot corrosion is recognized as a form of high temperature oxidation caused by sodium sulfate from sea water, condensing on turbine blades and vanes. Hot corrosion or sulphidation will only take place under the proper condition of temperature (760 – 980°C (1400 – 1800°F)) and hot gas pressure. These are metal temperatures, gas path temperatures will be higher. Approximation can be made for the metal temperatures, but care must be used since some hardware is air cooled. The engine parts should be resistant to both types of corrosion.

The Using Service should look into the aviation 3M database for current estimates of annual costs of corrosion induced maintenance actions to get a better feel for the severity of corrosion on engines of various applications.

REQUIREMENT LESSONS LEARNED

A COMNAVAIRLANT message in 1974 stated: “exorbitant maintenance man–hrs, 2.5 million documented in FY73, on corrosion control and are barely holding our own against deterioration effects.” A COMNAVAIRPAC message in 1974 also discussed the “major problem” of structural corrosion and cited huge man–hour expenditures on various aircraft for corrosion control and repair. This message also urged that the “best anti–corrosion materials/processes/methods, etc.” should be “specified in new hardware procurement contracts.” Two specific engine corrosion problems are on file: (1) a TF30–P–412A engine first stage fan turbine blade failed due to FOD at a crack which propagated rapidly under wet salt conditions and (2) T56 engine compressor second through thirteenth stage wheels were being scrapped at an increasing rate at overhaul because of corrosion pits in or near the dovetails. Also, the T56 had a history of turbine sulphidation problems and the T76 had magnesium gear box corrosion problems in the 60’s.

A point paper by LCDR M. Johns, NAVAIR, dated 8/10/90 stated that squadron HS–10 removed three SH–60F T700–GE–401C engines for low power. The cumulative operating hours were between 650 and 857 hours. Examination of disassembled engines revealed 1st stage GGT stage 1 blade tip and LE damage caused by hot corrosion.

ASME 70–GT–96 provided information on salt concentrations at various altitudes above salt water. J.C. Arribat presented information such as concentrations of 0.03 g/m^3 at 8 meters above sea water, with a total ingestion of 900 grams/hour for a helicopter was to be expected.

4.3.1.3 Corrosive atmosphere The requirements of 3.3.1.3 shall be evaluated by engine test, on a new or newly overhauled engine, in accordance with table X and figure 18. Prior to starting the corrosion susceptibility test, the engine shall be disassembled sufficiently to inspect the surface condition of all parts normally exposed to atmospheric conditions. Detailed photographic coverage of these parts shall be taken. Upon reassembly and after an initial calibration, including data required to establish compliance with transients of 3.2.1.2, the engine shall be subjected to the cycles of 3.3.1.3 for 24 hours each in accordance with table X. Should engine performance during the test deteriorate more than the amount specified in 3.3.1.3 from that determined during the initial calibration, water washing shall be accomplished in accordance with 3.7.14. Water wash shall be used only to correct for performance loss. External water wash is not permitted. If performance cannot be recovered after water washing, the engine shall be disassembled and inspected to determine the effect of the corrosion testing on performance loss. If the unrecoverable performance loss is determined to be caused by a problem not related to corrosion, the engine shall be repaired and reassembled and the test continued. During the test, the engine shall be subjected to internal inspections after every tenth cycle to detect any evidence of corrosion or progression of corrosion of internal parts. Additional inspection may be conducted with approval of the Using Service.

After completion of the corrosion susceptibility test, the cleaning procedure of 3.7.14, Wash system, shall be accomplished prior to recalibration of the engine. During recalibration the applicable transient performance specified in 3.2.1.2 shall be demonstrated. Following recalibration, the engine shall be disassembled and inspected for evidence of corrosion. Detailed photographs shall be taken of all parts which show evidence of corrosion. Metallurgical analyses that completely characterize the types of corrosion found and test specimen evidence shall be available for review.

The corrosion susceptibility test will be considered to be satisfactorily completed when upon recalibration: (1) the engine non–recoverable performance deterioration at steady–state measured temperatures or engine pressure ratio (EPR) at Maximum Continuous and above does not exceed the deterioration specified in 3.3.1.3, for the same value of measured temperature or EPR obtained during the initial calibration, (2) the engine exhibits not more than the increase in specific fuel consumption specified in 3.3.1.3, for the same values of measured temperature or EPR obtained during the initial calibration at Idle and above, (3) the transients conducted are in accordance with the specification requirements, and (4) the extent of corrosion evident at test completion satisfies the following criteria for all engine parts:

a. All internal parts exposed to gas path air, upon cleaning, shall show no impairment of their function due to corrosion. Minor corrosive attack is acceptable only when the part design criteria (e.g., fatigue resistance) are not affected.

b. All corrosion–protected parts shall indicate no effects of corrosion upon cleaning and stripping of the protective schemes. Recoating of parts designed for recoating shall restore such parts to as–new condition.

c. All other parts shall show no corrosion that affects component integrity or prescribed maintenance procedures.

EVALUATION RATIONALE (4.3.1.3)

Testing is required to evaluate the corrosion resistant properties of the engine.

EVALUATION GUIDANCE

If paragraph 3.3.1.3 is labeled “Not Applicable”, then this paragraph should be labeled “Not Applicable”.

Background:

An initial evaluation by analysis can be made of the materials used in the engine to determine if any are prone to corrosion. The engine contractor may want to demonstrate the corrosion properties of a material by the use of a “strip test”. It has been found that hot corrosion can occur in a wide range of temperatures 760 – 980°C (1400 – 1800°F). Therefore the engine should be operated over a wide range of turbine temperatures. This is to ensure all possible engine parts are exposed to the hot corrosion temperature range. A mission type of engine operation could be substituted for the generic temperature test if the range of temperatures are covered. A mission oriented test is designed to determine structural durability while a corrosion test is designed to determine the corrosion resistance properties of the engine under various environmental conditions in the shortest possible time. Conducting the test according to table X means conducting the test 24 hours a day, seven days a week. This is not always possible, because facility or maintenance problems can disrupt the test. The test schedule must be flexible to allow for any disruptions.

The previous corrosion test was designed in the sixties when engines were qualified with a 150–hour test. In the seventies the simulated mission type of engine operation replaced the 150–hour qualification test. The services and industry feel that the corrosion test should use some type of mission oriented engine operation. Engine life goals have increased over the years and it is felt that the operational phase of the test should be greater than 150 hours. Engine turbine temperatures have increased, so operating the engine at high power could reach the gas temperatures where sulphidation occurs.

The external air velocity was not specified in the previous test, only the salt concentration. A high air velocity across the external surface of the engine would deposit more salt than a low velocity air stream for the same period of time. A static engine may only be exposed to the outside winds. Therefore, a velocity of 4.6–6.1 m/sec (15–20 ft/sec) was chosen to represent an average wind velocity, which should not impose a large burden on a test facility.

On some corrosion tests conducted at NAWC AD TRENTON, the engine contractor requested the engine to be water washed if there was more than a 24–hour break in the test. This does not represent field operation of the engine. Fresh water is a valuable commodity on board aircraft carriers. It is rare to use it for water washing engines and external water wash is rarely practiced even on land bases. The test should simulate actual use of the engine so water wash should only be used to restore engine performance.

Dilution of the salt concentration is permitted to give better distribution and atomization of the salt spray when testing large engines with high air flows. Dilution has caused problems with the sodium monitor and the salt sampling instrumentation. On small engine tests, the dilution actually improved the performance of the engine. The 40:1 dilution is an arbitrary figure which is a compromise for the spray nozzle equipment used at NAWC AD TRENTON. Generating small droplets at very low flows is another challenge. Small droplet size is needed to effectively atomize the salt solution. The 25 micron droplet size is the smallest droplet size NAWC AD TRENTON equipment will generate. Smaller size droplets would be desirable.

The Army precedes their corrosion susceptibility test with two hours of sand ingestion in accordance with 4.3.2.4. This test is conducted on helicopter turboshaft engines with inlet filters or inlet particle separators to determine if the protective coatings are eroded.

EVALUATION LESSONS LEARNED

The corrosion test developed in the mid 60s for the general engine specification has uncovered numerous corrosion problems on such engines as the T58–GE–100, T400–CP–400, and YT700–GE–700. These tests have resulted in material changes to improve the life of engine parts in service.

3.3.1.4 Icing conditions The engine shall meet the requirements of this specification while operating in the meteorological conditions shown on figures 19a and 19b and table XI, with not more than (a) percent total loss in thrust/power and (b) percent total increase in specific fuel consumption at all operating conditions. After removal of the icing conditions, the engine shall perform the same as prior to encountering the icing conditions. No damage to the engine shall be allowed as a result of operating in the icing environment.

REQUIREMENT RATIONALE (3.3.1.4)

Service aircraft are intended to operate under all types of environmental conditions, therefore the engines must be capable of operating under conditions conducive to icing.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The number inserted should not be more than five percent.

(b): The number inserted should not be more than five percent.

Background:

Stratiform clouds occur at altitudes from 0.9 km (3,000 ft) to 1.8 km (6,000 ft). These clouds have a low to moderate liquid water content (LWC) and are characterized by figure 19a. Cumuliform clouds occur from 1.2 km (4,000 ft) to 7.3 km (24,000 ft) and have a LWC moderate to high. The characteristics of icing conditions in these clouds are depicted by figure 19b. Icing encounters in cumuliform clouds tend to be of relative short duration (about one minute). The icing conditions of this requirement are fairly consistent with encounters in cumuliform clouds, but the engine is required to operate for a much greater period than would be expected for these types of encounters.

Generally, some form of anti–icing system is incorporated into the engine to provide warm compressor bleed air to those surfaces which are susceptible to an ice buildup. These systems are intended to be turned on when an icing environment is encountered to prevent ice buildup on susceptible engine inlet components which could shed and cause engine failure, damage, surge, stall, or flameout. Ice buildup could also cause performance deterioration and/or turbine overtemperature due to air starvation and/or control schedule changes caused by engine sensor malfunctions.

Civilian aviation figures are located in Appendix C of FAR Part 25. Data sources for figures 19a and 19b are from NACA technical notes 1855 and 2569 published in 1949 and 1951, respectively. These data have been generally accepted as the aircraft/engine industry standard.

Most engines that have inlet struts or inlet guide vanes and a fixed bullet nose have been designed with an anti–icing system. There are engines that do not have inlet struts or inlet guide vanes and the air goes directly to the first stage blades. The bullet nose is part of the rotor and rotates with the blades. This type of engine usually does not have an anti–ice system and should not be given any special consideration but should be subjected to the same icing conditions.

REQUIREMENT LESSONS LEARNED

Early engine requirements called for altitude test in addition to sea level tests. The requirement for altitude tests was dropped, since it was felt that the most potentially serious icing problem of a jet airplane occurs at the ground and very low altitudes.

Navy Research Laboratory Report 9025 dated 12/30/86 concludes that some Navy aircraft have “shown worrisome susceptibility to ice damage” and others possess “a continuing history of icing related mishaps.” The report also cites a National Transportation Safety Board finding of 100 icing related accidents and 28 fatalities per annum in the general aviation community.

4.3.1.4 Icing conditions The requirements of 3.3.1.4 shall be evaluated by test as follows:

a. This part shall consist of two runs at each of several engine thrust/power settings under each of the conditions in Part 1 of table XI. The engine thrust settings shall be: Idle, 25, 50 and 75 and 100 percent IRP (Maximum Continuous). At each icing condition and at each thrust/power setting, the engine shall be operated for a period of not less than (a) minutes. During each period, at intervals after ice buildup, the engine shall be rapidly accelerated to Intermediate (Maximum Continuous) thrust/power to demonstrate acceleration response.

b. This part shall consist of a (b) minute run at Idle with no throttle movement, followed by an acceleration to Maximum thrust/power at the end of the period. During this run the engine shall be operated under the conditions listed in Part 2 of table XI. At each of the test thrust/power demand settings above, thrust/power shall be measured to evaluate performance loss, without any power lever movement.

If the engine incorporates an anti–icing system, the above tests shall be performed using the anti–icing system to demonstrate the requirements of 3.3.1.4.

EVALUATION RATIONALE (4.3.1.4)

Since aircraft are required to operate in icing conditions, there is a need to test the engine under icing conditions specified.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should establish the time based on mission analysis.

(b): The Using Service should establish the time based on mission analysis.

If altitude testing is desired as part of the test plan, the following is recommended to replace the sea level test in specification paragraph:

a. Runway hold and takeoff. This run shall be accomplished at an altitude of _____ m (_____ ft) or less at zero flight speed under table XI icing conditions. Run for _____ minutes at Idle power with maximum customer air bleed and then accelerate to and maintain Maximum _____ power for a minimum of _____ minutes.

b. Low altitude climb. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at Intermediate power setting with icing conditions established at Figure(s) 16 conditions for the first _____ followed by _____ minutes at figure(s) 19 conditions.

c. Low altitude descent. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at flight Idle power setting icing conditions established at figure(s) 19 conditions.

d. High altitude climb. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at Maximum Continuous power setting with icing conditions established at figure(s) 19 conditions.

e. High altitude descent. This run shall be accomplished at an altitude of_____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at Idle power setting with icing conditions established at figure(s) 19 conditions.

f. Low altitude loiter. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at Maximum Continuous power setting with icing conditions established at figure(s) 19 conditions.

g. High altitude loiter. This run shall be accomplished at an altitude of _____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at Cruise power setting with icing conditions established at figure(s) 19 conditions.

h. Low altitude landing loiter. This run shall be accomplished at an altitude of_____ m (_____ ft) at a flight speed equivalent to _____ Mach number, _____ ambient temperature. Run for _____ minutes at Idle power setting with icing conditions established at figure(s) 19 conditions followed by an acceleration to Intermediate/Maximum Continuous.

Background:

The Using Service, engine contractor, and airframe contractor should negotiate the blanks for flight conditions, thrust/power setting, and duration under which the engine should be tested.

Official anti–icing qualification tests are conducted at sea/level conditions in an icing tunnel where all icing conditions can be controlled. Some icing tests are conducted with the aircraft flying behind a tanker aircraft. The tanker sprays water at an altitude where the water will freeze. However, testing under these conditions is difficult to monitor and control.

For testing in an icing tunnel the following icing conditions should be used to demonstrate compliance with the requirements of 3.3.1.4. For this test, the engine should be operated under the free air conditions listed in table XI. For each test run, the liquid water content and droplet size, should be measured at a distance within five feet of the engine inlet face and still within the engine inlet duct. The liquid water content measured at this station should be corrected to the free air conditions as specified in table XI. This meteorological data should be recorded at suitable intervals during each test run. The method and procedure for collecting and determining the water droplet size and liquid content should be specified. The engine performance level (i.e., thrust or parameters correlated to thrust), speed and vibration should be continuously recorded during the test and high speed photographic coverage of the engine inlet should be provided. The base line for determining engine performance loss should be established by operating the engine with no customer bleed air or power extraction and under the inlet temperature conditions of tables Ia, Ib, and Ic with air between 80 and 100 percent relative humidity and zero liquid water content. The thrust/power loss and specific fuel consumption increase should be determined by comparison of engine performance when operating at the icing conditions defined in table XI with the aforementioned baseline values.

For engines that use oil as the anti–icing fluid the oil temperature should be maintained at the minimum operating oil temperature specified in the specification, or less, during all runs except at Idle condition where the oil should be maintained at the same temperature as the engine inlet air.

Engines that use compressor bleed air for anti–icing may have a variable capacity for anti–icing capability. At high power, the engine speed and pressure ratio are higher than at low power. This results in higher velocity and temperature air for anti–icing. Therefore more heat is delivered to the anti–ice surface at high power. This assumes a fixed flow area and a simple on/off valve to control the anti–ice air. Some engine contractors use a thermostatic valve which controls the anti–ice air flow. At high power it reduces the flow, while at Idle it is wide open. With time this type of valve tends to corrode and stick in the closed position.

Although the test conditions cannot be justified by actual data, they have resulted in reliable engines in service.

EVALUATION LESSONS LEARNED

Some contractors have interpreted that 5.0 percent power loss should only be applied to the Intermediate thrust/power demand setting since this is the highest available thrust condition. The contractor’s reasoning is that at any part power the throttle could be moved forward if the anti–icing/icing condition caused a thrust loss at the original throttle setting. In the case of a turboprop or turboshaft engine that runs at constant speed, the loss in power is automatically compensated by an increase in turbine temperature. These engines are not always run at Maximum power. Therefore, they have excess power to make up for small loses in power.

The T700–GE–400 Engine Model Specification E1221 permitted a loss of 8.5 percent delivered shaft horsepower available for conditions of figures 19a and 19b and a 5.5 percent loss relative to table XI at operating conditions above 50 percent Maximum Continuous power setting. The specification also provides a lengthy discussion relative to an actual power restriction and a time restriction on the engine, depending on the original power.

The F404–GE–400 Engine Model Specification CP45K006 permitted a “10 percent total loss in thrust commanded” and a 10 percent increase in fuel consumption above 50 percent Maximum Continuous power setting.

In 1959 the Navy conducted its first icing tests at NAPTC, Trenton, N.J. The F3H–2 aircraft powered by the J71–A–2E engine experienced a number of accidents when flying through rain or icing conditions. The test found that ice blockage of the PT2 sensor affected the fuel control.

In 1961 the Navy tested the J57–P–20 engine to determine its suitability for all weather application. On this engine only the inlet struts were anti–iced. The bullet nose was not supplied with anti–ice air. An alternator and constant speed drive was located in the bullet nose. The heat given off by these units was supposed to anti–ice the bullet nose. The tests showed that sufficient heat was not produced to prevent ice formation.

In the test of the J60–P–3A engine a thrust loss greater than the allowable 5% occurred at the lower engine speeds. The loss was as high as 23% after 5 minutes in icing conditions.

In 1966 testing of the J52–P–6A engine in icing conditions resulted in modification to the A–4E and A–6A aircraft flight manuals. The pilot was warned to use high power when entering icing conditions.

Tests of the J79–GE–15 engine found that modification to the anti–icing system was necessary to supply sufficient heat to the inlet guide vanes of the engine.

In 1968, the TF30–P–8 engine was the first turbofan engine tested under icing conditions at NAPTC. The tests revealed inadequate anti–icing for the inlet guide vanes and the inlet total pressure sensor.

An F404–GE–400 engine was flight tested in an F/A–18 aircraft that flew behind a tanker that sprayed water on the engine inlet duct. Ice accumulated on the inlet duct lip at the same flight conditions where field experience showed ice ingestion problems. When the aircraft pulled away from the tanker, ice shed off and caused engine damage that represented field experience.

3.3.2 Ingestion capability (hazard resistance)

4.3.2 Ingestion capability (hazard resistance)

3.3.2.1 Bird ingestion The engine shall continue to operate and perform during and after the ingestion of birds as specified in table XII.

REQUIREMENT RATIONALE (3.3.2.1)

Engines must be capable of ingesting birds encountered during missions without significant power loss, deterioration, or safety implications. The total weapon system mission environment must be studied to examine the probability of bird strike occurrence, bird sizing criteria, flocking densities, mission routing, training, etc., to determine the design criteria for bird ingestion capability requirements for an engine.

REQUIREMENT GUIDANCE

The following should be used to tailor table XII:

In the event specific weapon system bird strike criteria has not been established for the engine, the following birds vs inlet area criteria should be used. The inlet area to be used should be the aircraft inlet or engine inlet whichever is smaller.

The number of birds to be ingested should be based on inlet area as follows: one 100 gm (3.5 oz) bird per 300 cm^2 (46.5 in^2) of inlet area plus any fraction larger than 50 percent thereof, up to a maximum of 16 birds; one 1 kg (2.2 lb) bird per 1500 cm^2 (232.5 in^2) of inlet area plus any fraction larger than 50 percent thereof; one 2 kg (4.4 lb), regardless of the size of the inlet, provided the inlet is large enough to admit a 2 kg (4.4 lb) bird.

The 100 gm (3.5 oz) birds should be ingested at random intervals and be randomly dispersed over the inlet area. Birds 1 kg (2.2 lb) and larger should be directed at critical areas of the engine face.

The bird velocity and engine power setting for each condition should be as described below:

a. Birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb) (one at a time) ingested at a bird velocity equal to the takeoff flight speed, with the engine at Maximum takeoff thrust/power.

b. Birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb) (one at a time) ingested at a bird velocity equal to the cruise flight speed with the engine at Cruise power setting.

c. Birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb) (one at a time) ingested at a bird velocity equal to the descent flight speed with the engine at descent power.

d. For aircraft that have a low level, high speed mission requirement: birds weighing 100 gm (3.5 oz) (a maximum of sixteen at a time) and birds weighing 1 kg (2.2 lb) (one at a time) ingested at a bird velocity equal to the aircraft maximum sea level speed and the engine power setting required to achieve that speed.

e. Birds weighing 2 kg (4.4 lb) ingested at a bird velocity equal to the aircraft takeoff speed or low level operational airspeed, whichever is more severe, with engine power equal to that required for the flight condition.

For 100 gm (3.5 oz) birds, the engine should sustain performance 95 percent or greater of the initial thrust/power, and all damage to the blades and vanes should be blendable (within repair limits) with flight line type tooling. The 1 kg (2.2 lb) bird ingestion may cause some damage; however, it should not result in immediate engine shutdown, and post ingestion thrust/power levels should be 75 percent or greater of the initial thrust/power at the operating condition. Under condition e. above, no engine failure should occur which would result in damage to the aircraft or adjacent engines. No bird ingestion should prevent the engine from being safely shutdown.

Performance recovery times will vary as a function of the bird size, number of birds, and size of the engine. The performance recovery time after ingestion of the 100 gm (3.5 oz) bird(s) should occur in 5 seconds or less after the final volley of birds has been ingested. The performance recovery time after ingestion of the 1 kg (2.2 lb) bird(s) should occur in 5 seconds or less for small engines, 5–10 seconds for moderate sized turbofans/turbojets, and up to 5–15 seconds for large by–pass turbofans.

Turboshaft engines should recover within 5 seconds after an ingestion event with no less than 95 percent of the power prior to the ingestion event, and without exceeding any engine control limits.

Background:

Bird strike durability is a necessary safety design criteria to be used in all engines. Analyses should be conducted to determine the sensitivities to blade design, blade to stator spacing design, control system operation, and stator design for structure performance, and engine control. Bird strike tolerance can be enhanced by providing large axial clearances between blade and stators at the front of the engine.

An Aviation Week & Space Technology article, dated 14 September 1992, page 76, stated that a Rolls–Royce Trent test engine had ingested seven birds weighing up to 8 pounds, impacting the engine's wide cord blades. The blades showed generally minor plastic deformation but no loss of material. The bird strike requirement probably comes from the European Joint Aviation Authority.

REQUIREMENT LESSONS LEARNED

Results of findings for bird ingestion experience assembled by Aerospace Industry Association, "Bird Ingestion Experience for Aircraft Turbine Engines," 1979, generally supports the FAA part 33.77 criteria. Similar military studies have indicated that the engine may receive over 60 percent of the total aircraft strikes, and most strikes occur below 1.8 km (6000 feet) at takeoff, landing, and low level penetration speeds.

A GAO/NSIAD–89–127 report dated July 1989 states that from 1983 to 1987, military aircraft have collided with birds over 16,000 times. Many of these collisions caused only minor damage; however, the services lost six crew members, incurred $318 million in damages, and lost nine aircraft. During this period the Air Force lost six aircraft, the Navy lost two aircraft, and the Army lost one aircraft. A review of nine military jet engines developed since the early 1970's showed that the services lessened the MIL–SPEC requirements in engine model specifications for the sizes and the numbers of medium birds used in testing engines.

Recent studies (USAF/ASC & AIA) of bird ingestion data indicate that older specifications did not accurately reflect the sizes and the number of birds actually ingested. Prior to the publication of this specification, engine MIL–SPECS required medium birds sizes for ingestion tests of 1.5 pounds for the Air Force and 2 pounds for the Navy.

DOT Report DOT/FAA/CT–84/13 by Gary Frings, dated 9/84, states bird ingestion is a rare but probable event. For every one million aircraft operating hours, 230 bird ingestions (of all weights) will occur, on average. An average bird weighs 26 ounces. Most likely weight of birds in the areas of runways is 11 ounces. A small percentage (<3%) of ingestion events involved birds weighing 61 ounces. Ten percent of events involved ingestion of multiple birds into a single engine. One–half of one percent of events involved multiple engine ingestions. Five percent of all bird ingestions resulted in engine failure.

4.3.2.1 Bird ingestion The requirements of 3.3.2.1 shall be evaluated by analysis, demonstration, and test. The contractor shall specify in the pretest data the critical target area for bird ingestion.

EVALUATION RATIONALE (4.3.2.1)

A test with a complete engine is necessary to ensure all interactive effects of a bird strike are properly accounted for in the engine design. Analysis and component test (fan and/or compressor rigs) have been successfully used to predict loading and deflection criteria leading up to full–scale engine tests.

EVALUATION GUIDANCE

Background:

A development test program consisting of analysis and component test should be established to provide confidence in the design prior to formal engine ingestion tests. The number of birds, bird sizes, engine RPM's, bird velocities, performance criteria and amount of damage for the test engine should be based on the parameters of 3.3.2.1. Analysis should indicate the location for critical bird strikes at the front face of the engine and pass/fail criteria should be included in the test plan. The birds should be ingested in a random sequence and dispersed over the inlet area to simulate an encounter with a flock. Synthetic "birds" have not been used by the military services to date, but their use should not be precluded in future programs if sufficient information and justification is presented to the Using Service.

EVALUATION LESSONS LEARNED

Severe out–of–balance conditions creating high vibrations, along with rotating to static structural contact, have been encountered following bird ingestion events.

3.3.2.2 Foreign object damage (FOD) The engine shall meet the requirements of the specification for (a) inspection intervals of 3.4.1.7.4 without repair after ingestion of foreign objects which produce damage equivalent to a stress concentration factor (K_t) of (b) at the most critical locations of flow path components.

REQUIREMENT RATIONALE (3.3.2.2)

The high inlet air flow velocities associated with turbojet/turbofan engines create conditions where loose foreign objects may be ingested into the engine resulting in gas path damage. The engine must tolerate the ingested objects up to a specified level.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): A value of at least 2.

(b): A value of at least 3.

Background:

The two inspection interval requirement should agree with maintenance inspection and repair cycles requirements.

The stress concentration factor of 3 is similar to a notch in the fan/compressor blade or stator caused by impact damage from a small object.

Typical foreign objects to be considered are nuts, bolts, rivets, rocks, aircraft parts, shell casings, and tools. Therefore, structures subject to this type of damage must be capable of operating to the next subsequent depot interval to avoid immediate teardown when damage is detected and determined to be within acceptable limits.

Design criteria is required to establish capability of both fracture critical parts and durability critical parts to operate with damage present.

The Using Service should look into the aviation 3M database for current estimates of annual costs of FOD induced maintenance actions.

REQUIREMENT LESSONS LEARNED

Blades insufficiently designed for FOD considerations have caused in–flight shutdowns (IFSDs) and/or costly damage to the engine. FOD tolerance can be enhanced by use of more damage tolerant materials and a redistribution of the blade mass.

Foreign object damage is costing the Navy over $15M each fiscal year based on a conservative estimate in one source reference. Common culprits are “housekeeping” items, such as nuts, bolts, safety wire, screwdrivers, etc. This paragraph is aimed at this type of FOD problem (environmental factors such as ice, sand, water and birds, are covered in other specification paragraphs).

4.3.2.2 Foreign object damage (FOD) The requirements of 3.3.2.2 shall be evaluated by analysis, demonstration, and test.

Simulated foreign object damage shall be applied to the (a) critical stage blades at one or more sections of the (b) of the airfoil. The damage applied shall produce at least the stress concentration factor (K_t) of 3.3.2.2. Following the foreign object damage application, the damaged blades shall be tested to the life required in 3.4.1.5.1 and 3.4.1.5.2. At the completion of the test there shall be no evidence of blade failure or flaw sizes beyond values allowed by the in–service inspection flaw size of 3.4.1.7.3 as the result of the foreign object damage. Sufficient instrumentation for monitoring the structure of the engine shall be included in the test engine.

EVALUATION RATIONALE (4.3.2.2)

Analysis, demonstration, and test are required to ensure that the fan and compressor airfoils can meet the operational requirement of 3.3.2.2 and to establish accept/reject criteria for damage that is detected during flight line inspections.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The three first stage blades.

(b): The leading edge, or at stage(s) and blade location(s) where the highest stresses of 3.3.2.2 occur.

Background:

The locations selected for simulated FOD should be those most sensitive to FOD. Typically, these location are where a combination of steady–state and vibratory stress combine to make the lowest fatigue life and is known as the critical location. The addition of a stress riser, resulting from FOD, at this critical location further degrades the life of the airfoil. The engine test cycle should be in accordance with the endurance test. A successful engine test will demonstrate that the engine design is robust enough to safely meet the test requirements.

It is recommended that LCF and residual life analysis of 4.4.1.5 and 4.4.1.7, respectively be reviewed to ensure proper blade and stage locations are used in the test.

EVALUATION LESSONS LEARNED

Past engine programs have shown that FOD ingestion and subsequent engine repair is a problem aboard aircraft carriers, because it requires engine module rework or extra maintenance to blend out damage.

3.3.2.3 Ice ingestion The engine shall operate and perform in accordance with table XIII, during and after ingestion of hailstones and sheet ice at the take–off, cruise, and descent aircraft speeds. The engine shall not be damaged beyond field repair capability after ingesting the hailstones and ice.

REQUIREMENT RATIONALE (3.3.2.3)

Sufficient structural capability is needed to tolerate ingestion of environmentally generated ice (hailstones), ice shed from the airframe, and/or engine inlet generated ice.

Particles or chunks of ice can be dislodged or break off of inlet duct components such as cowl lips, boundary layer bleed wedges, inlet accessory covers, variable inlet spikes, etc., and cause compressor damage.

REQUIREMENT GUIDANCE

The following should be used for tailoring table XIII:

For inlet capture area of 0.065 m^2 (100 in^2) the engine should be capable of ingesting one 25 mm (1.0 in) diameter hailstone. For each additional 0.065 m^2 (100 in^2) increase of the initial capture area (0.065 m^2 (100 in^2)), supplement the first hailstone with one 25 mm (1.0 in) and one 50 mm (2.0 in) diameter hailstone. The engine shall be capable of ingesting a sheet of ice 75x230x6 mm (3x9x0.25 inches) or sized proportionally to the circumference of the inlet. Within 5 seconds after an ice ingestion event, the engine thrust/power should be at least 95 percent of the thrust/power immediately prior to the event. The hailstones and sheet ice should be between 0.80 g/cm^3 (50 lbs/ft^3) and 0.90 g/cm^3 (56 lbs/ft^3) specific gravity. Hailstones should be ingested at typical takeoff, cruise, and descent conditions. Sheet ice should be ingested at typical takeoff and cruise condition.

Background:

All–weather aircraft/engines should be designed to withstand potential hail conditions. An engine should also have some capability to ingest, without major damage, chunks or sheets of ice which may dislodge or break off of inlet duct components, such as cowl lips, inlet ramps or doors, inlet accessory covers, and airframe surfaces.

REQUIREMENT LESSONS LEARNED

Engines that have inlet guide vanes and lack adequate anti–icing/deicing provisions have had operational restrictions and/or increased maintenance workloads when exposed to icing conditions. The problems were eliminated by providing adequate anti–icing systems that reduced inlet case ice accumulation and resultant fan blade damage. Whereas the damage from inlet case ice shedding is a maintenance problem (blending and replacement of fan blades), ice shed from the airframe or environmentally generated ice can be a safety of flight concern if sufficient structural capability in the engine design is not provided.

Numerous incidents of axial–flow compressor damage caused by ice ingestion are on record. Some of these resulted in complete engine failure and disintegration of the engine.

The TF30–P–412 engine, on the Navy's F–14 aircraft, passed its FSD icing condition requirement by similarity. However, during fleet operations, engines experienced damage from ice ingestion events. The damage occurred when ice, formed on the inlet lip and duct walls during flight, was dislodged and ingested during descent in warm weather and/or during hard carrier landings.

Naval Research Laboratory Report 9025 dated 30 December 1986, states that about 525 mishaps related to ice ingestion have occurred on U.S. Navy aircraft from 1964 through 1984. It mentions that the F404–GE–400 (F/A–18 aircraft) often sustained compressor blade damage that was substantial or required an overhaul. The F404–GE–400 had not been tested for ice ingestion during qualification tests so the effects of ice ingestion were unknown when the engine entered service. It became apparent in field deployment that ice ingestion was a concern and this may have been determined earlier if an ice ingestion test had been conducted during development of the engine.

4.3.2.3 Ice ingestion The requirements of 3.3.2.3 shall be evaluated by analysis, demonstration, and test. The test procedure shall require the engine to run for (a) minutes following ice ingestion, before it is shut down for inspection. During the ingestion test, high speed photographic coverage shall be taken. Sufficient instrumentation for monitoring the structure of the engine shall be included in the test engine.

EVALUATION RATIONALE (4.3.2.3)

The engine icing test is needed to evaluate satisfactory engine performance in icing conditions throughout the aircraft flight envelope.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): A value of at least five minutes.

Background:

The procedures to be used for introduction of the hailstones and sheet ice at the engine inlet and the engine power settings and speed at which the hailstones and sheet ice are to be ingested should be specified in the pretest data. The temperature of the ice should be between –2.2 to 0°C (28 – 32°F). The test procedure for sheet ice ingestion should require the most severe ice velocities representing ice shedding off the aircraft inlet lip.

EVALUATION LESSONS LEARNED

After undergoing icing tests behind a tanker, an F/A–18 F404 engine sustained severe damage from ice when the aircraft descended to a warmer altitude and inlet ice was ingested by the engine.

3.3.2.4 Sand and dust ingestion The engine shall meet all requirements of the specification during and after the sand and dust ingestion event specified herein. The engine shall ingest air containing sand and dust particles in a concentration of (a) mg sand/m^3. The engine shall ingest the specified course and fine contaminant distribution for

_(b)_ and _(c)_ hours, respectively. The engine shall operate at Intermediate/Maximum Continuous thrust/power with the specified concentration of sand and dust particles, with no greater than _(d)_ percent loss in thrust/power, and _(e)_ percent gain in specific fuel consumption.

REQUIREMENT RATIONALE (3.3.2.4)

The operation of aircraft in sand and dust environments can result in serious erosion damage to engine parts. Sand and dust particles are highly abrasive and tend to erode the thin metal tips and trailing edges of gas turbine engine compressor blades and vanes.

Sand and dust ingestion is also needed to determine the effect on surface coatings.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

CONCENTRATION GUIDANCE			
	Fix–wing Aircraft	Helicopter	VSTOL
(a): Concentration (mg sand/m^3 air) Time (hours)	350	53	350
(b): Coarse sand	1	50	5
(c): Fine sand	1	54	5
(d): Thrust/power loss (%)	5	10	5
(e): SFC gain (5)	5	15	5

PARTICLE SIZE DISTRIBUTION GUIDANCE		
	Particle Size (Microns)	Cumulative % by Weight
Coarse sand*	1,000	100
	707	95–99
	500	89–93
	354	77–81
	250	60–64
	177	38–42
	125	18–22
	88	6–10
	63	1–5
Fine sand**	1,000	100
	500	85–90
	250	70–75
	125	50–55
	75	25–30
	<75	10–15

* Composition is crushed quartz (SiO_2).

** Composition is 60% quartz (SiO_2), 26% gypsum (hydrated $CaSO_4$), 12% calcite ($CaCO_3$), and 2% soluble salts. Soluble salts shall contain the identical weight fraction of the salts from table X. Soluble salts shall all be <75 microns in particle size.

Background:

This requirement is based upon a severe but realistic potential service ground environment. It is recognized, however, that the time the engine is subjected to the concentration of sand and dust particles is quite dependent upon the particular engine and airframe contribution.

The coarse/fine sand and dust particle size is representative of field deployment where pilots will train in the United States desert areas and be sent to the middle east desert as in Operation Desert Shield/Desert Storm.

Large particles are more likely to cause erosion on airfoils (blades/stators) while small particles are more likely to block cooling holes in the turbine and cause corrosion. The SiO_2 in both particle size distributions are likely to melt in the combustor discharge gases and be deposited on first stage turbine nozzle vanes. An evaluation should be made to determine the effect of different sand compositions on the combustor/HPT at elevated temperatures.

Notes on sand/dust concentration in guidance:

(1) The thrust/power loss above should be evaluated at constant turbine temperature for all engine classes. If evaluating at constant commanded power setting, the engine thrust/power loss will be less. This is because engines commanding constant fan speed tend to increase thrust as components deteriorate rather than decrease. SFC loss should be evaluated at constant thrust/power output.

(2) The ratio of thrust/power loss to SFC loss varies with engine cycle. The ratio of shaft power loss to SFC loss is as high as 3:1 for typical helicopter engines. The relative losses can be evaluated by calculating performance with changes in compressor efficiency.

Recent Army helicopter engine specifications have all used “C–Spec” coarse sand with a concentration of 53 mg/m^3, except for T55 which was 707 mg/m^3. The T800 specification added an additional 54 hr test for “AC Fine” sand. The T406 specification uses a sand finer than “AC Fine”. The usual time duration for the sand test is 50 hours, except for T406 which is 20 hours.

The deterioration factors for power and SFC loss may be approximately the same for higher pressure ratio engines. Helicopter engines, with lower pressure ratios, experience power loss more rapidly than SFC loss (at constant T41). The factors may be more equal for higher pressure ratio engines.

Coarse sand produces blade erosion and a near term loss of compressor efficiency. Fine sand causes plugging of cooling holes, deposits, and long term engine distress. Performance losses from fine sand are not immediate and may not be measurable during the duration of a sand test. Fine sand tests should be conducted to determine the engine’s vulnerability to airhole plugging and nature of deposits. Deposits formed also depend upon the composition of the sand.

Most of the arid regions of the world correspond to C–Spec coarse sand in size except regions in the Middle East. Saudi sand, in general, is close to C–Spec in size, very sharp particle shape, with high aluminum content. When mixed with water (i.e., waterwash), Saudi sand may form cement, which, when heated, may form porcelain. Saudi fine sand has been encountered as high as 15000 feet. Very fine sand is found in parts of Israel and some other nearby regions. The regions with very fine sand are not conducive to large scale military operations and may be insignificant for design purposes.

The performance loss due to coarse sand ingestion depends upon the total quantity of sand ingested, particle size, and engine power setting. Quantity of sand (per unit size of the engine) depends upon sand concentration in the air and duration of the test. (53 mg/m^3 for 50 hrs may be equivalent to 530 mg/m^3 for 5 hrs). Particle size ingested varies rapidly with hover height, for clouds created by rotor downwash. Particle size is largest near the surface and decreases rapidly with height. Power setting affects the effectiveness of the IPS (inlet particle separator). The IPS is most effective at high power settings where high airflows result in high velocities. Filtration is needed to be effective at all

power settings or with the finer particles. A helicopter on the ground at low engine power settings could conceivably experience more sand damage than when hovering a few feet above the ground at high power. A comprehensive sand test should; therefore, consider sand concentration level, composition, particle size in the bed, particle size in the cloud, hover height, time duration, and power setting. The present sand test (constant concentration, time, and power setting regime) forms a good benchmark for engine vulnerability to sand erosion, but may not be indicative of actual field experience.

While this requirement paragraph is oriented around thrust/power/SFC performance, it is stated that the engine has to meet all requirements of the specification. This includes life and performance effectiveness of special technology materials and features on the engine inlet/front face which may be vulnerable to sand erosion.

REQUIREMENT LESSONS LEARNED

The operation of aircraft in sand and dust environments has resulted in serious erosion damage to engine parts. Sand and dust particles are highly abrasive and tend to erode the thin metal tips and trailing edges of gas turbine engine compressor blades and vanes. Helicopter operations are most significantly impacted by the sand and dust problem, although fan blade leading edge damage has been experienced on the TF30 engine in the F–14 aircraft as a result of operation “near sandy, dry desert–like areas such as NAS Miramar”. One source pointed out that the “accelerated replacement of erosion damaged helicopter turbines in Southeast Asia was estimated to cost the U.S. Government about $150 million a year” (during the Vietnam war period). The finer particles of sand and dust have been known to clog turbine cooling passages and cause engine failures. Sand and dust has been encountered several hundred miles out at sea as well as over land masses.

Sand has been determined to be more detrimental during foreign operation due to the differences found in the sand composition. A lower melting point has been observed in foreign sand test samples which allows the sand to melt and adhere to the turbine airfoils. Spalling of the coating and base metal then occurs.

Due to the unexpected turbine blade failure and short life encountered in engines with gas temperatures in excess of 2000°F, special consideration in blade design should be taken into account. The soil analysis of Saudi Arabia and other middle eastern countries reveals that very high amounts of sulphur, calcium, and magnesium exist. A study has shown that an HPT blade failure occurred because of excessive sulphidation. At high temperatures, a flux or coating of calcium sulphate and magnesium sulphate attaches to the blade, attacks the base metal, and erodes it, causing failure.

Sand and dust ingestion and subsequent performance is a primary cause of engine removal on helicopters.

The following references are provided for the reader:

Particle distribution: “Development of the Lycoming Inertial Particle Separator”, H.D. Conners, presented at the Gas Turbine Conference & Products Show, Wash. DC, 17–21 Mar 1968.

Particle densities: ASME 70–GT–96, J.C. Arribat, ASME Gas Turbine Conference, Brussels, Belgium, 24–28 May 1970.

Particle distribution: SEA AIR 947, Issue 1, 2/71.

Blade damage including blade and vane erosion, secondary airflow deposits resulting in power reduction and stall margin loss. ASME 68–GT–37, G.C. Rapp and S.H. Rosenthal, presented at Gas Turbine Conference & Products Show, Wash. DC, 17–21 Mar 1968.

Particle distribution: “Evaluation of the Dust Cloud Generated by Helicopter Blade Downwash”, Sheridan Rogers, MSA Research, Proceedings of the 7th Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems, 25–27 Sep 67. Single helicopter dust concentration reached 16.2 mg/cu ft, takeoff and approach reach 40 mg/cu ft. Two helicopter levels of 64 mg/cu ft were reached.

Particle distribution: Kaman Report No. R–169, "Amount of Dust Recirculated by a Hovering Helicopter", dated 26 Dec 1969. Densities ranged from .29 gm/cu ft to 26 gm/cu ft.

4.3.2.4 Sand and dust ingestion The requirements of 3.3.2.4 shall be evaluated by analysis, demonstration, and test. During the engine test, the coarse sand and dust shall be ingested first, with the fine particle sand and dust ingested afterward. An engine disassembly and inspection shall be conducted between the coarse and fine sand tests as specified by the Using Service.

For turbofan and turbojet engines, the engine shall be tested at Intermediate thrust, with sand and dust ingested at the concentration levels and for the length of time specified in 3.3.2.4. During each hour of operation, at least one deceleration to Idle and acceleration to Maximum augmentation shall be made, with power lever movements of 0.5 seconds or less. If an anti–icing system is provided, ten periods of one minute operation of the anti–icing system shall be performed during the first test hour. During the entire test, maximum customer bleed air shall be extracted from the engine. The customer bleed air shall be continually filtered and the total deposits measured and recorded. Following the post–test performance check, the engine shall be disassembled to determine the extent of erosion, and the degree to which the contaminant may have entered critical areas in the engine. The test will be considered satisfactorily completed when the criteria of 3.3.2.4 have been met and the teardown inspection reveals no failure or evidence of impending failure.

For turboshaft and turboprop engines, the engine shall be tested at Maximum Continuous with sand and dust ingested at the concentration levels and the length of time specified in 3.3.2.4. With the coarser sand, the engine shall be tested for 50 hours at Maximum Continuous power. With the finer sand, the engine shall be tested for 9 hours at Maximum Continuous, 27 hours at Intermediate, and 18 hours at Maximum, for a total of 54 hours. The engine shall be tested with the IPS (Inlet Particle Separator) if it is an inherent part of the engine design. Engines with an IPS should perform additional testing at reduced power settings to determine the effectiveness of the separator and engine vulnerability to sand and dust. During each hour of operation, at least one deceleration to Idle and acceleration to Maximum Continuous/Intermediate/Maximum shall be made, with power lever movements of 0.5 seconds or less. If an anti–icing system is provided, ten periods of one minute operation of the anti–icing system shall be performed during the first hour. During the entire test, maximum customer bleed air shall be extracted from the engine. The customer bleed air shall be continually filtered, and the total deposits measured and recorded. If an engine internal washing system is provided, it shall be demonstrated during the test. Following the post–test performance check, the engine shall be disassembled to determine the extent of sand erosion, and the degree to which sand may have entered critical areas in the engine. The test will be considered satisfactorily completed when the criteria of 3.3.2.4 have been met and the teardown inspection reveals no failure or evidence of impending failure.

EVALUATION RATIONALE (4.3.2.4)

The effect of sand ingestion on the performance, bleed air quality, and internal air cooling system can only be evaluated by engine test.

EVALUATION GUIDANCE

Background:

This test could be combined with the AMT. If combined, the sand test should occur after post–AMT recalibration and endurance test completion, due to performance degradations of the engine.

The recommended text decreases the operational time in the extreme sand and dust environment from ten hours to two hours for turbofan/turbojet engines. Engine contractors have been unwilling in the past to guarantee their engines for ten hours (helicopter subjected to the severe Vietnam sand and dust environment typically used inlet filtration systems). Although the recommended text has reduced the time to two hours, the increased concentration in the recommended text actually imposes an even more stringent requirement on the engine. Undoubtedly, the time requirement will have to be negotiated with each engine contractor in specific future specification negotiations.

The sand concentration should be calculated with customer bleed air extraction. The A/I switch should be activated five times during each hour of sand ingestion at equally spaced intervals. The test should be conducted with a thrust bed and load cell measurement of thrust in lieu of calculating thrust by EPR. Disassembly and inspection between the coarse and fine sand tests should be conducted for 100 lb/sec air flow or smaller engines.

EVALUATION LESSONS LEARNED

None.

3.3.2.5 Atmospheric liquid water ingestion The engine shall start and operate satisfactorily with (a) percent of the total airflow weight in the form of water (liquid and vapor), with (b) percent of the liquid water entering the inlet through a segment equivalent to one–third of the inlet area.

REQUIREMENT RATIONALE (3.3.2.5)

Large amounts of water in the engine inlet affect engine performance, resulting in stall, rotor speed rollback, main or afterburner combustor blowout. Two sources of water are rainstorms and aircraft nose wheel spray generated when passing through standing water on runways during takeoff and landing.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of at least 5 percent.

(b): A value of at least 50 percent.

Background:

Ingestion requirements which concentrate half of the liquid water in one–third of the inlet area are an attempt to simulate the concentration that often occurs because of installation effects.

Consistent with the requirement for all–weather aircraft, the engine must be capable of operating in severe rain conditions without flaming out or stalling. There is also a potential for water to be ingested during takeoff and landing situations where standing water exists on the runway surface.

As of December 1992, the FAA was at the very beginning of the regulatory process to change this requirement (Part 33) and may take years to establish new regulations.

The 5.0 percent water–to–air mixture tolerance requirement for engines is somewhat arbitrary, although MIL–STD–210 supports it with data for the highest recorded rainfall in a one minute period at sea level (1.23 in/min (6.4 percent water–to–air mixture)). At 19,700 ft, the highest recorded rainfall rate for one minute was 1.7 in/min (17.2 percent water–to–air mixture). The water–to–air ratios are reduced significantly, however, when the 0.5 or 0.1 percent extremes are considered in MIL–STD–210. For example, at 19,700 ft a water–to–air mixture of 2.6 percent will be exceeded on an average of only 0.1 percent of the time in the rainiest area of the world during the rainiest month.

Although the probability of exceeding a 5.0 percent water–to–air mixture on a worldwide basis is extremely low, a finite probability does exist. The extreme values in MIL–STD–210 represent those recorded officially at weather stations throughout the world – it is likely that higher rainfall rates have occurred at locations where weather stations do not exist. Another interesting point is that, “It is unlikely, however, that one minute precipitation rates from storms elsewhere will greatly exceed rates measured in the United States since it is the world center of intensely precipitating thunderstorms.”

Tabulated rainfall information was obtained from MIL–STD–210 for the worldwide extreme and 0.1 percent extremes between sea level and 45,900 ft altitude. The worldwide extreme water–to–air mixture is 10.5 percent at 45,900 ft and 1.9 percent for the 0.1 percent extreme.

REQUIREMENT LESSONS LEARNED

Several aircraft have encountered engine water ingestion problems in ground operation during takeoff or landing through water. Problems were reduced through the use of nose wheel chine tires or nose wheel deflectors.

Incidents have been recorded where engines were damaged or aircraft were lost because of water ingestion. Fuselage runoff water supposedly drained into the J71 engine inlets which led to the loss of several F–3H aircraft. In 1972, an F–111A aircraft experienced a dual engine stall (TF30–P–3 engine) due to "severe precipitation" at 16,500 ft. Another F–111A experienced engine damage from runway water ingestion. Water ingestion was suspected as the possible cause of a TF30–P–412A engine stall in an F–14A on 11 Sept 1976 (runway water).

In a paper entitled "Engine/Airframe Interface Considerations for Future Open–Ocean ASW Aircraft", G. Ward, LTV Aerospace Corp., Proceedings of the 7th Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems, 25–27 Sep 1967, the presenter recommended that military aircraft engines be designed to operate satisfactorily with ingested water levels up to 5% of total airflow.

4.3.2.5 Atmospheric liquid water ingestion The requirements of 3.3.2.5 shall be evaluated by analysis, inspection, and test. The engine shall start and operate satisfactorily from ground Idle to Maximum power with water ingested at varying rates of (a) to (b) percent with no airbleed, (c) to (d) percent with air bleed, and with a minimum of (e) percent (up to the limit of the engine) of total inlet mass flow in the form of water entering the engine inlet through a segment equivalent to one third the inlet area. Idle–max–Idle throttle transients shall be conducted at each test condition. The engine shall be operated at each condition for five minutes. During the test, the effects of the water ingestion on engine performance shall be noted and recorded.

Following the performance check, the engine shall be disassembled sufficiently for inspection. This test shall be considered to be satisfactorily completed when adequate clearances were maintained, no damaging or detrimental rubbing occurred during the test, the performance has not deteriorated, and the gas–flow path parts show no damage.

EVALUATION RATIONALE (4.3.2.5)

The effects of water ingestion must be determined.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of 2 percent.

(b): A value of 3 percent.

(c): A value of 3.5 percent.

(d): A value of 5 percent.

(e): A value of 5 percent.

Background:

The water ingestion test should be conducted at sea level static conditions.

Flight tests to evaluate the effects of the engine installation on water ingestion should be conducted behind an aircraft water spray test vehicle. Tanker flight tests may be conducted at the same time as tanker icing tests; however, spray nozzle configurations may differ for water and ice.

EVALUATION LESSONS LEARNED

None.

3.3.2.6 Armament gas ingestion (AGI) The engine shall meet all the requirements of the specification during and after AGI, except as follows.

If any performance loss occurs during AGI, the engine shall recover within (a) seconds and without pilot intervention, to the thrust/power available immediately prior to the AGI event. In addition, when an AGI signal is present, engine thrust/power shall never fall below (b) percent of the thrust/power available immediately prior to the AGI.

REQUIREMENT RATIONALE (3.3.2.6)

AGI may cause a rapid rate of pressure and temperature change and attendant airflow distortions to the engine of combat aircraft. The engine's tolerance to such ingestion should be determined to avoid compressor stall, surge, engine flameout, excessive fan/compressor blade stresses, or other mechanical damage.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of at least 5–10 seconds.

(b): A value of at least 93 percent.

If a performance loss occurs, the engine should recover full performance in 5–10 seconds depending on the type of gas ingested and length of gas ingestion. This performance loss is assumed to occur without an AGI signal. When an AGI signal is provided by the aircraft, the engine should lose no more than 7% thrust/power for 5–10 seconds depending on the type of gas ingested and length of gas ingestion. The engine should recover to full performance within 5 seconds following AGI signal cancellation.

The following is recommended to be transferred verbatim into the specification paragraph: The engine operating condition(s) and armament gas ingestion condition(s) and characteristic(s) shall be specified by the engine contractor using information provided by the airframe contractor. The conditions and characteristics shall include: engine power setting, pressure altitude, Mach number, gas ingestion duration, mass flow rate, propellant type and missile position.

Background:

For turboshaft engines, the engine operating and armament gas ingestion conditions specified by the contractor should follow the guidelines below:

The engine operating conditions should be at the Maximum Continuous power setting, zero Mach number and armament gas conditions should be with rocket firing duration of 0.1 second with the rocket positioned 4.6 meters from the engine inlet face with the rocket exhaust perpendicular to the engine face and on the engine center line. Characteristics for demonstration should be as follows:

a. Altitude 1.2 kilometers (4,000 ft)

   Rocket mass flow rate 12.6 milligrams per second

   Propellant characteristics Non–aluminized double base

b. Same as a above except propellant characteristics will be aluminized composite.

c. Altitude Sea level

Rocket mass flow rate 36.3 kilograms per second

Propellant characteristics Non–aluminized double base

d. Same as c. above, except propellant characteristics will be aluminized composite.

In the case of gun gas ingestion, define the critical conditions from the aircraft's design and mission, and for each of these conditions define the engine power setting, altitude, Mach number, maximum firing duration, and quantity of gas ingested.

In establishing rocket gas ingestion conditions, consider the type of rocket, rocket mass flow rates, firing durations, rocket plume characteristics versus altitude and Mach number, and rocket flight paths for the various loading configurations.

All engine control system accommodations for AGI should be specified when utilized in the design. Such control accommodations should be capable of being actuated by the armament firing trigger system in the aircraft cockpit and should be a normally–off type design. All corresponding aircraft requirements (e.g., signals and interface) should also be specified. Further, the timing of the airframe signal to the ingestion event should be considered and integrated into the airframe/engine control algorithms.

Avoiding armament gas ingestion is desired, and most successful airframe/engine integrations have adopted this philosophy. Engine contractors should work closely with the weapon system contractor during preliminary design phase, and efforts should be directed at avoiding/minimizing armament gas ingestion.

REQUIREMENT LESSONS LEARNED

During the A–10 development program, gun gas ingestion from the PAUSE–8 mm gun caused the TF34–100 engine to experience compressor stalls resulting in engine flameout or overtemperature in some instances. Extensive testing of different baffles and gas stripper configurations was required to develop a satisfactory solution. Periodic water washing of engines to remove gun gas combustion products was necessary for field operations.

In the past, weapon systems such as the SPARROW and SIDEWINDER missiles, had engine operating conditions, AGI conditions and characteristics specified by contractors as follows:

a. Engine power setting – Intermediate. This represents the power setting at which most air to air rockets would be fired. Idle represents power setting for air to ground rockets

b. Altitudes of 11 km (36,089 ft) and 2.3 km (7,500 ft) representing the maximum altitudes where either guided and/or unguided rockets could be fired.

c. Mach number – maximum and minimum aircraft flight speed where a rocket could be fired.

d. Rocket mass flow rates of 30 and 80 lb/sec. 30 lb/sec was the highest flow rocket in the 1972/73 time period and 80 lb/sec was the flow rate for a rocket under consideration.

e. Rocket firing duration of 2.0 seconds – this represents the maximum time interval during which the engine inlet is exposed to the rocket exhaust plume (i.e., maximum time interval during which the engine can ingest rocket gases as the aircraft travels through the rocket plume).

f. Rocket position of 4.5 m (15 ft) – selected because this position was considered a realistic position.

g. The Using Service provided Propellant types.

4.3.2.6 Armament gas ingestion (AGI) The requirements of 3.3.2.6 shall be evaluated by test.

EVALUATION RATIONALE (4.3.2.6)

Testing should be conducted to determine the impact of armament gas ingestion on engine performance.

EVALUATION GUIDANCE

Background:

This test should be conducted on the aircraft during flight tests.

EVALUATION LESSONS LEARNED

Estimates for various engine inlet temperature rates range from 550°C (1000°F) to 5500°C (10,000°F) per second for an armament gas ingestion period of one–tenth second.

3.3.2.7 Steam ingestion The engine shall meet all specification requirements without stall, surge, flameout, mechanical/structural damage, or permanent performance loss, when the total inlet massflow is at least two percent steam, and has a minimum temperature ramp (increase) of (a) °C/sec ((a) °F/sec) with a (b) °C ((b) °F) temperature rise in a region less than (c) percent of the engine inlet flow area. After a steam ingestion event, the engine shall meet all requirements of the specification.

REQUIREMENT RATIONALE (3.3.2.7)

Steam from catapult launching causes an adverse environment for aircraft engine operation. Steam ingested by gas turbine engines can cause compressor surge by the distortion effect on the compressor inlet temperature and specific humidity, combined with the effect of ingesting water.

REQUIREMENT GUIDANCE

This requirement and the evaluation of 4.3.2.7 should be labeled “Not Applicable” for weapon systems not utilizing steam catapults.

For carrier–based aircraft, the following should be used for tailoring the specification paragraph:

(a): A maximum value of 2840°C/sec (5150°F/sec).

(b): A value of 100°C (212°F) .

(c): A value of 25 percent.

Background:

The results of tests indicate that during actual catapult the free leakage steam follows a flow pattern that concentrates the steam ingestion in 25% of the inlet on some aircraft. On aircraft with serpentine inlets, the inlet flow patterns may concentrate the steam ingestion into an area greater than 25% of the total inlet area or may cause concentration in several smaller areas less than 25%. When the steam ingested is not concentrated the engine is capable of ingesting over 3 times the quantity of steam without any compressor surge occurring.

REQUIREMENT LESSONS LEARNED

In 1966 during the initial land based catapult testing of the A–7A aircraft powered by the TF30–P–6 turbofan engine, it experienced compressor surge by steam ingestion. This is a single engine aircraft with a low chin inlet, and it was capable of ingesting a large quantity of steam. The older degraded catapults leaked more steam than the new ones. An

improvement to the steam catapult serving arrangement was incorporated on all the carriers which solved the problem for the A–7A steam ingestion problem. Testing experience from A–7A/TF30 revealed a temperature ramp of 2840°C/sec (5,150°F/sec) and a total temperature rise of 100°C (212°F). Reference: Catapult Steam Ingestion Tests of Three Turbofan Engines in the A–7 Aircraft, W. B. Mallett and R. F. Parcells.

In 1988 F–18/404 aircraft started to experience engine stalls on catapult launches. This is a twin engine aircraft and there were times when both engines would stall with fireballs exiting the exhaust.

Data taken on an F404–GE–400 engine during catapult shots recorded 100°C (212°F) temperature rises within 0.2 seconds. No exact data on temperature range and temperature rises where determined because low response thermocouples were used.

4.3.2.7 Steam ingestion The requirements of 3.3.2.7 shall be evaluated by test. Steam shall be introduced into the engine inlet as 0.5, 1.0, 1.5, and 2.0% of the total mass flow. Initial temperature of the steam shall be 245°C (475°F) and gradually raised to obtain a total temperature rise of 100°C (212°F) in the 25% region of the inlet that received the steam.

EVALUATION RATIONALE (4.3.2.7)

A test is required to determine if the engine can operate satisfactorily while ingesting steam.

EVALUATION GUIDANCE

This evaluation and the requirement of 3.3.2.7 should be labeled “Not Applicable” for weapon systems not utilizing steam catapults.

Background:

Considerations for referencing other paragraphs or aircraft tests:

The Using Service should determine if the results of 4.3.2.5 Atmospheric Liquid Water Ingestion and 4.3.2.6 Armament Gas Ingestion can be substituted for this test. Another alternative would be an aircraft/engine test with a catapult that can vary the steam leakage. This test should be done on the aircraft during flight tests.

EVALUATION LESSONS LEARNED

Steam Ingestion is a problem that is a function of aircraft inlet design, compressor stall margin, and steam leakage or condition of the catapult. The first incident was reported in 1966, but after the catapult service was improved, there were no incidents reported until 1988.

The only documented laboratory steam ingestion test was by Pratt & Whitney on an aircraft in 1966. This was presented at the Seventh Annual National Conference on Environmental Effects on Aircraft and Propulsion Systems at Princeton, New Jersey on September 25–27, 1967. Its title is “Steam Ingestion by Aircraft Gas Turbine Engines,” by Nicholas R. Tomassetti.

3.3.3 Electromagnetic environmental effects (E^3) The engine shall meet the electromagnetic environmental effects requirements specified in this section.

REQUIREMENT RATIONALE (3.3.3)

An integrated design approach to E^3 is needed to achieve compatible operation of the engine and compatible operation of the engine with the aircraft and its electromagnetic environment. Electromagnetic fields are emitted from friendly and hostile ships, aircraft, missiles, satellites, and ground transmitters as well as from high altitude nuclear detonations

and lightning. The engine must be protected from these intentional or non–intentional electromagnetic sources. Electromagnetic compatibility between the electrical and electronic systems and subsystems of aircraft, aircraft carrier deck, transient electromagnetic fields, and lightning electromagnetic environments is essential for proper and safe operation of the engine. Electrical and electronic equipment performance may be degraded, or the equipment may malfunction, as a result of internally generated interference or susceptibility to external fields and voltages.

REQUIREMENT GUIDANCE

Background:

An E^3 program is needed to provide maximum confidence that the engine will operate in the overall electromagnetic environment of the aircraft. Compliance with these requirements alone will not guarantee operation on the aircraft or with the external environment. A parallel effort is required by the airframe or integration contractor to demonstrate that the stress levels specified for the engine electronics are consistent with the aircraft electromagnetic environment. Through the systems engineering process, requirements are allocated to the engine subsystem through transfer functions which translate external environmental factors (lightning, EMP, and RF) to engine level stresses based on the aircraft architecture. If there are any conflicts, it may require tailoring of the MIL–STD–461 limits or special considerations in the aircraft installation.

REQUIREMENT LESSONS LEARNED

Past use of these E^3 requirements has resulted in successful operation of engines with the aircraft in many varied electromagnetic environments.

4.3.3 Electromagnetic environmental effects (E^3) The requirements of 3.3.3 shall be evaluated by analysis, demonstration and test.

EVALUATION RATIONALE (4.3.3)

E^3 designs need to be evaluated to demonstrate compliance with the appropriate requirements.

EVALUATION GUIDANCE

Background

For EMI, the requirements of MIL–STD–461 should be evaluated with the testing techniques and testing methods established in MIL–STD–462. The contractor should prove by test that the engine electrical and electronic systems and subsystems will operate satisfactorily within an EMI environment, and within safe limits, as established in the contract. MIL–STD–462 establishes general techniques for use in the measurement and determination of the electromagnetic emission and susceptibility characteristics of electronic, electrical, and electromechanical equipment and subsystems.

EVALUATION LESSONS LEARNED

There were problems found on the 701C/401C Digital Electronic Control Unit (DECU), during EMI testing performed at government facilities.

3.3.3.1 Electromagnetic interference (EMI) All engine electrical and electronic equipment and subsystems shall meet their performance requirements when subjected to the susceptibility requirements of (a) requirement (b) and shall comply with the emission requirements of (a) requirement (c). These requirements shall be met for engine operation in all environmental conditions within the operating envelope for all control system operating modes.

REQUIREMENT RATIONALE (3.3.3.1)

The engine will be subjected to various electromagnetic environments from sources within the aircraft as well as those external to the aircraft which may degrade or interrupt engine performance. The engine may also be a source of interference to other equipment on the aircraft.

REQUIREMENT GUIDANCE

The Using Service should review the requirements of the latest version of MIL–STD–461 for applicability to unique EMI environments. MIL–STD–461 should be tailored as specified in paragraph about Tailoring of requirements of MIL–STD–461 for the specific engine application.

The following should be used to tailor the specification paragraph:

(a): MIL–STD–461.

(b): The Using Service should specify the specific design requirements of MIL–STD–461 that are applicable. The applicable susceptibility requirements of MIL–STD–461 for all engines are CS101, CS114, CS115, CS116 and RS103. If the engine is intended for Army use, then RS101 is also required.

(c): The Using Service should specify the specific design requirements of MIL–STD–461 that are applicable. The applicable emission requirements of MIL–STD–461 for all engines are CE102 and RE102. If the engine is intended to be installed on Navy aircraft with Anti–Submarine Warfare capability or intended for Army use, CE101 and RE101 are also required.

Background:

MIL–STD–461 has been coordinated and approved by the tri–services. MIL–STD–461 may not adequately address certain unique environments for all possible applications, therefore MIL–STD–461 may need tailoring as specified in paragraph about Tailoring of requirements.

For engines that have safety critical (full authority) electronics, MIL–STD–461 defaults to design requirements for CS114 and RS103 which represent hardening levels that have worked successfully in most aircraft applications. Included are concern with external environments such as ground based and carrier deck radars and communications equipment. These two requirements may require some tailoring of the levels after review of the intended operational environment.

Two particular areas that may require tailoring are for the indirect effects of lightning for composite airframes and electromagnetic pulse (EMP) requirements. CS115 and CS116 specify transient requirements that are adequate for most applications. However, the low frequency content of lightning waveforms together with the conductivity characteristics of composites can result in substantial currents and voltages internal to the airframe. The lightning, EMP and external RF environment areas, in particular, require coordination with airframe contractors to determine the flow down of internal stress levels (MIL–STD–461) from the external lightning, EMP, and RF environments (MIL–STD–1818, MIL–STD–1795, and ADS–37) imposed upon the entire aircraft system. Both MIL–STD–1818 and MIL–STD–1795 contain additional information on how the external aircraft environments can be transferred into equipment level specifications, such as this engine specification. Also, RTCA DO–160 provides general waveform and amplitudes for use in composite airframes for commercial aircraft applications, which may be applicable for military use. If the engine wiring harnesses are intended to be installed on the aircraft in a relatively electromagnetic protected area, then it is expected that the levels specified in MIL–STD–461 are usually adequate without tailoring. If the harnesses are to be installed in an open area, such as in a non–conductive nacelle, then the limits may need to be tailored. A possible alternative to imposing higher susceptibility levels (may exceed test capability) is to require that the engine components meet the requirement with the overall shields removed from the wiring harness. The use of overall shields in the actual installation then provides additional protection.

REQUIREMENTS LESSONS LEARNED

Past use of E^3 requirements has resulted in successful operation of aircraft and engines in many varied electromagnetic environments. There have been EMI problems encountered with some engine installations in the past, but in those cases, the EMI requirements were either inadequate for the particular application or waivers were given to the requirements. Application of effective EMI requirements and continual monitoring of the design development by the Using Service has resulted in compatible operation of engines on most aircraft.

There is a documented case where ground support equipment was affected on the aircraft carrier deck as a TF34 was being trimmed on an S–3A.

4.3.3.1 Electromagnetic interference (EMI) The requirements of 3.3.3.1 shall be evaluated by test in accordance with the following procedures of (a) .

EVALUATION RATIONALE (4.3.3.1)

Standardized electromagnetic interference testing methods, procedures, and test equipment are mandatory to ensure consistent and repeatable test results. Evaluation by analysis or similarity are not acceptable options.

EVALUATION GUIDANCE

(a): The Using Service should specify MIL–STD–462.

For aircraft that will be exposed to high level of EMI, the Using Service may want to add a Ground Plane Interference test as shown on figure 20.

Background:

Each EMI design requirement in MIL–STD–461 has a companion test procedure in MIL–STD–462. Demonstration tests in accordance with MIL–STD–462 should be conducted on test articles which simulate that of the flight configurations. Interconnected wiring, physical arrangement, and mounting methods should fully emulate the aircraft installation.

EVALUATION LESSONS LEARNED

None.

3.3.3.2 Intrasystem electromagnetic compatibility EMC The engine shall meet the applicable EMC requirements of MIL–STD–1818, MIL–E–6051, and ADS–37.

REQUIREMENT RATIONALE (3.3.3.2)

The engine contractor must design into the engine sufficient hardening to allow the engine to be able to be installed in the intended aircraft and be electromagnetically compatible with the aircraft.

REQUIREMENT GUIDANCE

Background:

The Using Service must specify the applicable requirements from the indicated documents, based on the intended aircraft.

Any specific additional hardening measures that should be accomplished by the integrator, should be identified by the engine contractor and specified in the corresponding ICD.

A proactive approach must be taken to ensure that the new engine is not designed in a “vacuum” and that considerations of the intended platform are made.

REQUIREMENT LESSONS LEARNED

Experience has shown that compliance with electromagnetic interference (EMI) standards, alone, is not always sufficient to ensure EMC, especially for programs that have allowed EMI waivers for some of the equipment already on the aircraft or aircraft that have unique emissions or susceptibility characteristics.

4.3.3.2 Intrasystem electromagnetic compatibility EMC The requirement of 3.3.3.2 shall be evaluated by test on the engine as installed on the aircraft.

EVALUATION RATIONALE (4.3.3.2)

The engine must be tested on the intended platform to ensure the requirements of 3.3.3.2 have been met. If this testing is not done by the engine contractor, the engine contractor should support the integration contractor in the conduct of the test.

EVALUATION GUIDANCE

Background:

The engine should meet the applicable EMC requirements of MIL–STD–1818, MIL–E–6051, and ADS–37 as applicable.

EVALUATION LESSONS LEARNED

None.

3.3.3.3 Intersystem electromagnetic compatibility EMC The engine as installed in the aircraft, shall meet the requirements of the specification when exposed to the applicable external radiated environment of MIL–STD–1818, MIL–HDBK–235, and ADS–37.

REQUIREMENT RATIONALE (3.3.3.3)

The engine must be able to operate in the radiated environment external to the aircraft. The engine contractor must design into the engine sufficient hardening to allow the engine to be installed in the intended aircraft and operated in the intended electromagnetic environment of the aircraft.

REQUIREMENT GUIDANCE

Background:

Any specific additional hardening measures that must be accomplished by the integrator, should be identified by the engine contractor and specified in the corresponding ICD. The Using Service should specify the applicable requirements from the indicated documents, based on the intended aircraft for the engine.

REQUIREMENT LESSONS LEARNED

Based on experience with H–60 series aircraft engines, when the engines are not designed to operate in the intended operational environment of the aircraft, numerous problems are often found, many with flight safety consequences. The solutions to these problems are often time consuming and expensive. The experience with these aircraft have shown that the integrator cannot sufficiently protect a system that does not have sufficient hardening already designed into it.

4.3.3.3 Intersystem electromagnetic compatibility EMC The requirement of 3.3.3.3 shall be evaluated by test on the engine as installed on the aircraft.

EVALUATION RATIONALE (4.3.3.3)

The engine must be tested on the intended platform to ensure the requirements of 3.3.3.3 have been met.

EVALUATION GUIDANCE

Background:

The Using Service should provide the specific contractor support requirements for the testing. Typically, this testing is performed by a Government test activity based on a test plan generated by that activity. The contractor provides the technical expertise on the engine and its operations and assists in the troubleshooting of any problems encountered during the testing.

EVALUATION LESSONS LEARNED

None.

3.4 Integrity The engine shall meet the integrity requirements specified in this section.

REQUIREMENT RATIONALE (3.4)

Integrity requirements are needed to achieve a maximum level of operational readiness.

REQUIREMENT GUIDANCE

The requirements of all applicable integrity programs should form a part of this specification.

Background:

Integrity programs are needed to maintain emphasis on meeting structural/mechanical/avionic integrity requirements throughout the operational life of the engine.

REQUIREMENT LESSONS LEARNED

Engine qualification testing has often failed to uncover engine structural/mechanical/avionic problems. Many problems did not appear until after several years of operational service. Creation of ENSIP was the result of numerous structural integrity problems with gas turbine engines of the 60’s and 70’s.

4.4 Integrity The requirements of 3.4 shall be evaluated by analysis, inspection, demonstration, and test.

EVALUATION RATIONALE (4.4)

Integrity evaluations must be conducted on the engine system, subsystem, and its parts and components to ensure life and durability requirements are satisfied.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.4.1 Structural integrity The engine shall meet the structural integrity requirements specified herein and in the following paragraphs. The integrity requirements contained below shall apply to all parts, components, externals, and expendables as applicable.

REQUIREMENT RATIONALE (3.4.1)

Structural integrity requirements must be met to ensure engine durability and airframe safety.

REQUIREMENT GUIDANCE

Background:

As of this writing (Jan 1994), MIL–STD–1783 is being rewritten. There may be an effort to create a new engine structures requirement document, which would contain essentially the same content as the paragraphs below. If the new document is developed, it should be referenced in the main specification paragraph, and the paragraphs below should contain any modifications, additions, and/or exceptions to it, as appropriate. An Engine Structural Integrity Program (ENSIP) should be conducted in accordance with MIL–STD–1783.

REQUIREMENT LESSONS LEARNED

During the 1960's and 1970's, a large number of structurally related safety and durability problems occurred in Air Force engines. A Scientific Advisory Board recommended a structural integrity program be developed, based on lessons learned, that would be similar to the one used for aircraft structures (ASIP). One of the lessons learned was that it is unrealistic (and dangerous) to assume defect–free structure in safety–of–flight parts. The Engine Structural Integrity Program (ENSIP) was developed to reduce the magnitude of durability problems associated with military gas turbine engines.

4.4.1 Structural integrity The requirements of 3.4.1 shall be evaluated by analysis, inspection, demonstration and test.

EVALUATION RATIONALE (4.4.1)

The requirements of 3.4.1 must evaluated to ensure compliance with ENSIP and the specification requirements.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.4.1.1 Design service life The engine shall have a design service life of at least (a) when subject to the design usage of 3.4.1.2.

REQUIREMENT RATIONALE (3.4.1.1)

The service life must be established since it is one of the primary design goals.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The design service life should be determined by the Using Service based on the mission need statement. The units (e.g., cycles, mission hours, flight hours, etc.) for design service life should be determined by the Using Service. The following information should be used as a guide to determine the design service life.

		Service Life			
System Category	Parts	Flight (hours)	Ground Run (hours)	Flight (missions)	Ground Runs (missions)
Fighter/Attack	Cold Parts	4,000	400	3,000	200
	Hot Parts	2,000	200	1,500	100
Bomber	Cold Parts	10,000	1,000	2,500	200
	Hot Parts	4,000	500	1,250	100
Cargo	Cold Parts	30,000	3,000	9,000	1,000
	Hot Parts	15,000	1,500	4,500	500
Trainer	Cold Parts	18,000	5,400	13,500	1,500
	Hot Parts	9,000	2,700	6,750	750
Helicopter	Cold Parts	6,000	TBD	TBD	TBD
	Hot Parts	6,000	TBD	TBD	TBD

Background:

None.

REQUIREMENT LESSONS LEARNED

Structural life requirements are the most difficult primary design goals to fulfill. This was the case on many past engine development programs due to a lack of adequate usage parameter definitions. Although requirements for mission profiles, mission mix, and flight hours have often been defined accurately, the important usage parameters that govern cyclic life (major throttle cycles other than the start–stop excursion, time at or above Intermediate power, dwell times, etc.) have not been accurately defined. As a result, operational data has revealed usage parameters not accounted for in design and has resulted in significant reduction in life limits for critical parts and the associated need for redesign and spare parts. Therefore, it is important that realistic design usage information be identified at the outset of the development program for use in design, analysis, and test. The information below presents design service life requirements that have been used in the past.

		Service Life			
System Category	Parts	Flight (hours)	Ground Run (hours)	Flight (missions)	Ground Run (missions)
Fighter: F–22(F119)	Cold Parts	4,000	1,350	2,938	35
	Hot Parts	2,000	675	1,469	18
Bomber: B–2 (F118)	Cold Parts	10,000	N/A	1/	1/
	Hot Parts	4,000	TBD	1/	1/
Cargo: C–17 (F117)	Cold Parts	30,000	TBD	8,516	TBD
	Hot Parts	15,000	TBD	4,258	TBD
Trainer: T–1A (JT15D–5)	Cold Parts	9,000	12,600	2,760	TBD
	Hot Parts	4,500	6,300	1,380	TBD
Turboshaft T800	Cold Parts	6,000	N/A	N/A	N/A
	Hot Parts	6,000	N/A	N/A	N/A

1/ Information classified.

4.4.1.1 Design service life The requirements of 3.4.1.1 shall be evaluated by analysis, inspection, demonstration, and test.

EVALUATION RATIONALE (4.4.1.1)

Design service life requirements must be evaluated to ensure the desired levels of damage tolerance, durability, functional capability, operability, performance, reliability, and strength are attained.

EVALUATION GUIDANCE

Background:

Evaluation is generally accomplished by analysis and test. The process of conducting an evaluation program is the responsibility of the contractor and should be an integral part of the tailored integrity program.

EVALUATION LESSONS LEARNED

None.

3.4.1.1.1 Hot parts Hot parts shall have a usable life of (a) times the design service life specified in 3.4.1.1. Hot parts and their lives shall be listed in table XIV.

REQUIREMENT RATIONALE (3.4.1.1.1)

Hot parts life must be specified in order to achieve logistic and economic effectiveness.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value between one–half (1/2) and one (1) times.

A tabular listing of hot parts and their lives should be provided by the engine manufacturer.

Background:

Hot parts include all parts exposed to the hot gas stream such as the combustor liner, turbine blades and vanes, and exhaust nozzle.

Helicopters designed to operate in environments with minimum maintenance facilities available should require longer hot parts lives for readiness purposes, and be equal to cold parts life.

REQUIREMENT LESSONS LEARNED

In the past, time to cracking did not necessarily become the life limit for a hot part, therefore, the design of hot section parts have been based on evaluation of wear, LCF, creep, stress rupture, oxidation/erosion, and sulphidation.

Cooled turbine vanes were designed such that cooling air would outflow in the presence of cracking or other distress that extends through the thickness. Dual compartment positive outflow design has been needed to minimize erosion rates subsequent to the occurrence of thermal mechanical fatigue cracks thereby maximizing total usable life of the airfoil.

Turbine vanes have also been designed for positive retention so that vane segments would not fall into the gas flow path and cause secondary damage subsequent to total burn–through or severance of an airfoil.

4.4.1.1.1 Hot parts The requirement of 3.4.1.1.1 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.1.1)

Hot parts life must be evaluated to ensure compliance with the requirement of 3.4.1.1.1.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

A sensitivity analysis shall be conducted (on selected hot parts) to identify the effect on parts lives resulting from a range of usage parameters (above and below the design points).

Background:

Failure modes (e.g., LCF, creep, stress rupture, etc.) analyses should be conducted by the contractor to establish design stress levels and lives for engine hot parts based on the design usage.

Usage parameters to be considered in the sensitivity analysis should include airspeed, altitude, ambient temperature, partial throttle cycles, and dwell times at Minimum and Maximum power levels.

Evaluation of hot part lives should be attained as part of the required mission endurance testing. Evaluation of hot parts lives should also be accomplished via the other evaluations conducted in 4.4.1.5 thru 4.4.1.11. Pass/fail criteria (i.e., allowable post–test part condition) should be established for all hot parts life testing. Pass/fail criteria for hot parts life testing should be quantified by defining the post–test condition in terms of dimensional tolerances and wear limits.

EVALUATION LESSONS LEARNED

Improper definition of allowable post–test condition of hot parts has been a shortfall in most engine development programs.

3.4.1.1.2 Cold parts Cold parts shall have a usable life of (a) times the design service life specified in 3.4.1.1. Cold parts and their lives shall be listed in table XV.

REQUIREMENT RATIONALE (3.4.1.1.2)

Cold parts life must be specified in order to achieve logistic and economic effectiveness.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of at least one, minimum.

A tabular listing of cold parts and their lives must be provided by the engine manufacturer.

Background:

Parts not listed as hot parts in 3.4.1.1.1 are considered to be cold parts.

HCF problems affect those parts that are subjected to aero–induced and vibratory loading. All rotating parts that contact static and/or other rotating parts are susceptible to wear. Although creep is a phenomena that is typically associated with hot parts, cold and hot section disks have been known to creep due to high centrifugal stresses and the thermal environment.

REQUIREMENT LESSONS LEARNED

Cold parts have been typically designed to an LCF requirement that assured cracking would not occur prior to reaching the required durability limit. The incidence of LCF failures has been reduced after many years of emphasis on designing against LCF. However, other failure modes have become increasingly bothersome (e.g., HCF, wear, and creep).

4.4.1.1.2 Cold parts The requirement of 3.4.1.1.2 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.1.2)

Cold parts lives must be evaluated to ensure compliance with the requirement.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

A sensitivity analysis shall be conducted (on selected cold parts) to identify the effect on parts lives resulting from a range of usage parameters (above and below the design points).

Background:

Failure modes (e.g., LCF, HCF, creep, etc.) analyses should be conducted by the contractor to establish design stress levels and lives for engine cold parts based on the design usage.

Usage parameters to be considered in the sensitivity analysis should include airspeed, altitude, ambient temperature, partial throttle cycles, and dwell times at Minimum and Maximum power levels.

Evaluation of cold part lives can be attained as part of the required mission endurance testing. Evaluation of cold parts lives should also be accomplished via the other evaluations conducted in 4.4.1.5 thru 4.4.1.11. Pass/fail criteria (i.e., allowable post–test part condition) should be established for all cold parts life testing. Pass/fail criteria for cold parts life testing should be quantified by defining the post–test condition in terms of dimensional tolerances and wear limits.

EVALUATION LESSONS LEARNED

Improper definition of allowable post–test condition of cold parts has been a shortfall in most engine development programs.

3.4.1.1.3 Expendables The minimum life without replacement of all expendable parts and components shall be equal to the minimum maintenance–free operating period. Expendable parts, components, and their lives shall be listed in table XVI.

REQUIREMENT RATIONALE (3.4.1.1.3)

It is necessary to specify the minimum life of expendable parts and components since their failure or degradation will affect life cycle cost, maintainability, and functional readiness of the engine and subsystems.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The contractor should provide a tabular listing of expendables with their respective functional lives.

Background:

Expendable parts are those normally replaced at maintenance or overhaul, such as minor hardware, O–rings, and gaskets. Expendable components include starters and ignitors.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.1.3 Expendables The requirement of 3.4.1.1.3 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.1.3)

Functional life of expendables must be evaluated to insure practical and economical maintenance intervals.

EVALUATION GUIDANCE

Background:

Evaluation of expendable parts and components lives can be attained as part of the durability test program. Expendables will typically be replaced during an AMT.

Analyses are not always a practical means of evaluating the required lives of expendables.

EVALUATION LESSONS LEARNED

None.

3.4.1.1.4 Bearings The mainshaft and gearbox bearings shall have B1.0 lives equal to at least the design service life of the engine. A list of bearings and their lives shall be presented in table XVII.

REQUIREMENT RATIONALE (3.4.1.1.4)

Main shaft and engine gearbox bearings life must be specified to ensure that the bearings meet engine durability requirements.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The contractor should provide a tabular listing of the bearings and with their respective lives.

Background:

Special attention to life testing should be taken when specifying high DN (diameter x rpm) (2.5×10^6) or unusual mounting configuration bearings, e.g., outer race rotating bearings or in shaft bearings in which both races rotate.

REQUIREMENT LESSONS LEARNED

Predictions of bearing performance and/or life have not always been reliable, especially those bearings whose operating conditions exceed recent engine experience or whose designs are unusual. For example, bearing performance analytical models and limited life tests did not have the capability to predict or reveal roller dynamic instability which occurred in some high DN engine roller bearings. More extensive life testing may have uncovered this potentially catastrophic bearing failure mode.

4.4.1.1.4 Bearings The requirement of 3.4.1.1.4 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.1.4)

Analyses and tests are required to determine the lives of the engine bearings.

EVALUATION GUIDANCE

Background:

A bearing life analysis and bearing tests should be conducted.

The bearing life presentation should clearly identify whether the Weibull slope in the analysis is assumed or whether it is the slope acquired from rig testing. Evaluation should be accomplished early enough in the program to allow for redesign and requalification tests and to make the necessary inputs into the spares provisioning programs.

It is critical that analytical and empirical evaluation of rotor thrust balance occur in IFR and FFR milestones. Rotor thrust balance reports should show agreement between analytical and test data. Rotor thrust evaluation tests should be conducted on appropriate configurations. The intent of the test and analysis is twofold, (1) to ensure peak thrust loads are consistent with thrust bearing life requirements, and (2) to ensure rotor thrust crossovers occur only transiently and in the absence of significant radial loads. The latter is needed to preclude skidding damage to ball bearings.

EVALUATION LESSONS LEARNED

None.

3.4.1.1.5 Components Engine components shall have a usable life of (a) times the design service life specified in 3.4.1.1. Engine components and their lives shall be listed in table XVIII.

REQUIREMENT RATIONALE (3.4.1.1.5)

Engine components life must be specified in order to achieve logistic and economic effectiveness.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of one, minimum.

A tabular listing of engine components and their lives must be provided by the engine manufacturer.

Background:

Engine components include: Fuel pumps, engine controls, jet nozzle and actuators, anti–icing valves and the temperature sensing system.

REQUIREMENT LESSONS LEARNED

Air Force engine related mishap data from 1976–1987 shows a high incidence of failure/malfunction of engine components and externals. During that time period, engine component and external failures/malfunctions accounted for no less than 35% of the total engine related mishaps.

4.4.1.1.5 Components The requirement of 3.4.1.1.5 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.1.5)

Engine components lives must be evaluated to ensure compliance with the requirement.

EVALUATION GUIDANCE

Background:

Failure modes analyses should be conducted by the contractor to establish design lives and stress levels when subject to the design usage. Evaluation of engine component lives can be attained as part of the required mission endurance testing. Evaluation of engine component lives should also be accomplished via other evaluations in this specification.

EVALUATION LESSONS LEARNED

Insufficient qualification testing (methods and duration) has resulted in the fielding of engine components that were not capable of meeting the desired operational life.

3.4.1.2 Design usage The design usage shall include: (a) .

REQUIREMENT RATIONALE (3.4.1.2)

Design usage such as engine envelope, aircraft missions and mission mix, usage parameter environments, etc., must be defined since they affect engine life, engine structural performance and service operability.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The design usage should include:

1. Missions and Mission Mix
2. Usage parameters
3. Externally applied forces
4. Operating envelope
5. Engine attitude limits
6. Ambient temperature extremes
7. Icing environment conditions
8. Corrosive atmosphere conditions
9. Noise environment conditions
10. Customer bleed air extraction, loaded accessory pads and power take–off usage
11. Engine performance retention characteristics

Generic usage parameters are specified in the information below.

System Category	Parts Classification	Type I Cycles	Type III Cycles	Type IV Cycles	Augmentor Lights or Thrust Reversing	Augment Time or Vector Cycles (hrs)	Time at IRP and Above (hrs)
		0–max–0	Idle–max–Idle	Cruise–Int–Cruise			
Fighter	Cold Parts	3,200	20,000	24,000	17,000	200	800
	Hot Parts	1,600	10,000	12,000	8,500	100	400
Bomber	Cold Parts	2,700	30,000	30,000	16,000	250	1,80
	Hot Parts	1,350	15,000	15,000	8,000	15	90
Cargo	Cold Parts	10,000	14,000	TBD	N/A	N/A	6,300
	Hot Parts	5,000	7,000	TBD	N/A	N/A	3,000
Trainer	Cold Parts	15,000	150K	150K	TBD	TBD	3,600
	Hot Parts	7,500	75,000	75,000	TBD	TBD	1,800
Helicopter	Cold Parts	15,000	N/A	N/A	0	0	N/A
	Hot Parts	15,000	N/A	N/A	0	0	N/A

Background:

The design usage should include, but not be limited to, the defined missions and mission mix, design duty cycle, usage parameters, nozzle usage, environmental (external, internal, and installation) conditions, unique flight conditions, and the non–operating environment. The internal environment is specified in 3.4.1.2.1. The engine operating envelope specified in 3.2.2.1 is presented on figures 10. The engine operating envelope is determined by the engine contractor. The missions and mission mix of 3.4.1.2 are presented in tables VI. Unique flight conditions are discussed in 3.2.2 and presented on figure 9. Vibration and dynamic response characteristics are specified in 3.4.1.8. Design usage should also include the external environmental conditions specified in 3.3, which covers atmospheric conditions and engine ingestion capability such as bird, ice, water, steam, sand, and dust.

The typical rate of performance deterioration should be based on the performance program and performance deterioration model. The contractor should address deteriorated engine conditions as part of the design practice and account for it in the life predictions.

Engines with control systems that maintain minimum thrust levels, by increasing engine temperature and speed, will decrease the potential parts lives by exposing the engine to increased thermal and mechanical stresses.

The ability of engine hot parts to meet design life requirements can be significantly reduced due to engine uptrim or other conditions that result in hot gas stream temperatures higher than that of a production engine. To account for the impact on hot parts life, by operation at increased temperatures, margins of 30°F to 70°F above production acceptance (non–degraded) maximum steady–state gas temperature have been imposed by the procuring activity during the design of hot parts to assure that design life goals will be met. AIA PC Project 338–2A members made a consensus recommendation in 1982, that the procuring activity not establish a specific temperature margin since this number will vary with engine type and application. For analysis purposes, the F100–PW–229 was designed for 1/3 life at nominal production performance and 2/3 life at full deterioration levels in order to provide full life, even with deteriorated engines.

REQUIREMENT LESSONS LEARNED

One of the major shortcomings in past engine development programs has been inadequate definition of the design usage parameters critical to engine durability. Although requirements for mission profiles, mission mix, and flight hours have often been defined accurately, the important usage parameters that govern cyclic life (major throttle cycles

other than the start–stop excursion, time at or above Intermediate power, dwell times, etc.) have not been accurately defined. As a result, operational data has revealed usage parameters not accounted for in design and has resulted in significant reduction in life limits for critical parts and the associated need for redesign and spare parts. Therefore, it is important that realistic design usage information be identified at the outset of the development program for use in design, analysis, and test.

Past program specific usage parameters are specified in the information below.

System Category	Parts Classification	Type I Cycles	Type III Cycles	Type IV Cycles	Augmentor Lights or Thrust Reversing	Augment Time or Vector Cycles (hrs)	Time at IRP and Above (hrs)
		0–max–0	Idle–max–Idle	Cruise–Int–Cruise			
Fighter F–22 (F119)	Cold Parts	2,973	20,503	22,074	20,239	186 (109)	684
	Hot Parts	1,487	10,252	11,037	10,165	93 (54)	342
Bomber B–2 (F118)	Cold Parts	2,371	7,113	N/A	N/A	N/A	TBD
	Hot Parts	948	2,844	N/A	N/A	N/A	TBD
Cargo C–17 (F117)	Cold Parts	8,516	25,840	17,178	12,700	N/A	TBD
	Hot Parts	4,258	12,920	8,589	6,350	N/A	TBD
Trainer T–1A JT15D–5	Cold Parts	28,000	20,288	3,817	N/A	N/A	885
	Hot Parts	14,000	10,140	1,909	N/A	N/A	443
Helicopter	Cold Parts	15,000	TBD	TBD	0	0	TBD
	Hot Parts	15,000	TBD	TBD	0	0	TBD

4.4.1.2 Design usage The requirements of 3.4.1.2 shall be evaluated by analysis, inspection, demonstration, and test.

EVALUATION RATIONALE (4.4.1.2)

Design usage is specified in 3.4.1.2 but is evaluated in tests specified throughout the specification.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.4.1.2.1 Internal environment The internal environment shall be specified herein for the conditions of 3.4.1.2.

REQUIREMENT RATIONALE (3.4.1.2.1)

The internal environment must be established to accurately assess structural integrity.

REQUIREMENT GUIDANCE

Background:

The contractor should define the internal environment including thermal, pressure, vibration, and dynamic loading. The internal environment should be specified at various radial and axial locations, as necessary, to completely characterize engine operating conditions.

REQUIREMENT LESSONS LEARNED

Inadequate characterization of the internal environment has been a shortcoming of many engine programs.

4.4.1.2.1 Internal environment The requirement of 3.4.1.2.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.2.1)

Evaluation of the internal environment is necessary to ensure the actual operating conditions are accounted for in the design.

EVALUATION GUIDANCE

Background:

An analysis should be performed to establish the internal environment. Instrumented engine tests should be conducted to substantiate the internal environment analysis. As a minimum, engine tests should be performed for both steady–state and transient conditions to measure internal gas stream, cooling flow cavities and metal temperatures, and pressure distributions.

EVALUATION LESSONS LEARNED

None.

3.4.1.2.2 Externally applied forces The engine shall function satisfactorily and no deformation shall occur during or after exposure to the externally applied forces indicated in the Design Load Diagrams of figure 2.

REQUIREMENT RATIONALE (3.4.1.2.2)

Engines must be designed to withstand the externally applied forces which may result from flight maneuvers, landings, and takeoffs.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph: The contractor should provide design load diagrams on figure 2.

Background:

Externally applied forces include: loads produced by take–off, landing, in–flight maneuvers, gusts, vibration, installation, and crash conditions. Limit loads should be based on a weight factor consisting of the dry weight of the engine, increased by the specific weight allowed for all engine mounted components and operating fluids. In installations where airframe externals are supported by the engine, the weight of these externals should also be included in the weight factor.

Engines are subjected to “g” forces and forces caused by aircraft roll, pitch, and yaw while in flight. Aircraft used on carriers are exposed to significant “g” loads resulting from catapult launch or carrier arrest landings. The “g” forces presented on figure 2 may be too severe for some given applications. A judgement must be made whether to reduce the requirements for a particular less severe application. Figure 2 considers the extremes for all manned aircraft.

REQUIREMENT LESSONS LEARNED

Aircraft such as the F–6, A–4, A–6, A–7, F–4 and F–8 have recorded load factors up to 7g's, and F–16s go up to 9g's. Aircraft maneuver requirements have increased so that particular weapon systems must now be capable of 8 to 10 g's.

The F404–GE–400 engine model specification CP45K006 of 15 November 1975 used a maneuver load spectrum which showed various values of positive and negative "g's" and cycles per 1000 hours for each "g" value. The maximum static load requirement was reduced to 1.4 times the landing loads rather than the 1.5 times requirement.

4.4.1.2.2 Externally applied forces The requirement of 3.4.1.2.2 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.2.2)

There is a need to evaluate the engine by analysis and test its capability to withstand the external forces to which it may be subjected due to flight maneuvers, landings and takeoffs.

EVALUATION GUIDANCE

Background:

Stress and deflection data should be obtained at critical locations as determined by analysis and preliminary stress coating tests. Engine cases and mounts should be subjected to a static rig test. The static rig test, utilizing the applicable engine static structure, should be conducted to demonstrate the capability of the engine and its supports to withstand maximum externally applied forces specified in 3.4.1.2.2 without permanent deformation of any component and 1.5 times those forces without failure of any component. The loads should be applied separately and in combination.

EVALUATION LESSONS LEARNED

None.

3.4.1.3 Material characterization Material properties shall be based on (a).

REQUIREMENT RATIONALE (3.4.1.3)

This requirement is needed to ensure adequate material capability by limiting the statistical variation of the properties to conventional aerospace design practices.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The intent is to base material properties including elongation on minimum material capability, therefore it should be stated that material properties will be based on minus three sigma (–3μ) values with a fifty percent (50%) confidence level or minus two sigma (–2μ) values with a ninety–five percent (95%) confidence level. An alternative is to state that material properties will be based on B0.1 values. The confidence level for B0.1 is assumed to be 50 percent. Typical (B50) properties can be used to characterize fracture toughness and crack growth rate. Another alternative is "A Basis" from MIL–HDBK–5 which uses properties for 99% exceedance with 95% confidence (equivalent to 3.129 sigma properties).

Background:

All material properties will have an associated statistical variation (i.e., worst, average, best). For this reason there is no single value that describes a material's capability. The value to be used will typically be stated as some variation from the mean. –3μ material properties are used in conventional aerospace design practices. It accounts for the

statistical variation in material properties and represents the properties which will be exceeded by 99.87 percent of the population. B0.1 values represent properties which will be exceeded by 99.9 percent of the population. Use of values other than those stated must be approved by the Using Service.

Past engine program means of defining minimum material LCF capability include: (a) defining LCF properties based on –3 below the average life data, where average life data is plotted as a log–normal failure distribution (i.e., the logarithm of cycles to failure is normally distributed); (b) defining LCF properties based on –3 below the average life data, where average life data is plotted as N^c versus log (N is number of cycles to failure and ^{c} is the value that forms a straight line relationship between N^c and log strain); (c) defining LCF properties based on B0.1 life data, where the B0.1 value is determined from the Weibull plot using abscissa values determined by the ratio of number of actual cycles to number of predicted cycles to failure (N_A/N_P); (d) defining LCF properties based on B0.1 life data, where the B0.1 value is determined from the Weibull plot using abscissa values determined by the number of cycles to failure.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.3 Material characterization The requirement of 3.4.1.3 shall be evaluated by analysis, test and inspection.

EVALUATION RATIONALE (4.4.1.3)

Material properties must be evaluated to ensure appropriate selection of materials.

EVALUATION GUIDANCE

Background:

A material characterization plan should be prepared and existing material data should be presented. Final definition of structural capability should be based on the testing of specimens fabricated from “as–manufactured” parts.

The contractor should submit a material characterization plan that identifies and schedules each of the tasks and interfaces in design, material selection, and testing. The tasks to be identified in the plan should include:

1. Design analyses depicting the environment (i.e., stresses and temperatures) of each part with a correlation to the operating envelope conditions.

2. Preparation of a parts listing, with the corresponding material and manufacturing process.

3. Identification of mechanical properties needed for each material/part.

4. The source of material data.

5. Number of tests conducted for each material property curve needed for each part.

6. Quality control actions for materials and processes.

7. Risk assessment for the use of any advanced materials and processes.

Existing material property data obtained through earlier tests can be used during the initial design only when the manufacturing processes are similar (e.g., same methods of forging, heat treating, machining, etc.). However, final definition of structural capability must be based on material property curves generated by testing specimens cut from “as–manufactured” parts to evaluate material properties relative to different locations on the part. The number of tests conducted for each curve or condition must be adequate to establish “confidence” with the material’s structural capability.

Many durability problems can be traced to the selection of unsuitable materials. The need to reduce engine weight has forced gas turbine engine contractors to make compromises in the selection of materials or higher risk designs. Characterization and choice of materials should be closely monitored by the procuring activity. Material selection should be reviewed under a risk assessment or management plan.

EVALUATION LESSONS LEARNED

None.

3.4.1.4 Parts classification All engine parts, components, controls and externals and expendables shall be classified for criticality in table XIX.

REQUIREMENT RATIONALE (3.4.1.4)

Parts, components and expendables must be classified to assure the appropriate design requirements are applied.

REQUIREMENT GUIDANCE

Background:

For all applications, the classifications should be fracture critical, durability critical, and durability non–critical. Fracture critical parts may be further classified as safety and mission critical. The engine manufacturer should provide the classification summary. The format of table XIX indicating the Using Service classifications should be provided by the Using Service to be filled in by the contractor.

A failure mode and effects criticality analysis (FMECA), a safety/hazard assessment or other engineering analysis should provide the basis for classification.

REQUIREMENT LESSONS LEARNED

It has been difficult to determine, in advance, engine parts or components whose failure would have resulted in secondary failure which would have lead to loss of aircraft or essential mission capability. Therefore, continued evaluation has been required as the subsystem was being defined. An example is the high pressure hydraulic or pneumatic pressure vessel, which by itself, may not be safety–of–flight critical, but if located on or close to primary airframe structure could precipitate a failure of the airframe if it were to fail.

The number of critical parts can be significantly larger for an engine model used in a single engine configuration as opposed to a dual engine configuration. For example, the number of critical parts for the F100 engine in the F–15 and F–16 is 45 and 70, respectively excluding externals.

4.4.1.4 Parts classification The requirement of 3.4.1.4 shall be evaluated by analysis and inspection.

EVALUATION RATIONALE (4.4.1.4)

Evaluation of classification is necessary to ensure the appropriate design requirements are applied.

EVALUATION GUIDANCE

None.

Background:

A Failure Modes and Effects Criticality Analysis (FMECA), in addition to other engineering analyses, could be conducted to evaluate the results. Parts should be classified as fracture critical, safety critical, mission critical, durability critical, or durability non–critical, depending on the application (USA, USAF, USN). The parts classification summary for the same engine may vary with application or use. For example, single engine vs multi–engine aircraft will have different lists.

EVALUATION LESSONS LEARNED

None.

3.4.1.5 Durability The engine shall be durable for ___ times the design service life when subjected to the usage of 3.4.1.2.

REQUIREMENT RATIONALE (3.4.1.5)

Durability requirements must be applied since engine durability is a primary design requirement. Durability requirements must be applied to minimize cracking or other structural or material degradation which could result in functional impairment or excessive in–service maintenance problems.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

A value of at least one should be put in the blank.

Background:

A durability control plan should be prepared to identify and schedule each of the tasks and interfaces in the functional areas of design, material selection, manufacturing control, and inspection of engine parts and components. The tasks to be identified in the plan should include:

1. Design concepts/material/weight/performance/cost trade studies.

2. Life analysis, development testing, and proof of compliance testing tasks.

3. List that identifies parts, locations, and special controls required to meet life requirements (e.g., material specification controls, quality assurance requirements, etc.).

4. Zoning of drawings for parts to identify critical locations and associated quality control requirements, defect locations, orientation, inspection method, and acceptance standards. The use of alternate procedures for identification of critical locations, etc., on drawings may be proposed.

5. Basic materials data.

6. Identification and control of variables that affect properties in the material procurement and manufacturing process specifications.

7. Traceability requirements on all tiers of procurement, processing, fabrication and assembly for durability critical parts. Serialization or time coding requirements for tracking operational exposure of individual parts.

8. Quality control requirements during manufacture. Subcontractor, vendor and supplier quality control requirements during manufacture. Identification of procedures for certifying and monitoring subcontractor, vendor and supplier inspection, and quality control.

9. Nondestructive inspection requirements for use during depot and base level inspections including supporting manuals (technical orders) and equipment needs.

Durability of all engine parts and components is obtained by proper material selection and control, control of stress levels, detail design, and use of protection systems.

Most of the tasks contained in the Durability Control Plan have been accomplished by engine manufacturers in past development and production programs. However, the durability requirements established here impose tighter controls and more interface involvement between the functional areas.

REQUIREMENT LESSONS LEARNED

The J85–21 engine had a compressor blade flutter problem (high cycle fatigue) which was discovered after the engine had successfully passed qualification testing and after engine production started. Structural failures during F100 engine qualification testing threatened the existence of the weapon system program and raised serious questions about military engine procurement techniques. Fatigue failures in the TF41 engines have caused loss of life and aircraft. Although the Navy F–14 aircraft has two engines, TF30 engine problems resulted in loss of life and aircraft. Cracking of expensive turbine blades in the TF30 produced serious maintenance support problems through excessive replacement rates during engine overhaul.

4.4.1.5 Durability The requirements of 3.4.1.5 shall be evaluated by a strength and life analysis, inspection, demonstration, and part, component, and full–scale engine tests.

EVALUATION RATIONALE (4.4.1.5)

Attainment of durable parts is achieved through application of detail requirements in the functional areas of design, materials selection, manufacturing control, and inspection. The strength and life analysis report is one of the most important reports submitted by the engine contractor. Special inspections conducted periodically during the durability test programs are essential to preclude part failure and loss of the development engine.

Definitive criteria is needed to judge the success of qualification tests. Problems will likely be discovered during teardown inspection of these engines prior to attaining the required test durations. Therefore, criteria is needed to establish course of action in the event problems occur and to define follow–on actions (i.e., remaining test duration, problem cause, corrective action, operational implications, qualification tests for redesign, etc.).

EVALUATION GUIDANCE

The following inspection requirements and success criteria should be transferred verbatim into the specification paragraph:

The engine contractor shall specify inspection procedures, in addition to 4.10.3.5.3, for the endurance test engines. Inspection requirements shall include in–service design inspections developed in accordance with the durability and damage tolerance requirements of this specification. Inspection requirements shall also duplicate the expected field maintenance concept for the engine. Special inspections to monitor the status of critical parts shall be included.

Each structural problem (failure, cracking, yielding, wear, erosion, etc.) discovered during endurance testing inspections shall be analyzed to determine cause, corrective actions, and operational implications, including development schedule and cost impacts.

Background:

A Strength and Life Analysis report should be prepared containing an analysis which defines: (l) the lives in equivalent LCF cycles for all appropriate parts, (2) the LCF duty cycle for the individual component tests and any full scale engine testing, (3) the cool down time between cycles, and (4) the total number of cycles to demonstrate the equivalent of the LCF life requirements. The LCF lives of all the appropriate parts and the mission hours equivalency for the selected features in a given test mode should also be defined. The predicted burst speeds of critical rotating disc components should be specified.

For parts subject to fatigue (thermal or mechanical) the report should identify the allowable size of cracks which would be cause for failure. The contractor should provide a specific description of the expected failure mode of each part listed.

The Strength and Life Analysis report should define all variables (e.g., material properties, manufacturing and assembly processes and tolerances, and operating temperatures, pressures and stresses, etc.) that significantly affect the durability and life of the engine. The report should define the sensitivity of parts to variations in these variables and

other design assumptions. The limits to variation of these variables should be specified. The report should define how design, processing, manufacturing, assembly, quality control, testing, etc., will be used to assure that all engine durability and life affecting variables fall within these limits.

The report should also contain an analysis to establish the accelerated mission oriented engine duty cycle and the resulting equivalences for selected features for the AMT/endurance test. These engine duty cycles should be constructed such that the AMT/endurance test produces hot parts damage equivalent to at least the hot parts life of 3.4.1.1.1. The accelerated engine duty cycles should be derived from the mission cycles of table VI. A composite accelerated duty cycle (or cycles) is an acceptable alternative to individual AMT/endurance test duty cycles. Consistent with the engine duty cycle of table XX, a random sequence schedule for the mission duty cycles should also be established with the approval of the Using Service.

Both a pictorial and tabular definition of the accelerated engine duty cycles should be included in the report and in the model specification as table XX. There should also be a detailed description of the methodology used to develop the duty cycles which lists the test acceleration factors and shows appropriate damage factors.

A fixed level of inlet air temperature and pressure should be established for each AMT/endurance test duty cycle. The inlet air temperature and pressure level determined for each duty cycle should produce the damaging effects on the engine consistent with the actual Mach number and altitude variation defined for each mission in table XX and the ambient temperature distribution of figure 4. The strength and life analysis should also provide the methodology and data utilized to define the engine parts lives (B0.1, B1.0, B10 and B50). Revised or updated reports should include actual data utilized from the development program such as inspection data (NDI), etc.

Stress analysis should include such items as engine cases, discs, vanes, blades, mounts, combustion liners, bearing supports, gears, brackets, and tubing.

A strength and life analysis should be performed and a report submitted prior to completion of IFR. The report should be updated by change pages by completion of FFR, ISR, and OCR. A revised report should be issued after completion of OCR.

If desired, this paragraph should include a requirement to evaluate problems or failures that occur during endurance testing (i.e., loss of mission capability or requiring maintenance actions) relative to contract/specification reliability and maintainability goals and to use test results to validate the Life Cycle Cost (LCC) models.

EVALUATION LESSONS LEARNED

Complete failure of test engines has occurred in past development programs due to undetected damage growth. Many of these failures could have been prevented by use of detail inspection requirements (methods and intervals) developed through the durability and damage tolerance control plans and experience obtained from earlier development testing (e.g., previously observed deterioration and distress).

3.4.1.5.1 High cycle fatigue (HCF) life Engine parts shall not fail when subject to the maximum attainable combined steady–state and vibratory stresses. The vibratory or HCF stress shall be restricted to (a) percent of the material capability. All engine parts shall have a minimum High Cycle Fatigue (HCF) life as follows: (b) .

REQUIREMENT RATIONALE (3.4.1.5.1)

Experience has shown that engine structural components operating under combined steady and vibratory stress conditions must be designed to ensure resistance to HCF cracking.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): forty

(b):

MATERIAL	LIFE (cycles)
Ferrous, Nickel–Base Superalloys	10^7 cycles
Titanium	10^9 cycles
Other Materials	3×10^7 cycles

Background:

The vibratory or HCF maximum stress, ratioed to the worst location and for the worst condition, should be restricted to a value of 40 percent of that allowed by the minimum value material allowable due to the sensitivity of high cycle stresses to damping variability, part–to–part resonance variation, unknown excitations, etc. The 40% of maximum stress criteria is based on RMS values. Other percents (using non–RMS stresses), up to 60%, may be used if the maximum stress is based on maximum instantaneous stress. An alternative design approach to achieve margin is to limit the steady stress such that significant levels of vibratory stress (e.g., 70–105 mPa (10–15 Ksi) peak–peak) will not exceed the minimum value material allowable. For airfoils having FOD/DOD damage tolerance requirements of K_t=3, the alternating stress should be limited to 40% of the minimum unnotched HCF material allowable or 100% of the K_t=3 minimum notched HCF material allowable, whichever is less.

For materials which do not have a discrete endurance limit knee on the stress versus cycles to failure curve, the engine manufacturer should present in the vibration and stress analysis report the method of achieving adequate high cycle fatigue life. Complications exist with the concept of specifying that all parts be designed to some discrete specified endurance limit. Some of these are:

(1) Prior stressing at a higher stress can cause a lowering of the endurance limit.

(2) Stress cycling at gradually increased cyclic stress can result in an increased endurance limit (this is known as coaxing).

(3) Interactions between LCF and HCF can result in either increased or decreased lives depending upon the magnitude of the loads, the order of the loading, and the material.

(4) Installation, handling and environmental sensitivities can result in significantly higher steady–state and vibratory stresses which will reduce or even have negative margins for HCF capability. Such an example would be external parts which may be sensitive to all of the above. Realistic levels of stress due to these sensitivities should be included when assessing HCF capability.

Parts which are subjected to LCF loads in addition to HCF loads should be designed considering the effect of LCF damage on the material HCF life.

REQUIREMENT LESSONS LEARNED

USAF experience with the F100 and F110 programs has shown that titanium airfoils can be very sensitive to vibratory stresses. Failures of fan and compressor airfoils, resulting from vibratory stresses, have caused domestic object damage and catastrophic failure. Controlling this problem has required significant additional effort by field units. Testing performed as part of these investigations has shown that the fatigue strength continues to decline past 3×10^7 cycles. Based on the results of this testing, the life requirement of titanium materials should be set at 10^9 cycles. USAF experience and materials data has shown that 10–15 Ksi peak–peak to be the maximum vibratory stress desirable to meet the HCF life requirements.

4.4.1.5.1 High cycle fatigue (HCF) life The requirements of 3.4.1.5.1 shall be evaluated by analysis and test. A stair–step test shall be conducted before and after the specified engine test(s). The engine shall undergo an increasing speed stair–step run consisting of (a) periods of (b) duration each, at equal increasing rotational speed increments beginning at Idle and continuing up to and including the maximum rotor speed.

EVALUATION RATIONALE (4.4.1.5.1)

The HCF life must be evaluated to avoid HCF failures.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 25 periods or as determined by analysis. In the event significant peak vibration points exist at any conditions between Idle and maximum rotational speed, the number of increments chosen may be altered at the option of the Using Service, to increase the amount of running time obtained at the peak vibration points up to an amount not to exceed 50 percent of the total time of the run.

(b): One hour duration or as determined by analysis.

Background:

The engine contractor should provide an HCF analysis. The analysis should be provided at the initiation of EMD to ensure appropriate selection of materials and design configurations. The analysis should identify resonant mode shapes, frequencies, and those stresses produced by a forced response, resonance, and flutter. As a minimum, the analysis should be performed on all support frames, rotating and static airfoils, and engine cases and heatshields. Additional evaluation can be conducted via bench testing of parts to confirm the resonant mode shapes, frequencies, and stresses identified in the analysis. The HCF analysis should be substantiated/correlated with full–scale engine testing.

The source of HCF full–scale engine test data should include all accelerated mission/endurance and altitude tests. The additional tests required in 4.4.1.8 should allow further evaluation of the HCF life of parts. Sufficient instrumentation should be provided on all HCF test engines to obtain resonant mode shapes, frequencies, and stress levels.

EVALUATION LESSONS LEARNED

None.

3.4.1.5.2 Low cycle fatigue (LCF) life Engine parts shall have a minimum LCF life of (a) times the number of (b) cycles that correspond to the parts life of 3.4.1.1.1 for hot parts and 3.4.1.1.2 for cold parts.

REQUIREMENT RATIONALE (3.4.1.5.2)

Low Cycle Fatigue is one of the most severe and costly problems encountered in service. This requirement is an attempt to reduce or eliminate LCF failures.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): a value at or between 1 and 2

(b): the Using Service should specify LCF or TAC

Background:

Low cycle fatigue is the damage generated in a material by cyclic load reversals, which cause repeated plastic deformation. Rotating parts are subjected to mechanical and thermal stresses during engine operation. These stresses are represented as cyclic loading on engine parts. A full LCF cycle occurs each time the engine is started, run to Maximum power, then shutdown. Partial LCF cycles occur during throttle changes. The severity depends upon the degree of change in engine speed and temperature that accompanies the change. Miner's rule may be used for the summation of LCF cycles to determine the cumulative damage. Cumulative damage from LCF cycles will, eventually, cause a crack to initiate.

Predicting LCF life is not an exact science and must be determined on a statistical basis. Traditionally, the minimum LCF life of a part is defined as the B0.1 life to crack initiation. This means that 1–in–1000 parts will develop an LCF initiated crack during the minimum required life of that part.

LCF design procedures must be used to assure that generalized cracking will not occur prior to reaching the required design service life. LCF design and analysis procedures, if properly applied, can generate a component configuration that will experience minimal cracking by proper material selection and control of stress levels, detail design to eliminate or optimize stress concentrations, and control of component surface finish and correlation with LCF material allowables. Special attention should be placed during detail structural design to minimize stress concentrations. Those stress concentrations that are present in the design should be optimized to produce the largest cyclic life possible. Other factors which effect LCF life are material properties, subsurface material flaws, surface flaws, operating temperatures, and part finishes.

REQUIREMENT LESSONS LEARNED

Most LCF problems did not surface until a number of years after qualification and acceptance by the Using Service.

4.4.1.5.2 Low cycle fatigue (LCF) life The requirement of 3.4.1.5.2 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.5.2)

The LCF lives must be evaluated to preclude the occurrence of part failures.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

For engines not using the AMT of 4.9.1.1.1 to evaluate LCF, the following shall be conducted:

For ISR and OCR, the length of the LCF engine test shall be equivalent to 1/4 of the cold section life, or 1/4 of LCF life, or 3750 cycles, whichever is longer. For IFR and FFR, the LCF test time may be reduced to 1/4 the test time for ISR and OCR, or 1000 cycles, whichever is longer.

The LCF test cycle shall be in accordance with table XXI and the following:

(1) inlet test conditions corresponding to tables I and II condition(s) selected by Using Service.

(2) power settings run to initial shaft power levels from tables I and II condition(s) selected by Using Service.

(3) engine controls operating with maximum acceleration/fuel flow schedules and transient load change commands of 1/2 second or less.

(4) output shaft speed operating at maximum operating values from tables I and II condition(s) at high power settings decreasing to the minimum self–sustaining speed at idle.

(5) with customer bleed and power extraction.

Background:

LCF engine testing should be required for engines not tested with AMT duty cycles. These include mainly turboprop and turboshaft engines using the time–at–temperature durability tests from 4.9.1.1.2. For helicopter engines, the LCF engine test is considered more rigorous and demanding than missionized schedules or schedules simulating partial cycles. LCF engine tests add thermal cycling and fatigue effects and have uncovered problems not found in endurance test runs. The LCF test time should be for 1/4 of the engine LCF design life or 1/4 of cold section design life, whichever is longer.

A Strength and Life Analysis should be prepared and a report should be submitted to the Using Service for approval.

A minimum of three sets of fracture, safety, mission and durability critical engine parts, identical to the parts list and configuration of the IFR milestone engine should be subjected to LCF testing as specified below to verify the LCF requirements of 3.4.1.5.2. LCF testing should be conducted in accordance with the requirements of “zero failure verification testing”, as referenced in AFWAL–TR–83–2079. This method outlines the number of units to be tested and the amount of time to be accumulated on each unit without failure to verify the life. The underlying failure distribution should be assumed to be Weibull. A confidence level of 90% should be assumed. The test plan should meet all the requirements of “zero failure verification testing” in order to verify the minimum LCF lives.

The LCF testing in the subparagraphs below should constitute the full effect of strain generated by centrifugal, pressure, and aerodynamic forces as well as thermally generated strains. Dwell times at thrust settings should be sufficient to accomplish stabilization of strains equivalent to those encountered in a service mission cycle. The actual number of cycles, duty times and length of cool down time should be based upon the Strength and Life Analysis Report.

a. One set of fracture, safety, mission and durability critical engine parts should be subjected to official full scale engine AMT/endurance testing which produces LCF damage equivalent to at least one–half the cold parts lives specified in 3.4.1.1.2. These same parts should then be subjected to further testing, either by individual component tests (spin pit) until the minimum LCF lives required by 3.4.1.5.2 are evaluated, or by continued testing in an engine.

b. Two sets of fracture, safety, mission, and durability critical engine parts should be subjected to LCF testing until the minimum LCF lives required by 3.4.1.5.2 are evaluated. The LCF testing should be achieved in any manner appropriate to the full scale development program (e.g., full scale engine tests, individual component tests or combinations thereof). Component testing of these selected parts, except for the combustor, should be performed either with high temperature and loads appropriate for simulating engine maneuver load conditions, or with loads adjusted for material properties at the test temperature. The combustor should be tested only at high temperature conditions.

All repairs and parts replacement should be recorded and reported. Test substantiation (i.e., full scale engine testing) of critical parts may require other parts to receive damage greater than their LCF life. Part replacement or repair may then be accomplished in order to continue the test. The LCF test should be considered successful and the minimum lives of 3.4.1.5.2 verified, if no units fail within the prescribed test time. Failure is defined as generation of a crack size of 3.4.1.7.3. Test times are derived in accordance with AFWAL–TR–83–2079. The test time for an HPT disk is determined in the following example:

ASSUMPTIONS:

1. The failure distribution is Weibull.

2. The shape parameter (ß) is 3.0.

3. The confidence level is 90%.

The Weibull distribution can be expressed as:

$$R(t) \quad e^{(t\ )}$$

The reliability goal for the HPT disk is 99.9% at 4000 LCF cycles.

$$R(4000) = 0.999$$

This is equivalent to the goal of having a characteristic life of 39,990 LCF cycles.

$$\frac{t}{[\ \ \ln R(t)]^{1}}$$

$$\frac{4000}{[\ \ \ln(0.999)]^{1\ 3}}$$

Characteristic Life (eta) = 39,990 LCF cycles

From table 5.1 in AFWAL–TR–83–2079, we can determine the characteristic life multiplier. Three sets of hardware will be tested. Therefore, the characteristic life multiplier for a sample size of 3 and ß = 3.0 is 0.916.

$$(0.916)39{,}990 = 36{,}630$$

Hence, we must test three units for 36,630 LCF cycles without a failure to ensure that the reliability goal of 99.9% at 4000 LCF cycles is met. Failure is defined as generation of a crack size per 3.4.1.7.3.

EVALUATION LESSONS LEARNED

Most LCF tests on the complete engine were not conducted for the LCF life of the parts. This would be a very long and expensive program, even when it is accelerated. The lack of statistical significance of one data point (a single engine) justifies running several identical parts in spin pits. An increase in sample size provides an increase in the statistical significance. Disks, which are expensive to replace, have been tested in a spin pit with as many as five samples. Spin pit testing has certain disadvantages. Specifically, it only simulates the centrifugal loading. It does not simulate the vibratory, aero–elastic, thermal or loads from adjacent disks. Of those loads, simulating the thermal stresses imposed on disks is the most difficult.

3.4.1.5.3 Creep The engine static and rotating parts shall not creep to the extent that it results in unsatisfactory engine operation. Part creep shall not affect disassembly and reassembly of the engine.

REQUIREMENT RATIONALE (3.4.1.5.3)

Growth must be minimized on static and rotating parts to ensure that acceptable engine operation and maintenance is not impaired, and part replacement is not required. Parts must be able to be reused with used or new hardware.

REQUIREMENT GUIDANCE

None.

Background:

Metallic parts, when subjected to elevated temperature and stress simultaneously, will grow in the direction of the stress until rupture occurs. Reasonable shop practices including minor machining or plating repairs is permissible to maintain build tolerances.

Two options that the contractor may utilize in designing the engine and parts are: (1) Design stresses should not exceed the value for 0.1 percent creep strength at the operating metal temperatures and times at temperature based on the design usage. (2) Design stresses should not exceed values associated with utilizing greater than 50 percent of the minimum stress rupture life based on the design usage.

Creep usually occurs at temperatures near the high end of the material's usable temperature range. The phenomena of creep may occur on any part in the engine since it is material dependent and thus a function of material selection and part metal temperature. Assembly, disassembly and operating problems could occur unless creep (of all affected parts) is accounted for in design of the engine.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.5.3 Creep The requirements of 3.4.1.5.3 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.5.3)

Creep and stress rupture analysis is required to establish stress levels that will assure critical parts can operate satisfactorily for the design service life when exposed to the design usage. Final evaluation of creep and stress rupture capability must include extended operation of the full engine.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph: A parts growth analysis shall be performed.

Background:

Analytical prediction of creep and stress rupture life, as a function of design usage, should be accomplished on each creep limited part. Design operating stresses should be established based on past experience that indicates a high probability that satisfactory creep and stress rupture life can be achieved (e.g., 0.1 percent plastic creep life, 0.005 inch diametrical rim growth, 50 percent rupture life, etc.). Creep status of all rotors should be based on a per wheel and not a per blade basis.

Several tests will be run during the engine qualification phase (e.g., operability, vibration and thermal surveys, accelerated mission tests, endurance tests, etc.) and the inspection of critical parts from these engines to evaluate creep life is adequate. These tests should be conducted as early as possible on critical parts (e.g., turbine disk rim lugs and turbine airfoils) to evaluate the parts growth analyses.

Inspection requirements for test engines should include measurement of critical dimensions (e.g., snap and rim diameters, bolt circles, bores) prior to test and at each scheduled inspection interval. Evaluation of dimensional growth as a function of test time should be conducted and included as a part of the interpretation and evaluation of test results.

The scope of engine tests needed to evaluate creep and stress rupture should be identified in an appendix to this specification.

EVALUATION LESSONS LEARNED

None.

3.4.1.6 Strength The engine shall meet all the requirements of the specification during and after exposure to limit loads, singly and in combination. The engine shall not experience catastrophic failure when subjected to ultimate loads, singly and in combination.

REQUIREMENT RATIONALE (3.4.1.6)

Limit and ultimate loading must be addressed since the engine may be exposed to limit and ultimate loading conditions during operation.

REQUIREMENT GUIDANCE

Background:

Limit and ultimate loading requirements are specified in 3.4.1.6.1. Specific guidance for strength requirements is contained in subsequent subparagraphs.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.6 Strength The requirements of 3.4.1.6. shall be evaluated by structural analysis and part, component, and full–scale engine tests.

EVALUATION RATIONALE (4.4.1.6)

Structural analyses and tests are required to evaluate that the engine and its components can meet strength requirements.

EVALUATION GUIDANCE

None.

Background:

The contractor should describe the extent of structural analysis to be performed. Design analysis methods used to demonstrate ability to meet operational requirements should be identified. The schedule for analysis should be identified.

Structural modeling techniques should be described for each class of components (e.g., static structures, disks, shafts, airfoils). It is recommended that the analysis approach employ direct utilization of the thermal heat transfer model required to establish stress as a function of flight conditions. Final analysis should utilize finite element breakups or comparably precise methods to establish stress concentrations and gradients at structural discontinuities (bolt holes, rim slots and posts, radii, blade shrouds and dovetails, etc.).

The contractor should describe the extent of component strength tests performed. Specific tests to be used to demonstrate the ability to meet operational requirements should be identified. The schedule for these tests should also be identified.

Strain gauges should be utilized during component strength test to verify analysis methods relative to nominal stresses and peak surface stresses at concentration details. It is recommended that strain gauges be utilized on each component strength test of static structure. Strain gauge data on rotating structures should be obtained from core and full–scale engine testing.

Stress coat techniques should be used to assist in establishing locations for strain gauges. Photoelastic modeling and test is suggested for better understanding of maximum stresses and gradients at complicated structural details (e.g., blade dovetails, rim slots).

EVALUATION LESSONS LEARNED

None.

3.4.1.6.1 Factors of safety The following factors of safety shall be applied to design usage induced loads to establish limit and ultimate conditions: (a) .

REQUIREMENT RATIONALE (3.4.1.6.1)

Sufficient factors of safety must be determined to assure adequate safety margins exist in designs.

REQUIREMENT GUIDANCE

The following table should be used to tailor the specification paragraph:

(a):

LOAD TYPES	LIMIT	ULTIMATE	
		a	b
Externally applied loads	1.0	1.0	1.5
Thermal loads	1.0	1.0	1.5
Thrust loads	1.0	1.2	1.0
Internal pressures	1.5	1.0	2.0
Aircraft flow field loads	1.0	1.0	1.5
Crash loads	N/A	1.0	1.0

NOTES:

(1) For all castings, a factor of safety of 1.33 shall be applied to the limit and ultimate load factors specified above, unless the castings have been fully characterized.

(2) Two combinations (a & b) shall be used for establishing ultimate loading conditions required in other paragraphs in this section.

REQUIREMENT LESSONS LEARNED

Catastrophic failures of cast parts and pressure vessels have occurred due to porosity and poor manufacturing processes. These parts were designed with a 1.5 factor of safety for ultimate load conditions. A more graceful (i.e., non–catastrophic) failure would have occurred if the hardware had been designed with a 2.0 factor of safety.

4.4.1.6.1 Factors of safety The requirements of 3.4.1.6.1 shall be evaluated by the analyses and tests.

EVALUATION RATIONALE (4.4.1.6.1)

Factors of safety must be evaluated to assure adequate safety margin.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

The test should be conducted in accordance with 4.4.1.6.

Background:

Strain gauges and other instrumentation should be used during tests to evaluate analysis methods. It is recommended that tests be conducted progressively to ultimate load conditions.

EVALUATION LESSONS LEARNED

None.

3.4.1.6.2 Blade and disk deflection The blades and disks shall not contact any static parts of the engine other than seals and shrouds. During all phases of engine operation including surge and stall occurrences. Seals and clearances shall remain effective under all internal and external operational loads.

REQUIREMENT RATIONALE (3.4.1.6.2)

Sufficient rigidity must be provided so that the engine can operate to the limit loads and repeated loads that occur within the flight envelope without detrimental damage.

REQUIREMENT GUIDANCE

None.

Background:

High thrust bearing loads cause rotors to shift thereby increasing clearances or causing detrimental damage to static or rotating hardware. This happens more in the turbine where high temperature increases creep and thermal stress. The resultant rub opens the blade clearances or can damage hardware where rubs were not intended to occur.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.6.2 Blade and disk deflection The requirements of 3.4.1.6.2 shall be evaluated by the analyses and tests.

EVALUATION RATIONALE (4.4.1.6.2)

Structural analysis and test are required to evaluate that blade and disk deflection does not result in contact with any static parts of the engine other than seals and shrouds. Blade and disk rigidity must be evaluated to ensure flight safety against blade failures and titanium fires.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

The test should be conducted in accordance with 4.4.1.6.

EVALUATION LESSONS LEARNED

X–ray photography has been used to determine that seals and clearances are effective under all operational loads.

3.4.1.6.3 Containment The engine shall radially contain a fan, compressor or turbine blade failure at the blade root section in the fillet below the platform which is released within the engine, and shall not result in uncontained engine fire, at any engine operating speed. Engine components with rotating parts shall contain any failed rotating part at any engine operating speed.

REQUIREMENT RATIONALE (3.4.1.6.3)

For safety of personnel and equipment, it is imperative to contain all rotor system failures within the engine.

REQUIREMENT GUIDANCE

Background:

Initial design and sizing of parts and components should include criteria to provide containment. Containment design procedures should be based on past experience and test data that establishes material containment capabilities. Initial procurement documentation should specify if containment is to be provided by the engine static structure or by shielding material in the airframe structure. Containment material/features can be tailored or distributed non–uniformly to minimize/optimize weight for using vehicle application.

REQUIREMENT LESSONS LEARNED

The consequences of engine component failures increase significantly when the failed parts are not contained. For example, there have been many cases of uncontained blades that penetrated the airframe and severed airframe fuel tanks and lines, hydraulic lines, etc., resulting in loss of the aircraft. Excessive weight penalties have occurred for containment provisions for larger mass items such as disks and seals; therefore, past designs have provided containment for only blades. Uncontained rotor failures result in numerous aircraft losses, mission aborts, and fatalities.

4.4.1.6.3 Containment The requirements of 3.4.1.6.3 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.3)

Evaluation of engine and component containment capability is necessary to avoid the occurrence of uncontained failures.

EVALUATION GUIDANCE

Background:

The engine contractor should perform a blade containment analysis which relates the released blade kinetic energy to the energy required for containment. The containment analysis should be provided at the initiation of EMD to provide confidence that the design will contain certain failures. The analysis should be substantiated/correlated with containment tests. Prior failures on identical structures can be discussed in the analysis and used for substantiation of the containment test. Containment tests are necessary since many uncertainties exist with the various containment analysis procedures (e.g., dynamic considerations of pressure loading versus ballistic loading, effectiveness of containment structure due to varying geometry, material capability forces, etc.).

The tests should be conducted at or above the maximum allowable rotor speeds and maximum operating temperatures. The blades selected for the test (those blades determined to be the most critical) should be modified to fail at a predetermined speed. The test should be considered satisfactorily completed when all damage is contained.

The engine contractor should also provide component containment analyses and/or conduct component testing. The analyses/tests should be conducted to ensure all engine components utilizing rotating parts will contain any rotating part failure at maximum transient speed.

EVALUATION LESSONS LEARNED

Early use of containment criteria and analysis can avoid a redesign. For safety of personnel and equipment it is very desirable to have all blade failures contained within the engine.

3.4.1.6.4 Blade out Subsequent to a single blade failure, with resulting secondary loss of another blade in the same stage at maximum allowable transient speed, the engine shall not experience uncontained fire; catastrophic rotor, bearing, support, or mount failures; overspeed conditions; leakage from flammable fluid lines, or loss of ability to shutdown the engine. The design conditions for a single blade failure with secondary loss as stated above shall be as follows: (a) .

REQUIREMENT RATIONALE (3.4.1.6.4)

The engine must possess adequate structural integrity after blade loss so that a stable time period exists without uncontained catastrophic destruction to allow time for pilot awareness and appropriate action. In addition, to the requirement for containment of the failed blade, the secondary failure modes that would result in catastrophic failure must be avoided.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): a. Blade loss loads for conventional blades should be based on the imbalance equivalent to fracture in two blade attachments at the minimum neck section above the outermost retention feature.

b. Blade loss loads for integrally bladed rotors should be based on the imbalance equivalent to liberation of two airfoils including the fillet material down to the rotor rim diameter.

Blade out conditions should also address the possibility of interactive blade/disk vibration modes resulting from imbalance or acoustics.

Background:

Design for blade failures should include the fan, compressor and turbine rotors individually. A single blade failure results in blade out loads equivalent to two blades out due to subsequent secondary damage. Furthermore, adequate damping must be provided so that a single blade failure does not cause engine operation at a critical speed which would cause further failures of other hardware. Blade out loads are needed for proper design of aircraft engine mounts.

REQUIREMENT LESSONS LEARNED

Blade failures have caused uncontained fire and catastrophic rotor failure in compressors that use titanium for blades and cases. An imbalance condition caused by loss of blades can lead to bearing and rotor support failure. Interactive blade/disk vibration modes have resulted in the failure of a blade retainer and subsequent loss of a fan rotor assembly.

4.4.1.6.4 Blade out The requirements of 3.4.1.6.4 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.4)

Structural analyses and tests are required to evaluate that blade failure does not cause the engine to experience uncontained fire; catastrophic rotor, bearing, support or mount failures; overspeed conditions; leakage of flammable fluids or loss of ability to shut down the engine.

EVALUATION GUIDANCE

Background:

Evaluation of blade out requirements should include analyses of the fan, compressor and turbine sections of the engine. Evaluation of the most critical rotors should be accomplished by an engine test. Blade out testing could destroy a complete engine. Hence, it is may be permissible to perform part testing instead of full–scale engine testing. This may be done in conjunction with the containment evaluation.

Failure should be assumed to occur at the maximum transient rotor speed (i.e., the maximum normal operating speed plus adjustments to account for deterioration, control and measurement tolerance, engine–to–engine variations, and Idle–to–Intermediate acceleration overshoots).

EVALUATION LESSONS LEARNED

Evaluation by engine test during development is very rare. As part of FAA certification, the GEAE manufactured CF6 was required to demonstrate compliance with blade out requirements via a full–scale engine test.

3.4.1.6.5 Overspeed/overtemperature The engine shall meet all the requirements of the specification during and after the following overspeed and overtemperature conditions:

a. Engine rotor speeds of (a) percent of the maximum allowable steady–state speed at the maximum allowable turbine temperature or first stage turbine rotor inlet gas temperature limit of 3.2.2.11 for five minutes.

b. (b) temperature of at least (c) °C ((c) °F) in excess of the maximum allowable temperature or above the first stage turbine rotor inlet gas temperature limit of 3.2.2.11, and at maximum allowable steady–state rotor speed for five minutes.

c. Engine component rotor speeds of (d) percent of the maximum allowable steady–state speed for five minutes.

d. Engine component fuel, lube and hydraulic inlet temperatures of (e) °C ((e) °F) above the maximum allowable operating fluid temperature at the maximum allowable steady–state speed for five minutes.

e. For turboprop/turboshaft engines, the power turbine shaft speed shall not exceed (f) percent of the transient speed limit, or the predicted speed attained following loss of load with the engine at Intermediate power and the power turbine running at the highest rated speed, whichever is greater. The predicted maximum speed shall be specified herein.

REQUIREMENT RATIONALE (3.4.1.6.5)

These requirements are needed to provide an operational margin for rotor structural integrity while allowing continued use. Requirement a. provides for an overspeed condition while operating at normal maximum temperature conditions. Requirement b. provides for an overtemperature condition while operating at normal maximum allowable speed conditions.

These requirements a. and b. are needed to provide an operational margin for engine rotor structural integrity while allowing continued use (should not yield) as opposed to the requirement of 3.4.1.5 which is intended to represent the ultimate strength of the material (should not burst). Requirements c. and d. are specified to provide an operational margin for component rotor structural integrity while allowing continued use.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of 115 percent, minimum

(b): insert HPT blade metal, HPT inlet, measured, or gas

(c): a value between 42°C (75°F) to 45°C (81°F)

(d): 115 percent

(e): 25°C (45°F)

(f): A value of 115 percent, minimum

Background:

The temperature description (e.g., HPT blade metal, HPT inlet, measured, gas, etc) should be consistent with the performance rating temperature description.

REQUIREMENT LESSONS LEARNED

Past experience indicates that the engine can overspeed or exceed design temperature due to control system malfunctions or other engine operating anomalies.

4.4.1.6.5 Overspeed/overtemperature The requirements of 3.4.1.6.5 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.5)

Overspeed/overtemperature of the engine must be evaluated to ensure structural integrity of rotating parts.

EVALUATION GUIDANCE

Background:

An analysis depicting the overspeed and overtemperature capability of the engine should be provided. Overspeed and overtemperature tests should be conducted to substantiate/correlate the analysis.

For the overspeed test, all rotors should be subjected to engine operation for a stabilized period of at least five minutes duration at (value specified in 3.4.1.6.5) percent of maximum allowable steady–state speed at the engine's maximum allowable temperature. Following the test, parts and assemblies should be within allowable dimensional limits and there should be no evidence of imminent failure. If a cold spin pit is used for hot flow components, speed should be added to compensate for temperature effects (with Using Service approval).

Upon successful completion of the overspeed test, the same engine should be operated at a (HPT blade metal, HPT inlet, measured, gas) temperature of at least (value specified in 3.4.1.6.5) in excess of the maximum allowable temperature and at no less than maximum allowable steady–state speed for five minutes. Following the test, parts and assemblies should be within allowable dimensional limits and there should be no evidence of imminent failure.

For engines with more than one rotor system, the test may be performed separately for each rotor system. Rig tests may be performed with Using Service approval.

EVALUATION LESSONS LEARNED

None.

3.4.1.6.6 Disk burst speed The minimum loaded disk burst speed of the complete disk assembly shall be (a) percent when the disk is subjected to the maximum temperature gradient and maximum material temperature that will occur for that part.

REQUIREMENT RATIONALE (3.4.1.6.6)

This requirement assures adequate margin against the risk of a disk burst in service.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 115 to 122 percent of the maximum allowable steady–state rotor speed or 5 percent above maximum transient rotor speed, whichever is greater

Background:

The 122 percent represents a factor of safety of 1.5 (centrifugal stresses vary as the square of speed). The loaded disk burst requirement is necessary since stresses on the disk are obviously greater when it is loaded with blades. The material properties and stress distributions are more severe when subjected to the maximum temperature gradient and maximum temperature conditions for the part.

Proposed values for disk burst speeds may provide a compromise between crack growth capability and tensile strength.

Generally, “damage tolerant” materials provide better crack growth capability but lack high tensile strength (burst) capability.

REQUIREMENT LESSONS LEARNED

The FAA uses 120 percent of maximum allowable steady–state rotor speed. The USN and USAF have allowed minimum disk burst speeds of 115–117 of maximum allowable steady–state rotor speed. For titanium fan disks, the USAF has required minimum disk burst speeds of 130 percent of maximum allowable steady–state rotor speed.

4.4.1.6.6 Disk burst speed The requirements of 3.4.1.6.6 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.6)

The required disk burst speed must be evaluated to prevent the occurrence of a catastrophic failure.

EVALUATION GUIDANCE

Background:

The Strength and Life Analysis of 4.4.1.5 should include a detailed evaluation of the operating environment and stress levels seen by each engine disk. The analysis should provide an initial evaluation of the burst capability of each disk. This information should be substantiated/correlated with disk burst spin pit testing.

Disk burst testing should be conducted to evaluate whether the burst margin requirement of 3.4.1.6.6 can be met with a minimum tensile strength disk (based on the minimum properties specified in 3.4.1.3). Disk burst testing should be conducted on all engine disks. As a minimum, disk burst tests may be conducted on the most limiting rotor (disk with the minimum burst capability) of each module.

Disks should be operated at burst speeds no less than those of 3.4.1.6.6 while exposed to the maximum temperature gradient and maximum material temperature that would occur for that part. These conditions should be maintained for a minimum of 30 to 60 seconds. The test should be considered successfully completed if there is no evidence of imminent failure.

Since the blades may actually fail before the disk, substitute blades(dummy blades) may be used in lieu of actual disk blades during the evaluation.

EVALUATION LESSONS LEARNED

None.

3.4.1.6.7 Output shaft torque limits For turboprop and turboshaft engines the maximum allowable steady–state delivered shaft torque (mechanical) limit shall be at least (a) percent greater than the rating value.

REQUIREMENT RATIONALE (3.4.1.6.7)

A limit is required to provide a margin of torque to prevent catastrophic component failure.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 20 percent

REQUIREMENT LESSONS LEARNED

Past experience indicates that the engine output shaft torque can increase because of a malfunction or other anomalies. Some engine contractors have asked for a deviation. They felt they have a good control of output shaft torque and torque limits could be lower. Other systems (helicopters) with torque limits depend upon pilot action for limit observance.

4.4.1.6.7 Output shaft torque limits The requirements of 3.4.1.6.7 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.7)

Evaluation by analysis and tests are needed to assure the engine will not be degraded by shaft torque operation up to the limit.

EVALUATION GUIDANCE

Background:

The test can be conducted on the total engine or only on the effected component. The torque should be conducted for a minimum time of 5 minutes.

EVALUATION LESSONS LEARNED

None.

3.4.1.6.8 Output shaft speed limits For turboprop and turboshaft engines the maximum allowable steady–state delivered shaft speed (mechanical) limit shall be at least (a) percent greater than the value of 3.2.1.1. The shaft shall be able to operate at this speed for at least (b) and function satisfactorily thereafter. Following loss of load, the output shaft speed shall not exceed the maximum shaft speed predicted with the engine at Intermediate power and the output shaft running at the highest rotor speed of 3.2.1.1.

REQUIREMENT RATIONALE (3.4.1.6.8)

An output shaft speed limit is required to provide a margin of speed to prevent catastrophic failure.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of 15 percent, minimum.

(b): A value of 5 minutes, minimum.

REQUIREMENT LESSONS LEARNED

Past experience indicates that the engine output shaft speed can increase because of a malfunction or other anomalies. Engine contractors have asked for a deviation on the speed requirement since they believe the newer control systems will prevent any overspeed.

4.4.1.6.8 Output shaft speed limits The requirements of 3.4.1.6.8 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.8)

Evaluations by analyses and tests are needed to prevent catastrophic component failure.

EVALUATION GUIDANCE

Background:

This test can be conducted in conjunction with the rotor overspeed test. The test should prove structure integrity and parts are not stressed to yield.

EVALUATION LESSONS LEARNED

The evaluation was conducted by analysis or similarity on some engines in the past.

3.4.1.6.9 Pressure vessel/case All engine cases and pressure loaded parts and components shall withstand the ultimate loading conditions defined in 3.4.1.6.1. The cases must remain intact, although permanent deformation and distress, requiring repair or replacement, is permitted. Engine cases shall not fail due to combustion process burning or erosion.

REQUIREMENT RATIONALE (3.4.1.6.9)

Pressurized vessels, cases and components require internal and external load safety margins to preclude failure (e.g., burst and hazardous venting conditions). Also, pressurized vessels, cases and components must be protected from degradation caused by combustion processes or erosion.

REQUIREMENT GUIDANCE

Background:

The pressure vessels should be designed to meet the ultimate load capability of 2.0 times the maximum operating pressure plus 1.5 times the maneuver loads plus 1.5 times the thermal loads. The engine should withstand the combined affects of these loads without catastrophic failure.

REQUIREMENT LESSONS LEARNED

Experience has shown that if a case is designed to a factor of safety of two, problems of rupture, LCF and burn–through are reduced. The TF39 burner case was initially designed close to yield and exhibited problems in service due to LCF and burn–through. After the engine was redesigned to eliminate these problems, the factor of safety on rupture was checked and found to be approximately two. In the F101 program a factor of safety of two was mandated. During testing of this engine it was found to be capable of tolerating a burn–through without rupture. The F110 program required a factor of safety of two. Subsequently, the F110 has experienced several burn–throughs without rupture.

4.4.1.6.9 Pressure vessel/case The requirements of 3.4.1.6.9 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.6.9)

The engine pressure vessels and gas pressure loaded components must be capable of withstanding the combined operating ultimate loads without catastrophic failure to assure that the engine has been designed with appropriate safety margins, and that it can operate satisfactorily for the required design usage and service life.

EVALUATION GUIDANCE

Background:

The engine contractor should provide an analysis of all pressure loaded parts and components. The analyses should show that all pressure loaded parts and components can meet the requirements of 3.4.1.6.9 when constructed with minimum strength materials, as defined in 3.4.1.3. The analyses should be substantiated/correlated with pressure vessel/case testing.

All pressure loaded parts and components should be tested to at least two times the maximum operating pressure in combination with the external ultimate loads based on the external loads encountered during engine operation. These tests should be conducted at the maximum allowable temperature or at a test pressure adjusted to account for the differences between operating and test temperatures.

The above tests are qualification tests to demonstrate that the design meets the strength requirements. It is also recommended that the production acceptance/quality control requirements include proof pressure test of each pressure loaded component to 1.33 – 1.50 times maximum operating load pressure. Proof tests of each article are required to screen detrimental porosity, crack/void, below blueprint tolerances or other detrimental anomalies that would reduce the life of the component.

EVALUATION LESSONS LEARNED

None.

3.4.1.6.10 Pressure balance The engine thrust bearings shall provide sufficient thrust load to ensure satisfactory bearing operation during the design service life.

REQUIREMENT RATIONALE (3.4.1.6.10)

This loading requirement is necessary to assure against problems due to inadequate engine pressure balance design. Transient engine loads on output shaft components shall be “taken–out” by engine structure.

REQUIREMENT GUIDANCE

Background:

The contractor should assure that his practices for bearing design include requirements for pressure balance to assure that load and direction are adequate to achieve satisfactory bearing operation. The requirement is not intended to restrict the thrust load to any one direction or even any specific minimum magnitude but rather to provide a pressure balance system which maintains bearing loads sufficient to assure adequate bearing life. It is necessary that under any steady–state operating condition a minimum thrust be maintained on the thrust bearing in order to prevent skidding. Standard design practices are to design the pressure balance system to provide the minimum required bearing thrust load at the worst (minimum load) steady–state condition and then check the highest thrust load on the bearing by adding the change in engine generated loads to this minimum load. If the thrust load is required to be unidirectional, the maximum bearing load will be greater than if the thrust load is allowed to pass through zero during some transient condition. Restricting the contractor to unidirectional loads could, therefore, result in shorter bearing lives than if bearings were allowed to be null loaded. Trade studies should be performed on sizing bearings to take unidirectional loads vs allowing crossover. Null loaded bearings can lead to rotor vibrational problems within the operating envelope which can lead to bearing failure or reduced life. If crossover is allowed to occur, the loss of loading on the bearing can result in transient rotor dynamic vibration. Transient rotor dynamic vibration in the operating range may result in pilot discomfort and distraction, and cause durability problems.

Engine/airframe system interactions can be excited by these types of vibration and cause similar problems. These interactions are typically difficult to predict and are usually not found until late in the integration effort.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.6.10 Pressure balance The requirement of 3.4.1.6.10 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.10)

The engine pressure balance system must be evaluated to ensure compliance with the requirement of 3.4.1.6.10.

EVALUATION GUIDANCE

Background:

The engine contractor should provide an analysis of the engine pressure balance system. The analysis should show that loads imposed on the engine bearing(s) are of sufficient magnitude to assure adequate bearing operation without skid damage. The analysis should be substantiated/correlated with engine testing.

An engine should be suitably instrumented and tested to demonstrate that the loads imposed on the engine bearing(s) are of sufficient magnitude to assure adequate bearing operation without skid damage at all power settings throughout the engine operating envelope. This test should be conducted in an altitude test cell to simulate altitude and ram conditions representative of operational use.

EVALUATION LESSONS LEARNED

None.

3.4.1.6.11 Gyroscopic moments The engine shall meet all the requirements of the specification at maximum allowable steady–state engine speeds when subjected to the rotational velocities and accelerations within the flight envelope and the gyroscopic moment conditions specified below:

a. A steady angular velocity of (a) radians per second around any axis in a plane perpendicular to the rotor axis, combined with a $\pm$ 1g vertical maneuver load for a total of (a) seconds.

b. A steady angular velocity of (b) radians per second in any axis in a plane perpendicular to the rotor axis for a cyclic life of 10^7 cycles at all load factor conditions within the flight envelope.

REQUIREMENT RATIONALE (3.4.1.6.11)

Engine flight loads are increased due to rotations and accelerations that occur during aircraft rolling, pitching and yawing maneuvers. The engine must be designed to resist these loads at the limiting conditions.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): this requirement is principally a spin departure criteria requiring a high angular velocity for a short period of time. Angular velocities as high as 3.5 radians per second for fighter aircraft and 1.5 radians per second for bomber and cargo aircraft for 15 seconds are appropriate. For rotary wing aircraft, angular velocities of 2.5 radians per second for 30 seconds is appropriate. The 15–second and 30–second durations are considered to be cumulative exposure time.

(b): this requirement is principally a maximum angular velocity that may be experienced numerous times for long periods such as tight turns or numerous gust induced nacelle oscillations of a pylon mounted engine. For fighter aircraft this angular velocity value is approximately 1 radian per second in pitch or yaw. The angular velocity should be

the vector sum of the angular velocities in pitch and yaw. Therefore, it is recommended that a steady angular velocity of 1.4 radians per second in any axis in a plane perpendicular to the rotor axis for a cyclic life of 10^7 cycles at all load factor conditions within the flight envelope. For rotary wing aircraft, a steady angular velocity of 0.9 radians per second in any axis in a plane perpendicular to the rotor axis for a cyclic life of 10^7 should be used.

Alternately, for rotary wing aircraft, use 0.4 radians per second for 24% of duty cycle life, 0.9 radians per second for 20% of duty cycle life, and 1.4 radians per second for 1% of duty cycle life.

REQUIREMENT LESSONS LEARNED

A USAF study showed that a bearing load could be increased by as much as 15 times by a 3.5 rad/sec aircraft turn rate. The force on an individual rotor blade could be three times the magnitude of the corresponding aerodynamic force. Gyroscopic forces are cyclic in nature, thus tending to reduce the cyclic life of rotor blades. The vector sum of pitch and yaw rates is very important from the standpoint of the gyroscopic moments produced on the engine.

The fighter/attack aircraft and the lighter weight engines with tighter clearances will make the gyro requirement even more essential. Simulation data from an F4 aircraft produced yaw rates of less than 0.5 rad/sec and pitch rates less than 1 rad/sec. Intentional departure stall maneuvers have been used as a “last chance” evasive action against SAM’s. This action results in high yaw rates and gyro loads. Yaw rates of 2.7 rad/sec were demonstrated years ago, in flight, during accelerated departure stall maneuvers, with the A–7 aircraft. Yaw rates of 3.1 and 3.2 rad/sec were measured on F–4 and F–15 aircraft. A recent mishap involving the F/A–18 aircraft was reported as yawing in excess of 200 degrees per second (3.5 rad/sec). The aircraft had violently departed controlled flight due to asymmetrical positioning of the leading edge flaps. Testing of the F109–GA–100 engine on the gyro rig showed significant mechanical interference occurring at 2 rad/sec. The manufacturer was able to correct the problem without major redesign.

4.4.1.6.11 Gyroscopic moments The requirements of 3.4.1.6.11 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.6.11)

Aircraft are exposed to gyroscopic moments during normal operation and the ability of their engines to withstand those conditions must be evaluated.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph: Prior to installation on the test stand, the engine shall be assembled with special emphasis placed on measuring and recording clearances between blades and cases and radial and axial rotor clearances. Rub probes shall be installed around compressor and turbine cases at symmetrical locations and at blade tip locations as designated in the pre–test data. Instrumentation shall be sufficient to permit measurement of rotor deflection and shift under gyroscopic loads. Strain gage instrumentation shall be provided to measure stresses at critical locations. Sufficient instrumentation of the oil system shall be provided to evaluate the oil system’s ability to scavenge and function properly during the test.

The engine shall be operated with an inlet configuration and exhaust nozzle as specified in the pre–test data. Prior to the test, the engine shall be subjected to a performance calibration.

The test shall be conducted with the gyroscopic rig operated in incremental steps of 0.5 rad/sec from 0.5 rad/sec up to and including 3.5 rad/sec. At each step, the engine shall be operated as follows:

a. Idle for one minute.

b. Accelerate from Idle to maximum allowable rotor speed in 30 seconds.

c. Dwell at maximum allowable rotor speed 10 seconds or time sufficient to record data.

d. Decelerate from maximum allowable rotor speed to Idle in 30 seconds.

e. Stop rig and engine for visual check of rub.

NOTE: At gyro loads above 1.5 rad/sec, snap accelerations and decelerations may be made to reduce time exposure. The total time at 3.5 rad/sec gyro load shall not exceed the time specified in 3.4.1.2.2.

The above test shall be conducted with the gyroscopic rig rotating in one direction and then the test shall be repeated with the rig rotating in the opposite direction. At the completion of the test, the engine shall be subjected to a post–test performance calibration and then disassembled for inspection.

The test shall be satisfactorily completed when, in the judgment of the Using Service: (1) The post test calibration reveals no significant loss in performance, (2) the engine and its systems operated properly during the test, (3) structural loads were within acceptable limits, and (4) teardown inspection reveals no evidence of excessive blade rubbing or evidence of impending failure.

Background:

The engine contractor should provide a gyroscopic moments analysis. As a minimum, the analysis should discuss engine mounts, bearings, and bearing support structure capability while exposed to the gyroscopic moment conditions specified in 3.4.1.6.11. The analysis should be substantiated/correlated with a gyroscopic moment engine test.

EVALUATION LESSONS LEARNED

None.

3.4.1.7 Damage tolerance Fracture/safety and mission critical engine parts shall be damage tolerant with defects resulting from material quality, manufacturing processing, and handling damage during the periods specified in 3.4.1.7.4.

A Damage Tolerance Control Plan shall be prepared to identify and schedule each of the tasks and interfaces in the functional areas of design, material selection, manufacturing control, and inspection of fracture, safety, and mission critical parts. The tasks to be identified in the plan include:

a. Design concepts/material/weight/performance/cost trade studies.

b. Damage tolerance analysis, development testing and proof of compliance testing tasks.

c. Parts list that identifies fracture, safety and mission critical parts, location,s and special controls required to meet damage tolerance requirements (e.g., material specification controls, quality assurance requirements, etc.).

d. Zoning of drawings for fracture, safety and mission critical parts to identify critical locations and associated quality control requirements, defect locations, orientation, inspection method, and acceptance standards. The use of alternate procedures for identification of critical locations, etc., on drawings may be proposed.

e. Basic materials fracture data (e.g., K_{IC}, K_C, K_{ISCC}, da/dn).

f. Identification and control of fracture toughness and crack growth rate properties in the material procurement and manufacturing process specifications.

g. Traceability requirements on all tiers of procurement, processing, fabrication, and assembly for fracture critical components. Serialization or time coding requirements for tracking operational exposure of individual components.

h. Quality control requirements during component manufacture. Identification of procedures for certifying and monitoring subcontractor, vendor, and supplier inspection and quality control. Nondestructive inspection

requirements for use during depot and base level inspections including supporting manuals (technical orders) and equipment needs.

REQUIREMENT RATIONALE (3.4.1.7)

The requirement protects fracture, safety, and mission critical parts from potentially degrading effects of handling damage and/or material, manufacturing, and processing anomalies which could result in premature engine failures and loss of aircraft. This process ensures proper material choices, control of operating stress levels, use of fracture resistant design concepts, manufacturing and process controls, and the use of reliable inspection methods during production and in–service maintenance. Attainment of engine damage tolerance is achieved through application of detail requirements in the functional areas of design, materials selections, manufacturing control, and inspections.

REQUIREMENT GUIDANCE

Background:

Damage tolerance requirements should be applied to all fracture, safety, and mission critical engine parts. Damage tolerance requirements should not, in general, be applied to durability critical and durability non–critical parts. Damage tolerance requirements may be applied to durability critical parts to (1) identify components that are sensitive to manufacturing variables and pre–damage which could cause non–economical maintenance (e.g., blades), or (2) aid in establishing economic repair time or other maintenance actions.

Damage tolerance can be achieved by proper material selection and control, control of stress levels, use of fracture resistant design concepts, manufacturing and processing controls, and the use of reliable inspection methods.

The requirement for a Damage Tolerance Control Plan and the associated schedule should be contained in the contract. Required actions in areas that are identified during development should be incorporated as a part of the production engine manufacturing, quality control, and life management requirements. Engineering changes involving fracture, safety, or mission critical parts should be classified as Class I in accordance with MIL–STD–973.

Crack growth analysis should be performed as an integral part of detail design of fracture, safety, and mission critical parts. Initial flaws (cracks) should be assumed in highly stressed locations such as edges, fillets, bolt holes, and blade slots. Imbedded defects (cracks) should also be assumed at large volume locations such as a disk live rim, web, and bore. Growth of these assumed initial flaws as a function of imposed stress cycles should be calculated. Total growth period from initial flaw size to part failure is thus derived. Trade studies on (1) inspection methods and assumed initial flaw sizes, (2) stress levels, (3) material choice, and (4) structural geometry can be made until the safety limit is sufficiently large such that the need for in–service inspection is eliminated or minimized. Damage tolerance design procedures that account for distribution of variables (i.e., probabilistic analysis) that affect growth of imbedded defects are permitted.

The design objective should be to qualify parts as in–service noninspectable to eliminate the need for depot inspections prior to achieving one design lifetime. As a minimum, parts should be depot or base level inspectable for the hot parts life specified in 3.4.1.1.1.

The imposition of damage tolerance requirements will add weight to fracture, safety, and mission critical parts which impacts the thrust–to–weight potential. It is necessary to minimize this impact by using fracture mechanics life prediction systems and probabilistic design analyses which are not overly conservative.

REQUIREMENT LESSONS LEARNED

Recent examples of optimized part designs following the ENSIP/DTD guidelines include the F109–GA–100, F100–PW–220 ILC, F100–PW–229 IPE, F110–GE–129 IPE, and F119–PW–100 ATF engine designs. These design configurations have shown that damage tolerance requirements can be met with small or modest increases in overall engine weight, will have little impact on engine performance, and will provide greatly improved engine durability while significantly reducing weapon system life cycle cost.

Most of the tasks to be contained in the Damage Tolerance Control Plan have been accomplished by engine manufacturers in past development and production programs. However, the durability and damage tolerance requirements established here impose tighter controls and more interface involvement between the functional areas.

4.4.1.7 Damage tolerance The requirements of 3.4.1.7 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.7)

A damage tolerance analysis is needed to support damage tolerance design concepts, material selection, maintenance requirements, and cost and weight impacts. Damage tolerance tests are required to support material selection and trade studies, obtain early evaluation of allowable stress levels and environments, and to verify analysis procedures and damage tolerance characteristics.

EVALUATION GUIDANCE

Background:

Damage tolerance analysis should be conducted early in the program. Early analysis will enable identification of structural sensitive areas which do not meet the desired crack growth intervals and design changes can be introduced early with minimum impact. Emphasis on conducting early analysis will minimize occurrence of deficiencies in later development and proof of compliance testing and facilitate meeting important full–scale test milestones. Damage tolerance analysis should be conducted on each part classified as fracture, safety, or mission critical in 4.4.1.4.

Each location of the part should be surveyed to determine the most critical locations for the assumed initial flaw considering such features as edges, fillets, holes, blade slots, and other high stressed areas. Stress/environment spectra should be developed for each part and locations to be analyzed. Damage tolerance analysis that addresses imbedded defects can be based on probabilistic methods that account for the distribution of variables. Average (B50) fracture mechanics properties can be used in the crack growth and residual strength calculations.

Stress intensity should be based on the structural geometry and assumed flaw geometry. The critical stress intensity should be based on the required residual strength load level and temperature conditions that exist at the component locations being analyzed. Limiting stress intensity should include consideration of the allowable stress intensity accounting for the effect of vibratory stresses. Certain minimum levels of vibratory stress (e.g., 10 ksi) should be assumed to exist to identify sensitive components.

The requirement for a damage tolerance analysis and the associated schedule should be contained in the contract specifications.

Early testing should include tests of simple specimens and small elements. Specimen and element tests should be conducted on representative structural details and materials. Final evaluation of damage tolerance should be by part and full–scale engine testing. The full–scale engine test may be required to demonstrate proof of compliance under realistic environments.

Parts should either be preflawed or contain natural flaws. Parts can be preflawed by scratching, etching, electro–discharge machining (EDM), or tackwelding of the surface. Some cycling of parts may be required to “grow” the preflaw to the necessary size prior to actual testing.

Test results should be correlated with predictions of flaw growth intervals and critical flaw sizes. Parts previously tested to evaluate low cycle fatigue should be utilized for damage tolerance testing. The damage tolerance test program should be of sufficient scope to evaluate the fracture, safety, and mission critical parts.

Deletion of evaluation of certain fracture, safety, and mission critical parts can be proposed based on similarity of materials and structural configurations and demonstrated knowledge of the applied stresses.

Electro–discharged machined preflaws are “clean” (sharply defined). Tackweld preflaws are “dirty” (difficult to determine crack length and to differentiate crack lines from thermal cracks).

EVALUATION LESSONS LEARNED

None.

3.4.1.7.1 Residual strength The residual strength shall be equal to the maximum stress that occurs during design usage conditions. Residual strength requirements shall be established for all damage tolerant designed parts and components. Associated static and dynamic loading conditions for these parts and components shall include: (a) .

REQUIREMENT RATIONALE (3.4.1.7.1)

The load carrying capability of fracture, safety, and mission critical parts, with “damage” present, must remain above some minimum value during part design service lives.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): a. Maximum limit maneuver loading

b. Maximum pressure loading

c. Maximum speed loading

d. Maximum temperature effects

Background:

The engine contractor should provide the residual strength requirements for all parts. The minimum residual strength for each part (and location) should be equal to the maximum stress that occurs during design usage conditions. Control system overshoot and engine deterioration should be included. Burst margin overspeed conditions should be excluded.

Analytical studies have shown that not every part location will be limited by a crack growing to a calculated critical stress intensity equal to the material’s fracture toughness. Some part locations will in fact be life limited by cracks growing to a predicted vibratory threshold μK_{th} HCF.

Where $K_{max\ allowable}$ LCF = μK_{th} HCF/(1 – R)

and R = ($_{steady}$ – $_{vibratory}$) / ($_{steady}$ + $_{vibratory}$)

$_{steady}$ = maximum operating stress neglecting vibratory stress

$_{vibratory}$ = peak to peak vibratory stress

and μK_{th} HCF = f(R, temp)

Overspeed residual strength requirements need not be considered for those part locations limited by cracks reaching a calculated vibratory threshold. One overspeed cycle occurring at a crack size equal to the vibratory threshold creates less damage (change in crack size) than additional LCF/HCF crack growth from the vibratory threshold to a maximum stress intensity (K_{CRIT}) defined by the material fracture toughness.

For those locations not limited by vibratory stress concerns, the part’s maximum allowable crack size should be limited to a size that will survive the maximum design stress that occurs on the last cycle of the calculated safety limit.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.7.1 Residual strength The requirements of 3.4.1.7.3 shall be evaluated by the analyses and tests.

EVALUATION RATIONALE (4.4.1.7.1)

Evaluation of residual strength by analyses and tests are required to ensure less than critical size flaws will not grow and cause failure due to the application of the required residual strength load.

EVALUATION GUIDANCE

The following should be used to the tailor the specification paragraph:

The test should be conducted in accordance with 4.4.1.7.

Background:

Analyses should assume the presence of flaws in the most unfavorable location with regard to geometry stress and material properties and should show that at the end of the required damage tolerance operational period, the strength requirement can be met for this flaw configuration and the required load.

EVALUATION LESSONS LEARNED

None.

3.4.1.7.2 Initial flaw size The assumed minimum initial flaws sizes that exist in a part as a result of material, manufacturing, and processing anomalies shall be as specified below:

Inspection Method	Material	Flaw Type	Flaw Size (depth X length)

Assumed initial flaw sizes based on production NDI methods shall have a probability of detection (POD) of _(a)_ at the lower bound _(b)_ percent confidence level (CL). Assumed initial flaw sizes based on the intrinsic material defect distribution shall encompass _(c)_ percent of the defect population when a factor of safety of _(d)_ on residual life is used.

REQUIREMENT RATIONALE (3.4.1.7.2)

This requirement is necessary to establish the probable flaw size that can exist in a part after manufacture. Damage tolerance, as applied in ENSIP, assumes the presence of a flaw in each fracture, safety, and mission critical part, at the highest stressed location with an orientation most unfavorable with respect to the stress field. Tolerance to these initial flaws must be designed into the part.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The contractor should provide the values in the table.

(a): 90%

(b): a value of 50% or 95% depending of method of inspection

(c): 99.9%

(d): factor of safety of 2

Background:

Flaw sizes are intended to represent the probable damage that can exist in a part after manufacture. The initial flaw sizes are used to determine the safe crack growth period (i.e., safety limit). Initial flaw sizes should be proposed by the manufacturer and subject to approval by the Using Service.

Assuming an NDI method reliability of 90%POD/95%CL, the minimum initial flaw sizes for crack growth analysis should be:

Inspection Method	Material	Flaw Type	Flaw Size (depth X length)
Manual	All	Surface	0.035 X 0.070 inches
Manual	All	Corner	0.035 X 0.035 inches
Manual	All	Imbedded	0.035 inch diameter
Semi–Auto	All	Surface	0.020 X 0.040 inches
Semi–Auto	All	Corner	0.020 X 0.020 inches
Semi–Auto	All	Imbedded	0.035 inch diameter
Automated	Ti	Surface	0.010 X 0.020 inches
Automated	Ni	Surface	0.007 X 0.014 inches
Automated	Ti	Corner	0.010 X 0.010 inches
Automated	Ni	Corner	0.007 X 0.007 inches
Automated	Ti	Imbedded	0.025 inch diameter
Automated	Ni	Imbedded	0.032 inch diameter

Assuming an NDI method reliability of 90%POD/50%CL, the minimum initial flaw sizes for crack growth analysis should be:

Inspection Method	Material	Flaw Type	Flaw Size (depth X length)
Automated	Ti	Surface	0.010 X 0.020 inches
Automated	Ni	Surface	0.005 X 0.010 inches
Automated	Ti	Corner	0.010 X 0.010 inches
Automated	Ni	Corner	0.005 X 0.005 inches
Automated	Ti	Imbedded	0.017 inch diameter
Automated	Ni	Imbedded	0.020 inch diameter

The initial flaw size detectability requirement of 90%POD/95%CL should be used for all manual and semi–automated NDI methods. The 90%POD/50%CL requirement can be used for some automated NDI methods. The 90%POD/95%CL was originally used because manual FPI was the most common method of inspection. With the introduction of enhanced automated eddy current inspection systems, the POD/CL requirement was changed to 90%POD/50%CL to reflect the reduced/removed operator variability. Demonstration of flaw size detection reliability should be required.

It is recommended that initial design and sizing of components be based on 0.038 cm (0.015 in) depth by 0.08 cm (0.03 in) length surface flaws and 0.038 cm (.015 in) by 0.038 cm (.015 in) corner cracks. Initial design should also account for 0.038 cm (.015 in) diameter imbedded flaws in large surface areas. The imbedded flaws in weldments should have a diameter equal to 20% of the thickness of the weld. These initial flaw assumptions provide some margin when analysis and test results indicate that stresses are higher than anticipated.

REQUIREMENT LESSONS LEARNED

Fluorescent Penetrant Inspection

Eddy Current

Ultrasonic

Radiographic

Magnetic Particle

i i i

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 90%

(b): 50% or 95% depending on the method of inspection

Background:

Although this paragraph establishes a requirement that applies to a post–EMD activity, the information (given up–front) is relative to the logistic requirements for the engine. The probable flaw sizes assumed to exist in a part after completion of a depot, intermediate or base level inspection must be consistent with nondestructive inspection (NDI) capability used during in–service inspections. It is not essential for the assumed flaw sizes following depot, intermediate or base level inspections to be the same as those following production inspections as long as inspectability is insured. However, in–service inspection flaw sizes should be larger than or equal to those flaw sizes detectable through current NDI methods. Flaw sizes for in–service inspectable flaws and in–service noninspectable flaws should be based on the NDI methods incorporated into the life management plan. The reoccurring inspection interval will be based on the assumed flaw size after completion of the initial depot, intermediate, or base level inspection.

REQUIREMENT LESSONS LEARNED

See Lessons Learned of 3.4.1.7.2.

4.4.1.7.3 In–service inspection flaw size The requirements of 3.4.1.7.3 shall be evaluated by analysis, inspection, and test.

EVALUATION RATIONALE (4.4.1.7.3)

Evaluation of in–service inspection flaw sizes is necessary to ensure the flaw sizes specified in 3.4.1.7.3 can be detected with available NDI equipment.

EVALUATION GUIDANCE

Background:

The Using Service should able to demonstrate the necessary NDI reliability. The NDI reliability should provide for a Probability of Detection (POD) at the lower bound Confidence Level (CL) consistent with 3.4.1.7.3. Although this evaluation paragraph seems to apply only to a post–EMD activity, it actually provides information (up–front) relative to logistic requirements. In–service inspection flaw sizes should be larger than or equal to those flaw sizes detectable through current NDI methods. The engine contractor should include the in–service inspection requirements (methods and intervals) in the life management plan. Logisticians may not be able to provision for detection of unreasonable flaw size values (smaller than current NDI capability).

In–service inspection flaw size detection capability should be demonstrated on parts with a significant amount of prior engine operation time. This will provide a better understanding of in–service inspection flaw size capability.

EVALUATION LESSONS LEARNED

None.

3.4.1.7.4 Inspection intervals The frequency of inspection shall be as follows:

a. In–service noninspectable: Equal to (a) times the service life specified in 3.4.1.1.

b. Depot, intermediate, or base level inspectable: Equal to (b) times the service life specified in 3.4.1.1.1.

REQUIREMENT RATIONALE (3.4.1.7.4)

These requirements are intended to provide the minimum information necessary to show that basic maintenance functions have been considered in the design of the engine.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 1

(b): 1

Background:

Inspection intervals should be compatible with the overall weapon system maintenance plan. The inspection intervals may be proposed by the contractor and approved by the Using Service. The in–service noninspectable period should be at least one times the design service life specified in 3.4.1.1. The minimum depot, intermediate or base level inspection interval should be equal to the hot parts life of 3.4.1.1.1. The inspection intervals, when accepted by the Using Service, should be contained in the contract specification.

Parts are usually designated “in–service noninspectable” because (1) inspection capability precludes detection of flaws (i.e., parts contain imbedded flaws or unaccessible flaw regions) or (2) the part is not intended to be inspected during its design life. Parts designated as depot, intermediate, or base level inspectable are classified as such because inspection capability exists such that they can be readily inspected (i.e., surface flaw inspection).

REQUIREMENT LESSONS LEARNED

The design objective of damage tolerance is to qualify parts as “in–service noninspectable”. However, the weight penalty incurred to achieve a flaw growth interval sufficiently large to preclude the need for in–service inspections may be prohibitive on some parts.

4.4.1.7.4 Inspection intervals The requirements of 3.4.1.7.4 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.7.4)

Evaluation of inspection intervals is required to ensure the flaw growth interval of 3.4.1.7.5 is of sufficient duration to preclude failure between inspections.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

The test should be conducted in accordance with 4.4.1.7.

Background:

Inspections after engine testing should provide data that substantiates the flaw growth interval of 3.4.1.7.5. This data should be compared to the time between inspections to ensure that the appropriate inspection interval has been chosen.

EVALUATION LESSONS LEARNED

None.

3.4.1.7.5 Flaw growth The initial flaw sizes specified in 3.4.1.7.1 shall not grow to critical size and cause failure of the part due to the application of the required residual strength load within:

a. (a) times the in–service noninspectable interval defined in 3.4.1.7.4, for in–service noninspectable flaws.

b. (b) times the depot, intermediate, or base level inspection intervals specified in 3.4.1.7.4, for depot, intermediate, or base level inspectable flaws.

REQUIREMENT RATIONALE (3.4.1.7.5)

The flaw growth interval (i.e., safety limit) must be specified to ensure the assumed initial flaw will not grow to critical size and fail due to application of the required residual strength load.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): two

(b): two

Background:

The flaw growth interval is also known as the safety limit. It is recommended that the flaw growth intervals be twice the inspection intervals specified in 3.4.1.7.4. Flaw growth interval margins, other than two, can be used when individual assessments of variables (i.e., initial flaw size, da/dN, K_{IC}, etc.) that affect flaw growth can be made (e.g., to account for observed scatter in crack growth during testing). In treating variables which can affect the calculation of the flaw growth interval, the following should be considered:

a. The beneficial effects of interference fasteners, cold expanded holes, shot peening, overload spinning, and other stress enhancement procedures may be used in achieving compliance with the flaw growth requirements. These beneficial effects must be verified and the extent of life "crediting" must be approved by the procuring activity.

b. Damage in a primary structure may result in load increases in the secondary structure. The analysis of such secondary structures should account for this.

c. Continuing damage should be assumed at critical locations where the initial damage assumption does not result in failure of the part (e.g., the case of a free surface at a bolthole). The following assumptions of initial damage and location should be considered with the limiting condition used to establish safety limits and inspection intervals:

(1) When the primary crack and subsequent growth terminates prior to component failure, an initial flaw equal to or greater than .015 inch surface length should be assumed to exist at the opposite location after the primary crack has terminated. The stress gradient assumed at the opposite location should be based on the boundary conditions that exist when crack growth has terminated at the primary location. The safety limit for this condition should be the sum of the crack growth at the primary location and at the opposite location.

(2) Growth of an assumed initial flaw at the location opposite the primary location should be evaluated as an initial condition.

d. The effects of vibratory stress on unstable crack growth should be accounted for in establishing the safety limit. Threshold crack size should be established at each individual sustained power condition (Idle, Cruise, Intermediate) using the appropriate values of steady stress and vibratory stress. The smallest threshold crack size should be used as a limiting value in calculating the safety limit if it is less than the critical crack size associated with

the material fracture toughness. An analytical approach to defining the effects of vibratory stress is based on a maximum stress intensity allowable, $K_{max\ allowable}$ LCF, which is predicted from appropriate material high cycle fatigue vibratory threshold μK_{th} HCF properties at steady–state operating conditions. This relationship is as follows:

Assume $K_{max\ allowable}$ LCF = μK_{th} HCF/(1 – R)

where, R = ($_{steady}$ – $_{vibratory}$) / ($_{steady}$ + $_{vibratory}$)

$_{steady}$ = maximum operating stress neglecting vibratory stress

$_{vibratory}$ = peak to peak vibratory stress

and μK_{th} HCF = f(R, temp)

μK_{th} HCF vs R–ratio material property curves used in this evaluation at various temperatures should be developed during material characterization as necessary.

e. Galling/fretting limits (i.e., permissible depth of surface damage) for fan/compressor blade–to–disk contact surfaces should be defined based on $K_{max\ allowable}$ LCF.

REQUIREMENT LESSONS LEARNED

Since average fracture properties have been used in analysis, parts made from materials with scatter factors greater than two have failed prior to their inspection interval. Thus, for materials with large scatter factors (i.e., greater than two), factors of safety greater than two, on residual life, should be considered.

4.4.1.7.5 Flaw growth The requirements of 3.4.1.7.5 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.7.5)

Evaluation of flaw growth is necessary to ensure that initial flaws will not grow to critical size and cause failure due to the application of the required residual strength load.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

The test should be conducted in accordance with 4.4.1.7.

Background:

Analyses should demonstrate that the assumed initial flaws will not grow to critical size for the usage, environment, and required damage tolerance operational period. The analyses should account for repeated and sustained stresses, environments, temperatures, and should include the effects of load interactions. Analysis methods should be evaluated by test, utilizing engine and rig testing.

EVALUATION LESSONS LEARNED

None.

3.4.1.7.6 Composites Composite parts shall be damage tolerant with defects resulting from material quality, manufacturing processing, and handling damage.

REQUIREMENT RATIONALE (3.4.1.7.6)

Damage tolerance of composites must be specified since they are a special type of fracture, mission and safety critical part. This paragraph is needed to establish a means for composites to comply with damage tolerance requirements.

REQUIREMENT GUIDANCE

Background:

Damage tolerant design of organic matrix composite (OMC) parts is extremely complex. Composites exhibit near–linear stress–strain characteristics up to failure, while most metals display some ductile deformation. Hence, composites are less tolerant to overload than metals. Composites generally exhibit good resistance to tension fatigue and are susceptible to local delaminations resulting from compression fatigue.

Because of the multi–phase nature of the materials used in composites, a substantially higher number of defects may exist in a composite part than would occur in a metallic part.

Handling damage to composites includes scratches, gouges, delamination, and fiber breakage. In these instances, delaminations and fiber breakage are usually the result of impact damage. The extent and type of damage resulting from impact on composites depends on the energy involved in the impact.

Defects in composites due to manufacture are usually of two types: (1) those produced during the preparation and production of the composite and (2) those produced during machining, processing, and assembly of the final component.

Some composite materials are known to absorb moisture and lose strength over time.

REQUIREMENT LESSONS LEARNED

Damage tolerance requirements for composite parts have been derived from the work done for the Aircraft Structural Integrity Program (ASIP).

4.4.1.7.6 Composites The requirements of 3.4.1.7.6 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.7.6)

Tests and analyses must be performed to evaluate the damage tolerance of fiber composites.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph: Composite parts should be subjected to impact damage equivalent to 100 ft–lbs (74 N–m), using a one inch (2.54 cm) diameter spherical impactor, then operated for the periods specified in 3.4.1.7.4.

Background:

Composite parts should be subjected to impact damage to evaluate damage tolerance to handling/maintenance induced impact damage. The contractor should propose other means of evaluating the damage tolerance of OMC parts.

EVALUATION LESSONS LEARNED

Methods for evaluation of damage tolerance of composites is derived from the work done for the Aircraft Structural Integrity Program (ASIP).

3.4.1.8 Vibration and dynamic response The engine shall be free of vibration and dynamic response that would cause the engine to operate unsatisfactorily and not meet life requirements of 3.4.1.1.

REQUIREMENT RATIONALE (3.4.1.8)

Absence of damaging vibration and dynamic response is essential to maintain the desired mechanical and structural integrity of the engine. This requirement is intended to apply to rotating equipment and to other critical components in a vibratory environment on the airframe.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

Some of the sources of vibration which have been identified and studied by researchers are; (1) asymmetric stiffness properties of shafts or bearing supports, (2) internal friction in shafts or other rotating parts, (3) aerodynamic or gas flow excitation, (4) excitation from fluids trapped within rotor shafts, (5) rubbing friction between radial surfaces or rotor and stator housings, (6) excitations from other rotating components in the same engine structure, and (7) excitation from gears at the gear tooth mesh frequencies.

Small turboshaft engines are more susceptible to rotor vibration problems. An increase in engine performance has been obtained through higher shaft speeds. This has heightened the importance of rotor dynamics considerations in the development of the small engine.

Rotor vibration experienced on multi–shaft engines have resulted directly from rotor imbalances existing in the numerous critical speeds inherent in light–weight, high–speed engines.

Campbell frequency diagrams have been required to assure frequency mistuning margins against vibration and fatigue failures. This method has not always predicted blade flutter. The damage caused by blades failing in vibration fatigue has caused catastrophic results in life and aircraft. When determining the total stress of an engine part, contractors typically allow 25% – 50% of yield stress for stress due to vibrational loads. The more conservative values contractors have used are between 30–35%.

Airseal and spacer failures were experienced by an engine. As part of the investigation it was learned that transient engine operational cavity temperatures caused acoustic excitations and mechanical resonances to coincide and caused damaging vibrations. Another engine's front frame experienced cracking during development testing that was attributed to the fan blade tip exceeding supersonic speed. The shock wave off the blade tip provided the stimulus to excite natural frequencies of the frame strut panels. Since Mach number is related to temperature, the fan speed at which the blades went supersonic varied with inlet temperature conditions. The stresses in the struts peaked when the fan blade tip reached supersonic speeds and coincided with the natural frequency of the strut panels.

4.4.1.8 Vibration and dynamic response The requirements of 3.4.1.8 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.4.1.8)

Analysis is required to establish equipment and subsystem rigidity and to evaluate equipment and subsystem design to minimize the potential vibration/dynamic response problems. Testing is needed to evaluate the vibration and dynamic response characteristics of the engine.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

The following shall be conducted:

a. A vibration and stress analysis shall be accomplished on all fan, compressor and turbine stage blade, vane, shaft and disk designs, and other parts and components where high vibration and stress occur.

b. Dynamic models shall be developed for major engine parts and components. The vibratory stress distribution and the various modes of vibration including complex modes shall be obtained.

c. Acceleration spectrograms shall be provided at the highest vibrational level in the operating envelope and at designated engine performance points.

d. Bench tests shall be conducted on various parts and components, as appropriate, to evaluate/correlate with analyses and models to establish material/part/component stress capabilities.

e. A vibration and stress test shall be conducted.

Background:

The critical speeds, excitation frequencies, various modes of vibration including complex modes, and stress values for the vibratory stress distributions and nodal patterns should be determined and correlated with the strength and life analysis. The report should include the results of actual engine operation or where that is impractical by component test and analytical study. Analysis of the data should include the measured and referred stress values at high stress areas on the cases, blades, vanes, discs, shafts, spacers, engine mounts, and other instrumented parts. Equations and sample calculations for all analytical methods used should be included in the report. The data should show the effects on stress levels due to vibration throughout the operating range of the engine. The report should show modified Goodman and Campbell diagrams for each component design. Plots of excitation frequency vs rotor speed showing the primary orders of excitation and the modes of vibration should be plotted with points noting stress, measured and referred. A summary of all critical speeds should be defined in the report.

Analytical dynamic computer models should be developed to accomplish dynamic analysis of engine components and the assembled engine as a dynamic system. The models should be updated and verified throughout the development program as hardware and test information becomes available.

A dynamic model of the total engine system including rotor(s), bearings, frames, cases, and engine supports should be developed to analyze maneuver loads and synchronous and non–synchronous vibration responses. The model should include the effects of shear deformations, rotary inertia, multi–shaft gyroscopic influences, bearing speed effects on stiffness and damping, anisotropic bearing supports, and speed/frequency influences on supporting structural stiffness. Stiffness values used to represent flanges, splines, couplings, joints, and tapered elements of the structure/rotor should be supported/verified by test. The model should be capable of handling imbalanced distributions resulting from bowed rotors or assembly of shaft components including residual imbalance plus angular or offset misalignments of these shaft sections. Vibration amplitudes, clearances and bearing loads should be determined at each critical speed with the most adverse magnitude and phase relations of the imbalance associated with the critical speed mode shape. Parametric studies of design changes should be conducted to determine ways of altering any detrimental dynamic modes which exist in the operating range. Results from the models will be used to guide imbalanced engine testing. The models will be updated and verified throughout the development of the engine as hardware and test information becomes available.

Dynamic models should also be developed for major engine components such as blades, vanes, bladed disk assemblies, seals, shafts, spacers, engine mounts, cases, and other components where high vibration can occur. The vibratory stress distribution and the various modes of vibration including complex modes should be obtained. Campbell diagrams (plots of excitation frequency vs rotor speed) showing the primary orders of excitation and the modes of vibration should be prepared . Goodman diagrams (plots vibratory stress vs steady stress) should be prepared to evaluate to part and component. The analytical studies should be verified/correlated with data obtained from actual engine operation and testing.

A resonance search test should be conducted to evaluate/correlate with dynamic analysis and to uncover any large amplitude vibratory modes that could adversely affect safety and durability.

Acceleration spectrograms (plots of acceleration/amplitude vs frequency) should be provided at the highest vibrational level in the operating envelope and at designated engine performance points. Spectrograms should be generated from each transducer shown on the engine configuration and envelope figure. Critical parts and components should be identified on each spectrogram. Each spectrogram should cover the frequency range of 5 Hz to 10 kHz and present acceleration data in peak g’s.

Acceleration spectrograms are a diagnostic tool which can be used to determine the cause of an unacceptable vibration level. Points selected for spectrogram analysis should provide substantiation of the overall velocity limit defined in 3.4.1.8.1 The spectrogram analysis should also establish a detailed signature of critical parts and components at guaranteed performance points for future comparisons. The frequency range required for the spectrograms is considered to be the range that, in general, would capture all useful vibratory information and is also consistent with the state–of–the–art in accelerometer frequency response.

The test engine should be excited by a shaker in mutually perpendicular planes throughout a frequency range sufficient to cover engine operation. Strobe lights and movable accelerometers should be utilized to determine maximum response locations. Resonance frequencies should be determined and compared with potential sources of excitation and with the analysis.

Instrumented engine tests should be conducted to evaluate the dynamic response of individual parts and components as well as the total engine system. Instrumentation should include accelerometers, strain gauges, and proximity probes in the vertical and horizontal planes. Tests should be conducted as early as possible on a core engine to determine vibratory stresses and to investigate flutter boundaries. Subsequent tests should be conducted on the full engine. Ram conditions, aircraft inlet conditions, simulated fan distortion, compressor bleed and nonbleed, power extraction, off–nominal guide vane schedules, stalls, and other important variables should be simulated during each test as appropriate. Effects of rotor imbalance up to the maximum allowable should be evaluated.

A sufficient number of blades and vanes in each stage should be instrumented with strain gauges to obtain continuous data and to determine worst case value of stress. Disks and other components subject to high vibratory stress should be instrumented. Strain gauges should be mounted at locations of maximum stress as indicated by the analysis. Sufficient instrumentation should be installed at appropriate locations on main bearings to permit measurement of bearing loads, cage rotation, and rotor deflections. External components such as fuel controls, fuel pumps, valves, plumbing lines, etc., should be instrumented at appropriate locations with accelerometer vibration equipment. Inlet or exhaust systems and other components that are mounted directly on or supported by the engine in the aircraft installation should be mounted in the same manner for these tests. The tests should check all critical engine speeds where analysis indicates significant stress or vibratory conditions occur on each component. Vibration and stress measurements should be made during all engine operating modes and should include but not be limited to conditions of maximum inlet distortion, stall, limits of variable geometry travel if applicable, maximum compressor air bleed and power extraction, maximum inlet pressure and temperature capabilities of the engine and combinations thereof. The engine tests should include sufficient dwell time at each of the critical speeds above Idle where the response in the rotor system dynamic evaluation testing shows peak values. Overall true RMS velocity measurements and acceleration spectrograms should be obtained for each accelerometer mounted on the engine core and external accessory components.

Effects of rotor imbalance up to the maximum allowable amplitude should be evaluated. The rotor(s) should be imbalanced with the most adverse weight placement for the lowest critical speed and a phase angle predicted by analysis of residual imbalance. Magnitudes of total imbalance should be large enough to overcome typical residual imbalances to reach maximum levels found in similar engines prior to overhaul and to reach field vibration limits. The engine should be run through the operating range to Maximum power. The procedure should be repeated for the other critical speeds below maximum speed if these critical speeds lead to maximum response at any point in the engine. Modes above and close to the maximum speed should be checked with the imbalance distribution required to excite these modes. If required, the phase of imbalance distributions will be changed to help determine residual imbalance.

Speeds at which response peaks occur will be correlated with computer model predictions. Measured and predicted amplitudes and clearances should agree when adjusted for residual imbalance. Model flexibilities and damping should be adjusted to obtain agreement between measured and predicted critical speed and response magnitudes.

Bench tests should be conducted on blades and other parts as appropriate to verify/correlate with the analysis and to establish material/component stress capabilities.

EVALUATION LESSONS LEARNED

Engine contractors have said that it is difficult to build an engine with a maximum imbalanced rotor for vibration tests. They have said it is difficult to predict the total imbalance of the rotor when the engine is completely assembled.

The engine rotors are supported by bearings mounted on relatively stiff supports. The vibration loads are transmitted to the engine via the bearing supports. These loads and deflections can be reduced by soft–mounting the bearings through "squirrel cages" and/or squeeze–film bearings.

Vibration problems appear to occur more frequently in rotors with a cantilevered disk. The overhung rotor configuration is characterized by a disk having a shaft bearing only on one side. There has been a long history of dynamic–related problems associated with overhung rotors.

Small turboshaft and turboprop engines with their high speeds and front drive are more susceptible to vibration problems. The vibration analysis of blades was usually made prior to engine buildup to preclude the possibility of resonance problems. Experimental testing was always feasible unless the hardware was non existent or unavailable.

Aerodynamic blade flutter or vibration was not uncovered by test more by lack of environmental definition than from blade characteristics. Flutter is a function of pressure and temperature which makes its prediction a very complex problem. On the compressor map, many areas of flutter can be depicted (e.g., subsonic and supersonic stalled flutter, subsonic and supersonic unstalled flutter, choke flutter, etc). The areas are so numerous that it seems impossible to avoid flutter. The Campbell diagram shows how close some rotor speeds come to resonant vibration with certain harmonics of rotor speed. This can all change with the ambient environment.

3.4.1.8.1 Vibration limits The location and maximum permissible engine vibration limit of each vibration transducer shall be as specified in table VIII.

REQUIREMENT RATIONALE (3.4.1.8.1)

The intent of this paragraph is to specify the vibration limits which are utilized as a diagnostic tool and monitoring engine health. Vibration limits are also needed to establish baseline performance of developmental engines to determine when design changes are warranted.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The Using Service should indicate in the format of table VIII that maximum permissible engine vibration limits be specified as an overall true RMS velocity (or displacement) limit or an overall average velocity (or displacement) limit.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.8.1 Vibration limits The requirements of 3.4.1.8.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.8.1)

Evaluation is required to ensure compliance with the requirements of 3.4.1.8.1.

EVALUATION GUIDANCE

See EVALUATION GUIDANCE of 4.4.1.8.

EVALUATION LESSONS LEARNED

See EVALUATION LESSONS LEARNED of 4.4.1.8.

3.4.1.8.2 Critical speeds The engine shall be free of resonance conditions at all shaft rotational speeds that would cause the engine not to meet all the requirements of the specification. Critical speeds existing below the engine operating range shall be at least (a) percent below Idle. Critical speeds existing above the maximum operating speed shall be at least (b) percent above the maximum allowable transient shaft rotational speed. The natural frequencies of the mounting system with the engine installed shall be at least (c) percent below Idle shaft rotational speed(s) in all damaging modes of vibration which can be excited by the rotor imbalances.

REQUIREMENT RATIONALE (3.4.1.8.2)

Resonance conditions should be avoided so that amplified response and structural failures do not occur. Margin is required between engine speeds and resonance speeds due to the variation that can occur in speeds due to Mach number, deterioration, ambient conditions, and combinations, thereof.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 20 percent

(b): 20 percent

(c): 20 percent

Background:

It is recommended that a margin of at least 20 percent be specified for damaging resonance conditions that exist above maximum allowable transient shaft operating speed or below Idle speed. Damping and appropriate balancing should be provided so that any critical speed existing below maximum operating speed shall be traversed safely with smooth engine operation including the blade out conditions of 3.4.1.6.4. It is also recommended that the natural frequencies of the mounting system with the engine installed should be 20 percent below Idle rotor speeds in all detrimental modes of vibration.

REQUIREMENT LESSONS LEARNED

Steady–state engine operation at a critical speed has quickly consumed the high cycle fatigue life of parts due to vibratory resonance.

4.4.1.8.2 Critical speeds The requirements of 3.4.1.8.2 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.8.2)

Evaluation is required to ensure compliance with the requirements of 3.4.1.8.2.

EVALUATION GUIDANCE

See EVALUATION GUIDANCE of 4.4.1.8.

EVALUATION LESSONS LEARNED

See EVALUATION LESSONS LEARNED of 4.4.1.8.

3.4.1.8.3 Blade, disk, and static structure vibration The engine shall meet all the requirements of the specification when subjected to engine induced natural frequencies and resonances.

REQUIREMENT RATIONALE (3.4.1.8.3)

Determining the frequency and relative amplitude of various sources (forcing functions) will provide design information on frequencies to be avoided in components as well as information to determine fatigue resistance and vibration associated wear problems.

REQUIREMENT GUIDANCE

Background:

Sources of forcing functions which can cause vibration and expected frequency spectra should be identified. Blades, vanes and disks should be designed so that low order and known excitation orders (e.g., blade passing orders for vanes) are voided in the sustained power operating ranges (i.e., Idle, Cruise, and Intermediate power). It is recommended that a 10 percent margin with worst tolerance on component resonance frequency be maintained. Proposed design changes should be fully evaluated for their effect and/or response to possible excitation orders. The effects of inlet distortion, flow disturbances propagated upstream from obstructions, wakes from obstructions in the flowpath, scheduling tolerances associated with the variable geometries, and interactions between blades and vanes should be such that damaging excitation should not occur in the operating range.

REQUIREMENT LESSONS LEARNED

Blade failures have been experienced on several occasions due to lack of margin between a resonance or critical condition and a sustained power operating condition.

4.4.1.8.3 Blade, disk, and static structure vibration The requirements of 3.4.1.8.3 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.8.3)

Evaluation is required to ensure compliance with the requirements of 3.4.1.8.3.

EVALUATION GUIDANCE

See EVALUATION GUIDANCE of 4.4.1.8.

EVALUATION LESSONS LEARNED

See EVALUATION LESSONS LEARNED of 4.4.1.8.

3.4.1.9 Catastrophic failure The engine shall incorporate the following fail–safe features or capabilities to eliminate catastrophic failure: (a) .

REQUIREMENT RATIONALE (3.4.1.9)

For safety of personnel and equipment, it is very desirable to contain all rotor system failures within the engine.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a):

a. A main rotor shaft bearing or lubrication system failure should not cause parting or decoupling of the shaft(s).

b. A control, accessory, and external subsystem rotating shaft bearing or lubrication system failure should not cause parting or decoupling of the shaft which would propagate to failure of attached rotating parts.

c. In the event of any type bearing failure, the structures supporting the rotating masses should minimize the probability of gross misalignment of the engine rotating parts.

d. All areas of the rotor that could puddle oil should have appropriate drains.

e. Disks should be protected by having blades fail first under overspeed/overtemperature malfunctions.

Background:

The contractor should list all fail–safe designs intended to eliminate catastrophic failure. Initial design and sizing of components should include criteria to provide containment.

Item (e) above may not be possible with current day metallurgy on some helicopter engines. Some helicopter engines fail disk first with a requirement to not fail catastrophically.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.9 Catastrophic failure The requirements of 3.4.1.9 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.4.1.9)

Fail–safe designs intended to eliminate catastrophic failure must be evaluated to ensure effectiveness.

EVALUATION GUIDANCE

Background:

The contractor should propose the method of evaluation for all fail–safe designs intended to eliminate catastrophic failure.

EVALUATION LESSONS LEARNED

None.

3.4.1.10 Sustained acceleration Components shall function satisfactorily during and after sustained acceleration, in accordance with MIL–STD–810, and no permanent deformation shall occur.

REQUIREMENT RATIONALE (3.4.1.10)

The components must be designed to withstand the forces induced by the external forces applied to the vehicle in which they are mounted.

REQUIREMENT GUIDANCE

Background:

The different requirements imposed by fixed wing and rotary wing installation must be considered.

REQUIREMENT LESSONS LEARNED

The forces induced by acceleration can cause: permanent deformations and fractures that disable or destroy equipment, broken fasteners or mounting hardware that cause equipment to become projectiles, short or open circuits on electronic boards, changes in inductances and capacitances, relays to open or close, actuators and other mechanisms to bind, seals to leak and pressure and flow regulators to charge valves.

4.4.1.10 Sustained acceleration The requirement of 3.4.1.10 shall be evaluated by test.

EVALUATION RATIONALE (4.4.1.10)

Tests must be performed to evaluate component operation after they have had forces induced on them by the external forces applied on the vehicle.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph: Components shall undergo an acceleration test in accordance with MIL–STD–810. The test item shall be subjected only to the operational tests in accordance with the procedure for the fixed wing or rotary wing vehicle category, as applicable. The value of "A" shall be 4g consistent with figure 2. Test time may be increased beyond the minimum specified, if necessary, to determine proper operation.

EVALUATION LESSONS LEARNED

None.

3.4.1.11 Shock Components shall function satisfactorily after experiencing a shock, as induced in accordance with MIL–STD–810.

REQUIREMENT RATIONALE (3.4.1.11)

The components must be designed to withstand the relatively infrequent, non–repetitive shocks, or transient vibrations induced by the external forces applied to the vehicle in which they are mounted.

REQUIREMENT GUIDANCE

Background:

Shock conditions are specified by MIL–STD–810, method 516. The forces induced by shock (e.g., sudden impact) can cause: permanent deformations and fractures that disable or destroy equipment, broken fasteners or mounting hardware that cause equipment to become projectiles, short or open circuits on electronic boards, changes in inductances and capacitances, relays to open or close, actuators and other mechanisms to bind, seals to leak and pressure and flow regulators to charge valves. The different requirements imposed by fixed wing and rotary wing installation must be considered.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.11 Shock The requirement of 3.4.1.11 shall be evaluated by test.

EVALUATION RATIONALE (4.4.1.11)

Tests must be performed to evaluate component operation after they have had external forces induced on them.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph: Components shall be subjected to an impact test in accordance with MIL–STD–810 for helicopters and fixed wing aircraft not stationed on aircraft carriers. Tests shall be conducted under room ambient conditions. A crash hazard test for flight equipment shall be conducted for components with a peak acceleration of 40 g's for a duration of 6–9 ms and a cross over frequency of 45 Hz. The shock pulse, amplitude, and duration are from MIL–STD–810B. Carrier based fixed winged aircraft shall be subjected to Procedure IX, which is a new category for catapult launched and arrested landing aircraft. Components shall be subject to pre– and post–test calibration or functional checks as applicable, but need not be operated during the test. Post–test inspection procedure shall be in accordance with MIL–STD–810.

EVALUATION LESSONS LEARNED

None.

3.4.1.12 External surface foreign object damage The engine's external surface, protruding from the airframe, shall meet all requirements of this specification, without repair, after FOD that produces damage equivalent to a stress concentration factor (Kt) of (a) .

REQUIREMENT RATIONALE (3.4.1.12)

The outer surfaces of thrust vectoring nozzles, which are external of the airframe surface, may be susceptible to hits by birds and ice while vectoring takes place.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The airframe systems specification should be checked for the required stress concentration factor for FOD tolerance, due to hail and bird strikes, on airframe outer surfaces. The airframe specification requires outer surfaces to be damage resistant from impacts of ice and birds at specific masses and velocities. The stress concentration factors can be derived from this. The engine should be more resistant to FOD, bird, ice, etc. than aircraft structure (except for canopy) as it is needed for flight operation.

REQUIREMENT LESSONS LEARNED

None.

4.4.1.12 External surface foreign object damage The requirement of 3.4.1.12 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.4.1.12)

None.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.4.2 Mechanical equipment and subsystems integrity The engine's mechanical equipment and subsystems shall meet the integrity requirements specified (a) .

REQUIREMENT RATIONALE (3.4.2)

None.

REQUIREMENTS GUIDANCE

(a): AFGS–87249 should be used to tailor the requirement.

Background:

MIL–STD–1798 should also be used as guidance for tailoring the requirements.

REQUIREMENT LESSONS LEARNED

None.

4.4.2 Mechanical equipment and subsystems integrity The requirements of 3.4.2 shall be evaluated by analysis, inspection, demonstration, and test.

EVALUATION RATIONALE (4.4.2)

Evaluation is required to ensure compliance with requirements of 3.4.2.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.4.3 Avionic/electronic integrity The engine shall meet the electronic system integrity requirements specified (a).

REQUIREMENT RATIONALE (3.4.3)

None.

REQUIREMENTS GUIDANCE

(a) MIL–A–87244 should be used to tailor the requirement paragraph.

REQUIREMENT LESSONS LEARNED

None.

4.4.3 Avionic/electronic integrity The requirements of 3.4.3 shall be evaluated by analysis, inspection, demonstration, and test.

EVALUATION RATIONALE (4.4.3)

Evaluation is required to assure compliance with requirements of 3.4.3.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.5 Reliability and maintainability (R&M) The engine shall exhibit characteristics of reliable operation and be designed such that it can be retained in, or restored to a working and usable condition within a specified period of time, when the maintenance is performed in accordance with prescribed procedures and resources. The engine shall be subject to the following:

a. The R&M definitions of Section 6 and MIL–STD–721.

b. The scheduled maintenance intervals of (a) .

c. The “Mission and Mission Mix” profile of table VI.

REQUIREMENT RATIONALE (3.5)

Reliability requirements must be specified because safety, mission capability, and cost are impacted by the reliability of the engine. Maintainability can have a major impact on operational support cost if it is not adequately addressed during development.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The scheduled maintenance interval should be specified by referencing the maintenance portions of the contract and the required operational capability document.

Background:

R&M requirements should be established by starting with operational needs, including those portions of the needs which are not within the control of the engine manufacturer. Values will be assumed for those portions of the needs which are not controllable (e.g., FOD rate). The assumed values for that portion of the operational need which is not controllable will be subtracted from the total operational needs to establish the specification requirements. Non–controllable failures may be reduced by upgrading specification requirements in appropriate areas. When the Using Service determines that these requirements are adequately covered by another contract provision, they may be deleted from the specification.

REQUIREMENT LESSONS LEARNED

Past engine programs experienced spare parts procurement problems with logistics impacts. Scheduled maintenance intervals and “non–life limited” parts lives were shown to be different in the field, than the values before field introduction and at ISR. This impacted program cost and mission capability.

4.5 Reliability and maintainability (R&M) The requirements of 3.5 shall be evaluated by analysis, inspection, demonstration, and test.

EVALUATION RATIONALE (4.5)

Engine reliability and maintainability must be evaluated.

EVALUATION GUIDANCE

Background:

Procedures and test methods for reliability and maintainability should include the following:

a. Identification of reliability and maintainability test program tasks in accordance with MIL–STD–781 and MIL–STD–471.

b. Identification of the contractual status of an integrated reliability program and an effective maintainability program.

c. Identification of the life cycle, mission, and environmental profiles which represent in service usage.

d. Detail description of hardware configuration, equipment, and system operation under test.

e. Performance parameters to be monitored during the test.

f. Provision of reliability and maintainability test data and reports to the Using Service.

g. Specification of alternate failure categories.

Effort should be taken to include demonstrations and tests for evaluation of reliability and maintainability.

EVALUATION LESSONS LEARNED

Where reliability and maintainability were evaluated by analysis only, past engine programs had cost overruns because of poor assumptions in the analysis.

3.5.1 Reliability The engine shall meet the reliability requirements of the following paragraphs in accordance with MIL–STD–785.

REQUIREMENT RATIONALE (3.5.1)

A reliability program is needed for the prevention, detection, and correction of design deficiencies, weak parts, and workmanship defects. The reliability program should include an appropriate mix of reliability engineering and accounting tasks.

REQUIREMENT GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph: The contractor shall identify and control Reliability Critical Items (RCIs), those items which require “special attention” because of their potential impact on system reliability. This effort shall be conducted in accordance with Task 208 of MIL–STD–785 and shall be included in the Failure Modes, Effects, and Criticality Analysis (FMECA). The RCI identification shall include new and modified hardware whose individual failure may result in a Category I or II hazard. The hazards shall be categorized in accordance with MIL–STD–882.

Background:

The reliability program should (1) improve operational readiness and mission success of the major end item, (2) reduce item demand for maintenance manpower and logistic support, and (3) provide essential management information and reduce overall program and scheduling cost in accordance with the Using Service. To establish an acceptable and cost effective reliability program, appendix A of MIL–STD–785 should be used to initially identify those tasks which typically are included in an effective reliability program for the particular acquisition phase involved.

Reliability critical items (RCIs) should be chosen and identified considering their impact on safety, mission capability, and costs. Selected RCIs should be limited to controls and externals unless a particular program justifies selection of another item. They should be chosen as items for which improvements in reliability growth would be adequately

achieved if additional testing is conducted, above and beyond normal qualification testing, including that received via engine (e.g., ASMET) testing. The FMECA should be used to aid in selecting RCIs for safety considerations. The FMECA alone should not be used to select safety critical items (SRUs). Also, the impact of reliability on in–flight shutdown should be considered in selecting RCIs for safety reasons. The impact of reliability on life–cycle cost should be considered while selecting RCIs for cost reasons.

The FMECA should be conducted on all hardware incorporated into the engine and the engine components (WRAs) in accordance with Task 204 of MIL–STD–785 employing the procedures of Tasks 101, 102, and 103 of MIL–STD–1629. If applicable, existing engine data should be used as a baseline for the FMECA. The FMECA should be performed to the work unit code levels necessary to account for the impact of the new and modified hardware. The analysis should be carried down to the SRA/function level for the electronic equipment and performed for typical mission profile conditions and relate all associated failure modes (down to SRA and piece part level), effect (up to the higher indenture levels, including the subsystem and weapon system level) and severity levels (categories I–IV) for each indenture level. The overall system analysis should be carried down to the subsystem/functional level for typical mission profile conditions.

CDRLs should be required for the Reliability Status Report, in which updates and failures of RCIs relevant for reliability accounting will be reported, and the FMECA.

In a development specification, requirements and goals should be used so the contractor will make best efforts to surpass requirements. This will benefit the Using Service by showing what is available in production. In a production specification, requirements should be used because the contractor only needs to meet the requirements.

REQUIREMENT LESSONS LEARNED

In the past, a reliability program has resulted in the redesign of electronic controls, correcting problems identified during Quality Assurance Provisions (Evaluations) before ISR.

4.5.1 Reliability The requirements of 3.5.1 shall be evaluated by inspection, analysis, demonstration, and test.

EVALUATION RATIONALE (4.5.1)

An IRTP during the development, qualification, and production of the engine is required to avoid duplication of test effort and to ensure that deficiencies are not overlooked.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

An Integrated Reliability Test Program (IRTP), in accordance with MIL–STD–781 and as specified by the Using Service, shall be prepared by the Contractor.

Background:

The IRTP should contain a description of the test plans selected for use, the decision risks, and the environmental test conditions and shall be keyed to the program life cycle phases as described in MIL–STD–781 and MIL–HDBK–781.

EVALUATION LESSONS LEARNED

None.

3.5.1.1 Reliability quantitative requirements The engine reliability shall be as defined in tables XXIIa and XXIIb.

REQUIREMENT RATIONALE (3.5.1.1)

The requirement for quantitative reliability must be specified for safety, mission capability, and cost reasons. The designer must take reliability considerations into account, or these three key factors will be adversely impacted.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: The Using Service should provide a formatted table for the contractor to complete.

Background:

Reliability parameters are chosen to induce improvements to key issues as follows:

Parameters	Primary Impact	Secondary Impact
1. Mean engine flight hours between engine removal (MEFH/EF)	Cost	Mission Capability
2. Mean engine flight hours between engine failures (MEFH/F)	Safety	Mission Capability
a. In–flight shutdown rate (events/1000 EFH)	Safety	Mission Capability
3. Mean engine flight hours between maintenance actions (MEFH/MA)	Cost	Mission Capability
4. Mean engine flight hours between maintenance actions unscheduled (MREFH/MAU)	Cost[1]	Mission Capability

[1] Impacts fuel costs in addition to hardware and manpower costs.

The Using Service should consider data provided by the contractor from other programs (e.g. operational data for a specific component) for determining initial reliability. After converting reliability data to the equivalent engine flight hours, the subassembly reliability should be established by analysis of the data using appropriate techniques as those described in MIL–STD–781 Appendix A or AFWAL–TR–83–2079.

The contractor should perform a math model allocation and design reliability prediction of the engine and the engine components (WRAs) of the engine in accordance with MIL–STD–785 using Table VI mission and mode of operation. Any existing engine predictions and allocations may be used as a baseline for these predictions, if applicable. The modeling and predictions shall be performed in accordance with Tasks 101, 102, 201, 202, method 2001 of MIL–STD–756 (Similar Item Method) for mechanical equipment and method 2005 (Stress Analysis) for electronic engine components. These analyses are should be done prior to the start of the Critical Design Review.

The Weapon Replaceable Assembly (WRA)/Line Replaceable Unit (LRU) Removal Rate takes frequency of occurrence into account in its definition, so engine manufacturers should position the WRA/LRUs to facilitate quick removal and replacement for WRA/LRUs with lower reliability levels. If it is not possible to locate a WRA/LRU in a location to allow rapid replacement time, the engine manufacturer should conduct reliability growth test to improve the reliability of the WRA/LRU, so that its frequency of replacement is less. The current reliability of the item and its criticality with respect to mission capability and safety should also be taken into account. In addition, the engine manufacturer should ensure that it is not necessary to remove a higher reliability WRA/LRU to gain access to a lower reliability WRA/LRU.

The selection of the value assigned to engine removal rate for exceedance of performance deterioration should be based on a life cycle cost analysis, taking into account the cost of maintenance required when an engine is removed for performance deterioration and increased fuel cost if the engine is not removed.

Each engine program should generate a specific assessment plan which details the data collection and data analysis methods to be used in each portion of the engine program. During development, the contractor's data collection system will be used. During flight test, the government data collection system will be used unless special provisions are made in a program to fund for contractor data collection.

Severity factors should be based upon mission profiles. If mission profiles change, reliability requirements should be adjusted accordingly. The effects of individual failure modes should be studied in terms of impact to in–flight shutdown.

Weight penalties and weight reduction implementation may adversely affect the component and system reliability. The Using Service, if it desires to introduce a weight reducing incentive or prior to invoking weight penalties on the contractor, should participate in an intensive trade–off study which considers weight, reliability, cost (both acquisition and LCC), and repairability.

REQUIREMENT LESSONS LEARNED

The sophistication of reliability analysis techniques has greatly improved over time. On some past engine development programs reliability considerations were not as exact as performance (i.e., thrust). Insufficient reliability growth testing during development has resulted in failures during operational usage and has required Component Improvement Programs for corrective action. Insufficient reliability requirements have resulted in increased maintenance.

Past engine programs had contractors calculate wrong parameter values because they assumed certain engine failures/behavior to be scheduled and non–engine caused. This gave parameter values that were better than actual.

Contractors have included some engine caused events the non–engine caused category for In–Flight Shutdowns (IFSD) such as: in–flight stall events (within aircraft and engine flight envelopes), engine over–temperature (within flight and operating envelopes), engine over–temperature beyond limits during airstarts (within airstart envelope requirements), engine variable geometry cycling, and engine low oil pressure readings (known not to be A/C system caused). The Using Service recognized these as engine caused. This biased the engine–caused IFSD rate to be lower than actual. Contractors have not listed/shown shop visit rate/maintenance events that are not engine–caused. This did not let the Using Service determine if some should be engine–caused.

4.5.1.1 Reliability quantitative requirements The requirements of 3.5.1.1 and tables XXIIa and XXIIb shall be evaluated by test and analyses in table XXIIIa. Progress toward meeting any operational reliability requirements shall be assessed with data from the contractor's Failure Reporting and Corrective Action System (FRACAS) during the development phase, but from the Using Service data collection system during the operational phase, when service data becomes available.

EVALUATION RATIONALE (4.5.1.1)

Reliability tests are necessary to establish that reliability projections are correct.

EVALUATION GUIDANCE

Background:

Data may be gathered on components or subassemblies provided that the data may be correlated to expected operational reliability levels. When it is possible to associate a severity factor(s) to the failure modes expected, it is possible to use data gathered during a test for which severity factors are other than one to project reliability levels. After reliability data is available, it is possible to compute subassembly level reliability values by using applicable techniques such as those of MIL–STD–781 and AFWAL–TR–83–2079.

Data gathered on a component or subassembly should be considered pertinent to reliability determination only when a severity factor(s) can be established for the test. To find events per equivalent engine flight hours (EFH), multiply test events per test hour by the corresponding severity factor. For failure modes for which it is known that the severity factor is significantly different than one, table XXIIIb provides a proposed format for collected data.

General comments:

a. Testing should be conducted during a development program to identify failure modes which will have corrective action for each malfunction before the production phase. Accelerated mission tests should be run on engines. Reliability tests and other development bench tests, which incorporate environmental stresses should be run on components.

b. Severity factors should be carefully chosen because excessive severity factors can result in unrealistic failure modes. Imposed tests should not require failure modes which cannot be operationally and/or ESS induced.

c. Enough time is desirable between each phase of the multi–step development process to permit fixes to be developed and incorporated into the design for failure modes which were identified in the previous phase. When sufficient time is not available, milestone adjustments may be required to permit fixes of failures of inviolable requirements. Failure modes of requirements which are not inviolable may be corrected in ensuing milestones providing the final target is met.

d. The amount of effort expended during an engine development program to design or achieve improvements in reliability via test analyses and fix, for cost consideration, should be justified on the basis of life–cycle cost (LCC) analyses. Such analyses should take into account the impact on all programs in which a given part or subassembly will be used. Life–cycle cost analysis should account for many factors, including: (1) initial engine or component cost, (2) spare cost, (3) maintenance costs at all levels of maintenance, (4) level of engine modularity.

e. Where safety and/or mission capability are impacted by the reliability of a part or subassembly, higher levels of reliability should be justified.

f. Test time which accumulated during tests for which the severity level cannot be established for failure modes should not be considered pertinent to reliability determination. Examples of data gathered on subassemblies for which the data should not be considered pertinent to reliability determination are as follows:

Pump tests operated at over–speed conditions, elevated temperatures, and with excessive fluid contamination.

g. Subassemblies scheduled for a significant amount of test time at conditions which would prohibit the test time to be considered pertinent to reliability determination should be excluded for use in establishing reliability. The reason for this is that a large portion of the component's life is likely to have been expended during a time when the test time will not be considered pertinent.

h. The following development tests and analyses should be considered for inclusion in table XXIIIa:

(1) Reliability Development/Growth Tests conducted on subassemblies and engines.

(2) Reliability Qualification Tests per MIL–STD–781, Task 301.

(3) Failure modes effects and criticality analyses in accordance with the appropriate contract provision.

i. During development, data analysis will be based on the cumulative method utilizing a method similar to Duane Theory or another agreed upon reliability growth analysis method. The AMSAA model from MIL–HDBK–189 is also recommended. The development program should be divided into phases. Otherwise, the early failures will prevent the reliability growth curve from reaching its goal. Severity factors will be taken into account in analyses.

Severity factors will be based upon mission profiles. If mission profiles change, reliability requirements should be adjusted accordingly.

j. Guidance to consider related to the parameters of table VI.

k. The goals at 500K hours should be an improvement over the goals at 100K hours as well as the requirements.

EVALUATION LESSONS LEARNED

Considerable cost has been saved by using the severity level concept to allow the use of all pertinent reliability data to establish reliability levels. If all testing were conducted at a severity level equivalent to actual operation, the test cost would be prohibitive for a comparable equivalent test time.

In the past, if an engine development program was rushed and time was not permitted for fixes to be developed and incorporated into the design, reliability growth during development has not proceeded optimally. Because of this, the multi–step development process was initiated.

The severity levels of many components during the endurance test and engine performance testing have been low. This has been especially true for controls and externals. The RCI concept has been applied to these components to permit enough equivalent test time to identify failure modes.

3.5.2 Maintainability The engine shall meet the maintainability requirements of the following paragraphs in accordance with MIL–STD–470.

REQUIREMENT RATIONALE (3.5.2)

The purpose of the Maintainability Program is to improve operational readiness, reduce maintenance manpower needs, reduce life cycle cost, and provide data essential for management.

REQUIREMENT GUIDANCE

Background:

The Maintainability program should: (a) establish procedures which will identify the means by which maintainability engineering contributes to the design of the system or equipment acquisition including its fault detection and diagnostics subsystem at all maintenance levels of application and; (b) be consistent with the criticality of the mission, the severity of the requirements, the complexity of the design, commonality, and the manufacturing techniques as required by the Using Service. The program should include, but not be limited to, maintenance concept, maintenance environment, skill levels of personnel, level(s) of maintenance to be demonstrated, and modes of operation for test (including configuration and missions), and tailored to fit program needs and constraints, including readiness, mission success criteria, life cycle costs including manpower, personnel and training, Logistics Support Analysis and testability/diagnostics considerations.

To establish an acceptable and cost effective maintainability program, Appendix A of MIL–STD–470 should be used to initially identify those tasks which typically are included in an effective maintainability program for the particular acquisition phase involved.

REQUIREMENT LESSONS LEARNED

None.

4.5.2 Maintainability The requirements of 3.5.2 shall be evaluated by analysis, demonstration, and test. Evaluation of qualitative and quantitative maintainability requirements shall be performed in accordance with MIL–STD–471, and the Maintainability Test Program shall be prepared by the Contractor and approved by the Using

Service. Evaluations shall be conducted using production hardware, field tools and support equipment, protective clothing (e.g., chemical gear, cold weather gear, etc.) and by the Using Service's fifth (5th) percentile female stature through ninety–fifth (95th) percentile male stature.

EVALUATION RATIONALE (4.5.2)

Procedures and test methods for maintainability evaluation provide qualitative assessment of various integrated logistic support factors. This includes technical manuals, personnel, tools and test equipment, maintenance concepts, and making provisions for their effect on quantitative and qualitative demonstrated maintainability parameters.

EVALUATION GUIDANCE

Background:

The Using Service should provide guidance to the Contractor as to the relationship between system life cycle phases (C&E, Dem/Val, E&MD, production and deployment) and the maintainability evaluation phases. Coordination of the maintainability evaluation with other required demonstrations should be accomplished whenever possible to avoid unnecessary duplication of effort. The use of mockups or development hardware, in the early phases of E&MD, should be allowed for maintainability demonstrations since production hardware would not be available.

EVALUATION LESSONS LEARNED

Past demonstrations of maintainability have used unrealistic procedures. A separate ratchet, socket, and extension was used for each nut run up on all threaded studs during replacement of all controls and externals. The nuts were pre–placed in all sockets before the demonstration began. This resulted in unrealistic maintenance times.

3.5.2.1 Maintainability quantitative requirements Maintainability requirements shall be as specified in table XXIV. These values shall be derived from the system maintainability allocations and shall be based on the planned weapons system utilization rate and mission mix.

The engine shall have the following diagnostic capabilities:

a. WRA/LRU fault isolation of installed engine components for at least (a) percent of all failures shall be within (b) minutes.

b. Percent of success which diagnostics tests correctly isolates faulty subassemblies shall be (c) percent.

c. Percent of success which diagnostic tests correctly diagnoses fault by retest shall be (d) percent.

REQUIREMENT RATIONALE (3.5.2.1)

Maintainability considerations, depending on how well they are addressed during development, have a major impact on operational support costs and also affect the mission capability, because they affect the ability of maintenance crews to turn around aircraft between sorties. The degree of difficulty and length of time, required to perform maintenance, affects war making effectiveness.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph

(a): A value from 70% to 75%.

(b): A maximum value of 15 minutes (this includes time for hookup of all needed diagnostic equipment, external and internal to the aircraft, and to perform all diagnostic testing)

(c): A value from 90% to 95%.

(d): A value from 95% to 100%.

Background:

The R&M terms defined and specified in MIL–STD–721 and OPNAVINST 4790.2 should be used in this requirement. Maintainability index should be kept low, so that support costs will be kept low and mission capability will be improved. Special attention should be given to keeping task times low at the organizational level because of their direct bearing on aircraft turnaround time and thus on mission capability. The maintainability index, MMH/EFH, is a very useful parameter to control cost, since cost is a function of total maintenance time. The maintainability index is a versatile parameter, since it can be varied by changing task frequency, which is affected by reliability, or task times. Engine replacement time and WRU/LRU replacement time are key maintainability considerations at the organizational maintenance level and increase mission capability by allowing aircraft to be turned around more quickly between sorties. Rapid and accurate diagnostic capability is needed to reduce maintenance time, thus reducing maintenance costs. The need for diagnostics should be identified starting with known failure modes as identified via reliability considerations. Diagnostics should be able to isolate failures down to the WRA/LRU level within 30 minutes for at least 95% of all known failures. For ultimate maintainability, all known failures should be isolated to the faulty WRA/LRU within a reasonable amount of time. Special attention should be given to making diagnostic techniques reliable. Attempts should be made to monitor the conditions that exist when a failure occurs. Intermediate shops and, if necessary, depot facilities should be equipped to fault isolate using techniques which simulate flight conditions.

REQUIREMENT LESSONS LEARNED

The Army made significant maintainability improvements in the T700 engine by paying attention to maintainability considerations. Many improvements are now requirements in the specification. Some of these considerations were modular construction, ease in removal for repair, and no fix time before overhaul.

Significant reductions in “cannot duplicate rate” of electronic components on the F100 engine program were achieved when CERT–type techniques were used to fault isolate problems.

4.5.2.1 Maintainability quantitative requirements The requirements of 3.5.2.1 shall be evaluated as follows:

a. Total maintenance man–hour/EFH shall be determined via demonstration and analysis, and shall be based on the following:

(1) Frequency of failure based on the most up–to–date reliability predictions, results of development testing, and like–item field data where available.

(2) Maintainability demonstration task times as determined utilizing Using Service personnel and procedures for task times that are demonstrated. The number and skill level of personnel shall be consistent with current Using Service maintenance policy.

(3) Maintainability predictions for task times that are not demonstrated.

b. Mean time of repair shall be measured via demonstration.

c. Engine replacement time shall be measured via demonstration.

d. WRA/LRU replacement times shall be measured using established Using Service procedures.

e. Diagnostics:

(1) WRA/LRU fault isolation of installed engine components shall be evaluated by demonstration. This demonstration shall be based on failure modes which can be simulated. At least one failure mode shall be selected from each WRA/LRU which comprises at least the top 80 percent of the total WRA/LRU removal rate. Also, the top 30 failure mode drivers shall be included in the demonstration.

(2) Percent of success which diagnostic's tests correctly isolates faulty subassemblies shall be evaluated by demonstration. Data called for in table XXV shall be documented during the demonstration.

(3) Percent of success in which diagnostic tests correctly diagnose fault by retest shall be evaluated by analysis of data gathered during development, documenting the number of times that the diagnostic testing properly isolated problems or gave false readings.

EVALUATION RATIONALE (4.5.2.1)

Quantitative requirements for maintainability, maintainability indices, must be evaluated to confirm that the engine meets operational requirements.

EVALUATION GUIDANCE

Background:

It is recognized that maintainability requirements will be assessed for compliance with the specification at the time of the functional configuration audit. However, it is important that the contractor conduct incremental maintainability demonstrations on prototype hardware throughout the development program to assure that the ultimate design will be capable of meeting maintainability requirements. It is to be understood that the status of the results of incremental maintainability demonstrations will be reviewed throughout the development program, especially at the time of design reviews.

Maintenance demonstrations should be conducted under the conditions which are expected in the operational environment. Expected weather conditions should be considered if a large percentage of the maintenance tasks will be performed under adverse conditions. Where portions of the maintenance actions are expected to occur under different environmental conditions, multiple maintainability demonstrations should be conducted. The effects of varying task times should be weighed depending on the proportionate period of time that each of the varying environments is expected to be encountered.

Engine replacement time should be demonstrated using either an installation mockup or an aircraft. For a new centerline engine, a development mockup should be used to evaluate WRA/LRU accessibility.

As a minimum, failure modes observed during development should be analyzed during the diagnostics and effectiveness demonstration. Specific demonstration procedures should be established based on the automatic test equipment (ATE) system under consideration.

The maintainability index, MMH/EFH, is a function of task time frequency. The most up–to–date reliability demonstrations, accounting for measured failure rates, provide the most accurate failure frequencies. Real or simulated failures must be used to evaluate the diagnostic and fault isolation capabilities.

EVALUATION LESSONS LEARNED

In those instances in which maintainability considerations were not demonstrated, operational maintainability and mission capability suffered. The T700 program proved to benefit significantly in this area because of the effects of conducting maintainability demonstrations. During this program the Army required three demonstrations for maintainability. Each demonstration was an improvement on the previous one. They also required a full–time human engineering engineer present at all times.

3.5.2.1.1 Excluded maintenance functions All excluded maintenance and repair functions shall be listed in table XXVI.

REQUIREMENT RATIONALE (3.5.2.1.1)

The Contractor needs to specify all maintenance functions that are excluded from the maintenance index calculation.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: The contractor should provide the listing in table XXVI. If the Using Service prefers to have the list provided separately from the specification, then make sure the list of excluded maintenance and repair functions is in the CDRL. This paragraph should then be labeled “N/A”.

Background:

The following maintenance and repair functions should be excluded from the maintenance index calculation and added to the list:

a. Technical Order and Time Compliance Technical Order compliance when related to upgraded engine performance or other changes in specification requirements.

b. Removal and installation of non–engine supplied equipment to perform engine maintenance (e.g., access panels, parts installed on the engine or nacelle by the weapon system Contractor). [Excluded maintenance functions for non–engine supplied equipment to access engine components should be reviewed on a case–by–case basis to prevent shifting the maintenance accountability burden to the airframe side or losing accountability altogether.]

c. Engine fuel servicing.

d. Engine removal/replacement in the aircraft except when required due to:

(1) chargeable engine failures/malfunctions.

(2) engine design which requires removal to perform routing maintenance on such items as ignitors, fuel filters, oil filters, etc.

e. All maintenance tasks resulting from failures or discrepancies that are not chargeable to the engine.

f. All maintenance tasks required to position ground equipment, and/or move the aircraft.

REQUIREMENT LESSONS LEARNED

Some engine maintenance requirements were specified in the contract, such as for the T700–GE–401, rather than the engine model specification.

4.5.2.1.1 Excluded maintenance functions The requirements of 3.5.2.1.1 shall be evaluated by inspection and analysis.

EVALUATION RATIONALE (4.5.2.1.1)

Inspection and analysis of the excluded maintenance functions are needed to determine if any pertinent engine maintenance was excluded.

EVALUATION GUIDANCE

Background:

Logistic and maintenance personnel of the Using Service should review the excluded maintenance functions.

EVALUATION LESSONS LEARNED

None.

3.5.2.2 Maintainability qualitative requirements The engine shall meet the qualitative maintainability requirements designated in table XXVII.

REQUIREMENT RATIONALE (3.5.2.2)

When only quantitative requirements are specified and qualitative requirements are not considered, maintainability is adversely affected. Qualitative requirements are needed also.

REQUIREMENT GUIDANCE

The following should be used to tailor table XXVII:

Maintainability should be influenced fRom the tailored application of MIL–STD–470, MIL–STD–471, MIL–STD–1472, and MIL–STD–1800, as appropriate. The maintainability data developed from the maintainability program will be used as source data for MIL–STD–1388 as applicable. The list below should be reviewed to develop the requirements for table XXVII:

1) The contractor should design the engine to minimize required skills, tools, and support equipment.

2) Parts of the engine requiring routine service, checking, adjustment, or frequent replacement should be made readily accessible without removal of other parts or components, and without disassembly of the engine.

3) Provisions for access to the location of WRA/LRUs should be based on reliability predictions, i.e., the least reliable WRA/LRU on the engine should have the best accessibility.

4) WRA/LRUs should be no more than one deep. Removal of the WRA/LRU should not require the removal of unaffected WRA/LRUs.

5) Shimming or measuring of WRA/LRUs should not be required for WRA/LRU replacement.

6) No adjustments, balancing, matching, or rigging should be required to install or replace WRA/LRUs on an assembled engine.

7) Plumbing or electrical removal should not be required for WRA/LRU access or removal.

8) Captive fasteners should be used for WRA/LRU mounts/connections. No fasteners smaller than 1/4 inch diameter should be used for WRA/LRU mounts or for organizational level maintenance. Individual WRA/LRUs should have the same style and size fastener for mounting and the same style and size fasteners for mating.

9) Positive securing clamps, nuts, bolts, connectors, and keyed interfaces should be provided to prevent misalignment and improper installation of WRA/LRUs, plumbing, and cables. The design should permit removal without exceeding capacity/capability of required tools after the engine has been in service.

10) Fluid carrying systems subject to disassembly for WRA/LRU or engine replacement should be designed to minimize system drainage or spillage on the maintainer.

11) Features should be provided to prevent functionally different components with the same or similar design from being improperly installed.

12) Electrical connectors should be designed to prevent improper mating connectors.

13) On wing fan trim balance capability should be provided when applicable.

14) Safety wire, cotter pins, and bent tab washers should not be used in securing connectors, fasteners, etc. at the intermediate and organizational levels and should be kept to a minimum at the depot level. Any safety wire, cotter pins, and bent tab washers used at this level should be confined to those areas out of the reach of organizational level personnel.

15) The engine design should allow centrally located engine–to–airframe disconnects, subject to survivability/vulnerability requirements.

16) The design should not require engine–to–engine, engine–to–aircraft, or accessory/installation rigging. System alignment or complicated operational checks should not be required.

17) The design should allow for adequate access, space, and tool clearance around fasteners and connectors to provide adequate clearance for component removal and installation at all levels of assembly; and to provide proper seating of tools and uniform torque application.

18) WRA/LRU modules/SRUs should be interchangeable with like items without match balancing.

19) The design should provide provisions for rapid repair such as external tubing repair versus whole line replacement.

20) Placement of engine furnished electronic equipment in the airframe rather than on the engine should be considered to avoid hostile environment posed by the engine.

21) Sensor check/replacement should not require engine rig or trim. Any sensor which requires more than 20 minutes to replace while the engine is installed in the aircraft should not be mission critical.

22) Aircraft leveling should not be required for installed engine servicing or quantity status reading.

23) Fluid level quantity indicators, if part of the engine design, should be clearly visible for accurate reading using Using Service approved red or white flashlight or natural daylight illumination sources.

24) Connecting devices should be selected to prevent improper installation while minimizing the number of different parts in accordance with MIL–STD–965.

25) No routine/scheduled maintenance should be needed for field washing, cleaning, or drying of engine components. However, the engine should incorporate provisions for water wash compressor cleaning if required.

26) The design should include provisions for the incorporation and use of pyrometers to check temperatures on all turbine blades during depot level testing.

27) Engine operation after installation in the aircraft should not be required except for leak check and control.

28) Disconnect points for external plumbing should coincide with module/SRU flange/disconnect locations.

29) The engine hydraulic system should include dedicated fittings conveniently located on the engine for each access and attachment of ground test equipment when installed in the aircraft.

30) The color red should not be used on the engine for any identification means without an adjacent symbol readily visible under redlight conditions.

31) Components subject to wear, requiring repair/rework to restore dimensioned features, shall provide a dimensional baseline datum which is not destroyed or degraded in service or during repair/rework.

32) The engine configuration should permit installation in the aircraft with as–received components and connectors mounted on the engine.

33) Estimated maintenance and repair periods, to be provided, should not be construed as final determinations. These should be established via the Logistic Support Analysis Process.

REQUIREMENT LESSONS LEARNED

Previous engine designs which violated the provisions of the above list experienced increased support costs and reduced mission capability.

In the past, many engines sustained FOD when nuts, bolts, and fasteners were left on the ramp. Therefore, captive fasteners are a desirable maintenance feature.

4.5.2.2 Maintainability qualitative requirements The requirements of 3.5.2.2 shall be evaluated by demonstration and inspection.

EVALUATION RATIONALE (4.5.2.2)

Evaluation of qualitative maintainability must be accomplished by demonstration and inspection.

EVALUATION GUIDANCE

The following should be used to tailor table XXVII:

Qualitative maintainability should be demonstrated in an operational environment or on an installed engine mockup. For each of the requirements listed in table XXVII, an evaluation method should be noted. Most of the requirements can be evaluated by demonstration and inspection, while a few may also include some amount of engine operation or test as part of the demonstration (e.g., fan balancing should be evaluated by demonstration of the balancing task, and operating the engine to demonstrate the task has been properly performed).

EVALUATION LESSONS LEARNED

When qualitative maintainability requirements were not adequately analyzed early during development, operational support costs went up and mission capability was reduced.

3.5.2.3 Maintenance All maintenance events, scheduled and unscheduled, shall be completed at the lowest maintenance level feasible and limited to the least number of actions possible.

REQUIREMENT RATIONALE (3.5.2.3)

Having engine maintenance to fit within the existing framework is needed so it can be cost effective in terms of manpower, tooling, and implementation. Completing scheduled and unscheduled maintenance events at the lowest maintenance level feasible is needed for cost effectiveness in terms of labor rates and transportation costs, and it helps to maintain a higher readiness rating and reduces the “spares” pipeline.

REQUIREMENT GUIDANCE

Background:

The US Navy maintenance posture consists of three maintenance levels. Level one is the simplest (squadron level) and level three is the most detailed (depot). At all times carrier–based maintenance should be considered as the core of the USN maintenance posture. OPNAVINST 4790.2 explains the US Navy policy in detail. The US Air Force and US Army maintenance posture contains two levels of maintenance, squadron and depot.

The number of scheduled maintenance events, frequency, and completion times should be reduced over the current field engines.

REQUIREMENT LESSONS LEARNED

None.

4.5.2.3 Maintenance The requirements of 3.5.2.3 shall be evaluated by demonstration, analysis, and test. A reliability centered maintenance analysis in accordance with MIL–STD–2173 shall be the basis for establishing preventative maintenance requirements (scheduled maintenance). All maintenance actions (scheduled and unscheduled) shall be evaluated through demonstrations conducted as specified in 4.5.2.

EVALUATION RATIONALE (4.5.2.3)

The reliability centered maintenance analysis is meant to be an objective view of the maintenance requirements such as preventative maintenance, scheduled on–condition tasks, scheduled rework tasks, scheduled discard tasks, safe life limits, economic life limits, scheduled failure finding tasks, and age exploration.

EVALUATION GUIDANCE

Background:

The number of scheduled maintenance events, frequency, and completion times should be reduced over the existing field engines. Any maintenance event should be evaluated during each phase of the program when the opportunity arises (Altitude testing, Endurance testing, Flight testing, etc.).

EVALUATION LESSONS LEARNED

None.

3.5.2.3.1 Modules The structural components which define each engine module, and the length of time to remove and replace each module, shall be specified in table XXVIII. All modules shall be capable of removal and replacement at the (a) maintenance level. Disconnects for all external plumbing and wiring associated with a module shall be located as near the front and back planes of the module as possible.

REQUIREMENT RATIONALE (3.5.2.3.1)

Modular concept is needed for easy replacement of major engine sections and to reduce or eliminate the requirement to send the complete engine back to the depot for maintenance.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should provide the maintenance level.

Background:

Replacing major engine sections at the organizational or intermediate maintenance level can either reduce or eliminate the requirement to send the complete engine back to the depot for maintenance when an internal engine repair is necessary. This approach will tend to improve the availability of the engine (and weapons system) for service use and reduce labor time and cost at the field and intermediate maintenance levels.

Modules should be completely interchangeable among engines. This interchangeability should be possible for any combination of modules at any time during their required life. Complete engines should be interchangeable in multi–engine aircraft (i.e., right side to left side). Typical engine modules are: fan, compressor, turbine, augmentor/nozzle, etc.

REQUIREMENT LESSONS LEARNED

In the past, an internal engine part failure often required that the complete engine be shipped back to the depot to effect the repair. This tied up two sets of hardware: all the satisfactory components of the faulty engine and another set of duplicate components in the replacement engine shipped into service. The modular concept has reduced the need for a second set of satisfactory duplicate components.

4.5.2.3.1 Modules The requirements of 3.5.2.3.1 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.5.2.3.1)

Inspection and demonstration are required to evaluate the modular concept.

EVALUATION GUIDANCE

Background:

Performing demonstrations on engine mock–ups or zero time engines will not result in representative remove and replacement times because interface wear between modules which results in alignment difficulties will not exist.

Maintenance personnel of the Using Service should evaluate the proposed modular concept.

EVALUATION LESSONS LEARNED

Past engine programs have shown that field engines have longer module replacement times than the contractor showed on development engines due to field environment effects.

3.5.2.3.2 Maintenance, inspection and repair cycle The estimated maintenance, inspection and repair periods for the total engine and each module shall be specified in table XXIX, together with the estimated time in manhours required to perform these functions.

REQUIREMENT RATIONALE (3.5.2.3.2)

The Contractor needs to provide the information necessary to show that basic maintenance, inspection and repair functions have been considered in the design of the engine.

REQUIREMENT GUIDANCE

The following is recommended to be transferred verbatim into table XXIX of the specification:

The table below shall be used for estimated maintenance, inspection, and repair periods for the total engine and each module.

a. Time to remove and replace modules, assuming engine uninstalled and including the time to remove and refit other modules and externals as necessary:

Module	Total Time	Elapsed Time

b. Removal and replacement times of externally mounted engine components with no allowance being made for aircraft access:

Item	Total Time	Elapsed Time

c. Estimated module inspection period:

Module	Module Inspection Interval	Mean Time Between Repairs	
		Initial	Mature

d. Depot overhaul times (estimated maintenance manhours for engine/module rework):

Module	Rework Time	Repair Time

Background:

The contractor should show balance between engine performance, reliability, and maintainability relative to the life cycle cost and ultimate service usage of the engine.

The maintenance and repair cycles of the engine components should coincide or be some multiple of each other to eliminate duplicate and frequent engine removals (from Guidance 4.5.2.3.2). The time between scheduled inspections should be lengthened or eliminated where possible to reduce the life cycle cost.

REQUIREMENT LESSONS LEARNED

The following is a table of expected values, based on USN experience, for current engines types:

Engine Type	Module	Average FHRS–1st Failure	Hot Section Inspection
Turbofan	Fan	500	None
	Compressor	430	None
	Combustor	525	None
	HPT	720	None
	LPT	490	None
	Augmentor	385	None
Turboshaft	Cold Section	400	None
	Accessory	580	None
	Hot Section (Combustor, G.G.)	875	None
	Power Turbine	710	None
Turboprop	Gearbox	824	
	Power Section	1450	
	Torque Meter	Insufficient Data	

The following is a table of USAF expected values for the F100–229:

	Design TACs	Life TAC*	Inspection TACs	Interval TAC*
Cold Parts (Durability Critical)	8,000	8,600	4,000	4,300
Cold Parts (Fracture Critical)				
In–service Inspectable	8,000	8,600	4,000	4,300
In–service Non–inspectable				
Deterministic	16,000	17,200		
Probabilistic	8,000	8,600		
Bearings (based on average bearing and engine installation tolerances)				
Ball (B1 value based on a material factor = 25)	4,000 EOH		4,000	4,300
Roller (B1 value based on a factor = 6)material	4,000 EOH		4,000	4,300
Hot Parts (Durability Critical)	4,000	4,300		
Hot Parts (Fracture Critical)	4,000	4,300	4,000	4,300
Augmentor Module (excl. duct)	1,200	(total operation hr)	1,600	1,720
Ignitor Plugs High Energy	2,000	2,150	2,000	2,150
Low Energy	1,000	1,075	2,000	2,150
Augmentor	2,000	2,150	2,000	2,150
Expendable Items	4,000	4,300	4,000	4,300

* Includes ground operation and test cell/troubleshooting time.

4.5.2.3.2 Maintenance, inspection, and repair cycle The requirements of 3.5.2.3.2 shall be evaluated by demonstration.

EVALUATION RATIONALE (4.5.2.3.2)

There is a need to perform the maintenance, inspection, and repair cycles to demonstrate functionality.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.5.2.3.2.1 Maintenance inspection techniques Engine design shall permit maximum use of non–destructive inspection techniques and multi–purpose test and inspection equipment. Inspection provisions, including access envelopes, shall be shown on the engine configuration and envelope figure.

Provisions for 360 degree inspection of the installed engine shall be made for the fan, compressor, combustor, and turbine sections of the engine. A positive means of slowly rotating the rotor system shall be provided to facilitate inspection. The access port size shall be a minimum of (a) mm ((a) inch). The same tool shall be used for removal

and replacement of port covers, plugs, and associated fasteners. The location of access ports shall, as a minimum, permit inspection of the following locations: (b) . Access ports shall be accessible without removing other components.

REQUIREMENT RATIONALE (3.5.2.3.2.1)

An effective maintenance philosophy requires the detection of a potential failure condition. To be cost effective the inspection techniques should not require disassembling the engine. The borescope inspection method is the most common non–destructive inspection technique used, so a sufficient number, size, and location of access ports needs to be stated so an adequate internal inspection of the engine can be accomplished.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Borescope ports should be large enough to accept the largest borescope used by the maintenance group. The Using Service should indicate the borescope port size. A typical size for large engines is 10 mm (0.4 inches). For small engines, the borescope access port size should be a minimum of 5 mm (0.2 inches).

(b): The Using Service should indicate the location of access ports or the areas to be seen through inspection. Typical areas of interest are:

a. Fan

b. Compressor inlet guide vane leading edge.

c. Compressor inlet guide vane trailing edge and compressor first stage rotor leading edge.

d. Compressor last stage rotor trailing edge and combustor inlet.

e. Combustor liner and fuel nozzle faces.

f. Combustor outlet and turbine first stage vane leading edge.

g. High pressure turbine first stage vane trailing edge and first stage rotor leading edge.

h. High pressure turbine last stage rotor trailing edge and low pressure turbine first stage vane leading edge.

i. Low pressure turbine first stage vane trailing edge and first stage rotor leading edge.

j. Power turbine first stage vane trailing edge and first stage rotor leading edge.

The number of access ports should be kept to a minimum. They produce areas of high stress which eventually produce cracks in cases. Access ports are a source of leakage and maintenance discrepancies.

REQUIREMENT LESSONS LEARNED

In the past, some engines had a limited number of access ports. Therefore, complete 360 degree coverage of non–rotating parts was impossible. This prevented detection of impending failures.

Some past engines had small borescope access ports (10.16 mm, 0.40 inch) which required special borescopes for inspection. This led to higher costs for logistics support.

Past fan engines have required a set of special tools to remove and install borescope plugs, port covers and fasteners in access ports, which have caused purchase of the special tool sets and have increased logistics support costs.

4.5.2.3.2.1 Maintenance inspection techniques The requirements of 3.5.2.3.2.1 shall be evaluated by inspection and demonstration on an installed engine.

EVALUATION RATIONALE (4.5.2.3.2.1)

Maintenance inspection techniques must be demonstrated by the Using Service personnel using procedures and tools/equipment available in the field.

EVALUATION GUIDANCE

Background:

Each inspection method developed for the engine should be employed during routine inspections of development engines. Any deficiencies that are discovered should receive design attention as early as possible so that improvements can be made prior to engine flight operation.

EVALUATION LESSONS LEARNED

Past demonstrations by the contractor used procedures and tools that were never used in the field.

3.5.2.3.3 Tools Required tooling shall be held to a minimum. The engine shall be capable of being maintained with a set of standard hand tools, as identified by the Using Service. Where provisions for standard hand tools are not feasible, the design shall provide, wherever possible, for special tools and test equipment that are available and in use on other in–service engines, as identified by the Using Service. Any requirement for new special tools, fixtures, and test equipment shall be subject to specific approval of the Using Service. Cadmium–plated tools are permitted with specific approval of the Using Service.

REQUIREMENT RATIONALE (3.5.2.3.3)

The use of standard hand tools is needed to minimize the cost of tooling for the engine. Required tooling must be kept to the absolute minimum for economy and ease of maintenance.

REQUIREMENT GUIDANCE

Background:

Cadmium plated tools may be prohibited if titanium materials exist in the engine. However, the Air Force has not encountered a detrimental problem when cadmium tools are used in titanium at temperature of 600°F or less. If titanium is used above 600°F with cadmium, liquid metal embrittlement may occur.

The Air Force has discovered that iron oxide may cause corrosion in titanium.

REQUIREMENT LESSONS LEARNED

Non–standard tools have been quite expensive, and logistics/provisioning problems have occurred in service when an inadequate number of each special tool was purchased to support maintenance and repair efforts. An engine designed to be maintained with standard tools has been simpler for personnel to maintain. It has taken less time to train personnel to maintain an engine using standard tools rather than special tools.

NADEP, Norfolk, found in the early 70's that "cracking of stressed titanium can occur due to contact with solid metals such as cadmium". The NADEP "banned the use of cadmium plated tools while working on titanium hardware" and coated the tools with black iron oxide.

4.5.2.3.3 Tools The requirements of 3.5.2.3.3 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.5.2.3.3)

The use of hand tools can only be evaluated by inspection and demonstration.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.5.2.4 Battle damage repair The capability for battle damage repair shall be described. Combat damage repair design techniques shall utilize tools and materials found at organizational and intermediate levels.

REQUIREMENT RATIONALE (3.5.2.4)

The primary purpose of combat damage repair is to allow aircraft to continue flying combat missions. A secondary purpose and absolute necessity is to allow ferrying of the damaged aircraft to rear areas for more lasting repairs.

REQUIREMENT GUIDANCE

Background:

Survivability is measured by the capability of the aircraft to accomplish its mission in a hostile environment. The ability of the subsystems to retain mission essential performance following a hit is a major contribution to survivability. The ideal repair resolution is to fly the airplane with existing engine damage if possible. Where damage would jeopardize flying safety or mission accomplishment, repairs must be authorized that can be completed within a short time with tools, men, material, and spares on hand.

REQUIREMENT LESSONS LEARNED

Aircraft operating in high–threat, high–intensity conflicts have received frequent damage from combat. War games conducted with computer models have indicated that during the early days of a conflict many aircraft will be returning from combat damaged.

4.5.2.4 Battle damage repair The requirements of 3.5.2.4 shall be evaluated by analysis and demonstration.

EVALUATION RATIONALE (4.5.2.4)

Battle damage repair techniques must be evaluated.

EVALUATION GUIDANCE

Background:

Specimens from the engine component ballistic tests should be used to demonstrate battle damage repairs. Emergency repair methods, materials and restrictions for non–catastrophic battle damage should be analyzed. The report should include an analysis which evaluates the effects of repairs on engine operability, reliability and maintainability (e.g., performance, life, and oil and fuel consumption, etc.).

Structural repair or patching of advance engine materials, especially composites, must be demonstrated since the data base may be very limited. Repair of cooling liners in afterburner augmentor/tail pipes especially needs attention to enable puncture repair and preserve cooling path.

EVALUATION LESSONS LEARNED

None.

3.5.3 Human performance and human engineering The engine shall be designed to be maintained by at least the central ninety percent (90%) of the maintainer population (5th percentile female stature through 95th percentile male stature) wearing a full complement of personal protective equipment, under all environmental conditions (including day/night), defined in the specification and all operational scenarios at operating bases and deployed locations in accordance with MIL–STD–1472, MIL–STD–1800, and MIL–H–46855.

REQUIREMENT RATIONALE (3.5.3)

The application of human engineering is necessary in engine systems, subsystems, equipment, and facilities within the limits of time, cost, and performance trade–offs.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The Using Service should select the applicable requirement between MIL–STD–1472 and MIL–STD–1800.

Background:

Human engineering should be applied in order to achieve mission success through integration of the human into the system, subsystem, equipment, and facility, and achieve effectiveness, simplicity, efficiency, reliability, safety of system operation, training, and maintenance, and foster design standardization within and among engines.

REQUIREMENT LESSONS LEARNED

None.

4.5.3 Human performance and human engineering The requirements of 3.5.3 shall be evaluated by demonstration and analysis.

EVALUATION RATIONALE (4.5.3)

Evaluation of human engineering is necessary to achieve required performance by operations and maintenance personnel.

EVALUATION GUIDANCE

Background:

During the maintenance demonstration the contractor should use the minimum and maximum physical requirements for the personnel. They should be clothed for battle field conditions and extreme low temperatures.

EVALUATION LESSONS LEARNED

Maintenance personnel dressed in chemical protective clothing with arctic gear for cold protection had problems performing maintenance task.

3.6 Combat survivability The engine shall be designed to sustain mission performance in the intended combat threat environment. If the engine is intended for a single aircraft or class of aircraft, the engine shall be optimized for the threat environment described in the System Threat Report (STR) defined in DODI 5000.2.

If the engine is partly or wholly intended for a general class of aircraft with combat missions (e.g. patrol/anti submarine warfare, combat reconnaissance, combat transport, strike fighters/attack, assault helicopter, or other), the engine shall be optimized for the general threat environment described in the threat assessment for those missions or classes of combat.

REQUIREMENT RATIONALE (3.6)

Combat survivability must be required. An aircraft engine will affect, and in many cases dominate, critical survivability characteristics of the aircraft systems.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: Section 3.6 should be classified when tailored for a particular weapon system. A classified appendix with the same numeration, paragraph titles, and sequence of Section 3.6 should be attached to or referenced by the RFP/model specification. In the event a classified appendix is utilized, the paragraph numbers and titles in Section 3.6 should be retained but type "See Classified Appendix __" in the text portion of each paragraph.

Background:

This paragraph and its subparagraphs may be classified in accordance with the contract DD form 245 or other government provided classification guidance and shall be in a separable classified appendix (paragraph number and arrangement should remain the same).

Critical engine survivability characteristics should be specified for intended aircraft systems mission performance requirements, which might include observable reduction, vulnerability, degree of benign failure, fail–safe feature or tolerance to other threats effects such as fuel ingestion or blast pressure transients. The tri–services agree that engine performance related to threat weapons capabilities must be classified, although the degree of classification may vary.

REQUIREMENT LESSONS LEARNED

Observable and ballistic test experience in the V–22 osprey (T406), F–22 (F119), and other engine programs, demonstrate that the critical engine survivability characteristics are capable of considerable design optimization, provided that appropriate design/performance requirements are specified. Development tests and evaluations (DT&E) need to contribute to the design process and should be conducted as early as possible in development. These tests and evaluations should be directed toward specific design adaptation prior to IFR.

4.6 Combat survivability Survivability development test and evaluation (DT&E) shall be conducted on the engine to evaluate critical survivability characteristics at specified development milestones.

EVALUATION RATIONALE (4.6)

Considerable DT&E is required to help define the achievable engine survivability characteristics and their contribution to aircraft system mission effectiveness.

Optimizing the engine in combination with the total aircraft system and knowledgeable allocation of critical survivability between airframe and engine will produce the most effective combat system. Survivability DT&E is required to provide information to designers seeking to incorporate survivability requirements with the lowest penalties.

To ensure that ongoing DT&E is accomplished, progress toward achievement of critical survivability characteristics must be measured and reported at appropriate engine design milestones. Survivability DT&E is also required to determine that specification requirements have been met and to measure the actual survivability characteristics that have been achieved.

EVALUATION GUIDANCE

Background:

The individual survivability evaluation requirements for each critical survivability characteristic or requirement should be specific and detailed in the subparagraphs below, and should be based on engine missions or mission types. The description of the overall evaluation procedure and critical milestones, as applicable, should be specified in this section. Tests required in the survivability program should reference MIL–STD–2069 or MIL–STD–1799.

EVALUATION LESSONS LEARNED

A considerable data base of engine survivability evaluations has been available from individual services and repositories such as the Survivability and Vulnerability Information Analysis Center (SURVIAC), which is sponsored in part by the joint technical coordinating group on aircraft survivability (JTCG/AS), and is located at the Wright–Patterson Air Force Base in Ohio.

3.6.1 Susceptibility The engine shall not exceed the detectable signatures and observable characteristics as defined by the Using Service.

REQUIREMENT RATIONALE (3.6.1)

Survivability of the aircraft in combat often depends on the level of its observable signature and characteristics. Survivability may be enhanced by altering the signature so as to defeat one or more of the critical events that must occur in the detection, fly–out or “end–game” portion of a weapon system guidance or enemy tactics.

The engine is often the dominant contributor to air vehicle signatures, particularly in the forward and aft direction. Thus, in some aircraft/engine system combinations, survivability will be highly dependent on engine design. In these cases the signature becomes a critical system characteristic of the air vehicle as defined in DODI 5000.2

REQUIREMENT GUIDANCE

Background:

Engine signature requirements should be determined from the aircraft system mission effectiveness and survivability trade studies. Alternative analyses should be conducted by the appropriate cooperative combinations of government, airframe contractor and engine contractor to determine the most effective and practical combination of critical signature characteristics to be allocated to airframe and engine. The Using Service must provide detailed mission scenarios based on the validated system threat report defining realistic threat encounter conditions envisioned in the mission need statement or equivalent document. Similar trade studies should be conducted for engines being developed for a number of different aircraft with varying mission–threat–encounter conditions.

The Using Service/contractor should consider the range of electromagnetic emissions (EME) for purposes of enemy weapon cueing. The EME should not exceed the ranges of other observables by the air vehicle.

REQUIREMENT LESSONS LEARNED

Combat experience gained with the F–117 in Panama, and in the Persian Gulf, demonstrates that aircraft with specially tailored observable signatures can be enormously effective when used properly. Classified range tests involving a variety of combat aircraft types also bear out that a proper mix of susceptibility reduction features can have greatly enhanced effectiveness.

4.6.1 Susceptibility The requirements of 3.6.1 shall be evaluated by analyses and tests.

EVALUATION RATIONALE (4.6.1)

Engine signature characteristics are often major contributors to combat aircraft signatures. They may have a considerable impact on system performance and on system survivability and mission effectiveness. Susceptibility reduction design achievements must be measured and evaluated at appropriate design review milestones.

EVALUATION GUIDANCE

Background:

An overall signature and observable characteristics evaluation program should be established by the Using Service describing DT&E analyses, tests, and schedules. This evaluation program must be constructed to provide all of the technical information needed to determine the progress of the design at design review milestones and to measure achieved characteristics at completion.

EVALUATION LESSONS LEARNED

Signature prediction codes and war gaming models used to evaluate signatures, though extremely sophisticated, have been supplemented with extensive actual testing. Experience in classified programs abound with examples where surprises occurred when full scale models or actual hardware were tested on ranges. Experience clearly shows that well designed and timed evaluation programs, using appropriate tests and not relying totally on analyses, reduce risk.

3.6.1.1 Noise The near and far field noise levels for an uninstalled engine, vectored or non–vectored, operating at sea level static shall not exceed the levels of figures 21a, 21b, and 21c.

REQUIREMENT RATIONALE (3.6.1.1)

Engine acoustic signature noise represents a health hazard to personnel, an annoyance to civilian communities, and an observable which can be used to detect combat and search and rescue aircraft.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: The contractor should provide levels in figures 21a, 21b, and 21c. The figures should include the full range of vectored positions, if applicable, including those for take–off and landing.

Background:

Noise reduction, while desirable, should remain secondary to performance requirements. No noise limits are defined here since such limits could place constraints on necessary performance. Engine noise limits should be defined on a program–by–program basis. Combat aircraft are usually not required to meet any standards, unless imposed for detection reduction purposes. Transport type aircraft should be required to meet flight noise requirements set forth in the Federal Aviation Regulations (FAR) Part 36. This requirement for transport aircraft is outlined in Air Force Regulation (AFR) 80–36. Transport aircraft are exempt from FAR 36 only if it is determined that compliance would inhibit mission success.

Engine noise limits should include consideration of crew environment, aircraft structural requirements, maintenance, and ground operation. Noise limits should be established to preclude health hazards to personnel. Personnel noise limits should be specified by the government in accordance with AFR 161–35 noise exposure limits. Aural signature must be controlled so that noise is not detectable by threat weapons.

REQUIREMENT LESSONS LEARNED

Aircraft engine noise has caused hearing loss. Noise from aircraft engines being tested or operated in the vicinity of populated areas produced complaints from the public and in some cases, resulted in restricted operation.

4.6.1.1 Noise The requirements of 3.6.1.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.6.1.1)

Test measurement of engine noise is required to demonstrate specification limits are not exceeded by the actual engine.

EVALUATION GUIDANCE

The blanks must be filled in (this is guidance); the S&V community is requested to give us recommended numbers.

The following is recommended to be transferred verbatim into the specification paragraph: A noise survey shall substantiate that limits will not be exceeded. The engine shall be mounted on an outdoor test stand with a minimum clearance of seven feet between the lowest part of the engine and the ground. Microphones shall be located in relatively flat terrain, free of excessive ground absorption characteristics. There shall be no obstructions that significantly influence the engine noise field.

The weather shall be free of precipitation with relative humidity between __ and ___ percent, ambient temperature between __°C (__°F) and __°C (__°F), wind velocity less than __ kilometers per hour (__ miles per hour) and no temperature inversions or anomalous wind conditions. No post–test engine recalibration shall be required. The signal level shall be at least __ dB greater than the background noise level in each third octave band in the frequency range of interest.

a. Data presentation. The engine noise level data shall be presented as follows and in accordance with the format shown on the referenced figures:

(1) Equal overall sound pressure level contours (dB reference of 0.0002 micro bars) as shown on figures 21a and 21b.

(2) One–third octave frequency spectra from a center frequency of 20 Hz to 10 kHz at the positions A through G specified on figure 21a shall be tabulated. All near field sound pressure levels shall be based on measurements at head level where personnel are located during engine maintenance or other operation. In addition, one–third octave band sound pressure levels from a center frequency of 20 Hz to 10 kHz in the form of uncorrected data (e.g., uncorrected for weather, terrain, etc.) shall be tabulated for all table Ia, Ib, and Ic performance points and for __ positions. The __ positions shall be based on measurements located __ meters (___ feet) radially from the center of the engine's exhaust plane at ten degree increments starting at zero degrees directly in front of the engine and ending at 180 degrees directly behind the engine.

(3) Equal sound pressure level contours plotted for octave bands with center frequencies of ___, ___, and____ Hz shall be shown on figure 21a for the Maximum and Maximum Continuous power of table Ia, Ib, and Ic only.

(4) Equal perceived noise level (tone–corrected) contours in __ PNdB increments shall be shown on figure 21b for all table Ia, Ib, and Ic performance points.

(5) Three narrow band spectrum plots from 20 Hz to 10 kHz for the inlet and exhaust noise shall be shown in figures. These narrow band spectrum plots shall be presented in a spectral density format (e.g., sound pressure spectrum level) which normalizes the data to account for changing analysis bandwidth. (Position and power setting for the narrow band data shall be determined for an examination of the third–octave spectra).

(6) The total acoustic power (dB ref _____ watts) generated at the Maximum power condition.

(7) The contractor shall provide sufficient data in the report to enable prediction of the engine noise signatures at any power setting. The required data shall include rotor speed, number of blades, hub to tip ratio, diameter, discharge total temperature, pressure ratio, exit velocity, exhaust mass flow, and ambient temperature and pressure.

b. Measurement system. The acoustical measurement system shall consist of approved equipment equivalent to the following:

(1) A microphone system with frequency response compatible with measurement and analysis system accuracy as stated in c. below.

(2) Tripods or similar microphone mountings that minimize interference with the sound being measured.

(3) Recording and reproducing equipment with characteristics, frequency response, and dynamic range compatible with the response and accuracy requirements of c. below.

(4) Acoustic calibrators using sine wave or broadband of known sound pressure level. If broadband noise is used, the signal shall be described in terms of its average and maximum RMS value for a non–overload signal level.

(5) Analysis equipment with the response and accuracy requirements of d. below.

c. Sensing, recording and reproducing equipment. The sound produced by the engine shall be recorded in such a way that the complete data history is retained. A magnetic tape recorder is acceptable.

(1) The characteristics of the system must comply with recommendations given in International Electromechanical Commission (IEC) 179 with regard to the microphone and amplifier characteristics.

(2) The response of the complete system to a sensibly plane progressive sinusoidal wave of constant amplitude must lie within the tolerance limits specified in IEC 179 over the frequency range of interest.

(3) The equipment must be acoustically calibrated using facilities for acoustic free field calibration and electronically calibrated as stated in paragraph d. below.

d. Analysis equipment. A frequency analysis of the acoustical signal shall be performed using one–third octave filters complying with the recommendations given in IEC 225.

(1) The analyzer indicating device must be analog, digital, or a combination of both. The preferred sequence of signal processing is:

(a) Squaring the one–third octave filter outputs.

(b) Averaging of integrating and linear to logarithmic conversion.

(2) The indicating device must have a minimum crest factor capacity of three and shall measure, within a tolerance of $\pm$1.0 dB, the true root–mean–square (rms) level of the signal in each one–third octave band. If other than a true rms device is utilized, it must be calibrated for nonsinusoidal signals. The calibration must provide means for converting the output levels to true rms values.

(3) The amplitude resolution of the analyzer must be at least ___ dB.

(4) Each output level from the analyzer must be accurate within $\pm$___ dB with respect to the input signal, after all systematic errors have been eliminated. The total systematic errors for each of the output levels must not exceed $\pm$___ dB. For continuous filter systems, the systematic correction between adjacent one–third octave channels may not exceed ___ dB.

(5) The dynamic range of the analyzer must be at least 55 dB in terms of the difference between full–scale output level and the maximum noise level of the analyzer equipment.

(6) The complete electronic system must be subjected to a frequency and amplitude electrical calibration by the use of sinusoidal or broadband signals at frequencies covering the range of interest of known amplitudes covering the range of signal level furnished by the microphone.

If broadband signals are used, they must be described in terms of their average and maximum rms values for a non–overload signal level.

(7) Narrowband analysis shall be conducted with a maximum bandwidth of ___Hz in the __ to ___Hz frequency range, __ Hz in the ___ to ___ Hz frequency range, and __ Hz in the ___ to ___ Hz frequency range.

e. Noise measurement procedures. In order to ensure uniform practices relative to acoustic testing of engines, the following procedures shall be required:

(1) Immediately prior to and after each test, a recorded acoustic calibration of the system shall be made in the field with an acoustic calibrator to check system sensitivity and provide an acoustic reference level for the analysis of the sound level data.

(2) For the purpose of minimizing equipment or operator error, field calibrations shall be supplemented with the use of an insert voltage device to place a known signal at the input of the microphone, just prior to and after recording engine acoustic test data.

(3) The ambient noise, including both acoustical background and electrical noise of the measurement system, shall be recorded and determined in the test area with the system gain set at levels which will be used for aircraft engine noise measurements.

f. Reporting and correcting measured data. Data representing physical measurements or corrections to measured data shall be recorded in permanent form and included in the final report. Estimates must be made of the individual errors inherent in each of the operations employed in obtaining the final data.

(1) Measured and corrected sound pressure levels must be presented in one–third octave band levels obtained with equipment conforming to the standards described in previous paragraphs.

(2) The type of equipment used for measurement and analysis of all acoustic, meteorological, and engine performance data shall be reported.

(3) The following atmospheric environmental data shall be measured at representative microphone positions.

(a) Air temperature in degree Celsius (Fahrenheit) and relative humidity in percent.

(b) Maximum, minimum, and average wind velocity in knots and wind direction relative to engine centerline.

(c) Atmospheric pressure in kPa (psf).

(d) Comments on local topography, ground cover, and events that might interfere with sound recordings shall be reported.

(4) All data shall be corrected to the following:

(a) Sea level pressure of 101.325 Kpa/ (2116 psf).

(b) Ambient temperature of 25°C (77°F) (ISA + 10°C).

(c) Relative humidity of 70 percent.

(d) Zero wind.

On an engine with vectoring exhaust nozzles, the noise survey shall include all vectored positions from the centerline in increments of 5 degrees or less, as determined by the Using Service. This may require a change in the engine mounting on the outdoor test stand.

EVALUATION LESSONS LEARNED

None.

3.6.1.2 IR radiation The propulsion system contribution to the total platform signature shall be specified herein. The engine shall not exceed the maximum IR signature levels of (a), while meeting the requirements of 3.3, after one hot section life.

REQUIREMENT RATIONALE (3.6.1.2)

Combat survivability depends on keeping IR levels below which the enemy can detect and utilize, or use to target or defeat (kill) the aircraft.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should determine and specify the maximum IR signature levels.

The following should be transferred verbatim into the specification paragraph, to be filled in by the contractor, and placed in model specification in a classified appendix:

The following shall be the absolute (non–contrast) IR signatures for the uninstalled engine:

a. Azimuth angles: 0°; 10°; 20°; 30°; 45°; 60°; 90°; 120°; 135°; 150°; 160°; 165°; 170°; 175°; 180°. An extension of the centerline forward of the engine shall define the 0° azimuth and 0° elevation position. The 180° azimuth corresponds to the centerline aft of the engine. The 0° azimuth, 0° elevation angles, and centerline are defined as being in a plane parallel to a level ground plan. If the radiation pattern is symmetrical about the centerline, a polar plot with notation indicating symmetry may be used.

b. Elevation angles: 0°; 5°; 10°; 15°; 20°; 30°; 40°; 60°; 90°; 135°; 160°; 170°; and 175° (above and below horizontal).

c. IR spectral regions: IR spectral intensities (Watts/Steradian–Micrometer) shall be computed between 1 and 15 micrometers with a spectral resolution of at least 5 to 50 cm^{-1}. In addition, broadband (integrated) intensities (Watts/Steradian) shall be computed for the following filter bands of 100 percent transmittance: 1–3, 3–4, 4–4.7, 3–5, and 8–14.

d. Attenuation ranges: Spectral and broadband intensities defined in paragraph c. shall be evaluated at source (zero range), 0.1, 2, and 10 km (0.66, 1.2, and 6.2 miles ranges along each viewing angle specified in paragraphs a and b. The atmosphere will correspond to the 1976 U.S. Standard atmosphere at the 23 km (14 miles) visibility condition.

e. IR signature display: Spectral source and attenuated intensities shall be plotted as a function of wavelength between 1 and 15 micrometers for viewing aspects to be specified by the Using Service. Broadband intensity in the bands specified in paragraph c shall be plotted in polar coordinates to provide source and attenuated polar plots in the engine azimuth and elevation planes.

f. Altitudes: Sea level 4 km (13K–ft) and 11 km (36,089 ft).

g. Engine power settings: Maximum, Intermediate, and Maximum Continuous.

h. Transient IR data for engine decelerations from IRP to Idle and acceleration from Idle to IRP at conditions specified by the Using Service.

Background:

Engine manufacturers have developed and validated a considerable array of analysis tools to account for cumulative build up of exhaust system and plume IR signature contributions to the aircraft signature budget. In the aft hemisphere of aspect angles, exhaust system nozzle and plume radiation are the dominant contributors to the aircraft IR signature pattern. The air–to–air IR missile threat is a short range engagement encounter where plume radiation in the forward hemisphere and exhaust system cavity and plume radiation in the aft hemisphere determine lock–on and missile lethality envelopes.

Lowering an aircraft signature to the level that it can not be detected is not always practical. Jam–to–signal ratio then becomes important where a combination of reduced signature and jamming or deception is used to defeat the threat.

REQUIREMENT LESSONS LEARNED

Infrared has been the most effective anti–aircraft missile guidance technique for the past three or more decades. IR missiles have had an extremely high success rate in killing and damaging aircraft.

4.6.1.2 IR radiation The requirements of 3.6.1.2 shall be evaluated by analysis, demonstration, and test. Contrast spectral radiant intensity measurements shall be acquired over the 1.5 to 14 micron region of the IR spectrum. The propulsion system shall be integrated into the airframe structure so that measured signature data accounts for IR radiation emitted and reflected from internal flow path components, exhaust nozzle convergent and divergent flaps, and external expansion ramp surfaces. Plume radiation shall be measured.

EVALUATION RATIONALE (4.6.1.2)

A data base of propulsion system signature measurements is required prior to range evaluation and acceptance testing on the aircraft to provide confidence criteria with respect to the degree of convergence that can be expected between measured and predicted aircraft propulsion system signature levels.

IR characteristics must be measured for airframe integration, engine–airframe design trade offs, and combat tactics development.

EVALUATION GUIDANCE

Background:

Prediction codes can be measured against a signature data base acquired from an instrumented engine test that provides exhaust/gas plume temperature correlation. Peak uninstalled engine IR radiation patterns should be measured. The IR signature should be measured as total (hot parts and reflection and plume), apparent (attenuated), and absolute (non–contrast) radiation for the uninstalled engine. The IR measurement test plan should include IR measurement installation description, measurement equipment, and calibration procedures. Data should be acquired for sea level static test conditions on an outdoor test stand at the ranges, aspect angles, and engine throttle settings specified by the Using Service. The Using Service engine program manager should arrange for any airframe–supplied exhaust system package to be available for the test. This should be specified in the applicable contract.

EVALUATION LESSONS LEARNED

None.

3.6.1.2.1 IR suppression system An infrared suppression system shall be required. The absolute (non–contrast) intensity limits and IR detection limits against specific threats shall be as defined by the Using Service for the uninstalled engine with and without suppression.

a. A description of the system shall be provided, including the method of actuation if ON–OFF capable, operating limitations in the suppression mode, and fail–safe provisions.

b. The detailed effects of IR suppression system operation on thrust, SFC, and other performance parameters shall be included in the engine performance computer programs of 3.2.1.1 and 3.2.1.2.

REQUIREMENT RATIONALE (3.6.1.2.1)

Some weapon systems require the use of an IR suppression system on the engine. A description of the suppression system and its affect on engine performance needs to be specified. Some aircraft, especially rotary wing, operate close to weapons that use IR radiation signature as their targeting technique. Those aircraft need IR suppression.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

For aircraft engines that do not require IR suppression, delete the text of the paragraph and put "Not Applicable" following the paragraph number.

Background:

When applicable, IR suppression should be required in the development specification. If the contractor can show in the evaluation tests that engine IR signatures are low enough not to require a suppression system, this requirement should be deleted in subsequent model specifications. IR suppression minimizes the observability of the engine. The tradeoff must balance increased engine weight and aircraft drag, which negatively impacts air vehicle performance, against increased mission effectiveness in the hostile environment. The uninstalled engine is defined as when the engine is not physically in the aircraft.

REQUIREMENT LESSONS LEARNED

In the past, trainers have been used in combat (e.g., T–37). This required them to be fitted with IR suppressors. Engine signature suppression has resulted in more weight, less power, increased specific fuel consumption, and reduced life. Airframe suppression techniques could minimize system penalty and provide equivalent protection. Lowest penalties occur when the combined air vehicle/engine performance requirements are allocated early in development and the designs optimized as a system. IR suppression techniques include thermal barrier coating on hot section parts, reduction of plume temperature by mixing additional air with engine gas, and installation of devices to shield hot surfaces from view. Infrared guided missiles have been responsible for the majority of aircraft losses over the past three or more decades.

4.6.1.2.1 IR suppression system The requirements of 3.6.1.2.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.6.1.2.1)

The evaluation of a proposed IR suppression system must be determined by analysis and test.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

For aircraft engines that do not require IR suppression, delete the text of the paragraph and put "Not Applicable" following the paragraph number.

Background:

The IR signature for the uninstalled engine with and without suppression should be determined according to 3.6.1.2 and 4.6.1.2. An analysis should be coordinated with the airframe contractor to minimize system penalties and provide sufficient protection. A computer program is not the best method of determining if a suppression system is needed; actual IR measurements should be used.

EVALUATION LESSONS LEARNED

Errors in measuring IR signature have been made because of background interference from the sun. This made it necessary to determine IR signature at night for measuring the effects of a suppression system.

3.6.1.3 UV radiation The following shall be the absolute (non–contrast) UV signatures for the uninstalled engine: (a) .

REQUIREMENT RATIONALE (3.6.1.3)

More information is required to determine the extent UV radiation from non–afterburning engine exhaust plumes can be exploited, and what must be provided to prevent detection. Afterburning propulsion systems emit so much IR energy that UV emission would not be a driving consideration from a detection point of view. An extensive data base of rocket engine UV measurements and predictions does exist and the aerospace community can provide relevant information concerning source emission characteristics, atmospheric transmission, and background effects.

REQUIREMENT GUIDANCE

If a UV Radiation requirement is not needed, delete the text in the specification paragraph, and insert "Not Applicable" after the paragraph number.

The following should be transferred verbatim into the specification paragraph, with blanks to be filled in by the contractor and placed in the model specification in a classified appendix:

(a): The following shall be the absolute (non–contrast) UV signatures for the uninstalled engine:

a. Azimuth angles: 0°; 10°; 20°; 30°; 45°; 60°; 90°; 120°; 135°; 150°; 160°; 165°; 170°; 175°; 180°. An extension of the centerline forward of the engine shall define the 0° azimuth and 0° elevation position. The 180° azimuth corresponds to the centerline aft of the engine. The 0° azimuth, 0° elevation angles, and centerline are defined as being in a plane parallel to a level ground plan. If the radiation pattern is symmetrical about the centerline, a polar plot with notation indicating symmetry may be used.

b. Elevation angles: 0°; 5°; 10°; 15°; 20°; 30°; 40°; 60°; 90°; 135°; 160°; 170°; and 175° (above and below horizontal).

c. UV spectral regions: UV spectral intensities (Watts/Steradian–Micrometer) shall be computed between 0.1 and 0.4 micrometers with a spectral resolution of at least 10 cm^{-1}. In addition, broadband (integrated) intensities (Watts/Steradian) shall be computed for the following filter bands of 100 percent transmittance by the 0.2 to 0.4 micrometer bandpass.

d. Attenuation ranges: Spectral and broadband intensities defined in paragraph c shall be evaluated at source (zero range), 0.1, 2 and 10 Km (0.06, 1.2, and 6.2 miles) ranges along each viewing angle specified in paragraphs a and b. The atmosphere will correspond to the 1976 U.S. Standard atmosphere at the 23 km (14 miles) visibility condition.

e. UV signature display: Spectral source and attenuated intensities shall be plotted as a function of wavelength between 1.0 and 0.4 micrometers for viewing aspects to be specified by the Using Service. Broadband intensity in the bands specified in paragraph c shall be plotted in polar coordinates to provide source and attenuated polar plots in the engine azimuth and elevation planes.

f. Altitudes: Sea level 4 km (13,000 ft) and 11 km (36,089 ft).

g. Engine power settings: Maximum, Intermediate, and Maximum Continuous.

Background:

It is not clear that UV radiation is a non afterburning air breathing propulsion system observable that can be exploited by sensor hardware to provide detection information. There is no tri–service data base of non–afterburning UV measurements that document the source signature characteristics of this emission. For afterburning air breathing propulsion systems the UV emission would not be a driving consideration from a detection point of view since afterburning IR emissions would dominate the long range detection scenario. The requirement to supply a report specifying the propulsion system UV signature should be waived until a government/Industry panel can review the situation and more recommendations concerning the specification issues. Depending on EGT, there may be no need to quantify UV radiation from a non–afterburning gas turbine engine cycle.

REQUIREMENT LESSONS LEARNED

UV radiation is generated under plasma conditions where free radicals exist (i.e., H^+, OH^-) in excited energy states. Non–afterburning jet engine plumes exhibit frozen plume chemistry with no molecular dissociation in the products of combustion. In general, only engines with afterburners provided significant UV radiation.

The engine in the ATF/NATF program has increased exhaust nozzle temperatures above previous turbojet/turbofan designs (e.g., F100, F110, and F404). This has put more emphasis on checking UV signatures in present and future engines designed with higher exhaust nozzle exit temperatures.

4.6.1.3 UV radiation The requirements of 3.6.1.3 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.6.1.3)

UV levels should be measured for airframe integration, engine–airframe design tradeoffs, and combat tactics.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

If UV radiation is not required, delete the text of the paragraph and put “Not Applicable” following the paragraph number.

Background:

The UV test plan should include calibration procedures and standard UV radiation source. UV absolute (non–contrast) attenuated signatures should be measured for an uninstalled engine. The UV radiometer and spectrometer/interferometer shall be calibrated before and after the UV test and the data recorded. UV radiation test plan should include a description of the standard UV radiation source, UV measurement equipment, and calibration procedures.

EVALUATION LESSONS LEARNED

None.

3.6.1.4 Smoke The engine shall not emit detectable smoke contrails or smoke puffs during any transient or steady state operation when using any primary or alternate fuel as specified by the Using Service. The maximum smoke number shall not exceed __(a)__.

REQUIREMENT RATIONALE (3.6.1.4)

Visible or detectable exhaust smoke contrails or smoke puffs are observables and can be used to detect combat aircraft.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The contractor should specify the smoke number limit from the smoke number invisibility criterion of figure 22. (corresponding to the nozzle diameter and viewing angle).

Background:

The smoke parameter “nd”, should be employed where “d” is the diameter of the engine and “n” is the maximum number of engine exhaust streams through which an observer could possibly sight. The invisibility criterion graph is used in AFAPL–TR–76–64.

REQUIREMENT LESSONS LEARNED

For mixed flow fan engines, the smoke number is usually much lower than for other engines. However, this does necessarily mean that plume visibility is less since plume diameter is also a factor. The fuel control and combustor components/system interaction, or lack of interaction, has been the primary source of smoke/smoke puffing. Statistics have shown that for aircraft with visibly high smoking engines, a tenfold increase in combat aircraft losses have occurred.

4.6.1.4 Smoke The requirements of 3.6.1.4 shall be evaluated by analysis, demonstration, and test in accordance with ARP 1179.

EVALUATION RATIONALE (4.6.1.4)

Smoke emission must be measured to provide values for visible signature suppression.

EVALUATION GUIDANCE

Background:

Smoke level should be determined at four engine power settings: Idle, 75 percent Maximum Continuous, Intermediate, and Maximum and minimum afterburning (and corresponding settings for turboshaft/prop engines). Prior to sampling, a 10–minute stabilization run will be made at that power setting. No post–test performance check is required. Visual or photographic means should be used to determine if smoke puffs are present during throttle transients. Fuel used should be the primary fuel or fuel in accordance with table XXX.

EVALUATION LESSONS LEARNED

Post qualification redesign has been needed to correct visible smoke conditions which began apparently due to some subtle change in the manufacturing process of combustion systems. Smoke measurements using the principle of ARP 1179 have occasionally led to the erroneous conclusion that the exhaust gas of an engine is invisible. A good understanding of this method is needed to prevent these erroneous conclusions.

3.6.1.5 Gaseous emissions The engine shall not produce by–product exhaust emissions greater than the levels specified on figures 23a and 23b when using any primary or alternate fuel as specified by the Using Service.

REQUIREMENT RATIONALE (3.6.1.5)

Emissions (carbon monoxide (CO), hydrocarbons (HC), nitrogen oxides (NOX)) represent hazards to personnel and the environment. These emissions can also be observable at some conditions.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

The contractor should specify the emission limits.

Background:

Emission limits should be in compliance with the Federal Clean Air Act Amendments of 1970 as follows:

a. CO and HC levels must be below levels which result in an Idle combustion efficiency of 99.5 percent for engines with an Idle pressure ratio above 3:1.

b. CO and HC levels must be below levels which result in a combustion efficiency of 99 percent for engines with an Idle pressure ratio below 3:1.

c. NOX levels must be less than 50 percent of the uncontrolled level of figure 23a.

d. For engines using water injection, NOX levels must be less than 25 percent of the uncontrolled level of figure 23a.

The contractor should supply the emission limits in the format of figures 23b and tabulation of pollutant emission index vs. thrust/power setting (Maximum, Intermediate, Maximum Continuous, 75 percent Maximum Continuous, 25 percent Maximum Continuous, and Idle).

REQUIREMENT LESSONS LEARNED

In past programs, as combustor exit temperatures have increased, NOX has appeared as visible, yellow gas. With advancing technology, certain gaseous emissions have proven to be a reliable signature for threats.

4.6.1.5 Gaseous emissions The requirements of 3.6.1.5 shall be evaluated by analysis, demonstration, and test in accordance with ARP 1256.

EVALUATION RATIONALE (4.6.1.5)

Gaseous emission levels must be demonstrated to show conformance with the Clean Air Act of 1970 and limit the possibility of future threats.

EVALUATION GUIDANCE

Background:

Engine exhaust gas should be analyzed for non–visible contamination using equipment, instrumentation, and procedures mutually agreeable to the contractor and the Using Service. ARP 1256 has been used but alternative procedures may be satisfactory. Regardless of the procedure, measurements should be made at various power settings including Maximum and Idle.

The fuel used should be any primary or alternate fuel in accordance with table XXX or as specified by the Using Service. As a cost saving measure, this test should be run in conjunction with the altitude test of 4.9.

EVALUATION LESSONS LEARNED

Past engine programs have successfully tested gaseous emissions in engine altitude test cells during performance and operability tests.

3.6.1.6 Fuel streaming/vapor puffing The engine shall not emit visible, unburned fuel (fuel streams or vapor puffs) at any power setting at or above Flight Idle or during any transient or steady–state operation when using any primary or alternate fuel as specified by the Using Service.

REQUIREMENT RATIONALE (3.6.1.6)

Fuel streaming/vapor puffing is a fuel control/augmentor problem which, if visible, is an observable and can be used to detect combat aircraft.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

This requirement applies primarily to augmented aircraft and may be waived, if not applicable, at the discretion of the Using Service. If this requirement is not used, put "Not Applicable" following the paragraph number.

Background:

Vapor puffing can occur during the activation of augmentation but is usually more evident during throttle chops out of A/B. Vapor puffing is more frequent when JP–5 fuel is used. Special attention should be given to the upper left hand corner of the flight envelope where vapor puffing is more likely to occur. Vapor puffing is more frequent with JP–5 and JP–8 fuels.

REQUIREMENT LESSONS LEARNED

The F404 utilizes a vapor puff relay which, after an augmentor chop, keeps the core speed and temperature up in order to dissipate any fuel remaining in the flow path. The same type of relay, used on the F110, was ineffective.

4.6.1.6 Fuel streaming/vapor puffing The requirements of 3.6.1.6 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.6.1.6)

Fuel streaming/vapor puffing should be measured to provide values for visible signature suppression.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

The contractor should specify the method or procedure to be used. If this evaluation paragraph is not to be used, put "Not Applicable" following the paragraph number.

Background:

An analysis using an IR transmissometer for measuring the level of hydrocarbons (unburned fuel) in the flow path should be used during the tests for gaseous emissions. It can be difficult to determine the level of unburned fuel which represents a visible stream or vapor puff. The contractor should specify this level of visually detectible unburned fuel. Visual or photographic inspection, during flight test, is also a feasible means of detecting fuel streaming/vapor puffing.

The fuel used should be any primary or alternate fuel in accordance with table XXX or specified by the Using Service.

EVALUATION LESSONS LEARNED

While not common, vapor puff testing, using an IR transmissometer, has been performed on the F110 at NAWC AD Trenton NJ.

3.6.1.7 Water vapor contrails The engine shall not emit water vapor contrails during any transient or steady state operation when using any primary or alternate fuel as specified by the Using Service.

REQUIREMENT RATIONALE (3.6.1.7)

Contrails are observables and can be used to visually detect low observable designed aircraft at extremely long ranges.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

This requirement applies to low observable aircraft and may be waived if not applicable at the discretion of the Using Service. Since a by–product of combustion is water, contrail suppression is difficult to accomplish and any required hardware may present a weight penalty. This requirement should only be considered for applications requiring stringent adherence to low observables to achieve mission objectives. If this requirement is not used, put "Not Applicable" following the paragraph number.

REQUIREMENT LESSONS LEARNED

Water vapor contrail suppression hardware has proven to be a heavy addition to the engine and the application may not always justify the weight penalty. Contrails have been visible as low as 14,000 ft.

4.6.1.7 Water vapor contrails The requirements of 3.6.1.7 shall be evaluated by analysis and test as follows: (a) .

EVALUATION RATIONALE (4.6.1.7)

Water vapor contrails are observables and can be used to detect low observable designs.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The contractor should specify the test method or procedure to be used. If this evaluation paragraph is not to be used, put "Not Applicable" following the paragraph number.

Background:

Since water vapor contrails are highly dependent on atmospheric conditions, visual or photographic inspection during flight test is perhaps the best means of detecting contrails. The fuel used should be any primary or alternate fuel in accordance with table XXX or as specified by the Using Service.

EVALUATION LESSONS LEARNED

Water vapor contrails have not been successfully tested during engine tests in test cells.

3.6.1.8 Radar cross section (RCS) With all RCS reducing features, materials, and devices installed, the engine static RCS signature levels shall not exceed the weapons system interface control document budget allocation for the propulsion system contribution to the aircraft inlet and exhaust nozzle returns. Budget levels shall not exceed the values defined in table XXXI given in sector averages in median and mean (dBsm units) and linear mean (meters squared units). Sector average shall be computed by adding all the 0.05 degree data (using 512 integrations per data point). For each sector, the linear mean in meters squared shall be computed by taking the anti log of each of the 0.05 degree dBsm points and dividing the sum by the number of data points. Engine dynamic RCS signature return shall be

presented in the array format shown in table XXXI. The array shows the RCS magnitude of the rotating components as a function of frequency and engine rotation and represents the mean dynamic return computed over a full 360–degree rotation of the first stage fan blades with scattering from all stationary elements eliminated. Dynamic returns due to rotating fan module components shall not exceed the weapons system interface control document budget allocation for this contributor to the total aircraft inlet return.

REQUIREMENT RATIONALE (3.6.1.8)

RCS is an observable which can be used to detect and track aircraft. Dynamic RCS data is required in addition to static RCS backscatter since the propulsion system includes rotating blade structures (fore and aft face) that may be visible to the radar.

REQUIREMENT GUIDANCE

Radar signature levels should be partly or completely specified by the Using Service.

The amount of RCS degradation following environmental and foreign object ingestion (sand, bird, and ice) should be specified and quantified. The Using Service should reference the applicable hazard ingestion paragraphs, e.g. sand, bird, and ice, and the amount of allowable degradation for each hazard.

Degradation of the hot parts RCS signature should be specified by the Using Service.

Background:

RCS data should be provided for discrete frequencies from 800 Mega Hz to 18 Giga Hz (L, S, C, X, Ku bands) over the sector and in the format specified in 3.6.1.8. In addition the raw 0.05 degree data should be smoothed using the 50th, $75^{th,}$ and 90th percentile medians processed over a one degree windows with a one degree slide. The dynamic signature data presented in the format shown in table XXXI differs from the static signature since the spectral response of the former is spread across the band–width of the receiving radar and has a response that varies with integration time.

Median values (arithmetic average over 100 degrees) of RCS should be reported. The ten degree intervals over which the median values are obtained should extend, as a minimum, over the angular range of $\pm$60 degrees in both azimuth and elevation, as measured from the engine inlet centerline for the forward hemisphere and at the engine exhaust centerline for the aft hemisphere.

Where variable exhaust nozzle systems are used or IR suppression devices are incorporated in the nozzle system, the manufacturer should determine RCS values for these devices in each mode appropriate to system operation. Special provisions for reducing RCS should be described. Engine signature suppression is heavily dependent on the engine installation, and integration of the inlet and exhaust nozzle into the airframe. Therefore, responsibilities of the airframe builder and engine manufacturer must be clearly defined. The bookkeeping system to define airframe and engine contribution must be established.

Significant degradation of stealth characteristics following foreign object ingestion is undesirable.

Where the Using Service chooses to specify RCS only for certain primary coordinate axis or as a series of required and desired levels the engine contractor should propose the unspecified portions or best effort total RCS of the engine over the frequencies bands of 1 to 18GHz (L,S,C,X, & Ku), or lower if required, in dBsm and displayed in the form of global plots. A global plot (polar plot) provides azimuth display on the angular scale, frequency displayed on the radial scale and RCS displayed as color levels.

Static returns (i.e., non rotating propulsion systems contributors) are the traditional radar backscatter signals emanating from aircraft platform stationary scattering centers including propulsion system inlet and exhaust cavities.

Dynamic returns (i.e., rotating propulsion system inlet and exhaust cavities) impart periodic modulations to the backscatter pattern distinct from the intrinsic RCS in the absence of rotation. These time varying modulations are generally termed dynamic or doppler modulation spectra. The dynamic data set is compiled by subtracting from the total RCS return associated with discrete positions of the fan rotor the coherent mean of the response associated with a full 360 degree sweep of the fan rotor. This coherent mean subtraction effectively implements a zero doppler filter which eliminates contributions from stationary scatterers.

The Contractor should provide in the model specification actual RCS data.

Engine cavity termination RCS treatment may result in smaller weight and performance penalties when evaluated on a system basis (i.e. traded off against other inlet duct treatments and severe offset geometries). System penalty is minimized when a combination of aperture shaping and treatment, duct geometry and treatment, and engine front frame shaping and treatment are exploited in concert to provide the best system solution.

Component testing should be done during engine development to quantify the damage encountered during ingestion to determine inspection and/or maintenance intervals for the affected parts.

REQUIREMENT LESSONS LEARNED

Radar guided threats have demonstrated high kill probability against aircraft with high RCS signature.

4.6.1.8 Radar cross section (RCS) The requirements of 3.6.1.8 shall be evaluated by analysis, demonstration, and test. Static and dynamic measurements shall be evaluated by test on an uninstalled engine including engine inlet device (if required), and any engine components which may contribute to the RCS signature. RCS measurement shall be acquired either on a full scale inlet fixture or a “zero” duct length fixture with traceable correlation to the full scale inlet. RCS measurement shall be made on full scale engine hardware. Conic cut data shall be taken using either swept or stepped frequency techniques. The RCS measurement for engine nose and tail shall be 0, plus and minus 60 degrees in azimuth as measured from the engine inlet or aft centerline. Data shall be measured 0, plus and minus 10, 20, and 30 degrees in elevation. In addition, median values for each frequency band, for 0, plus and minus 10, 30, and 60 degrees about the nose and tail shall be measured. RCS degradation shall be determined after each individual ingestion or durability test and shall not be cumulative.

EVALUATION RATIONALE (4.6.1.8)

RCS characteristics must be measured for airframe integration, engine–airframe design tradeoffs and combat tactics development.

EVALUATION GUIDANCE

Background:

Components that contribute to the RCS signature may include the following: front frame, first stage rotor and vane assembly, and exhaust system. Static and dynamic RCS data should be obtained at nominal frequencies and bandwidths over the 800 Mega Hz to 18 Giga Hz frequency range and over the angular sectors $\pm 60^\circ$ in both azimuth and elevation as measured from the engine centerline at the inlet and exhaust. Static RCS data should be provided for discrete frequencies as a function of fixture azimuth and elevation angles with first stage fan rotor blades in a fixed position. The 50th and 90th percentiles processed over one degree window with one degree slide should be used to evaluate RCS backscatter over the sector of interest. Static RCS signature level, when measured in the “zero” duct length fixture, should not exceed the interface control document budget allocation for propulsion system contribution to the aircraft inlet(s) return. Dynamic RCS data should be obtained for discrete frequency bandwidths and fan blade rotor rotation steps that satisfy Nyquist criteria limits. The dynamic response should be evaluated for total energy content over the entire spectral response (MTI type processing) and as a function of spectral content (Pulse doppler processing) using the data obtained from coherent, wide–band scattering measurements of targets with rotating structures. All official endurance testing should be performed with RCS reducing features and materials installed.

The RCS of the engine inlet device (if required), engine front frame, and 1st stage rotor and vane assembly should be measured in a "zero" duct length fixture to isolate the engine front face hardware contribution to the total inlet return. The "zero" duct length fixture should have a traceable correlation with full scale inlet measurements to establish an RCS budget for the engine hardware. RCS measurement should be conducted on full scale hardware at an outdoor test range or comparable in–door facility. The RCS measurement test plan should include a description of the measurement equipment, measurement facility, and calibration and test procedures.

EVALUATION LESSONS LEARNED

RCS measurement on engines and engines with aircraft inlets installed have shown that math models do not adequately predict actual measurements from hardware.

3.6.1.9 Radar absorbent materials (RAMs) and coatings The ingestion requirements of 3.3.1.3 and 3.3.2.4 combined shall not deteriorate the RCS signature level for any radar absorbent material (RAM) and coating by more than (a) dBsm.

REQUIREMENT RATIONALE (3.6.1.9)

Sand and salt ingestion can change the surface characteristics of some radar absorbent materials and coatings and the engine RCS signature levels.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service must determine the maximum allowable change in RCS signature caused by environmental effects in the field.

Background:

The maximum contribution, from the engine, must be determined separately from the maximum contribution by the aircraft inlet, and the exhaust systems.

REQUIREMENT LESSONS LEARNED

Sand ingestion has shown that erosion of coating can change RCS signatures.

4.6.1.9 Radar absorbent materials (RAMs) and coatings The requirements of 3.6.1.9 shall be evaluated with engines fitted with all the radar absorbent devices, RAMs, and RAM coatings specified for production, and that have satisfactorily completed the sand ingestion and corrosion tests of 4.3.2.4 and 4.3.1.3, respectively.

EVALUATION RATIONALE (4.6.1.9)

The change in RCS signature levels can be determined from the engines used in the salt corrosion and sand ingestion tests. These engines must be fitted with all the RAM's and RAM coatings specified for production. This will show the cumulative effect of sand and salt environments on changing RCS signature levels.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

Analysis and computer model simulation techniques have not been developed to adequately substitute for engine test results on effects of sand ingestion and corrosion on RCS signatures.

3.6.2 Vulnerability The engine shall be designed to control, reduce, tolerate, and resist finite degradation caused by interaction with weapon effects (e.g., enemy or friendly nuclear effects, chemical and biological weapons, ballistic threats, or other area weapons) as specified below.

REQUIREMENT RATIONALE (3.6.2)

Aircraft engine vulnerability can, and should be, controlled. The effects of engine degradation or failure due to threat weapons phenomena vary from loss of aircraft and perhaps pilot for any engine failure of a single engine aircraft, to potential loss of aircraft with two or more engines, due to engine failure effects propagating to other engines or flight or mission critical components of the aircraft.

REQUIREMENT GUIDANCE

Background:

The engine can be a major contributor to, or even dominate, the vulnerability of the combat aircraft system. Aircraft vulnerability requirements should combine requirements for low penalty, smart vulnerability reduction features, and carefully allocated vulnerability reduction requirements derived from government, engine contractor, and airframe contractor mission effectiveness, survivability, and system engineering trade studies. Such studies and the resulting requirements should be based on mission–threat–encounter condition scenarios derived from the applicable validated System Threat Assessment Report (STAR) and the mission scenarios envisioned in the Mission Need Statement (MNS).

The requirements below should ensure that engines being developed for multiple aircraft, or class of missions, take into account all appropriate mission–threat–encounter conditions expected.

REQUIREMENT LESSONS LEARNED

Extensive ballistic data has been collected and is available from individual services at the Survivability & Vulnerability Information Analysis Center (SURVIAC) located at Wright–Patterson Air Force Base, Ohio.

4.6.2 Vulnerability The requirements of 3.6.2 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.6.2)

Measuring an engine's response to threat mechanisms must be evaluated at each milestone design review, milestone completion, and the end of full scale development, for achievement of specific requirements and to measure total achievement of engine vulnerability reduction and control of secondary damage propagation from the damaged engine.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

The Using Service should introduce an overall vulnerability evaluation program by providing conditions and requirements which apply to all subparagraphs under 4.6.2.

EVALUATION LESSONS LEARNED

Vulnerability data has been obtained in two primary ways on the test range and in combat. Combat data is particularly elusive since serious design flaw data is generally lost with aircraft which don't return to base. On one particular aircraft, combat damage analysis, all hits received on returned damaged aircraft were plotted on a single outline of the aircraft for the purpose of determine the likelihood of hitting various parts of the aircraft. As an accidental result

researchers noticed that one particular rectangular area in the belly of the fuselage toward the aft edge of the wings showed no hits on returned aircraft. On further examination they discovered that the blank area on the diagram corresponded with a bay on the aircraft which contained fuel pumps and fuel controls. It took dozens of hits during combat sorties and a considerable number of lost aircraft and crews to discover a correctable design flaw. Full evaluation of the vulnerable response of an engine to threat mechanisms during and after development is an operational requirement.

3.6.2.1 Ballistic weapons The engine, including externally mounted components, shall tolerate ballistic weapon threats as follows: (a) .

REQUIREMENT RATIONALE (3.6.2.1)

The engine must be capable of avoiding and/or withstanding a combat hostile environment defined for the weapon system.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should establish the ballistic requirements.

Background:

The ballistic requirements should include such things as threat weapon, kill type, penetration condition (velocity, direction, fragment weight), engine operating condition, effect on engine operation, etc. Fuel ingestion tolerance, loss of lubrication power and duration requirements, lube level sensing or lube loss rate sensing, and catastrophic failure containment should be specified in this paragraph or included by cross–reference to other paragraphs in this specification.

The engine design should incorporate enhancement features which reduce vulnerability. Such features include component redundancy, component location, component shielding, suppression of active and passive damage and minimizing vulnerable components, and battle damage repair. These survivability enhancement features should be incorporated during the early stages of development. Engine requirements allocated from an air vehicle system specification should be included in the engine model specification.

Consideration of engine components toughness, ductility, and rupture resistance is vital to combat damage resilience. Attention to lubrication, fuel and control systems has a large effect on propulsion system overall survivability. Survivability biased control algorithms, to provide proper engine control after combat damage, incorporated into FADEC engine controls should be considered.

REQUIREMENT LESSONS LEARNED

None.

4.6.2.1 Ballistic weapons The requirements of 3.6.2.1 shall be evaluated by a combination of analyses, transfer of data from other programs, and ballistic testing of replicas, components, systems, major assemblies and running engines under conditions that simulate realistic combat. The evaluation shall be structured to provide design information at times supportive of program design milestones, evaluate achievement of specification requirements, and evaluate the actual engine ballistic hardness achieved at a measure of expected combat performance. The test evaluation shall be tailored to support intended present, or future use of the engine in systems covered under the Live Fire Test Law (Title 10 USC 2366) and as discussed by DODI 5000.2 and DODD 5000.2M.

EVALUATION RATIONALE (4.6.2.1)

Ballistic damage effects are complex and dynamic and must be evaluated through testing, unless applicable experimental results are available.

EVALUATION GUIDANCE

As an option, the paragraph may require tests to be conducted in accordance with the survivability program and plan as called out by MIL–STD–2069 or MIL–STD–1799.

Background:

Tests to provide design information should also be conducted where the experimental base is lacking. Live Fire Test and Evaluation (LFT &E) is a process of analyses, data gathering, and live fire test that begins as early as possible in the acquisition cycle preferable during demonstration and validation to gather design data and reduce risk, and proceeds during Engineering and Manufacturing Development (EMD) and ISR to evaluate achievement of specification compliance and to measure expected combat performance under realistic conditions. During EMD it is desirable to test before the end of IFR to allow enough lead time for redesign and again after the hardware configuration has been established. After hardware configuration (after IFR) has been established engines dedicated for live fire testing should be selected from those having completed development or flight testing to maximize use of development hardware.

EVALUATION LESSONS LEARNED

Prior ballistic testing of aircraft and engines has provided valuable design feedback to development programs resulting in improved combat effective systems.

3.6.2.1.1 Vulnerability – components and externals Components and externals shall confer vulnerability reduction to the engine by means of:

a. The size, location, and packaging of engine components and externals including the interconnecting elements (e.g., piping, wires, and cables) shall minimize the vulnerable area to the maximum extent allowed by other requirements of this specification.

b. Non–flight–critical parts and externals, including the interconnecting elements (e.g., piping, wires, and cables), shall be positioned to protect flight–critical engine parts from ballistic damage to the maximum extent allowed by other requirements of this specification.

c. Components and externals shall exhibit minimum vulnerability by means of redundancy, separation, shielding, fault tolerance, damage recognition, analysis, and automatic control algorithm reconfiguration.

REQUIREMENT RATIONALE (3.6.2.1.1)

This requirement reduces the vulnerability of the engine by the selection and location of components and externals.

REQUIREMENT GUIDANCE

Background:

The location and packaging of engine components and externals, to minimize a vulnerable area, should also consider the engine's maintainability requirements. Accessibility and removal of externals should be considered when positioning these parts.

REQUIREMENT LESSON LEARNED

Some external components, controls, and externals are redundant to the engine and have "back up" redundant features. Redundant components, controls, and externals have prevented penetration of the main engine by absorbing the impact energy of incoming ballistic threats. The F404 engine has experienced this and allowed the engine to continue operation for the next mission, after replacement of the external component or part.

4.6.2.1.1 Vulnerability – components and externals The requirements of 3.6.2.1.1 shall be evaluated by analysis and Live Fire Test and Evaluation (LFT&E).

EVALUATION RATIONALE (4.6.2.1.1)

Live Fire Test and Evaluation (LFT &E) is required to determine the vulnerability reduction achieved by the select placement of engine parts.

EVALUATION GUIDANCE

Background:

This requirement should be conducted in conjunction with 4.6.2.2.

EVALUATION LESSONS LEARNED

Experience has shown that a combination of ballistic test data and vulnerability analysis is the most accurate and economical way to determine engine ballistic tolerance. Engines that have been tested include the V–22.

3.6.2.1.2 Static structure No case or gas pressure loaded non–rotating component, except airfoils, shall collapse or rupture when subject to its maximum operating pressure and a round hole of (a) diameter placed anywhere on the structure.

REQUIREMENT RATIONALE (3.6.2.1.2)

It is not practical to build engine cases and static structures with the capability to repel bullets and warhead fragments; in fact, increasing demands for weight reduction dictate construction which is increasingly vulnerable to penetration. Engines will seldom fail from mere penetration and the resulting overboard leakage. However, it is critical that the engine's response to penetration be graceful, i.e. it should not cease operation or release parts, even internally, which could augment or cascade the damage.

"Static structure" includes internal non–rotating parts, in addition to external cases. Engine failure could result from removal of load–carrying material, stress concentration, or unintended airflow through the hole. The requirement is not intended to apply to bearings, lube and fuel lines, or holes aligned such that the entire load path is severed. Cases should have enough ductility and/or thickness to tolerate without pressure–loading caused rupture, buckling, or crack propagation, holes which nominally produce elastic stress concentration factors of 3.0.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The hole diameter should not be less than that corresponding to a 14.5 mm armor–piercing round, which makes a minimum 0.6 inch diameter hole, if not yawed to the direction of flight. The 23 mm API is a more common threat, and makes a minimum 1 inch diameter hole. Warhead fragments and high–explosive projectiles can make holes generally up to the equivalent of 1 inch diameter. In general, the hole should be assumed to have its axis perpendicular to the surface it penetrates.

Background:

Provided that it within the allowances of the structural integrity requirements, stationary airfoils should be made of ductile materials which will allow the airfoil to be bent 90° through the airfoil chord axis at the root with a mean bend radius of five times the nominal airfoil root maximum thickness (disregarding any root fillet) without separating or fracturing. Naval Air Warfare Center China Lake has provided the input for this requirement.

REQUIREMENT LESSONS LEARNED

Testing at NAWCWD China Lake has indicated that engines generally continue to produce acceptable performance despite multiple hits by ballistic threats, if the original damage is not amplified by brittleness in the structure which allows release of engine structural debris into the flow path. Most engines tested to date use vane and blade materials sufficiently ductile that, although ballistic threats and foreign objects mangle and destroy the aerodynamic function, the damaged airfoils usually do not release and cause cascading failures.

4.6.2.1.2 Static structure The requirements of 3.6.2.1.2 shall be evaluated by analyses, demonstration, and test.

EVALUATION RATIONALE (4.6.2.1.2)

Analysis and demonstration are required to determine the vulnerability of a pressure loaded component when it is pierced.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

Analyses may be presented to satisfy this requirement, except where confidence in analytical tools is lacking, in which case, ballistic firings, or static tests of parts with drilled holes, should be required as negotiated with the Using Service.

Background:

Tests should be in accordance with 4.6.2.1.

For further information regarding this paragraph contact Naval Weapons Center, Survivability Engineering Building Code 3183.

EVALUATION LESSONS LEARNED

There has been experience in testing and in analysis to establish tolerance of engines to ballistic damage.

3.6.2.2 Directed energy weapons (DEW) effects The engine shall meet the specific directed energy weapons vulnerability requirements of (a) .

REQUIREMENT RATIONALE (3.6.2.2)

Directed energy weapons threats have been predicted by the threat community for the next century.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service Survivability/Vulnerability community should provide a requirement for the blank.

REQUIREMENT LESSONS LEARNED

The use of advanced directed energy weapons has not been seen in combat. Penetration/destruction of drone air vehicles in flight by DEW has been demonstrated.

4.6.2.2 Directed energy weapons (DEW) effects The requirements of 3.6.2.2 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.6.2.2)

It appears that potential adversaries are developing advance directed energy weapons for future use.

EVALUATION GUIDANCE

Background:

Directed energy weapons vulnerability evaluation should be determined by the nature of the threats predicted for the life of the engine.

EVALUATION LESSONS LEARNED

There are past events where foreign and U.S. aircraft flight control systems have been disrupted by suspected microwave beams.

3.6.2.3 Nuclear weapons effects The engine shall have the following operational capabilities and balanced hardness in nuclear environment that includes blast/gust, dust, thermal, transient radiation effects, and electromagnetic pulse: (a) .

REQUIREMENT RATIONALE (3.6.2.3)

Future aeronautical vehicles may have to operate in a strategic or tactical nuclear warfare environment. Engines must be optimized to permit mission completion within that environment.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should specify the engine operating capabilities and balanced hardness for a particular nuclear environment.

Background:

Nuclear requirements for an engine should be allocated from requirements derived from the air vehicle mission threat encounter conditions or from generic air vehicle mission threat encounter conditions for the engine's intended class or type of aircraft. All potential missions in which exposure to enemy or friendly Nuclear Weapon Effects is expected must be considered including all expected yield and detonation conditions. For example, an air vehicle which has a very high altitude mission profile or which may have to provide close air support in a tactical nuclear environment would very likely be exposed to gamma missions or disruptions of electronic signals. Aircraft for which base escape is an issue or which must support tactical operations at low altitude would have to deal with nuclear detonation generated dust. Nearly all aircraft expected to operate in a nuclear environment will have to deal with some degree of blast, gust, flash and thermal effects. All aircraft with missions which are mission essential to the Using Service will have to resist electromagnetic pulse. Specifications for engines should match expected air vehicle requirements and reduce or eliminate single point weaknesses. The Using Service program manager must ensure that the Interface Control Document (ICD) includes the air vehicle mission threat encounter conditions and aircraft degree of protection provided, for each user's mission requirements and environments. The most stringent users mission requirements should dictate the final requirement if practical.

Nuclear contamination includes only radioactive particles and induced gamma activity resulting from nuclear detonation. It does not include primary effects such as blast, initial radiation, thermal radiation and electromagnetic pulse. Decontamination of the engine to remove dust particles can be accomplished by water washing.

REQUIREMENT LESSONS LEARNED

Nuclear hardening programs on many past and present aircraft indicate that careful tailoring of the requirement is crucial. Analysis concerning mission threat encounter mission completion and parametric trade studies which calculate the cost/benefits and performance/benefits in relation to mission effectiveness are an absolute must for specification requirement derivation. Dust and ash raised by nuclear weapons will have an extremely deleterious effect on engines equivalent to volcanic ash, if ingested. Engines intended for such mission encounters must be designed to withstand the environment.

4.6.2.3 Nuclear weapons effects The requirement of 3.6.2.3 shall be evaluated by the analyses, demonstrations and tests.

EVALUATION RATIONALE (4.6.2.3)

Testing and analyses are required to accurately determine the blast/gust, dust, thermal, transient radiation, and EMP characteristics of the engine.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.6.2.4 Chemical/biological agent effects The engine shall survive contamination, from chemical and biological agents, and decontamination without losing mission capability as follows: (a) . The engine shall be capable of being decontaminated within 30 minutes using the procedures and decontamination agents defined by (b) . Engine materials shall be compatible with the decontamination agents. The engine shall be capable of being operated and maintained by personnel in nuclear, biological and chemical (NBC) protective ensembles. The engine shall withstand the chemical/biological environments defined by the applicable System Threat Report (STR).

REQUIREMENT RATIONALE (3.6.2.4)

Chemical/biological weapons are widely available and have been used in recent warfare so the engine must be capable of sustaining combat in a chemical/biological environment.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should specify engine mission capabilities in a chemical/biological contamination environment.

(b): The Using Service should specify the decontamination procedures, agents, and protective ensembles.

Decontamination should be in accordance with Army Regulation (AR) 70–71.

Background:

Much of the information regarding chemical warfare requirements is classified. The materials to which the engine may be exposed during decontamination operations are as follows:

a. MIL–D–50030 – used to neutralize chemical warfare agents on surfaces.

b. MIL–D–12468 – used to neutralize chemical warfare agents on surfaces.

c. Ethylene Oxide – used to neutralize bacteriological agents.

d. Water – incorporated as a matrixing–dissolving fluid and as steam.

e. Isopropyl Alcohol – used as a solvent to wash away contaminants.

f. Additional types of decontamination fluids may be used as the threat changes. The possibilities should be researched when the specification is prepared.

All three services should utilize the U.S. Army Nuclear and Chemical Agency guidance in letter DAMO–SW, subject: Department of Army Approved Quantitative Contamination Survivability Criteria, dated 24 Oct 91.

The characteristics of NBC Contamination Survivability (NBCCS) are:

a. Decontaminatability. The engine should be capable of being decontaminated as defined in FM 3–5, NBC decontamination, to reduce the hazard to a level where personnel operating under Mission Orientated Protective Posture (MOPP) level 0 suffer no physical impairment or injury.

b. Hardness. The engine should be capable of full, unimpaired operation after exposure to NBC contamination and the procedures and agents required to decontaminate the system. Surfaces should be constructed of materials that are resistant to NBC agents and decontaminants. The contractor should be responsible for providing data proving that selected materials are resistant to NBC agent and decontaminants. They can acquire the required data through either the Chemical–Biological Information Analysis Center or through material testing as required. The engine design should minimize entrapment of agents and decontaminants. Design guidance is available from MIL–HDBK–784.

c. Compatibility. The engine should be capable of being operated, maintained, and resupplied by persons operating at the Mission Orientated Protective Posture (MOPP) Level 4. Equipment should be designed to minimize performance degradation under NBC conditions.

REQUIREMENT LESSONS LEARNED

NBC contamination can have direct effects upon equipment items by degrading critical properties (e.g., physical, chemical, mechanical, electrical, optical, or thermal) of the material's construction. In addition, materials or procedures used to decontaminate equipment items may also reduce component and system capabilities. Indirect effects may arise if it is not possible to decontaminate to levels which are not hazardous to unprotected personnel. Decontamination of a surface may be hindered because materials absorb the contaminant or the contaminant becomes trapped in cracks or crevices. The heat associated with normal engine operation in the airframe should substantially reduce the degree of residual contamination.

4.6.2.4 Chemical/biological agent effects The requirements of 3.6.2.4 shall be evaluated by analysis, demonstration, and test. Material compatibility shall be shown by coupon tests or existing data, as applicable. Prior to any exposure to agents or decontaminants, the critical functions/properties of the engine shall be demonstrated to measure engine performance and material properties after the decontamination.

EVALUATION RATIONALE (4.6.2.4)

NBC decontamination must be evaluated.

EVALUATION GUIDANCE

Background:

Decontamination procedures may be conducted concurrent with the maintainability demonstration of 4.5.2. Long term effects should be determined by the compatibility tests. A test should be conducted to evaluate engine compliance to the AR 70–71, or equivalent, requirement. All chemical agent resistant coatings should be evaluated by documentation.

EVALUATION LESSONS LEARNED

None.

3.6.2.5 Fuel ingestion effects The engine shall be capable of ingesting fuel via the inlet duct under the conditions specified in the following subparagraphs.

REQUIREMENT RATIONALE (3.6.2.5)

Engine/airframe vulnerability to inlet fuel ingestion effects and adverse engine reaction to fuel ingestion can be minimized by contractor design methods. The airframe developer bears the responsibility of protecting the engine from fuel leakage to the extent possible by eliminating common tank/duct walls, and where this cannot be done, providing the lowest possible fuel leakage rates upon penetration of the common wall by specified threats. This paragraph requires the engine contractor to specify techniques to increase the engine's tolerance to fuel ingestion caused by combat damage, tank failures, and aerial refueling accidents.

REQUIREMENT GUIDANCE

Background:

Given predicted airframe leakage of fuel into the engine inlet, the engine contractor should define the swirl displacement angles for core and fan flow paths which would be experienced by entrained fuel. Prediction of burn through zones are straightforward when the swirl displacements of both the fan–duct–entrained ingestant and core–captured ingestant are known for various throttle settings. The Joint Technical Coordination Group on Aircraft Survivability (JTCG/AS) has funded project VP–2–01, Fuel Ingestion Tolerance Specification (FITS), starting in FY92. The objectives of the project are to "develop specifications and related test techniques by which engine can be qualified to demonstrate adequate tolerance of fuel ingestion (primarily quick dump)." VP–2–01 was funded in part because of the ongoing development of the Tri–Service General Engine Specification and the Navy's F414 engine. A related follow–on project scheduled to start in FY93, VP–3–01, Detection of Fuel Ingestion to Save Engines (DEFENSE), will develop means by which imminent fuel ingestion can be detected. Control activities and other design features necessary for minimization of the engine's vulnerability to both quick–dump and steady–flow ingestion will also be developed. For further information contact NAWC China Lake for fuel ingestion recommendations (1992).

REQUIREMENT LESSONS LEARNED

The Joint Live Fire program has completed numerous realistic tests to assess the vulnerability of U.S. and foreign aircraft to inlet fuel ingestion. Experimental determination of swirl displacements and burn through zones during JLFT have led to recommendations for selective application of thermal protection materials.

4.6.2.5 Fuel ingestion effects The requirements of the subparagraphs of 3.6.2.5 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.6.2.5)

One result of steady–flow fuel ingestion is burn through of the engine cases upon ignition of the ingested fuel. It is important to be able to predict and test the locations of these burn through zones so that thermal protection materials may be applied at only the vulnerable locations in the engine compartment (the materials are heavy and expensive.) Testing is the only viable means for accurately evaluating an engines tolerance to fuel ingestion.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph: The analysis and tests required for evaluation should be specified by the Using Service.

Background:

JTCG/AS project VP–2–01 and VP–3–01 (see 3.6.2.5) will define the recommended test approaches and techniques for use in this paragraph. Actual qualification testing can be done in contractor or Government facilities as appropriate. Tests should be conducted early in the development program since engine vulnerability discovered later in Live Fire Testing will make design fixes harder, if not impossible, to implement. Several methods of both detection and mitigation have been tested, and some successful mitigating techniques have been demonstrated as recently as December 1991 at the Weapons Survivability Laboratory, Naval Air Warfare Center, Weapons Division, China Lake, California. Minor design changes, based on test results could enable engines to survive undamaged in even the worst–case quick–dump fuel ingestion events. Projects sponsored by the JTCG are applicable to all the Services. Steady–flow fuel ingestion can be detected and control actions taken to mitigate damage. VP–2–01 will develop the test techniques required to quantify an engine's ability to withstand inlet fuel ingestion, as no such "standard" tests yet exist. For further information regarding fuel ingestion, contact Gary R. Burgner or G. Philip Dixon, NWC China Lake (1992).

Swirl angles are easily measured by injecting dye at the engine face and observing its track through the engine. Some compressor and turbine design codes output swirl displacement, but these may be inaccurate or not capable of accounting for swirl in the combustor, fan duct, or exhaust ducting, so testing should be done.

EVALUATION LESSONS LEARNED

Recent testing of the F100, F402, F404, and certain foreign engines indicate that airframe/engine vulnerability to fuel ingestion can be significantly reduced with minor design changes (with minimum penalty in weight, cost, or reliability). On the F404 engine, HP compressor blade failures caused by fuel ingestion resulted in titanium fires, and design changes were implemented late during engine development at extremely high cost.

3.6.2.5.1 Fuel ingestion – steady–flow The engine shall be capable of detecting and mitigating, by engine fuel control action, the effects of steady fuel flow via the inlet duct with the fuel and plume characteristics and flow rates as follows: (a) . The swirl displacement angles for core and fan flow paths shall be specified by the contractor.

REQUIREMENT RATIONALE (3.6.2.5.1)

Engines vary in their ability to tolerate the ingestion of steady fuel leakage through a wound in duct walls, their tolerance subject to functions of engine size and bypass ratio. For small engines, and low bypass ratios, wounds resulting from small fire can cause engine case burn throughs with subsequent damage to flight–critical airframe–mounted components and loss of the aircraft. Swirl angles are required to predict the locations of engine case burn throughs and where airframe thermal protection may be required.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The survivability/vulnerability community should be consulted for the most recent correlations between threat types and leakage flow rates.

In addition, this requirement should specify the acceptable mitigation requirements and levels of engine performance and/or damage for various ingestion flows and conditions.

REQUIREMENT LESSONS LEARNED

Analyses and demonstrations have indicated that engine control activity effected by sufficiently intelligent control systems can mitigate the effects of inlet fuel ingestion. Swirl angles are easily measured by injecting dye at the engine face and observing its track through the engine.

4.6.2.5.1 Fuel ingestion – steady–flow The requirements of 3.6.2.5.1 shall be evaluated by test. Fuel shall be ingested at flow rates in steps of (a) with cooling periods of (b) between injections. Each fuel ingestion condition shall be conducted with the engine operating at (c). The engine shall be inspected for damage at periodic intervals. The fuel injection mechanism shall simulate airframe fuel system behavior and characteristics.

EVALUATION RATIONALE (4.6.2.5.1)

This is a potentially destructive test, in that structural overheating and thermal distortions will likely occur. No other reliable method of determining actual tolerance has been developed. It is important that the injection mechanism allow the ingestant flow rate to adjust to engine reactions to demonstrate effectiveness of the mitigation techniques.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a), (b), (c): The fuel flow rate step increase and cooling period, and engine operating requirements during the test, should be established by the Survivability and Vulnerability experts within the Using Service.

EVALUATION LESSONS LEARNED

The Joint Live Fire Program sponsored by DDR&E (T&E) (Live Fire Test) has suggested injection techniques which replicate aircraft fuel system behavior. Swirl angles are easily measured by injecting dye at the engine face and observing its track through the engine.

3.6.2.5.1.1 Fuel ingestion – cooling air contamination Contamination of engine internal cooling air from fuel ingestion shall not degrade engine life nor engine parts life below that specified in 3.4.1.1.1 and 3.4.1.1.2.

REQUIREMENT RATIONALE (3.6.2.5.1.1)

Turbine or bearing coolant air extracted from the compressor, if contaminated by ingested fuel, can accelerate engine failure by releasing heat at locations where cooling is needed. Contaminated cooling air can cause failure of turbine blades by releasing heat where the system was designed to transfer heat away from the turbine section. The desirability of keeping fuel out of cockpit conditioning air and other engine bleed air destinations is obvious.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

This requirement may be waived if the Using Service determines that inlet fuel ingestion is not a threat.

Background:

Bleed air extraction location should be determined very early in the design stage. Changes during qualification testing could be very difficult.

REQUIREMENT LESSONS LEARNED

None.

4.6.2.5.1.1 Fuel ingestion – cooling air contamination The requirements of 3.6.2.5.1.1 shall be evaluated by inspection and analysis.

EVALUATION RATIONALE (4.6.2.5.1.1)

Compliance can be determined by reviewing the engine design.

EVALUATION GUIDANCE

Background:

The cooling air requirements of 3.6.2.5.1.1 should be evaluated by inspection of the engine drawings.

EVALUATION LESSONS LEARNED

Ingested fuel has been generally concentrated at the fan case and core case walls by centrifuging in the fan and compressor. This has been clearly responsible for more rapid thermal erosion and failure at turbine nozzle outboard ends than at inboard ends, as seen in ingestion tolerance testing of the F100 and F404 engines. In some tests of engines which extract coolant air at the core case, as opposed to the rotor hub, the failure of the turbines was suspected to have been accelerated by contamination of the coolant air by ingested fuel. Drawing uncontaminated cooling air from the hub, instead of the fan and compressor blade tip, (where the ingested fuel tends to concentrate) has allowed the cooling air system to perform as designed even with fuel ingestion present. The J79 and F110 engine is an example of a design that successfully extracted required bleed air from the hub, and passed the JLFT.

3.6.2.5.2 Fuel ingestion – transient ("quick dump") The engine shall ingest transient fuel without permanent damage or performance degradation under operating conditions as follows: (a) . Engine instability is permitted during the event, but normal engine function shall be reestablished within (b) seconds of the ingestion event with no pilot input.

REQUIREMENT RATIONALE (3.6.2.5.2)

Impact of small arms fire on tank/duct common walls can carburet inlet duct air in what is known as "quick–dump" fuel ingestion. The event happens so fast that engine control activity probably cannot be initiated quickly enough to provide mitigation. However, when the stoichiometric amount of fuel has been mixed in the inlet duct, the potential for damage is maximized.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

The Survivability and Vulnerability, and Performance community of the Using Service should review this requirement and determine operating and inlet pressure and temperature conditions, and establish the maximum time for recovery to initial operating condition following any engine instability caused by the fuel ingestion event.

(a): IRP/Maximum continuous, standard day.

(b): 5 to 10 seconds.

Background:

The values specified above should be selected based on the operating envelope and control scheme. The greatest potential for engine damage occurs at the highest air density condition in the compression machinery. The most recent correlations between threat types and quick–dump parameters, and the latest thinking on engine–affected mitigation, using the current technology is necessary.

REQUIREMENT LESSONS LEARNED

None.

4.6.2.5.2 Fuel ingestion – transient ("quick dump") The requirement of 3.6.2.5.2 shall be evaluated by analysis and test. The engine shall ingest transient fuel, without permanent damage or performance degradation, of a stoichiometrically carbureted column of inlet duct air at a stagnation temperature of (a) °F and stagnation pressure of (b) psia at (c) thrust/power. The test shall subject the engine to an injection of fuel which effectively carburetes the complete inlet air column for a length corresponding to the amount of air contained between engine face and exhaust nozzle exit.

EVALUATION RATIONALE (4.6.2.5.2)

Testing is necessary because analysis is incapable of modeling the complex phenomena involved in "quick–dump" ingestion.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): A temperature value of 59°F.

(b): A value of 22.4 psia.

(c): The condition of Intermediate thrust (Maximum Continuous power).

Background:

The inlet air temperature and pressure specified correspond to flight at Mach 0.8 sea level, cold day, which is a typical stores–delivery condition where threat impact is likely, and it is a condition at which engine pressures (and damage potential) are near maximum values. A Joint Technical Coordinating Group on Aircraft Survivability (JTCG/AS) project starting in FY92 will further develop a test techniques required to evaluate 3.6.2.5.2

EVALUATION LESSONS LEARNED

Most engines exhibit damage modes and severity which suggest that minor improvements could significantly reduce vulnerability to quick–dump fuel ingestion. The Joint Live Fire Program tested the F100, F404, and F402 engines recently for fuel ingestion tolerance, and the J52, TF30, TF41, and certain foreign engines have been tested under other sponsorship. The F100 test report is available from the office of WL/FIVS, Wright–Patterson AFB. The F402 report is being written at NWC, China Lake. In general, these engines exhibit damage which could be precluded by minor redesign and strengthen certain critical parts such as IGV actuation linkages and afterburner liners. The F100 and F404 recovered very quickly from the ingestion event and continued to operate stably with minor thrust degradation for substantial periods. Modern engines already have control logic to provide good surge recovery and relight if necessary. Advanced control systems being developed will have some quick–dump fuel ingestion, although thrust may be degraded. A Joint Technical Coordinating Group on Aircraft Survivability project starting in FY92 will develop the test techniques to evaluate this requirement.

3.7 Subsystems

4.7 Subsystems

3.7.1 Anti–icing/de–icing system The anti–icing/de–icing system, the system actuating method and system limitations shall be described. Operation of the anti–icing/de–icing system shall not result in a loss of performance greater than that specified in 3.3.1.4. Within (a) seconds following termination of the anti–icing/de–icing system, engine performance shall return to the level prior to system actuation.

The anti–icing/de–icing system shall allow (no/no detrimental) accumulation of ice on any engine part while operating under the icing condition specified in 3.3.1.4. Operation of the anti–icing/de–icing system throughout the operating envelope shall not damage the engine. The performance and operability effects of anti–icing/de–icing system operation in a non–icing environment shall be specified. The anti–icing/de–icing system shall provide a signal for indicating that the anti–icing/de–icing system is operating. The fail–safe mode for the ice protection system shall allow safe return to base in icing conditions and non–icing conditions without damage to the engine. If an inlet particle separator (IPS) is used, ice formation on the IPS shall not be permitted. Limited ice accumulation shall be permitted in the IPS scavenge system if it does not affect IPS operation. The service life of the anti–icing/de–icing system shall be specified if it is other than one cold part life.

REQUIREMENT RATIONALE (3.7.1)

An anti–icing/de–icing system must be required since accumulation of inlet ice affects safety of flight. Inlet ice can produce stalls, loss of thrust, rapid increase in turbine temperatures and compressor blade damage due to ice ingestion.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value up to and including five seconds.

Background:

The Using Service should ensure that the affected paragraphs correctly identify that either an anti–icing or de–icing system is used. An anti–icing system prevents ice from forming, therefore there should be no ice build up. A de–icing system sheds ice that has already formed, therefore there should be no detrimental ice build up or damage when the ice sheds once the system has been activated.

Icing conditions can occur at ambient temperatures well above freezing. This is due to the decrease in static pressure in the inlet area which causes a corresponding temperature drop. When the temperature falls below freezing, ice starts to accumulate on the sides of the inlet, on the inlet struts, IGVs and the first stage compressor blades.

The critical periods for aircraft are during Ground Idle and low power landings when there may be insufficient anti–icing/de–icing bleed air and during takeoffs when the inlet temperature drop is the greatest.

Anti–icing/de–icing systems should be designed to minimize the amount of bleed air or other source of energy (electrical, mechanical, chemical), avoid thermal stress and fatigue of the engine anti–iced structure and not cause engine operational problems.

Anti–icing/de–icing should be provided at the front of the engine for anti–icing/de–icing a customer–supplied inlet dome or accessory section. Details of the connection should be shown on the engine configuration and envelope figure. If compressor bleed air is used for this purpose, it should be considered part of the quantity specified in 3.1.1.15.

For electrical anti–icing/de–icing systems, the engine should be capable of simultaneous operation of the anti–icing/de–icing system and all other engine electrical systems.

A means of actuating the anti–icing/de–icing system automatically is preferred. Manual override should be provided if the system is automatically actuated.

Turbofan/turbojet engines operate at higher air speeds (mach number) than turboshaft/turboprop engines. The operation at higher speeds results in a higher inlet temperatures (T_2). An anti–icing/de–icing system in a fail–on position may produce bleed air temperatures up to 1240°F or higher. The combination of higher inlet temperatures (T_2) and constant bleed air flow may induce fan/strut integrity problems due to thermal damage. An

anti–icing/de–icing system in a fail–off position and the aircraft encountering an icing environment, may result in loss of the aircraft due to engine failure from ice buildup on the leading edge of the fan frame struts and bullet nose. Consequently, the contractor should provide a system that allows safe return to base without damage to engine regardless of the design [fail–on or fail–off].

The water content of the air and the droplet size should be measured within one to two inlet diameters of the engine inlet face and within the engine inlet.

Anti–icing/de–icing systems are complicated by the use of relatively low temperature materials in the frames and inlet structures for which there are few substitutes available to meet other weapon system requirements. Normal candidates are: engine bleed air, electrical heating, pneumatic boot/pulse de–icers, and hot oil/fluid recirculation. Life of these systems should be specified if other than one cold parts life. Failed–open valves could lead to failure of these parts unless other actions are taken, such as power level reductions, etc. Meanwhile, failed–closed valves could lead to unacceptable ice buildup, if the system is in use at the time. The broader wording recommended allows for the design of fault accommodation to be approached on a system level. It is quite possible that the valves (if indeed a hot air system is employed) need to be designed so that they fail in different ways depending on their state at the time of failure. Also, for non–hot air systems, there still has to be consideration of preferred failure modes.

REQUIREMENT LESSONS LEARNED

On a hot air anti–icing/de–icing system, when anti–icing/de–icing was required for Ground Idle operation, design considerations were made to determine which stage of compression was to be used for adequate heat transfer to the anti–iced parts. Engine anti–ice systems were prone to thermal distress in thin sheet metal structures, such as inlet guide vanes. The usual design problem encountered in hot air anti–icing/de–icing systems was to provide adequate anti–icing/de–icing capability when the engine was operating at low approach power conditions without overheating the system under takeoff conditions.

An engine has experienced a significant adverse impact on the anti–ice valve because the actual bleed air temperature was higher than what was predicted by the original design analysis.

Moisture buildup and contamination from cleaning procedures and environment have caused valve malfunctions. Valve orientation and design should minimize moisture buildup and tolerate contamination as much as possible.

Some engines utilize modulating valves to minimize the amount of anti–ice flow at high power. One engine switched to a modulating valve design because inlet struts were cracking due to thermal stresses from the anti–ice flow at high power in flight test engines.

The J–52 has had problems preventing anti–ice build–up under mid and low power conditions. However, in cases where the pilot had to throttle up to 75% power or higher for anti–icing purposes while on icy runways it has been difficult or impossible to stay on the runway. Also, due to ice–build up, the engine has experienced flameouts on the deck because of the constrictions in the engine airflow.

The B–1 has experienced a problem of engines building–up with ice which can break free and destroy the engines.

4.7.1 Anti–icing/de–icing system The requirement of 3.7.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.1)

The anti–icing/de–icing system must be evaluated by engine test in an icing environment.

EVALUATION GUIDANCE

Prior to the engine test, the anti–ice system should be bench tested to characterize the temperature, pressure, and flow of the anti–ice system at different power settings.

Flight conditions that should be considered in establishing icing test requirements are listed in 4.3.1.4

Background:

Engine performance in icing and non–icing conditions should show compliance with power loss and fuel consumption requirements.

During testing, power, rotor speeds, and vibrations should be continuously recorded. High speed photographic coverage of the inlet and face should be provided.

EVALUATION LESSONS LEARNED

In most tests conducted in the past, the most severe test condition usually was running the engine at Idle followed by an acceleration to high power. Ice accumulated at Idle power and was ingested when the engine accelerated to high power conditions.

Engine incorporating IPS have complicated flow paths requiring use of bleed air and oil cooling for anti–icing/de–icing protection. Protecting the entire IPS, including the scavenge system was difficult and adversely affected the engine design.

3.7.2 Control system The control system shall be redundant, fault tolerant, and automatically control the engine to meet all specification requirements. The engine control system, airframe/engine control system integration, all subcomponents, interfaces, and related functional operability and performance shall be described. The control system shall provide __(a)__ percent fault coverage for any single control system malfunction or failure. The control system malfunctions or failures that cannot be fault isolated shall be specified.

For engines with afterburner, augmentor ignition shall be automatically activated when the control system receives the appropriate airframe throttle position signal.

Control system interpretation for all airframe demanded engine thrust/power requirements shall be specified herein.

Regions of control limiting functions as a function of altitude, Mach number, and thrust/power setting for standard, hot, and cold day atmospheres of figure 4, shall be shown in the format of figure 24. The limiting values used to establish the various regions shall be specified on the figure.

Reprogramming capability shall be provided at the LRU level installed in the aircraft or as required by the Using Service.

For turboshaft/prop applications, the engine control system shall provide for load sharing and torque matching for multiple turboshaft engines.

REQUIREMENT RATIONALE (3.7.2)

The control system architecture and all interfaces to the engine control system, engine and integrated control modes, and related functional engine capability require a description that shows compliance with engine operational, environmental, and mission requirements.

There are regions in the envelope, for example low altitude and high Mach number, where engine pressures may exceed the maximum design pressure of the compressor case. Therefore, engine limiting must provide protection in this region of the flight envelope.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The Contractor should specify at least 95 percent fault coverage.

Background:

The engine control system should optimize engine performance and operability regardless of harsh atmospheric conditions. Implementation of closed loop control logic should accurately schedule fuel during engine starting, acceleration, deceleration, steady–state operation, and augmentor sequencing through engine shutoff. If practical, augmentor fuel scheduling should be closed loop. Limits should be imposed on critical engine parameters to avoid exceeding aerodynamic, thermodynamic, and structural engine limits to ensure engine protection, safety, and durability. Engine limit considerations should consist of stall margin protection, pressure, temperature, engine speed, and blow–out protection during rapid decelerations. In addition, the controller should provide flameout detection and corrective action, stall/surge detection and recovery, automatic trim/engine deterioration compensation, fault detection and isolation, and should automatically clear hung starts and prevent hot starts.

Modulation of thrust from cutoff to Maximum power by means of unrestricted thrust/power demand excursions should reduce pilot workload, increase combat maneuverability, and mission effectiveness.

Engine control modes should be implemented to reduce pilot workload by providing linear correlation between power input and power output. Integrated control modes should be considered to improve aircraft operation such as carrier approach, landing, wave–off, aerial refueling, loiter, and combat.

Dual redundant FADEC systems by definition should provide main engine, A/B and nozzle control. The dual units or channels can both be active or one can be active and the other provide passive control. Both units or channels should be fault tolerant and should provide control unit switching in the event of a single failure without engine degradation. The mission reliability of the engine may drive the design of the control system.

Therefore, the requirement does not preclude the use of a triplex system or other designs that provide fault tolerant systems and enhance fault accommodation. There should not be any single point software failure in redundant electronic controls.

The description of the control system should identify the functional capability of each sub–component including the FADEC unit(s), electrical harness, sensors, and the actuation, fuel management, and ignition systems. The system should be described as a function of power input to identify engine variable geometry modulation and fuel scheduling during engine starting, steady–state and transient operation, A/B sequencing, and shutdown. In addition, inputs from the airframe such as gas ingestion, FOD, icing, weight on wheels, distortion index, and mach number should be described to identify functional capability.

The digital data bus, MIL–STD–1553 MIL–STD–1773 or others, should provide the integration of the engine aircraft controls and transmit data from the engine control to the engine monitoring system. The type and amount of data transmitted should be in the format required by the engine monitoring system. The design of the control system, hardware and software, dedicated to conditioning and transmitting data to the engine monitoring system should be partitioned from normal control functions to eliminate fault propagation.

The contractor should complete figure 24. There are times when the engine contractor must limit pressure and/or temperature of the engine in a given region of the flight envelope. This is done for structural or aerodynamic reasons. Therefore, a pressure limiter is used in this region of the flight envelope. At supersonic flight speed, air flow through the aircraft inlet duct needs to be controlled between an upper and lower limit to prevent supersonic flow anomalies such as “inlet buzz”.

Intermediate rated power turbine temperature can be down rated at sea level, increased at altitude up to a given flight Mach number, and then reduced again at the high Mach number.

Overtemperature and overspeed is limited under normal operating conditions. These limiting functions usually lose all authority under abnormal conditions and cannot prevent overspeed and overtemperture.

The engine must not sustain damage due to inadequate limiting of critical engine parameters.

The control system should be designed to accept and condition electrical power from engine driven alternators and uninterrupted power from the aircraft power bus. This provides pre–flight back–up capability and control system interrogation of built–in test (BIT) to determine “go” or “no go” control status without starting the engine.

The electronic control should allow for connector programming to minimize the cost associated with a field software retrofit.

REQUIREMENT LESSONS LEARNED

Increased complexity of engines has placed higher demands on control system functions. Full Authority Digital Electronic Controls (FADECs) have proved to be highly flexible and were quickly modified to solve unexpected design problems during development stages.

Every study and development program sponsored by the government with major engine manufacturers has indicated that hydromechanical fuel control systems cannot provide the engine performance and operability required by advanced engines and aircraft. Engine and aircraft control requirements have exceeded the capability of hydromechanical control systems and hydromechanical controls should not be considered for use as a primary control system for growth and future propulsion systems.

Integration of the flight, air inlet, and engine control system have demonstrated significant improvements in aircraft performance. Exploitation of the technology is essential to comply with the requirements of future aircraft systems.

Existing engines such as the T700 have had operational problems with proper control of the engine when they are coupled into a common gear box. Past specification did not cover load share requirements.

In past engine programs, FADEC systems have controlled the following variables: main fuel flow to combustor, augmentor fuel flow, ignition, variable geometry control, exhaust nozzle control, temperature control, fuel thermal bypass flow, and anti–icing/deicing system.

4.7.2 Control system The requirements of 3.7.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.2)

The control system must be evaluated to assure that it controls the engine as required.

EVALUATION GUIDANCE

Background:

Real time dynamic engine simulations should be used to evaluate FADEC hardware operability and performance requirements to allow early detection and correction of problems prior to actual engine test. This test should simulate sensor and actuator functional and engine operability capabilities and provide evaluation of control laws and logic during steady–state and transient performance. Engine limits and operational logic such as automatic trim, engine deterioration compensation, fault detection, isolation and accommodation, control switching, limiting of speed, temperature and pressure, the prevention of flameouts, i.e. blowouts during rapid deceleration and hot and hung starts should be evaluated.

Control system and engine testing should ensure critical engine parameter limiting throughout the entire envelope at standard day, cold day, and hot day conditions.

The integration of the FADEC system, including the fuel delivery system, should be bench tested and evaluated in a closed loop system interfaced to a real time dynamic engine simulator to demonstrate total engine capability. Thrust/power should be modulated throughout the test to evaluate steady–state and transient conditions between start–up and Maximum augmentation at sea level, altitude, and selected Mach number conditions. The test demonstration and analysis should ensure proper engine operation without engine damage.

Engine control modes and integrated control modes, if applicable, should be tested by a flight simulator to evaluate aircraft and engine performance and operability including loiter, takeoff and landing, aerial refueling and other control modes required by the Using Service. Prior to engine delivery, the control system should undergo engine environmental tests to ensure the structural integrity of the control and operational capability, such as vibration, extreme temperature, and EMI.

Aircraft nacelle/engine bay environmental conditions and engine skin temperatures subject the control units to high temperature at high Mach number conditions. Component testing does not address these conditions. To properly evaluate the control system operability, structural and thermal design, and cooling effectiveness, the control units should be subjected to simulated nacelle environmental atmospheric and aircraft flight conditions. This could be accomplished by a Combined Environmental Reliability Test (CERT). This test should identify control failures caused by design flaws and/or inadequate component specification requirements, and will project the reliability (MTBF) at maturity (1×10^6 engine flight hours). The number of test hours should conform to the number of flight hours the engine will be in service.

The structural integrity analyses required in 3.4 should state which engine structures require regions which limit engine functions. The performance presentation of 3.2.1.1 should show how the control affects engine performance when it is limiting engine functions. The altitude test requirements of 4.9.1.2 should be used to demonstrate the regions of control limiting functions, which should also be demonstrated during the component tests of 4.9.2.

EVALUATION LESSONS LEARNED

Real time dynamic simulations have identified control system problems and fixes prior to engine test. Fixes incorporated during engine test were accomplished through the data bus or an engine connector on–line.

Timing of the CERT is important. On an Air Force program, the CERT was accomplished late and revealed problems that had already been experienced in the field. The CERT should be done before full production.

Engine contractors claim it was very difficult to overspeed or overtemperature an engine with an electronic control. However, during a test conducted at NAWCADTRN, the production configuration FADEC on a Navy engine allowed the engine to exceed its speed limits.

For the Navy: The fuel control limiting functions were designed with a built–in tolerance. This tolerance was determined by the engine contractor. When the tolerances were very close, the pilot did not have a smooth power transient when he was flying on the edge of the region. On the other hand, if the tolerances were too great, then there was an unnecessary loss of performance or overstressing of engine structures. A test should demonstrate the extremes of the limiting functions as past testing has shown.

3.7.2.1 Control system performance The control system fault accommodation shall detect failures that shall result in fail–operational, fail–degraded, and fail–safe engine operation, during and after control system malfunction(s) or failure(s). The engine control system shall provide fully modulated thrust/power response, and prevent exceedance of aerothermodynamic and mechanical limits when subjected to unrestricted thrust/power demand inputs. Thrust/power signal input versus engine thrust/power output shall be linear between Idle and Intermediate/Maximum Continuous. The relationship between control input and power output shall be shown on figure 25. For thrust reversers, the relationship between reverse thrust and thrust demand input signals shall be linear and free of abrupt changes.

For turboshaft/turboprop engines, the system shall allow the selection of any output shaft speed between _(a)_ and _(a)_ percent, inclusive, at all powers equal to and greater than no load. The system shall have the capability to automatically match the output shaft torque of both engines of a twin engine installation within _(b)_ percent of the torque available from a single engine at Maximum Continuous Power.

REQUIREMENT RATIONALE (3.7.2.1)

Engine operational capability must be required after any single, dual, or multiple failure.

Modulation of engine thrust/power from start up through Maximum power and shutdown by means of unrestricted throttle movement must be linear between power input and thrust/power output.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Typical values for this requirement on several engine programs are from 70 to 110 percent.

(b): A maximum value of five (5) percent.

Background:

The requirement of fail–operational should provide full engine operational capability after any first failure or control system malfunction. Switching logic should require transfer to the redundant unit. Fault detection, isolation and accommodation regarding hardover, out of range, and soft (latent) failures should be described. After the second failure (fail–degraded) the level of engine thrust remaining should be plotted showing thrust remaining as a function of input and output sensors and effectors. Critical control unit malfunctions such as switching circuitry and logic and the impact on the main engine, A/B, and nozzle control should be described. A failure modes and effects analysis should be conducted to identify the probability of failures and impact on engine performance. Single, dual, and triple failures should be evaluated to determine the impact on engine performance. The Contractor should provide the above information in the FMECA CDRL.

A non–integrated control system places limitations on the pilot's ability to make unrestricted throttle movements due to the possibility of engine stall. The FADEC system integrated with the airframe control system should allow unrestricted movement of the engine throttles throughout the flight envelope. The FADEC response to a throttle transient should increase or decrease fuel flow at a rate appropriate to the aircraft operating condition. In the event of engine stall or rapid increase in engine turbine temperature the FADEC should take recovery action or delay the transient until it is safe to continue. Unrestricted power input should require integration of flight/propulsion controls providing automatic flight and engine envelope limitations to effectively increase combat maneuverability and reduce pilot workload.

Closed loop control modes should provide: accurate and responsive control of thrust and linear correlation between power demand and power output, repeatable starting, precise fuel scheduling, control of critical engine parameters, and overspeed and blowout/stall protection. Closed loop control eliminates adjustments for FADEC trim for engine tolerance or engine deterioration and should provide improved engine operability by providing corrective actions in the event of flameout, hung, and hot starts.

With closed loop control (feedback control), some output quantity is measured and then compared with a desired value, and the resulting error is used to correct the systems output. Without feedback, it is often difficult to build a controller that will be suitably accurate under all conditions.

A major advantage of closed loop control is that the error in the controlled quantity is substantially less sensitive to system–parameter variations than an open–loop control, and therefore precise knowledge of system characteristics is not required in order to achieve precise control. This is very important when the system is a large high powered device that is naturally subject to substantial variations.

Integrated control modes should be implemented to optimize aircraft performance and operational capability, i.e., integration with the FADEC and aircraft inlet control should provide the capability to eliminate or significantly reduce air flow instability (buzz) by modulating ramp and bleed door positions with engine corrected air flow to allow an increase in the stability corridor. In addition, reducing SFC while maintaining the same level of thrust at cruise conditions should be considered.

Thrust augmentor step limits should be compatible with engine/aircraft operation. Abrupt changes occurring during initiation and termination of augmentor sequencing should be minimal but sufficient for pilot awareness. Step changes in thrust should be minimal when initiating and terminating augmentor operation. Thrust step change should be small enough not to disorient the pilot, but sufficient enough to let the pilot know the augmentor is lit. It also assures a smooth transition which will prevent a pressure spike from traveling up the fan duct and causing a compressor stall.

REQUIREMENT LESSONS LEARNED

In past engines, abrupt thrust changes occurring in the system have resulted in stalls and flameouts. Augmentor induced stall, caused by hard augmentor lights, have caused fan stall in turbofan engines, which have resulted in shutdown.

Large thrust increases at augmentor light–off have resulted in pilot disorientation, death, and loss of aircraft. Conversely, pilots have, at times, not realized that the augmentor was lit, resulting in reduced augmentor life and increased fuel consumption.

During some aerial refueling exercises, aircraft at Intermediate thrust settings experienced abrupt thrust changes during augmentor initiation. This caused unacceptable closing rates between the tanker and the aircraft.

Engine/airframe integrated control laws/modes have been evaluated and demonstrated by most engine and airframe manufacturers. These control modes have improved SFC during cruise conditions.

Both F–18 and the F–14B/D have had performance problems at entry into service because of non–linear control system performance.

4.7.2.1 Control system performance The requirements of 3.7.2.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.2.1)

Control system performance must be evaluated to ensure satisfactory engine operation under all expected flight conditions.

EVALUATION GUIDANCE

Background:

Control System Testing:

Fail–operational, fail–degraded, and fail safe operation should be tested and a failure modes and effects analysis should identify the impact of FADEC, sensor and effector failures, or malfunctions on engine operation. Automatic engine and flight control limiting as an integrated system should be tested to evaluate unrestricted power modulation. Linear correlation between power demand and thrust output should be evaluated. Control laws/logic should be tested and implemented to eliminate pressure spikes and reduce abrupt thrust step changes during augmentor sequencing.

Engine Testing:

Engine testing should consist of sea level static and selected flight conditions throughout the envelope demonstrating performance and operability and should demonstrate fail–operational, fail degraded, and fail–safe engine operation. The engine should conform to the same parts list as the endurance test engine. Control system performance should be demonstrated and evaluated in conjunction with the engine tests required in 4.2. Engine starting, steady–state, and transient operation should demonstrate consistent engine performance at the same simulated flight conditions.

During the high and low temperature qualification test, consistent starting times should be demonstrated at each test condition and the control should prevent hot and hung starts.

Fault detection, control unit switching (transfer), and fault accommodation should be demonstrated to assure smooth engine control transfer with no engine degradation. Control system faults should be simulated covering control system inner and outer loops. Hardover, soft, and out of range sensor failures should be simulated and compared to the results of the failure modes and effects analysis. The fail–degraded and fail–safe test should consist of those failures providing the highest and lowest levels of thrust. Fail–safe operation should demonstrate the capability to shutdown with no engine damage and/or minimum engine performance.

Qualification testing of the engine control system should include all integrated airframe signals (which requires the airframe flight control interaction algorithms). Historically, some of this testing has been left to the airframer to complete.

Successful engine performance and operability is dependent on the thoroughness of FADEC system test and evaluation. Schedule changes, logic changes and hardware fixes should be incorporated and re–evaluated prior to engine test. Minor adjustments could be made on line, i.e. starting schedules, while the engine is running, which significantly reduces down time and cost. For critical engine tests, experience has shown that the control system should be the intended production configuration to provide valid test results.

EVALUATION LESSONS LEARNED

FADEC development programs have demonstrated full operational capability after any first sensor and actuator failure and fault accommodation after the second and third failures which resulted in performance degradation less than 10 percent.

3.7.2.1.1 Backup control (BUC) A BUC shall be considered part of the control system. A BUC in conjunction with a redundant control system shall be required for single engine aircraft. During transfer from the primary mode to the backup mode and during operation, the BUC shall provide, as a minimum, a safe return to base. The description of the BUC, and the capabilities and limitations of the engine when operating with the BUC shall be specified. Transfer from the primary system to the BUC and back to the primary system shall be possible at any power setting without engine instability or change to the control system input. Transfer from primary control to BUC shall be (a) after system malfunction or failure. Thrust/power demand input signal relationships shall be identical to those of the primary control system. The BUC shall use a separate metered fuel flow path in single engine applications. The BUC shall prevent the engine from exceeding any limits.

REQUIREMENT RATIONALE (3.7.2.1.1)

BUC must be required wherever EMI/EMP threat may cause failure of the primary electronic control system. Control capabilities and limitations, when operating in the backup control mode, must be established to ensure compatibility with mission and safety requirements.

The requirement to transfer between FADEC and BUC without changing the power input to match power improves overall safety and ease of operation. The requirement results in a control system where power in the BUC and FADEC modes are similar in each mode at the same power setting.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Either “automatic” or “manual” should be inserted depending upon results of system trade studies for cost, complexity, and weight.

Background:

A backup to the control system should be required for multi–engine aircraft if the loss of an engine affects flight safety, the aircraft will be subjected to a combat environment in which survivability becomes a large factor, or when mission requirements dictate that power on all engines be maintained. Single engine aircraft not using a BUC of dissimilar technology must be justified to the Using Service.

The contractor should specify the description of the BUC, and the capabilities and limitations of the engine when operating on the BUC. The BUC capability, compatibility, and limitations that require description should include but not be limited to compatibility with the FADEC, pilot notification, PLA modulation from Idle to any thrust/power setting, and flight envelope (temperature and pressure) limitations.

Dual, multi, or single engine aircraft should require redundant FADEC systems to provide fail–operational capability and increased mission reliability which cannot be accomplished with single strand control systems. Either FADEC unit can act as a primary control with the standby unit providing active or passive engine control. A dissimilar backup control should be considered for applications where EMI/EMP is generated by either friendly or hostile forces (e.g., submarine patrol A/C that are bombarded with EMI/EMP from surface ships). The BUC design of dissimilar technology provides protection against common mode failures and possible aircraft loss. The entire control system architecture and internal design should provide enhanced survivability by reducing the vulnerability of the system.

Jamming, EMI, EMP, software, and circuit malfunctions may shutdown both FADEC units which may result in aircraft loss.

An evaluation should be conducted to establish backup control functions and design architecture. The control system (BUC and FADEC) requires compatibility to effectively perform switching functions. Upon primary system failure, it is preferred that switching be automatic with an indication to the pilot. The transfer should occur without stall or flameout. System check–out should be conducted prior to flight to assure the system is working at full capability. The BUC may be required to accommodate gun and rocket gas ingestion. To enhance combat survivability, a reduction of control system vulnerability should be provided. This can be accomplished by effectively separating the primary control units in order to minimize the probability of kill of both controllers by a single hit. Composite airframes provide little protection against the electromagnetic environment, friendly or unfriendly, increasing the vulnerability of the control system.

For turbofan/jet applications the BUC should have the following to ensure safe return to base capability:

a. It should be possible to modulate power between Idle and 80 – 100 percent of the Intermediate power available on a standard day from sea level to 10 km (32,800 feet) altitude. The low end to the power range (Idle) should provide low enough forward velocity on the ground so that the brakes can stop the aircraft. The top of the power range should provide sufficient power to complete a take–off maneuver safely. The 10 km (32,800 foot) altitude is high enough to clear any mountain.

b. It should be possible to prevent engine operation beyond established limits. There should be no overspeed or overtemperature beyond the stated transient limits and no burner or compressor instability during or following a power input change or a change of the output shaft load.

REQUIREMENT LESSONS LEARNED

Experience has shown that it is particularly difficult to design and incorporate a backup control for application on an existing engine design due to space limitations, interference problems and the inability to detect a primary failure and accomplish a transfer. The backup control system must be sufficiently sophisticated so that it does not require undue attention of the pilot during starting and operation.

The automatic transfer requirement has been usually matched with an altitude bias in the backup system. This type of system can be found on various J52 and TF30s. The F402–RR–406 system has no altitude bias and can only allow transfer without matching at high power at sea level. The pilot must be very careful not to cause a surge during normal throttle handling.

Hydromechanical fuel controls have provided BUC in the manual mode and have required excessive pilot workload to stay within engine limits.

4.7.2.1.1 Backup control (BUC) The requirements of 3.7.2.1.1 shall be evaluated by engine and component analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.2.1.1)

Backup control operation must be demonstrated by engine and component tests.

EVALUATION GUIDANCE

Background:

The BUC should be analyzed and tested for single point failure modes.

EVALUATION LESSONS LEARNED

Past engine tests have shown some problems in transfer to the BUC from the main control. These problems were fixed prior to first flight.

3.7.2.2 Control system adjustments Control designs requiring mechanical adjustment of the engine control system shall not be permitted except upon approval by the Using Service.

REQUIREMENT RATIONALE (3.7.2.2)

Control designs requiring mechanical adjustments should not be allowed. Control adjustments result in increased maintenance workload, engine ground running and potential for misadjustment.

REQUIREMENT GUIDANCE

In the event external adjustment is permitted by the Using Service and a remote trimmer is required, the following should replace the specification paragraph:

Adjustment features shall accommodate AS 848 and shall be self–locking under all operating conditions. All other adjustments shall be protected to prevent tampering. Once adjusted, no further adjustment shall be necessary to provide within–limits operation throughout the complete environmental conditions and operating envelope of the engine. All adjustment locations shall be shown on the engine installation and interface drawing.

Background:

State–of–the–art fuel controls may not require adjustments. All recent designs are digital electronic. Digital electronic systems, using closed loops, can eliminate adjustments for Idle, Maximum speed, fuel type, temperature, and pressure. In the past, hydromechanical designs were not able to meet the requirement due to the use of open loop control mode design.

FADEC systems should not need adjustments. FADECs can provide automatic trim, compensates for engine deterioration, and may not require fuel specific gravity adjustments.

If allowed by the Using Service, external adjustments to the controls should be limited to adjustments which can be made with the system installed on the engine and with reference to the operating characteristics on the ground. The adjustments should be clearly marked, accessible, and adjustable with the engine running. Once adjusted, no further adjustment should be necessary for within–limits operation. External adjustments may be necessary to achieve peak engine performance and economy. Adjustments are made on the ground with the engine in the aircraft or test stand and while the engine is running.

Incorrect control settings often reduce operational capability and cause extensive engine damage. Self–locking adjustments are required to ensure that vibration, maneuver loads, etc., will not change control settings while the engine is running. The fail–safe characteristic of the self–locking requirement exists to reduce risks associated with other methods of securing adjustments. Protection against tampering is required to prevent unauthorized adjustments.

Adjustments may be permitted because of reasons peculiar to a specific design, only during the development program, or only at overhaul. A variety of engine parameters may be considered for control system operation and adjustment. Some examples are described below:

a. Fuel grade adjustment. No adjustment should be required when operating with the primary fuels specified in the specification.

b. Idle speed adjustment. Idle rotor speed should be adjustable within a range of $\pm$5 percent of the specified idle speed. This adjustment should be independent of and should not affect the maximum speed setting.

c. Maximum speed adjustment. Maximum rotor speed should be adjustable within a range of $\pm$5 percent of the intermediate rated value. If the engine has more than one rotor and each rotor speed is adjustable, the specification should specify the adjustment ranges and how the adjustments are made.

d. Measured temperature adjustment. Maximum measured temperature should be adjustable within a range of $\pm$5 percent of the intermediate rated value.

REQUIREMENT LESSONS LEARNED

A common result of incorrect control settings has been combustor and turbine damage. Engine life can be significantly reduced, and the resulting damage can be extensive and expensive to repair.

The worst case example of an adjustable fuel control required the time between maintenance actions to be less than 30 hours. This control was replaced by a digital system requiring no adjustments.

4.7.2.2 Control system adjustments The requirements of 3.7.2.2 shall be evaluated by engine and component test and demonstration.

EVALUATION RATIONALE (4.7.2.2)

Engine and component tests must be conducted to evaluate engine and control response to external adjustments.

EVALUATION GUIDANCE

Background:

The evaluation of self–locking adjustments should be accomplished during engine testing. Adjustment positions should be marked prior to testing and checked at the end of the test prior to recalibration. The existence of tamper–protected adjustments should be accomplished by component examination.

EVALUATION LESSONS LEARNED

None.

3.7.2.3 Overspeed protection system The overspeed protection system shall prevent all rotors from exceeding their overspeed limits. The rotor speeds at which the overspeed protection occurs shall be specified.

For turboshaft and turboprop applications, the measurement of the power rotor system speeds shall be sensed directly with no intermediate mechanical devices such as gears, flexible shafts, or clutches.

REQUIREMENT RATIONALE (3.7.2.3)

There is a need for an overspeed protection system for safety purposes. The device needs to directly sense the rotation of the power turbine to prevent any erroneous signals.

REQUIREMENT GUIDANCE

Background:

Loss of power turbine load may result in rotor overspeed. Propeller blades and helicopter rotor blades should not be stressed by the power turbine rotor going overspeed. Overstressed blades have caused the loss of aircraft. The safety elements required of the overspeed device requires a system that directly senses rotor system rotational speed.

Small engines used for turboshaft and turboprop applications have engine rotors that can withstand as much as 150% overspeed. Yet the propeller or helicopter blade system that the engine drives can only take up to 120% overspeed. It is important to interface with the total craft to determine the minimum overspeed that the total system can withstand. Some overspeed devices not only use a point actuation, but also use a rate speed increase to limit the rotor speed.

REQUIREMENT LESSONS LEARNED

An engine overspeed incident has occurred in a Navy test cell with a “packaged” bill of materials FADEC on the engine. A fuel metering valve was stuck full open and the engine control system had no overspeed protection device to stop the engine from overspeeding. This occurred with a fully operational FADEC.

4.7.2.3 Overspeed protection system The requirements of 3.7.2.3 shall be evaluated by analysis, demonstration, and test.

For turboshaft/turboprop applications the engine shall be set up in a test cell with a shaft power absorber that can be instantaneously decoupled from the engine. The power turbine output shaft speed shall be adjusted to the highest speed with the engine operating at Maximum Continuous power setting. After reaching a stabilized condition, the shaft power absorber shall be instantaneously disconnected from the engine. This test shall be considered satisfactorily completed if the power turbine overspeed logic prevents destructive overspeed of the power turbine and no destructive overspeed of engine rotors.

EVALUATION RATIONALE (4.7.2.3)

The rotor speed at which the overspeed protection occurs, needs to be demonstrated on an engine test.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.7.3 Fuel system The engine fuel system shall be described herein. Provisions shall be made for installing separate flowmeter(s) for combustor and augmentor fuel flows for test and diagnostic purposes.

REQUIREMENT RATIONALE (3.7.3)

A description of the fuel system must be provided for proper evaluation of the physical and functional characteristics. This requirement also provides for installation provisions for a flowmeter transmitter to obtain fuel flow rates.

REQUIREMENT GUIDANCE

Background:

The flowmeter transmitter will provide fuel flow rate data needed during acceptance testing, engine condition monitoring, post overhaul testing, and for pilot cockpit monitoring.

REQUIREMENT LESSONS LEARNED

None.

4.7.3 Fuel system The requirements of 3.7.3 shall be evaluated by inspection.

EVALUATION RATIONALE (4.7.3)

The physical and functional characteristics of the fuel system must be evaluated to assure compatibility with the engine and application.

EVALUATION GUIDANCE

Background:

All physical features of the fuel system should be examined and compared to the specification and installation drawing. Functional characteristics should be evaluated by demonstration and test.

EVALUATION LESSONS LEARNED

The effect of the fuel system on engine aircraft integration should be evaluated. Insufficient evaluation has led to cost and schedule impacts, continuing flameouts, and operability problems.

3.7.3.1 Fuels The engine shall meet the requirements of the specification while using the fuels as required in the following paragraphs.

REQUIREMENT RATIONALE (3.7.3.1)

This requirement identifies the primary, alternate, restricted, and emergency fuels, as applicable. Fuels must be identified so that the engine can be designed to operate and perform using the selected fuels.

REQUIREMENT GUIDANCE

The following should be used for tailoring the primary, alternate, and restricted fuel specification paragraphs that follow in this section: Table XXX contains four options for choice of fuels. The Using Service should select one option and delete the others from table XXX.

Background:

The options in table XXX are listed in order of preference. The information below explains the rationale for choosing Options:

Option 1 is the most desirable since requiring all of the fuels listed in table XXX affords the following advantages:

a. The engine/aircraft will be capable of unrestricted operation on any common aviation turbine fuel found world–wide greatly simplifying the logistics necessary to support emergency operations.

b. Operations out of commercial airfields are simplified since the aviation jet fuel normally supplied will be usable.

c. Foreign Military Sales (FMS) attractiveness is increased since several foreign Military Organizations continue to operate on JP–4 fuel.

d. The inclusion of Wide–Cut fuels means that the engine's performance will not be compromised if fuel or crude oil supply shortages require military operations to be converted to the more easily refined Wide–Cut fuels.

e. There is no need to have an alternate or restricted fuel entry since the engine will operate without restrictions on any of the aviation turbine fuels commonly used world–wide.

Option 2 is the second choice. It moves Wide–cut fuels (JP–4 and Jet B) from the Primary Fuels category to a "Restricted fuels" category. This change can significantly simplify the airframe and engine design since the high volatility of this type of fuel will not have to be accommodated. Moving wide–cut fuels will negate advantages c, d, and e above; however, USAF and Army are currently in the process of converting all of their operations world–wide to JP–8 fuel. Conversion is scheduled to be complete by 1996. The current thinking among fuel logisticians is that once the conversion takes place it will be very unlikely that the Air Force and Army will reconvert to JP–4 even in an extremely tight oil supply scenario. If the wide–cut fuels are placed in a restricted category, the engine manufacturer must identify the operating restrictions which will be placed on the engine (and aircraft) such as high fuel temperature, when operating on a wide–cut fuel.

Option 3 is the third (and much less desirable) choice. It breaks out JP–4 and fuels without additives FSII and CI/LI into an alternate or restricted fuels category depending on whether the lack of additives effects long term durability or operational performance respectively.

Option 4 is the least desirable choice. It moves JP–4 and all fuels which do not contain fuel system icing inhibitor (FSII) into the restricted fuel category since the lack of FSII can impose altitude or other operational restrictions on the aircraft.

The requirement for JP–4 in either the primary of alternate fuel listings drives the system design to incorporate larger, less efficient pumping and fuel pressurization systems. There is also significant loss in flexibility of combustor fuel nozzle design required to address low power combustion instability attributed to higher volatility JP–4. Weight savings are realized due to reductions in boost pumps, inducers, tank pressurization systems, and heat exchangers.

REQUIREMENT LESSONS LEARNED

During the first few months of Desert Shield, Navy and Marine Corps aircraft were often required to use commercial Jet A–1 without additives since these materials and the required injection facilities were not available in–country. Future conflicts could require even longer periods of commercial jet fuel use if additive and fuel supplies are effected. For this reason it is recommended that commercial jet fuels be included among the "Primary Fuels".

During the mid 70's, the Navy experienced problems with TF–30 fuel controls and pumps that were related to the use of low lubricity fuels (hydrotreated JP–5 being delivered in the Mediterranean). These engine components were modified to make them less sensitive to low lubricity fuels. At the same time corrosion inhibitor/lubricity improver additives were added to JP–5 to prevent similar (potential, unidentified) problems from surfacing in other engines and aircraft components. It is prudent to develop and qualify equipment such that it is not dependent on this additive since commercial fuels do not contain it.

Several studies of the fuel being delivered to aircraft by Navy ships has shown that a significant percentage of the fuel (>5%) does not contain minimum use levels of FSII. In addition, commercial turbine fuels do not contain FSII. Engine/airframe fuel systems should be designed for unrestricted operation on fuel without FSII.

Antioxidant additives are included in Military fuels to insure stability of the fuel during long term storage (>6 months). SDA is injected in JP–4 and JP–8 fuels to improve static charge dissipation during fuel handling operations. No evidence has ever been found that these additives effect engine performance or operation.

Design, development, and qualification testing of the engine should be performed primarily on Jet A without additives since engine manufacturer will have to design the engine for operation on the worst case fuel.

The TF34 engine experienced leakage at seals when it transitioned from JP4 to JP8. Leaks disappeared when engine used JP–4 fuel. Leaks were caused by the high percentage of low aromatics in certain batches of JP–4 fuel.

It is good practice to avoid gasolines (MIL–G–5572 and MIL–G–3056) due to handling and availability problems.

4.7.3.1 Fuels The requirements of 3.7.3.1 shall be evaluated by test as required in the evaluation paragraphs below.

EVALUATION RATIONALE (4.7.3.1)

The specific type(s) of fuel, including the types of additives, if any, must be specified herein to be designated for use in tests throughout the specification.

EVALUATION GUIDANCE

Background:

The scope of the evaluation test program can be reduced by choosing the relevant fuel properties and testing the worst case extremes rather than all fuels. For example, combustor hot section durability to be qualified based on fuel that burns with highest luminosity (e.g., JP–5, JP–8, or specially prepared worst case fuel) not all fuels, and operability to be qualified using fuels that cover the range of selected properties relevant to the tests to be conducted.

EVALUATION LESSONS LEARNED

None.

3.7.3.1.1 Primary fuel The primary fuels shall be as specified in table XXX. The engine shall meet the requirements of the specification when using any of the fuels conforming to and having any of the variations in characteristics permitted by the specifications listed in table XXX. The engine shall meet all requirement of the specification after transition between fuels.

REQUIREMENT RATIONALE (3.7.3.1.1)

Utilization of established fuel specifications insures wide availability and constant supply.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: table XXX should reflect the option chosen for the engine application. Refer to guidance of 3.7.3.1.

REQUIREMENT LESSONS LEARNED

None.

4.7.3.1.1 Primary fuel Unless noted otherwise within the specification, engines shall be evaluated using the worst case fuel(s) below: (a) .

EVALUATION RATIONALE (4.7.3.1.1)

Conducting various engine tests with the appropriate specific fuel insures satisfactory operation with the fuels listed.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should select one or more of the following:

– MIL–T–5624 Grade JP–8/JP–5 ST fuel without FSII and lubricity improver.

– MIL–T–5624 Grade JP–5 without FSII and lubricity improver

– ASTM D 1655 Jet A without FSII and lubricity improver

– MIL–T–83133 Grade JP–8 or Jet A–1 without FSII and lubricity improver

– MIL–T–5624 Grade JP–5 with additives

– MIL–T–83133 Grade JP–8 with additives

Background:

The following provides background for selecting a test fuel:

– A Specific Test (ST) fuel is preferred since JP–8/JP–5 ST fuel is specially formulated to be the worst case fuel in terms of engine durability, emissions, cold temperature start, and altitude relight characteristics. Note: The use of wide–cut primary and restricted fuels can be critically limiting factors in hot fuel tests because of their high volatility, therefore high temperature operation and performance tests in Section 3.2 should be specifically written to use either JET–A or JP–4 fuel.

– JP–5's general volatility characteristics make it slightly worse than Jet A or JP–8 in terms of engine durability, emissions, cold temperature start, and altitude relight characteristics.

– ASTM D 1655 Jet A is commercial jet fuel without additives which is readily available throughout the US.

– Fuels containing FSII and lubricity improver are listed at the bottom since the presence of these additives may mask design problems during testing.

Utilization of established fuel specifications ensures wide availability and constant supply.

Use of JP–5 and JP–8, for example, can be expected to result in higher combustor liner temperature and lower life because of their reduced hydrogen content relative to JP–4.

EVALUATION LESSONS LEARNED

Cold day ground start and altitude relight capabilities have been observed to vary significantly with fuel types. In general, JP–8 and JP–5 fuels have reduced capabilities in these areas compared to JP–4 because of reduced fuel atomization and volatility.

JP–4, JP–5, or JP–8 have potentially differing durability impacts.

3.7.3.1.2 Alternate fuel The alternate fuel(s) the engine shall be capable of using shall be specified in table XXX. Any limitations on the engine when using alternate fuels shall be specified herein.

REQUIREMENT RATIONALE (3.7.3.1.2)

Alternate fuels must be specified so the Using Service knows which fuels are needed when primary fuels are not available.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: Table XXX should reflect the option chosen for the engine application. Refer to guidance of 3.7.3.1.

If no alternate fuel is required in the Option selected, delete the text in the paragraph and insert “Not applicable”.

Background:

Characteristics of NATO and commercial fuels (such as NATO F–35 characterized in NATO STANAG 3824 F&L and Jet A or Jet A–1) should be taken into consideration during design of the engine. Variations in light off capability and lubricity have been problems in the past when fuel changes were required.

The engine should be capable of starting and operating satisfactorily when using alternate fuels. The operating limits, thrust outputs, and thrust transients specified in the specification should not be adversely affected when using alternate fuels. There should be no adverse effect on aircraft range.

Commercial jet fuels differ only slightly from the fuels prescribed in 3.7.3.1 and are suitable for most flight operations. The option for designating alternate fuels permits the Using Service to tailor for specific conditions as well as accommodate the use of shale and coal derived fuels which deviate from specification fuels.

REQUIREMENT LESSONS LEARNED

The conversion from JP–4 to Jet A–1, JP–8 or JP–5 fuel may increase aircraft maintenance and degrade systems safety due to increased fuel leaks. This is due to the high swell properties of JP–4 and the permanent set taken by the seals (especially fluorosilicone) after prolonged periods of operations with JP–4 fuel. The introduction of Jet A–1, JP–8 or JP–5 that have low swell properties will result in seal shrinkage and the leakage of seals. These problems are further aggravated by transition from high ambient ground temperature to low flight temperatures.

4.7.3.1.2 Alternate fuel The requirements of 3.7.3.1.2 shall be evaluated by test.

EVALUATION RATIONALE (4.7.3.1.2)

If required, alternate fuels must be used in engine tests.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

If 3.7.3.1.2 is not required, delete the text in the paragraph and insert “Not applicable”.

Background:

Alternate fuels testing is required only when option 3 of table XXX is selected. Alternate fuels are essentially identical to primary fuels with the exception that they do not contain additives.

Durability tests should be conducted with the fuels of 4.7.3.1.1.

Some fuels shown in 4.7.3.1.1 contain FSII and lubricity improver additives, therefore the use of these fuels in AMT will result in higher durability.

EVALUATION LESSONS LEARNED

None.

3.7.3.1.3 Restricted fuel The restricted fuel(s) the engine shall be capable of using shall be specified in table XXX. Any limitations on the engine when using restricted fuels shall be specified herein.

REQUIREMENT RATIONALE (3.7.3.1.3)

None.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: Table XXX should reflect the option chosen for the engine application. Refer to guidance of 3.7.3.1.

If no restricted fuel is required in the Option selected, delete the text in the paragraph and insert “Not applicable”.

REQUIREMENT LESSONS LEARNED

None.

4.7.3.1.3 Restricted fuel The requirements of 3.7.3.1.3 shall be evaluated by test.

EVALUATION RATIONALE (4.7.3.1.3)

Specific tests must be conducted to define the operating restrictions imposed by the use of “restricted fuels”.

EVALUATION GUIDANCE

This paragraph is only necessary if restricted fuels are included in 3.7.3.1, otherwise this paragraph should be labeled “The use of restricted fuel is not required.”

Background:

Wide–cut fuels can effect hot fuel operations and altitude relight envelopes. The specific test paragraphs where these tests are established should state that the fuel to be used will be JP–4 or Jet B.

EVALUATION LESSONS LEARNED

None.

3.7.3.1.4 Emergency fuel The engine shall be capable of operating for a time period of at least (a) hours from sea level to (b) (km) ((b) ft) altitude, throughout a range from (c) to (c) percent of IRP/Maximum continuous thrust/power, and at no greater than (d) percent of the specific fuel consumption of 3.2.1.1 when using automotive or marine diesel fuels (VV–F–800, MIL–F–16884, and MIL–G–3056). Any other limitations on the engine when using emergency fuels shall be specified herein.

REQUIREMENT RATIONALE (3.7.3.1.4)

This requirement identifies fuels which can be used in the engine but which may cause significant damage to the engine or other systems under extended use.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The number of hours should be based on the number of missions required, i.e., at least one mission (USAF/USN) or at least two missions (USA).

(b): 6.1 km, (20,000 ft) turboshaft(helicopter), 10.7 km, (35,000) ft for all others, minimum.

(c): A range from Idle to 90 percent.

(d): A value of 120 percent maximum.

Background:

The minimum operating time period, altitude and power ranges and maximum specific fuel consumption limit for which the engine should function satisfactorily when using the selected emergency fuels should be established by the Using Service based on mission requirements or a mission operational trade study. If applicable, operating limitations, special inspection or maintenance actions required as a result of using an emergency fuel should be described in the specification.

Automotive diesel fuels should be considered as an emergency fuel due to their wide availability.

REQUIREMENT LESSONS LEARNED

None.

4.7.3.1.4 Emergency fuel The requirements of 3.7.3.1.4 shall be evaluated by test.

EVALUATION RATIONALE (4.7.3.1.4)

This evaluation is to ensure the engine's capability to accomplish typical missions with emergency fuel.

EVALUATION GUIDANCE

Background:

Evaluation should determine all significant engine performance limitations using designated emergency fuels. The engine may require overhaul after completing any of the typically stated sorties.

Function and operation of the engine should be tested for compliance with design requirements, using the endurance test procedures as closely as possible, except as modified below:

a. The test should consist of at least 10 hours of engine operation.

b. Separate tests should be conducted for each fuel specified.

c. Engine calibration using primary fuel and recalibration using emergency fuel should be conducted to identify performance deterioration levels as defined in the pretest data.

d. Engine disassembly and inspection should be limited to the extent necessary to perform inspections of the hot section, including fuel nozzles.

e. Any additional operating limitations, special instructions, or maintenance actions required as a result of using emergency fuels should be specified in the specification.

f. This test will be considered satisfactorily completed, when the engine continues to meet the requirements specified and there are no part failures or impending failures which might compromise safety of flight.

Diesel fuels should conform to VV–F–800, should have a sulfur content of 1.00 percent =0.05 by weight, and Grade DF–2 (Oconus) shall have a kinematic viscosity of at least 6.0 centistoke at 20°C (68°F).

Tertiary butyl disulfide should be added to the diesel fuel, if necessary, to obtain the required sulfur content.

It is good practice to avoid gasolines (MIL–G–5572, MIL–G–3056, and ASTM D 439) due to handling problems.

Automotive type fuels generally have low lubricity characteristics which may adversely affect the life of fuel system components.

EVALUATION LESSONS LEARNED

None.

3.7.3.2 Fuel system performance

4.7.3.2 Fuel system performance

3.7.3.2.1 Fuel contamination The engine shall meet specification requirements when supplied with fuel contaminated in an amount up to the extent specified in table XXXIIa, and when supplied with (a) liters ((a) gallons) of fuel, contaminated per table XXXIIb, with the filter in full bypass. Engines for land based rotor craft and engines with unassisted (suction) fuel systems shall meet specification requirements with fuel contaminated in amounts specified in table XXXIIc, with fuel filter not in bypass mode.

REQUIREMENT RATIONALE (3.7.3.2.1)

Contamination may be introduced into the fuel during maintenance, storage, or transfer systems from the atmosphere, through the aircraft vent systems or due to deterioration of materials used in the aircraft fuel systems. Engine development and fuel systems design requires consideration of contaminated fuel during long term continuous usage or usage for short periods of highly contaminated fuel.

The results of an Army survey support the table XXXIIa requirements. Land based rotor craft operate in dirty air and fuel environment and need to withstand higher fuel contamination. Also, helicopters typically have unassisted fuel systems and less total filtration.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The Using Service should require a volume of fuel equivalent to at least that required to complete one mission.

The Using Service should specify the most appropriate table for the application.

Background:

Engines should be tolerant of contaminated fuel. Engine filtration systems may be necessary to eliminate or reduce contamination levels for servo systems, pumps, nozzle systems and actuators. Tables XXXIIa and XXXIIb should be used for design and test purposes when specific aircraft system interface data has not been defined or is unavailable. The fuel system should be able to handle all of the fuel used by the engine during the mission that uses the most fuel with the filter in full bypass. The intent is to ensure that the aircraft can get home after the filter has gone into bypass. The engine should operate satisfactorily with the filter in bypass.

REQUIREMENT LESSONS LEARNED

One engine was initially designed to accommodate or pass through fuel with contaminate particles up to the size passed through the refueling system. Experience has shown, however, that large particles, such as bits of safety wire which were introduced to critical parts of the engine fuel system, resulted in stuck fuel nozzle, a torched turbine vane, and a turbine failure caused by response to a one–per–rev pulse.

Fuel contamination was a significant problem on land–based helicopters in Desert Storm.

4.7.3.2.1 Fuel contamination The requirements of 3.7.3.2.1 shall be evaluated by test.

EVALUATION RATIONALE (4.7.3.2.1)

The ability of the fuel system to handle contaminated fuel must be evaluated by component and engine test.

EVALUATION GUIDANCE

Background:

Test assemblies containing components normally in contact with fuel should be supplied with contaminated fuel downstream from the fluid tank with at least the amount of contaminant specified in table XXXIIa and XXXIIb. The solid contaminant should not be recirculated. During the test the fuel filters, if furnished for the engine fuel system, should be serviced as recommended by the engine manufacturer, but at intervals representing a volume of fuel equivalent to not less than that required in 3.7.3.2.1.

EVALUATION LESSONS LEARNED

None.

3.7.3.2.2 Fuel system performance with external assistance The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

Normal Operation:

a. Fuel temperature: From a minimum of (a) °C, ((a) °F) when using worst case primary fuel or (b) centistoke, whichever is worse, to a maximum of no less than (c) °C, ((c) °F).

b. Fuel pressure: From a minimum of (d) kPa, ((d) psi) above the true vapor pressure of the fuel to a maximum of (e) Kpa, ((e) psig) with a vapor/liquid volume ratio of zero (0).

c. Fuel pressure surges: The fuel system shall accept inlet pressure surges up to (f) kPa, ((f) psig) occurring at a maximum rate of change of up to (g) kPa, ((g) psi) per second. Engine induced pressure surges at the fuel inlet flange shall not exceed (h) kPa, ((h) psi) above the fuel inlet pressure during augmentor initiation or termination.

REQUIREMENT RATIONALE (3.7.3.2.2)

This requirement provides information to the airframe contractor so that the aircraft system (e.g., boost pumps, filters, lines, heat exchangers, valves, etc.) can be properly sized. The requirements of this paragraph tend to lead to an optimized sizing of the overall aircraft fuel pumping system. Specifying the minimum and maximum fuel pressures is to ensure that the airframe contractor provides a reasonable fuel pressure to obtain a vapor/liquid ratio of zero (0).

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): the Using Service should specify the coldest temperature of figure 4

(b): The Using service should select the centistoke value based on the temperature in (a).

(c): The maximum hot soak temperature of figure 4

(d) – (h): The contractor should provide the design parameter values to define the system capabilities and limitations.

Background:

Past engine programs have used the following values:

(d): 34 kPa (5 psi)

(e): 345 kPa (50 psig)

(f): 689 kPa (100 psig)

(g): 13.79 MPa (2,000 psi)

(h): 689 kPa (100 psi)

The minimum fuel temperature should be found on figure 4 for the sea level cold atmosphere. This represents the coldest temperature the fuel will be at in service when the engine is initially started. The maximum fuel temperature should be that which would permit the airframe contractor to use the fuel as a cooling medium. For high altitude, hot fuel conditions this requirement will be more difficult to achieve. Minimum fuel pressure value is necessary for aircraft designs in general to meet almost all operating conditions. Systems have been designed to operate at a minimum pressure of 14 to 21 kPa (2 to 3 psi) above the TVP.

The maximum fuel pressure must be coordinated between the aircraft and engine manufacturers to assess the impact of high pressure operation on system strength and weight. This pressure represents the no flow pressures from aircraft tank–mounted boost pumps and takes into consideration the fuel feed requirements, aircraft weight and line size trade–offs. Fuel systems have exceeded the 345 kPa (50 psig) required in previous specifications. The value should not be considered a state–of–the–art limitation. Previous operating pressures of 414 kPa (60 psi) and proof pressures of 827 kPa (120 psi) were consistent with the aircraft fuel system structural design requirements. Ideally the airframe contractor attempts to design the aircraft system to the lowest pressure that will meet the engine's V/L = 0 requirement. The CF6 fuel system operates up to a maximum inlet gage pressure of 517 kPa (75 psig) and surge peaks up to 1724 kPa (250 psi). The aircraft induced pressure variations have been compensated for by the pump inlet structure design.

A minimum pressure of 34 kPa (5 psi) above the true vapor pressure of the fuel should be maintained to assure the fuel (mainly JP–4) will not boil or the system may become vapor–locked at elevated temperatures.

This paragraph defines the fuel properties at the fuel inlet interface. From a structural integrity aspect the fuel inlet should be designed to withstand the start–up surges of the aircraft system or the emergency shutdown induced surges.

REQUIREMENT LESSONS LEARNED

The earlier design requirements that the fuel supplied to the engine at no less than 103 kPa (15 psia) was criticized as unrealistic. This design condition in some applications required the design and installation of oversize fuel boost pumps and did not take into account consideration the margin of pressure above the vapor pressure of 5 psi plus the true vapor pressure of the fuel (a design condition that is feasible within present state–of–the–art pump design). Operating pressures at the engine interface of 517 kPa (75 psi) have been designed with surge pressures up to 1724 kPa (250 psi), with resulting higher weight to provide structural integrity.

A study of the F–15 aircraft Life Cycle Costs (LCC) related to this subject showed that by operating the engine with 93°C (200°F) fuel rather than 57°C (135°F) fuel (an old limit), the aircraft ram air coolers would not be required at a savings of $200M. Trade studies of this type should be performed on all new engine/aircraft programs. Fuel at temperatures of 149 – 177°C (300 – 350°F) may begin to suffer stability problems, causing deposits and rapid nozzle performance deterioration. Other disadvantages are rising fuel temperatures at low fuel quantities and operation of the ECS during ground maintenance.

High fuel pressure is associated with pump shaft seal leakage and if the designer is unaware of the high pressures, the pump may be improperly designed.

The YF120 flight test engines for the ATF competition allowed fuel inlet temperatures up to 235°F (based on JP–4 and fuel pressurization). However due to significant aircraft system and engine heat loads and the use of fuels as the heat sink medium, on integrated fuel thermal management approach was adopted. Critical limiting parameters were sensed

and a modulating fuel bypass control was incorporated to allow maximum heat rejection without comprising system limits. Protection was provided for fuel nozzle temperature (coking), fuel inlet temperature (vaporization), engine hydraulic fluid and lube oil temperature, as well as specific aircraft system temperatures.

4.7.3.2.2 Fuel system performance with external assistance The requirements of 3.7.3.2.2 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.3.2.2)

The adequacy of the fuel system must be evaluated by test observations and experience gained during the qualification testing phase, including component tests, low and high temperature engine tests, alternate fuel and emergency fuel tests.

EVALUATION GUIDANCE

Background:

During the endurance and altitude tests the fuel system is required to operate at the pressure and temperature extremes specified for the engine. Observations of the fuel system function and performance throughout these tests should be recorded and included in the test reports for use in verifying the fuel system. Evaluation should be obtained by the fuel pump altitude proof test, the component tests, the low and high temperature engine tests, and the alternate and emergency fuel tests.

EVALUATION LESSONS LEARNED

None.

3.7.3.2.3 Fuel system performance with no external assistance The fuel system shall supply the required amount of fuel at the required pressure and temperatures for operation, including starting and augmentation (if applicable), with the following conditions at the fuel inlet connection:

a. Fuel Temperature: From a minimum of (a) °C ((a) °F) when using worst case primary fuel or (b) centistoke, whichever is worse, to a maximum of no less than (c) °C ((c) °F).

b. Fuel Pressures: From a minimum of (d) above the true vapor pressure of the fuel to a maximum of (e) .

c. Vapor/liquid volume ratio from zero (0) to (f) .

REQUIREMENT RATIONALE (3.7.3.2.3)

The fuel system must supply fuel at the required pressures and temperatures when it is designed as a suction system.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The Using Service should specify the coldest temperature of figure 4.

(b): The Using service should select the centistoke value based on the temperature in (a).

(c): The maximum hot soak temperature of figure 4.

(d)–(f): The contractor should fill in the value.

Background:

For aircraft with a suction fuel system, the fuel system must provide the required amount of fuel up to the highest altitude required in 3.2.2.1, the ambient temperatures of figure 4, and the ground and air start requirements of 3.7.11. The maximum fuel temperature should include the effects of a wing tank being heat soaked by the sun in conjunction with a low fuel condition in the aircraft when the fuel in used as a cooling medium (heat sink).

Suction feed systems are desired on helicopters and other low altitude mission aircraft to provide improved crash worthiness and increased safety against ballistics that may result in fuel leakage and a fire.

REQUIREMENT LESSONS LEARNED

The T406, T700, and T800 were examples of engines designed with a suction fuel system. Most suction systems in Army experience have not worked well and have continuing problems with flameouts and operability due to aircraft/engine fuel system layout, refueling configuration, and fuel aeration (bubbles).

4.7.3.2.3 Fuel system performance with no external assistance The requirements of 3.7.3.2.3 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.3.2.3)

The adequacy of the fuel system to function as a suction feed, either normally or in emergency situations, must be verified by specific component tests.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

Fuel System Suction Test

Prior to completion of FFR testing, a fuel system suction test shall be conducted. This test shall include a simulation of temperature soakback on engine starting, and operation at the extremes of fuel and air temperature and altitude environment of the engine fuel system. Fuel system performance parameters shall be demonstrated with the designated primary fuel that is most critical for that parameter. Performance parameters that are critical due to vapor and air evolution shall be demonstrated with JP–4, if JP–4 is designated as a primary or alternate fuel. The test shall be conducted within the guidelines established in (4.9.1.2, Altitude Test). Demonstration test points shall include: (1) sea level, (a) fuel temperature, and (b) surrounding air temperature; (2) 1.2km (3937 ft), (c) fuel temperature, and (d) surrounding air temperature; and (3) 3.0 km (9842 ft), (e) fuel temperature, and (f) surrounding air temperature. Soakback testing shall be accomplished on the first two test conditions. For soakback testing, the engine shall be operated for 30 minutes before shutdown. A restart shall be attempted after soak periods of 15 and 30 minutes.

Background:

Bench testing should be used to demonstrate the capability of the fuel system to operate without external assistance at the worse case conditions.

A fuel pump test should be run with the vapor/liquid (V/L) ratio at the pump inlet from 0 to the section 3 limit. Additionally, a fuel pump cavitation test should be conducted on the portion of the fuel system from the engine fuel inlet to the primary and augmentation pumps (as applicable), with the V/L simulating aircraft fuel system conditions at the maximum fuel flow rates and the lowest inlet fuel pressure for that condition.

The following should be used to tailor the above paragraphs:

(a): 68°C (154°F)

(b): 70°C (158°F)

(c): 68°C (154°F)

(d): 70°C (158°F)

(e): 55°C (131°F)

(f): 38°C (100°F)

EVALUATION LESSONS LEARNED

The T700 was the first engine specifically designed for a suction fuel system. The problems experienced on the UH–60 and AH–64 indicated that the design must consider actual phases of liquid separated from fuel phases in addition to high V/L ratios. The T800 was the first attempt to include this type of vapor and fuel separation (bubbles). The resulting fuel system design will handle large bubbles and has significant dry lift capability. These capabilities are not in any other current Army engine.

3.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor When the vapor/liquid ratio is greater than (a) at the fuel inlet, it shall not be necessary for the fuel system to supply fuel for engine operation. However, after the conditions of 3.7.3.2.2 or 3.7.3.2.3 have been re–established, it shall be possible to accomplish normal starts and operate satisfactorily thereafter. Vapor relief provisions shall be incorporated to vent excess vapor during pump priming.

REQUIREMENT RATIONALE (3.7.3.2.4)

Following conditions of excessive fuel vapors, the engine must operate satisfactorily after reestablishment of normal operating conditions has occurred.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): For JP–5/JP–8 fuels the V/L ratio should be 1.0 and for JP–4 fuels 0.45. The V/L requirement in this paragraph must be coordinated with the V/L requirements in 3.7.3.2.2 and 3.7.3.2.3.

Background:

When the maximum V/L ratio of 3.7.3.2.2 or 3.7.3.2.3 is exceeded, the fuel system must clear the excess vapor or air from the system when the inlet conditions of 3.7.3.2.2 or 3.7.3.2.3 have been re–established, and the engine must be able to be restarted and operate satisfactorily.

The V/L = 0.45 requirement is somewhat inconsistent with the fuel properties of JP–4. The RVP for JP–4 ranges between 14–21 kPa (2–3 psi). At a fuel temperature of 57.8°C (136°F) and with a RVP of 21 kPa (3 psi), JP–4 may boil at 7.2 km (23,500 feet) and with a RVP of 14 kPa (2 psi) may boil at 9.8 km (32,000 feet) (the specific distillation characteristics of the JP–4 fuel also have an effect on the pressure at which the fuel will boil).

In systems using fuel as an airframe system cooling medium, fuel pressurization may be required continuously to prevent total system failure due to vaporization in the A/C plumbing and heat exchangers, particularly with JP–4 and when actual failure scenarios are considered. In general, such systems will require a “prime reliable” aircraft boost system.

Without venting, a severe vapor lock problem could occur in the fuel system quickly if the aircraft incurred a boost pump failure above 9.1 km (30,000 feet) (for a constant aircraft fuel system pressure drop, V/L ratio increases rapidly as altitude increases). Venting the fuel system as the altitude decreases below 9.1 km (30,000 feet) would be expected to permit the engine to start and operate in accordance with 3.7.3.2.2 and 3.7.3.2.3.

REQUIREMENT LESSONS LEARNED

None.

4.7.3.2.4 Fuel system performance under conditions of excessive fuel vapor The requirements of 3.7.3.2.4 shall be evaluated by test.

EVALUATION RATIONALE (4.7.3.2.4)

The requirements of this paragraph must be evaluated by test.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

FOR TURBOFAN/TURBOJET APPLICATIONS:

Fuel Pump Cavitation. For the fuel pump cavitation test, the portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet and augmentation pump inlet, if applicable, shall be simulated in the test assembly. This shall include lines, fittings, filters, and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the engine fuel system downstream of the pump which might have an effect on the pump. Prior to the start of this test, the system shall have had fluid passed through it at Intermediate power engine fuel flow for two hours contaminated with at least twice the amount of contaminant specified in table XXXIIa and XXXIIb. Unweathered, clean fluid may be used to conduct the test. Fuel pump cavitation performance shall be demonstrated with the designated primary fuel that is most critical for vapor and air evolution. The primary and secondary pumps shall be operated for 47 hours at intermediate rated speed and at the maximum flow and discharge pressure required by the engine at sea level standard altitude absolute condition and ram pressure ratio of 1.15. Pressure on the fuel tank will be maintained at 20 inches Hg absolute during the cavitation test. An additional restriction will be introduced ahead of the engine fuel inlet to provide the required vapor/liquid ratio. The fluid vapor/liquid ratio at the engine inlet shall be maintained at not less than 0.45 and the fluid temperature shall be at least 43°C (110°F). Augmentation pumps shall be operated at the above conditions at least 12 hours, continuously. Except as stated above the test procedure shall be in accordance with ARP 492.

Fuel Pump Altitude Test. The portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet and augmentation pump inlet, if applicable, shall be simulated in the test assembly. This shall include lines, fittings, filters, and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the fuel system downstream of the pump which might have an effect on the pump. All independent and separately replaceable fuel pumps shall be operated for five hours at flow and pressure conditions corresponding to those required by design maximum engine performance at 30,000 feet altitude conditions. Performance parameters that are critical due to vapor and air evolution shall be demonstrated with JP–4, if JP–4 is designated as a primary or alternate fuel. The fuel pump shall be operated with pressure on the fuel tank maintained at 4.37 psia (30,000 ft. altitude) and an engine fuel inlet temperature of 43°C (110°F). The fuel vapor/liquid ratio at the engine fuel inlet shall be maintained at not less than 0.45. Except as stated above, the test procedure shall be in accordance with ARP 492. During this test the fuel pump dry lift capability defined in 3.7.3.2.5 shall also be evaluated.

The fuel pump altitude test shall be considered to be satisfactorily completed when, in the judgment of the Using Service, the pump performance during the course of testing and the performance deterioration, determined from calibration runs, conform to the requirements established by the applicable design specification and do not adversely affect engine performance; the pump dry lift capability as defined in 3.7.3.2.5 has been demonstrated; and the component teardown inspection does not disclose parts failure or impending failures.

FOR TURBOSHAFT/TURBOPROP APPLICATIONS

Fuel Pump Cavitation. The portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet and augmentation pump inlet, if applicable, shall be simulated in the test assembly. This shall include lines, fittings, filters, and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the engine fuel system downstream of the pump which might have an effect on the pump. Prior to the start of this test, the system shall have had fluid passed through it at Maximum Continuous power engine fuel flow for five hours contaminated with at least as much contaminant specified in table XXXIIa and XXXIIb. Unweathered, clean fluid may be used to conduct the test. Each test cycle shall consist of 45 minutes at the maximum speed and flow required by the engine at the altitude corresponding to the fuel tank pressure followed by 15 minutes at the minimum Idle speed and flow required at the same altitude. The test shall consist of two parts described below.

(1) For part one, the fuel shall conform to MIL–T–5624, grade JP–4. A restriction shall be introduced ahead of the engine fuel inlet to provide a fuel pressure at the engine fuel inlet during the maximum speed and flow portion of each cycle of not more than 7.0 kPa (1.0 psi) above the true vapor pressure of the fuel or not more than the fuel pressure which corresponds to a fuel vapor to liquid (V/L) ratio of 1.0, whichever is the higher pressure. The restriction shall be maintained during the Idle speed and flow portion of each cycle. Part one shall consist of:

a. 20 cycles with pressure on the fuel tank maintained at 101 kPa (14.6 psi) and with engine fuel inlet temperature of 71°C (160°F).

b. 20 cycles with pressure on the fuel tank maintained at 94.2 kPa (13.7 psi) and with engine fuel inlet temperature of 71°C (160°F).

c. 20 cycles with pressure on the fuel tank maintained at 64.2 kPa (9.3 psi) and with engine fuel inlet temperature of 71°C (160°F).

d. 20 cycles with pressure on the fuel tank maintained at 94 kPa (13.6 psi) and with engine fuel inlet temperature of 57°C (135°F).

During the last 10 of the cycles below (e–f–g), the vapor shall be introduced continuously as vapor only (bubbles) equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet followed by liquid fuel only equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet.

e. 20 cycles with pressure on the fuel tank maintained at 78.8 kPa (11.4 psi) and with engine fuel inlet temperature of 57°C (135°F).

f. 20 cycles with pressure on the fuel tank maintained at 48.3 kPa (7.0 psi) and with engine fuel inlet temperature of 57°C (135°F).

g. 20 cycles with pressure on the fuel tank maintained at 82.0 kPa (11.9 psi) and with engine fuel inlet temperature of 57°C (135°F).

During the last 10 of the cycles below (h–i–j–k–l), the vapor shall be introduced continuously as vapor only (bubbles) equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet followed by liquid fuel only equivalent to 300 mm (12 in.) of fuel line of the same area as the engine fuel inlet.

h. 20 cycles with pressure on the fuel tank maintained at 67.0 kPa (9.7 psi) and with engine fuel inlet temperature of 43°C (109°F).

i. 20 cycles with pressure on the fuel tank maintained at 37.3 kPa (5.4 psi) and with engine fuel inlet temperature of 43°C (109°F).

j. 20 cycles with pressure on the fuel tank maintained at 76.0 kPa (11.0 psi) and with engine fuel inlet temperature of 29°C (84°F).

k. 20 cycles with pressure on the fuel tank maintained at 61.5 kPa (8.9 psi) and with engine fuel inlet temperature of 29°C (84°F).

l. 20 cycles with pressure on the fuel tank maintained at 37.3 kPa (5.4 psi) and with engine fuel inlet temperature of 29°C (84°F).

(2) For part two, the fuel shall conform to MIL–T–83133, grade JP–8. A restriction shall be introduced ahead of the engine fuel inlet to provide a fuel pressure at the engine fuel inlet during the maximum speed and flow portion of each cycle of not more than 35 percent of the fuel tank pressure or 15.0 kPa (2.2 psi) absolute, whichever is the higher pressure. This restriction shall be maintained during the Idle speed and flow portion of each cycle. The fuel temperature at the engine fuel inlet shall be maintained to provide a fuel viscosity of not less than 12 centistoke. Part two shall consist of:

a. 20 cycles with pressure on the fuel tank maintained at 100 kPa (14.5 psi) absolute.

b. 20 cycles with pressure on the fuel tank maintained at 60 kPa (8.7 psi) absolute.

c. 20 cycles with pressure on the fuel tank maintained at 30 kPa (4.4 psi) absolute.

Except as stated above the test procedure shall be in accordance with ARP 492.

Fuel Pump Altitude Test. The portion of the engine fuel system from the fuel inlet of the engine to the engine fuel pump inlet shall be simulated in the test assembly. This shall include lines, fittings, filters and other items, as applicable, between the engine fuel inlet and fuel pump inlet, as well as any elements of the fuel system downstream of the pump which might have an effect on the pump. All independent and separately replaceable fuel pumps shall be operated for:

a One hour at the flow, pressure, and speed corresponding to those required by design maximum engine performance at sea level, 35°C (95°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure not more than the minimum specified in 3.7.3.2.3, followed by one hour at the flow, pressure, and speed corresponding to those required by design Idle engine performance at sea level, 35°C (95°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure as established by the line restriction determined above.

b One hour at the flow, pressure, and speed corresponding to those required by design maximum engine performance at 7.0 km (22966 ft), –10°C (14°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure not more than the minimum specified in 3.7.3.2.3, followed by one hour at the flow, pressure, and speed corresponding to those required by design Idle engine performance at 7.0 km (22966 ft), –10°C (14°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure as established by the line restriction determined above.

c. One hour at the flow, pressure, and speed corresponding to those required by design maximum engine performance at 9.0 km (29528 ft), –25°C (–13°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure not more than the minimum specified in 3.7.3.2.3, followed by one hour at the flow, pressure, and speed corresponding to those required by design Idle engine performance at 9.0 km (29528 ft), –25°C (–13°F) conditions with a fuel temperature of at least 57°C (135°F) and a fuel inlet pressure as established by the line restriction determined above.

The fuel shall be in accordance with MIL–T–5624, grade JP–4. Except as stated above, the test procedure shall be in accordance with ARP492. During this test the fuel pump dry lift capability defined in 3.7.3.2.5 shall also be evaluated. The fuel pump altitude test shall be considered to be satisfactorily completed when, in the judgment of the Using Service, the pump performance during the course of testing and the performance deterioration, determined from calibration runs, conform to the requirements established by the applicable design specification and do not adversely affect engine performance; the pump dry lift capability as defined in 3.7.3.2.5 has been demonstrated; and the component teardown inspection does not disclose parts failure or impending failures.

EVALUATION LESSONS LEARNED

None.

3.7.3.2.5 Fuel pump priming The fuel pump shall be self priming and be capable of a dry lift of (a) meter, ((a) feet) at an absolute pump inlet pressure of (b) kPa, ((b) psia), using the fuels of 3.7.3.1 with the Reid Vapor Pressure (RVP) not to exceed (c) kPa, ((c) psi), and a fuel temperature not to exceed (d) °C ((d) °F).

REQUIREMENT RATIONALE (3.7.3.2.5)

The fuel pump must be capable of priming itself and starting the engine with air in the fuel lines.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The vertical distance from the fuel tank to the inlet.

(b) – (d): The contractor should provide the numbers.

Background:

The requirements of 3.7.3.2.5 need not apply to operation with emergency fuel.

For fixed wing aircraft, a self–priming or suction capability is needed to provide limited starting and operating capability with no external assistance (boost pressure) from the airframe fuel system. A gravity feed aircraft fuel system with sufficient pressure head would negate this requirement.

The fuel pump should be capable of priming itself when subjected to a dry lift of the vertical distance from the fuel tank to the inlet or of at least 0.3 meter (1 foot) at an absolute pump inlet pressure of 30.4 kPa (9 in Hg) with the Reid Vapor Pressure not to exceed 21 kPa (3 psi), and at a fuel temperature not to exceed 43°C (110°F).

The Army allows initial priming of the fuel system by an externally supplied and driven pump for cases where the engine is high above the fuel tank. For large vertical distances an external means may be necessary to supply fuel to the engine.

The fuel pump may have to be vented if air or vapor created after the failure of the aircraft boost pump cannot be pumped through the fuel system.

The requirements of this paragraph need not apply to operation with emergency fuel.

Engine fuel lines between the inlet and the fuel pump should perform satisfactorily and should not collapse at standard sea level conditions when the minimum fuel pressure is applied.

REQUIREMENT LESSONS LEARNED

None.

4.7.3.2.5 Fuel pump priming The requirements of 3.7.3.2.5 shall be evaluated by test.

EVALUATION RATIONALE (4.7.3.2.5)

The capability of the fuel pump to prime itself must be evaluated by specific engine and component tests.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph:

Fuel pump priming may BE evaluated during the fuel pump altitude and cavitation tests.

EVALUATION LESSONS LEARNED

None.

3.7.3.2.6 Fuel lubricity The engine shall operate satisfactorily when using fuel with a lubricity value equivalent to a 0.75 – 0.85 mm (0.030 – 0.033 inch) Wear Scar Diameter (WSD) as measured by the Ball–On–Cylinder Lubricity Evaluator (BOCLE).

REQUIREMENT RATIONALE (3.7.3.2.6)

The engine must operate satisfactorily with a low lubricity fuel.

REQUIREMENT GUIDANCE

Background:

The requirement paragraph applies to commercial fuels with no lubricity additives. Military specification fuels have a lubricity value lower than 0.42 mm.

Current and future fuels are being derived from lower quality crude oil sources to improve fuel availability. These crude oils require higher levels of refining which are detrimental to fuel lubricity. Poor lubricity fuels have a BOCLE WSD diameters as high as 0.85 mm (0.033 inch). This value was determined to be a realistic value through extensive research at United Technology Hamilton Standards Test Facility. Pump failures during accelerated mission tests and bench type endurance runs have revealed a variety of failure modes.

REQUIREMENT LESSONS LEARNED

This requirement resulted from fuel component problems on several engines (J52 Fuel Control, TF30 Fuel Pump, etc.) caused by low lubricity fuel acquired from several refineries. Adding a corrosion inhibitor to the fuel increases the lubricity of the fuel. Hydro–mechanical controls had numerous problems with low lubricity fuels.

4.7.3.2.6 Fuel lubricity The requirement of 3.7.3.2.6 shall be evaluated by analysis, demonstration and test.

EVALUATION RATIONALE (4.7.3.2.6)

The operation with a low lubricity fuel is evaluated by a component test using a simulated endurance cycle.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

SAE AIR 1794 provides guidance for a BOLCE test method. SAE ARP 1797 provides guidance for test facility setup. In the near future an ASTM test method will be developed for fuel lubricity testing. Fuel systems assemblies or components normally requiring fuel for self–lubrication shall undergo a simulated mission operational test for at least

100 hours. If necessary, the cycle shall be modified to give a minimum of 50 throttle excursions per hour. The fuel shall be in accordance with MIL–T–5624, grade JP–5 degraded to provide a lubricity of 0.75 – 0.85 mm (0.030 – 0.033 inch) Wear Scar Diameter (WSD) as measured by the Ball–On–Cylinder Lubricity Evaluator (BOCLE).

EVALUATION LESSONS LEARNED

None.

3.7.3.2.7 Fuel system performance with water saturated fuel The engine shall meet the requirements of this specification when supplied with fuel saturated with water in accordance with MIL–F–17874 at worst case fuel pressure and temperature conditions.

REQUIREMENT RATIONALE (3.7.3.2.7)

Prevention of ice build–up in the fuel filters and screens is necessary for acceptable system safety.

REQUIREMENT GUIDANCE

Background:

When no fuel anti–icing additive is used, a fuel anti–icing heat exchanger type system should be incorporated. If an air–fuel heat exchanger is used, the contractor should specify provisions required for the cooling air. The fuel anti–icing heat exchangers should be installed upstream of the fuel filter and fuel control. It should be a pilot manually selected type, incorporating a warning feature for cockpit indication during icing conditions. A means should be provided to indicate when the fuel anti–icing system is in operation. Operational requirements should determine to a large extent whether a heat exchanger system is used. A fuel system icing inhibitor should be incorporated into locally available fuel. The inhibitor chemical composition, use, and handling should conform to MIL–I–27686 and MIL–I–85470. When the fuel system design does not incorporate an anti–icing heat exchanger and an alternate commercial fuel (such as Jet Al) is being used, engine operation should be limited to areas of the flight envelope that preclude freezing of moisture in the fuel system. Methods of fuel system anti–icing should perform so as not to affect the fuel system components or engine operation and should provide complete fuel icing protection.

If a fuel heater is supplied, it should be described in the specification.

REQUIREMENT LESSONS LEARNED

Army helicopters have not experienced fuel icing problems in the field, consequently fuel de–icing is not a requirement on Army engines.

Fuel anti–icing additives have the beneficial side effect of reducing micro–biological contamination in the fuel tank(s).

4.7.3.2.7 Fuel system performance with water saturated fuel The requirements of 3.7.3.2.7 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.3.2.7)

The characteristics of the fuel anti–icing system must be evaluated by test.

EVALUATION GUIDANCE

The following should be transferred verbatim in the specification paragraph:

Engines that are exposed to water saturated fuel shall be tested as follows:

The engine shall be supplied with water saturated fuel to demonstrate that icing does not adversely affect engine operation. The fuel inlet temperatures shall be set at –3.2 to –1.2°C (26 – 30°F), –10.4 to –8.4°C (12 – 17°F) and the minimum temperature specified. Fuel shall conform to MIL–T–5624, grade JP–5 and should not contain an anti–icing inhibitor. At each inlet temperature, fuel which has been initially saturated with water at 27°C (80°F) and having 0.75 cc of free water per gallon in accordance with MIL–F–17874, shall be supplied to the engine. The engine shall be operated for three (3) hours (longest ferry mission) at each fuel inlet temperature using the AMT cycle. This test shall be considered satisfactorily completed when there has been no deterioration in the steady–state performance or loss of the capability to perform a transient during or after the three–hour operation.

Background:

Periodic activation of the fuel anti–icing system during the endurance test is recommended to demonstrate the system durability in the engine environment.

EVALUATION LESSONS LEARNED

None.

3.7.3.2.8 Fuel filter Fuel filtration on the engine shall be provided, as needed, for the engine to meet the requirements of the specification. The fuel filter shall consist of cleanable/non–cleanable filter(s), and shall be made of a material that does not cause fuel contamination due to media migration. The filtration ratings of the filter elements, contaminated per table XXXIIa, shall be as follows:

Capacity		Filtration Microns or Mesh	Capacity Liter (gal) to Impending Bypass	Liter (gal) Impending Bypass to Bypass
Type	Location			

The filter assembly shall incorporate a pressure relief bypass and shall prevent the discharge of filter contaminant through the bypass. The filter assembly shall emit a signal for (a) indication of impending and actual filter bypass. The filter assembly shall have (b) indications of both impending and actual filter bypass. The indications shall remain visible until manually reset. A drain valve shall be provided at the low point of the filter assembly to drain the filter element cavity and to purge the fuel system of air following filter element replacement. The filter shall be removable without spills or draining of the fuel lines. Fuel pressure surges during operation within the limits specified in 3.7.3.2.2 shall not cause false activation of the impending or actual bypass indicators.

REQUIREMENT RATIONALE (3.7.3.2.8)

Fuel filtration is necessary to ensure successful engine operation in a contaminated fuel environment.

REQUIREMENT GUIDANCE

The following should be transferred verbatim into the specification paragraph:

(a): Remote (This requirement should be consistent with the EMS and R&M requirements.)

(b): Visual (This requirement should be consistent with the EMS and R&M requirements.)

The fuel filter impending bypass indicator shall be manually reset, externally, without disassembly of the filter.

The following should be used for tailoring the table:

Filter capacity from impending bypass to bypass mode should be sufficient to allow completion of at least two combat missions.

Background:

Fuel filter requirements should be tailored to include the following system design and envelope requirements: fuel flow rate, capacity, mesh size, media, fuel viscosity, and scheduled servicing intervals (removals). Desirable features to be considered in filter selection are: visual impending and actual bypass indication (popout button), remote impending and actual bypass indication (to cockpit and/or monitoring system), internal or external reset for indicator, and bowl removal shutoff valve.

Redundant indications for filter impending bypass may be unnecessary, depending on the system design.

The type of element material must be compatible with the fuel to prevent adverse reactions. The micron rating must be specified to give an indication of the size of the particles passing through the filter. The capacity of the filter element must be specified since it will determine the frequency of element removal and cleaning.

The fuel filtration may be supplemental or accomplished by airframe filters. Fuel capacity should be based on anticipated maintenance procedures and practices. The manufacturer must size the filter using the dirty fuel specified in conjunction with the amount of fuel used during the mission that consumes the most fuel. The definition of dirty fuel is provided for standardization. The pressure relief bypass is necessary to prevent system damage in the event of excessively high pressure due to a plugged filter element. The pressure relief feature must not allow debris on the element to be washed through the filter housing to prevent contamination of the whole fuel system. The filter assembly will provide visual indication of impending and actual bypass when the differential pressure across the element exceeds specified values. Visual indication of impending and actual bypass alerts the ground crew that the filter element must be changed. The remote indication provides bypass information to the monitoring system or cockpit.

The manual reset requirement is necessary to prevent maintenance personnel from resetting the indicator without removing, cleaning or replacing the filter element, which would defeat the purpose of the warning indicator.

One disadvantage to manually resettable visual indications is that it may also rule out improved monitoring algorithms which may be based on fuel flow and temperature as well as pressure.

REQUIREMENT LESSONS LEARNED

A fuel strainer located between the A/B fuel–oil cooler and the A/B fuel control was originally positioned horizontally. When serviced, contaminants which had collected near the strainer lip would contaminate the A/B fuel control and require its removal and return to depot for repair. By repositioning the fuel strainer vertically, servicing could be accomplished with minimal chance of contaminating the A/B fuel control.

4.7.3.2.8 Fuel filter The requirements of 3.7.3.2.8 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.3.2.8)

The fuel filter requirements must be evaluated to ensure the filter will remove the required level of contaminants.

EVALUATION GUIDANCE

Background:

Fuel filter durability and maintenance features should be demonstrated during the engine endurance test.

EVALUATION LESSONS LEARNED

None.

3.7.3.2.9 Fuel flow limit The maximum and minimum fuel flow shall be specified herein and in table VIII.

REQUIREMENT RATIONALE (3.7.3.2.9)

All limits must be specified so the Using Service knows the full range of the engine's capabilities and for use by the airframe contractor. Fuel flow limits affect fuel system design and performance limits for Maximum power and Idle.

REQUIREMENT GUIDANCE

Background:

The control system tolerances should be taken into account when the limits are specified. Minimum fuel flow often defines Idle power at high altitude.

REQUIREMENT LESSONS LEARNED

None.

4.7.3.2.9 Fuel flow limit The requirements of 3.7.3.2.9 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.3.2.9)

All engine limits must be evaluated by test.

EVALUATION GUIDANCE

Background:

Test results should be adjusted for worst case tolerance in the engine and the control system.

EVALUATION LESSONS LEARNED

None.

3.7.4 Electrical system

4.7.4 Electrical system

3.7.4.1 Electrical power The primary electrical power for the engine shall be redundant and supplied by the engine and shall be described herein. Electrical equipment shall accept externally supplied power complying with MIL–STD–704 (a) .

REQUIREMENT RATIONALE (3.7.4.1)

Engine supplied electrical power, independent from airframe electrical power, is required to ensure continued engine operation in the event of airframe electrical power generation failure or interruption. This makes the engine more reliable. It also makes the engine more survivable in the event of external damage.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The Using Service should specify the version of MIL–STD–704 to match the version used on the aircraft.

Background:

The engine should accept uninterrupted power from the airframe if power is provided as defined in MIL–STD–704.

The primary electrical power for the engine control system, engine ignition system, and any other engine electrical accessory should provide for safe operation at all engine speeds throughout the operating envelope of the engine.

An external electrical power failure should not result in loss of electrical power to the engine or the engine control system.

Unmonitored electrical power characteristics have resulted in unpredicted operation of the devices being powered. Undervoltage or overvoltage conditions can adversely affect electrical circuitry function, performance and reliability.

REQUIREMENT LESSONS LEARNED

Numerous false fault flags occurred on an engine monitoring system due to an interrupted power input.

4.7.4.1 Electrical power The requirements of 3.7.4.1 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.4.1)

Electrical power provision must be evaluated by component and engine tests.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.7.4.1.1 Generator/alternator The engine generator/alternator shall provide the required electrical power from the specified minimum ignition exciter speed to the maximum engine rotor speed. The generator/alternator performance shall not degrade during and after continuous operation at a speed equivalent to (a) percent of the maximum allowable steady–state engine speed under full electrical load. The generator/alternator performance shall not degrade following the overspeed of 3.4.1.6.5, with no electrical load. The generator/alternator housing shall completely contain all damage if a mechanical failure should occur when operating at or below maximum transient rotor speed.

REQUIREMENT RATIONALE (3.7.4.1.1)

Electrical power is required over the entire operating speed range of the engine. Containment of all generator/alternator damage within the housing is a safety requirement to prevent damage to the adjacent accessories or possible catastrophic engine failure due to severing of fuel/oil lines or major control functions.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor should fill in the blank.

Background:

Because of possible control system degradation the generator/ alternator must have sufficient design margin to provide electrical power up to the maximum allowable transient rotor speed. Continuous operation of the generator/alternator at the maximum overspeed condition without load ensures the generator/alternator integrity is equivalent to the main rotor requirements.

REQUIREMENT LESSONS LEARNED

This requirement was waived for the F402 engine. The negotiated General Electric F404 and T700 engine specifications revised the 115 percent speed maximum speed requirement upward to 135 and 125 percent, respectively, and eliminated the generator/alternator containment requirement. The T700 engine specification included a statement that the minimum design burst speed was equivalent to 190 percent of the engine maximum allowable transient overspeed. The contractor also tested the units to 200 percent speed without failure.

4.7.4.1.1 Generator/alternator The requirements of 3.7.4.1.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.4.1.1)

The generator/alternator requirements must be evaluated.

EVALUATION GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph:

In addition to any other tests required by this specification, the following shall be conducted.

a. Overspeed. The generator/alternator shall be tested in accordance with the overspeed requirement of 4.4.7.5, with no electrical load. At the completion of the test there shall be no evidence of mechanical or electrical damage or impending failure.

b. Load test. The generator/alternator should be operated at a speed corresponding to the main ignition exciter speed under the full ignition load for (a) hour(s), and at a speed corresponding to the percent of the maximum allowable rotor speed under full rated electrical load for (b) hour(s). During this test, the generator/alternator shall be subjected to the maximum component limiting temperature. At the completion of the test there shall be no evidence of mechanical or electrical damage or impending failure.

c. Containment. The generator/alternator shall be tested at the maximum transient rotor speed in a manner to cause a mechanical failure of the rotor system. All damage shall be contained within the generator/alternator housing.

Background:

(a): The value should be determined by the Using Service.

(b): In the past, a one hour test has been used for the load test under full electrical load.

EVALUATION LESSONS LEARNED

None.

3.7.4.2 Alternate/external electrical power The engine shall automatically accept airframe provided electrical power in accordance with MIL–STD–704 (a) for both backup use and for ground power during engine–off checkouts. Power transients allowed by MIL–STD–704 (a) shall not result in engine performance and operability degradation. The alternate/external power supply shall be described.

REQUIREMENT RATIONALE (3.7.4.2)

An alternate electrical power source is required to ensure continued engine operation to provide a safety of flight redundancy.

The engine electrical requirements must be specified so that the aircraft power source will supply the engine with the proper alternate electrical power.

The MIL–STD–704 version must be tailorable to ensure engine/airframe power compatibility.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The Using Service should specify the version of MIL–STD–704 to match the version used on the aircraft.

Background:

Power interrupts exceeding the requirements of MIL–STD–704 should not permanently or temporarily affect performance or operability including any transients.

The engine manufacturer should specify the form and quantity of electrical power required. The engine should accomplish satisfactory air starts and should operate safely at all rotor speeds when it is utilizing alternate power.

REQUIREMENT LESSONS LEARNED

There have been cases in which an engine equipped with a supervisory electronic control powered by the engine mounted alternator has encountered chronic and almost unsolvable series of alternator connector failures. This caused the engine to be susceptible to augmentor anomalies and stall. If an alternate electrical power source had been available, these anomalies could have been averted.

The T700–GE–401 specification states that some control functions are lost and no ignition capability is available when the engine is operating on an external power source.

4.7.4.2 Alternate/external electrical power The requirements of 3.7.4.2 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.4.2)

Operation with alternate electrical power must be evaluated by test.

EVALUATION GUIDANCE

The following is recommended to be transferred verbatim into the specification paragraph:

The following test shall be conducted as follows:

a. Operate the engine at Intermediate/Maximum Continuous power with full customer bleed air and power extraction until thrust/power is stabilized, then switch off the primary electrical power to the control system. The engine shall automatically switch to the alternate electrical power and stabilize at Intermediate/Maximum Continuous power. Decelerate to part power and switch back to the primary electrical power. The engine shall not exceed any of its limits nor exhibit performance/operability degradation during the test.

b. Interrupt the primary electrical power to the control system at the most critical point in an acceleration and a deceleration. During the time the primary electrical power is off and during the switching, the engine shall not exceed any of its limits nor exhibit performance/operability degradation.

c. With the engine operating at Idle and also Maximum augmentation, switch off the normally supplied external electrical power and check for limitations or loss of function.

d. With the engine off, the alternate electrical power on, and the primary electrical power off, start and accelerate to Intermediate/Maximum Continuous power. Operate at this power until the engine is stabilized, decelerate to Idle and stabilize then shutdown. The engine shall not exceed any of its limits nor exhibit performance/operability degradation during the test.

Background:

Transfer to and from the alternate electrical power should be conducted at critical operating conditions. Evaluation of the operating characteristics of the engine when external electrical power is removed can be done on a bench or systems rig test before the engine qualification test is conducted. If these tests are used, the variable geometry and other similar control functions need to be fully evaluated to ensure no engine performance/operability problems are encountered.

EVALUATION LESSONS LEARNED

None.

3.7.4.3 Electrical connectors and cables Electrical connectors shall comply with MIL–C–38999, Series III. Electrical connectors and cables shall have sufficient spare pins in the connectors and wires in the cables to permit future growth and shall be repairable. The quantity of pins and wires is subject to Using Service approval. Electrical connections located in close proximity to each other shall be made physically non–interchangeable. At the sea level cold atmospheric temperature of figure 4 it shall be possible to flex electrical cable and conductors during routine maintenance without damage to these items and to connect or disconnect electrical connectors using normal maintenance procedures. Electrical connectors shall perform satisfactorily in the engine bay environment.

For turboshaft applications, electrical connectors shall be MIL–C–38999, series (a) type (or equivalent metric) and shall also comply with the series III, paragraph on External bending moment of MIL–C–38999. The engine shall have not more than three electrical connectors for connection to the airframe.

REQUIREMENT RATIONALE (3.7.4.3)

The functional, durability and maintainability requirements for the electrical connectors and cables must be established to ensure suitable operation and life. Physically interchangeable connectors in close proximity can and do get mixed which may render the system inoperative.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Series IV

Background:

The cable environment should include heat, vibration, spilled fluids, sand, dust, and other engine bay environmental conditions. The minimum ambient temperature should be the sea level cold atmospheric temperature. Self–locking connectors should be considered for all engine applications. Spare pins and wires should be considered for future growth.

REQUIREMENT LESSONS LEARNED

Connectors which were physically the same but connected to different systems were, on occasion, inadvertently mixed causing an engine malfunction. Some electrical insulation became embrittled at low temperatures causing electrical shorts leading to engine system failures.

A certain attack helicopter experienced at least two in–flight dual engine shutdowns caused by water intrusion into the wiring harness.

4.7.4.3 Electrical connectors and cables The requirements of 3.7.4.3 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.4.34.7.4.3)

The electrical connector and cable requirements must be evaluated by component test, demonstration, and inspection.

EVALUATION GUIDANCE

Background:

Demonstration of removal and installation of connectors and cables should be done on an engine at the specified ambient temperature of 3.7.4.3. Non–interchangeability of connectors should be evaluated by engine inspection.

EVALUATION LESSONS LEARNED

None.

3.7.4.4 Electronic components Electronic components shall meet the requirements of MIL–E–5400, Class 1 equipment and the reliability program requirements of MIL–STD–785. Electrical parts shall be derated in accordance with (a) .

All integrated circuits shall be qualified in accordance with MIL–M–38510. Semiconductor devices shall be qualified in accordance with MIL–S–19500.

REQUIREMENT RATIONALE (3.7.4.4)

The purpose of this requirement is to ensure that electronic components meet aerospace electronic equipment standards.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Electrical and electronic parts should be derated in accordance with MIL–HDBK–217, NAVSEA–0967, or NAVSEA TE–000–AB–GTP–010(NAVAIR).

Background:

MIL–A–87244 could be used in lieu of this paragraph. Junction temperatures of semi–conductor devices under the worst case conditions should not exceed 110°C (230°F), otherwise they become unstable and deteriorate rapidly. This will cause faulty electric signals to be transferred between electronic components.

REQUIREMENT LESSONS LEARNED

None.

4.7.4.4 Electronic components The requirements of 3.7.4.4 shall be evaluated by analysis, demonstration, and test. The following test are required: (a) .

EVALUATION RATIONALE (4.7.4.4)

The requirements of 3.7.4.4 must be evaluated.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Appropriate tests in MIL–STD–781, tasks in MIL–STD–785, and detailed test specifications should be specified by the Using Service.

Background:

MIL–STD–781 and MIL–STD–785 outline general guidelines for reliability testing procedures and programs, including Environmental Stress Screening, Sneak Analysis, growth testing, and FMECA. In addition to citing specifications and standards, the Using Service should specify test details such as length of test, temperature, pressure, etc. The test program should be conducted on a statistically sufficient number of units. The minimum operating time on the lowest time unit should not be less than half of the operating time of the highest time unit.

EVALUATION LESSONS LEARNED

None.

3.7.4.5 Electrical grounding The grounding system for all electrical and electronic components of the engine shall be compatible and consistent with the aircraft grounding scheme. All components shall be grounded to the engine system. Power returns shall be isolated internally from the chassis of each component. A separate pin shall be provided on each power connector as a current return for each source of power supplied by the alternate/external electrical power system.

a. Safety grounding. A pin on each power connector which is connected internally to the component chassis shall be provided to permit internal grounding of the component through wiring. No circuit shall utilize ground wire as a power return.

b. Shield grounding. Cable shields shall be terminated at each end to the back shell of connectors that are conductive and provide bonding of shields 360 degrees peripherally around the connector.

REQUIREMENT RATIONALE (3.7.4.5)

All components must be grounded to the engine system to prevent ground loops and common ground returns for signal and power circuits, provide effective shielding for signal circuits, reduce EMI effects and protect personnel from electrical hazards.

REQUIREMENT GUIDANCE

Background:

For safety grounding, a wire of minimum length (generally wire length should not be greater than two feet), and not connected internally to the component chassis, should be used on a pin on each power connector.

Safety grounding of electrical systems/electronic components should be in accordance with the E^3 requirements of the Using Service.

REQUIREMENT LESSONS LEARNED

None.

4.7.4.5 Electrical grounding The requirements of 3.7.4.5 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.7.4.5)

Grounding of electrical components must be evaluated for safe and proper operation.

EVALUATION GUIDANCE

Background:

Evaluation methods should be submitted by the contractor in the E^3 Control Plan. The Using Service should ensure that a evaluation plan for the proposed grounding schemes is included as a CDRL item. Proper performance of the engine electronics during engine functional testing and electromagnetic interference testing helps to demonstrate the adequacy of the grounding scheme.

EVALUATION LESSONS LEARNED

None.

3.7.4.6 Electrical bonding The internal and external bonding requirements shall be in accordance with (a) . For composite materials, bonding shall be accomplished at impedance levels consistent with the materials in use. Bonding provisions shall be compatible with other requirements imposed on the system for corrosion control.

REQUIREMENT RATIONALE (3.7.4.6)

Internal and external bonding protection is essential for proper and safe engine operation.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): Either MIL–B–5087 or MIL–STD–1818 should be specified.

Background:

Good electrical bonding practices have long been recognized as a key element of successful system design. Since electrical bonding involves obtaining good electrical contact between metallic surfaces, and corrosion control often tries to avoid electrical continuity between dissimilar materials, it is necessary to ensure that both disciplines are properly considered. Electrical bonding by the use of jumper and/or metal surface contact and electrical isolation by the use of insulating surface and insulating clamps is a mandatory safety requirement.

AFSC DH 1–4 and SAE ARP 1870 provide additional guidance for bonding methods. All electrical and electronic components which produce electromagnetic energy should be installed in a manner that provides a continuous low–impedance path from the equipment enclosure to the structure.

Electrical bonding measures should be implemented to manage electrical current paths and control voltage potentials to ensure the required electrical system performance is achieved and to protect personnel.

Bonding provisions should be in accordance with MIL–B–5087 and compatible with other requirements imposed on the system for corrosion control.

REQUIREMENT LESSONS LEARNED

Electrical bonding has provided vital protection in many instances of lightning strikes. One aircraft type has encountered 18 strikes in ten years without failure of the electronic control. In one case, the strike was so severe that the engine inlet was blown off and the engine kept on running.

4.7.4.6 Electrical bonding The requirements of 3.7.4.6 shall be evaluated by inspection and test.

EVALUATION RATIONALE (4.7.4.6)

A test of the DC resistance of each bond is the only reliable method of evaluating the requirement.

EVALUATION GUIDANCE

Background:

The adequacy of the internal and external electrical bonding should be evaluated by inspections and component functional operation during engine and component tests. In the event of an electrical failure, all bonding features should be thoroughly inspected to ensure that they are intact, (i.e., not fused, burned, or otherwise deteriorated). Further evaluation should be obtained by the EMI testing conducted on the engine electrical and electronic components in accordance with MIL–STD–461 and MIL–STD–462. All test observations, experiences, and results should be recorded and included in the test reports.

EVALUATION LESSONS LEARNED

Problems with bonding methods have become apparent during EME testing.

3.7.4.7 Ground isolation The engine/aircraft interface shall not use aircraft structure for power or signal return. The isolation between power returns/signal returns and structure shall be at least 1 megohm at DC.

REQUIREMENT RATIONALE (3.7.4.7)

This requirement ensures current return paths are established to control ground loops and minimize electromagnetic coupling. Provisions should be taken to ensure that there is sufficient isolation resistance between grounds.

REQUIREMENT GUIDANCE

Background:

This requirement is necessary to prevent grounding of components/parts in a series circuit such that static discharge or lightning strike to one component/part will not go through another on its way to being grounded out. A series circuit for grounding may cause unnecessary loss of both components/parts because of a faulty ground route taking place.

REQUIREMENT LESSONS LEARNED

None.

4.7.4.7 Ground isolation The requirements of 3.7.4.7 shall be evaluated by inspection and test.

EVALUATION RATIONALE (4.7.4.7)

Ground isolation must be evaluated to ensure electromagnetic coupling is minimized and to provide the airframe contractor the flexibility in grounding the engine to the airframe structure.

EVALUATION GUIDANCE

Background:

Inspection of engineering drawings showing the isolation between component and structure may satisfy the evaluation criteria. Evaluation methods should be submitted by the contractor in the E^3 Control Plan. The Using Service should ensure that an evaluation plan for ground isolation is a CDRL requirement.

EVALUATION LESSONS LEARNED

None.

3.7.4.8 Potting compounds The use of potting compounds shall not be permitted except where specifically approved by the Using Service. When approved, the potting compounds and their application(s) shall be described.

REQUIREMENT RATIONALE (3.7.4.8)

Potting materials are required in some electrical and electronic components to protect the internal parts from environmental damage, (e.g., temperature, vibration, corrosion, etc.).

REQUIREMENT GUIDANCE

Background:

The potting materials and processes used in electrical components for environmental protection should not deteriorate the chemical, physical, and electrical properties of the components. Potting materials in repairable components should be easily removable/replaceable. Water absorbing potting compounds should not be used for electrical soldering structural supports. For additional guidance refer to AFGS–87219.

REQUIREMENT LESSONS LEARNED

Potting compounds can significantly influence the reliability and maintainability of an electrical component. In one case, epoxy used to stabilize the windings in a wetted stepper motor were eventually found to create corrosion of the motor wires after the epoxy was exposed to fuel. In another component that required frequent maintenance, the potting compound had to be removed by technicians using dental picks. This was extremely costly and time consuming.

4.7.4.8 Potting compounds The requirements of 3.7.4.8 shall be evaluated by demonstration, inspection, and test. Removable potting compounds shall be removed and replaced following the engine and component endurance test. The components shall then be functionally tested to ensure that no damage exists as a result of potting compound removal and replacement.

EVALUATION RATIONALE (4.7.4.8)

The evaluation is needed to ensure the potting compound will not degrade or cause deterioration of the electrical component in the engine operating environment. The tests also ensure that the removal of the compound for repair does not further damage the internal components.

EVALUATION GUIDANCE

Background:

The evaluation should be conducted on the engine endurance test that has the greatest number of thermal cycles. This ensures that the compound will be thermally exercised so that the ability to resist cracking may be known.

EVALUATION LESSONS LEARNED

There have been problems that result from the use of potting compounds that cannot be found during engine and bench testing. These problems have shown up after hundreds of thousands of fleet hours of operation, and may result in serious support problems, possible fleet grounding and large redesign and retrofit activities.

3.7.5 Ignition system(s) The engine ignition system(s) shall be electrically self–sufficient and shall not require power external to the engine. A means shall be provided to deactivate the engine ignition system(s), and it shall be described herein.

The engine ignition system(s) shall light the combustor(s) under carbon and water fouling conditions.

a. Main ignition system. The main engine ignition system shall be described herein to include power source, ratings in terms of stored energy level and delivered energy level to each ignitor, in joules per spark and frequency of spark, and type (continuous, manual, automatic, etc). For single engine applications the ignition system shall be redundant, shall provide a minimum of two separate ignitors and two independent exciter output circuits. Each separate ignitor and circuit shall release sufficient energy for all ground and air starting requirements.

b Augmentor ignition system. If applicable, the augmentor ignition system shall be described herein.

REQUIREMENT RATIONALE (3.7.5)

The main ignition system must be capable of supplying sufficient energy for starting and relighting the engine under all environmental and operating envelope conditions. Two separate ignitor systems are needed for redundancy in a single engine application.

REQUIREMENT GUIDANCE

Background:

Each ignitor and associated exciter output circuit should be independent to ensure proper operation with any other exciter output open–circuited or short–circuited. In areas of the flight envelope where there is a high risk of a combustor blowout, one of the ignitor systems should be capable of continuous operation to permit successive starting attempts without delay. Continuous duty operation is desirable as a preventive measure against flameout. The ignition system should have provisions to allow its external activation and termination which will provide safety during maintenance by eliminating a lethal shock hazard.

The contractor should describe the augmentor ignition system (ignitor plug, fuel slug from main combustor, etc). Augmentor ignition is controlled by the engine control system and additional requirements are contained in 3.7.2.

REQUIREMENT LESSONS LEARNED

Experience has shown that continuous ignition requires the use of a low energy system to provide long ignitor life. The combination of one high and one low energy ignition will usually suffice. Capacitive discharge has proven to be superior to inductive discharge. Past experience has shown that a minimum energy level, discharged in 10 to 100 microseconds at a frequency of at least one spark per second is essential to satisfactory operation.

Augmentor ignition requires the application of correct fuel–air ratio and air velocity in the vicinity of the ignitor. Experience has shown that a pilot burner may be required to meter air flow and fuel flow in combination with low air velocity adjacent to the ignitor to ensure reliable augmentor ignition throughout the flight envelope.

4.7.5 Ignition system(s) The requirement of 3.7.5 shall be evaluated by tests including fouling tests. The test procedure and schedule shall be as follows: (a) .

EVALUATION RATIONALE (4.7.5)

The ignition system(s) must be evaluated.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The self–contained ignition system or component test assemblies should be tested in accordance with the following 4–hour schedule(s). For the purpose of these tests the minimum and maximum voltages and frequencies should correspond to those extreme conditions permitted on the engine for satisfactory functioning of the ignition

system. The ignition system should be tested with the same number of ignitors as used on the engine. The ignitors should be installed in a chamber. The chamber pressure and air flow should be regulated to simulate the internal engine pressures and flows from minimum windmilling pressure and flow to the maximum pressures and flows the ignitors will see in the engine flight envelope. For the high temperature test, the complete ignition system should be placed in a suitable chamber to simulate, prior to initiating the test, conditions of 95×5 percent relative humidity at the maximum component limiting temperature of 3.2.2.12.1.

Procedures:

(1) High temperature endurance. The ignition system should be operated for 300 hours of cycling in accordance with appropriate schedule outlined below and at the maximum component limiting temperature given in 3.2.2.12.1. A 2–hour shutdown should follow each cycle at the temperature conditions corresponding to the requirements of 3.2.2.12.1(c). At the conclusion of testing, checks should be made of insulation resistance, over–voltage capability, ignitor output energy, and spark rate.

(2) Room temperature endurance. The ignition system should be operated at an ambient temperature between 16°C (60°F) and 38°C (100°F) for 100 hours of cycling in accordance with the appropriate schedule outlined below.

(3) Low temperature endurance. The ignition system should be tested for 24 hours at an ambient temperature equal to the sea level cold atmospheric temperature of figure 4. The system should be operated for 12 hours of cycling, in accordance with the appropriate schedule outlined below, followed by a 10–hour minimum inoperative soaking period, and a final 12 hours of cycling.

Schedules:

a. Continuous duty ignition system. The continuous duty start and augmentor ignition systems should be cycled simultaneously according to the schedules specified.

For the start ignition system, each cycle should consist of the following:

		ON	OFF
(1)	NOM voltage	1 hour	20 minutes
(2)	MAX voltage	40 minutes	20 minutes
(3)	MIN voltage	30 minutes	20 minutes
(4)	NOM voltage	40 minutes	10 minutes

For a continuous duty augmentor ignition system, each cycle should consist of the following:

		ON	OFF
(1)	NOM voltage	5 minutes	20 minutes
(2)	NOM voltage	5 minutes	5 minutes
(3)	NOM voltage	5 minutes	40 minutes
(4)	MAX voltage	5 minutes	15 minutes
(5)	MAX voltage	5 minutes	5 minutes
(6)	MAX voltage	5 minutes	25 minutes
(7)	MIN voltage	5 minutes	10 minutes
(8)	MIN voltage	5 minutes	30 minutes
(9)	MAX voltage	5 minutes	5 minutes
(10)	MAX voltage	25 minutes	15 minutes

b Intermittent ignition system. Intermittent ignition systems should be cycled according to the following schedules:

CONDITIONS	ON	OFF	REPEAT
(1) MIN voltage	30 seconds	4.5 seconds	4X
(2) MAX voltage	30 seconds	4.5 minutes	4X
(3) MAX voltage	5 minutes	5 minutes	1
(4) NOM voltage	5 minutes	5 minutes	1
(5) MIN voltage	5 minutes	5 minutes	2X
(6) NOM voltage	1 minute	2 minutes	7X
(7) MIN voltage	5 minutes	5 minutes	2X
(8) MIN voltage	5 minutes	4 minutes	1
(9) MAX voltage	30 seconds	4.5 minutes	6X
(10) MIN voltage	5 minutes	5 minutes	1
(11) MAX voltage	5 minutes	5 minutes	1
(12) OFF 10 minutes			
(13) MAX voltage	1 minute	4 minutes	4X
(14) OFF 10 minutes			
(15) NOM voltage	5 minutes	5 minutes	2X

EVALUATION LESSONS LEARNED

None.

3.7.5.1 Ignition system fouling

4.7.5.1 Ignition system fouling

3.7.5.1.1 Carbon fouling The spark ignitors of the ignition system shall provide sparking performance with spark gaps completely covered, filled, or bridged with an application of graphite petrolatum compound, MIL–T–5544. With the minimum power input of 4.7.5, and these carbon fouling conditions, the sparking rate shall not be less than the minimum design value.

REQUIREMENT RATIONALE (3.7.5.1.1)

The ignition system must be capable of igniting while fouled with carbon.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.7.5.1.1 Carbon fouling The requirement of 3.7.5.1.1 shall be evaluated by test.

EVALUATION RATIONALE (4.7.5.1.1)

The ignition system assembly spark ignitors must be tested to evaluate ignitor performance under carbon fouling conditions.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.7.5.1.2 Water fouling The spark ignitors of the ignition system shall provide sparking performance when thoroughly drenched with water or frosted over. With the minimum power input of 4.7.5, and the water fouling conditions, the sparking rate shall not be less than the minimum design value.

REQUIREMENT RATIONALE (3.7.5.1.2)

Engine flame–out in extreme environmental conditions may introduce moisture/water to the spark ignitors. Spark ignitors under these conditions must be capable of providing minimum performance necessary to start the engine consistently.

REQUIREMENT GUIDANCE

None.

REQUIREMENT LESSONS LEARNED

None.

4.7.5.1.2 Water fouling The requirement of 3.7.5.1.2 shall be evaluated by test.

EVALUATION RATIONALE (4.7.5.1.2)

The ignition system assembly spark ignitors must be tested to evaluate ignitor performance under water fouling conditions.

EVALUATION GUIDANCE

Background:

The spark ignitors should be positioned in a manner simulating the mounted position in the engine. The minimum power/voltage input should be supplied in accordance with 4.7.5. The spark ignitors of the ignition system test assembly should be thoroughly drenched with water to simulate extreme atmospheric conditions.

EVALUATION LESSONS LEARNED

None.

3.7.6 Engine monitoring system (EMS) The EMS and the equipment which make up the system shall be described herein. If a ground support system is required, it shall include all the (a) support equipment needed to achieve the overall EMS requirements. Hardware and software shall be provided to monitor and record the aircraft and engine parameters defined in tables VIIa and b during flight and ground operation, and to provide data in an organized format for ground analytical condition checkout capability, parts life tracking, warranties, and scheduling maintenance actions. The malfunction of any EMS hardware or software shall not affect engine performance or operability. The EMS shall be compatible with existing engine maintenance systems. The engine monitoring system shall be completely functional after failure of any other subsystem of the engine, including engine control and electrical systems.

REQUIREMENT RATIONALE (3.7.6)

An EMS is required to monitor and record parameters for maintenance and performance trending.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The blank should be completed with organizational, intermediate, or depot, depending on which levels of maintenance will retrieve and/or use the data. It may be completed by the Using Service to drive the design if a firm maintenance concept has been established, or the contractor may complete it to allow greater flexibility in optimizing the design.

Background:

The EMS should be an integral part of the engine maintenance system and should be compatible with both the aircraft data collection and ground support systems. The EMS data may or may not be retrieved through the aircraft system, and the description should state whether or not the aircraft system is used. All aircraft/engine EMS interfaces should be described in the interface section 3.1.1.11 of this specification. The contractor should describe the equipment included as part of the engine monitoring system (i.e., engine data collection box and sensors, and if applicable, the aircraft data collection system and ground support system, data transfer system [removable memory card, computer port]). Detailed descriptions of the EMS equipment are not required, but a sufficient system level description is needed so that the Using Service and contractor understand the system's capabilities.

The EMS should include installed airborne equipment, ground equipment, flight line support equipment, intermediate level support equipment, JOAP, etc., and depot support equipment, technical publications, visual/manual inspections and personnel required to accomplish the complete task. Obviously, collecting data is a small part of the systems approach required to develop an effective EMS. Some of the details to be considered in the design are the sensors, signal conditioners, data processor, data storage, crash survivability, data exchange rate(s), engine or airframe mounting, ground based processing station, and programming language.

Some of the parameters that should be monitored are rotor speeds, gas path pressures, and temperatures, accelerometers for vibration measurement, FADEC signals and aircraft parameters (velocity, angle of attack, yaw, missile launch, pressure altitude, etc.). Tables VIIa and VIIb should be the tables where the monitored parameters and their accuracies will be shown. Two tables are recommended, one each for the engine and aircraft. The contractor should fill in the table. Tables VIIa and VIIb should be coordinated with ILS and life management personnel to establish data format and requirements (low cycle fatigue cycles, full thermal cycles, multiple categories of partial thermal cycles, multiple categories of minor thermal cycles, etc.). The engine maintenance concept (i.e., two or three level) needs to be chosen before the EMS is designed so the contractor can properly design the system. Additional guidance can be found in ARP 1587.

There has been considerable proliferation of EMS ground support equipment. It is generally recognized that the airborne electronic engine monitoring equipment will be somewhat application unique, but the ground equipment required to extract, transmit, process, and distribute the data can be adaptable to the application.

EMS program software algorithms and instrument range, system accuracy, time response, sample rates, and electrical characteristics for each parameter, should be provided to the Using Service via CDRL.

REQUIREMENT LESSONS LEARNED

The USAF has had numerous bad experiences with EMS that did nothing more than collect data. The data that were collected could not be reduced into useful information, and some of the more important parameters were not collected.

F/A–18C/D and AV–8B engine monitoring system designs were frozen at an early stage of development because they were incorporated into mission computers. This has provided zero growth capability for the EMS software in these aircraft. More importantly, it has prevented operational lessons learned from being incorporated into these aircraft. In the F/A–18E/F, the MIL–STD–1553 data bus used for EMS parameters does not have a fast enough data rate to support real time computations of engine health based on state–of–the–art engine modeling techniques. This severely limits the future growth capability of the In–Flight Engine Condition Monitoring System (IECMS) on this aircraft.

4.7.6 Engine monitoring system (EMS) The requirement of 3.7.6 shall be evaluated by inspection, analysis, and test.

EVALUATION RATIONALE (4.7.6)

EMS drawings and electrical diagrams and the engine maintenance concept need to be reviewed to ensure the EMS design will comply with the requirements.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

The accuracies of the temperature parameters in tables VIIa and VIIb shall be evaluated by independent engine test, and by the calculated relationship between the actual measured temperature and the temperature output signal. The above calculations shall be provided herein.

Background:

The monitoring system should be installed on all engines that are tested during development, to ensure all problems have been identified and solved prior to the official qualification testing.

EVALUATION LESSONS LEARNED

None.

3.7.6.1 EMS fault detection/isolation The EMS shall provide (a) percent fault detection and isolation of all faults. The system's functions required to achieve this level of detection/isolation faults shall be described. The on–board engine diagnostics shall provide fault detection/isolation to the faulty LRU/WRA to meet the reliability and maintainability requirements as specified in 3.5. On–board engine diagnostics system false alarms shall be considered as failures and shall be used in the reliability and maintainability calculations per 3.5. The mission critical equipment and functions of the on–board diagnostic system which monitor mission and safety critical parameters shall be defined in the applicable contract. The on–board engine diagnostics system shall not cause a failure of any other mission or safety critical system.

REQUIREMENT RATIONALE (3.7.6.1)

The EMS fault detection/isolation capabilities and the equipment and maintenance efforts necessary to accomplish these functions need to be specified.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor should specify between 90–100 percent, depending on the system. For ground based fault detection systems the fault detection rate should be 95–100 percent.

Background:

The contractors list and description of the functions to detect/isolate faults may include: the on–board engine diagnostics, inspections, results from processed EMS data, and troubleshooting (automated, computer assisted and manual).

For higher percentage coverage of fault detection and isolation, more maintenance functions should be listed. That is, if 90 percent coverage is specified, then the list may only have on–board engine diagnostics. If 100 percent coverage is required, then the list may have all of the features described above.

As the Using Services reduce their number of personnel, the importance of the EMS will increase. The goal is to have the on–board system identify any faulty LRU/WRA and have maintenance personnel only conduct minor troubleshooting tasks on the ground. A high percentage of the fault detection and isolation should be accomplished through automated processes (both in flight and on the ground) to save maintenance work hours. Evaluating the problem in the field through the EMS can also save depot rework hours and other supply costs by reducing the number of "cannot duplicate" occurrences. The EMS should be compatible with existing engine maintenance systems; Core Automated Maintenance System (CAMS) and Comprehensive Engine Management System (CEMS) for the USAF; Aircraft Engine Management System (AEMS), Naval Aviation Logistics Data Analysis (NALDA), Aviation Maintenance–Material–Management System (Aviation–3M) and Engine Component Tracking System (ECOMTRAK) for the USN.

REQUIREMENT LESSONS LEARNED

None.

4.7.6.1 EMS fault detection/isolation The requirement of 3.7.6.1 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.7.6.1)

The EMS must be functionally evaluated to ensure its usefulness as a maintenance, performance trending, and warranty tool will be fulfilled.

EVALUATION GUIDANCE

Background:

Testing should be performed throughout the development cycle and should include qualification testing, test cell testing, and flight testing. The flight testing should evaluate the effectiveness of the entire system including airborne equipment, ground equipment, and data output product usage. Flexibility of the EMS during flight test should be maintained so that changes can be made to the system to ensure it is a useful maintenance tool.

EVALUATION LESSONS LEARNED

Many systems have been thoroughly tested from a functional standpoint and adequate hardware testing has also been performed. However, the software area has been neglected, resulting in problems after the EMS has been fielded. These problems were due to "bugs" that were not discovered during functional testing but surfaced when the EMS became operational. These types of problems occurred specifically on the F101 Engine Cycle Counter (ECC) and the KC–135R Turbine Engine Monitoring System (TEMS). Thorough evaluation of the software is necessary to prevent this from happening.

Past experience has shown that during flight test, necessary and desired changes to the EMS do occur in every system. Reasons for some of these changes are as follows:

(1) Some of the EMS functions originally chosen do not adequately aid maintenance.

(2) Excessive and “non–usable” data is being gathered.

(3) More information about the engine is available resulting in additional required functions being added to the EMS.

(4) Several trend points are originally programmed with the “best” point being chosen based on flight test results. Due to these known changes in the EMS, it is recommended that the EMS be as flexible as possible during flight test. This can be implemented via “flexible” software that can be reprogrammed quickly during flight test. The final EMS production software can then take advantage of flight test results and can be made “permanent”.

3.7.6.2 On–board engine diagnostic functions The on–board engine diagnostics system shall perform the following functions:

a. Acquire data to determine engine performance status and performance trends, and mission profile data (altitude, Mach number, alpha, beta, etc.).

b. Acquire data for parts life tracking.

c. Acquire limit exceedance data (a) seconds prior to the event until (b) seconds after.

d. Acquire pilot initiated data records.

For carrier based aircraft, the on–board engine diagnostics system shall perform:

e. Takeoff thrust/power check and provide a signal to the airframe. The thrust/power shall be within (c) percent of the actual net thrust/power. The EMS shall be able to determine engine mission capability and provide a signal to the airframe.

REQUIREMENT RATIONALE (3.7.6.2)

A listing of the diagnostic functions needs to be specified to ensure the functions are consistent with the overall monitoring system requirements.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of at least 15 seconds.

(b): A value of at least 15 seconds.

(c): For turbofan/jet applications: a maximum value of 4 percent.

For turboprop/shaft engines: a maximum value of 2.5 percent.

Depending on the complexity of the monitoring system, the following may be transferred verbatim into the specification paragraph:

– Provide flight line indication of on–board diagnostic system failures (self–test).

– Provide oil quantity indications/readings at a readily accessible location and at the servicing location for the engine oil. A reliable oil quantity indication/reading shall be retained after engine shutdown and electrical power is off. The reading shall be retained for status display until the engine is restarted, the engine is replaced, or maintenance is performed on the oil system.

– Provide failure identification to the individual sensor or associated circuit.

– Provide propulsion data and advisory status to the aircraft in accordance with the ICD.

– Acquire vibration data to provide indication of engine mechanical health.

– Oil and/or fuel filter impending and actual bypass. Acquire quantitative oil debris parameters including small, large and total debris. Oil debris analysis should include hit trending (debris accumulation rate) along with total and individual particle counts.

– Transmit engine data to aircraft flight data recorder for use in characterizing mission usage.

– Hydraulic fluid working pressure and temperature indications.

Background:

The following further defines the information that the contractor should provide the Using Service at design and technical reviews:

a. The algorithm and parameters used to measure and calculate engine performance trending should be reviewed with the contractor at Design Reviews or other technical interchange meetings.

b The structures engineers should review the proposed list of parts life tracking parameters and the algorithms used to calculate cycles and/or life.

c. All parameters that exceed their limits should be recorded by the EMS, and any mission or safety critical parameter whose limit has been exceeded should also be displayed in the cockpit. Limit exceedance may also include engine stalls or blowouts. For stall events, the FADEC may provide a signal to the EMS that instructs the EMS to record the appropriate parameters. Combustor blowout events may be recorded in a similar manner.

d. The pilot should have the capability to initiate a data record any time during the mission. The pilot initiated data should include all parameters normally recorded by the EMS unless the Using Service has a unique set of parameters that it wants recorded.

e The pilot must be able to check takeoff thrust/power level and to be able to assess whether the engine is mission capable, especially for carrier operations. Adequate thrust/power is essential for successful launch from a carrier, and notification of a component/system failure which would jeopardize completion of a mission should be provided. The Using Service should check the ICD/airframe specification for implementation of GO/NO–GO indication in the cockpit.

f The EMS needs to diagnose and report any malfunctions of itself.

g Oil quantity is a safety critical parameter and should be recorded. It is desired to have the oil quantity available with the engine not running without having to open any access panels to reduce aircraft turnaround times.

h. The on–board diagnostic system may not be sophisticated enough to pinpoint a failure to a specific part or component. However, the on–board system can, as a minimum, find a failure in an electrical circuit (i.e., wiring, connector, or sensor) if it cannot fault isolate to a specific sensor or component.

j. The EMS may be required to provide data to the aircraft for either the aircraft monitoring system or for cockpit display of engine parameters. The ICD or other interface document should list which parameters need to be provided to the aircraft.

k. The EMS may need to provide data to the aircraft's flight data recorder in lieu of a monitoring system. The EMS design should be flexible enough to accommodate data transfer to several aircraft systems.

l. Vibration data is needed to detect early bearing failures and/or imbalance conditions. The number of sensors proposed for the engine should be sufficient to detect compressor and turbine imbalance, impending bearing failures, and gearbox problems. Depending on the complexity of the on–board system, the vibration frequency analysis and Fast Fourier Transform analysis may or may not be part of the on–board system. Analysis of the data should be part of the EMS, but whether it's part of the on–board or ground support system should be made on a system level cost effectiveness basis. All turboshaft and turboprop engines should have gearbox mounted sensors.

m. An oil and/or fuel filter impending and actual bypass signal may be needed. This feature would help reduce aircraft turnaround times since no access panels would have to be opened for the inspections.

n. The EMS should also maintain the engine serial number, FADEC S/N, FADEC software version number, EMS S/N, and EMS software version number. Crash survivability may or may not be an EMS requirement. New weapon systems may be required to store data in a crash survivable component and this may require a similar component on the engine. Conversely, the aircraft may only require that certain parameters be provided to the aircraft for storage in the aircraft's system.

REQUIREMENT LESSONS LEARNED

None.

4.7.6.2 On–board engine diagnostic functions The requirement of 3.7.6.2 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.7.6.2)

The EMS data must be reviewed and analyzed during all engine testing to ensure the EMS is recording the proper data accurately and that the data can be retrieved easily.

EVALUATION GUIDANCE

Background:

It is important that the EMS be tested on the qualification engines to ensure the system will work properly. An important function of the tests will be to establish thresholds for determining out of range faults, verifying when false faults occur, determining if signal noise is present, and revealing the other plethora of problems that may occur. Flight testing is needed since that is when the true EMS environment and usage will occur.

EVALUATION LESSONS LEARNED

None.

3.7.7 Optical systems Engine subsystems that use optical systems, including fiber optic system(s), shall be identified herein. The fiber optic system shall be described herein. Optical system hardware shall operate satisfactorily in the engine bay environment.

The optical system shall be powered by the engine's primary electrical power system. Switching from the primary to the alternate/external electrical power system shall not affect the operation of the optical system. Optical sensors (e.g., A/B flame detectors, optical pyrometers) shall maintain accuracy and not lose calibration during the life of the engine module in which it is installed.

The optical system components shall be in accordance with the following component specifications: (a) .

REQUIREMENT RATIONALE (3.7.7)

The Using Service needs to know if a fiber optic system is being used on the engine and for which systems. A description of the system is also needed. Optical systems (pyrometers and flame sensors) are used by engine controls to schedule engine functions and must be powered independently of the aircraft. If the engine is required to operate on alternate/external electrical power, then the optical systems must function during switching to this power. An optical sensor must not be changed during the life of the module it is installed on, so maintenance is kept to a minimum.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

the Using Service should reference the following specifications, as applicable:

(a):	MIL–A–24726	Attenuators
	MIL–M–24731	Multiplexers, Demultiplexers, Freq. Division
	MIL–T–24735	Transmitters, Analog
	MIL–R–24737	Receivers, Analog
	MIL–C–24621	Couplers, Passive
	MIL–T–24721	Transmitter, Digital
	MIL–R–24720	Receiver, Digital
	MIL–M–24736	Multiplexers, Demultiplexers, Time Division

By 1994 the above documents should have been superseded by other documents from SAE Subcommittee AS3–B for optical application in flight and engine controls.

Reminder: Section 2 must be updated if the references above are used in the main paragraph.

Background:

The use of fiber optic components or subsystems in place of electrical or electronic components or subsystems to perform engine control functions, such as engine monitoring and data communications, is permitted, subject to the approval of the Using Service.

The Using Service should ensure that specifications referenced in this section are current because of the dynamic changes in fiber optic technology.

In a fiber optic engine control system optical components perform functions that are normally performed by electrical or electronic components. For example, fibers replace wires as the signal transmission medium of a data link. The overall functionality of the control system is left unchanged. The performance requirements for the overall engine control system should therefore be the same whether the system contains fiber optic components or electrical components, or both.

In the simplest configuration a fiber optic data link consists of a single transmitter, a single fiber and a single receiver. The transmitter utilizes a light emitting diode (LED) or laser diode (LD) to convert an electrical signal to an optical signal. The optical signal is transmitted through the fiber to a receiver, which utilizes a detector to convert the optical signal back into an electrical signal. System modularity requirements require the addition of fiber optic connectors, which facilitate repair and maintenance.

Fiber networks of increasing complexity contain additional fiber optic components. Passive Couplers allow optical signals to be divided (or combined) among two or more pathways without the need to convert them to electrical signals. Multiplexers allow several distinct signals to simultaneously occupy a single fiber element. These include time–division and frequency–division multiplexers, which are analogous to the electrical devices of the same name. Additionally, wavelength–division multiplexers and demultiplexers combine and divide signal channels that are distinguishable by their wavelengths. To control signal levels fiber optic attenuators are used. When damage to optical fibers occurs, repair rather than replacement may be elected, in which case fiber optic splices are implemented to join two fiber ends together.

The use of fiber optics offers significant advantages over electrical wires in several areas. Examples are: (1) weight reduction, (2) electrical bandwidth, (3) electromagnetic environment performance, among others. On the other hand, fiber optics have some unique disadvantages as well. For example, light sources are sensitive to temperature with respect to output power and bandwidth drift. LEDs typically do not produce high output power, while the higher powered laser diodes generally require temperature stabilization, have a shorter life, and may raise eye safety concerns. Optical fibers are subject to microcracking due to mechanical stresses and bending losses. Connectors are not always repeatable in mating and unmating with respect to insertion loss, are easily contaminated, and susceptible to rough handling by untrained personnel. These issues and others are addressed by the requirements and evaluation methods set forth in this specification.

In order to realize certain advantages that fiber optic systems have over electronic systems (for example, high bandwidth, EMI immunity, low volume and weight) control system designs that include fiber optic components need to be developed. The performance and reliability of the engine control system should be paramount in this development; hence, the engine control system requirements must be met at all stages of this development. Indeed, the control system performance requirements should guide this development. Along with advantages, fiber optic components may bring with them inherent weaknesses that could compromise the performance or reliability of the engine control system. A comprehensive set of requirements and evaluation procedures for fiber optic systems that take into account these weaknesses will ensure that the integrity of the engine control system is maintained.

Transmitters:

Optical transmitters interface with the electrical system and supply the optical signal energy for all fiber optic engine control and data communication system functions. Therefore, they may be considered as both electrical and optical components. Maintaining certain characteristics of the optical transmitters within strict design tolerances is essential for proper operation of the engine control system.

The referenced fiber optic transmitter general specifications (MIL–T–24735 and MIL–T–24721) are comprehensive documents that may be applied to fiber optic transmitters in a variety of applications.

The key differences between shipboard and aircraft applications pertain to the section on environmental requirements, specifically to the temperature extremes and fluid exposure. Laser diodes are particularly sensitive to high temperatures, such that they typically require thermoelectric cooling to sustain reliable operation. The temperature ranges that the transmitter is expected to experience under both operating and nonoperating conditions should be specified by the engine manufacturer.

The performance requirements specified for the transmitter will depend to a large extent on its role in the system. The peak emission wavelength, spectral width, and power levels will depend on the type of fiber used (e.g., single mode or multi–mode), which, in turn, will depend on bandwidth and power requirements of the data link. These parameters should be considered carefully in the early stages of system design. The electro–optical properties of the transmitter, on the other hand, are dependent on its role as an electronics component, affecting the overall system transfer function. In this regard the signal–to–noise ratio (SNR), frequency range, distortion, and group delay of the optical transmitter should be included as parameters in the electronics system analysis.

Receivers:

Like Optical Transmitters, Optical Receivers can be considered as both optical and electrical components. As with optical transmitters, the optical, electro–optical, and electrical performance parameters, along with their respective tolerances are defined during the fiber optic system design process. The fiber optic engine control system manufacturer should specify these performance requirements, to include spectral responsibility, dynamic range, sensitivity, signal–to–noise ratio, frequency ranges, signal distortion, group delay, and output signal interface requirements.

Multiplexers/demultiplexers:

The engine control system manufacturer should specify as part of the control system design the physical and performance requirements of all multiplexers and demultiplexers used. These are to be identified as Frequency–division or time–division multiplexers (FDM or TDM), demultiplexers (FDDM or TDDM), or multiplexers/demultiplexers (MULDEMS). The referenced specifications (MIL–M–24731 and MIL–M–24736) are identified as specific to shipboard applications, but in fact are sufficiently generic as to be readily adopted for aircraft applications as well.

Frequency–division multiplexers and demultiplexers:

The engine manufacturer should specify as part of the control system design the classification of the frequency–division multiplexers and demultiplexers. The designations used to classify these units are Function, Multiplexed Signal Type, Channel Configuration, Analog Subchannel Class, Digital Subchannel Class, and Style. The Function designation defines the unit as a FDM, FDDM, or a MULDEM. The Multiplexed Signal Type designation defines the signal outputs and/or inputs as electrical or optical. The Channel Configuration designation defines the number of input or output channels. The Analog Subchannel designation defines the highest required analog signal subchannel bandwidth. The Digital Subchannel Class designation defines the highest equivalent digital data rate. The Style designation defines the package configuration. For more details on these designations see MIL–M–24731.

The engine manufacturer should specify quantitative reliability requirements expressed in mean–time–between–failures (MTBF), mean–cycle–between failures, or other quantitative measure of reliability. The engine manufacturer should specify performance requirements on transmitters, receivers, and connectors built into the multiplexers as part of the multiplexer design. Also specified should be dimensions, weight, solderability, source of prime power, and interface parameters.

Time–division multiplexers/demultiplexers:

The engine manufacturer should specify as part of the system design the classification of the time–division multiplexers and demultiplexers. The designations used to classify these units are Function, Multiplexed Signal Type, Channel Configuration, Analog Subchannel Class, Digital Subchannel Class, and Style. The Function designation defines the unit as a TDM, TDDM, or a MULDEM. The Multiplexed Signal Type designation defines the signal outputs and/or inputs as electrical or optical. The Channel Configuration designation defines the number of input or output channels. The Analog Subchannel designation defines the highest required analog signal subchannel bandwidth. The Digital Subchannel Class designation defines the highest equivalent digital data rate. The Style designation defines the package configuration. For more details on these designations see MIL–M–24731.

The engine manufacturer should specify quantitative reliability requirements expressed in mean–time–between–failures (MTBF), mean–cycle–between failures, or other quantitative measure of reliability. The engine manufacturer should specify performance requirements on transmitters, receivers, and connectors built into the multiplexers as part of the multiplexer design. Also specified should be dimensions, weight, solderability, source of prime power, and interface parameters.

Attenuators:

The engine manufacturer should specify the design and performance requirements of all fiber optic attenuators used in the fiber optic system. Fiber optic attenuators are classified according to Type, Port Construction, and Package. The Type designation defines whether the level of optical power reduction is fixed or variable. The Port Construction designation defines the method of interfacing attenuators to the fiber optic cables. The Package designation defines the type of packaging as either stand alone or printed circuit board mounted. For more details on these designations see MIL–A–24726.

The specification of attenuator requirements should include design and performance requirements on the fiber optic pigtails and connectors integral to the attenuator package.

REQUIREMENT LESSONS LEARNED

None.

4.7.7 Optical systems The requirements of 3.7.7 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.7)

Component requirements unique to fiber optic engine control systems must be evaluated to ensure proper operation of the engine control system under extreme operating conditions.

EVALUATION GUIDANCE

Background:

The specification documents referenced in 3.7.7 and in following paragraphs are sufficiently comprehensive and generic to be applicable in their totality to the various components of the fiber optic control system. Some were developed with shipboard applications in mind. They are augmented here by the appropriate paragraphs to include the severe operating conditions unique to aircraft engine applications. These conditions include, most notably, temperature extremes, vibration, and chemical exposure conditions. Shipboard documents may be used until aircraft fiber optic specifications are written because no set of documents exists for airborne applications.

A fiber optic engine control system utilizes components that operate on different physical principles than electronic components and thus may behave very differently under extreme operating conditions.

Transmitters:

The most critical area in the evaluation of optical transmitters is that of reliability, especially for laser diodes in high temperature environments or high power applications. Lifetimes of laser diodes are in the range of 40,000 to 50,000 hours, at room temperature, while those of LEDs may exceed 200,000 hours. The advantage of laser diodes over LEDs is the higher power and efficiency that they afford.

For LEDs and LDs, spectral peak output power shifts approximately 0.25 nm per °C. Laser diodes generally require thermoelectric cooling, which itself is subject to reliability problems at elevated temperatures.

Receivers:

The engine manufacturer should specify the ranges of temperature (both operating and nonoperating) and vibration, and the nature of the chemical exposure that the optical receiver is expected to undergo while in service. This information will be used to determine test conditions during the evaluation of performance under the effects of temperature extremes, thermal shock, and nuclear radiation resistance.

Several noise components of the optical receiver are functions of temperature. Thermal noise is the dominant noise source at 25°C, and increases slowly with temperature. Shot noise (the statistical fluctuation of photon arrival rate) and diffusion current noise are usually small at room temperature, but increase rapidly with temperature. Temperature thus affects the SNR, and hence the dynamic range and sensitivity. Nuclear radiation may lead to ionization–induced photocurrents and likewise reduce the SNR. In addition, nuclear radiation may induce transient absorption and luminescence in the fiber pigtails attached to the receiver. These are temporary effects that usually recover soon after the radiation field is removed; however, the recovery rate is a function of temperature (faster at higher temperatures) and the luminescence intensity may, in some cases, be high enough to permanently damage a detector element.

Optical Multiplexers/Demultiplexers:

The performance requirements for optical multiplexers and demultiplexers include specifications of the limits of tolerable performance degradation due to manufacturing tolerances and adverse mechanical and environmental conditions. These limits on performance degradation must be adequate to ensure reliable overall performance of the fiber optic system under adverse conditions. The performance requirements should also be specified in the engine specification. Compliance with these requirements should be evaluated by tests in which the appropriate performance parameters are measured under simulated adverse conditions. The methods for simulating these conditions and for measuring the performance parameters are described in the referenced documents.

EVALUATION LESSONS LEARNED

None.

3.7.7.1 Fiber optic cables Fiber optic cables shall comply with MIL–C–85045, and shall have sufficient spare fibers to permit future growth. The quantity of fiber is subject to Using Service approval. At the sea level cold atmospheric temperature of figure 4, it shall be possible to flex fiber optic cables during installation and routine maintenance without damage. Installation of a fiber optic cable shall not induce a change in optical transmittance that exceeds (a) dB. Repairs to fiber optic cables shall not degrade performance beyond the limits of this paragraph.

REQUIREMENT RATIONALE (3.7.7.1)

Fiber optic cables are perhaps the most critical components of a fiber optic system since they convey the signals. These components must meet the minimum requirements of MIL–C–85045 in order to ensure the integrity of the system.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor shall provide the maximum allowable change in optical transmittance induced by installation. Generally, accepted values should be 0.5 dB (multi–mode fibers) or 0.2 dB (single mode fibers).

Background:

Cabling of optical fibers allows several to be combined into a single modular unit, simplifying installation, maintenance, and repair. The fibers within a given cable need not be related to the same functional unit, nor even be of the same class. The fiber optic engine control system manufacturer should specify the class (single mode or multi–mode) of each fiber and the total number of fibers used in each cable of the fiber optic engine control system. It should also specify the design, construction, and physical dimensions of the complete cable. The engine manufacturer should specify the operating and storage temperature ranges to which the fiber optic cable will be exposed during its service life, as well as methods employed for routing of cables within the airframe.

Specific optical performance requirements will depend on the system design. The engine control system manufacturer should specify the maximum allowable attenuation rate, the maximum change in optical transmittance of each fiber, and far end crosstalk between any two fibers. The change in optical transmittance of a cable due to mechanical stresses in installation may lead to unsatisfactory performance of the optical system. The referenced document (MIL–C–85045) contains default values for the attenuation rate and change in optical transmittance requirements, if these are not specified.

REQUIREMENT LESSONS LEARNED

In some commercial applications problems have been encountered relating to excessive loss in cables due to tight bend radii and improper cable clamping.

4.7.7.1 Fiber optic cables The requirements of 3.7.7.1 shall be evaluated by analysis, demonstration, and test. All cables shall be tested for change in optical transmittance in accordance with EIA–455–20 following installation on the engine.

EVALUATION RATIONALE (4.7.7.1)

Compliance of fiber optic cables to the specified performance requirements must be evaluated by tests.

EVALUATION GUIDANCE

Background:

MIL–C–85045 is the approved document covering all types and classes of fiber optic cable for military applications. Cables intended for use in aircraft are designated as application A (Airborne). The equipment manufacturer should specify the appropriate specification sheet for each cable, depending on, for example, the environment in which it is used or on particular aspects of its design. Only the tests called for by the specification sheet are to be performed. The conditions for those tests are listed in the specification sheets.

EVALUATION LESSONS LEARNED

None.

3.7.7.2 Fiber optic connectors Fiber optic connectors shall be in accordance with (a) . Multiple terminus optical connectors shall have sufficient spare terminals to permit future growth. Optical connections located in close proximity to each other shall be made physically noninterchangeable. At the sea level cold temperature of figure 4, it shall be possible to connect or disconnect optical connectors using normal maintenance procedures, without damage to the fiber cables or connectors. Connectors shall be resistant to engine bay contamination and to the bleed air contamination of 3.1.1.15.1.

It shall be possible to remove these contaminants from the ends of the transmissive elements without causing damage to them. During maintenance operation in which a connector is disconnected, the procedure for removing contaminants from the ends of the transmissive elements before reconnection shall be described.

REQUIREMENT RATIONALE (3.7.7.2)

Fiber optic connectors allow the fiber optic system to be designed with a degree of modularity, which enhances serviceability. However, connectors potentially represent a weak link in a signal channel because of contamination, damage, or lack of repeatability. Standardized requirements for the design and construction of connectors are needed to minimize the potential for these types of problems.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor should provide the connector type and applicable military specification.

Background:

Typical connector types the contractor should specify are MIL–C–83526 circular, environmental resistant, hermaphroditic, MIL–C–83522 single terminus fiber optic, and/or MIL–C–28876 circular plug and receptacle style.

Fiber optic connectors provide the means for separable interfaces among components of the fiber optic system. The choice of connector used in a given application is driven primarily by the design of the components being connected. The engine manufacturer should specify as part of the design the type of connector to be used, all materials,

dimensions, adhesives and sealing compounds, special features, such as keys, keyways, coupling rings, etc., any special tools necessary, any requirements for these tools, and methods for maintaining the cleanliness of the termini. The system manufacturer should also identify the potential for future growth that might impact each of the connectors. The Using Service should specify the contaminants that the connectors are expected to be exposed to during their service life.

MIL–C–38999 contains numerous pertinent requirements that should be applied to fiber optics systems.

REQUIREMENT LESSONS LEARNED

In some commercial aircraft applications contamination has been a severe problem, particularly in cases of connectors having recessed termini that tended to accumulate contaminants. In many cases removal of contaminants from the ends of the transmissive elements has not been 100% satisfactory.

4.7.7.2 Fiber optic connectors The requirements of 3.7.7.2 shall be evaluated by analysis, demonstration, and test. In addition, the mating durability test of EIA–455–21 shall be performed on connectors that have been contaminated with the contaminants specified in 3.7.7.2 and subsequently cleaned.

EVALUATION RATIONALE (4.7.7.2)

Contamination, damage, and lack of repeatability can have severe deleterious effects on the function and reliability of a fiber optic connector. These effects can only be characterized through tests that accurately simulate the operational conditions within which the connector is designed to operate.

EVALUATION GUIDANCE

Background:

The documents referenced in 3.7.7.2 contain evaluation methodologies for fiber optic connectors. They do not include, however, a test of the combined effects of contamination and repeated mating and unmating. For this test, a representative sample of connectors should be exposed to the specified contaminants in a manner that necessitates subsequent removal of the contaminants from the ends of the transmissive elements. This procedure is followed by the mating durability test as described in EIA–455–21A.

EVALUATION LESSONS LEARNED

None.

3.7.7.3 Fiber optic passive couplers Fiber optic passive couplers shall comply with MIL–C–24621.

REQUIREMENT RATIONALE (3.7.7.3)

Standardized requirements for fiber optic passive couplers are mandatory to ensure compatibility with the fiber optic system and reliability in an aircraft environment.

REQUIREMENT GUIDANCE

Background:

The fiber optic system manufacturer should specify the classification of all fiber optic passive couplers used in the optical system as well as where they are used. The coupler classification contains designations of Type, Class, and Style. The Type designation defines the wavelength of operation and method of interfacing to external fiber optic cables. The Class designation defines the optical power distribution made by the coupler (either uniformly distributive or non–uniformly distributive). The Style designation defines the style of packaging. These designations are described in more detail in MIL–C–24621.

The engine manufacturer should specify the maximum temperature range (both operating and non–operating) and the possible fluid exposure that the coupler will undergo during its service life.

REQUIREMENT LESSONS LEARNED

None.

4.7.7.3 Fiber optic passive couplers The requirements of 3.7.7.3 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.7.3)

Standardized testing methods for fiber optic passive couplers are mandatory to ensure compatibility with the fiber optic system and reliability in an aircraft environment.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

Fiber optic passive couplers shall be tested in accordance with MIL–C–24621.

Background:

MIL–C–24621 provides extensive guidance related to quality assurance provisions for fiber optic passive couplers.

EVALUATION LESSONS LEARNED

None.

3.7.7.4 Fiber optic high altitude radiation resistance All fiber optic systems shall not have degradation in performance due to radiation effects at altitudes above _(a)_ km (_(a)_ ft).

REQUIREMENT RATIONALE (3.7.7.4)

The fiber optic systems must be capable of operating at high altitudes without loss of performance due to exposure to radiation from outer–space. The earth's atmosphere provides an effective barrier against radiation emanating from space. At high altitudes the buffer effect caused by the earth's atmosphere is reduced and aircraft operating in this region are exposed to higher levels of radiation.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The Using Service should insert 18.2 km (60,000 ft) in the blank.

Background:

High altitude surveillance aircraft may operate as high as 30.5 km (100,000 ft). At this altitude (Gamma, Alpha, etc.) rays emanating from the sun or planetary interactions may be of sufficient intensity to degrade performance of fiber optic systems.

REQUIREMENT LESSONS LEARNED

None.

4.7.7.4 Fiber optic high altitude radiation resistance The requirement of 3.7.7.4 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.7.7.4)

The most accurate method of determining the performance capability of the optical system at high altitude conditions is through flight test.

EVALUATION GUIDANCE

Background:

Evaluation of the optical system to withstand the radiation effects encountered in a high altitude environment should be accomplished by flight tests. Analysis should be used as a secondary measure.

EVALUATION LESSONS LEARNED

None.

3.7.8 Lubrication system The engine lubrication system physical and functional characteristics shall be described herein.

Any coke and sludge build–up shall not prevent the engine from meeting the specification requirements. A signal for continuous remote indication of the oil pressure shall be provided. If the lubrication and hydraulic systems share a common reservoir, the lubrication system shall be protected from total depletion in the event of a hydraulic failure.

REQUIREMENT RATIONALE (3.7.8)

A description of the lubrication system must be provided for proper evaluation of the physical and functional characteristics.

There is a need to assure the characteristics of the oil system are not conducive to coking and sludge formation. The formation of these substances can block breather tubes and oil filters.

REQUIREMENT GUIDANCE

Background:

The mechanism of oil deposit formation is related to the oil wetted surface area by the residence time of the oil, surface temperature, and local hot spots.

REQUIREMENT LESSONS LEARNED

Thermal soakback on ground shutdown is the primary cause for oil coking and sludge build–up on helicopter engines. New improved oil formulations have reduced the problem of coking and sludge formation.

4.7.8 Lubrication system The requirements of 3.7.8 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.8)

The lubrication system must be inspected to ensure it complies with the description. Engine tests are needed to assure the vents and oil filters will not be obstructed by carbon buildup and oil sludge.

EVALUATION GUIDANCE

Background:

There is no one specific engine test for evaluating oil vents and filters for coking and sludge formation. Any and all engine tests and flight tests should be considered for evaluation of vents and oil filters. Any abnormal rise in oil pressure should be investigated to be sure it is not oil vent related. Look for the presence of sludge when changing filters, particularly if the filter is in a bypass mode. All discrepancies should be noted on the engine log sheet or test log.

All physical features of the lubrication system should be examined and compared to the specification and installation drawing. Functional characteristics should be evaluated by demonstration and test in accordance with the subparagraphs of this section.

The lubrication system should be self–contained unless special considerations dictate otherwise. Check valves used in the lubrication system should be designed to preclude reverse installation.

Special provisions such as oil preheaters or oil dilution should not be required for starting and operating.

The complete lubricating system, including oil cooler(s) and oil reservoir, should be furnished as part of the engine. The oil system should not provide a function for other aircraft accessories or components and should be designed to prevent contamination of oil by seepage of fuel or other fluids from aircraft air–oil and fuel–oil heat exchangers, fluid filled bellows, and seals.

EVALUATION LESSONS LEARNED

Field problems such as engine failure and bearing failures have occurred when the engine lube system also services aircraft accessory gearboxes, constant–speed drives, generators, and hydraulic actuators.

Field problems such as contamination have occurred when fuel–oil heat exchangers, fluid–filled bellows, and seals for compressor bearings leak in the lubrication system.

Oil leaks at compressor bearing seals have caused contamination of the cabin pressurization air, resulting in eye irritation to the pilots. Internal puddles of oil in the hot section will cause heavy varnish or carbon deposits and can be a potential fire hazard.

Reverse installation of check valves has caused internal rupture of the fuel–oil heat exchanger.

Engines that were used for high flight Mach number aircraft were more susceptible to coking and sludge formation.

3.7.8.1 Lubrication oil The engine shall meet the requirements of the specification when using the following lubricating oil(s): (a) .

REQUIREMENT RATIONALE (3.7.8.1)

Lubricating oil selection is critical to the design and important for compatibility with logistical support of field operations. The oils must be specified.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): Lubricating oils conforming to MIL–L–7808 and MIL–L–23699 should be used as standard lubricants. Special lubricants may be required for unusual and/or non–standard situations.

Background:

The oil selected must be able to withstand the operational environment for the full oil change period without loss of properties and capability to lubricate and dissipate the engine heat.

MIL–L–23699 is not recommended for use below –40°C (–40°F). MIL–L–7808 is recommended as the standard oil for use down to –54°C (–64°F). The limitation with MIL–L–23699 oil is based on the fact that MIL–L–7808 (lighter oil) originally had a new oil viscosity limit of 13,000 centistoke at –54°C (–65°F). In contrast, MIL–L–23699 was limited to 13,000 centistoke at –40°C (–40°F) for new oil.

MIL–L–7808 and MIL–L–23699 oils have very similar performance characteristics except that MIL–L–7808 oil has less load carrying capability, lower viscosity, and is more volatile (higher vapor pressure). Therefore, MIL–L–7808 has superior cold start capability, but MIL–L–23699 oil consumption is lower and its high temperature performance is better. The Navy selected MIL–L–23699 because it had even better oxidation and thermal stability than MIL–L–7808.

Selection of oil type should be based on factors such as bearing design, mission cycle mix, and operating environment and should be in accordance with MIL–STD–838.

Use of specially developed lubricants may be needed in order to achieve design objectives but must be incorporated into logistic planning early in the design phase.

REQUIREMENT LESSONS LEARNED

The J58 turbojet (SR–71 engine) requires the use of a specially developed lubricant MIL–L–87100 lubricating oil.

4.7.8.1 Lubrication oil The requirement of 3.7.8.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.8.1)

Lubricating oil must be evaluated to ensure it is able to operate at the conditions defined in the specification.

EVALUATION GUIDANCE

Background:

The contractor may procure high quality oil for use during development testing. The government agency responsible for procuring oil stocks will probably buy for the field based on the lowest priced oil. While both products may be on the (Qualified Products List) QPL, the oil used by the contractor may be substantially superior to that available in government supply for field operation.

EVALUATION LESSONS LEARNED

None.

3.7.8.1.1 Oil pressure and temperature limits The maximum and minimum operating oil pressure limits shall be (a) and (b), respectively. The maximum transient and maximum allowable steady–state oil temperature limits shall be (c) and (d), respectively. The above limits shall also be listed in table VIII. The maximum and minimum oil pressures during starting and initial operation at the sea level cold atmospheric temperature of figure 4 or (e) centistoke oil viscosity, whichever is worse, shall be (f) and (g) respectively. Minimum and maximum oil pressures during starting and initial operation shall not persist for more than (h) minutes.

REQUIREMENT RATIONALE (3.7.8.1.1)

The engine oil system must effectively lubricate the engine within the specified limits.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a) – (d): The contractor should fill in the value.

(e): The Using Service should select the centistoke oil viscosity which should depend on the sea level cold day temperature of figure 4.

(f) – (h): The contractor should fill in the value.

Background:

The minimum and maximum oil pressure should be set by the oil system characteristics. The minimum oil temperature should be consistent with figure 4. The maximum transient and steady–state oil temperature should be determined by a combination of environmental requirements and the oil system capability. The engine contractor will provide the oil system characteristics and provisions for measuring the oil temperature. Minimum and maximum oil pressures during starting and initial operation should not persist for more than 2.5 minutes.

The oil system pressure in conventional designs is a function of oil temperature and may vary significantly, particularly during cold start and warm up. The allowable operating pressures should therefore be related to oil temperature. The acceptable range of pressures with the allowable range of temperatures should be specified. If the minimum specified oil pressure corresponding to the maximum allowable oil temperature is unacceptable at the minimum operating oil temperature, such should be identified in the specification and installation requirements.

The contractor should note the temperature as ‘bulk’ or ‘local’ temperature for clarity.

REQUIREMENT LESSONS LEARNED

Lube system designs that function well under static conditions, have failed to function satisfactorily under dynamic conditions of flight.

4.7.8.1.1 Oil pressure and temperature limits The requirement of 3.7.8.1.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.8.1.1)

The adequacy of the oil system must be evaluated by demonstration and test.

EVALUATION GUIDANCE

Background:

Component testing should be performed, as a minimum on the oil pump and the gearbox. Engine testing and subsequent inspections of oil wetted components should also be performed to assure the oil system is effectively lubricating those components.

At high operating turbine temperatures or high Mach number flight, oil can coke in internal engine passages which is detrimental to the lubrication system.

EVALUATION LESSONS LEARNED

During cold temperature starting tests, large oil pressure spikes have occurred. In some cases the pressure was great enough to crack a filter housing.

3.7.8.1.2 Oil consumption limits The maximum allowable oil consumption rate shall not exceed __(a)__ liters/hour (__(a)__ gallons/hour). Oil consumption limits shall also be specified in table VIII.

REQUIREMENT RATIONALE (3.7.8.1.2)

Oil consumption rates must be established so that the oil reservoir can be sized to meet mission requirements. Oil consumption limits are necessary for maintenance information and for pass/fail criteria in qualification and acceptance testing.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The contractor should provide the value.

Background:

Oil consumption rate should include all forms of oil loss from the engine. Oil seal design, higher operating temperatures, and leakage are means by which oil is lost from the lubricating system.

REQUIREMENT LESSONS LEARNED

Formerly, oil tanks were furnished by the airframe contractor. When this was the case, it was critical that the engine contractor not specify an unwarranted large oil consumption which eased the endurance test oil consumption requirement and drove the size of the airframe furnished tank to an unjustified large size and weight.

4.7.8.1.2 Oil consumption limits The requirement of 3.7.8.1.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.8.1.2)

Engine testing is needed to establish oil consumption rates.

EVALUATION GUIDANCE

Background:

Consideration should be given to the margin between average and maximum rates to address problems such as wear and seal manufacture variations.

EVALUATION LESSONS LEARNED

None.

3.7.8.1.3 Oil flow interruption/depletion The engine shall meet the requirements of the specification during and after operation at _(a)_ thrust/power for a period of at least _(b)_ seconds with no oil (only air) supplied to the oil pump inlet. The engine shall be capable of operating without oil for an additional _(c)_ minutes at _(d)_ thrust/power without rotor seizure.

REQUIREMENT RATIONALE (3.7.8.1.3)

Oil flow interruption can occur due to failure, maneuvers, combat damage, or improper servicing, so the interruption duration capability must be specified. The engine must operate satisfactorily without oil for the period of time specified in order to egress the hostile area, initiate engine shutdown, or any other required action. Most fighter/attack engines need the capability to withstand multiple brief oil interruptions without damage.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): Intermediate/Maximum Continuous.

(b): 30 seconds minimum

(c): 30 minutes minimum

(d): 30% IRP, except for turboshaft which should be a minimum of 75% Maximum Continuous

Background:

Twenty 15–second interruptions and ten 30–second interruptions are probably a good evaluation of bearings ability to tolerate repeated interruptions.

Three minutes at IRP is sufficient time to regain speed and altitude following an engagement. Thirty minutes is usually sufficient time to permit egress from heavily defended areas where combat damage is likely to be rendered (e.g., 100 nmi at a "limp home speed" of 200 ktas). Typical fighter/attack aircraft can maintain flight at moderate altitudes with thrust levels generally well under 30% of IRP SLS thrust. Thirty percent IRP SLS thrust is typically a comfortable operating condition for the engine, with low temperatures and rotor thrust loads. Since the main function of oil lubrication in rolling element bearings is heat removal, reduction of bearing loads and surrounding metal temperatures effected by throttle reduction should enable the bearings to survive for extended periods. A number of relatively low penalty design possibilities exist such as auxiliary lube from a shielded or protected sump.

An engine intended for use in a single engine aircraft should also be capable of operating for longer periods without oil until rotor seizure occurs, and a requirement describing this capability should be stated in this paragraph.

High agility fighter/attack aircraft may experience repeated transient negative "g" conditions which can interrupt oil supply.

REQUIREMENT LESSONS LEARNED

In some current and past Army helicopter programs, engines had to operate satisfactorily for a minimum of six minutes at 75% Maximum Continuous power after permanent loss of all oil supply to the engine oil pump, followed by 30 seconds of successful operation at Intermediate power before engine shutdown with no engine failure. If oil flow was restored, the engines had to operate satisfactorily for 30 minutes at 75 percent Maximum Continuous power.

The requirement for endurance of the engine without lubricating oil has permitted recovery of aircraft that would otherwise have been destroyed.

Fleet experience has shown some systems cause problems during high angle climbs, negative g and minimum oil level conditions. Operation down to the oil level defined as "unusable" is intended to prevent oil system pressure fluctuations that might occur in a marginal design when the oil level drops significantly.

The capability to operate without oil for a significant period, even at reduced thrust, would prevent most oil depletion aircraft losses.

4.7.8.1.3 Oil flow interruption/depletion The requirement of 3.7.8.1.3 shall be evaluated by test.

EVALUATION RATIONALE (4.7.8.1.3)

The tolerance of the engine to oil flow interruption must be evaluated by test.

EVALUATION GUIDANCE

Background:

Since thrust bearings tend to be most sensitive to oil–out, rig testing on one or both thrust bearings to demonstrate multiple interruption capability should be conducted.

EVALUATION LESSONS LEARNED

Experience has shown that sudden rotor seizure due to lack of lubrication can be violent enough to break the engine out of its mounts and cause significant damage to aircraft/test cells.

3.7.8.2 Lubrication system components and features

4.7.8.2 Lubrication system components and features

3.7.8.2.1 Oil reservoir Unvented oil reservoirs shall meet the pressure vessel requirements of 3.4. Deaeration features shall be specified herein.

The capacity of the oil reservoir shall be:

Usable Oil	_____liters	(_____Gallons)
Gulping Volume Oil	_____liters	(_____Gallons)
Unusable Oil	_____liters	(_____Gallons)
Expansion Space (equivalent oil)	_____liters	(_____Gallons)
Total Enclosed Reservoir Volume (equivalent oil)	_____liters	(_____Gallons)

REQUIREMENT RATIONALE (3.7.8.2.1)

Structural integrity of the oil reservoir is mandatory to avoid the fire hazards which can accrue from rupture or leakage thereof and to avoid oil depletion. Deaeration and oil baffling is required to minimize foaming and provide proper inlet flow.

The usable oil supply must be compatible with the oil consumption rate and the aircraft mission to ensure an adequate supply of oil.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The contractor should provide all the required oil reservoir capacities. Usable oil quantity should be a minimum of the longest aircraft mission at the maximum hourly oil consumption specified in 3.7.8.1.2.

Background:

Engine mounted oil reservoir(s) and mountings should be constructed of corrosion resistant material capable of withstanding, without permanent deformation or fatigue damage, the stresses imposed by reservoir pressurization, engine vibration, landing loads, and cyclic stresses imposed by variations in ambient pressure. The oil reservoir should withstand a differential proof pressure of 100 kPa (15 psi)(pressurized systems) positive and negative or twice the maximum reservoir operating differential pressure within the operating envelope, whichever is greater, without visible leakage or deformation.

The reservoir should satisfactorily withstand 10,000 cyclic pressure reversals at the maximum reservoir operating pressure without visible leakage or permanent deformation. The reservoir internal design should be suitably baffled or use other means to prevent oil sloshing, should contain provisions to facilitate deaeration of return oil, and should include an inverted breathing system to ensure satisfactory oil system operation for applications where inverted flight will be encountered.

High reservoir capacity should be considered for aircraft with long range, high sortie mission requirements and/or sealed installations. A low–level warning signal should be provided to the aircraft when there are two hours of usable oil left.

Expansion space above the maximum oil level is required to provide for thermal expansion, deaeration, foam buildup in the tank, and a free air space at the tank vent outlet to prevent oil spewing and tank over–pressurization.

Total capacity relates not only to the physical size of the reservoir, but also the quantity of oil available for removing heat. A small capacity reservoir causes the oil to be worked harder since for a given flow rate, the bulk quantity of oil sees more passes per minute through the lube system circuit. The usable and unusable quantities determine the true operating volume of oil and the minimum quantity required to sustain oil pressure. The gulping volume shows how much of the initial capacity will be pumped into the system on start–up.

REQUIREMENT LESSONS LEARNED

Several engines have had oil tank cracking problems. Inverted flight has caused oil tank over–pressurization. Low altitude, high Mach, and inverted (negative g) operation has been found to be the most critical condition for over–pressurizing oil tanks in certain engine designs. Redesign has been required even when high altitude inverted flight had been demonstrated. Internal oil tank tubing and baffling have failed due to resonant vibrations in the engine vibration environment.

Inadequate oil tank capacity has resulted in mission aborts and in–flight shutdowns. The usable oil quantity can be enhanced by proper design and test of the deaeration system in the reservoir.

Usable oil quantity equal to a minimum of 12 times the maximum hourly oil consumption was provided so that the aircraft could fly two 6–hour missions without replenishing the oil supply.

4.7.8.2.1 Oil reservoir The requirements of 3.7.8.2.1 shall be evaluated by analysis, inspection, and test.

EVALUATION RATIONALE (4.7.8.2.1)

The structural integrity and performance of the oil reservoir must be demonstrated.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

The filler cap and other fittings shall be installed, the tank mounted in a manner similar to that as found on the engine, and the entire test assembly shall be subjected to the following tests:

a. Cyclic Fatigue Test. The oil tank shall be cycled between the minimum and maximum differential pressure limits up to the design service life of 3.4. The differential pressure to be used in the cyclic fatigue test shall be the absolute value of the difference between external and internal pressure of the oil tank. The reservoir temperature shall be kept at the maximum oil operating temperature for a length of time in accordance with the missions and environments of the specification.

Throughout this cycling, no leakage or permanent deformation of the oil reservoir, filler cap, or fittings shall occur.

b. Proof Pressure Test. Upon successful completion of the cyclic fatigue test, the same oil reservoir assembly shall be subjected to a proof pressure test. The proof pressure shall be held for a minimum of six minutes with the oil reservoir at the maximum oil operating temperature.

No leakage or permanent deformation of the reservoir, filler cap, or fittings shall occur.

c. Valve Tests. If the oil reservoir assembly incorporates a pressurizing valve or pressure relief valve, the assembly shall be tested to demonstrate proper functioning.

Background:

The oil tank capacity requirements should be evaluated through bench testing prior to engine test. Demonstration should be accomplished by determining the quantity of oil that is required to fill the oil reservoir when it contains only the minimum amount of oil necessary to provide oil to the pump at the maximum operating speed of the engine (i.e., unusable oil plus the gulping volume specified.) This quantity of oil must be at least equal to the amount specified as usable oil.

EVALUATION LESSONS LEARNED

None.

3.7.8.2.1.1 Oil reservoir external features The oil reservoir shall contain features to determine the oil level and to service and drain the reservoir for attitudes of 3.2.2.2. An oil quantity signal __(a)__ not be provided to the airframe. The reservoir shall have a __(b)__ system and shall have features for overfill protection.

REQUIREMENT RATIONALE (3.7.8.2.1.1)

Oil level indicators are required to facilitate proper oil servicing. Oil quantity indicators may be required to warn the pilot of impending oil depletion.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor should put “shall” or “shall not” in the blank.

(b): The contractor should put “pressure” or “gravity fill” in the blank.

Background:

Oil servicing should be possible with the engine at attitudes ranging from 15 degrees nose–up to 20 degrees nose–down.

An indicating device should be provided on or in the oil reservoir to notify the maintenance personnel when the tank needs servicing. Graduated markings should be clearly legible and should be spaced such that overfill conditions can be easily recognized. Airframe/engine manufacturer coordination should be conducted to ensure oil checking/servicing can be performed easily. Sight gage glasses should be removable for cleaning of the glass and background. Consideration should be given to chemical effects which may cloud the glass or darken the background. A high contrast background is desirable.

A continuous oil quantity gaging system should be provided, incorporating provisions to interface with a quantitative cockpit readout instrument. A signal of the oil quantity should be provided to the airframe to give the pilot an indication of his oil loss rate; however, reliability and maintenance requirements should be given special attention if a gage is required.

The oil system servicing concept should be considered. Oil port caps should be of the quick–disconnect type with positive closure feature and should be oversized to permit operation with gloved hands. Caps should be attached to the fill port by a plastic covered steel cable. Gravity–fill filler caps should conform to MIL–C–38373. The caps and adapter should be such that water can not collect and drain into the reservoir.

The reservoir should be designed to prevent filling the expansion space when servicing. Servicing capability should be designed to prevent oil overflow of the reservoir from spilling on engine or aircraft components. The pressure fill connector should be in accordance with MS24476–1 and the overflow return connection in accordance with MS24476–2. A convenient method should be provided for servicing the oil reservoir for those aircraft installations where the reservoir is not readily accessible.

A conveniently accessible self–closing drain valve should be provided for draining the oil reservoir.

REQUIREMENT LESSONS LEARNED

Considerable difficulty has been encountered in attempts to devise a remote reading oil quantity system for gas turbine engines. This was largely due to the relatively small quantities of oil used and the short dwell times. Nucleonic continuous reading systems have been incorporated but difficulties have been encountered. Overfill scenarios can present potentially serious hazards. One such occurrence resulted in an in–flight lube pump failure which caused the engine rotor to seize forcing the pilot to make an unpowered emergency landing.

4.7.8.2.1.1 Oil reservoir external features The requirements of 3.7.8.2.1.1 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.7.8.2.1.1)

The oil reservoir external features must be evaluated to insure compliance with the lubrication system requirements.

EVALUATION GUIDANCE

Background:

The oil reservoir external features should be examined and operated to demonstrate their capability to display the quantity of oil in the tank, to service, and to drain the reservoir.

EVALUATION LESSONS LEARNED

None.

3.7.8.2.2 Oil drains Drain ports shall be provided at appropriate low points in the oil system for draining the oil and taking oil samples while the engine is installed in the aircraft.

REQUIREMENT RATIONALE (3.7.8.2.2)

Drains for completely draining the lubrication system must be provided so that the engine can be effectively and easily maintained.

REQUIREMENT GUIDANCE

Background:

If spectrometric oil analysis is to be part of the engine system maintenance plan, then a readily accessible, self–closing drain valve should be provided at an optimum location in the oil system for obtaining representative oil samples for analysis. The oil samples used for the oil spectrometric analysis must be free of contamination from other fluids or dirt. Since maintenance technicians must perform oil sample tests on a frequent basis, the design of the oil drain should be such that operator safety is ensured while also allowing ease of operation of the oil drain valve. The oil drain valve should be constructed to prevent accidental opening during routine maintenance operations or during severe engine vibration cycles. One oil drain should be placed in the lowest practical point in the oil system to ensure all oil is removed from the engine and sump during periodic maintenance oil changes. Complete drainage must be possible during varying climatic conditions as specified in MIL–O–19838.

The engine should be designed so that repair and maintenance can be done in the vertical position without oil spillage or leakage.

REQUIREMENT LESSONS LEARNED

Experience has shown that the oil spectrometric analysis is one of the best indicators of turbine engine condition. Some lubricating systems have incorporated oil sample drains that were inconvenient to use or had limited accessibility, preventing operating personnel from obtaining oil samples on a scheduled basis. Other oil sample drain valves have been located in such a manner that the samples contained an unrealistic quantity of sludge and contamination, thereby misrepresenting the actual condition of the engine. Self–locking drain valves reduced the probability of maintenance errors.

Spectrometric analysis has been ineffectual with fine filtration and has not detected imminent failures. Spectrometric analysis has been useful in development testing and as a post–qualification activity to determine oil sampling intervals in service.

4.7.8.2.2 Oil drains The requirements of 3.7.8.2.2 shall be evaluated by demonstration and inspection.

EVALUATION RATIONALE (4.7.8.2.2)

The adequacy of the oil system drains and the ease of servicing must be demonstrated.

EVALUATION GUIDANCE

Demonstration should be accomplished by inspection of the qualification engines and by the maintainability demonstration of 4.5.2.

EVALUATION LESSONS LEARNED

None.

3.7.8.2.3 Oil filters Oil filter type(s), micron size, and capacity shall be specified herein. Primary filter assemblies shall incorporate a pressure relief bypass, an impending bypass indicator, and a bypass indicator. The indications shall remain visible until manually reset. The time from impending bypass to bypass shall be specified herein.

REQUIREMENT RATIONALE (3.7.8.2.3)

Adequate oil filtration must be provided to reduce the wear of oil wetted parts of the engine. Filter operation must continue under conditions of severe wear, partial failure, and progressive failure.

REQUIREMENT GUIDANCE

The following should be transferred verbatim into the specification paragraph:

The oil filter impending bypass indicator shall be manually reset, externally, without disassembly of the filter.

Background:

The time from impending bypass to bypass should be sufficient to allow completion of two combat missions.

Oil filtration should be provided in the oil system in both the pressure and scavenge portions. The filter elements should be accessible and easily removed and the filter housing should have provisions for draining the element prior to its removal and a shutoff valve to prevent system drainage after its removal.

The filter assembly should incorporate a pressure relief bypass and be of a design which will prevent the discharge of filter contaminant throughout bypass. The filter assembly should provide visual indication of both impending and actual bypass, initiated. Manual indications should remain visible until manually reset. The manual reset requirement is necessary to prevent maintenance personnel from resetting the indicator without removing, cleaning, or replacing the filter element, which would defeat the purpose of the warning indicator. The impending bypass differential pressure level should be such that an impending bypass is detected through pre– and post–flight inspection thereby minimizing the likelihood of operating in bypass. Pressure surges should not cause false activation of the indicators. The filter assembly should also provide a signal for remote indication of impending bypass.

Scavenge side filtration may be accomplished by strainers or screens, preferably located at the outlet from each bearing compartment. Each strainer/screen should provide sufficient capacity to collect anticipated debris and to prevent clogging due to buildup of debris.

Additional design and installation requirements which should be considered for oil filters are as follow:

a. Last chance filters/screens to the individual main bearing oil jets should be accessible from outside the engine for removal/inspection/cleaning with no other disassembly required.

b. Lube flow filter areas should be established consistent with service periods. Reusable filters on the engine must be proven cleanable and inspectable with reasonably available equipment such as conventional solvents, caustics and brushes. It is recommended that the filter be located immediately downstream of the pressure pump discharge.

c. Filtering of the oil should be accomplished by external to internal flows to permit more rapid identification of contamination during visual inspection of the filter.

d. On filters having a bowl installed with a specific torque, the value should be permanently marked on the bowl.

e. “Slave filtration” is accomplished by either a cleanable permanent type filter or an external filter assembly which by–passes the engine filter. The “slave filtration” is used only on initial run or after maintenance to capture manufacturing or maintenance generated debris prior to engine use in the fleet. Slave filters should be clearly distinguishable from the engine filter.

Experience has shown that sequence filtration systems with a high efficiency three micron scavenge filter provides excellent protection and virtually eliminates bearing wear during normal operation. Placement in the scavenge system, upstream of the oil cooler, maximizes life due to higher oil temperature and consequent lower pressure. A coarse, cleanable element used in the lube circuit prevents contamination introduction from external sources from entering the engine.

REQUIREMENT LESSONS LEARNED

Engines have used a wide variation of filtration systems which was a cause for many maintenance and logistic problems. Some engines did not have a filter or a screen at the scavenge portion of the oil system, which allowed broken bearing tabs, seals, and washers to damage the oil scavenge pump.

4.7.8.2.3 Oil filters The requirements of 3.7.8.2.3 shall be evaluated by demonstration and inspection.

EVALUATION RATIONALE (4.7.8.2.3)

The suitability of the oil filters to meet the engine lubrication system requirements must be tested and demonstrated.

EVALUATION GUIDANCE

Background:

Oil filter performance should be evaluated by bench test prior to engine test. Engine filter elements should be inspected during engine tests to ensure that the oil filter configuration, filtration capacity, and servicing features are in accordance with the specification.

EVALUATION LESSONS LEARNED

None.

3.7.8.2.4 Oil debris monitor Oil debris monitors shall be installed in locations where wear particles in the oil would most likely be detected.

REQUIREMENT RATIONALE (3.7.8.2.4)

Oil debris monitors placed within the oil–wetted portions of the engine facilitate detection of impending failures of bearings, gears, pumps, and associated parts.

REQUIREMENT GUIDANCE

Background:

Oil debris monitors strategically located throughout the lubrication system serve as diagnostic maintenance aids to permit early detection of impending failures of oil–wetted parts.

The master chip detector should be exposed to the entire scavenge oil flow, along with magnetic plugs in individual scavenge lines for fault isolation.

Oil debris monitors should be designed for easy access by maintenance personnel and should be of a self–closing design with an electrical continuity check–out feature.

Allowance should be made for either visual (i.e., remove and check) or indicating (via immediate access or remote indicator) monitors. Development of reliable remote indicators should be encouraged. These monitors should be designed to minimize indications caused by normal wear particles (meaningless fuzz) and should be designed to provide a high probability of indicating impending engine failures.

REQUIREMENT LESSONS LEARNED

Oil debris monitors have been clearly responsible for detecting engine failures that were not detectable by the spectrometric analysis. Monitors are frequently the sole means of identifying particles too large to be held in suspension in the oil.

The self–closing feature permits removal of the monitor without loss of oil. The absence of a cockpit warning light in some aircraft is the result of many “false alarm” warnings caused by metal “fuzz” on the detector, resulting in mission aborts. For this reason development of improved monitors, such as the “burn off” type, is continuing.

4.7.8.2.4 Oil debris monitor The requirements of 3.7.8.2.4 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.8.2.4)

The capabilities of the oil debris monitor must be demonstrated to show that early detection of impending failures of oil wetted components is possible.

EVALUATION GUIDANCE

Background:

Oil debris monitor performance should be evaluated by bench test prior to engine test. Demonstration of the oil debris monitor capability should be accomplished by accumulating oil samples and periodically inspecting oil debris monitor throughout the qualification testing. Details on the nature of the deposits and the amount and rate of accumulation of deposits on the oil debris monitor should be recorded and included in the test reports for use in demonstrating the adequacy of the monitor configuration and location. Monitor data should be correlated with spectrometric oil analysis, oil filter inspection, and engine teardown results.

EVALUATION LESSONS LEARNED

None.

3.7.8.2.5 Oil coolers The type and number of oil coolers used in the oil system shall be specified herein.

REQUIREMENT RATIONALE (3.7.8.2.5)

A description of the oil coolers is needed to give an indication of the systems complexity, reliability, and maintainability.

REQUIREMENT GUIDANCE

The following should be transferred verbatim into the specification paragraph: Oil coolers shall incorporate a pressure relief bypass valve and features which shall permit disassembly of the cooler for cleaning and inspection.

Background:

The oil cooler should operate above a certain level of effectiveness for the life of the part. The basic method of cooling (i.e., fuel–oil or air–oil heat exchanger) and whether the cooler is an integral part of the engine or a removable unit should be identified.

The design features of the cooler, e.g., fixed or removable matrix, brazed/welded seams, “O”–ring sealing, type of metals used, etc., should be identified. These features relate to field experience with exchangers that have experienced seam cracking and seal leakage, and with desirable features such as ease of disassembly, cleaning, and repair.

REQUIREMENT LESSONS LEARNED

Modern engines employ single piece coolers for lighter weight and reduced leakage paths.

Experience with permanently assembled compact fin–plate designs with large area inlet screens has been very positive. These permanently sealed units can be easily backflushed during maintenance actions, without disassembling the cooler. Such an approach has reduced cost, time, and logistic support.

4.7.8.2.5 Oil coolers The requirements of 3.7.8.2.5 shall be evaluated by analysis, demonstration and inspection.

EVALUATION RATIONALE (4.7.8.2.5)

A description of the oil cooler must be provided for inspection. The adequacy of the oil coolers must be evaluated by analysis and test.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

If the heat exchanger assembly incorporates a bypass valve, regulator, or indicating feature, appropriate tests shall be conducted to demonstrate proper functioning. Heat exchangers for cooling or heating of engine fluids or components shall be subjected to the following tests.

a. Flow, Pressure, and Temperature Cycling Test. The heat exchanger shall be subjected to a flow, pressure, and temperature cycling test for one design life. The number of cycles and/or hours shall be as specified in 3.4.1.2. The fluid pressure and temperature cycling shall be mission oriented and be derived from the engine mission duty cycles contained in the specification. An appropriate number of cycles shall include the maximum delta pressure that will occur between fluid passages, i.e., highest fuel pressure with lowest oil pressure and lowest fuel pressure with highest oil pressure. Ambient temperature shall be varied during the test if it is a parameter that degrades cooler life. At the completion of the test there shall be no leakage or permanent deformation.

b. Heat Exchanger Proof Pressure. Upon successful completion of the flow, pressure, and temperature cycling test, the same heat exchanger should be subjected to a proof pressure test. Each fluid side of the heat exchanger shall be individually subjected to proof pressure for at least one/two time(s) and held two minutes for each pressure application. During the application of pressure to one side the other element shall be empty and at atmospheric pressure. There shall be no evidence of leakage into the dry side. Following this test, both sides of the heat exchanger shall be simultaneously subjected to their maximum working pressures for at least one/two time(s) and held two minutes for each pressure application. At test completion there shall be no evidence of leakage or permanent deformation.

c. Fan Duct Heat Exchangers. Tests shall be performed to determine the effect of fan duct heat exchangers on duct pressure drop, redistribution of airflows, and non–uniformity of duct exit temperature.

Background:

The analysis of the oil cooler should be in the form of a “Oil Flow and Heat Rejection” report requested by a CDRL.

The instrumentation requirements and sensor locations for evaluation of the oil system performance should be identified in the test plan by the engine contractor.

Analysis and demonstration should not only show that the heat exchanger provides adequate cooling of the oil but also point to any localized maximum oil temperatures and bearing temperatures in critical areas. The engine manufacturer should provide the heat rejected to the oil which should include the extent to which the oil is being “worked” and the potential for oil degradation. Oil flow should include flow rates to critical bearings and gears to assure that heat isn’t added from viscous churning of excessive oil. The oil flow and heat rejection data should represent the most adverse environment the engine oil system would encounter in service.

Heat exchangers located in the fan duct can cause a redistribution of airflows and warm streaks in the cooling flows for downstream components.

EVALUATION LESSONS LEARNED

In the past, some oil coolers did not perform adequately under extreme high ambient temperatures. External fuel–oil coolers were designed by airframe manufacturers using rating table SFCs in the model specification. The engines in service had much better SFCs. As a result the fuel–oil coolers were marginal on the ground in a hot environment. The pilots had to shut down the engines because hot oil limits were exceeded.

During vibration testing, an oil cooler experienced braze cracks because the test fixture was too stiff and was magnifying the vibration levels on the cooler. A mounting arrangement which closely matches the actual engine mounting configuration should be used to ensure accurate test results will be obtained.

3.7.9 Hydraulic system The hydraulic system shall be described. The hydraulic system fluid(s) shall be _(a)_. Nominal and maximum working pressure and maximum operating fluid temperature shall be _(b)_. Peak pressure (ripple or surge) shall not exceed _(c)_ percent of maximum system working pressure. The allowable pressure ripple shall be defined as less than or equal to _(d)_ Pa abs (_(d)_ psia) at _(e)_ hertz. The hydraulic system shall withstand a proof pressure of _(f)_ times the maximum working pressure. The total volume and the usable quantity of hydraulic fluid in the reservoir shall be _(g)_. Hydraulic systems which are integral to the fuel system shall meet the requirements of 3.7.3.

REQUIREMENT RATIONALE (3.7.9)

Hydraulic system performance and limits must be specified for safe system operation. Hydraulic fluid system operation must not create resonances in the tubing. A description of the hydraulic system must be provided for proper evaluation of the physical and functional characteristics.

Information on the hydraulic fluid reservoir is needed for evaluation of the design.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): The contractor should provide a fluid type that conforms to existing military specifications.

(b) – (e): The contractor should provide the values.

(f): A value of at least 1.5 to 2.0

(g): The contractor should provide both values.

Background:

The hydraulic system should be able to withstand operational stresses and vibration without failure and there should be no permanent deformation, collapse, or seam cracking when exposed to these stresses and vibrations.

System pumps should utilize a pressure regulating device and an independent means of limiting excessive pressure.

Designing the system with an integral unloading feature prevents heat build up in the hydraulic pump and reduces the power required by the system from the engine.

The hydraulic system should utilize hydraulic fluid (e.g. MIL–H–83282), engine fuel, or oil.

REQUIREMENT LESSONS LEARNED

None.

4.7.9 Hydraulic system The requirements of 3.7.9 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.9)

The physical and functional characteristics of the hydraulic system must be evaluated to assure compatibility with the engine and its application. Hydraulic system performance and limits must be evaluated.

EVALUATION GUIDANCE

The following should be transferred verbatim in the specification paragraph:

The hydraulic system testing may be conducted either in a complete system rig test or as installed on an engine substantially identical to the endurance test engine. The hydraulic system high pressure side shall be subjected to proof pressure for one/two successive times and held two minutes for each pressure application. Hydraulic test fluid shall be maintained at maximum system working temperature. The equipment shall be operated in its normal function between applications of the test pressure. At test completion, there shall be no leakage, excessive distortion or permanent set. Components which require varying test pressures in different elements may have these pressures applied either separately or simultaneously.

All physical features of the hydraulic system shall be examined and compared to the specification and installation drawing.

Functional characteristics shall be evaluated by demonstrations and tests. Test cycles shall be consistent with the environment and usage specified in 3.4.2. Tests shall be accomplished at both sea level and simulated altitude test conditions. Maximum loads on the system shall be based on an analysis of flight conditions. For further guidance, see MIL–H–87227.

The hydraulic fluid reservoir shall be subjected to a cyclic fatigue test.

EVALUATION LESSONS LEARNED

Engine testing at high Mach number, low altitude conditions frequently gave maximum loads on hydraulically actuated exhaust nozzles.

3.7.9.1 Hydraulic ground test provisions Ground test provisions shall be provided. Fluid leakage shall not occur during ground test. Self–sealing ground test couplings shall be provided with the hydraulic system. These couplings shall be secured with sealing caps when not used.

REQUIREMENT RATIONALE (3.7.9.1)

Ground test provisions are needed for easier troubleshooting capabilities.

REQUIREMENT GUIDANCE

Background:

Ground test points should be easily accessible and compatible with ground support test equipment. Ground test provisions should be shown on the installation drawing.

REQUIREMENT LESSONS LEARNED

Sealing caps have been required to prevent spillage of flammable fluid when self–sealing couplings were not used.

4.7.9.1 Hydraulic ground test provisions The requirements of 3.7.9.1 shall be evaluated by analysis, demonstration, inspection, and test.

EVALUATION RATIONALE (4.7.9.1)

Hydraulic test provisions must be evaluated for accessibility and sealing capabilities.

EVALUATION GUIDANCE

Ground test equipment compatibility should be tested concurrent with the maintainability demonstration. Ground test points should be easily accessible and compatible with ground support test equipment.

EVALUATION LESSONS LEARNED

None.

3.7.9.2 Hydraulic system fire and safety hazards The hydraulic system design shall minimize fire hazards caused by proximity of system plumbing and components to heat sources and electrical equipment. All hydraulic operated services which are essential to safe engine operation shall have redundancy or shall have provisions for emergency activation.

REQUIREMENT RATIONALE (3.7.9.2)

Since the hydraulic system contains a flammable fluid, there is a need to emphasize safety and to prevent fire hazards.

REQUIREMENT GUIDANCE

Background:

The system must be designed to prevent leakage from a removable joint since this could result in a fire. Redundancy is provided for safety of flight.

REQUIREMENT LESSONS LEARNED

Aircraft fires have resulted from leakage at hydraulic fluid system joints and faulty maintenance. Aircraft have been lost in spite of redundancy or emergency activation.

4.7.9.2 Hydraulic system fire and safety hazards The requirements of 3.7.9.2 shall be evaluated by analysis, demonstration, inspection, and test.

EVALUATION RATIONALE (4.7.9.2)

The hydraulic system design must be evaluated to assure it incorporates the necessary fire and safety provisions.

EVALUATION GUIDANCE

Background:

The evaluation should include a fire test of lines, fittings, and components. AS 1055 also specifies evaluation at the worst possible combination of fluid parameters encountered throughout the environmental conditions and operating envelope of the engine.

All hydraulic fluid leaks should be logged in the test log for all engine tests. Also, note maintenance needed to correct any leaks.

Provisions for emergency activation should be full–scale sea level and altitude engine testing.

EVALUATION LESSONS LEARNED

None.

3.7.9.3 Hydraulic system air removal Entrapped air shall not cause sustained loss of system pressure or degradation of system operational performance. There shall be provisions for bleeding air from the system at critical points for maintenance purpose.

REQUIREMENT RATIONALE (3.7.9.3)

Hydraulic system entrapped air must not affect the system’s operating characteristics.

REQUIREMENT GUIDANCE

Background:

Disconnection of lines or loosening of fittings is not an acceptable method for bleeding air because it is impossible to control or direct the bleed fluid. Reconnection of lines often causes entrapment of air into the system. The system must be designed to automatically scavenge free air from the system in order to minimize the need for manual bleeding.

Critical point for bleeding air from the system includes fluid filter locations because these filters are replaced during regularly scheduled maintenance.

REQUIREMENT LESSONS LEARNED

In the past, engines that tried to save weight by eliminating air removal devices usually required the devices once the engine was put into service.

4.7.9.3 Hydraulic system air removal The requirements of 3.7.9.3 shall be evaluated by analysis, demonstration and test.

EVALUATION RATIONALE (4.7.9.3)

The ability of the hydraulic system to remove entrapped air must be evaluated.

EVALUATION GUIDANCE

Background:

Entrapped air removal should be evaluated in a complete rig test. Engine testing should also be used to evaluate hydraulic system functional characteristics both at sea level and simulated altitude test conditions.

EVALUATION LESSONS LEARNED

None.

3.7.9.4 Hydraulic fluid filters Filter(s), type(s), location(s), and filtration rating(s) in microns (nominal and absolute particle sizes) shall be specified herein. Each filter element shall be accessible and easily removed. Each filter housing shall incorporate an automatic shutoff device to prevent fluid drainage during filter element removal. Each filter assembly shall supply indication of impending bypass and bypass. Each indication shall remain visible until manually reset. Whenever activation of the indicator is a result of the exceedance of the differential pressure limit, the indicator shall remain in the fault mode until removal, cleaning, or replacement of the filter element is accomplished.

Filters for the hydraulic system shall meet the following requirements:

Supply line

a. Filters shall be located such that all fluid from the engine pumps and Ground Test Equipment (GTE) pressure connections will be filtered prior to entering components of the system.

Return line

b. Filters shall be located such that all fluid will be circulated through the filter prior to entering the return line relief valve, pumps, or reservoir.

Other

c. Additional filters at critical component locations shall be provided where filter protection is necessary.

REQUIREMENT RATIONALE (3.7.9.4)

Filters are necessary to protect critical components such as hydraulic pumps, seal mechanisms, and valves from particulate contamination.

REQUIREMENT GUIDANCE

The following should be transferred verbatim to the specification paragraph:

The hydraulic filter impending bypass indicator shall be manually reset, externally, without disassembly of the filter.

Background:

The filter housings should have an automatic shutoff device because spilled fluid can be a fire hazard and maintenance nuisance.

Filters are normally located in the supply and return lines because in these locations the flow is reduced to a single line and the necessary filtering can be easily accomplished before the fluid reaches critical components in the system. Filters should be made of a material resistant to media migration.

Most filter assemblies incorporate an integral differential pressure activated pop–out device that will give visual warning by raising a red indicator when the differential pressure across the element exceeds a specified value. Once activated, the red indicator should remain extended until manually reset internally after filter removal.

The manual reset requirement is necessary to allow maintenance personnel to manually reset the indicator without removing, cleaning, or replacing the filter element, when activation of the indicator resulted from a pressure surge and not an exceedance of the differential pressure limit across the filter element.

REQUIREMENT LESSONS LEARNED

Pressure ripples occurring during field operation have caused false activation of indicators.

4.7.9.4 Hydraulic fluid filters The requirements of 3.7.9.4 shall be evaluated by analysis, demonstration, inspection, and test.

EVALUATION RATIONALE (4.7.9.4)

The physical and functional characteristics of the fluid filters must be evaluated to assure compatibility with the hydraulic system and its application.

EVALUATION GUIDANCE

Background:

Hydraulic fluid filters should be evaluated during all engine tests at sea level and altitude conditions. Maintainability features of the filters should be evaluated during the maintainability demonstration of 4.5.2.

EVALUATION LESSONS LEARNED

None.

3.7.10 Pneumatic system The functional and physical characteristics of the pneumatic system shall be described.

REQUIREMENT RATIONALE (3.7.10)

A description of the pneumatic system must be provided for proper evaluation of the physical and functional characteristics.

REQUIREMENT GUIDANCE

Background:

When a pneumatic system is utilized, the complete system should be furnished as part of the engine.

The pneumatic system must be a part of the engine so that the engine does not rely on aircraft systems. Pressures, temperatures, and airflows should be defined.

REQUIREMENT LESSONS LEARNED

None.

4.7.10 Pneumatic system The requirements of 3.7.10 shall be evaluated by analysis, demonstration, inspection, and test.

EVALUATION RATIONALE (4.7.10)

The physical and functional characteristics of the pneumatic system must be evaluated to assure compatibility with the engine and application.

EVALUATION GUIDANCE

Background:

All physical features of the pneumatic system should be examined and compared to the specification and installation drawing. Functional characteristics should be evaluated by demonstration and test.

EVALUATION LESSONS LEARNED

None.

3.7.10.1 Pneumatic air contamination Pneumatic system components requiring fan or compressor air shall operate satisfactorily with the inlet air contaminated as defined in 3.3.2.4.

REQUIREMENT RATIONALE (3.7.10.1)

Pneumatic system components must be able to operate when subjected to contaminated air.

REQUIREMENT GUIDANCE

Background:

Air filters may be used to meet this requirement.

REQUIREMENT LESSONS LEARNED

None.

4.7.10.1 Pneumatic air contamination The requirement of 3.7.10.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.7.10.1)

The pneumatic system must be evaluated to ensure satisfactory operation in a contaminated environment.

EVALUATION GUIDANCE

Background:

The pneumatic system components should be evaluated with the same contamination that is required for evaluation of the fan and compressor. The pneumatic system should be evaluated during the sand ingestion test of 4.3.2.4.

EVALUATION LESSONS LEARNED

None.

3.7.11 Starting The engine shall satisfactorily make ground and air starts, and shall restart under the conditions, limitations, and procedures, specified in this section. The engine shall start with inlet air at any temperature between and including the extreme hot and cold atmospheres of figure 4, with fuel inlet temperatures between and including the maximum fuel temperature specified in 3.7.3.1 and the minimum inlet fuel temperature either equal to the cold atmospheric temperature of figure 4, or that temperature which corresponds to a fuel viscosity of 12 centistoke, whichever is warmer. Starts shall be made with the primary fuel of 3.7.3.1.1 and oil(s) of 3.7.8.1, unless specified otherwise. A start shall be considered satisfactory when the engine rotor is accelerated from either rest or windmilling speed to Idle, when the minimum assist torque is provided by a starter or ram air, as specified in 3.1.1.19.1, and the engine stays within the engine starting and operating limits of figure 9.

REQUIREMENT RATIONALE (3.7.11)

The conditions under which an engine can be successfully started and the definition of a successful start need to be specified.

REQUIREMENTS GUIDANCE

Background:

It has been recognized that with a starter and starting system of constant capability, start times will be longer at both high and low inlet air temperatures than at standard day temperature. Low temperature starts take longer because of the increased viscosity of the oil and high temperature starts take longer due to the high temperature/low density air. The engine contractor should specify the effect of degraded equipment. The starts may be acceptable on a new engine, but may not be acceptable on a time worn service engine, or if a poorly performing APU, battery, or starter is used.

REQUIREMENTS LESSONS LEARNED

None.

4.7.11 Starting The requirements of 3.7.11 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.11)

The ability to start the engine using the specified procedure with minimum starter torque under various temperature and load conditions must be demonstrated.

EVALUATION GUIDANCE

Background:

Adequate testing should be accomplished to show that the starting system can meet all starting requirements. As a minimum, start testing should be conducted at the cold, standard, and hot ambient temperatures specified for both the ground and air start environments. Start system performance is highly dependent upon aircraft accessory drive loads and power extraction, and these loads must be simulated or analytically accounted for in all engine testing. Sufficient data that adequately defines both the fired and unfired torque characteristics of the engine as a function of ambient temperature should be collected. Analysis can then be made to determine the starting characteristics of the engine at conditions other than those that were tested.

EVALUATION LESSONS LEARNED

Previous altitude and ground starting tests did not use any load or power extraction. The aircraft contractor criticized the testing since it did not give realistic results. Analytically accounting for the power extraction does not give the confidence of an actual test. Starting information is included in the pilot's handbook and should be as accurate as possible.

3.7.11.1 Ground starts The engine shall start at the specified ambient temperatures and altitudes, in times equal to or less than those shown on figure 26a. The engine shall be capable of an unlimited number of consecutive start cycles, with a maximum time interval of (a) seconds between the completion of one cycle and the initiation of the next cycle. The wind condition envelopes for ground starting shall be shown on figure 26b. Additionally, special starting requirements apply as follows:

1. Hot temperature soak start. The engine shall start, after a (b) hour hot soak at the temperatures specified on figure 4, using (c) fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperatures shown on figure 4. Following shutdown, the engine shall start, after a resoak period of (d) minutes at the temperatures specified on figure 4, using (c) fuel, and with inlet air, fuel, and oil at the maximum hot ambient temperature shown on figure 4.

2. Cold temperature soak start. The engine shall start, using (e) fuel and (f) oil, after a cold soak for a period of (g) hours with inlet air, fuel, and oil at the minimum cold ambient temperature shown on figure 4.

REQUIREMENT RATIONALE (3.7.11.1)

Worldwide, all–weather operation requires the use of specified fuels and oil under extreme hot and cold ambient temperature conditions.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph:

(a): A value between 0 and 60 seconds.

(b): 10 hours or the time required to thermally stabilize the engine and its contained operating fluids.

(c): The fuel used should be JP–4, MIL–T–5624, or worst case primary fuel.

(d): A value between 15 and 30 minutes.

(e): The fuel used should be JP–5, MIL–T–5624, or worst case primary fuel.

(f): The oil should be either MIL–L–23699 for maximum cold ambient temperature at –40°C and above; and MIL–L–7808 for below –40°C.

(g): 10 hours or the time required to thermally stabilize the engine and its contained operating fluids.

Background:

Figure 26a should be constructed by the Using Service and should be consistent with the ambient temperature requirements of the engine as required by 3.2.2.1 and figure 4. The Using Service should indicate on figure 26a the maximum hot and cold ambient temperatures (consistent with figure 4) and indicate the hot soak, hot resoak, and cold soak temperatures vs. times vs. altitudes.

The contractor should provide curves on figure 26a specifying the engine starting times for the complete ambient temperature range at incremental altitudes up to the maximum ground starting altitude of 3.7.11.3. The Using Service should be aware that according to SAE AS 1606, starters should be capable of at least five consecutive start cycles or attempts. However, the engine should not be limited in the number of consecutive starts.

REQUIREMENT LESSONS LEARNED

Tail winds have adversely affected the starting performance of engines, particularly the high bypass ratio turbofans. Since cold weather conditions can affect compression/combustion interaction, the cold soak requirement has proven difficult for engine contractors to meet. Combustor stability and Sub–Idle (start) stall margin have seen deficiencies that have led to operational starting problems. During a rapid restart attempt, the unburned fuel can collect and cause torching and excess turbine temperatures.

4.7.11.1 Ground starts The requirements of 3.7.11.1 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.7.11.1)

Testing is the only suitable means of demonstrating that the engine is capable of satisfactorily operating under the environmental extremes specified.

EVALUATION GUIDANCE

Background:

The high/low temperature testing should be scheduled near the end of the qualification program so that the engine configuration tested is as close to the final production configuration as possible. Climatic testing is best accomplished in an environmental facility because it is not subject to the variability of the natural environment.

EVALUATION LESSONS LEARNED

High/low temperature testing has identified shortcomings in designs that were not possible to find through analyses or were overlooked in the design process. For example, O–rings were found to leak badly at low temperatures because they lost their resiliency and elasticity. Gearboxes have contracted at a faster rate than the gears therein, causing severe pitch line interference and almost immediate bearing failure. The initial start at an extreme low temperature will cause the oil pressure to “spike”. This rapid increase in pressure has occasionally ruptured the filter casing and cracked oil tubes.

3.7.11.2 Air starts Windmilling, spooldown, and starter assisted airstart envelopes and maximum start times, with customer extractions shall be defined on figure(s) 27. Air starts within figure(s) 27 shall be considered acceptable when the engine achieves Idle speed within the air start times specified and no starting or operating limits are exceeded. The engine shall be capable of consecutive start cycles, with a maximum time interval of (a) seconds between the completion of one cycle and the initiation of the next cycle. Starting time shall be measured from the initiation of the starting sequence to the attainment of (b) percent of stabilized uncorrected engine Idle rotor speed. During air starts the engine rotor load shall be at least (c) percent of the total maximum accessory drive load specified for the drives shown in table V.

For turboprop engines, the engine shall start at the post loiter altitude start envelope of figure 10 after being inoperative for at least (d) hours.

REQUIREMENT RATIONALE (3.7.11.2)

The engine starting characteristics must be sufficiently defined so that an adequate starting system can be provided. Restart time(s) are needed to meet operational requirements.

REQUIREMENT RATIONALE (3.7.11.2)

The engine starting characteristics must be sufficiently defined so that an adequate starting system can be provided. Restart time(s) are needed to meet operational requirements.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value from 0 to 60 seconds.

(b): A value from 95% to 100%.

(c): The contractor should fill in the blank. The recommended value is from 5 to 10 percent.

(d): The contractor should specify the loiter period.

Background:

Air starting is accomplished in flight either by windmill, spooldown, and/or starter assist. Figure 27 should be formatted by the Using Service and filled in by the contractor, and should include altitude, attitude, ambient temperature, bleed air, power extraction, turbine temperature limit, and start time for air starts. The ambient temperatures in the figure should include the temperature extremes shown on figure 4. Aircraft accessory drive loads and power extraction should be included in starting torque requirements. There may be aircraft electrical loads imposed during an air start attempt that the pilot may be unable to uncouple from the engine that would hinder an engine start. Torque corrections for altitude should be defined on a figure. The calculation methods of AIR 1087 should be used to calculate aircraft accessory drag torque during an engine start. For purposes of timing, the start should commence when torque is applied to the starter pad and is complete when Idle speed is reached. Voltage drops associated with electrical starts should have no effect on other engine/aircraft systems. Additional guidance is available in AIR 713, AIR 781, and ARP 949. Starts should be satisfactory from either a “cold iron” or a “hot iron” condition and from any time interval after shutdown. The engine starting envelope should be defined and consistent with the aircraft operational requirements. Spooldown start characteristics should be investigated throughout the engine speed range because experience has shown some engines are stall prone in certain rotor speed ranges during spooldown starts.

REQUIREMENT LESSONS LEARNED

Cold weather conditions can affect compression/combustion functions within the engine.

4.7.11.2 Air starts The requirements of 3.7.11.2 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.7.11.2)

Full–up engine testing must be conducted to evaluate the requirement.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.7.11.3 Starting limits The starting altitude for ground starts and air starts shall be up to (a) km, ((a) ft) and (b) km, ((b) ft), respectively. The maximum turbine temperature for ground and air starts shall be specified. Differences in the altitude starting limits for “hot” and “cold” engines shall be specified. Limits with maximum customer bleed air and maximum customer power extraction shall be specified. The conditions where engine starting is the most difficult shall be specified. For turboshaft and turboprop applications, the post loiter altitude start limit and the associated loiter period (i.e., time the engine is inoperative) shall be specified.

REQUIREMENT RATIONALE (NO TAG)

Field use of the engine requires information on the ground starting limitations at the extreme high and low temperatures specified. Information on air start limitations is also required by the pilot.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): For turbofan/turbojet/turboprop engines: 4.5 km (15,000 ft). For turboshaft engines: 5 km (16,404 ft.).

(b): For turbofan/turbojet/turboprop engines: 8 km (26247 ft).

For turboshaft applications, references to airstart and_b_should be deleted.

Background:

The contractor should provide details as required in the paragraph. A “cold” engine is defined as one which has been allowed to windmill until the low pressure turbine exit gas temperature is within 55°C (100°F) of the engine compressor inlet temperature before a start is attempted. A “hot” engine is defined as one where a start is attempted within 10 seconds after a flameout or shutdown.

The type of fuel and oil determine the extreme temperatures for starting on the ground. The use of customer bleed air and power extraction during air starts is more realistic. These will degrade the limitations for air start.

REQUIREMENT LESSONS LEARNED

The airframe contractor has wanted to see air start limitations with customer bleed and power extraction which is how the engine is used. Some engines require a starter assist during air starts.

4.7.11.3 Starting limits The requirements of 3.7.11.3 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.7.11.3)

Starting limits need to be confirmed by tests.

EVALUATION GUIDANCE

Background:

Tests should be conducted at a test facility that is capable of simulating exact conditions for air and ground starts. The facility needs a capability of operating at –60°F air temperature and the maximum sea level air flow of the engine. For air starts it must be capable of maintaining a constant pressure and temperature at the engine inlet while the air flow is increased during an engine acceleration. These tests should be conducted using the worst case fuel and oil combination (i.e., for cold temperatures JP–5 and MIL–L–7808 oil).

The test should require air starts and restarts with and without starter assist and at the worst case starting conditions.

EVALUATION LESSONS LEARNED

In the past, contractors only supplied starting limits for a “bare” engine, there was no customer bleed air or power extraction. The airframe contractors had to develop their own starting limits.

3.7.11.4 Starting procedure The starting procedure shall be simple and shall not require critical timing. The engine shall be capable of starting with the thrust/power demand at Idle or above and, after initiation of the starting sequence, being accelerated immediately to any power request. This shall be accomplished within specified engine starting limits and without stall. A description of the starting procedure shall be specified herein.

REQUIREMENT RATIONALE (3.7.11.4)

Actions required for ground and air starting must be established to ensure aircraft/engine compatibility.

REQUIREMENT GUIDANCE

Background:

The airframe specification and/or ICD should keep the starting procedure simple for the pilot. Starts should not require simultaneous manual operation procedures, actuation of switches, levers, or combinations thereof. If a special operating procedure or ground equipment is necessary to obtain satisfactory starts after normal shutdown, the requirement should be specified.

REQUIREMENT LESSONS LEARNED

Complex starting procedures have been responsible for aircraft loss and an increase in pilot workload.

4.7.11.4 Starting procedure The requirements of 3.7.11.4 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.11.4)

Starting procedures can be done only by actual demonstration and test.

EVALUATION GUIDANCE

Background:

Engine starting procedures should be defined prior to and used during engine ground starting evaluation. Ground test starting systems used for evaluation should reflect actual start systems as closely as possible to provide valid test results.

EVALUATION LESSONS LEARNED

None.

3.7.11.5 Automatic relight The engine shall incorporate an automatic rapid relight system which shall immediately re–establish engine operation (without any pilot action) after a main combustor flameout or augmentor blowout has occurred. The automatic relight system shall detect any flameout that may have occurred and initiate a sequence for automatic recovery anywhere in the automatic relight operating envelope shown on figure 10. The automatic relight system shall be described herein.

REQUIREMENT RATIONALE (3.7.11.5)

An automatic relight capability is essential in a single engine aircraft and may be desirable in other aircraft during critical phases of flight, such as takeoff. The automatic relight capability improves system operating reliability and reduces pilot workload.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: The contractor should fill in figure 10 showing the limitations of the automatic relight system in the engine operating envelope.

Background:

The contractor should provide a description that includes details of the system activation, response time, sensitivity, and the time duration in which the ignition remains on after activation.

The automatic relight system's recovery of a flameout/blowout should be based on aircraft mission requirements, and if required by the aircraft mission, the engine should incorporate one of the following automatic relight systems: 1) an engine initiated system that controls the primary ignition elements which are automatically energized when the core rotor deceleration rate exceeds a scheduled limit; 2) a continuous ignition system that provides ignition via constant ignitor firing; 3) a passive ignition system, through coating of the combustor liner, that maintains sufficient combustor temperature to allow automatic ignition.

The system should consider the rotor speed at which relight occurs to avoid stalls and stagnation from low power flameout events. Automatic relight lockout provisions may be required in stall sensitive regions of the engine speed range to prevent stalls.

The system description should be reviewed to determine if:

a. the system is sensitive enough to the measured parameters to generate the required signal before spool–down occurs.

b. the system will respond quickly enough to prevent rotor spool–down after a blowout/flameout signal is generated.

c. the ignition remains on long enough to ensure that stabilized combustion has occurred.

The automatic relight capability shown on figure 10 should be equal to or better than the airstart envelope. Continuous ignition or rapid relight ignition is useful in preventing or overcoming the flameouts that result from:

a. ingestion of exhaust products from gun or rocket firings

b. distortion of the airflow at the engine inlet face during aircraft maneuvers, particularly at high altitudes

c. ingestion of ice

d. adverse weather conditions

REQUIREMENT LESSONS LEARNED

None.

4.7.11.5 Automatic relight The requirements of 3.7.11.5 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.11.5)

Automatic relighting procedures can be done only by actual demonstrations and tests.

EVALUATION GUIDANCE

Critical flight operating conditions should be determined, and tests of the automatic relight system should be conducted at these conditions. The critical flight conditions may be ice, water or bird ingestion; rocket or gun firing; or they may be the envelope extremes of the relight envelope shown on figure 10.

EVALUATION LESSONS LEARNED

None.

3.7.12 Exhaust nozzle system The exhaust nozzle physical and operational characteristics, including emergency or manual provisions, shall be described.

Any exhaust nozzle system failure shall allow safe return to base. For variable exhaust nozzles, failure of one segment of a multi–segmented nozzle shall not prevent operation of the nozzle.

Failure of any single part of the actuating mechanism on a controllable nozzle shall not cause asymmetric thrust relative to the nozzle. The operational mode, and limits resulting from any single part failure in the actuating mechanism of a variable nozzle, shall be specified herein. Provisions shall be made to actuate the nozzle for ground maintenance checkouts, rigging, and adjustment without the engine running. Life of the engine exhaust system, special components, features, materials, and applied coatings shall be specified in table XIV.

REQUIREMENT RATIONALE (3.7.12)

Exhaust nozzles have an appreciable effect on engine capability and aircraft performance. This is based on factors such as afterburner characteristics, discharge coefficients, thrust coefficients, cooling requirements, and internal drag. Life of the exhaust system, components, and features must equal or exceed engine parts life to reduce the number of maintenance actions, cost, and aircraft down time.

REQUIREMENT GUIDANCE

Background:

If the exhaust nozzle is supplied by the airframe contractor, the performance, physical characteristics, loads, and method of attachment, installation, and removal should be defined.

The description should include, if applicable: The method of nozzle actuation, type and number of actuators, control sensing parameters, provisions for nozzle position feedback, cooling provisions and method of cooling employed, fail–safe characteristics, degree of vectoring (if directionally controlled), nozzle throat and exit area/shape deformations, external geometric contour and surface anomalies which may affect aerodynamic drag or gross thrust of the nozzle, and any other features unique to the particular design, including protective and low observable coatings and features.

Variable nozzle fail–safe and manual or emergency features including actuator redundancy should be compatible with the intended application and should be described.

Mission requirements should dictate the nozzle design criteria. Data to be used in the design process should include the tradeoffs between thrust, weight, size, drag, stability, external flow field, low observables, fuel puff, ultraviolet and infrared radiation/visual signature, and cooling requirements. Nozzle design and operation should include consideration of airflow match with upstream components to ensure sufficient engine stall margin. Engine/nozzle flow matching shall consider the effects of internal cooling flows, flow area effectiveness, and nozzle throat/exit area deformation.

The life of exhaust system component, features, and coatings should be at least as high as the engine hot parts life to avoid/minimize added down time and maintenance actions. Exhausts with special technology features and coatings may need additional, even frequent, maintenance actions. The exhaust system should be repairable at “I” or “O” level. Shelf life of stored or spare exhaust systems (metallic or composite structure or parts) should exceed ten years (problem, moisture pickup). Nozzle structure, ducts, and components shall be repairable (versus “throw–away”). Exhaust system life should be achievable in all engine operating environments (e.g., acoustics) with a specified duty cycle.

REQUIREMENT LESSONS LEARNED

None.

4.7.12 Exhaust nozzle system The requirements of 3.7.12 shall be evaluated by analysis, inspection, and test. Nozzle structural integrity shall be tested during operation at maximum ram pressure conditions. Actuator scraper and seal design shall be evaluated for effectiveness and wear from engine generated debris in a heated environment.

EVALUATION RATIONALE (4.7.12)

The exhaust nozzle system must be evaluated.

EVALUATION GUIDANCE

Background:

Engine tests, to evaluate exhaust nozzle matching and functional characteristics, should be accomplished at sea level and altitude flight conditions. Engine tests at envelope extremes should be used to ensure nozzle stability and freedom from vibration. Engine testing at maximum inlet pressure is desirable to demonstrate structural capability. Component test may be used to demonstrate maximum structural loads, mounting and airframe forces and external fields. External flow field effects on the nozzle should be considered in the nozzle test program. External exhaust nozzle wear debris has caused actuator leakage due to ineffective rod scraper design. Exhaust systems with high aspect ratio or non–round shapes have nozzle area deformations which affect the engine flow match, performance, and stall margin. Multi–function exhaust systems have added weight, complexity, and wear. Exhaust systems with unconventional construction or materials need backup approaches (some nozzle approaches have been abandoned in major programs). Life and repair of special coatings and features should be demonstrated. The effectiveness and integrity of the exhaust cooling system should be demonstrated.

EVALUATION LESSONS LEARNED

Accelerated Mission Tests (AMTs) typically do not reveal all nozzle wear modes due to the interaction between the airframe, engine and the external flow fields. Flight testing of variable geometry nozzles has uncovered unexpected nozzle vibrations and wear problems.

3.7.12.1 Exhaust nozzle external asymmetrical air pressure loads The engine shall meet the requirements of the specification when subject to the asymmetrical and all other air pressure loads resulting from the external flow field pressure distribution on the nozzle and exhaust systems. The limits shall be specified herein.

REQUIREMENT RATIONALE (3.7.12.1)

The airframe manufacturer must provide the external flowfield pressure distribution to assure the engine is compatible with the airframe. The engine must be compatible with the air pressure loads external to the airframe.

REQUIREMENT GUIDANCE

Background:

The requirement is intended primarily for engines with variable and/or vectoring exhaust nozzles. The fact that an exhaust nozzle is variable lends itself to deformation by asymmetrical air pressure loads. Limit information is needed to prevent any permanent deformation. Nozzle with high aspect ratio non–round shapes can have structural area deformations with changing pressure loads. These area/shape deformations can affect the engine flow match, performance, and stall margin.

REQUIREMENT LESSONS LEARNED

None.

4.7.12.1 Exhaust nozzle external asymmetrical air pressure loads The requirements of 3.7.12.1 shall be evaluated by analysis and demonstration for each engine and airframe combination.

EVALUATION RATIONALE (4.7.12.1)

The exhaust nozzle external asymmetrical air pressure loads must be evaluated to assure compatibility with the airframe and its mission.

EVALUATION GUIDANCE

Background:

The initial evaluation must be conducted by analysis. Later the engine contractor should demonstrate the load limits by simulated loading.

EVALUATION LESSONS LEARNED

On one vehicle program, the nozzle area deformation caused an SFC degradation of about 5% at low altitude due to engine downmatching. On another vehicle program, the nozzle exit had to be reconfigured to reduce an aft surface heating problem. The selection of the engine fan operating line with deforming nozzles should consider stall margin at the smallest area and performance degradations at the largest areas. The effective flow area of the nozzle throat and exit with large or elaborate cooling schemes can be hard to determine.

3.7.12.2 Vectoring nozzle The thrust vectoring nozzle system shall be described. The engine thrust vectoring system shall revert to the fail–safe mode after receiving a fail–safe/standby signal from the airframe. Nozzle feedback signals, if required, shall be provided to the airframe and shall include: _(a)_.

REQUIREMENT RATIONALE (3.7.12.2)

Thrust vectoring is considered an aircraft performance enhancement feature, and it can improve aircraft handling qualities. The fail–safe signal requirement provides a degree of pilot control and safety for the mission, aircraft, and equipment.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor should fill in the blank.

Background:

The thrust vectoring nozzle should be designed for the mission of the aircraft and its on board weapon systems, not designed for distinct engine types.

The nozzle manufacturer should provide a description of the vectoring nozzle, including the actuation system, including type and number of actuators, performance, loads, interface attachments, sensing parameters, weight, nozzle position feedback, cooling provisions, geometric contours, fail–safe mode, manual operation, emergency features, materials, and coatings that are used. The contractor who is responsible for the nozzle should be specified.

Nozzle failure modes and their effects should also be addressed. Mission requirements should dictate the vectoring nozzle criteria. Data used in the design process should include the tradeoffs between thrust, weight, size, drag, stability, external flow field, exhaust cooling flows, and infrared radiation/visual signatures. Vectoring nozzle design and operation should include consideration of airflow match with upstream components to ensure sufficient engine stall margin with deformed throat/exit areas.

The engine/nozzle should not be considered an independent aircraft subsystem but should be synthesized into the aircraft's flight control laws. This will permit development of a truly integrated and optimized aircraft/engine/nozzle control system. Failure of one segment of a multi–segmented nozzle should not prevent operation of the nozzle. Nozzle structural integrity must be ensured during operation at maximum ram pressure conditions.

A thrust vector nozzle operating in a "transparent to the pilot" design configuration has been determined to be a more safe, reliable, stable method of controlling the vectoring nozzle operation. Similar to the other primary flight control surfaces, the vectoring nozzle does not require, but should allow, the engine/nozzle control system to be commanded by the aircraft flight control computer. This allows a central location for all flight control commands and permits the vectoring nozzle to be an integral part of the flight control laws and control surfaces.

To allow successful integration to occur, several items need to be specified by the integrator:

a. Data bus transmission rate

b. Total time delay through the nozzle control unit

c. Nozzle actuator spectral characteristics

d. Expected failure coverage levels of nozzle failures

e. A definition of the failure hierarchy and failure annunciation

f. A definition of critical parameters used for inner–loop control

The thrust vectoring nozzle may or may not operate in the critical mode depending on the design. Critical mode of operation is the maneuvering of the aircraft using the thrust from the vectoring nozzle(s) only.

If nozzle signal feedback is required, the engine contractor should provide the following information based on the airframe contractor's requirement: feedback may include current pitch and yaw position, current pitch and yaw maximum limit, calculated gross thrust, engine/nozzle status words, and vertical parity check (VPC) word to monitor health of the system. A vectoring nozzle system should be prime reliable if vectoring operation is necessary for vehicle flight safety and control. The nozzle should be returnable to the neutral position for horizontal flight in event of control failure.

REQUIREMENT LESSONS LEARNED

During the YF–22 program, while flight testing the thrust vectoring nozzle performance at Edwards Air Force Base, the aircraft was approaching the runway to perform a go–around for photo session purposes. While in this attitude the test pilot had difficulty controlling the aircraft and a mishap followed. It was later determined the mode switch, which was located in the cockpit and regulated the position of the vectoring nozzle, was placed in the wrong position. The mode switch was only for the purpose of performing flight test maneuvers and was supposed to be used only in distinct flight regimes. The flight control laws and piloted simulators had been developed using generic engine/nozzle representations instead of specific representations.

4.7.12.2 Vectoring nozzle The requirements of 3.7.12.2 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.12.2)

The requirements of the thrust vectoring nozzle must be tested and demonstrated.

EVALUATION GUIDANCE

Background:

Engine tests, to evaluate nozzle matching and vectoring functional characteristics, should be accomplished at sea level and at altitude flight conditions. Engine tests at envelope extremes should be used to ensure nozzle stability and freedom from vibration. Engine testing at maximum inlet pressure is desirable to demonstrate structural capability. Component test may be used to demonstrate maximum structural loads, mounting and airframe forces, and external fields.

Flight testing should incorporate a double–check system to ensure proper execution of each element of testing and each flight test program should be flown on a piloted simulator prior to actual flight. Additionally, each flight test element affecting stability should be initiated at altitude, leaving altitude margin for recovery.

The thrust vectoring nozzle is unique to the aircraft for it is an item which is shared by both the airframe and the engine, and should be evaluated in the "installed" configuration.

Testing of engine/aircraft integrated control system rigs/hot bench should be conducted to evaluate the integrated system control laws, transfers, and failures.

The flight control computer should control the vectoring nozzle like an integral segment of the entire primary flight control surface thus making it a transparent command function to the pilot.

Nozzles which vector, transition, block, or reverse flow for V/STOL operation should be thoroughly analyzed and demonstrated on both a steady–state and transient basis for effects on performance, stall margin, and operability.

EVALUATION LESSONS LEARNED

None.

3.7.12.2.1 Vectoring nozzle angle and rate The vectoring nozzle shall have a maximum slew rate of at least (a) degree/sec at (b) flight condition. The nozzle shall provide a maximum vectored deflection angle of (c) degrees from the symmetric axial thrust position with an accuracy of (d) degrees over the entire deflection range. The nozzle shall vector without interruption, in a constant, steady movement.

REQUIREMENT RATIONALE (3.7.12.2.1)

The slew rate and maximum deflection angle must be specified to assure proper vectoring nozzle operability and for aircraft design information. The airframe contractor needs to be informed of the nozzle positioning accuracy.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The maximum slew/thrust vectoring rate should be negotiated by both the engine and airframe contractors to match vehicle mission and stability requirements. The maximum slew rate capability should approach that of the vehicle control surfaces while assuring vehicle stability.

(b): The contractor should provide the flight altitude and Mach number or flight dynamic pressure.

(c): The contractor should provide the angle.

(d): The contractor should provide the accuracy.

Background:

The specified value for the "maximum slew rate of at least" requirement should be the fastest rate the nozzle can move under any condition. The nozzle must be able to move faster than this rate under all conditions, but the specification value is the fastest rate that can always be provided by the engine.

Some uses and benefits of vectoring nozzles are: improved nose pointing at low speed, reduced size of aircraft control surfaces, improved combat maneuvering, improved control/trim with inoperative engine(s), and steering potential.

One of the vectoring nozzle goals is to improve upon the slew rate and develop a rate that will offer good stability and vectoring performance. This rate is dependent on the desired achievable envelope, the aircraft static instability, and aircraft performance considerations. Statically stable aircraft or aircraft with sizable pitching moments of inertia may require a higher maximum nozzle slew rate. Based on experience, the nozzle is seldom driven at its maximum slew due to fan/compressor instability (i.e., stall margin).

The flight condition value determines actuator loads and should correspond to a value of constant maximum "q" for vectoring operation such as lower right flight envelope extending to higher machs at higher altitudes.

Nozzles being developed at this date have a specification deflection of +20 degree pitch, and $\pm$20 degree yaw maximum for the axisymmetric nozzle but this is simply a mechanical range. The requirement driving this parameter is total side force generated, or maximum pitch/yaw acceleration desired. These parameters will determine aircraft fuselage requirements and engine case loads. The lower bound on side force is determined by controllability and the upper bound is determined by weight and CG limitations. The vertical force generated is a function of Gross Thrust x Sine (vector angle) plus induced effects. Engine size, aircraft/nozzle configuration, and desired performance will drive this requirement.

For the weapon system to function in a complete arena of battle scenarios, the vectoring deflection angle requirement should permit the aircraft to maneuver, without danger of a stall, spin, or an out–of–control situation, to meet the demands of the mission. The normal aircraft systems check should be built into the control system Built–In–Test (BIT) and on–ground panel–up checks. This will allow the pilot to verify that the vectoring capability is delivering the proper amount of vector angle on pilot command. This is a safety requirement as well as a functional test of the system. When the vectoring nozzle system begins to lose its accuracy, the components that makeup the vectoring system will begin to indicate a possible malfunction, or begin to show a future component problem. This requirement has been put in place to offer both an indication of a decline in component quality and to establish a safety margin for component wear. As of 1993, manufacturers were capable of producing nozzles with $\pm$1.0 degree position accuracy. Maximum internal nozzle performance is of secondary importance during vectoring operation, because the deflection and deflection–rate capability have priority over the nozzle performance.

The Using Service should ensure that the ICD addresses the maximum and minimum dynamic response characteristics of the vectoring nozzle. The frequency response shall be consistent at all power levels and vector positions. If the nozzle is designed to be an integral part of the flight control system, the response characteristics must not show a salient amount of lead/lag, or out of phase, relationship between the vectoring of the nozzle to the aircraft control commands.

The flight control computer has a design characteristic that causes the nozzle vectoring command to either lead, and/or lag, the aircraft's primary flight control surfaces command input. The computer processing time (signal response) is known as the "throughput" of the computer control signal. The lead/lag condition occurs at specific flight regimes. Until such time that the aircraft's flight computer throughput time can be improved to cause the response of the vectoring nozzle to match the response characteristics of the primary flight control surfaces, an out–of–phase relationship between these two features will undoubtedly occur in some flight regimes.

Slew rates have been conducted up to 60 degrees/sec, but even higher rates have been tested without any aircraft instability or performance difficulty.

The nozzle slew rate required should be specified at a given flight condition or flight dynamic pressure to avoid over/under–designing nozzle actuators. The nozzle actuation system should have sufficient control power to achieve the desired vectoring at required flight conditions.

Vectoring requirements for nozzle design should be stated as slew rate and delta force (in pounds rather than percent) at the most important flight conditions and including those at maximum flight dynamic pressure; otherwise, the result is overdesign with added weight and cost.

REQUIREMENT LESSONS LEARNED

During the early development of the Thrust Vectoring Nozzle, it was learned that the ±20 degrees pitch/yaw full deflection angle was only occasionally needed. To achieve rapid high AOA (Alpha), and/or yaw (Beta), it was only necessary to introduce 5–10 degrees of nozzle vectoring for short periods of time to assist the aircraft primary flight control surfaces.

The vectoring nozzle slew rate has been a constantly changing parameter with each new vectoring nozzle development. This rate continues to be increased. As of 1993 the rate is up to 60 degrees/sec. The engine manufacturers seem to be satisfied with this figure.

4.7.12.2.1 Vectoring nozzle angle and rate The requirements of 3.7.12.2.1 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.12.2.1)

The maximum thrust vectoring rate and vectoring angle are considered high priority characteristics for thrust vectoring and must be evaluated. The nozzle position accuracy must be evaluated to detect the level of component wear.

EVALUATION GUIDANCE

Background:

The testing of the slew rate should address the issue of nozzle and/or system deterioration. The systems check should evaluate whether degradation is present.

Since the thrust vectoring nozzle angle is essential in the makeup of the aircraft maneuverability and aircraft trim characteristics, this parameter must be tested to ensure that continuous proper angle positioning is achievable.

Similar to the primary flight control surfaces, the pilot should have the capability to test the vectoring nozzle angle prior to aircraft launch. This is a safety requirement as well as a functional test of the system.

EVALUATION LESSONS LEARNED

Engine–off testing of the thrust vectoring nozzle system has required a remote hydraulic power source.

3.7.12.2.2 Vectoring nozzle failure accommodation Failure accommodation for the nozzle system shall be described. Any system failure shall not preclude the nozzle from reverting to a fail–safe position.

REQUIREMENT RATIONALE (3.7.12.2.2)

For both safety of crew and equipment, the thrust vectoring nozzle must fail to a fail–safe position. Without this ability, the aircraft could possibly assume a vulnerable flight attitude and result in aircraft loss.

REQUIREMENT GUIDANCE

Background:

The contractor should describe the failure accommodation for each type and number of failures. That is, one electrical failure might not affect nozzle operation, but two electrical failures may result in degraded operation (i.e., no augmentor operation or reduced thrust) or one hydraulic failure may result in fail–safe accommodation.

For naval aircraft carrier operation, it is essential that the thrust vectoring nozzle deploy to a fail–safe nozzle position. This is because of the short/narrow runways on board the carrier, the aircraft is left with little maneuvering room to counter any abnormal thrust vectoring angle attitude. The landing approach and arresting of the aircraft would be unsafe if the nozzle was permitted to retain a vectoring angle other than the fail–safe or neutral position.

The weapon delivery capability would be in jeopardy without the ability to automatically, or manually, deploy to the fail–safe configuration.

REQUIREMENT LESSONS LEARNED

The Multi–Axis Thrust Vectoring (MATV) program aircraft, the Air Force F–16 VISTA, has the ability to revert to a fail–safe condition using the fail–safe/standby switch in any power setting, which avoids nozzle instability.

4.7.12.2.2 Vectoring nozzle failure accommodation The requirement of 3.7.12.2.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.12.2.2)

The nozzle failure accommodations must be evaluated to ensure the nozzle will operate safely in the event of any failures. The testing of the thrust vectoring nozzle failure mode(s) shall provide the Using Service with a valuable tool to evaluate worst case scenarios in vectored flight.

EVALUATION GUIDANCE

Background:

The contractor should provide a means to test the thrust vectoring nozzle's failure mode accommodation. This may be incorporated into the BIT system, if required by the Using Service.

EVALUATION LESSONS LEARNED

None.

3.7.13 Augmentation system If thrust/power augmentation system(s) are used, the system(s) shall be described in the subparagraphs below. This system shall be free of destructive operating modes and shall meet all requirements of the specification. The engine shall provide thrust/power augmentor operation, as required via the airframe signal, including initiation and termination, throughout the augmentor operating envelope. The amount of augmentation shall be controllable, and transition from non–augmented to augmented operation shall be free of large abrupt thrust/power changes. The augmentor shall not induce an unacceptable loss of engine operability, either in operation or during ignition or cancellation over the required envelope.

REQUIREMENT RATIONALE (3.7.13)

Augmentation, when necessary to fulfill mission requirements, offers a high thrust–to–weight ratio method for obtaining limited time high thrust ratings. Use of the system must not result in premature degradation of engine life.

REQUIREMENT GUIDANCE

Background:

Augmented engines typically increase thrust 40–80% and weight 20–25% relative to a conventional non–augmented engine. The added weight, cost, and complexity of the augmented engine is often preferable to oversizing a non–augmented engine for the same application. Developing an augmented derivative for increased thrust needs may also be less costly than developing a growth version of the original engine.

For engines which must operate at extreme altitudes, the use of an automatic relight system may be essential to prevent flameouts. Ignition sources should not induce fan or compressor stall. Pilot burner stabilized flame, catalytic, or pyrotechnic igniters may be used to provide a continuous re–ignition source to reduce the possibility of augmentor flameout.

The Using Service should ensure the aircraft system specification has a requirement that the engine should not cause degradation to the airframe due to augmentation destructive operating modes (rumble, screech, etc).

Development elapsed time for augmented exhaust systems with unconventional features can equal or exceed the elapsed time for turbomachinery. Test hours required may only be 1/4 to 1/2 of the hours for turbomachinery. Some future engine programs may be paced by the exhaust system.

REQUIREMENT LESSONS LEARNED

Successful augmentor designs have incorporated light–off detectors in the control system to prevent hard lights and flameout/relight stalls from occurring. Augmentor design and tuning has been highly iterative and early development activity is recommended.

4.7.13 Augmentation system The requirement of 3.7.13 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.13)

The operational characteristics of thrust augmentation devices must be evaluated by engine test.

EVALUATION GUIDANCE

Background:

Augmentors should be tested as part of the total engine in an altitude facility to demonstrate satisfactory augmentor initiation, operation, and termination throughout the augmentor flight envelope.

EVALUATION LESSONS LEARNED

Testing has shown that augmentor operational characteristics can be adversely affected by relatively small changes to the engine configuration and control system. Every effort should be made to ensure the test engine is representative of the intended production configuration.

3.7.13.1 Afterburner (fuel injection) system If utilized, the afterburner system shall be described. Initiation of the afterburner system shall be accomplished by placing the thrust/power demand into any position in the afterburner range. Termination of the afterburner system shall be accomplished by placing the thrust/power demand in any position other than in the afterburner range.

REQUIREMENT RATIONALE (3.7.13.1)

Automatic activation/de–activation of the augmentor is required to reduce pilot work load.

REQUIREMENT GUIDANCE

Background:

The description of the afterburner system should include components such as afterburner case, liner, flameholder/spraybar system, mixer/diffuser, centerbody, etc.

Augmentor/nozzle liners are used to distribute internal nozzle cooling flows and control combustion instabilities. The cooling flows are needed for material survivability. The forward portion of the augmentor liner should serve as a “screech liner” to control destructive combustion instabilities.

Flameholders should have sufficient blockage to provide stable burning throughout their operating envelope. In turbofan engines, mixing of the hot and cold streams prior to burning has improved high altitude combustion stability.

The augmentor should not exhibit either low (rumble) or high (screech) frequency instabilities at a level which induces unacceptable impacts on augmentor/engine performance, operability, and life. Rumble is instability phenomena most frequently encountered. This axial mode discrete frequency instability is sufficiently low in frequency to cause fan stalls (usually 40–100 Hz, which is less than one–per–rev frequency of the fan) and burner blowouts.

Augmentor flameholder blockage can be increased to improve combustion stability at a cost of increased pressure loss during dry operation of the engine.

REQUIREMENT LESSONS LEARNED

None.

4.7.13.1 Afterburner (fuel injection) system The requirement of 3.7.13.1 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.13.1)

The augmentor must be evaluated to ensure there are no destructive operating modes and that operability and performance requirements are met.

EVALUATION GUIDANCE

Testing should cover the augmentor operating envelope and include takeoff, key operating points, and low Mach number/high altitude flight conditions. Attention should be on the upper left envelope for operability and on the lower right envelope for highest loads.

EVALUATION LESSONS LEARNED

None.

3.7.13.2 Water injection system The water injection system, if utilized, shall be described.

REQUIREMENT RATIONALE (3.7.13.2)

Information concerning the water injection system is necessary to facilitate a more detailed understanding of the overall system operation.

REQUIREMENT GUIDANCE

Background:

The contractor should provide a description of the water injection system that includes such information as water injection system flow limits, accuracies, adjustments, temperature and pressure limits, and system limitations. Location of injection nozzles should also be specified (example: burner dilution air, turbine casing and vane cooling air, cooling air inside combustion chamber inner casing, etc.)

Thrust augmentation using a water injection system is normally used for takeoff. A water injection system is used to add mass and cool the air flow after the compression process. This in turn produces an increase in thrust or a decrease in specific fuel consumption. Demineralized water is required to minimize formation of internal engine deposits.

There are inherent disadvantages to the use of a water injection system, including increased weight, susceptibility to freezing conditions, and the non–availability of usable water.

Water injection into the combustor should be provided for lift power settings of V/STOL engines, if appropriate.

REQUIREMENT LESSONS LEARNED

Water injection has required specialized logistics servicing and intensive field maintenance. It is not practical at low temperatures. Water injection systems have been employed on V/STOL aircraft, such as Harrier, and many older engines.

4.7.13.2 Water injection system The requirement of 3.7.13.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.13.2)

The water injection system must be evaluated to ensure that the engine delivers the rated thrust/power within the various engine limits.

EVALUATION GUIDANCE

Background:

Water injection systems are usually tested at sea level takeoff conditions.

EVALUATION LESSONS LEARNED

None.

3.7.13.2.1 Water injection system fluid If a water injection system is utilized, the water solution used shall be specified herein. The water solution shall use demineralized or distilled water, shall be visually clear, and shall conform to the following purity standards:

Impurities	Quality
Total solids (max) ppm	35
Dissolved solids (max) ppm	25
Ph Factor	6.5 – 7.5

REQUIREMENT RATIONALE (3.7.13.2.1)

The purity of the water used for the water injection system must be specified. Deposits, coatings, and corrosion may form on the engine and affect its performance.

REQUIREMENT GUIDANCE

Background:

The purity specifications are related to the water quality of military demineralization and distillation equipment. These requirements may be tailored by the Using Service when the state–of–the–art of the equipment changes. This also is related to the possible deterioration of the engine's performance by leaving deposits or coating on the engines blades and stators.

REQUIREMENT LESSONS LEARNED

Model specification E1130–B of September 1971 for the TF34–GE–2 engine lists these same limits for the wash water to be used with the engine internal water wash system.

4.7.13.2.1 Water injection system fluid The requirements of 3.7.13.2.1 shall be evaluated by inspection and test.

EVALUATION RATIONALE (4.7.13.2.1)

The purity of the water for the injection system must be evaluated. Engine tests are needed to determine if deposits and coatings deteriorate the engine's performance, operability, and life.

EVALUATION GUIDANCE

Background:

The purity of the water should be checked if it is not provided by the source. The required engine tests and tear downs should determine if deposits form and affect engine performance.

EVALUATION LESSONS LEARNED

None.

3.7.14 Wash system The wash system shall be described. An integral wash system shall be provided with the engine. The wash system shall not allow standing water to accumulate in the engine. The composition of the washing medium shall be in accordance with MIL–C–85704. Wash system interface connections shall be compatible with all existing Using Service support equipment.

REQUIREMENT RATIONALE (3.7.14)

Integral wash systems at the engine inlet are needed for cleaning of internal engine parts to restore performance. Washing an engine by flushing cleaning solution into the air inlet may not be adequate, there may be a need to introduce the solution (directly) further down stream. An integral wash system improves maintenance and reduces the inventory of maintenance items; this is especially important for shipboard aircraft where storage is at a premium. Standing water may freeze and damage the engine upon ingestion.

REQUIREMENT GUIDANCE

Background:

The contractor should provide a description of the wash system that includes information such as required pressure, temperature, flow, and procedure to be used for field service.

The manufacturer is responsible for developing the wash procedure for field service maintenance. In certain theaters of operation, the availability of water may be limited. The effectiveness of the wash system should also take into account the economical use of water.

All military aircraft during normal service operation encounter unusual environmental contaminants such as sea, salt water, sand/dust particles, armament gases, engine exhaust gas plumes, etc. which are ingested by the engine. These contaminants often accumulate as deposits on various engine internal parts which can result in performance deterioration, corrosion, erosion, etc. Aircraft engines operating in a salt laden atmosphere often develop heavy internal deposits of sea salts. Installation of an integral wash system on the engine inlet permits spraying of a washing medium of a prescribed composition into the engine for cleaning of internal parts. This cleaning action will remove all or most of the contaminants thus restoring the engine to acceptable performance levels and reducing corrosion of internal engine parts.

Standing water in the plumbing of the wash system may cause damage from freezing. The system should be designed to avoid standing water, and a method of purging the water from the plumbing should be provided.

The wash system should be able to remove foreign matter accumulations (such as, residue from sand, dust, and salt laden air) from the inside of the engine in a timely and effective manner. The wash system can be used to restore performance during the calibration and recalibration of the endurance test.

REQUIREMENT LESSONS LEARNED

During Desert Shield/Storm, the deposit due to the ingestion of very fine sand/soil had varying effects on the performance, time between maintenance intervals, parts life, and overall life of many turbine engines. Depending on the temperature of the combustion, some of the sand/soil either glazed over or encrusted the combustion chamber and turbine section, resulting in loss of performance, corrosion, and erosion. Loose sand clogged nozzles (resulting in heat stresses) and threatened to compromise seals and bearings. In at least one aircraft type a special cleaning procedure had to be implemented, a cleaning wand had to be fabricated and was inserted via the engine ignitor plug opening into the combustion chamber where it introduced multiple streams of washing medium to remove loose sand from the combustor and stage one nozzle components. This procedure was used as often as every 20 hours when operating in heavy sand/brown–out conditions, was effective in minimizing the damage caused by fine sand accumulation, and extended the cleaning interval.

4.7.14 Wash system The requirement of 3.7.14 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.14)

When engine performance deteriorates due to contamination or after operation in an adverse environment, the wash system must be evaluated to see if it removes a suitable amount of foreign matter to restore performance and preserve engine life.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.7.15 Brake system Turboshaft or turboprop engines shall incorporate sufficient braking to prevent rotation of the propeller when exposed up to (a) km/hr ((a) mph) wind from any direction while grounded with the lower half of its projected area blanked off (and the engine shutdown), or while in flight with the propeller feathered at all airspeeds up to 0.5 Mach. The torque capacity of the output shaft brake shall be specified herein.

REQUIREMENT RATIONALE (3.7.15)

A brake system is needed to prevent auto–rotation of parts, which may induce excessive wear, as a result of improper lubrication. A rotating propeller on a parked turboprop aircraft may present a safety hazard to ground personnel.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): a value of at least 80km/hr (50 mph).

Background:

When the engine is shutdown there is not sufficient lubrication supplied to the bearings, gears and seals. If the rotating parts are allowed to move freely with inadequate oil pressure, the bearing surfaces may experience excessive wear, the oil seals may eventually fail, and leakages may result.

Propeller tiedowns are not desirable.

The Using Service should ensure that this requirement is not duplicated in the propeller specification.

REQUIREMENT LESSONS LEARNED

None.

4.7.15 Brake system The requirement of 3.7.15 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.7.15)

A demonstration and test is needed to show the propeller will not rotate under the required conditions.

EVALUATION GUIDANCE

Background:

The ideal test for this requirement is to use an aircraft for both flight test and ground test. For the ground test the wind speed specified in 3.7.15 from any direction should be simulated by a wind machine. If the evaluation is required before a flight test, the torque and drag produced by the feathered propeller at Mach .5 should be computed and the brake should be evaluated by a simulated torque and thrust load applied to the output shaft.

EVALUATION LESSONS LEARNED

None.

3.7.16 Negative torque limiter For turboshaft or turboprop engines, following an engine failure or malfunction, a means to automatically prevent excessive negative torque shall be provided. The engine shall return to normal operation when the malfunction is corrected. The negative torque limiter shall be described.

REQUIREMENT RATIONALE (3.7.16)

A limiter is needed to prevent excessive negative torque that may cause engine damage.

REQUIREMENT GUIDANCE

Background:

The description of the negative torque limiter should include the method used, the amount and duration of negative torque, and the shaft power absorber signal (torque meter, strain gauge, etc.).

Negative torque reduces the life of the gearbox since it loads the gears, bearings, shafts, etc. in the opposite direction. With the Using Service approval, the engine contractor should set the negative torque for the minimum damage to the gearbox. Typical helicopters do not generate negative torque as rotor blades cannot go into negative pitch and blades do not reverse direction, even in autorotation.

To avoid duplication, the Using Service should ensure that this requirement is not specified in the propeller or system specification.

REQUIREMENTS LESSONS LEARNED

None.

4.7.16 Negative torque limiter The requirement of 3.7.16 shall be evaluated by analysis and test.

EVALUATION RATIONALE (4.7.16)

A test is needed to evaluate whether the negative torque limiter will prevent excessive negative torque and return the engine to normal operation under the required conditions.

EVALUATION GUIDANCE

Background:

The test should include negative torque values of magnitude greater than the specified value (example: if the specified negative torque value is –50 ft–lbs, then the test should include negative values such as –60, –70 ft–lbs, etc.)

The negative torque tested should be increased to the specified value. The duration of the negative torque should be varied from below the specified time to include negative torque duration after limiter activation. This test procedure is to determine the response of the system.

EVALUATION LESSONS LEARNED

None.

3.7.16.1 Multiple power section engines For multiple power section turboshaft engines, a means shall be provided to automatically prevent an inoperative power section from absorbing power from the remaining operating power section(s) and the shaft power absorber at any flight condition or power setting. Inoperative engine(s) in the power section shall not result in the operating engine(s) exceeding design or operating limits. Similarly, failure(s) in the power absorber propulsor, gearbox/transmission, or drive system shall not result in the operating engine(s) or power absorber system exceeding design and operating limits. A manual override shall be provided. A signal shall be provided to identify the section that has been separated. Separation and reconnection of a power section shall not cause any adverse affects on the performance and durability of the engine(s) and power train, and shall not affect safety of flight.

Multiple power section engines shall provide for load sharing and torque matching required by the engine control system in 3.7.2.

REQUIREMENT RATIONALE (3.7.16.1)

In a multiple power engine application it is necessary to separate an inoperative power section from the operating section(s) to maintain safety of flight.

REQUIREMENT GUIDANCE

Background:

The requirement also applies to “twin–pack” engines, twin engine helicopters with common rotor(s), and any other aircraft with cross–connected engines and/or propulsors. Automatic activation is necessary because manual activation would impose an unwanted burden on the flight crew.

The inoperative engine should be able to be restarted and reconnected with the drive system at any flight condition and power setting, without affecting safety of flight or durability of the operating engine(s). If power from the operating engines is used for the restart, there should be enough power remaining to maintain safety of flight.

It should not be necessary to reconnect or decouple multiple power section engines at extremely low power condition which may affect safety of flight due to insufficient power necessary to sustain flight. The Using Service should ensure that the aircraft system specification defines the number of engines and amount of power necessary to sustain flight in a multi–pack engine design.

Helicopter engines are typically qualified as single engine installations whether used in single or multi–engine aircraft. The engine control system has logic for torque matching and operation with inoperative engine(s).

REQUIREMENT LESSONS LEARNED

None.

4.7.16.1 Multiple power section engines The requirement of 3.7.16.1 shall be evaluated by analysis, demonstration and test.

EVALUATION RATIONALE (4.7.16.1)

Component and engine tests must be conducted to assure proper operation of remaining operating power section(s) when individual power section(s) become inoperative.

EVALUATION GUIDANCE

Background:

A component test will demonstrate the feasibility of the method. Actual engine tests will give the assurance that the method is acceptable for the aircraft.

EVALUATION LESSONS LEARNED

A past aircraft with inter–connected engines and propellers, crashed when a propeller gearbox failed and the engines rapidly overtorqued the remaining gearboxes, popping off each propeller in turn.

3.7.17 Power absorber (engine supplied) The engine supplied power absorber shall be described herein. The power absorber shall include any remote engine supplied mechanism which absorbs power from the engine cycle.

REQUIREMENT RATIONALE (3.7.17)

The engine must be capable of operating with the power absorber so that engine operation can be evaluated.

REQUIREMENT GUIDANCE

Background:

The type of power absorber (i.e., propeller, remote fan, gas ducting, gearbox, transmission, etc) should be specified by the contractor.

REQUIREMENT LESSONS LEARNED

Dissimilar power absorber characteristics for engines with multiple vehicle application may result in development of separate or modified engine control systems.

4.7.17 Power absorber (engine supplied) The requirement of 3.7.17 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.17)

Operation of the power absorber and engine must be evaluated by test.

EVALUATION GUIDANCE

The intended power absorber should be used for the test if possible. When a dynamometer is used to simulate the power absorber, not all of the loads such as thrust and vibration are reproduced.

EVALUATION LESSONS LEARNED

None.

3.7.17.1 Feathering Engines using propellers shall withstand all loads and temperatures imposed by feathering of the propeller from any operating condition. All multi–engine applications shall incorporate auto–feather systems. If fuel shutoff is required, it shall be accomplished automatically by the control system. If engine/gearbox lubrication oil shut–off is required, it shall not occur until rotation has ceased.

REQUIREMENT RATIONALE (3.7.17.1)

The engine must be capable of withstanding the loads and temperatures imposed when the propeller is in the feathered position.

REQUIREMENT GUIDANCE

Background:

The direction of the loads (thrust and torsion) on the drive shaft of a forward pitched and a feathered propeller are in opposite directions. All drive train components forward of the brake(s) will experience loading on the back side during feathering.

When a propeller feathering command has been issued, the engine should be shutdown simultaneously either manually or via an airframe signal. If the engine is not stopped when the propeller is feathered, the engine may exceed its temperature limit or other damage may occur.

REQUIREMENT LESSONS LEARNED

None.

4.7.17.1 Feathering The requirement of 3.7.17.1 shall be evaluated by analysis, demonstration, and test.

REQUIREMENT RATIONALE (4.7.17.1)

Engine compatibility with manual feathering operation must be evaluated by test to ensure operability.

EVALUATION GUIDANCE

Background:

An analysis of the loads on the engine by a feathered propeller should be made. The results of the analysis should then be evaluated during flight tests.

EVALUATION LESSONS LEARNED

None.

3.7.17.2 Reverse pitch operation For engines using propellers, the operational capability and limits with the propeller in reverse pitch shall be described.

REQUIREMENT RATIONALE (3.7.17.2)

The engine must remain within limits when operating in the reverse pitch mode without adversely affecting engine durability and operability.

REQUIREMENT GUIDANCE

Background:

The contractor should provide the limits of engine and propeller operation which should include maximum rotor speed, maximum time to reverse pitch angle, and maximum reverse pitch angle.

The loads and temperatures are different when the propeller is used in reverse pitch. The direction of the thrust load on the output shaft is opposite for a propeller in forward pitch and one in reversed pitch, however the torsional loads remain in the same direction. The reversed pitch propeller will experience vibration that may be completely different, and the characteristics of the vibration may vary with the velocity vector. There is also the possibility of ingesting hot exhaust gases.

The Using Service should ensure the reverse pitch operation limits/requirements are in agreement with the limits/requirements of the propeller and/or aircraft system specification.

REQUIREMENT LESSONS LEARNED

None.

4.7.17.2 Reverse pitch operation The requirement of 3.7.17.2 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.17.2)

Operation in the reverse pitch mode must be evaluated to ensure there is no adverse impact on the engine during reverse pitch operation.

EVALUATION GUIDANCE

Background:

An analysis of the loads on the engine by operating a reversed pitch propeller should be made. This should be evaluated by flight or ground tests. Flight tests are preferred since the air flow will be affected by the airframe and the engine nacelle.

EVALUATION LESSONS LEARNED

None.

3.7.18 Gearbox The engine shall be equipped with a gearbox to provide mechanical power to components required for engine and aircraft operation and as a means to transmit torque to the engine for starting. The mechanical power characteristics at each pad shall be specified in table V. The gearbox lubricant shall be the same type as the engine lubricating oil.

The accessory gearbox shall be capable of simultaneous operation of all the drives when each drive is subjected to (a) times the maximum permissible torque or power rating specified for the individual drive. The steady state and transient peak load capabilities shall be specified herein.

REQUIREMENT RATIONALE (3.7.18)

Physical and performance characteristics of the gearbox must be established to ensure satisfactory operation of critical engine and aircraft accessories. The gearbox lubricant must be the same type as the engine lubricant for logistic purposes. Peak transient loads are needed to design engine power offtake structure and components.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a) A value of at least 1.25 times.

Background:

The gearbox should supply mechanical power to the following engine and aircraft components: generators, fuel, lubrication, and hydraulic pumps; and a means to transmit torque to the engine for starting. The configuration, location, envelopes, ratings, dimensions, and alignment requirements of all accessory pads and drives should be shown on the installation drawing.

The 1.25 value is specified for the purpose of growth which is necessary to facilitate use of a design on other configurations and to provide a margin of safety.

The Using Service should ensure that the following is specified in the aircraft system specification or ICD: “Power Take–Off (PTO) shear section failure and accessory drive shear section failure shall not cause airframe or engine damage. A shear section is desired to prevent the PTO shaft from flailing.” If the (PTO) shaft is supplied by the engine manufacturer, then this requirement should be placed in the engine specification.

The Using Service should check that the aircraft systems specification defines all the steady–state and transient requirements for PTO shaft power extraction from the engine and the proper limits are in the engine specification.

All drive splines should be positively lubricated by engine oil or provide for use of non–metallic spline adapters in accordance with MS14184 or MS14169. Use of non–metallic spline adapters in lubricated accessory pads are permitted. Nonmetallic spline inserts conforming to MS14169 or MS14184 should be considered in applications where excessive fretting may occur.

The gearbox design or the mounted accessories should incorporate shear sections that would allow disengagement of seized components, promoting safe engine operation, or safe engine shut down to prevent further secondary damage to the accessory gearbox and components. Replacement of shear sections after failure should not require removal of the gearbox from the engine nor the engine from the airframe.

Magnesium alloys shall only be used with the approval of the Using Service.

The gearbox housing should be designed to minimize upper surface areas that trap water, which can lead to the formation of salt deposits which causes or accelerates corrosion.

Table V pad ratings must be based on actual duty cycle, including PTO duty cycle from the airframe contractor. All steady–state and transient torque conditions, including one engine only requirements (if applicable), shall be reflected in the table V data.

PTOs are being used more often on modern engine designs in lieu of a single gearbox which mounts both aircraft accessories and engine components. This permits faster engine removal and installation as the aircraft systems are not disturbed. Also, to simplify engine removal, aircraft mounted accessories with an independent lubricating system are preferable to engine mounted aircraft accessories. The PTO drive should incorporate a shear section to control the point of failure in the event of an aircraft accessory gearbox lock–up.

A significant life cycle cost driver on many military aircraft has been spline wear.

REQUIREMENT LESSONS LEARNED

Failure of accessory gearbox drives has resulted in aircraft losses on single engine aircraft.

Provisions must be made to prevent a post–failure PTO shaft from damaging surrounding hardware as happened during one event when a flailing PTO shaft severed an actuator fuel line and flooded the engine.

Engines with long PTO shafts which have deflected excessively during hard carrier landings have caused operational problems with the aircraft. Transient peak loads was a problem for F404 tower shaft on one aircraft competition. Some aircraft power demands are really work or power times time. For example, raising a landing gear may be 500 hp–sec. The delta power demand is thus 500 hpx for one second or 50 hpx for ten seconds. Aircraft with short moment arms for control surfaces have high actuator loads and a high transient peak duty cycle.

4.7.18 Gearbox The requirement of 3.7.18 shall be evaluated by analysis, demonstration, and test.

EVALUATION RATIONALE (4.7.18)

Examination of the gearbox must be conducted to ensure that the gearbox characteristics conform to the installation drawing. Testing must be conducted to evaluate the operational and load carrying properties of the gearbox.

EVALUATION GUIDANCE

Background:

A gearbox endurance test using duty cycles developed from the engine missions specified in section 3.4 should be accomplished. During the endurance test, the dynamic interface characteristics of the actual aircraft installation should be simulated, including support stiffness and PTO alignment. The endurance test should be run on the gearbox with all accessories and components installed. During the test, all drives should be loaded to at least the maximum overhung moment rating of the drive and subjected to at least the maximum spline misalignment allowed. Elements or components of the engine lubrication system such as filters and oil reservoirs may be incorporated in the test rig to simulate system characteristics. Where applicable, pad seal drain leakage should be monitored throughout the test. The power extraction values for the various endurance test segments should be selected to ensure representation of time at maximum steady state and transient torque values expected in the duty cycle over the life of the gearbox. Some of these transients may be 2X steady torque values in magnitude, but are only a few seconds in duration.

Prior to endurance running, the gear box should undergo a static torque test wherein the input shaft is held stationary and all drive pads except the starter pad are simultaneously loaded to a static torque value equivalent to the maximum operating torque, at any drive speed, to which the drive pad is subjected during engine operation for a period of five seconds. With this loading maintained, each drive pad, singularly, should be subjected to 110 percent of its maximum shear torque rating for a period of five seconds. In an additional test, with all drive pads loaded to the maximum torque value the pad would be subjected to during starting, the starter drive pad should be loaded to 200 percent of the maximum starter torque or to 120 percent of the maximum starter shear torque rating (the maximum torque that can be transmitted prior to shear section failure), which ever is greater, in a direction which will not unload any other component of the accessory drive train for a period of 5 seconds.

Following the static torque test, the gearbox with all accessories and components installed, should be subjected to a vibratory scan and resonant search, while operating throughout the speed range from Idle to 115 percent maximum speed under varying loads including maximum rated pad loads. Turboshaft applications may use 130 percent of the maximum operating speed.

After the last maximum speed run, at least ten minutes of operation should be at maximum speed, maximum oil inlet temperatures, and all pads loaded to maximum torque values. This is to check the gears for scoring which will occur under these conditions. If the gears will be damaged by scoring, it will occur within the ten minutes.

During and following the test, scavenge oil filter and magnetic plug residue should be collected and separated into magnetic and non–magnetic material. This material should then be weighed, analyzed, and the results reported. At the completion of the test, the gearbox should be completely disassembled, inspected dirty then cleaned, and parts inspected. There should be no evidence of material defects, undue wear, or impending failure. Conformance with the fluid leakage requirements should be demonstrated during this test.

EVALUATION LESSONS LEARNED

In the past gearbox testing on the engine did not expose all the problems. Also gearbox qualification by similarity did not expose the component to all the loads.

Field experience has shown that gear resonance is a major problem on engines due to the high speed and light weight of the gears. Normal engine testing does not show that gears are free of damaging resonances.

Field experience has shown that PTO systems should be exposed to maneuver and landing loads during qualification testing to expose any potential problem.

3.7.19 Tubing/plumbing system The tubing and plumbing system shall function satisfactorily while meeting the requirements of this specification.

REQUIREMENT RATIONALE (3.7.19)

Tubing and tubing connectors must have the same durability and integrity characteristics as the engine.

REQUIREMENT GUIDANCE

Background:

ARP 994 should be used for guidance in the design of tubing installations. B–nut connections between lines carrying flammable fluid should include a positive or redundant locking feature to ensure nuts remain sealed and do not leak. Sealing compounds used to lock connections should be in accordance with 3.1.3.1 (adhesives and sealants).

REQUIREMENT LESSONS LEARNED

Tubing failures within 90 flight hours have been caused by a combination of resonant frequency, marginal strength loop clamps, handling damage and tubing cross–section which had excessive ovalness (out–of–roundness) at the tube bend.

A seizing compound has repeatedly caused problems, when used as a primary or redundant locking or sealing device, and precluded component disassembly.

4.7.19 Tubing/plumbing system The requirement of 3.7.19 shall be evaluated by analysis, inspection, and test. Natural frequencies of all tubing lines shall be examined and compared to dominant engine and installation operating speeds (gear pump speeds, blade passing frequencies, etc.).

EVALUATION RATIONALE (4.7.19)

Evaluation of the system must be accomplished to ensure system integrity. Natural frequencies of all tubing must not coincide with engine operating frequencies to ensure that undesirable resonant conditions are avoided.

EVALUATION GUIDANCE

Background:

Evaluation should include inspections for nicks, scratches, and out of round cross sections caused by tube bending and installation. The lines should also be evaluated for strength and rigidity.

The loading at the connections and through out the line should include the reactions caused by the mass of the fluid within the line and pressure transients in the fluid. Both the static loads and the dynamic loads due to mode of flight and pressure transients in the line should be accounted for.

EVALUATION LESSONS LEARNED

Handling damage and manufacturing marks (stake marks) have reduced tube structural capability. Past experience has also shown that stainless steel tubing, especially 1/4 inch tubes, are very susceptible to handling damage and high cycle fatigue failures at the brazed ferrules.

3.7.19.1 Tubing/plumbing identification Method(s) for identifying tubing and plumbing lines shall be described. All methods shall be visible in red light conditions.

REQUIREMENT RATIONALE (3.7.19.1)

For maintenance purposes, identification is required as often as necessary to track the line function, and fluid type, particularly on lines entering and emerging from closed compartments.

REQUIREMENT GUIDANCE

Background:

The engine manufacturer should provide tubing/plumbing identification. All lines should be identified by some device such as labels, markings, color coding, or a combination of these methods.

Identification devices should be legible and visible on the assembled engine in the installed aircraft, and should remain legible for the service life of the engine.

REQUIREMENT LESSONS LEARNED

None.

4.7.19.1 Tubing/plumbing identification The requirement of 3.7.19.1 shall be evaluated by inspection and demonstration.

EVALUATION RATIONALE (4.7.19.1)

Tubing/plumbing marking(s) must be evaluated to ensure adequate identification.

EVALUATION GUIDANCE

Background:

The best method of evaluating the tubing/plumbing identification is by inspection under field conditions. The tubing and plumbing should be inspected after all engine tests to ensure the marking(s) are still attached and are identifiable. The Navy does not allow markings that are colored red.

EVALUATION LESSONS LEARNED

None.

3.7.19.2 Tubing/plumbing clearances Clearance and supports shall prevent physical contact of tubing with other tubing, components, or structures. Clearance and support shall be adequate for wrenching, installation, and removal.

REQUIREMENT RATIONALE (3.7.19.2)

Adequate support and clearance should prevent failure of tubes due to contact, rubbing, or abrasion and should accommodate wrenches and other tools for connecting mating tubes.

REQUIREMENT GUIDANCE

In high vibration areas the support should isolate (damp) the tubing from the vibration. Enough clearance should be provided for removing tubing without removing additional components.

REQUIREMENT LESSONS LEARNED

Two examples of causes of field failures and aircraft mishaps are:

a. Chafing of electrical wiring caused arcing and burn–through of the rigid tubing.

b. Titanium tubing was worn through by an adjacent tube support loop clamp in a high vibration environment.

4.7.19.2 Tubing/plumbing clearances The requirement of 3.7.19.2 shall be evaluated by inspection, demonstration, and test.

EVALUATION RATIONALE (4.7.19.2)

Tubing/plumbing clearances must be evaluated to ensure durability and ease of maintenance.

EVALUATION GUIDANCE

Background:

Mock–ups and the maintainability demonstration of 4.5.2 should be used to determine if maintenance personnel have adequate clearance for installation and removal of components. Inspections during and after engine tests should also be performed to determine if any tubing comes in contact with other parts.

EVALUATION LESSONS LEARNED

Mockups were useful on the F119 program. An oil line to the no.1 bearing had to be rerouted to clear the fan IGV link, and an oil line to the lube/scavenge pump had to be moved to ensure adequate wrenching clearance.

3.7.20 Inlet particle separator (IPS) For turboshaft engines equipped with an engine supplied inlet particle separator, the IPS shall be described. The separator scavenge exhaust (exit) duct shall be located to prevent re–ingestion of separated particles and prevent interference or contamination with other systems. The engine shall meet all requirements of the specification while utilizing the IPS.

REQUIREMENT RATIONALE (3.7.20)

Some turboshaft engines require an inlet particle separator or air filtration to minimize ingestion of sand and dust.

Unacceptable maintenance intervals and excessive wear will occur on most helicopters if sand and dust ingestion is not minimized.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

The airframe and engine contractors should determine if the IPS is part of the airframe or engine system. If the IPS is provided by the airframe then the requirement should be deleted from this specification and included in the airframe system specification and/or ICD.

If an inlet particle separator is required to meet 3.3.2.4 sand ingestion requirements, it should be qualified with the engine and as an engine part. The engine contractor should provide a description of the engine supplied IPS system. This should include the number of filter elements, self–cleaning features, etc.

Background:

Separated particles should not be allowed to enter other engine systems including the engine air intake, engine and transmission compartments, engine and transmission oil cooler intakes, crew cabin, etc.

In the design of the particle separator, consideration should be given to the ease of replacing or repairing parts susceptible to extreme particle erosion.

Some turboshaft engines use a swirl type separator. There is no physical barrier, the separator is considered part of the engine so that all the airborne particles that enter the separator is considered to be ingested. The separated particles are expelled from the engine before the compressor stages. If barrier filters are provided by the airframe contractor, the engine manufacturer should be given the amount, concentration, and particle size distribution of particles exiting the filter in the inlet air stream. The Using Service should ensure that the performance of the airframe supplied IPS is specified in the aircraft system specification.

The advantages and disadvantages of the proposed inlet particle separator should be analyzed by the Using Service. The change in efficiency of the separator (which may decrease with time) and the pressure drop (which may increase as the separator deteriorates) should be kept to a minimum, and should not prevent the engine from meeting system level performance. For rotary wing aircraft operating from land base facilities, an engine IPS or an aircraft barrier filter is often required.

The Using Service should specify the sand and dust ingestion requirements and associated limits on performance and industry should decide how best to meet them. For example, materials may advance in the future such that IPS systems are not needed. Similarly, the sand could be removed/reduced by provisions in the aircraft inlet.

REQUIREMENT LESSONS LEARNED

Helicopter operations during Desert Shield/Storm were hampered by the sand/soil that was encountered in that region. During takeoff, landing, and in ground effect hover, wash from helicopter rotors caused the formation of dense clouds of sand and dust, and aircraft often operated in brown–out conditions. Sand/soil ingestion caused excessive wear on the airfoils of the first stage compressor and deposited a thin hard erosive coating on the turbine and combustion sections; sand/soil also accumulated on all internal parts. Operating regularly in brown–out conditions, utility type helicopters were suffering from loss of performance and the accelerated removal of engines. Special procedures were devised and implemented, and were successful in extending engine lives. On one aircraft type, the engines were designed with an integral particle separator, and the separator had been claimed to be 92% efficient in removing the larger particles from the airstream. This engine was able to function acceptably longer than other engines, but it was not immune from the same service problems. The addition of an fine sand test in 4.3.2.4 is an attempt to address problems with fine sand and “Saudi soil”.

4.7.20 Inlet particle separator (IPS) The requirements of 3.7.20 shall be evaluated by engine test.

EVALUATION RATIONALE (4.7.20)

The inlet particle separator needs to be evaluated by an engine test.

EVALUATION GUIDANCE

Background:

The inlet particle separator should be tested during the evaluation of the sand and dust ingestion requirement 3.3.2.4. The duration of the test should be as specified in 3.3.2.4. At the end of the test the separator efficiency and pressure drop should be noted.

EVALUATION LESSONS LEARNED

Erosion damage caused by fine sand is different from that of large sand. Fine sand has a greater tendency for plugging of cooling passages. In the past, particle separators had problems removing very fine sand particles, as was experienced during Desert Storm.

There is also evidence that the problems encountered during Desert Storm were caused not only by individual fine and course sand particles, but by larger “pebble like particles of coalesced sand”. These particles were very low in density, and therefore became readily airborne and more difficult (when compared to denser particles) to separate using centrifugal methods.

3.8 Software resources

4.8 Software resources

3.8.1 Software integrity The software integrity shall be in accordance with (a) .

REQUIREMENT RATIONALE (3.8.1)

A software development integrity program is necessary to assure a high quality software product.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The software development process, documentation, management, and control should be governed by DOD–STD–2167 and MIL–STD–1521, or if the Software Development Integrity Program (SDIP) is required, MIL–STD–1803 should be inserted.

Background:

MIL–STD–1803 references DOD–STD–2167 and MIL–STD–1521.

REQUIREMENT LESSONS LEARNED

None.

4.8.1 Software integrity The requirements of 3.8.1 shall be evaluated in accordance with (a) .

EVALUATION RATIONALE (4.8.1)

The design of software programs for the engine computer resource must be evaluated to ensure that all requirements have been met.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): DOD–STD–2168 should be referenced, unless MIL–STD–1803 is required in 3.8.1. In that case, MIL–STD–1803 should be inserted in the blank since it contains an SDIP evaluation section.

Background:

Evaluation of software design should be accomplished through the contractor’s quality assurance program in accordance with the referenced documents. Plans, procedures, and test results should be reviewed by the Using Service. The Higher Order Language (HOL) and structure may be evaluated by inspection. Real–time processing should be evaluated by analysis and demonstration and the Built–in–Test (BIT) by test.

EVALUATION LESSONS LEARNED

None.

3.8.2 Software performance and design The Contractor shall design, develop, test, document, and deliver operational, test, and support software programs to meet and maintain equipment performance specified herein. The detailed software requirements and computational components shall be described. All programming language shall be in accordance with MIL–STD–1815. If required by the weapon system contractor, the equipment shall provide an output over the multiplex busses which uniquely identifies the operational configuration of the equipment and software combination. The operational software shall perform all computations, data processing, and interfacing functions necessary for the engine to meet the performance and operability requirements of the specification.

REQUIREMENT RATIONALE (3.8.2)

Following proper software engineering techniques will insure a sound software product. This includes a complete detection of faults, timing analysis, and reduced supportability costs. The final product should result in effective and efficient software operation.

REQUIREMENT GUIDANCE

Background:

Software design requirements are included as part of the DOD regulations and includes digital engine controls, engine monitoring systems, and support equipment containing software. The use of assembly language must be approved by the Using Service.

REQUIREMENT LESSONS LEARNED

None.

4.8.2 Software performance and design The requirements of 3.8.2 shall be evaluated by inspection, analysis, demonstration, and test. The functional performance of the complete computer system shall be demonstrated after integration into the computer system hardware during all engine testing.

EVALUATION RATIONALE (4.8.2)

The design of software programs for the engine computer resource must be evaluated to ensure that all requirements have been met.

EVALUATION GUIDANCE

Background:

The design evaluation must be accomplished in the early stages of development and must be completed prior to the beginning of overall computer program testing.

Evaluation of software design should be accomplished through the contractor’s quality assurance program in accordance with DOD–STD–2168. Plans, procedures, and test results should be reviewed by the procuring agency. The Higher Order Language (HOL) and structure may be evaluated by inspection. Real–time processing should be evaluated by demonstration and the Built–in–Test (BIT) by test.

EVALUATION LESSONS LEARNED

None.

3.8.2.1 Built–in–test and inspectability Built–in–Test (BIT) shall be implemented to accomplish failure detection and failure isolation to the WRA and LRU without the assistance of support equipment at the organizational level. The testability of the engine system electronics shall be in accordance with MIL–STD–2084. The failure status provided by the BIT to the airframe, in accordance with the Interface Control Document (ICD), for determining mission capability, shall be specified herein. A nonvolatile record of BIT failure data shall be retained in memory. The equipment shall include provisions via data bus to allow both ROM and RAM contents of the subject computer to be inspected (monitored).

The BIT shall monitor both the thrust vectoring nozzle deflection rate and position and shall be capable of detecting failure of the thrust vectoring nozzle.

REQUIREMENT RATIONALE (3.8.2.1)

BIT reduces maintenance downtime, cost, complexity, and personnel requirements and provides the ground crew and pilot the capability to assess the operational usability of the system.

Nozzle rate and position, must be measured to determine nozzle vectoring repeatability and nozzle stability. These parameters must be recorded to detect and quantify degradation in the thrust vectoring nozzle system. The simulation of a thrust vectoring nozzle failure event provides a quick and easy means for the ground crew to assess the fail–safe nozzle position due to hydraulic/electrical failure, or battle damage to the vectoring system and its accompanying components.

REQUIREMENT GUIDANCE

The following should be used for tailoring the specification paragraph: Tasks 103, 104, and 105 of MIL–STD–2084 should be referenced. MIL–STD–2084 should be tailored by the contractor to the requirements of the engine control system (e.g., false alarm rate, isolation levels, test time).

Background:

BIT may operate in a periodic mode, an initiated mode, or in a combination of both modes. The periodic mode is the primary mode of BIT and is intended to provide maximum operational capability and to support rapid aircraft turnaround. Periodic BIT should automatically operate when energized with electrical power. The initial periodic BIT at power–up should be a one time comprehensive system performance check intended to evaluate the system's capability to perform its operational requirements. The initiated mode should accomplish that portion of the failure detection and failure isolation capability which the periodic mode is unable to achieve. Initiated BIT should begin operation upon receipt of an external command, may interrupt normal operation and require operator participation.

REQUIREMENT LESSONS LEARNED

None.

4.8.2.1 Built–in–test and inspectability The requirements of 3.8.2.1 shall be evaluated by demonstration and test. The equipment fault detection and isolation capabilities shall be evaluated per MIL–STD–2165.

EVALUATION RATIONALE (4.8.2.1)

There is a need to evaluate BIT and inspectability requirements.

EVALUATION GUIDANCE

The following should be used for tailoring the specification paragraph: Appendix A, Task 203 of MIL–STD–2165 should be referenced.

EVALUATION LESSONS LEARNED

None.

3.8.2.2 Computer reprogramming. The engine system shall have the computer capability for making software changes without requiring removal of memory hardware, and without altering the aircraft installed configuration. Safety design provisions shall be provided to prevent unauthorized reprogramming. A description of the physical reprogramming procedure shall be provided. Each software version shall be identified by a unique software version number. This number shall be verified during the loading process.

REQUIREMENT RATIONALE (3.8.2.2)

The ability to reprogram computer equipment via the WRA connector is needed to reduce maintenance workload and component removal rates.

REQUIREMENT GUIDANCE

Background:

Reprogramming capability may be through the airframe or engine WRA/LRU.

REQUIREMENT LESSONS LEARNED

Original V–22 reprogramming design required removal and disassembly of the FADEC which required excessive labor. F–18 and AV–8B upgrades to EMS software required block upgrades to the mission computer software. This complication has prevented EMS software upgrades that incorporate lessons learned from being implemented.

4.8.2.2 Computer reprogramming. The requirements of 3.8.2.2 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.8.2.2)

The computer equipment reprogramming via the WRA connector, and safety design provisions preventing unauthorized reprogramming, must be demonstrated and tested.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.8.3 Spare resources. Prior to the completion of the _(a)_ engine Milestone, the electronic fuel control and engine monitoring system shall utilize no greater than _(b)_ and _(c)_ percent, respectively, of the available throughput during the worst case (most heavily loaded) execution of the operational software for each minor frame. The memory reserve shall be at least _(d)_ percent for control systems, and at least _(e)_ percent for engine monitoring systems.

REQUIREMENT RATIONALE (3.8.3)

In order to be the most cost effective, computer reserve capability must be provided early in the development stages of engine software.

REQUIREMENT GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The term IFR, FFR, ISR or OCR (depending on milestone selected).

(b): A value from 33 to 50 percent (depending on milestone selected).

(c): A value from 33 to 50 percent (depending on milestone selected).

(d): A value of 50 percent (depending on the milestone selected).

(e): A value of 50 percent (depending on the milestone selected).

Background:

The required Milestone for memory and throughput reserve capacity should be late in the development program, to allow for as many software changes as possible. Memory and throughput reserve capacity should be incorporated in the design to avoid costly changes or additions to the control microprocessor and memory chips. It may be advisable to specify the reserve capacity for individual memory devices (i.e., EEPROM, RAM, etc.).

The 33 to 50 percent reserve capacity for both memory and throughput should assure reprogramming without disassembly of the control units. Deviation from this requirement requires justification.

REQUIREMENT LESSONS LEARNED

In a late 1980's USAF program, the engine control system had a requirement for 33 1/3% growth capability after ISR. The need for electronic fuel control computational capability has increased considerably over the years. The control has been used as an engine condition monitor to reduce engine power during exhaust gas ingestion, monitor stall margin, control the aircraft inlet ramps during supersonic flight, and control vector nozzle thrust.

4.8.3 Spare resources. The requirements of 3.8.3 shall be evaluated by demonstration and test.

EVALUATION RATIONALE (4.8.3)

There is a need to confirm the software reserve capacity to ensure expandability for the resolution of field problems.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.9 Engine system and controls and externals (C&E) evaluation.

Note: Section 3.9 shall not contain requirements. Section 4.9 shall contain evaluation requirements for engine systems and controls and externals that have not been specified in other sections of this specification.

4.9 Engine system and controls and externals (C&E) evaluation The engine system and C&Es shall be evaluated by tests in accordance with the following paragraphs.

EVALUATION RATIONALE (4.9)

Full–up engine tests must be conducted in order to assess engine performance, operability, and durability capabilities. These tests are conducted to evaluate particular requirements in Section 3. C&E tests must be specified in order to evaluate C&E performance and integrity.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.9.1 Engine systems tests.

4.9.1.1 Endurance tests. The endurance tests in the subparagraphs below shall be required for the class of engines specified herein.

EVALUATION RATIONALE (4.9.1.1)

Endurance tests include the Accelerated Mission Test and the Durability Test to evaluate test flight safety, engine life and effects due to thermal cycling and to simulate accelerated deterioration.

EVALUATION GUIDANCE

The following should be used to tailor section 4.9.1.1:

Section 4.9.1.1 requires an AMT for turbofan/turbojet engines and Durability Tests for all classes of engines. Turbofan/turbojet engines should be tested to evaluate full life requirements of 3.4.1. Additionally, those engines should be required to be tested in accordance with the procedures in the missionized Durability Test (4.9.1.1.2.a), to evaluate the effects of accelerated deterioration at higher than average turbine temperatures.

Turboprop and turboshaft engines should be tested in accordance with the missionized Durability Test, and the Durability Test (4.9.1.1.2.b). Turboprop and turboshaft engines should be tested in accordance with the procedures of 4.9.1.1.2.b, in addition to 4.9.1.1.2.a, for "time–at–temperature" testing of flight regimes and power settings not included in the missionized Durability Test, and incremental torque runs to insure vibration free operation.

Background:

The intent of durability tests at IFR and FFR is to ensure flight safety in limited and unlimited flight envelopes and developmental tests for production configurations. The durability tests for ISR and OCR should be sufficient to prove durability and life throughout the flight envelope and worthiness for low and high rate production.

EVALUATION LESSONS LEARNED

None.

4.9.1.1.1 Accelerated mission test (AMT).

For turbofan and turbojet engines the following tests shall apply:

For IFR:

An engine conforming to the IFR configuration shall be subjected to an AMT of at least (a) hours or equivalent cycles. The test shall consist of the engine accelerated mission duty cycles defined in figure 28, table XX, and table XXXIII.

A minimum of (b) starts shall be made on the engine. There shall be (c) false starts and (d) restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least (e) starts shall be regulated to provide shutdown periods between starts of (f) minutes increasing at (g) minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service.

The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled. The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

For FFR:

An engine conforming to the FFR configuration shall be subjected to an AMT of at least _(a)_ hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined in figure 28, table XX, and table XXXIII. A minimum of _(b)_ starts shall be made on the engine. There shall be _(c)_ false starts and _(d)_ restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least _(e)_ starts shall be regulated to provide shutdown periods between starts of _(f)_ minutes increasing at _(g)_ minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after engine operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service. The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

For ISR:

An engine conforming to the ISR configuration shall be subjected to an AMT of at least _(a)_ hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined in figure 28, table XX, table XXXIII. A minimum of _(b)_ starts shall be made on the engine. There shall be _(c)_ false starts and _(d)_ restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least _(e)_ starts shall be regulated to provide shutdown periods between starts of _(f)_ minutes increasing at _(g)_ minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after engine operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service. The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

For OCR:

An engine conforming to the OCR configuration, but containing the same cold parts used in the ISR test, shall be subjected to an AMT of at least _(a)_ hours or equivalent cycles. The test shall consist of the accelerated mission duty cycles defined in figure 28, table XX, and table XXXIII. A minimum of _(b)_ starts shall be made on the engine. There shall be _(c)_ false starts and _(d)_ restarts. All test starts shall be performed with a starter as specified in 4.10.3.3.4. At least _(e)_ starts shall be regulated to provide shutdown periods between starts of _(f)_ minutes increasing at _(g)_ minute intervals. Each of the regulated shutdown periods shall be preceded by immediate engine shutdown, without being held at Idle, after engine operation for a duration of not less than 30 seconds at IRP at the discretion of the Using Service. The remaining starts shall be conducted in accordance with the temperature stabilization procedure specified below, but otherwise the shutdown period need not be controlled.

The test shall be preceded and followed by the stair–step (HCF) test schedule as specified in 4.4.1.5.1.

The test operation for all engine milestones shall be conducted as follows:

(1) A fixed level of engine inlet air temperature, fuel inlet temperature, and pressure shall be set for each mission cycle as defined by the Strength and Life Analysis Report and defined in figure 28, table XX, and table XXXIII.

(2) Air passing over the engine shall correspond to the engine inlet air temperature for that mission condition, within facility constraints.

(3) Upon completion of each mission cycle, the engine shall be shutdown prior to the start of a subsequent cycle. During the shutdown period the engine shall be motored or windmilled until the (h) measured temperature stabilizes below (i) °C ((i) °F) when the engine is not rotating.

(4) Condition monitoring and diagnostic system installed hardware and software shall be in the production configuration The condition monitoring and diagnostic system shall be operated in a manner analogous to its proposed service operation throughout the test and the output data shall be correlated to observations made during the test.

(5) The customer power takeoff drives and customer bleed air extraction level shall be representative of the maximum required at that condition.

(6) The engine anti–icing/de–icing system shall be actuated at periods during the test to simulate mission operation.

(7) The exhaust duct, inlet duct and bleed air connections shall be loaded to the maximum allowable loads specified in 3.1.1.8 and in the pretest data of 4.10.4.1.1 respectively, at the discretion of the Using Service.

(8) Fuel inlet temperature shall be maintained at the maximum fuel inlet temperature specified in 3.7.3.2.2 and 3.7.3.2.3 for at least (j) percent of the duration of the mission oriented schedule.

(9) The thrust/power demand shall be advanced or retarded in 0.50 seconds or less unless otherwise specified. The time for changing thrust shall be charged to the duration of the lower thrust setting.

(10) The oil pressure, if adjustable, shall be adjusted at the beginning of the test to the minimum steady–state value specified in 3.7.8.1.1. No further adjustments shall be permitted during the test except when authorized by the Using Service.

(11) The total quantity of fluid leakage from the engine drains shall be recorded and shall not exceed values specified in 3.1.1.16.

(12) Special engine control system features (e.g., overspeed, overtemperature) shall be cycled through their functions periodically as specified in the pre–test data of 4.3.5.1.

(13) At the completion of the test, the engine oil drain provisions specified in 3.7.8.2.2 shall be demonstrated.

(14) If the engine provides special features (i.e., IR suppressors, fuel heaters, etc.), these items shall be activated during the test to simulate mission operation.

(15) This test is to be run using the distortion pattern(s) of 3.2.2.10.

(16) The test exhaust system (if different from vehicle flight exhaust) shall produce the same aerodynamics/thermodynamic cycle match as the flight exhaust of the primary using application.

EVALUATION RATIONALE (4.9.1.1.1)

Accelerated mission testing is needed to evaluate life, performance retention, and operability requirements.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

IFR

(a): two times the number of hours or equivalent cycles of the high time flight test engine in the IFR Milestone

(b): the number of starts should be based on the flight test schedule

(c): a value between or including 5% – 10% of the total number of starts

(d): a value between or including 5% – 10% of the total number of starts

(e): a value between or including 5% – 10% of the total number of starts

(f): a value of at least 5 minutes

(g): a value between or including 5 minutes and 15 minutes

(h): T5 or T4.5 (measured/harness temperature)

(i): 100°C (212°F)

(j): a value of at least 10%

FFR

(a): the number of hours equivalent to one hot section life

(b): the number of starts should be based on the flight test schedule

(c): a value between or including 5% – 10% of the total number of starts

(d): a value between or including 5% – 10% of the total number of starts

(e): a value between or including 5% – 10% of the total number of starts

(f): a value of at least 5 minutes

(g): a value between or including 5 minutes and 15 minutes

(h): T5 or T4.5 (measured/harness temperature)

(i): 100°C (212°F)

(j): a value of at least 10%

ISR

(a) the number of hours equivalent to one hot section life

(b): the number of starts should be based on the engine design life

(c): a value between or including 5% – 10% of the total number of starts

(d): a value between or including 5% – 10% of the total number of starts

(e): a value between or including 5% – 10% of the total number of starts

(f): a value of at least 5 minutes

(g): a value between or including 5 minutes and 15 minutes

(h): T5 or T4.5 (measured/harness temperature)

(i): 100°C (212°F)

(j): a value of at least 10%

OCR

(a) the number of hours equivalent to one hot section life

(b): the number of starts should be based on the engine design life

(c): a value between or including 5% – 10% of the total number of starts

(d): a value between or including 5% – 10% of the total number of starts

(e): a value between or including 5% – 10% of the total number of starts

(f): a value of at least 5 minutes

(g): a value between or including 5 minutes and 15 minutes

(h): T5 or T4.5 (measured/harness temperature)

(i): 100°C (212°F)

(j): a value of at least 10%

In table XX the column labeled “Max. Turbine Temp.” may either be deleted or made a reference value as directed by the Using Service.

Background:

A mission oriented cyclic endurance test, also known as an Accelerated Mission Test (AMT), should be conducted on flight test configuration as well as production configuration engines. The purpose of the AMT is to demonstrate that the engine meets the durability and life requirements for the Milestone. That is, the flight test configured engine should be tested to the life and durability required for flight test, and the production configured engine should be tested to the full life requirements of the specification. Normally the life required for flight test is much less than full engine life.

The AMT exposes durability and other structural problems so they can be addressed and solved before a large number of engines enter fleet service. The flight test AMT ensures that the engine is structurally safe to fly.

Each proposed aircraft mission should be analyzed to define the corresponding engine duty cycles for each mission. Traditionally for fighter aircraft there are 8–11 missions. The engine cycles for each mission are then analyzed and the non–damaging cycles are discarded as are the dwell times at part power and Idle. Because the lengthy non–damaging dwell times are eliminated, the number of engine cycles accumulated per AMT test hour is higher than the number of cycles accumulated per flight hour. This is the origin of the term “accelerated”. For example, with a “2:1 acceleration factor” the test time would be one–half of hot section life. Test time may also be expressed in equivalent TAC cycles. Engine durability and life can be evaluated in a shorter time using the AMT than if the specification mission cycles were tested exactly as they are in the specification. Dwell times at Intermediate power or other power settings that cause damage are retained. Augmentor light–offs and augmentor dwell times should be included to obtain the proper number of light–off cycles and amount of time at temperature on the augmentor hardware. To ensure the engine is fully evaluated during the AMT, aircraft bleed and horsepower should be extracted at levels appropriate for the test condition.

Engines with short period ratings (time limit <30 minutes) should include the appropriate time for short period power settings if they are above Intermediate power. The engine missions and engine life should be analyzed to ensure the AMT includes the appropriate number of starts, hot starts, and false starts. After the test engine has been stopped it should be motored or otherwise cooled to simulate the thermal cycle a field engine would experience after it has been stopped, allowed to cool, and restarted.

To analyze and count the engine cycles for each mission, and to ensure the engine is tested for the proper amount of time during the AMT, the following definitions should be used:

Total Accumulated Cycle (TAC)= No. of Type I cycles + (No. of Type III cycles)/4 + (No. of Type IV cycles)/40 + (No. of Minor cycles)/K

where:

A Type I cycle is an engine cycle from off to Intermediate power (or equivalent turboshaft engine power setting), and then back to off.

A Type III cycle is an engine cycle from Intermediate power (or equivalent turboshaft engine power setting), to Idle, and then back to Intermediate power.

A Type IV cycle is an engine cycle from Intermediate power (or equivalent turboshaft engine power setting), to cruise, and then back to Intermediate power.

A Minor cycle is an engine cycle parameter that will be defined by the Using Service based on the engine type and mission usage, where “K” is a value to be determined by the Using Service based on the amount of damage the minor cycle will impose on the engine.

EVALUATION LESSONS LEARNED

None.

4.9.1.1.2a Missionized durability test.

For turbofan, turbojet, turboprop and turboshaft engines, the following tests apply:

For IFR:

An engine conforming to the IFR configuration shall be tested to the missionized cycles of figure 28, table XX, and table XXXIV for (a) hours/cycles. The test shall be conducted so that the (b) turbine temperature shall be equal to (c) °C ((c) °F) at thrust/power settings of IRP/Maximum Continuous and above.

For FFR:

An engine conforming to the FFR configuration shall be tested to the missionized cycles of figure 28, table XX, and table XXXIV for (d) hours/cycles. The test shall be conducted so that the (e) turbine temperature shall be equal to (f) °C ((f) °F) at thrust/power settings of IRP/Maximum Continuous and above.

For ISR:

An engine conforming to the ISR configuration shall be tested to the missionized cycles of figure 29, table XX, and table XXXIV for (g) hours/cycles. The test shall be conducted so that the (h) turbine temperature shall be equal to (i) °C ((i) °F) at thrust/power settings of IRP/Maximum Continuous and above.

For OCR:

An engine conforming to the OCR configuration shall be tested to the missionized cycles of figure 28, table XX, and table XXXIV for (j) hours/cycles. The test shall be conducted so that the (k) turbine temperature shall be equal to (l) °C ((l) °F) at thrust/power settings of IRP/Maximum Continuous and above.

EVALUATION RATIONALE (4.9.1.1.1)

All engine classes require durability testing at operating temperatures that are higher than average to accelerate the effects of deterioration.

EVALUATION GUIDANCE

The following should be used to the tailor the specification paragraph:

(a): the Using Service should determine the number of hours

(b): T4.1 (TRIT), T4.5 (LTIT/FTIT), T5 (TET)

(c): the maximum allowable temperature of 3.2.2.11

(d): the Using Service should determine the number of hours

(e): T4.1 (TRIT), T4.5 (LTIT/FTIT), T5 (TET)

(f): the maximum allowable temperature of 3.2.2.11

(g): the Using Service should determine the number of hours

(h): T4.1 (TRIT), T4.5 (LTIT/FTIT), T5 (TET)

(i): the maximum allowable temperature of 3.2.2.11

(j): the Using Service should determine the number of hours

(k): T4.1 (TRIT), T4.5 (LTIT/FTIT), T5 (TET)

(l): the maximum allowable temperature of 3.2.2.11

For table XX the values in the column labeled “Max. Turbine Temp.” should be the same values as specified in blanks “c”, “f”, “i”, and “l” above.

EVALUATION LESSONS LEARNED

None.

4.9.1.1.2b Cyclic durability test.

For turboshaft and turboprop engines, the following tests shall apply:

For IFR:

An engine conforming to the IFR configuration shall be tested for _(a)_ hours of the durability cycles defined in table XXXIV. The engine shall be tested such that the _(b)_ turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuel/oil of _(c)_ and the shaft torque absorber characteristics of 3.7.17.

For FFR:

An engine conforming to the FFR configuration shall be tested for _(d)_ hours of the durability cycles defined in table XXXIV. The engine shall be tested such that the _(e)_ turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on one engine using the fuel/oil of _(f)_ and the shaft torque absorber characteristics of 3.7.17.

For ISR:

An engine conforming to the ISR configuration shall be tested for _(g)_ hours of the durability cycles defined in table XXXIV. The engine shall be tested such that the _(h)_ turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on two engines, using the fuels/oils of _(i)_, respectively, and the shaft torque absorber characteristics representative of different vehicle applications, as applicable.

For OSR:

An engine conforming to the OSR configuration shall be tested for _(j)_ hours of the durability cycles defined in table XXXIV. The engine shall be tested such that the _(k)_ turbine temperature shall be equal to the maximum operating or control limiting temperature for each power setting.

The durability test shall be performed on two engines using the fuels/oils of _(l)_, respectively, and the shaft torque absorber characteristics representative of different vehicle applications, as applicable.

EVALUATION RATIONALE (4.9.1.1.2b)

In addition to durability missionized testing required in 4.9.1.1.2a, all turboshaft and turboprop engines must also be tested for “time–at–temperature” durability and vibration free operation.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): 30 Hours

(b): T4.1 (TRIT), T4.5 (LTIT/FTIT), or T5 (TET), etc.

(c): the Using Service should reference either the primary fuels paragraph or the test fluids paragraph

(d): 60 Hours

(e): T4.1 (TRIT), T4.5 (LTIT/FTIT), or T5 (TET), etc.

(f): the Using Service should reference either the primary fuels paragraph or the test fluids paragraph

(g): 150 hours

(h): T4.1 (TRIT), T4.5 (LTIT/FTIT), or T5 (TET), etc.

(i): the Using Service should reference two fuel/oil requirements

(j): 300 hours

(k): T4.1 (TRIT), T4.5 (LTIT/FTIT), or T5 (TET), etc.

(l): the Using Service should reference two fuel/oil requirements

Background:

The minimum durability test time to obtain flight clearance in a limited flight test envelope for IFR is 30 hours. The minimum durability test time for flight clearance in the full flight envelope is 60 hours. The minimum durability test time to qualify a production configuration for ISR is 150 hours. The minimum durability test time to qualify for

unlimited production at OCR is 300 hours. These times correspond to 60 hours and 300 hours used on previous (US Army) PFRT and QT milestones. A 1000 hour AMT test may be substituted for the 300 hour durability missionized test at OCR.

Two engines should be tested for both ISR and OCR. These engines should have different fuels and oils (e.g., JP–4/JP–5) and different shaft torque absorber characteristics if intended for multiple vehicles. The torque absorber should represent the upper and lower range of rotor inertia, torque limit, and other characteristics for vehicles of primary interest. Multiple absorber characteristics are primarily for evaluating transient load conditions and developing a common control system.

The durability test in table XXXIV essentially tests time–at–temperature for all power settings at Maximum Continuous and above, incremental torque runs for critical vibrations from Maximum to Idle, and transients from no–load to Maximum and Maximum to no–load. Approximately 60% of the test time is at Maximum Continuous and above. Contingency and other short period emergency power settings are tested at the end of the durability test. Starts and restarts are included in the test.

The AMT option for OCR is to prove durability and production readiness in another manner; i.e., simulating use in an operational vehicle. This test is also useful to predict fleet experience and identify remaining problems before deployment.

LCF testing should be accomplished for all engines at each milestone, either by the AMT test of 4.9.1.1.1, the missionized durability test of 4.9.1.1.2a, or the LCF engine test of 4.4.1.5.2.

EVALUATION LESSONS LEARNED

Turboprop/turboshaft engines have been tested for many engine programs using the durability test cycle requirements of 4.9.1.1.2b. This method of testing has been considered appropriate and time proven for helicopters, small engines, and multiple vehicle applications (commercial and military). The durability test of table XXXIV also has some commonality with commercial FAA tests, an important consideration for this engine class.

4.9.1.2 Altitude test. The performance requirements of 3.2 shall be evaluated by test. An engine conforming to the same parts list and configuration as the endurance test engine of 4.9.1.1, in accordance with 4.10.3.2, shall be subjected to altitude tests at several selected conditions within the operating limits envelope specified in figure 9 (P_2/T_2). Pre–test data shall be in accordance with 4.10.4.1.1. The test points shall include the effects of power extraction, bleed air extraction, inlet recovery, inlet distortion, anti–icing/de–icing, and windmilling on engine performance and stability. No adjustments shall be made to the engine without approval of the Using Service. The altitude tests shall be accomplished using the fuels and oils of 3.7.3.1.1 and 3.7.8.1, respectively. Fuel temperature shall be varied over a range sufficient to encompass all anticipated engine operating environments. If the engine has a continuous ignition system, it shall be in operation at all times after the start sequence has been completed. Data to be taken and recorded during the test shall be as specified in 4.10.4. Overall true RMS velocity measurements and acceleration spectrograms shall be obtained for each transducer mounted on the engine case and engine gearbox case at the engine speeds and thrusts selected for the test. The points selected shall include at least the altitude representative points and the point in the complete environmental conditions and operating envelope where the highest engine vibrational levels are generated. Critical components of the engine shall be identified on each spectrogram.

The following altitude test procedures apply to each of the following engine qualification milestones.

For IFR:

a. Altitude Performance Points. The tests shall consist of no fewer than (a) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than _(b)_ flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of _(c)_ operating conditions and _(d)_ distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than _(e)_ core turnaround speeds as defined by the Using Service. A success rate of _(f)_ shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at _(g)_ altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at _(h)_ flight test condition(s) selected by the Using Service.

For FFR

a. Altitude Performance Points. The tests shall consist of no fewer than _(aa)_ flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than _(bb)_ flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power, and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of _(cc)_ operating conditions and _(dd)_ distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than _(ee)_ core turnaround speeds as defined by the Using Service. A success rate of _(ff)_ shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at (gg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at (hh) flight test condition(s) selected by the Using Service.

For ISR

a. Altitude Performance Points. The tests shall consist of no fewer than (aaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than (bbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power, and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of (ccc) operating conditions and (ddd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eee) core turnaround speeds as defined by the Using Service. A success rate of (fff) shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at (ggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at (hhh) flight test condition(s) selected by the Using Service.

For OCR

a. Altitude Performance Points. The tests shall consist of no fewer than (aaaa) flight altitude condition performance points selected by the Using Service. An adequate number of engine thrust/power settings as determined by the Using Service shall be selected for each specified altitude test condition to establish operating and performance characteristics at each flight altitude condition tested. Operation at each test point shall be of sufficient duration to stabilize the engine.

b. Transient Operation and Functional Test. The requirements of 3.2.2.5 and 3.2.2.6 shall be evaluated at no fewer than (bbbb) flight altitude conditions selected by the Using Service. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated. The time elapsed versus engine speeds, measured pressures and temperatures, fuel flows, thrust/power, and facility operating conditions shall be obtained for stability evaluation over the entire thrust/power range. Transients shall be repeated at selected flight conditions to explore control system lags, repeatability, and performance droop. Augmentor light–off limits of 3.2.2.1 shall be tested.

c. Inlet distortion. The requirements of 3.2.2.10 shall be evaluated by engine operation with simulated inlet airflow distortion at selected operating conditions chosen by the Using Service. Engine stability shall be evaluated and stall margin remaining defined by engine test at a minimum of (cccc) operating conditions and (dddd) distortion patterns selected by the Using Service that combine worst case engine destabilizing influences. Effects of bursts, chops, and bodies throughout the thrust/power range shall be demonstrated.

d. Starts and Restarts. Sufficient engine start and restart testing shall be accomplished to evaluate the starting requirements of 3.7.11 and 3.7.11.2. If the automatic relight feature of 3.7.11.5 is provided, it shall be evaluated. The automatic relight feature may be disabled to allow evaluation of restart requirements with the approval of the Using Service. Each spooldown airstart test condition shall be accomplished at no fewer than (eeee) core turnaround speeds as defined by the Using Service. A success rate of (ffff) shall be used to establish successful airstarting and restarting at any given condition.

e. Windmilling Test. The requirements of 3.2.2.7 shall be evaluated at (gggg) altitude and Mach condition(s) as selected by the Using Service.

f. Stopping. The requirements of 3.2.2.3 shall be evaluated each time the engine is stopped and at (hhhh) flight test condition(s) selected by the Using Service.

EVALUATION RATIONALE (4.9.1.2)

Altitude testing is required to evaluate engine performance and operating capability within the specified engine operating envelope for each milestone.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

Background:

The contractor should complete the blanks. The large range of values for the blanks takes into account the wide variety of engine types and vehicle applications for which the specification is written (i.e., trainer, transport, turboshaft/turboprop, and augmented turbofans). A fighter engine, for example, will have more flight conditions and distortion pattern than would a trainer engine.

The following are expected input for tailoring of the specification paragraph by the contractor:

IFR: (Determine performance/operability/safety in selected portion of flight envelope.)

(a): 10 – 20 flight altitude/Mach/inlet temperature combinations should be tested for performance. Rotorcraft applications should use a minimum of 3 – 5 conditions.

(b): 10 – 20 altitude/Mach conditions should be tested for transients and operability.

(c): 12–21 flight altitude/Mach number/distortion/power demand.

(d): 4 – 7 distortion patterns should be tested at the pattern's design flight condition and other minimum surge margin conditions.

(e): 2 – 4 core turnaround speeds should be identified.

(f): a success rate of 2 out of 3 or 3 out of 4 should be required.

(g): a minimum of one (1) altitude and Mach number condition should be tested for windmilling.

(h): 1 – 2 test condition(s) including a high flight Mach number, high/low inlet ram temperature, and high/low inlet ram pressure, as appropriate, should be tested.

FFR: (Determine performance/operability/safety throughout full flight envelope.)

(aa): 15 – 30 flight altitude/Mach/inlet temperature combinations shall be tested for performance. Vary number with engine type and application. Rotorcraft applications should use a minimum of 3 – 5 conditions.

(bb): 10 – 20 altitude/Mach condition points should be tested for transients and operability.

(cc): 3 operating conditions per distortion pattern

(dd): All distortion patterns specified by 3.2.2.10 that have not been tested should be tested at the pattern's design flight condition and other minimum surge margin conditions.

(ee): 2 – 4 core turnaround speeds should be identified.

(ff): A success rate of 2 out of 3 or 3 out of 4 should be required.

(gg): a minimum of one (1) altitude and Mach number for windmilling. (a flight condition different from the IFR condition should be used.)

(hh): 1–3 test condition(s) should be tested at high flight Mach numbers, high and low inlet ram temperatures, and high and low inlet ram pressures.

ISR: (Repeat altitude performance/operability testing for engines having changes in configuration, materials, exhaust system, etc. Omit altitude performance/operability testing, if the ISR configuration or other changes do not affect performance or operability.)

(aaa): 15–30 flight altitude/Mach/inlet temperature combinations should be tested for performance. Rotorcraft applications should use a minimum of 5 – 10 conditions.

(bbb): 15–30 altitude/Mach condition performance points should be tested for transients and operability.

(ccc): 3 operating conditions per distortion pattern

(ddd): All distortion patterns specified by 3.2.2.10 that have not been tested should be tested at the pattern's design flight condition and other minimum surge margin conditions.

(eee): 2 – 4 core turnaround speeds should be identified.

(fff): A success rate of 2 out of 3 or 3 out of 4 should be required.

(ggg): One (1) altitude and Mach number condition should be tested for windmilling (only if redesign of oil sump and bearing cavity areas has occurred). A flight condition different from the other than IFR and FFR conditions should be used).

(hhh): 1–2 conditions at high flight Mach number, high and low inlet ram temperature and high and low inlet ram pressure condition(s) should be tested.

OCR: (Altitude testing should not be needed in this Milestone as performance and operability should be completely defined in prior milestones. However, if numerous design changes are made during the ISR Milestone another altitude test may be required.)

(aaaa) – (hhhh): TBD

Background:

The intent of the qualification milestones is to have performance and operability of the engine defined throughout the flight envelope by FFR. Subsequent milestones, ISR and OCR, are to qualify the engine configuration for production. Engines which change configuration, materials, parts, devices, exhaust systems, etc. between milestones may need to repeat altitude testing on later milestones to reconfirm/determine performance. Flight envelope conditions should consider alternate flight vehicles/applications at the discretion of the Using Service.

The flight conditions selected should serve to define engine operating line performance and operability over a range of altitude, Mach, and inlet ram temperature conditions. The number of conditions can vary with engine type or vehicle application. The flight conditions should include, but are not limited to table II performance points.

The number of flight conditions shown for sub–paragraphs a. and b. (above) at each milestone are not cumulative, but are the recommended number for that particular milestone, taking into consideration the number of conditions tested for a previous milestone. For example, at IFR, 15 flight conditions may be tested. At FFR a trainer or turboprop engine may only need an additional 10 conditions, whereas a fighter engine may need an additional 15 or 20. The number of flight conditions at ISR may also be related to the number of configuration changes made between FFR and ISR. If the ISR engine is essentially the same as the FFR engine, little or no additional testing may be needed, whereas an engine with several major configuration changes may need testing throughout the entire flight envelope.

A similar reasoning was used for the distortion patterns. For IFR the patterns that most affect safety, or patterns that may occur during flight test should be tested. During FFR, the number of additional patterns that may need testing is related to the engine type, size of the flight envelope and/or the extent of flight testing that is proposed (i.e., severe maneuvers that generate high distortion). For ISR, most, if not all, of the patterns described/shown in the specification should be tested. Exceptions may be patterns that are benign or patterns that have been tested for IFR and FFR and no major engine configuration changes have occurred between FFR and ISR.

Transient performance tests should include the following:

(a) Accels from various low power settings to high power at low and high altitude.

(b) Decels from high power setting(s) to Idle at low and high altitude.

(c) Perturbations at various low power settings at low altitude.

(d) Other transients as needed for inflight maneuvering or vehicle control.

(e) Transients as needed for augmentor lightoff.

(f) Transients as needed for loaded/unloaded turboshaft rotors. Transient performance should be repeated at selected flight conditions to investigate control system lags, repeatability, and performance droop (e.g., helicopter rotor, cold engine launch, hot day takeoff, etc.).

The number of start and restart flight conditions (including spooldown airstarts) should continually increase through ISR. IFR tests may not include much airstart testing, and FFR tests may only contain a few points at low risk flight conditions. ISR tests should map out the full airstart region to generate a data base that will be used to develop the aircraft flight manual. For airstart tests not only will the flight condition change, but so will the turnaround speeds (i.e., 25%, 40%, and 50% core rotor speed).

The windmilling test may take a long time to accomplish depending on the specification requirement. Windmilling may not be required for IFR depending on the program and safety requirements at the test site. Some commands require that initial flights occur within gliding distance of the airfield in case the engine fails. Depending on the number of windmilling flight conditions required in the specification, each milestone test should occur at different flight conditions, as long as major configuration changes have not occurred.

The test to evaluate stopping will occur during each test session, but may only be critical at a high altitude high Mach condition where rotor seizure may occur. Depending upon the thoroughness of the flight test program, an in–flight stopping test may not be accomplished until FFR or even ISR. Flight test facility safety guidelines may require an in–flight stopping test at an early milestone.

Demonstration of engine operation with the control system in control failure modes should be accomplished.

EVALUATION LESSONS LEARNED

None.

4.9.1.3 Engine systems tests pass/fail criteria.

The test shall be satisfactorily completed when the following conditions are met:

a. The engine shall meet the pass/fail criteria of the test plan.

b. The engine shall meet the inspection criteria of 4.10.3.5.5.

c. The engine and engine components meet the service limits of 4.10.3.4.1.

d. There are no part failures or any parts failing integrity requirements of 3.4.

e. All failures and problems encountered during the test, and any additional testing that may be required has been analyzed and resolved to the satisfaction of the Using Service.(i.e., penalty runs)

f. The test engine and all initial (start of test) components are operated satisfactorily to the end of the test and an analysis utilizing engine calibration results establishes that the performance characteristics of 3.2 and the starting requirements of 3.7.11 have been met.

g. Inspection of engine parts and engine components throughout the test, including the final inspection, shows that the parts life requirements of the Milestone have been met.

h. Test results indicate compliance with the reliability and maintainability requirements specified in 3.5.

i. The engine weight meets the requirements of 3.1.2.1.

j. The engine has not exceeded any limits of table VIII.

EVALUATION RATIONALE (4.9.1.3)

Pass/fail criteria is needed to establish a condition by which to approve test results.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.9.2 Controls and externals (C&E) component tests.

4.9.2.1 C&E Component operational tests. Simulated mission operational tests to evaluate the life specified in 3.4.1.1 shall be conducted on all C&E systems and components. The ignition system shall be tested in accordance with 3.7.5. All C&E tests shall be conducted on test assemblies that have the same part number, consisting of groups of

related components arranged and interconnected to simulate their normal relationship and function on the engine. However, subassemblies or components of a system may be tested separately if such separation does not prevent simulation of the complete function of the components or subassemblies. If practical, components shall be mounted in their normal position as mounted on the engine. No adjustments shall be made subsequent to the component calibration.

EVALUATION RATIONALE (4.9.2.1)

Missionized testing must be accomplished to assure component subassemblies/components capability and reliability to satisfy weapon system mission requirements.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.1 C&E component calibration. Prior to the initiation of the C&E tests, each component which has input–output relationships shall be subjected to a calibration. The calibration shall be extensive enough to cover the engine steady–state and dynamic ranges of operation of the component and shall indicate conformance with the design tolerance range of the component. The engine control components shall be shown to conform to accuracy, stability, and response requirement(s) stated in the component test procedure. Each calibration shall be recorded. Prior to initiation of the calibration, the contractor shall develop a list of components that do not require calibration. This list shall be subject to approval by the Using Service. Components not requiring calibration shall be operated under normal operating conditions to demonstrate satisfactory functioning.

EVALUATION RATIONALE (4.9.2.1.1)

Components and their subassemblies must be calibrated to ascertain accuracies and establish proper input/output relationships prior to using these systems in a test.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.2 C&E component test procedures. Prior to the start of any testing, all components shall be cleaned of any corrosion–preventive compounds used for storage protection to simulate the condition of the part as installed on the engine. Test assemblies or components shall be subjected to operating loads simulating those encountered on the engine. Sufficient instrumentation shall be provided to indicate the performance of each component and to indicate that the functional relationships of components are maintained as required by the applicable test schedule. Functional checks shall be performed at the end of each test or group of tests and at other times, at the option of the Contractor, to indicate that no calibrated component has changed its calibration beyond allowable service limits and that the function of uncalibrated components is unimpaired. All components shall be tested with the primary operating fluids of 3.7.3.1.1, 3.7.8.1, and 3.7.9. All items driven by the engine gearbox shall be operated at maximum allowable axial and angular misalignment conditions at the drive pad. All components shall be subjected to the vibration test of 4.9.2.1.8.

EVALUATION RATIONALE (4.9.2.1.2)

Component subassemblies/components must be the correct part number intended for use during the appropriate milestone. The component should be tested in the as–installed condition and exposed to engine operating loads and temperatures. This is to ensure the component test simulates the engine operating environment.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.3 C&E component missionized tests. All engine components, excluding the ignition system, shall be subjected a simulated mission operational tests (or cycles) intended to produce the same equivalent component damage as would be achieved by real time tests based on the weapon system mission and mission mix of table VI. The simulated mission operational test cycle shall be defined by the contractor and the method of development shall be specified herein. The test shall include maximum and minimum fluid temperatures expected during engine operation and shall include maximum hot and cold soak temperatures of figure 4.

Components utilizing bleed air or requiring pneumatic input signals shall be subjected to air at pressure and temperature values corresponding to those occurring throughout the range of engine operation. A sufficient number of test cycles, interspersed throughout the test, shall include air contaminated as follows:

(1) The amount of engine lubricating oil shall be consistent with what has leaked into the flow path in previous engine testing.

(2) A salt concentration of 0.2 parts salt (NaCl) per million parts of air by weight (salt shall be introduced using 4.0 percent water solution).

(3) Distilled water to saturate the air at 52°C (125°F) at an ambient pressure of 14.7 psia.

(4) Sand and dust concentration and particle size distribution of 3.3.2.4.

EVALUATION RATIONALE (4.9.2.1.3)

Missionized testing for components is needed to preclude failures that may occur in service. Missionized testing more accurately simulates the expected operating environment than non–missionized tests. Pneumatic air C&E components must not be affected by contaminants in the gas path flow.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

Test cycles shall be consistent with the following requirements:

a. Components in test assemblies shall function in their normal sequence of operation on the engine.

b. Cycling shall be controlled by varying simulated inputs to the test assembly or component. Pilot–controlled inputs, such as power settings, shall be varied continuously as required by the test cycle and without regards to variables having reached the demanded values. Engine–supplied inputs shall be varied in their usual relations to component outputs.

c. Input variables substantially independent of other control inputs, such as altitude pressure, shall be cycled at a rate faster or slower than the basic functional cycle in order that every component shall eventually have accomplished each part of its function at each value of the independent variables.

d. When a component provides for a manual or automatic transfer from one mode of operation to another, the manual or automatic means shall be used to obtain transfer an appropriate number of times during the test.

e. Components designed to prevent the engine from exceeding its operating limits, but are not actuated in normal operation, shall be actuated an appropriate number of times with their input variables set to reach the necessary range of values.

Background:

Test cycles should be developed and the method used to create the cycles should be provided by the contractor for review and approval by the Using Service.

EVALUATION LESSONS LEARNED

Compressor discharge and combustor pressures used in hydromechanical fuel control schedules have experienced contamination problems that caused incorrect scheduling of fuel to the engine. Contaminated bleed air has caused compressor bleed actuator failures.

4.9.2.1.4 C&E component accelerated aging. Upon completion of the C&E calibrations, all components containing age–sensitive non–metallic parts shall be placed in an air oven and maintained in an ambient temperature not less than the sea level hot ground soak temperature of figure 4 for (a) hours. Components may be aged individually or in test assemblies.

EVALUATION RATIONALE (4.9.2.1.4)

Components with age sensitive parts must be subjected to an aging test to ensure they will meet the full life required in the specification.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): A value of at least 168 hours, longer time periods are preferred.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.5 C&E component recalibration, disassembly, and inspection. Upon completion of the preceding tests, component calibrations of 4.9.2.1.1 shall be repeated. Components not subjected to calibration shall be operated under normal operating conditions to demonstrate satisfactory functioning. During recalibration, the same fluids, inputs and operating cycles shall be used as in the calibration. Each recalibration shall be recorded. All components shall then be completely disassembled and inspected for indications of failure or excessive wear.

EVALUATION RATIONALE (4.9.2.1.5)

Component recalibrations are necessary to ensure that no component has changed its calibration beyond allowable service limits.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.6 C&E component environmental test. All engine electrical components or sub–components, including electrical connectors shall be subjected to environmental component testing in accordance with table XXXV and as noted herein. These tests may be conducted on test assemblies or individual components, either of which may be new or not previously subjected to any testing. When a single hermetically sealed component is used in a series of the tests below, the component need not be disassembled for inspection until the last test of such series has been completed. In addition, hermetically sealed components need not be subjected to the explosion–proof, sand and dust, and fungus tests. Prior to disassembly, a test to determine hermetic seal integrity shall be performed. Failure of the hermetic seals during any test shall disqualify that component. The components shall be inspected for defects or damage which may have been incurred during any of the tests performed.

EVALUATION RATIONALE (4.9.2.1.6)

Environmental testing must be conducted on components to ensure they will survive real world environments.

EVALUATION GUIDANCE

Background:

The same serial number component should be subjected to all the environmental component tests.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.7 C&E component sand and dust. All electronic components shall be subjected to a sand and dust test in accordance with MIL–STD–810. At the conclusion of the test, and prior to any cleaning of the components, four power supply transients in accordance with MIL–STD–704 shall be applied to the component. Criteria for passing the test shall be as defined in MIL–STD–810.

EVALUATION RATIONALE (4.9.2.1.7)

Components must be exposed to a sand and dust test to ascertain the ability of the component to resist dust particles which may penetrate into cracks, crevices, and joints and to assess the degradation of the component's performance, reliability, and maintainability.

EVALUATION GUIDANCE

Background:

Components may be operated during the test based on the use requirements of the specific test item.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.8 C&E component vibration. All components shall be subjected to a vibration test in accordance with MIL–STD–810. Components may be tested in test assemblies or as individual units. During the test the component shall be subjected to its maximum limiting component temperature specified herein.

EVALUATION RATIONALE (4.9.2.1.8)

Vibration testing must be performed to determine the resistance of components to vibrational stresses expected in its shipping and operating environments.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.9.2.1.9 C&E component test pass/fail criteria. The component tests shall be considered to be satisfactorily completed when, in the judgment of the Using Service:

a. During the tests, component performance and function were within established limits.

b. During the tests, there was no fluid leakage from any component other than that of a nature and rate specified in the specification.

c. During the tests, there was no hang–up or hesitation of any component.

d. Recalibrations indicate that no component has changed its calibration beyond allowable service limits.

e. The component teardown inspection shows no indication of failed, excessively worn, and distorted parts. Measurements shall be taken and compared with the contractor's drawing dimensions and tolerances or with similar measurements made prior to the test.

EVALUATION RATIONALE (4.9.2.1.9)

The success or failure of the test needs to be assessed in accordance with pass/fail criteria.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

3.10 Engine qualification. For the purposes of qualifying the engine during Engineering and Manufacturing Development (E&MD), the engine shall be evaluated in accordance with Section 4.10.

Note: Section 3.10 shall not contain requirements. Section 4.10 shall contain evaluation requirements for engine tests that have not been specified in other sections of this specification.

REQUIREMENT RATIONALE (3.10)

None.

REQUIREMENT GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10 Engine qualification. The engine shall be evaluated during each engine milestone by analysis, inspection, demonstration, and test in accordance with this section, and table XXXVI. Each evaluation in each milestone must pass its respective criteria and gain approval of the Using Service, prior to proceeding to the next milestone. All evaluations shall have quantified pass–fail criteria based on expected failure modes and this criteria shall be approved by the Using Service prior to initiation of each milestone.

EVALUATION RATIONALE (4.10)

This section and table XXXVI establishes the test methods, procedures, and criteria that must be satisfactorily achieved for each engine milestone in order for the contractor to qualify the engine and to allow him to proceed to the next milestone or production status.

EVALUATION GUIDANCE

Background:

Section 4.10 is for the following purposes: 1) to define the four qualification milestones in engine development, 2) to define what the contractor must accomplish in order to progress through each milestone, 3) to provide a table that clearly outlines types of evaluations required in each milestone, and 4) to provide generic evaluation test methodology to be used, where applicable, for tests throughout section 4.

EVALUATION LESSONS LEARNED

None.

4.10.1 Qualification milestones. The engine shall be evaluated in accordance with table XXXVI for satisfactory completion of each of the engine qualification milestones defined below.

EVALUATION RATIONALE (4.10.1)

Each milestone must be defined to establish the purpose of its phase in development qualification.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.1.1 Initial flight release (IFR). The IFR milestone shall evaluate safe flight capability in a specific portion of the flight envelope. The acceptability of the engine design for IFR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVI and the approval of the Using Service.

EVALUATION RATIONALE (4.10.1.1)

IFR is conducted to provide the Using Service and the contractor limited performance and durability evaluation of the engine for initial flight testing. The engine must meet limited performance, operability, or durability requirements prior to completion of this milestone.

EVALUATION GUIDANCE

The following should be used for tailoring table XXXVI and the appropriate section 4 paragraphs:

The initial flight release engine should complete a minimum durability demonstration which is a specified multiple of the anticipated flight test usage.

This milestone was used as a Demonstration/Validation milestone in a late 1980’s Navy program. The intent was to possibly waive this milestone as an initial development milestone if the Using Service felt that the engine configuration was mature enough to go directly into FFR.

4.10.1.2 Full flight release (FFR). The FFR milestone shall evaluate the engine's operational and performance capability throughout the entire flight envelope. The acceptability of the engine design for FFR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVI and the approval of the Using Service.

EVALUATION RATIONALE (4.10.1.2)

This milestone establishes the acceptability of the engine to power the aircraft throughout its full envelope. The engine is not required to meet full evaluation requirements for durability and reliability and is not required to be the final production configuration (the engine should be close to or identical with the final production configuration).

EVALUATION GUIDANCE

The following should be used to tailor table XXXVI and the section 4 paragraphs:

The engine should complete a limited durability demonstration which consists of a specified multiple of the anticipated flight test usage.

EVALUATION LESSONS LEARNED

None.

4.10.1.3 Initial service release (ISR). The ISR milestone shall evaluate the initial operational capability and production configuration of the engine. The acceptability of the engine design for ISR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVI and the approval of the Using Service.

EVALUATION RATIONALE (4.10.1.3)

This milestone establishes the acceptability of the engine for low rate production release.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.1.4 Operational capability release (OCR). The OCR milestone shall establish the production engine configuration for unlimited production release. The acceptability of the engine design for OCR shall be predicated on satisfactory completion of the evaluation requirements of section 4 in accordance with table XXXVI and the approval of the Using Service.

EVALUATION RATIONALE (4.10.1.4)

None.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.2 Evaluation responsibility. The contractor is responsible for the performance of all evaluations (inspections, analyses, demonstrations, and tests) as specified herein. The contractor may use any facility suitable for the performance of the evaluation specified herein, unless disapproved by the Using Service. The Using Service reserves the right to perform any of the evaluations set forth in this specification when necessary to ensure supplies and services conform to prescribed provisions. Specific evaluations required may be waived by the Using Service provided the contractor submits documentation with appropriate justification for approval.

EVALUATION RATIONALE (4.10.2)

The Department of Defense policy is to ensure that responsibility rests upon the contractors for controlling product quality.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

Parts have been accepted by similarity and Government test results have sometimes differed from those of the contractor.

4.10.2.1 Responsibility for compliance. All articles shall meet the requirements of sections 3, 4, and 5. The evaluations set forth in this specification shall become a part of the contractor's overall design and product evaluation program. The absence of any evaluation requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Using Service for acceptance comply with all requirements of the contract. Sampling inspections, as part of manufacturing operations, are an acceptable practice to ascertain conformance to the requirements, however, this does not authorize submission of defective parts, either indicated or actual, nor does it commit the Using Service to accept defective material.

EVALUATION RATIONALE (4.10.2.1)

The Department of Defense policy of quality assurance is predicated on the fact that responsibility rests upon the contractors for controlling product quality and for offering to the military services for acceptance only those items or lots of items that conform to all contractual requirements. This paragraph is a MIL–STD–961 requirement.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.2.2 Quality conformance inspection. Test articles and test apparatus shall be subject to inspection by authorized Government officials who will be given the necessary information to determine conformance with this specification.

EVALUATION RATIONALE (4.10.2.2)

There is a need for the Using Service to make an independent inspection of the test article and test apparatus.

EVALUATION GUIDANCE

Background:

The Using Service should determine the information, timing, and facilities required for the inspection.

EVALUATION LESSONS LEARNED

In past inspections the Using Service has found problems (wear, cracks, etc.) the contractor did not detect.

4.10.2.2.1 Analysis inspection. Computer programs and calculations used in all analyses shall be subject to review and approval by the Using Service.

EVALUATION RATIONALE (4.10.2.2.1)

There is a need to review the analyses methods to determine how the assumptions and methods are applied. This will give the Using Service a confidence factor for the analyses. The Using Service will be able to check the validity of the analysis.

EVALUATION GUIDANCE

Background:

The Using Service should use their best analytical experts to determine the validity of the analyses and possible errors in the analyses.

EVALUATION LESSONS LEARNED

In past stress analyses programs, improper assumptions were used which produced results significantly lower than actual test results, causing costly retrofits.

4.10.3 Manner of testing. Unless otherwise noted, the following testing methods, conditions, and procedures shall be required for evaluations for full engine system qualification in accordance with table XXXVI.

EVALUATION RATIONALE (4.10.3)

None.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.3.1 Test surveillance. Each test and demonstration herein shall be subject to witnessing by authorized Using Service representatives. At convenient times prior to the tests and during teardown inspections, the engine and components shall be examined to determine if they conform to all requirements of the contract and specifications under which they were built. At no time shall any part of the engine or component be disassembled, adjusted, cleaned, replaced, or removed without prior approval of the Using Service representative.

EVALUATION RATIONALE (4.10.3.1)

All tests and demonstrations need an authorized Using Service representative to insure that all the requirements for testing are carried out according to the model specification. The Using Service representatives are needed to do an independent examination of engine parts at the designated inspection periods. This also prevents any unauthorized tampering with the engine without Government approval.

EVALUATION GUIDANCE

Background:

The Using Service should not rely on the contractor for proper conduct of a test or demonstration. There should be a Using Service representative witnessing all tests and engine inspections. The Using Service can perform its own independent inspections, if appropriate.

EVALUATION LESSONS LEARNED

The engine contractor usually is more optimistic about reporting the results of a test or demonstration. There were times in the past where the test results were interpreted differently by the Using Service. On some occasions, engine parts have been replaced during a test. This means the parts were not fully tested as required.

4.10.3.2 Test article configuration and parts list. Prior to the initiation of IFR, FFR, ISR, and OCR, the contractor shall submit a proposed parts list, which identifies the specific configuration, including any software logic version, for each test article. This parts list shall define the proposed engine configuration for each respective milestone. The parts list and configuration of each milestone test article and the differences from the proposed milestone parts list and configuration shall be identified and justified in the pretest data of 4.10.4.1.1 and shall be included in the detailed test procedure. The test article configuration for each evaluation conducted in accordance with table XXXVI shall be the approved engine qualification milestone parts list and configuration. The evaluations shall be modified, as appropriate, due to changes incorporated in the approved parts list and configuration, when required by the Using Service. The parts list for the engine configuration which successfully completes the IFR, FFR, ISR, and OCR evaluations shall be the approved parts list for the respective engine model. The mixing of parts of the same or different design such as different vendors components and parts in a multiple assembly like a segmented stator assembly or different designed blades in a disc is not allowed without Using Service approval. All parts shall be considered as having zero time at the start of a test, as appropriate, unless otherwise specified by the Using Service. While all parts do not have to be new, any part which fails the pass–fail criteria during the test shall be cause for rejection of that part or test article as determined by the Using Service. Both hardware and software are considered to be test articles. Hardware configuration shall be identified by part number and associated drawings. Software configuration shall be identified by a specific source code listing and associated logic diagrams.

EVALUATION RATIONALE (4.10.3.2)

A test article configuration and parts list is needed for each milestone. This gives the Using Service a record of test article changes made during the development phase. The contractor is prohibited from mixing parts for comparison purposes because the tests and demonstrations are not meant for the contractor to perform parts development. Deviation parts list is required to tell how configurations differ at all milestones. This includes any software version changes.

EVALUATION GUIDANCE

Background:

The test article configuration and parts list should always include all the latest proposed parts for the particular milestone. There should not be any interim parts from previous models or milestones just to maintain the development schedule, without Using Service approval.

EVALUATION LESSONS LEARNED

In the past, when a contractor was behind schedule, he substituted or used parts that were not intended for the configuration. When these parts failed, the explanation used was that they were not the official parts. This is not a cost effective manner to conduct an engine development.

4.10.3.2.1 Test article. The particular article intended for a specific milestone test or demonstration shall be officially designated by the contractor prior to the start of the test.

EVALUATION RATIONALE (4.10.3.2.1)

There is a need to know a particular article configuration for a specific test. This allows the Using Service to review the test article and determine its suitability for the test.

EVALUATION GUIDANCE

Background:

The official designated test article for a particular test should be reviewed to determine if it is configured to pass the test at the expense of other performance characteristics. Future DEC logic will be easily programmable "on–wing". Therefore, source code listing(s) should be provided for all software and all deviations from these listing(s) and/or changes made to the software, prior to or during testing, should be specified.

EVALUATION LESSONS LEARNED

Some engines in the past were configured to do well in a particular test. For example, an engine combustor and fuel schedule can be configured to pass a smoke test, though at the expense of the windmill air start envelope. Also, engine parts can be selected within their manufacturing tolerance range to increase the engines capability to pass a given test. For example, proper selection of first stage turbine area can lower the compressor operating line to give additional stall margin at the expense of compressor efficiency.

Past accidental reprogramming of digital electronic control logic has occurred on breadboard controls in the test cell. Electrical control logic has been easily programmable in the test cell which has resulted in undocumented software changes.

4.10.3.2.2 Installation interfaces. Inlet, exhaust, and air bleed connections shall be representative of airframe/engine installed environments. The engine power takeoff (PTO) shall duplicate the load distribution defined in 3.7.19

EVALUATION RATIONALE (4.10.3.2.2)

The test article interfaces and environment must be representative of the installed/airframe environment.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.3.3 Test equipment.

4.10.3.3.1 Automatic recording equipment. Automatic steady–state and transient recording equipment shall be used to record data during the execution of those parts of the tests required to meet the objectives of each evaluation. Automatic continuous recording equipment shall be used for tests requiring the evaluation of test article variables versus time.

EVALUATION RATIONALE (4.10.3.3.1)

It is necessary to use automatic continuous recording equipment when a measured parameter changes with time. This is to verify transient operability requirements. Steady–state data is needed for calculations to verify test article performance requirements.

EVALUATION GUIDANCE

Background:

The word continuous should not be taken literally. Only analog equipment monitors continually. Digital equipment is considered continuous when it uses a high sampling rate. The sampling rate is determined by the type of test. A very high sampling rate may pick up extraneous measurements while another test may require a very high rate or electrical scanning circuitry.

EVALUATION LESSONS LEARNED

Most automatic recording equipment used presently is digital equipment. This equipment works with a scanning valve or electrical, scanning circuitry. More parameters result in a smaller sampling rate or longer time between consecutive samples. This method is fine during an engine acceleration that may take four or more seconds. A test that may measure engine parameters during an engine surge will require a very high sampling rate. This reduces the number of parameters that can be recorded.

4.10.3.3.2 Vibration measuring equipment and response characteristics. The vibration measurement and analysis equipment shall operate over a frequency band of at least 3 Hz to 10 kHz and produce acceleration spectrograms with a confidence level of 95 percent. For turbojet and turbofan engines the maximum allowable effective filter bandwidth of the spectrum analysis equipment shall be 30 Hz. For turboshaft and turboprop engines, the maximum allowable effective filter bandwidth of the spectrum analysis equipment shall be 3 Hz up to 1,000 Hz and 6 Hz above 1,000 Hz. The equipment shall be calibrated as a complete system. The system frequency response, when calibrated by applying a known sinusoidal motion to the pickup, shall not deviate by more than $\pm$3 dB from the known sinusoidal input at frequencies from 3 Hz to 10 kHz. If high bypass filters are required when measuring overall velocity levels, the filters shall not exceed 3 dB down at frequencies of 30, 70, or 110 Hz, as appropriate, with a roll–off of at least 18 dB per octave. High pass filters shall not be used to produce velocity/acceleration spectrograms.

EVALUATION RATIONALE (4.10.3.3.2)

Engine vibration measurement is one of the necessary parameters used to monitor the health of the engine. Response characteristics are needed to control the type of measurement equipment used.

EVALUATION GUIDANCE

Background:

A sudden increase in the level of vibration can signal an existing or an impending engine failure. A gradual increase in vibration level may mean parts wearing. The different filter characteristics for each engine class assumes turbofan and turbojet engines have larger physical core sizes and lower rotational speeds than turboshaft or turboprop engines.

EVALUATION LESSONS LEARNED

Monitoring the vibration level of engines during a test has saved many engines from catastrophic failure. Even pilots monitor engine vibrations by sensing the level through the airframe structure.

4.10.3.3.3 Test stand dynamic characteristics. The test stand shall not induce damaging or detrimental resonances into the engine at any test or operating condition.

EVALUATION RATIONALE (4.10.3.3.3)

Test stand natural frequencies must not damage the engine or adversely affect the test instrumentation accuracy.

EVALUATION GUIDANCE

Background:

The test stand should have enough mass to ensure that the test stand natural frequency is not within the starting to maximum rotor speed range. If the engine has multiple rotors, all rotors should be considered when analyzing the test stand characteristics.

EVALUATION LESSONS LEARNED

Rotor unbalance can be excited by a test stand where the natural frequency is in the rotor speed range. This has caused problems during the test.

4.10.3.3.4 Starter test Starting shall be performed with a starter that has torque characteristics within (a) percent of the minimum required torque shown on figure 3.

EVALUATION RATIONALE (4.10.3.3.4)

This requirement determines the minimum size of the starter used in testing by specifying its torque characteristics. A starter with "below specification" performance can bias test results.

A starter with more than 5% above minimum torque characteristics also biases the test results.

EVALUATION GUIDANCE

The following should be used to the tailor the specification paragraph:

(a) The contractor should provide a value between 0–5 percent. A low value is preferred.

EVALUATION LESSONS LEARNED

To prevent any starting problems it is necessary to ensure the starter torque characteristics meet the engine minimum requirement. The starter torque characteristics should not exceed minimum requirements, considerably, to prevent unsuccessful engine starts with field engines.

4.10.3.4 Test conditions

4.10.3.4.1 Oil servicing and consumption The oil system shall be serviced at the start of each test or demonstration. Any and all additions of oil added after initial servicing shall be recorded. Oil shall be drained only when authorized by the Using Service. The use of external oil filters shall not be permitted. Oil consumption shall be measured and the average consumption rate determined.

EVALUATION RATIONALE (4.10.3.4.1)

This requirement is needed to control the test conditions when servicing the engine oil system. Addition of oil is recorded for the purpose of oil consumption determination. It is important to control the conditions for servicing the oil in the engine and the fluid that is used. Each test should start with new oil since oil is monitored during the engine test to determine test article deterioration.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

Oil samples for an oil spectrum analysis are used to determine condition of engine parts that are exposed to the oil. Using clean new oil at the start of the test provides a base line for the analysis. Starting a test with used oil can bias results because used oil may not meet the specification set forth for oil characteristics. Past experience has shown that engine oil can be a source of concern with test article deterioration.

4.10.3.4.2 Fluid properties for test

4.10.3.4.2.1 Fuel properties Each test and demonstration shall be conducted using the primary and alternate fuels specified in 4.7.3.1.1 and 4.7.3.1.2, unless otherwise specified. If more than one primary fuel is specified, the fuel used shall be the fuel which makes the test the most difficult as determined by the Using Service.

EVALUATION RATIONALE (4.10.3.4.2.1)

The properties of fuel must be specified. Their properties directly affect the results of the test.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

JP–4 has a lower flashpoint than JP–5. JP–5 is more viscous than JP–4 at the same temperature. For starting and windmill relights, JP–4 has given better results than JP–5. These results also depend on fuel system design such as pressure, nozzle configuration, and fuel flow.

Low lubricity fuel has caused problems with fuel controls and pumps, yet the fuel was within the specification. Fuels high in sulfur content have caused problems with certain alloys and salt air.

4.10.3.4.2.2 Oil properties Each test and demonstration shall be conducted using the oils specified in 3.7.8.1. If more than one oil is specified, the oil used will be the oil that makes the test most difficult as determined by the Using Service.

EVALUATION RATIONALE (4.10.3.4.2.2)

The properties of oil must be specified. Its properties directly affect the results of the test.

EVALUATION GUIDANCE

Background:

The type of oil will determine the extreme temperature at which the engine can start and operate.

EVALUATION LESSONS LEARNED

MIL–L–23699 oil has been tested with JP–5 fuel and MIL–L–7808 has been used with JP–4 fuel, this is because of the oil viscosity properties at various operating temperatures. JP–4 and MIL–L–7808 are matched to give better starting capabilities at lower temperatures.

4.10.3.4.2.3 Hydraulic fluid properties Each test and demonstration shall be conducted using the hydraulic fluids specified in 3.7.9.

EVALUATION RATIONALE (4.10.3.4.2.3)

Physical and chemical properties of hydraulic fluid must be specified since their properties directly affect the results of the test.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.3.5 Test calibrations and procedures

4.10.3.5.1 Pretest evaluation

4.10.3.5.1.1 Pretest dry mass (weight) of engine Prior to initiation of the calibration, the dry weight of the engine, as specified in 3.1.2.1, shall be evaluated in accordance with 4.10.4.1.2. The dry mass (weight) shall be measured before the engine has been serviced with fuel and oil.

EVALUATION RATIONALE (4.10.3.5.1.1)

The engine must be weighed prior to initiation of the calibration to determine its dry weight compared to the specification requirements.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.3.5.1.2 Pretest power level torque For hydromechanical controls, prior to the test, power lever torque loads shall be measured for evaluation of 3.1.1.11.1.2.

EVALUATION RATIONALE (4.10.3.5.1.2)

Power lever torque must be measured in order to assure improper loads are not required to modulate the shaft of the hydromechanical control.

EVALUATION GUIDANCE

The following should be used to the tailor the specification paragraph:

This paragraph only applies to control systems that use mechanical interfaces. For systems using electronic interfaces (e. g., data bus, potentiometer) delete the text and insert “Not Applicable”.

EVALUATION LESSONS LEARNED

Past engine qualification tests have resulted in breakage of the control shaft throttle spindle. The power lever torque measurement was made to assure that the facility’s input to the throttle was not high enough to break the spindle and that the spindle was designed for the mechanical load.

4.10.3.5.2 Calibration

4.10.3.5.2.1 Control and engine monitoring system calibration Prior to the initiation of the engine calibration all components necessary for proper control and engine monitoring systems operation and scheduling shall be checked on the bench and on the engine to establish proper functioning and calibration over the applicable ranges of engine operating conditions. If thermocouples are used, the relationship between temperatures and output signal shall be in accordance with the National Bureau of Standards Monograph 125. The accuracy of the signals in relation to the actual measured temperature and transient time response characteristics shall be specified. All fuel nozzles and fuel carrying components of the engine control system shall undergo bench calibrations using fluid in accordance with MIL–C–7024.

EVALUATION RATIONALE (4.10.3.5.2.1)

Sensing system(s) require a calibration to ascertain accuracies before attempting to use these systems prior to test.

Fuel system calibration should be conducted to insure the engine fuel system is supplying the required amount of fuel at the required pressures for operation throughout the environmental conditions and operating envelope of the engine. This includes both starting and augmentation operation.

EVALUATION GUIDANCE

Background:

Thermocouples are presently the most common temperature sensing devices. They can be calibrated by either using the ice bath (thermistor (RTD) integrated circuit sensor) or the cold junction compensation (using thermistors) method of calibration.

The thermocouple designations E, J, T, and K all refer to specific calibration curves with established limits of error but do not refer to the specific wire alloy used. Any metal or alloy which fits the curve within the specified limits of error is classified according to that curve. The types E and T thermocouples are useful for low temperature measurement, while type K is suited for high temperature measurement.

The following fuel flow calibration limits may be used:

1. Acceptance Limit – Criterion of acceptance at bench test after calibration of any new or overhauled component. This criterion should apply before shipment from the component manufacturer or overhaul facility.

2. Audit Limit – Criterion of acceptance at any bench test run after the vendor or overhaul facility acceptance and before any significant engine test hours have been accumulated. It can also be the criterion of acceptance at bench test following any engine milestone tests.

3. Ultimate Service Limit – The extreme limit of component performance beyond which satisfactory engine operation cannot be assured. It is the criterion of acceptance at bench test following QT simulated operational component tests or equivalent component tests.

The acceptance criteria must be determined by test on new or overhauled components.

EVALUATION LESSONS LEARNED

None.

4.10.3.5.2.2 Engine calibration The engine calibration shall be accomplished to establish the performance characteristics of the complete engine. Prior to the beginning of the calibration the engine may be cleaned using the wash procedure specified in 3.7.14 and engine controls may be adjusted but shall not be readjusted throughout the calibration. Calibrations shall be made initially with no customer power extraction and no bleed air extraction other than that required for continuous engine operation. Data required in 4.10.4.1.3 and 4.10.4.1.4 shall be recorded. During calibration, conformance with the leakage and shutdown drainage requirement of 3.1.1.16 shall be demonstrated. The engine shall be calibrated using fuel per 3.7.3.1.1 and oil per 3.7.8.1. The following shall be obtained:

a. Data to establish compliance with 3.2.1 and sea level performance of tables Ia, b and c. The use of outside ambient temperature (OAT), to calibrate the engine in lieu of table Ia, b and c conditions is acceptable with Using Service approval. (a) power settings shall be included in the performance calibration. For turboprop/ turboshaft engines, sufficient data to establish the accuracy of the torque sensor signals shall be recorded.

b. Data to establish compliance with 3.2.2.5 Stability, 3.2.2.6 Transients and 3.7.11 Starting requirements at sea level static conditions. All transient performance data shall be used to update the transient status performance computer program of 3.2.1.2 and 3.2.1.3.

c. Repeat items “a” and “b” conditions with maximum permissible bleed air extraction for that condition. The maximum permissible bleed air extraction includes customer and anti–icing bleed air flow, if applicable.

d. Repeat item “c” with customer/accessory power extraction as specified by tables Ia, Ib, and Ic, except for starting.

e. Data required to characterize engine performance as a function of engine flight hours and cycles.

f. Data showing adequate distorted stall/surge margin remaining with (b) distortion pattern (screen) selected by the Using Service. In addition, data establishing the distorted stall/surge margin shall be determined at (c) engine thrust setting selected by the Using Service.

Additionally, for turboprop engines, the following shall be obtained with the engine fitted with the power absorbing device specified in the specification:

g. Data to establish compliance with 3.7.17.1 Feathering and 3.7.17.2 Reverse pitch operation and load/no load operation.

EVALUATION RATIONALE (4.10.3.5.2.2)

The engine must be calibrated to establish a baseline of performance characteristics. The engine must be fitted with the intended power absorber or equivalent and operated throughout its operating modes in order to establish the operability of the propulsion system.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor should provide the number of power settings. The power settings should include maximum and minimum augmentation, Intermediate Rated Power (IRP) and a sufficient number of part power settings, including Idle, to accurately characterize the performance for turbojet and turbofans.

For engines having short period ratings above IRP (example turboprop/shaft, V/STOL applications) those power settings should also be tested. Examples are contingency, maximum, combat, etc. Thrust augmentation by other means such as water injection should also be tested.

(b): The contractor should provide the number of screens. One to two screens is recommended.

(c): The contractor should provide the number of thrust/power settings. One to three thrust settings are recommended, especially at IRP/Maximum Continuous, and a few part power settings where the surge margin “pinch point” may occur.

Background:

The calibration will probably take place at sea level using non–conditioned air (ambient pressure and temperature) at the contractor’s facility. If an altitude facility is used, then several flight conditions could be tested instead of just one sea level static point. Having conditioned air available at an altitude facility offers more test flexibility and better baseline data can be obtained.

For turboprop engines, the manual feather and reverse pitch calibration should be done with the specification power absorber (propeller). If that propeller has not been specified or is not available, a different power absorption mechanism, such as a club prop should be used. The full range of propeller travel from feathering to reverse pitch should be tested.

Stall/Surge margin check may be accomplished with the use of Bodie Runs.

Bodie Run. This run should consist of 180 minutes of thrust transients as follows:

(1) Twelve periods of four minutes at Idle thrust position and twelve periods of 30 ±3 seconds at the Maximum thrust position for a total of 54 minutes.

(2) Ten periods, each consisting of three minutes of operation, making a total of 30 minutes. Each period should be run as follows: with the thrust/power demand in the Idle thrust position, advance toward Intermediate thrust, and as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), decrease the thrust/power demand to the Idle thrust position.

(3) Ten periods, each consisting of three minutes of operation with the final period followed by an additional two minutes at Idle making a total of 32 minutes. Each period should be run as follows: with the thrust/power demand in Idle thrust position, advance toward Maximum thrust, and as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), decrease the thrust/power demand to the Idle thrust position. (For afterburning engines, as soon as the engine reaches the condition of minimum surge margin after lighting the afterburner, decrease the thrust/power demand to the Idle thrust position.) (If the engine does not have a Maximum rating, Intermediate thrust should be substituted for Maximum thrust.)

(4) Ten periods, each consisting of three minutes of operation with the final period followed by an additional two minutes at Maximum thrust making a total of 32 minutes. Each period should be run as follows: with the thrust/power demand in the Maximum thrust position, retard toward Idle, as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), increase the thrust/power demand to the Maximum thrust position. (For afterburning engines, as soon as the engine reaches the condition of minimum surge margin after lighting the afterburner, increase the thrust/power demand to the Maximum thrust position.) (If the engine does not have a Maximum thrust rating, Intermediate thrust should be substituted for Maximum thrust.)

(5) Ten periods, each consisting of three minutes of operation with the final period followed by an additional two minutes at Intermediate thrust making a total of 32 minutes. Each period should be run as follows: with the thrust/power demand in the Intermediate thrust position, retard toward the Idle thrust position, and as soon as the engine reaches the condition of minimum surge margin (i.e., rpm or gas temperature), increase the thrust/power demand to the Intermediate thrust position.

EVALUATION LESSONS LEARNED

These bodie tests in the calibration phase are used primarily to see if the engines is a “staller” before other tests are run.

4.10.3.5.2.2.1 Customer bleed air analysis calibration Prior to and during the (a) test, a customer bleed air analysis shall be performed. The customer bleed air shall be sampled from each bleed air outlet during an Intermediate/Maximum Continuous power run. A sample of air entering the engine inlet shall be taken at the same time the bleed air samples are obtained. The samples shall be properly identified and processed through laboratory analysis to determine whether the contaminant levels are within the limits specified in 3.1.1.15.1.

EVALUATION RATIONALE (4.10.3.5.2.2.1)

See 4.1.1.15.1, Customer bleed air contamination.

EVALUATION GUIDANCE

The following should be used to the tailor the specification paragraph:

(a): the contractor should specify which test will be used, i.e., altitude or endurance.

Background:

The Using Service should be aware that engine cleaning compounds can be a substance that could cause bleed air contamination. Analysis should be completed to detect its presence.

The test should be conducted at power settings which will generate the highest levels of contaminants. Results, methods, and apparatus should be detailed in the test report.

Acrolein is difficult to detect during the customer bleed air analysis. The Using Service should request that the engine contractor provide the method(s) used to detect this substance.

EVALUATION LESSONS LEARNED

None.

4.10.3.5.3 Inspection, maintenance and calibration The following inspections, maintenance and calibrations shall be performed:

a. Using established procedures, inspections shall be performed at intervals equivalent to the inspection periods specified in 3.4.1.7.4. Inspections and partial disassembles at other times during the test are permitted when warranted by performance deterioration and after approval from the Using Service is obtained.

b. Prior to and following each inspection interval, an abbreviated engine calibration may be performed. If performance deterioration exceeds 3.2.1.4, the engine may be cleaned. If cleaning is necessary, the washing system and procedure specified in 3.7.14 shall be used prior to initiation of the next test segment.

c. Maintenance shall be performed on a basis equivalent to the maintenance procedures specified in 3.5.2.3. Parts replacements during the test, except for consumables defined by maintenance instructions, shall not be permitted without approval of the Using Service. A record of all consumable and nonconsumable parts replaced during the test shall be maintained by the contractor. A failure analysis report shall be prepared for any failures resulting in the replacement or repair of non–consumable parts. Non–consumable parts replaced during the test shall require retest equivalent to the severity and test hours of the full engine test.

EVALUATION RATIONALE (4.10.3.5.3)

Inspections are necessary to ensure no imminent failures are present. Maintenance should be done in accordance with the established procedures to ensure the proposed reliability and maintainability projections are met.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.3.5.4 Recalibration After completion of the tests specified in 4.9.1, a recalibration in accordance with the requirements of 4.10.3.5.2.2 and 4.10.3.5.2.2.1 shall be made on the engine. The recalibration may be preceded by a run during which the cleaning procedure of 3.7.14 may be applied. The fuel and oil used shall be the same as those used during the initial calibration.

After completion of the engine recalibration, all components necessary for proper control and engine monitoring systems operation and scheduling shall be rechecked to establish proper functioning and calibration in accordance with 4.10.3.5.2.1.

The performance of these components shall meet their design service limits.

EVALUATION RATIONALE (4.10.3.5.4)

See 4.10.3.5.2.1 thru 4.10.3.5.2.2.1

EVALUATION GUIDANCE

Background:

See 4.10.3.5.2.1 thru 4.10.3.5.2.2.1

EVALUATION LESSONS LEARNED

None.

4.10.3.5.5 Engine disassembly and inspection For tests requiring engine disassembly the Using Service shall be notified of the inspection commencement date prior to each inspection. Prior to cleaning, the engine parts shall be given a "dirty" inspection for abnormal conditions. The dirty inspection shall be completed before any parts are cleaned. Parts shall then be cleaned and a "clean" inspection shall be performed. Part measurements shall be taken, as necessary, to determine excessive wear and distortion. Those measurements shall be compared with the manufacturer's drawing dimensions and tolerances and with similar measurements made prior to the test. Non–Destructive Inspection (NDI) techniques shall be in accordance with 3.4.1.7. During the clean inspection a visual examination and condition assessment shall be conducted. The contractor shall maintain all data and results of non–destructive inspection tests. All NDI inspection data and findings shall be made available for Using Service review during the inspections. All data of 4.10.4 shall be made available for Using Service review upon request.

As a result of the inspection, parts shall be categorized as follows:

Cat. A. No repair required.

Cat. B. Repair required before further use.

Cat. C. Outside repair limits.

EVALUATION RATIONALE (4.10.3.5.5)

All engine parts must be inspected to assure durability and safety requirements of the specification are met. Engine test data must be available for review during inspection to aid in diagnosing any inspection findings.

EVALUATION GUIDANCE

Background:

The Using Service should ensure that the contract requires at least 21 days notification of a scheduled inspection.

An example of some of the data the Using Service may want to review during the inspection is listed below.

a. Inspection forms filled out by the contractor listing all deficiencies and NDI data and findings.

b. Tabulation of all parts not meeting pass/fail criteria and reasons for failure.

c. The detailed configuration list of 4.10.3.2.

d. Test summary, test logs, and list of test events.

e. Spectrometric oil analysis results.

f. Chip detector/oil debris analysis and results.

g. Tabulation of all parts which require replacement prior to the test completion.

EVALUATION LESSONS LEARNED

Advanced NDI techniques have revealed fatigue and stress cracks that normally may have gone undetected. It is common for NDI inspections to detect flaws consisting of micro–cracks which could propagate.

4.10.4 Data and reporting Data and reporting for tests conducted in this specification shall be as required below.

EVALUATION RATIONALE (4.10.4)

Proper data and information must be recorded in order for the Using Service to evaluate requirements.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

None.

4.10.4.1 Types of data Data shall be recorded during tests in accordance with the following subparagraphs.

EVALUATION RATIONALE (4.10.4.1)

Whenever a test is conducted there is a need to specify the minimum data required to be able to assess the results of the test. The following requirements will specify when the data will be obtained and what is required as a minimum.

EVALUATION GUIDANCE

Background:

The following subparagraphs will give guidance to the types of data that are required. The Using Service should verify that this item is listed in the CDRL thereby insuring that the contractor is required to submit the information to the Using Service.

EVALUATION LESSONS LEARNED

None.

4.10.4.1.1 Pretest data Pretest data shall be made available to the Using Service for approval prior to the initiation of each test or demonstration. The data shall include:

a. Detailed test and maintenance plans. The detailed test and maintenance plans to be used by test and maintenance personnel during the conduct of the test. Test procedures shall be in accordance with the test plan.

b. Parts list and configuration of test article. Parts list and configuration of the test article, and justification for any differences shall be in accordance with 4.10.3.2.

c. Service/wear limits. Service limits to be used by maintenance personnel in the field and repair facilities.

EVALUATION RATIONALE (4.10.4.1.1)

Before the test, there is a need to require a test procedure and test article configuration. Standardized test plans and article configurations are needed to ensure adequate testing on appropriate test articles.

EVALUATION GUIDANCE

Background:

The test and maintenance plans should be reviewed to determine if the test will meet the requirements in the specification. The parts list and configuration must be reviewed to determine if the test will be conducted on the test article described in the model specification. The Using Service should not accept contractor justification of parts substitution without a thorough analysis.

EVALUATION LESSONS LEARNED

Contractors have substituted parts during evaluation tests.

4.10.4.1.2 Preliminary data Photographs and other pertinent data shall be obtained and recorded at the time the test article is prepared for the test. For engines, dry mass (weight) and center of gravity shall be pertinent data. If the mass (weight) is measured after the engine has been serviced with fuel and oil and subsequently drained, the dry mass (weight) may be calculated by subtracting the mass (weight) of the residual fluids specified in 3.1.2.2 from the measured engine mass (weight).

EVALUATION RATIONALE (4.10.4.1.2)

A record of the pertinent data is needed to track the progress of the test article development though each milestone.

EVALUATION GUIDANCE

Background:

Preliminary data should be submitted as part of the test reports.

EVALUATION LESSONS LEARNED

It has been a challenge to maintain specification weight throughout development.

4.10.4.1.3 Data – steady–state During operation at each specified steady–state condition and after performance stabilization, as a minimum data specified in table XXXVII shall be recorded.

EVALUATION RATIONALE (4.10.4.1.3)

The engine must reach a fully stabilized condition before steady–state data is recorded.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph: The contractor should fill in the table.

Background:

Steady–state data can only be taken after performance stabilization, which can take as long as ten minutes of engine operation.

The stabilization time periods have been dependent on the size of the engine and the magnitude of rotor speed change. In the past, data was recorded before stabilization which resulted in faulty performance determination.

4.10.4.1.4 Data – transient As a minimum, data specified in table XXXVII shall be recorded.

EVALUATION RATIONALE (4.10.4.1.4)

Transient data is needed in order to evaluate transient operability and performance of the engine. Transient performance response also affects engine life.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph: The contractor should fill in the table.

Background:

It is better to obtain more than one sample of transient data since the accuracy is usually less than steady–state data. Transient runs should be repeated at selected conditions to investigate control system lags, repeatability, and performance droop.

EVALUATION LESSONS LEARNED

As the name implies, transient data is a record of the dynamic response of a given change in an engine performance parameter.

Since it is a dynamic response, it does not always repeat exactly each time. Therefore, several samples may be needed to determine the average response.

4.10.4.1.5 Starting data During each start, data shall be recorded to evaluate requirements of 3.1.1.19.1, 3.7.11.3, and 3.7.11, as a minimum.

EVALUATION RATIONALE (4.10.4.1.5)

Starting data is recorded to keep a history of the engines health. This requirement specifies when and what starting data should be recorded.

EVALUATION GUIDANCE

Background:

The maximum turbine temperature and time to reach Idle speed should be recorded during the start. Include time, temperature and rotor speeds vs starting event (i.e., initiation of start, starter cutout, stabilized engine speed). Note flat spots on acceleration curves.

EVALUATION LESSONS LEARNED

Any increase in turbine temperature or time to reach Idle speed indicates a change in engine characteristics. Comparisons should be done at the same inlet conditions. Colder inlet temperature will require higher turbine temperature and longer time to reach Idle speed and could cause a hot start.

4.10.4.1.6 Accreditable test time Total test time for a period less than (a) minutes shall not be credited to the official run time, except when shorter periods are a test requirement.

EVALUATION RATIONALE (4.10.4.1.6)

This requirement prevents the engine contractor from counting minimal test periods as accreditable test time.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a) The contractor should provide the time interval.

Background:

Most required test runs last longer than 15 minutes. Tests should be credited for the exact time of run. For tests started and stopped in less than 15 minutes for reasons other than required by the test plan, no time credit should be given.

EVALUATION LESSONS LEARNED

In past engine tests, contractors have added many periods of less than 15 minutes run time to accumulate significant number of total test hours. These test hours do not represent a significant accomplishment in the test plan.

4.10.4.1.7 Miscellaneous data The date, test title, test article designation, and serial number shall be recorded on each page of the log sheet.

EVALUATION RATIONALE (4.10.4.1.7)

There is a need to identify each page of the log with all the pertinent data that will identify a particular test and test article.

EVALUATION GUIDANCE

None.

EVALUATION LESSONS LEARNED

Past engine test records have had missing pages, because there was no identification to show where to re–insert any unattached pages.

4.10.4.1.8 Test notes Notes shall be added to the log sheets to record all incidents and irregular functioning of the test article and equipment and the corrective measures taken. All test article software changes shall be recorded on the log sheets.

EVALUATION RATIONALE (4.10.4.1.8)

A place is needed to log all unusual occurrences during the test to assure compliance with the specification.

EVALUATION GUIDANCE

Background:

Most log sheets are pre–printed. They do not allow for any unusual incidence. The test notes are the place to record any irregular engine problems (such as, leaks, vibrations, and “flat spots” in acceleration) or facility problems.

EVALUATION LESSONS LEARNED

It is a rare test that is conducted without any irregular incidents occurring. The notes of these incidents will be used after the test to evaluate the results of the test.

4.10.4.2 Accuracy of data For all test article calibrations and tests or demonstrations, the reported data shall have a steady–state accuracy within the tolerances specified herein. The accuracy of transient data and the corresponding instrument calibration methods shall be described in the test plans. All instruments and equipment shall be calibrated

as necessary to ensure the required degree of accuracy is maintained. No credit or debit shall be permitted for measurement uncertainty to establish compliance with the requirements of this specification. The contractor shall minimize measurement uncertainty errors. The instrument calibration methods shall be submitted to the Using Service for approval prior to the initiation of the test. The "as tested" accuracies as defined by the test facility shall be included in the test report.

EVALUATION RATIONALE (4.10.4.2)

The contractor should insert the accuracies in the specification.

Since accuracy of data is a factor in the interpretation of test data, it needs to be stated in the specification, test plan, and test report.

EVALUATION GUIDANCE

The following should be transferred verbatim into the specification paragraph:

DATA ITEM	TOLERANCE
Rotor speed(s)	±0.2 percent of the value obtained at maximum rating
Metered thrust	±0.5 percent of the value measured – for minimum augmented rating and above
	±0.5 percent of the value measured at Intermediate rating – for Intermediate rating and below
Fuel Flow	±0.5 percent of the value measured – for Intermediate rating and above
	±0.5 percent of the value measured at Intermediate rating – for all values below Intermediate rating.
Airflow	±1.0 percent of the value measured for Intermediate rating and above
	±1.0 percent of the value measured at Intermediate rating – for all values below Intermediate rating
Temperatures	±1.0° up to 200°C
	±3.0° between 200°C and 800°C
	±4.0°C above 800°C
Engine Weight	±1.0 lbs or ±0.1 percent of the weight being determined, whichever is greater
Vibration Velocity	±5.0 percent of specified engine limit during 4.4.1.8 vibration survey, vibration scan, resonant search, and vibration & stress test
Vibration Velocity	±10.0 percent of specified engine limit for all other tests
All other data	±2.0 percent of the value obtained at maximum rating

Background:

The Using Service should use the accuracy characteristics of those instrumentation and data systems used in the test cell performing the qualification tests. Steady–state accuracy is easier to control than transient accuracy.

EVALUATION LESSONS LEARNED

None.

4.10.4.2.1 Measurement uncertainty analysis A detailed measurement uncertainty analysis addressing all elemental precision and bias errors shall be performed in accordance with AEDC–TR–73–5 or equivalent approved by the Using Service. As a minimum, the analysis shall cover the three categories of errors: calibration errors, data

acquisition errors, and data reduction errors. Analysis will be performed for the following measured parameters: airflow, fuel flow, torque, temperature, pressure, rotor speeds, (Ng & Np), SFC, output power, engine mass, and vibration.

EVALUATION RATIONALE (4.10.4.2.1)

The estimation of uncertainty is an essential requirement for all measurements. Without such an estimate, interpretation of the measurement may be difficult or impossible. Estimates of uncertainty can strongly influence the methods used to test engines, components, and the interpretation of results.

EVALUATION GUIDANCE

Background:

Methods, procedures, and documentation that substantiates the uncertainties in measuring and calculating airflow, net thrust, and thrust specific fuel consumption during steady–state testing conditions should be provided. Analysis information should include, but not limited to, the following:

a. All basic measurands and calculated parameters based on these measurands should have the uncertainty reported as combination of precision index and bias as specified in the handbook. All three values, uncertainty, precision index, and bias, should accompany each measurand and calculated parameter in the uncertainty documentation.

b. A hierarchy of uncertainty should be provided for all the pressure transducers, thermocouples, flow meters, load cells, and areas, or whatever equipment is used to calibrate and/or measure the performance parameters used in the calculation of airflow, net thrust, and thrust specific fuel consumption. The hierarchy should trace the uncertainty from the in–place measuring device back to the primary measurement standards traceable to the National Bureau of Standards.

c. The set of performance calculation equations used to calculate airflow, net thrust, and thrust specific fuel consumption should be provided. The partial derivative functions determined from the performance equations should also be given.

EVALUATION LESSONS LEARNED

None.

4.10.4.3 Atmospheric data

4.10.4.3.1 Barometer reading The barometric reading shall be corrected for temperature and shall be recorded at intervals of (a) hours, but not exceeding three hours.

EVALUATION RATIONALE (4.10.4.3.1)

All the engines gas pressures are dependent on the barometric pressure.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a): The contractor should provide the time interval.

Background:

The barometric pressure should be measured in the vicinity of the test and its accuracy should be comparable to that used for measuring gas pressure in the engine. The standard should be a mercury barometer with all its corrections.

EVALUATION LESSONS LEARNED

Some test facilities have been known to use the local airport or weather station for the information, which has led to inaccurate data. Weather stations do not update these readings as frequently as needed. Others have used an aneroid barometer of questionable accuracy. Aneroid barometers require frequent recalibration.

4.10.4.3.2 Relative humidity Relative humidity data readings shall be recorded at intervals of __(a)__, but not exceeding three hours if required for performance calculations. The relative humidity shall be measured in the vicinity of the test article. A wet and dry thermometer or a sling psychrometer shall be used for these measurements.

EVALUATION RATIONALE (4.10.4.3.2)

Relative humidity affects quality of the air that is used by the engine and can affect engine performance.

EVALUATION GUIDANCE

The following should be used to tailor the specification paragraph:

(a) The contractor should provide the time interval.

Background:

Relative humidity data should be recorded more frequently on days when weather conditions are rapidly changing or the temperature is hot (>80°F). Relative humidity should be measured where the engine is operating, not in the control room where the humidity may be different.

EVALUATION LESSONS LEARNED

None.

4.10.4.4 Fluid data Samples of fuel, oil, and hydraulic fluid, shall be taken at the start and completion of each test or demonstration. The sample shall be analyzed for physical and chemical properties to determine conformance with applicable fluid specifications. The results of these analyses shall be included as part of the test report.

EVALUATION RATIONALE (4.10.4.4)

Samples of fuel, oil, and hydraulic fluid need to be taken at the start and at the completion of the test to confirm that they meet specifications all through the test.

EVALUATION GUIDANCE

Background:

Fuel, oil, and hydraulic fluid samples are taken more often during some tests, but only the analysis conducted at the start and completion of the test are required in the test report.

EVALUATION LESSONS LEARNED

Fluid heat exchangers have been known to leak internally. Fluid analyses will indicate internal mixing of fluids due to leakage. In the past, off–specification fluids have caused abnormal engine wear.

50. PACKAGING

50.1 Preparation for storage and shipment The engine shall be prepared for storage and shipment in accordance with MIL–E–5607. The level of preservation, the type of shipping container, and the furnishing of engine historical records shall be in accordance with contract requirements. A packing list shall be furnished with each engine and shall include all components and tools which are not installed, but are shipped with the engine.

RATIONALE

Proper storage, preservation and shipping is required to avoid needless engine damage.

GUIDANCE

The words of paragraph 5.1 should appear in the specification.

Background:

Missile storage is critical due to possible long shelf life requirements.

LESSONS LEARNED

NADEP Alameda cited reduction gearbox corrosion on engines that were stored in improper storage containers.

60. NOTES

60.1 Intended use. Engines covered by this specification are intended for air vehicle propulsion.

RATIONALE

This paragraph specifies information relative to the use of the engine.

GUIDANCE

Background:

This specification cannot be used for contractual purposes without supplemental information relating to the performance requirements of a specific gas turbine engine for an air vehicle system. All tailored specifications should specify the aircraft, if known. The aircraft turbine engines covered by this specification are intended for propulsion of attack, bomber, cargo/transport, special electronic installation, fighter, helicopters, interdiction, tanker, observation, patrol, antisubmarine, trainer, or utility type aircraft.

LESSONS LEARNED

None.

60.2 Acquisition requirements Acquisition documents will be tailored based on the information in the tailoring handbook and peculiar program requirements.

RATIONALE

This paragraph provides a connection to all of the options the contracting officer has in preparing an acquisition specification.

GUIDANCE

The words of this paragraph should appear as is in the acquisition specification.

LESSONS LEARNED

None.

60.3 Consideration of data requirements Data requirements should be considered when this specification is applied in a contract. The applicable Data Item Descriptions (DIDs) shall be reviewed in conjunction with the specific acquisition to ensure that only essential data, as determined by the Using Service, are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition.

RATIONALE

This paragraph introduces the DIDs that describe the information the contractor is required to submit to the Using Service. The DIDs will be referenced in the CDRL which is a part of the contract.

GUIDANCE

The following DIDs should be considered for inclusion in the contract.

Reference Para	DID Number	DID Title
3.1.1.1 3.1.1.2 3.1.1.3	DI–DRPR–80651 or DI–DRPR–81000	Engineering Drawings Product Drawings and Associated Lists
4.2.1.5b 3.2.1.1b	DI–S–30591	Data and/or Analysis Summary
3.2.1.1 3.2.1.2	DI–MISC–80672	Gas Turbine Engine Steady State and Transient Performance Presentations for Digital Computer Programs
3.5.1b	DI–R–7080	Reliability Status Report
3.7.2.1b	DI–R–7085 or DI–RELI–80687	Failure Modes, Effects and Criticality Analysis Report Failure Modes, Effects and Criticality Analysis Report
4.7.4.5b 4.7.4.7b	DI–T–37089	Equipment Test Plan
4.7.8.2.5b	DI–GDRQ–80567	Subsystem Design Analysis Report

To ensure correct contractual application of the data requirements, a Contractor Data Requirements List (CDRL), DD Form 1423, must be prepared to obtain the data, except where DOD FAR Supplement 27.475–1 exempts the requirements for a DD Form 1423.

Background:

The Using Service should verify that all the required DIDs are referenced in the CDRL.

LESSONS LEARNED

The following DIDs for CDRLs have been selected in past programs for the CDRLs. Other unique data requirements have been added, as applicable:

DID Number	DID Title
	Model specification
	Engine characteristics summary
DI–MISC–80672	GTESS PPD
DI–MISC–80672	GTET PPD
	Parts list
	Pretest data
	Preliminary data
	Test reports
	Summary reports
	Engineering evaluation reports
	Structural load analysis
	Heat rejection and cooling analysis
	Radar cross section analysis
	Performance computer programs
	Turbine temperature correlation

	Performance verification report
	Residual operating fluid weight
	Noise level analysis
	Adhesive and sealant compounds
	List of corrosion protective treatments
	Electromagnetic interference and susceptibility analysis
	Matched and selected fit parts
	Vibration and stress analysis
	Pressure balance analysis
	Strength and life analysis
	Creep analysis and parts growth report
	Changes in vendors or fabrication process list
	Module replacement limits report
	Repair procedures and wear limits report
	Potting compounds list
	Relative damage chart
	Oil flow and heat rejection report
	Wear rate analysis report
	Infrared radiation analysis report
	Ultraviolet radiation analysis report
	Wash procedure
	Inlet airflow distortion report
	Safety wire list
	Survivability analysis and report
	Susceptibility analysis
	Vulnerability analysis
	Battle damage repair analysis
	Material and process documents
	Electronic component analysis
	Damage tolerance control plan and analysis
	True rms velocity limits
	Acceleration spectrograms
	Engine performance verification
	Customer bleed air
	Engine vibration survey
	Starting torque
	Maintenance test
	Verification of correction factors

[Note: The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12–L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current DIDs are cited on the DD Form 1423.]

60.4 Definitions Definitions shall be used as specified below:

a. General definitions

Accessories. Items of engine–mounted equipment not furnished by the engine contractor or airframe mounted equipment, which are required for aircraft operation or as auxiliaries for engine operation.

Alternate Fuel. Fuel on which the aircraft can be flown without operational restrictions but which can have long term durability or maintainability impact if used for continuous operation (multiple flights). Alternate fuels are used only on an occasional or intermittent basis.

Article. An individual item or particular unit which may be a component (e.g., fuel pump) or a system when considered as a whole (e.g., engine).

Automatic Ignition. Similar to manual ignition except the engine control activates the system. No pilot action is needed. In addition to the benefit and usage shown for the manual ignition system, the automatic system may be activated during decelerations at high altitudes, especially in the upper left corner of the flight envelope, or during rapid maneuvers of the aircraft.

Capture Area. Capture area is the projected physical area of the aircraft inlet.

Cold engine. (for starting only) A ‘‘cold” engine is defined as one which has been allowed to windmill until the low pressure turbine exit gas temperature is within 55°C (100°F) of the engine compressor inlet temperature before a start is attempted.

Cold Parts. Those parts not listed as hot parts.

Control Limiting Temperature. The maximum value built into the engine control to limit the operating value at a commanded power setting and flight condition.

Continuous Ignition. A low energy level, low spark rate (relative to the system used to ignite the combustor during an engine start) ignition system that is constantly fired regardless of operating condition or power setting. No pilot action is needed to activate the system.

Critical Size. The crack size where unstable growth occurs.

Damage Tolerance. The ability of the engine to resist failure due to the presence of flaws, cracks or other damage for a specified period of unrepaired usage.

Damage Tolerant. Resistant to failure due to the presence of flaws, cracks or other damage for a specified period of unrepaired usage.

Depot, Intermediate, or Base Level Inspectable Structure. Structure is depot, intermediate, or base level inspectable if the nature and extent of damage will be detected utilizing one or more selected NDI procedures.

Derating. Using an item in such a manner that the stresses applied during operation are lower than the stresses the item was designed to withstand.

Design Service Life. The life defined by the Using Service (used for design purposes) that the engine is expected to attain during operational service.

Design Usage. The engine usage conditions (used for design purposes) specified in 3.4.3.

Deteriorated Minimum Engine. Deteriorated minimum engine is the lowest thrust/power output and highest SFC after operation for the time period and environment specified in 3.2.1.4.

Diagram. A diagram is a sketch or outline giving only interface or mounting information for a part or part of an assembly without all the details of the item shown (i.e., diagrams show bolt hole patterns, flanges, electrical plug sockets, etc.). A diagram shows the relationship between the parts of a whole assembly. It does not show many details.

Dissimilar Control Technology. Other than digital electronic technology, e.g. hydromechanical, fluidic, analog.

Domestic Object Damage. Domestic object damage (DOD) is caused by engine parts coming loose and striking the engine. See foreign object damage (FOD).

Drawing. A drawing is a sketch showing major dimensions, clearances, and details of a part or assembly. It can show every detail you need to manufacture the part, assembly, or engine. Materials of composition are shown on the sketch.

Durable. Resistant to cracking, corrosion, deterioration, thermal degradation, delamination, wear, etc., for a specified period of time.

Durability. The ability of the engine to resist cracking, corrosion, deterioration, thermal degradation, delamination, wear, etc., for a specified period of time.

Durability Critical. Failure results in a significant economic impact to the system but will not necessarily impair flight safety or mission capability.

Durability Non–Critical. Failure results in a minor economic impact to the system but will not impair flight safety or mission capability.

Duty Cycle. A composite derived from the mission profiles and mission mix.

Dynamic Response. Response due to forced functions. Forced functions include: surge, stall, flutter, and resonance.

Emergency Fuel. Fuel which significantly limits the overhaul life of the engine and imposes operational restrictions on the aircraft as well.

Engine. The complete propulsion system (including all parts, components, and expendables) delivered by the engine contractor.

Engine Component. Items of equipment, furnished as part of and qualified with the engine, whose size, conformation, and dynamic and static characteristics are essential to attain the engine performance specified in the engine specification. Fuel pumps, engine controls, jet nozzle and actuators, anti–icing valves, and the temperature sensing system or devices are included in this category. Components may require separate qualification, calibration, and adjustment.

Engine Part. A piece, or two or more pieces joined together, that are not normally subject to disassembly without destruction of the designed use.

Fail–To–Trail. A failure mode where the thrust vectoring nozzle is automatically positioned into the zero vectoring angle. This is also known as the axisymmetrical position.

Fail Operational. The ability to detect any single or multiple failure(s) and to continue to operate the engine after detection and still be capable of meeting all defined mission and performance requirements.

Fail Degraded. The ability to continue to operate the engine after any single or multiple failure(s) and retain a defined minimum operational capability as defined in the ICD and the definition of levels of Flying Qualities.

Fail Safe. The ability to continue to operate the engine safely or safely shut–down following the failure or partial failure of one or more elements.

Flaw Growth Interval. The predicted typical propagation life required to grow an assumed initial flaw to its critical size.

Flight Hours. The time duration beginning at takeoff and ending at landing.

Foreign Object Damage. Foreign object damage (FOD) is caused by non–engine parts striking the engine. See domestic object damage (DOD).

Fracture Critical. Failure will result in probable loss of the aircraft or degradation in mission capability. There are two categories under fracture critical; safety critical and mission critical.

Frequency Response. The steady–state output of the system to input sinusoids of varying frequency. The output for a linear system can be completely described in terms of the amplitude ratio of the output sinusoid to the input sinusoid. The amplitude ratio, gain, and phase are functions of the frequency of the input sinusoid.

Global Plot. A global plot (polar plot) provides azimuth display on the angular scale, frequency displayed on the radial scale and RCS displayed as color levels.

Hot engine. (for starting only) A ‘‘hot” engine is defined as one where a start is attempted within 10 seconds after a flameout or shutdown.

Hot Parts. Parts exposed to the hot gas stream.

Impending Failure. The condition of a part where loss of performance or function would take place if the part was continued to be used for its part life required by the specification.

In–Service Noninspectable Structure. Structure is in–service noninspectable if either damage size, accessibility, or maintenance requirements preclude inspection.

Inspection. The physical examination of an article, item, drawing, or part.

Installed Configuration. When the engine is tested in the airframe of the aircraft, this is known as the installed configuration.

Latent Failure. Any undetected condition which prevents the part or component from performing its intended function. This includes undetected conditions between missions and during missions.

Limit Load. The maximum load expected to be encountered when operated for the design service life at design usage conditions. The factor of safety associated with this load is defined as the limit load.

Loaded Disk. The condition where all blades are attached to the disk.

Maximum Allowable Temperature. The maximum allowable gas temperature, as specified for either steady–state or transient condition, is the limit beyond which operation of the engine is not allowed.

Maximum Operating Temperature. The highest temperature the engine can obtain in normal operation at a commanded power setting considering all effects due to engine–to–engine variation, deterioration, installation factors, control limits/tolerances, environment, and operating point.

Manual Ignition. Similar to continuous ignition except the ignitor is fired only when the system is activated by the pilot via switch. This system also uses a low energy level, low spark rate ignition system. The benefit of this system is the ignitor lasts longer than a continuous ignition ignitor since it’s not used as often. This system would be used in instances where a flameout risk is high (i.e., rain or icing conditions and/or during takeoff or descent) or during a carrier approach under any condition.

Minimum Engine. ‘‘Minimum engine” is a performance level having the lowest thrust/power output and highest SFC at or over a time period and environment specified by the Using Service and considering all effects due to control system, engine–to–engine variation, operating time/deterioration, flight point, and operating condition.

Mission. Period beginning with the start of engine prior to flight and ending with the engine shutdown at the completion of the flight. Specific missions include: familiarization, air combat maneuvers, navigation, and air–to–ground weapons.

Mission Critical. Failure will generate a significant operational impact by degrading mission capabilities to the extent of creating an indirect safety impact on the weapon system or results in less than Level II handling qualities.

Mission Mix. The frequency each mission is flown in a finite period of time (e.g., 5% for familiarization and 25% for air combat maneuvers).

Mission Profile. A representation of a specific mission in terms of flight conditions and usage parameters.

New Minimum Engine. New minimum engine is the lowest thrust/power output and highest SFC when the engine is new.

Non–Operating Environment. The environment to which the engine is exposed during logistics/storage/maintenance/transportation activities.

Operational Life. The life of the engine when exposed to the operational usage.

Operational Usage. The usage that the engine is exposed to during actual service operation.

Primary Flight Control Surfaces. Movable aircraft surfaces which control the direction of flight. In most cases, they are airfoils that control the pitch, yaw, and lateral stability of the aircraft in flight. Rudders, ailerons, elevators, or stabilators are considered primary flight control surfaces.

Primary Fuels. Fuels on which the engine is designed to operate continuously without restrictions.

Residual Strength. The load carrying capability of a part at any time during the design service life. Residual strength accounts for the presence of damage and the growth of damage as a function of design usage and must maintain limit load capability.

Restricted Fuel. Fuel which imposes operational restrictions on the aircraft when used.

Safety Critical. Failure will result in the probable loss of the aircraft or hazard to personnel due to direct part failure or by causing other progressive part failures.

Safety Limit. The predicted typical propagation life required to grow an assumed initial flaw to its critical size.

Satisfactorily. The words ‘‘satisfactorily” or ‘‘satisfactory” as used in this specification in conjunction with words or terms relating to operation or performance of the engine described in this model specification shall mean: Under the condition specified, throughout the entire operating envelope, the engine operating characteristics and performance are not affected, and the operating and physical limits shown in the specification are not exceeded and no permanent deformation or other damage to the engine occurs.

Satisfactory. The words ‘‘satisfactorily” or ‘‘satisfactory,” as used in this specification in conjunction with words or terms relating to operation or performance of the engine described in the model specification, shall mean: Under the condition specified, throughout the complete environmental conditions and operating envelope, the engine operating characteristics and performance are not affected, and the operating and physical limits shown in the specification are not exceeded and no permanent deformation or other damage to the engine occurs.

Small Engines. A small engine is an engine with fan inlet corrected flow requirements of 100 pps or less or compressor inlet corrected flow of 20 pps or less.

Smoke Puff. A smoke puff is visually detectable unburned fuel droplets that are produced when a combustor and/or afterburner is initiated (light–off) or terminated (shutdown).

Stall. Stall is an engine compression system flow instability which does not result in the loss of engine control. A condition in the engine compression system where one or more blades or stages is operating with separated flow.

Steady–state Stability. Steady–state stability is the condition at a constant power setting in which thrust fluctuations do not exceed a stated margin.

Strength. The ability of the engine to withstand limit loading conditions while maintaining operational capability.

Surge. Surge is the loss of engine control resulting from an engine compression system flow instability.

Surge Margin. Surge margin is the calculated and/or demonstrated tolerance of the engine to adverse operating conditions while maintaining the required steady–state and transient performance capability.

Throughput. The time required for a computer circuit to process the input signal.

Transient. Transient include augmentor light–off and termination, as applicable, as well as thrust/power demand increase and decrease.

Transient Stability. Transient stability is the condition following a transient in which thrust fluctuations do not exceed a stated margin.

Transparent to the Pilot. Any event the pilot can not detect. This includes any significant difference of aircraft handling ability or aircraft stability that may or may not occur by having the thrust vectoring nozzle assembly attached or not having a thrust vectoring assembly on the aircraft.

Ultimate Load. The maximum load that a material, part, component, or engine must withstand without catastrophic failure.

Unweathered fuel. Fuel with chemical and physical properties that have not been degraded by environmental effects such as temperature and pressure cycles.

Usable Life. Life without repair or replacement.

Usage Parameters. Those parameters defining engine operation (e.g., ambient temperature, rotor speed, power/temperature dwell times, A/B lights, time at A/B, etc.).

b. Reliability definitions. Reliability definitions shall be listed herein and consistent with MIL–STD–721 when possible.

RATIONALE

These definitions are used to clarify the terms in the specification.

GUIDANCE

Background:

This definition list is organized into general and reliability definitions. These definitions should be included in the acquisition specification. Definitions as listed should not be changed but new definitions may be added.

In the past, contractors have applied liberal interpretations to these definitions.

60.5 Symbols, subscripts, abbreviations, and acronyms The symbols, subscripts, abbreviations and acronyms used in this specification are listed below and shall be in accordance with AS 681 and ARP 755.

A/B	Afterburner
A/C	Aircraft
ACGIH	American Conference of Government Industrial Hygienists
ADS	Automated Data System
AEDC	Arnold Engineering and Development Center
AEMS	Aircraft Engine Management System
AFB	Air Force Base
AFR	Air Force Regulation
AFSC	Air Force System Command
AGI	Armament Gas Ingestion
AIA	Aerospace Industries Association
AMS	Aerospace Material Specification
AMT	Accelerated Mission Test
AOA	Angle of Attack
APC	Aircraft Power Control
API	Armor Piecing Incendiary
APU	Auxiliary Power Unit
ARD	Aerospace Resource Document
ARP	Aerospace Recommended Practices
AS	Aerospace Standards
ASC	Aeronautical Systems Center
ASCII	American Standard Code for Information Interchange
ASIP	Aircraft Structural Integrity Program
ASME	American Society of Mechanical Engineers
ASMET	Accelerated Simulated Mission Engine Test

ASTM	American Society for Testing and Material
ASW	Anti Submarine Warfare
ATE	Automatic Test Equipment
ATF	Advanced Tactical Fighter
Aviation 3M	Aviation Maintenance–Material–Management System
AVIP	Avionics Integrity Program
BIT	Built in Test
BOCLE	Ball–on–Cylinder Lubricity Evaluator
BPS	Bytes Per Second
BUC	Backup Control
C&E	Controls and Externals
CAMS	Core Automated Maintenance System
CDRL	Contractor Data Requirements List
CEMS	Comprehensive Engine Management System
CERT	Combined Environmental Reliability Test
CFE	Contractor Furnished Equipment
CG	Center of Gravity
CL	Confidence Level
COMNAVAIRLANT	Commander Naval Air Atlantic Fleet
COMNAVAIRPAC	Commander Naval Air Pacific Fleet
CONUS	Continental United States
CR	Component Removal
CSCIs	Computer Software Configuration Items
dBsm	Decibels per sq meter
DDR&E	Deputy Director of Research and Engineering
DECU	Digital Electronic Control Unit
DEM/EVAL	Demonstration/Evaluation
DEW	Directed Energy Weapons
DID	Data Item Description

DOD	Domestic Object Damage
DT&E	Development Tests and Evaluation
DTD	Damage Tolerant Design
E&MD	Engineering and Manufacturing Development
ECC	Engine Cycle Counter
ECOMTRAK	Engine Component Tracking System
ECP	Engineering Change Proposal
ECS	Environmental Control System
EDSA	European Distribution System Aircraft
EEPROM	Erasable Electronically Programmable Read Only Memory
EFH	Engine Flight Hours
EGT	Exhaust Gas Temperature
EIA	Electronic Industries Association
EM	Electromagnetic
EMC	Electromagnetic Compatibility
EMD	Engineering and Manufacturing Development
EME	Electromagnetic Environment
EMI	Electromagnetic Interference
EMP	Electromagnetic Pulse
EMS	Engine Monitoring System
EMT	Elapsed Maintenance Time
EMV	Electromagnetic Vulnerability
ENSIP	Engine Structural Integrity Program
EOH	Engine Operating Hours
EPR	Engine Pressure Ratio
ER	Engine Removal
ESS	Environmental Stress Screening
FADEC	Full Authority Digital Engine Control
FDDM	Frequency Division Demultiplexer

FDM	Frequency Division Multiplexer
FER	Failure Engine Removals
FETT	First Engine to Test
FFR	Full Flight Release
FITS	Fuel Ingestion Tolerance Specification
FMECA	Failure Modes, Effects, and Criticality Analysis
FOD	Foreign Object Damage
FRACAS	Failure Reporting and Corrective Action System
FSD	Full Scale Development
FSII	Fuel System Icing Inhibitor
GdB	Giga decibel
GFE	Government Furnished Equipment
GGT	Gas Generator Temperature
HC	Hydrocarbon
HCF	High Cycle Fatigue
HEMP	High Energy Electromagnetic Pulse
HOL	Higher Order Language
HPT	High Pressure Turbine
IAS	Indicated Airspeed
ICD	Interface Control Document
IEC	International Electrotechnical Commission
IECMS	In–Flight Engine Condition Monitoring System
IFR	Initial Flight Release
IFSD	In–Flight Shut Down
IGV	Inlet Guide Vanes
ILS	Integrated Logistics Support
IPS	Inlet Particle Separator
IR	Infrared
IRP	Intermediate Rated Power

IRTP	Integrated Reliability Test Program
ISA	International Standard Atmosphere
ISR	Initial Service Release
JBD	Jet Blast Deflector
JLFT	Joint Live Fire Test
JOAP	Joint Oil Analysis Program
JTCG/AS	Joint Tactical Coordination Group on Aircraft Survivability
LCC	Life Cycle Costs
LCF	Low Cycle Fatigue
LD	Laser Diode
LE	Leading Edge
LED	Light Emitting Diode
LFT&E	Live Fire Test and Evaluation
LHV	Lower Heating Value
LI	Lubricity Improver
LO	Low Observable
LPT	Low Pressure Turbine
LRU	Line Replaceable Unit
LWC	Liquid Water Content
MA	Maintenance Action
MATV	Multi Axis Thrust Vectoring
MAU	Maintenance Actions Unscheduled
MECSIP	Mechanical Equipment Structural Integrity Program
MEFH	Mean Engine Flight Hours
MMH/EFH	Maintenance Man Hours/Engine Flight Hours
MNS	Mission Need Statement
MOPP	Mission Oriented Protective Posture
MTBF	Mean Time Between Failures
MTBMA	Mean Time Between Maintenance Action

MTTR	Mean Time to Repair
MULDEM	Multiplexer/Demultiplexer
NACA	Nation Advisory Committee for Aeronautics
NADEP	Naval Air Depot
NALDA	Naval Engine Logistics Data Analysis
NAPTC	Naval Air Propulsion Test Center
NATF	Navy Advanced Tactical Fighter
NAVAIR	Naval Air Systems Command
NAVSEA	Naval Sea Systems Command
NAWC	Naval Air Warfare Center
NAWCWD	Naval Air Warfare Center Weapons Division
NBC	Nuclear, Biological and Chemical
NBCCS	NBC Contamination Survivability
NDI	Non–destructive Inspection
NMC	Not Mission Capable
NOX	Nitrogen Oxide
OCR	Operational Capability Release
OEI	One engine inoperative
OMC	Organic Matrix Composites
OPNAVINST	Naval Operations Instruction
PFR	Preliminary Flight Release
PLA	Power Lever Angle
PNdB	Perceived Noise Level
POD	Probability of Detection
PPM	Parts Per Million
PTO	Power Takeoff
QAP	Quality Assurance Provision
QEC	Quick Engine Change
QPL	Qualified Products List

R&M	Reliability and Maintainability
RAM	Random Access Memory
RAM	Radar Absorbent Material
RCI	Reliability Critical Items
RCS	Radar Cross Section
RFP	Request for Proposal
RMS	Root Mean Square
RVP	Reid Vapor Pressure
SAE	Society of Automotive Engineers
SCN	Specification Change Notice
SDIP	Software Development Integrity Program
SFC	Specific Fuel Consumption
SHP	Shaft Horsepower
SI	International System of Units
SLS	Sea Level, Static
SNR	Signal to Noise Ratio
SRA	Service Replaceable Assembly
SRU	Service Replaceable Unit
STAR	System Threat Assessment Report
STR	System Threat Report
SURVIAC	Survivability and Vulnerability Information Analysis Center
TAC	Total Accumulated Cycles
TBO	Time Between Overhauls
TDDM	Time Division Demultiplexer
TDM	Time Division Multiplexer
TEMS	Turbine Engine Monitoring System
TJ/TF	Turbojet and Turbofan
TLV	Threshold Limit Value
TP/TS	Turboprop and Turboshaft

TSFC	Thrust Specific Fuel Consumption
USA	United States Army
USAF	United States Air Force
USC	United States Code
USN	United States Navy
UV	Ultraviolet
V/L	Vapor–Liquid Ratio
V/STOL	Vertical/Short Takeoff & Landing
VLF	Very Low Frequency
VPC	Vertical Parity Check
WOW	Weight of Wheels
WRA	Weapon Replacement Assembly
WRA/LRU	Weapons Replacement Assembly/Line Replaceable Unit
WSD	Wear Scar Diameter

RATIONALE

This is a quick and easy reference for all symbols, subscripts and abbreviations.

GUIDANCE

The following should be used to tailor the specification paragraph:

This list may be organized into convenient categories at the contractors discretion.

LESSONS LEARNED

None.

60.6 Material safety data sheet Contracting officers shall identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED–STD–313. The pertinent Government mailing addresses for submission of data are listed in FED–STD–313.

RATIONALE

This paragraph is required by MIL–STD–961, paragraph on Material Safety Data Sheets.

GUIDANCE

When the specification describes products containing hazardous material, material safety data sheets are required. The words of paragraph 6.6 should be used in the specification.

LESSONS LEARNED

None.

60.7 Subject term (key word) listing.

Engine, aircraft

Gas turbine engine, aircraft

Propulsion

Turbofan

Turbojet

Turboprop

Turboshaft

RATIONALE

This paragraph is required by MIL–STD–961, paragraph on Subject term (key word) listing.

GUIDANCE

The acquisition specification and model specification are not available during retrieval searches. Only the general specification is available during retrieval searches. Retain paragraph number and title and indicate N/A for the acquisition specification.

LESSONS LEARNED

None.

60.8 Metrication The engine shall be of metric design in accordance with Public Laws 94–168 and 100–418. The engine shall conform to the SI Standard described in American Society for Testing and Material (ASTM) E380, Standard for Metric Practice (also numbered American National Standards Institute Z210.1) and Institute of Electrical & Electronics Engineer Standard 268, as listed in the DOD Index of Specifications and Standards. All new metric items (requiring a new part identification) or any new design practice created/designed to replace any customary system of measurement shall be integrated into the SI Metric System of Measurement in accordance with Public Laws 94–168 and 100–418. The specification shall contain both metric and English units. The metric units shall be primary. English units shall be provided within parenthesis.

RATIONALE

Public law 94–168 declares that the policy of the United States shall be to coordinate and plan the increasing use of the metric system. Public law 100–418 designates the metric system of measurement as the preferred system of measurement for United States trade and commerce and requires that all Federal agencies use the metric system of measurement by the end of FY 1992 to the extent economically feasible. In conformance with Public Law 100–418, the Department of Defense (DoD) Directive 4120.18 requires that all DoD activities adopt the metric system of measurement in all their activities consistent with security, operational, economical, technical, logistical, and safety requirements, and shall place emphasis on developing metric specifications and standards to support the development of defense systems, equipment, and materials.

GUIDANCE

The words of 6.8 should appear in the specification.

LESSONS LEARNED

Rolls–Royce Pegasus F402, Adour F405 and Army T800 engines are metric engines presently in use by DOD.

60.9 Responsible engineering office The Air Force office responsible for the development and technical maintenance of this specification is ASC/ENFP, 2335 Seventh St, Ste 6, Wright–Patterson AFB, OH 45433–7809. Requests for additional information or assistance can be obtained from this office by writing to the above address or by calling DSN 785–9591, Commercial (513) 255–9591. Any information relating to Government contracts must be obtained through the contracting officers. The responsible Navy engineering office is the Naval Air Systems Command, AIR–536, Arlington, VA 22243–5360, DSN 664–3290, Commercial (703) 604–3290. The responsible Army engineering office is the Aviation Troop Command, ATCOM (AMSAT–R–EPT), 4300 Goodfellow Boulevard, St. Louis, MO 63120–1798, DSN 693–1644, Commercial (314) 263–1644.

RATIONALE

This paragraph provides a point of contact for each service to provide technical guidance.

GUIDANCE

The following should be used to tailor the specification paragraph:

Take this paragraph out when writing the model specification.

LESSONS LEARNED

None.

60.10 Streamlining For acquisitions, the required portions of this specification's tier reference documents, will be limited to the portion(s) referenced in the applicable paragraphs.

RATIONALE

This requirement comes from MIL–HDBK–800, paragraph on Streamlining amendment/revision to a procurement specification. This paragraph presents the results of the streamlining effort on the general engine specification.

GUIDANCE

The following should be used to tailor the specification paragraph:

The words of paragraph 6.10 should appear in the acquisition and model specifications.

LESSONS LEARNED

The streamlining effort has reduced the number of tiered documents in past specifications to a manageable level.

TABLE Ia. Performance at standard day, 15°C (59°F), sea level, static conditions[1/] with _____ nozzle.

Power Setting[2/]	Minimum Thrust N (lbf)		Maximum SFC (lbm/hr/lbf)		Max Engine Rotor Speed (rpm)		Maximum Gas Temp[3/] °C(°F)		Maximum Measured Temp[4/] °C(°F)		Total Airflow[5/] +/–___% (lbm/sec)	
	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]
Max. Aug. (if applicable)												
Min. Aug. (if applicable)												
Int.												
90% Int.												
75% Int.												
Idle	(Max)	(Max)	[6/]	[6/]	(Min)	(Min)						

NOTES:

[1/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.1.

[2/] Power settings and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air/power extraction).

[3/] Defined at the first stage turbine rotor inlet location (Contractor shall parenthetically insert, in column heading, the station designation).

[4/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[5/] Total engine airflow.

[6/] Maximum fuel consumption – lbm/hr.

[7/] Deteriorated engine performance is specified for ___ TACs (or hours) of use.

TABLE Ib. Performance at non–standard hot ambient, sea level, static conditions[1/] with _____ nozzle.

Power Setting[2/]	Minimum Thrust N (lbf)		Maximum SFC (lbm/hr/lbf)		Max Engine Rotor Speed (rpm)		Maximum Gas Temp[3/] °C(°F)		Maximum Measured Temp[4/] °C(°F)		Total Airflow[5/] +/–___% (lbm/sec)	
	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]
Max. Aug. (if applicable)												
Min. Aug. (if applicable)												
Int.												
90% Int.												
75% Int.												
Idle	(Max)	(Max)	[6/]	[6/]	(Min)	(Min)						

NOTES:

[1/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.1.

[2/] Power settings and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air/power extraction).

[3/] Defined at the first stage turbine rotor inlet location (Contractor shall parenthetically insert, in column heading, the station designation).

[4/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[5/] Total engine airflow.

[6/] Maximum fuel consumption – lbm/hr.

[7/] Deteriorated engine performance is specified for ___ TACs (or hours) of use.

TABLE Ic. Performance at non–standard cold ambient, sea level, static conditions[1/] with _____ nozzle.

Power Setting[2/]	Minimum Thrust N (lbf)		Maximum SFC (lbm/hr/lbf)		Max Engine Rotor Speed (rpm)		Maximum Gas Temp[3/] °C(°F)		Maximum Measured Temp[4/] °C(°F)		Total Airflow[5/] +/–___% (lbm/sec)	
	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]	New	Det.[7/]
Max. Aug. (if applicable)												
Min. Aug. (if applicable)												
Int.												
90% Int.												
75% Int.												
Idle	(Max)	(Max)	[6/]	[6/]	(Min)	(Min)						

NOTES:

[1/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.1.

[2/] Power settings and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air/power extraction).

[3/] Defined at the first stage turbine rotor inlet location (Contractor shall parenthetically insert, in column heading, the station designation).

[4/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[5/] Total engine airflow.

[6/] Maximum fuel consumption – lbm/hr.

[7/] Deteriorated engine performance is specified for ___ TACs (or hours) of use.

TABLE Ia. Turboprop/turboshaft engine performance at standard day, 15°C (59°F), sea level static conditions[6/] with _____ nozzle.

Power setting[1/]	Max. gas generator speed(s) (rpm)		Max. gas temp.[2/] (°C(°F))		Max. measured temp.[3/] (°C(°F))		Total airflow[4/] +/–___% (lb/sec)		Max. SFC (lbm/hr/kw)		Min. shaft power kw (shp)		Output torque N–m (lb–ft)		Min. output shaft speed (rpm)		Resid. jet thrust N (lbf)	
	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]
Contingency (if applicable)																		
Maximum (if applicable)																		
Intermediate																		
Maximum Continuous																		
90% Maximum Continuous																		
No load (if applicable)																		
Idle	Min.	Min.							[5/]	[5/]	Max	Max						

NOTES:

[1/] Parameters and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air).

[2/] Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

[3/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[4/] Total engine airflow.

[5/] Maximum fuel consumption – lb/hr.

[6/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.1.

[7/] Deteriorated engine performance is specified for ____ TACs (or hours) of use.

TABLE Ib. Turboprop/turboshaft engine performance at non–standard hot ambient sea level static conditions[6/] with _____ nozzle.

Power setting[1/]	Max. gas generator speed(s) (rpm)		Max. gas temp.[2/] (°C(°F))		Max. measured temp.[3/] (°C(°F))		Total airflow[4/] +/–___% (lb/sec)		Max. SFC (lbm/hr/kw)		Min. shaft power kw (shp)		Output torque N–m (lb–ft)		Min. output shaft speed (rpm)		Resid. jet thrust N (lbf)	
	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]
Contingency (if applicable)																		
Maximum (if applicable)																		
Intermediate																		
Maximum Continuous																		
90% Maximum Continuous																		
No load (if applicable)																		
Idle	Min.	Min.							[5/]	[5/]	Max	Max						

NOTES:

[1/] Parameters and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air).

[2/] Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

[3/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[4/] Total engine airflow.

[5/] Maximum fuel consumption – lb/hr.

[6/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.1.

[7/] Deteriorated engine performance is specified for ____ TACs (or hours) of use.

TABLE Ic. Turboprop/turboshaft engine performance at non–standard cold ambient sea level static conditions[6/] with _____ nozzle.

Power setting[1/]	Max. gas generator speed(s) (rpm)		Max. gas temp.[2/] (°C(°F))		Max. measured temp.[3/] (°C(°F))		Total airflow[4/] +/–___% (lb/sec)		Max. SFC (lbm/hr/kw)		Min. shaft power kw (shp)		Output torque N–m (lb–ft)		Min. output shaft speed (rpm)		Resid. jet thrust N (lbf)	
	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]	New	Det[7/]
Contingency (if applicable)																		
Maximum (if applicable)																		
Intermediate																		
Maximum Continuous																		
90% Maximum Continuous																		
No load (if applicable)																		
Idle	Min.	Min.							[5/]	[5/]	Max	Max						

NOTES:

[1/] Parameters and columns shall be added or deleted as required by the Using Service (e.g., customer bleed air).

[2/] Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

[3/] Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

[4/] Total engine airflow.

[5/] Maximum fuel consumption – lb/hr.

[6/] The engine performance values shown shall be from the engine performance computer program of 3.2.1.1.

[7/] Deteriorated engine performance is specified for ____ TACs (or hours) of use.

TABLE II. Performance at altitude conditions[4/] with _____ nozzle.

Power setting[1/]	Alt. (ft)	Amb. temp.	Mach num.	Minimum net thrust N (lbf)		Maximum SFC (lb/hr/lb)		Maximum engine rotor speed (rpm)		Maximum gas temp.[2/] (°C(°F))		Maximum measured temp.[3/] (°C(°F))		Total airflow (lb/sec) +/–___%		Cust. bld. air	Cust. pwr. ext.
				New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]		

NOTES:

1/ Power settings shall be as required by the Using Service to cover the operating envelope and to be compatible with mission requirements.

2/ Defined at the first stage turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

3/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

4/ The engine performance values shown shall be from the engine computer program of 3.2.1.1.

5/ Deteriorated engine performance is specified for _____ TACs (or hours) of use.

TABLE II. Turboprop/turboshaft performance at altitude conditions[4/] with _____ nozzle.

Power setting[1/]	Alt. (ft)	Amb. temp.	Mach num.	Max. gas gen. speed (rpm)		Max. gas temp.[2/] (°C(°F))		Max. meas. temp.[3/] (°C(°F))		Total airflow (lb/sec) +/–__%		Max. SFC (lb/hr/kw)		Min. shaft power kw (shp)		Output torque N–m (lb–ft)		Output shaft speed (rpm)		Resid. jet thrust N (lbf)		Cust. bld. air	Cust. pwr. ext.
				New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]	New	Det[5/]		

NOTES:

1/ Power settings shall be as required by the Using Service to cover the operating envelope and to be compatible with mission requirements.

2/ Defined at the first stage high pressure turbine rotor inlet location (contractor shall parenthetically insert, in column heading, the station designation).

3/ Contractor shall parenthetically insert, in column heading, the measurement plane station designation.

4/ The engine performance values shown shall be from the engine computer program of 3.2.1.1.

5/ Deteriorated engine performance is specified for _____ TACs (or hours) of use.

TABLE III. Control system interface.

TABLE IV. Instrumentation system interface.

Parameter	Signal characteristics	Interface description

TABLE V. Gearbox pads and drives.

Accessory or Component	Type of Drive 1/	Ratio Pad to Rotor rpm 2/	Rotation Direction Facing Pad	Torque n–mm (lbf–in)			Overhung Moment n–mm (lbf–in)	Maximum Spline Mis–alignment mm (in)
				Maximum Continuous	Overload 3/	Static		

NOTES:

1/ Give the type of drive including AND or MS number and type.

2/ Ratio of speeds based on 100 percent rotor speed of _____ RPM.

3/ Specify duration and frequency of overload.

TABLE VI. Design duty cycle summary.

Cold Parts									
	Time (hrs)	TAIAA (hrs)	Type I cycles	Type III cycles	Type IV cycles	A/B lts	A/B time (hrs)	Vector cycles	Other cycles
Flight operations									
Ground operations									
Test cell trouble–shooting etc									
TOTAL									

TAC = Type I + Type III/4 + Type IV/40 + Kx (other cycles)

Hot Parts									
	Time (hrs)	TAIAA (hrs)	Type I cycles	Type III cycles	Type IV cycles	A/B lts	A/B time (hrs)	Vector cycles	Other cycles
Flight operations									
Ground operations									
Test cell trouble–shooting etc									
TOTAL									

TABLE VIIa. Recorded and monitored aircraft parameters.

Commands/requests:		
TT2 probe heater request		
Nozzle angle request		
Nozzle exit area request		
Engine IBIT request/in progress		
Pilot initiated data request		
Engine fault reset		
Engine minimum airflow request		
Bleed flow request		
Aircraft sensed data:		
Angle of attack		
Angle of sideslip		
Calibrated airspeed		
Mach number		
Pressure altitude		
Static pressure – ambient press		
Axial acceleration		
Lateral acceleration		
Normal acceleration		
Time (data, hours, minutes, seconds)		
Missile ‘‘armed” index		
Missile ‘‘launched” index		
Weight on gear (nose, left and right)		
Shutoff valve closed – fire (nacelle)		
Fuel supply pressure		
Fuel density		
Fuel temperature at density meter		
Emergency fuel usage		
Cross–engine inlet temperature		
Cross–engine inlet pressure		
A/C boost pumps on–line		
Calculated torque (+/–)		
Maneuvering index		

TABLE VIIb. Recorded and monitored engine parameters.

Rotational speeds:		
Low rotor		
High rotor		
Gas path pressures:		
Turbine exhaust		
Inlet		
Main burner		
Fan duct		
Gas path temperatures:		
Inlet		
Turbine blade		
Exhaust gas		
Oil system:		
Lube debris indication		
Temperature		
Pressure		
Level		
Filter delta press		
Fuel system:		
Fuel filter delta press		
GG fuel temperature		
Actuator pressure		
GG throttle valve delta press		
Effector positions:		
Miscellaneous:		

TABLE VIII. Limits

Parameter	Steady State		Transient (Maximum)
	Minimum	Maximum	
Rotor Speed			
Turbine Temperature			
Burner Pressure			
Airflow			
Fuel Flow			
Vibration			
Etc			

TABLE IX. Component limiting temperature.

Component	Maximum Ambient Air Temperature	Limiting Surface Temperature	Time Limit for Temperature	Limiting Cooling Fluid Inlet Temperature
Fuel Pump				
Oil Pump				
FADEC				
Actuators				
Etc.				

TABLE X. Schedule of salt spray injection endurance cycles.

Phase No. 1/	Duration of Phase (hours)	Test Engine Operation	Salt Solution Parts per Billion (PPB)	Engine Ambient Air	
				Temperature	Relative Humidity
1	3 2/, 3/	Operating	200±40 4/	10°C min. 50°F min.	73% min.
2	2 5/	Not operating	0	Atmospheric	Atmospheric
3	7 5/	Not operating	200±40 4/, 6/	10°C min. 50°F min.	73% min.
4	12 5/	Not operating	0	43±5°C 109±9°F	90% min.

TABLE X. Schedule of salt spray injection endurance cycles – Continued.

NOTES

1/ Engine inlet and exhaust openings shall remain open for all phases of the test cycle.

2/ During shutdown, while the engine is decelerating from idle, the salt solution shall continue to be sprayed into the engine until the rotor has come to rest.

3/ An engine operating cycle shall be selected from the mission mix which has at least six power levels with dwell times of at least 5 minutes per level. The power levels shall roughly be equally spaced throughout the cycle. If a suitable cycle cannot be identified, then the cycle of figure 17 shall be used. The cycle shall be constructed so that the metal temperature in the hot gas flow path is between 760°C and 980°C (1400°F and 1800°F). This shall be accomplished by varying the turbine inlet temperature from 760°C (1400°F) to the maximum temperature.

4/ Salt solution ingested by the engine shall conform to that specified in Note 7/ below, and shall be regulated to provide a concentration of 200±40 PPB by weight of salt in air during those phases of each cycle which requires shalt ingestion. A salt sampling system shall be employed to determine the concentration level during each cycle of operation.

5/ The duration of phases 2, 3, and 4 can vary up to 10 percent of the time for the respective phase for any particular cycle. However, the total time for phases 1 – 4 (1 cycle) shall be 24 hours. Down time shall be considered as phase 2. Down time shall include time for engine facility maintenance, weekends, holidays, etc. The total down time shall not be less than 200 hours for 100 cycles. The maximum down time is negotiable. The minimum duration for phases 3 and 4 is 700 and 1200 hours, respectively.

6/ The test facility shall provide the flow of salt–laden air through the engine gas flowpath(s) and over the external surfaces of the engine. The external flow velocity shall be between 4.6–6.1 m/sec (15–20 ft/sec).

7/ The basic salt formulation shall be composed of the following materials dissolved with sufficient distilled water to make one liter of salt solution. Additional distilled water may be added to the salt solution, as required, to provide a uniform salt aerosol profile across the face of the engine. The maximum dilution shall not exceed 40:1. The aerosol droplet size shall not exceed 25 microns.

Chemical Designation	Quantity per liter of stock solution
NaCl (c.p.)	23 grams
$Na_2SO4\ 10H_2O$	8 grams
Stock solution (see note 8/)	20 milliliters

8/ The stock solution shall be composed of the following materials dissolved with sufficient distilled water to make one liter of stock solution:

Chemical Designation	Quantity per liter of stock solution
Kcl (c.p.)	10 grams
Kbr	45 grams
$MgCl_2\ 6H_2O$ (c.p.)	550 grams
$CaCl_2\ 6H_2O$ (c.p.)	110 grams

TABLE XI. Sea level anti–icing conditions.

	Part 1			Part 2
Ambient Temperature	–21 to –19°C (–5 to 3°F)	–10 to –9°C 1/ (14 to 16°F)	–6 to –4°C (22 to 24°F)	–6 to –4°C (22 to 24°F)
Velocity	0 to 31 m/sec (0 to 60 knots)	0 to 31 m/sec (0 to 60 knots)	0 to 31 m/sec (0 to 60 knots)	0 to 31 m/sec (0 to 60 knots)
Altitude	0 to 150 m (0 to 500 ft)	0 to 150 m (0 to 500 ft)	0 to 150 m (0 to 500 ft)	0 to 150 m (0 to 500 ft)
Mean Effective Drop Diameter	20 ±5 microns	20 ±5 microns	20 ±5 microns	30 ±5 microns
Liquid Water Content (Continuous)	1.25 to 1.75 gm/m^3	2.85 to 3.35 gm/m^3	2.85 to 3.35 gm/m^3	0.6 to 0.8 gm/m^3

NOTE:

1/ This condition applies only to turbofan engines.

TABLE XII. Bird ingestion.

Bird Size	Number of Birds	Bird Velocity	Thrust/ Power Setting	Percent Thrust/ Power Retention	Thrust/ Power Recovery Time	Damage
100 gm (3.5 oz)		Takeoff =				Blendable
100 gm (3.5 oz)		Cruise =				Blendable
100 gm (3.5 oz)		Low Level Hi–Speed =				Blendable
100 gm (3.5 oz)		Descent =				Blendable
1 kg (2.2 lbs)		Takeoff =				Minor
1 kg (2.2 lbs)		Cruise =				Minor
1 kg (2.2 lbs)		Low Level Hi–Speed =				Minor
1 kg (2.2 lbs)		Descent =				Minor
2 kg (4.4 lbs)		Takeoff or Low Level Hi–Speed =				Contain Failure

Each line of the table must be satisfied to comply with the requirements of 3.3.2.1.

TABLE XIII. Ice ingestion.

Dimensions	Number	Velocity km/hr (mph)	Thrust/ Power Setting	Percent Thrust/ Power Retention	Thrust/ Power Recovery Time	Specific Gravity g/cm^3 (lbm/ft^3)
Hailstone Diameter mm(in)	Hailstones					
Sheet Ice mm x mm x mm (in x in x in)	Sheet Ice					

Inlet area: _____ m^2 (_____ in^2)

TABLE XIV. Hot parts.

TABLE XV. Cold parts.

TABLE XVI. Expendable parts.

TABLE XVII. Bearing lives.

Bearing	Type (Roller or Ball)	Life (Hours)

TABLE XVIII. Components.

TABLE XIX. Parts classification.

Engine Part or Component	Safety Critical	Mission Critical	Durability Critical	Durability Non–critical

TABLE XX. AMT Test/_____ cycles.

Endurance Test Mission Profiles	No. of Missions								
	Total Missions	Heated Inlet °C (°F)			Ram Pressure kPa (psi)				Max. Turbine Temp.
		≥ X°	≥ Y°	≥ Z°	101.3 (14.7)	A	B	C	
Mission #1									
Mission #2									
Mission #3									
Mission (etc.)									
Composite									
Ground Test									

Note: the blanks with “X”, “Y”, “Z”, “A”, “B”, and “C” should be completed by the contractor and is based on the temperatures and pressures for the mission. When added together, the number of missions under the Heated Inlet and Ram Pressure columns should equal the value in the Total Missions column.

TABLE XXI. LCF engine test cycle turboprop/turboshaft engines.

Time (min)	Event
0.5	Start engine
2.0	Run at idle
0.1	Accel to maximum power
2.5	Run at maximum power
0.1	Decel to idle
3.0	Run to idle
0.1	Accel to max continuous
2.5	Run to max continuous
0.1	Decel to idle
2.0	Run to idle
2.1	Shutdown and cool down
15.0	Total

Notes:
1. Transient power commands are in 1/2 second or less.
2. Power settings are initial rating values of output thrust or power or specified by Using Service.
3. Engine control and fuel flow schedules are set for maximum acceleration.
4. Output shaft speeds are rated values at high power settings and minimum self sustaining speed at idle.
5. With customer bleed and power extractions.

TABLE XXIIa. Reliability quantitative requirements.

Reliability Requirements	Requirement
1. Mean engine flight hours between engine removal (MEFH/ERC)	
2. Mean engine flight hours between engine failures (MEFH/F) a. In flight shutdown rate (events/1000 EFH) b. In flight power loss rate (events/1000 EFH)	
3. Mean engine flight hours between maintenance actions (MEFH/MA)	
4. Mean engine flight hours between maintenance actions unscheduled (MEFH/MAU)	
5. ‘‘Combined” mean time between failure for RCI’s	

TABLE XXIIb. Reliability quantitative requirements.

	Checkpoint Values	Requirement	Operational Goals	
	IFR FFR ISR OCR	at FCA	100K EFH	500K EFH
a. Mean time between maintenance inherent (EFH)				
b. Combined mean time between failure for selected RCIs (EFH)	N/A N/A N/A N/A		N/A	N/A
c. Line replaceable unit removal rate (events/10^3EFH)				
d. Total controllable engine removal rate (events/10^3EFH)				
e. In–flight shutdown rate (events/10^3EFH)				
f. In–flight power loss rate (events/10^3EFH)				
g. Engine removal rate for exceedance of performance deterioration (event/10^3EFH) (if applicable)				
h. Combined engine removal rate (all causes) operational goals	N/A N/A N/A N/A	N/A		
i. MTBM total operational goal		N/A		
j. Mean time between essential maintenance action (MTBEMA) (USA only)				
k. Mission affecting failure (MAF) (USA only)				
l. Mean time between mission abort (MTBMA) (USA only)				

TABLE XXIIIa. Reliability quantitative requirement.

a. Accelerated mission tests	
b. Engine performance (e.g., altitude) tests	
c. Bench qualification tests or subassemblies	
d. FMECA	
e.	
f.	
g.	
h.	
i.	
j.	
k.	
etc.	

NOTE: The results of other engine and component level tests shall be counted providing that the contractor gains agreement with the Government, prior to conducting the tests, that they are pertinent to reliability determination.

TABLE XXIIIb. Reliability quantitative evaluation.

Subassembly: _____
Failure Mode:

Development test involving subassembly			Data Pertinent to Reliability		(A) Test(1) severity factor to failure mode	(B) Test time actual (H–)	(C) Equivalent test time of test (Hr) (A) x (B)	(D) Equivalent test time to date (C) to date	(E) Number failures during test for the failure mode	(F) Total failures to date for failure mode (E) to date
No	Title	Date	Y	N						

NOTES:

(1) Test severity factor is based on the mIssion profile and mix (i.e., severity factor is 1 for the mission profile and mix); Reference 3. If the mission profile and mix changes, the provisions of this table will have to be re–evaluated, because severity factors may change.

(2) Equivalent test time and number of failures experienced during a test are identified as zero for reliability determination, if it is felt that the test severity is significantly different than 1 (i.e., 0.75 $\times$ severity 1.25), and it is not considered possible or practical to establish a test severity factor.

TABLE XXIV. Maintainability quantitative requirements.

	Milestones					Operational Goals	
	IFR	FFR	ISR	OCR	Reqmnt	100K EFH	500K EFH
1. Total maintenance manhour per engine flight hour scheduled (MMH/EFH scheduled)							
Organizational (O–level)							
Intermediate (I–level)							
Depot (D–level)							
2. Total maintenance manhours per engine flight hour unscheduled (MMH/EFH unscheduled)							
Organizational (O–level)							
Intermediate (I–level)							
Depot (D–level)							
3. Total maintenance manhour per maintenance action (MMH/MA)							
Organizational (O–level)							
Intermediate (I–level)							
Depot (D–level)							
4. Total engine flight hours per maintenance action (MTBMA) (EFH/MA)							
Organizational (O–level)							
Intermediate (I–level)							
Depot (D–level)							
5. Mean time to repair (MTTR)							
Organizational (O–level)							
Intermediate (I–level)							
Depot (D–level)							
6. Engine replacement time							
(1) Total man hours (min)							
(2) Elapsed maintenance time (EMT) (CLOCK–min)							
7. Weapon replacement assembly (WRA)/line replaceable unit (LRU) replacement time–engine installed (minutes)							
(1) Weighted average							
(2) Not to exceed							
Diagnostics							
8. % of success that diagnostic testing can isolate faulty WRA/LRU assemblies within ____ minutes.							

TABLE XXIV. Maintainability quantitative requirements – Continued.

	Milestones					Operational Goals	
	IFR	FFR	ISR	OCR	Reqmnt	100K EFH	500K EFH
9. % of success that diagnostic testing can isolate faulty replaceable subassemblies within WRA/LRU assemblies.							
10. % of success that diagnostic testing correcting diagnosed fault by retest.							

TABLE XXV. Maintainability quantitative requirement.

Systems	Requirements	
	Percent ATE fault isolation to faulty replaceable subassemblies within WRA/LRU assemblies	Percent retest OK to specified subassemblies

TABLE XXVI. Excluded maintenance functions.

TABLE XXVII. Maintainability qualitative checklist.

Qualitative	Requirement

TABLE XXVIII. Modules

Module Removal	
Module Name	Removal Time (hrs)

TABLE XXIX. Maintenance, inspection, and repair cycles.

Maintenance, inspection, and repair cycles		
Item	Time or interval	Elapsed time or repair time

TABLE XXXa. Option 1 (see 3.7.3.1).

The altitude relight envelope for each of the two major fuel types (kerosene and wide–cut) shall be identified.

Primary fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL–T–5624 Grade JP–5	DERD 2452 AVCAT/FSII	F–44	None	None	High Flash Kerosene
MIL–T–83133 Grade JP–8	DERD 2453 AVTUR/FSII	F–34	ASTM D 1655 Jet A–1 with Additives[2]	None	Kerosene
None	None	F–35	ASTM D 1655 Jet A–1 w/o Additives[3]	DERD 2494 AVTUR	
None	None	None	ASTM D 1655 Jet A with Additives[2]	None	
None	None	None	ASTM D 1655 Jet A w/o Additives[3]	None	
MIL–T–5624 Grade JP–4	DERD 2454 AVCAT/FSII	None	ASTM D 1655 Jet B with Additives[2]	None	Wide–Cut
None	None	None	ASTM D 1655 Jet B w/o Additives[3]	DERD 2468 AVCAT	

Notes: 1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL–T–5624:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL–I–25017)
   b. fuel system icing inhibitor (FSII) (MIL–I–85470 or MIL–I–27686)
   c. an approved antioxidant (AO) material listed in para 3.3.1.1 of MIL–T–5624N
   d. an approved static dissipator additive (SDA) listed in para 3.3.5 of MIL–T–5624N
3. Commercial fuels do not normally contain any additives.

Alternate fuel(s): None required.
Restricted fuel(s): None required.
Emergency fuel(s): See requirement in 3.7.3.4.

TABLE XXXb. Option 2 (see 3.7.3.1).

Primary fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL–T–5624 Grade JP–5	DERD 2452 AVCAT/FSII	F–44	None	None	High Flash Kerosene
MIL–T–83133 Grade JP–8	DERD 2453 AVTUR/FSII	F–34	ASTM D 1655 Jet A–1 with Additives[2]	None	Kerosene
None	None	F–35	ASTM D 1655 Jet A–1 w/o Additives[3]	DERD 2494 AVTUR	
None	None	None	ASTM D 1655 Jet A with Additives[2]	None	
None	None	None	ASTM D 1655 Jet A w/o Additives[3]	None	

Notes: 1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL–T–5624:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL–I–25017)
   b. fuel system icing inhibitor (FSII) (MIL–I–85470 or MIL–I–27686)
   c. an approved antioxidant (AO) material listed in para 3.3.1.1 of MIL–T–5624N
   d. an approved static dissipator additive (SDA) listed in para 3.3.5 of MIL–T–5624N
3. Commercial fuels do not normally contain any additives.

Alternate fuel(s): None required.

TABLE XXXb. Option 2 (see 3.7.3.1) – Continued.

Restricted fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL–T–5624 Grade JP–4	DERD 2454 AVCAT/FSII	None	ASTM D 1655 Jet B with Additives[2]	None	Wide–Cut
None	None	None	ASTM D 1655 Jet B w/o Additives[3]	DERD 2468 AVCAT	

Notes: 1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.

2. The following additives must be injected into the fuel at the concentrations specified in MIL–T–5624:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL–I–25017)
   b. fuel system icing inhibitor (FSII) (MIL–I–85470 or MIL–I–27686)
   c. an approved antioxidant (AO) material listed in para 3.3.1.1 of MIL–T–5624N
   d. an approved static dissipator additive (SDA) listed in para 3.3.5 of MIL–T–5624N
3. Commercial fuels do not normally contain any additives.

Emergency fuel(s): See requirement in 3.7.3.1.4.

TABLE XXXc. Option 3 (see 3.7.3.1).

Primary fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL–T–5624 Grade JP–5	DERD 2452 AVCAT/FSII	F–44	None	None	High Flash Kerosene
MIL–T–83133 Grade JP–8	DERD 2453 AVTUR/FSII	F–34	ASTM D 1655 Jet A–1 with Additives[2]	None	Kerosene
None	None	None	ASTM D 1655 Jet A with Additives[2]	None	

Notes: 1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL–T–5624:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL–I–25017)
   b. fuel system icing inhibitor (FSII) (MIL–I–85470 or MIL–I–27686)
   c. an approved antioxidant (AO) material listed in para 3.3.1.1 of MIL–T–5624N
   d. an approved static dissipator additive (SDA) listed in para 3.3.5 of MIL–T–5624N
3. Commercial fuels do not normally contain any additives.

Alternate fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
None	None	F–35	ASTM D 1655 Jet A–1 w/o Additives[1]	DERD 2494 AVTUR	Kerosene
None	None	None	ASTM D 1655 Jet A w/o Additives[1]	None	

Notes: 1. Commercial fuels do not normally contain any additives.

TABLE XXXc. Option 3 (see 3.7.3.1) – Continued.

Restricted fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL–T–5624 Grade JP–4	DERD 2454 AVCAT/FSII	None	ASTM D 1655 Jet B with Additives[2]	None	Wide–Cut
None	None	None	ASTM D 1655 Jet B w/o Additives[3]	DERD 2468 AVCAT	

Notes: 1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL–T–5624:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL–I–25017)
   b. fuel system icing inhibitor (FSII) (MIL–I–85470 or MIL–I–27686)
   c. an approved antioxidant (AO) material listed in para 3.3.1.1 of MIL–T–5624N
   d. an approved static dissipator additive (SDA) listed in para 3.3.5 of MIL–T–5624N
3. Commercial fuels do not normally contain any additives.

Emergency fuel(s): See requirement in 3.7.3.1.4.

TABLE XXXd. Option 4 (see 3.7.3.1).

Primary fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
MIL–T–5624 Grade JP–5	DERD 2452 AVCAT/FSII	F–44	None	None	High Flash Kerosene
MIL–T–83133 Grade JP–8	DERD 2453 AVTUR/FSII	F–34	ASTM D 1655 Jet A–1 with Additives[2]	None	Kerosene
None	None	None	ASTM D 1655 Jet A with Additives[2]	None	

Notes: 1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL–T–5624:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL–I–25017)
   b. fuel system icing inhibitor (FSII) (MIL–I–85470 or MIL–I–27686)
   c. an approved antioxidant (AO) material listed in para 3.3.1.1 of MIL–T–5624N
   d. an approved static dissipator additive (SDA) listed in para 3.3.5 of MIL–T–5624N
3. Commercial fuels do not normally contain any additives.

Alternate fuels: None required.

TABLE XXXd. Option 4 (see 3.7.3.1) – Continued.

Restricted fuels

Military			Commercial[1]		Fuel Type
US	UK	NATO Code	US	UK	
None	None	F–34	ASTM D 1655 Jet A–1 w/o Additives	DERD 2494 AVTUR	Kerosene
None	None	None	ASTM D 1655 Jet A w/o Additives	None	
MIL–T–5624 Grade JP–4	DERD 2454 AVCAT/FSII	None	ASTM D 1655 Jet B with Additives[2]	None	Wide–Cut
None	None	None	ASTM D 1655 Jet B w/o Additives[3]	DERD 2468 AVCAT	

Notes: 1. Once treated with the additives listed in note 2 below, the commercial fuel listed in this column is virtually identical to the corresponding Military fuel.
2. The following additives must be injected into the fuel at the concentrations specified in MIL–T–5624:
   a. corrosion inhibitor/lubricity improver (CI/LI) (MIL–I–25017)
   b. fuel system icing inhibitor (FSII) (MIL–I–85470 or MIL–I–27686)
   c. an approved antioxidant (AO) material listed in para 3.3.1.1 of MIL–T–5624N
   d. an approved static dissipator additive (SDA) listed in para 3.3.5 of MIL–T–5624N
3. Commercial fuels do not normally contain any additives.

Emergency fuel(s): See requirement in 3.7.3.1.4.

Definitions:

Primary fuels are the fuels on which the engine is designed to operate continuously without restrictions.

An alternate fuel is one on which the aircraft can be flown without operational restrictions but which can have long term durability or maintainability impact if used for continuous operation (multiple flights). Alternate fuels are used only on an occasional or intermittent basis.

A restricted fuel is one which imposes operational restrictions on the aircraft when used.

An emergency fuel is one which significantly limits the overhaul life of the engine and imposes operational restrictions on the aircraft as well.

TABLE XXXI. Radar cross section signature.

See classified appendix – To be published.

TABLE XXXIIa. Fuel contaminants for continuous operation.

Contaminant	Particle Size Microns	Quantity mg/liter (g/1,000 gal)
Ferroso–Ferric Iron oxide (Fe_3O_4, Black color) Magnetite	0–5	0.264 (1.0)
Ferric Iron Oxide (Fe_2O_3, Hematite)	0–5	1.321 (5.0)
Crushed Quartz	300 – 420	0.132 (0.5)
	150 – 300	0.132 (0.5)
Prepared dirt conforming to AC Spark Plug Co., Part No. 1543637 (coarse Arizona road dust)	Mixture as follows: 0 – 5 (12%) 5 – 10 (12%) 10 – 20 (14%) 20 – 40 (23%) 40 – 80 (30%) 80 – 200 (9%)	0.528 (2.0)
Cotton Linters	Staple below 7 (USDA) as ground in a Wiley mill and screened through a 4 mm (0.157 in) screen	0.0264 (0.1)
Salt water prepared by dissolving salt in distilled water or other water containing not more than 200 parts per million of total solids	4 parts by weight of NaCl 96 parts by weight of H_2O	0.01 percent by volume entrained

TABLE XXXIIb. Fuel contaminants for equivalent mission time.

Contaminant	Particle Size Microns	Quantity mg/liter (g/1,000 gal)
Ferroso–Ferric Iron oxide (Fe_3O_4, Black color) Magnetite	0–5	0.396 (1.5)
Ferric Iron Oxide (Fe_2O_3, Hematite)	0–5	7.66 (29.0)
Crushed Quartz (turbojet, turbofan & turboprop engines)	300 – 420 150 – 300	0.264 (1.00) 0.264 (1.00)
Prepared dirt conforming to AC Spark Plug Co., Part No. 1543637 (coarse Arizona road dust)	Mixture as follows: 0 – 5 (12%) 5 – 10 (12%) 10 – 20 (14%) 20 – 40 (23%) 40 – 80 (30%) 80 – 200 (9%)	2.11 (8.00)
Cotton Linters	Staple below 7 (USDA) as ground in a Wiley mill and screened through a 4 mm (0.157 in) screen	0.0264 (0.1)
Salt water prepared by dissolving salt in distilled water or other water containing not more than 200 parts per million of total solids	4 parts by weight of NaCl 96 parts by weight of H_2O	0.01 percent Volume entrained

TABLE XXXIIc. Fuel contamination.

Contaminant	Particle Size (Microns)	Quantity (gm/1,000 gal)
(a)		
Iron Fe_3O_4	0–5	14.0
Iron Fe_2O_3	0–5	14.5
Iron Fe_2O_3	5–10	1.5
Crushed Quartz	1000–1500	0.25
Crushed Quartz	420–1000	1.75
Crushed Quartz	300–420	1.00
Crushed Quartz	150–300	1.00
Prepared Dirt (Arizona road)	0– 5 (12%) 5–10 (12%) 10–20 (14%) 20–40 (23%) 40–80 (30%) 80–200 (9%)	8.00
Cotton Linters	Staple below 7	0.10
Napthenic Acid	–	0.03% by fol.
Salt Water	–	0.01% by vol.
(b)		
Iron Fe_3O_4	0–5	0.90
Iron Fe_2O_3	5–10	4.73
Prepared Dirt (Arizona road)	Same as above	1.90
Cotton Linters	Staple below 7	0.02

NOTES:

(a) Helicopter engines shall operate satisfactorily with these amounts with filter not in bypass, use for first 5 hours of the 50–hour fuel system test.

(b) Use these amounts for last 45 hours of the 50–hour fuel system test.

TABLE XXXIIIa. Accelerated mission duty cycles.

Endurance test cycle profiles	TOTALS PER MISSION								
	Type I cycle	Type III cycle	Type IV cycle	A/B lights	A/B time (min.)	Int time (min.)	Hot time (min.)	Total time (min.)	TAC per cycle
Mission #1									
Mission #2									
Mission #3									
Mission (etc.)									
Composite									
Ground Test									

NOTE: For turboprop and turboshaft engines, the ‘‘A/B lights” and ‘‘A/B time” columns should be deleted. The ‘‘Hot time” column includes time at intermediate and above augmented for turbofan and turbojet engines, and includes the contingency ratings for turboshaft and turboprop engines.

TABLE XXXIIIb. Accelerated mission duty cycle.

Endurance test mission profiles	TOTAL AMTs FOR _____ HOURS								
	# AMT missions per ___ hrs	Total TAC	Type I cycle	Type III cycle	Type IV cycle	A/B lights	A/B time (hrs)	TAIAA (hrs)	Total run time (hrs)
Mission #1									
Mission #2									
Mission #3									
Mission (etc.)									
Composite									
Ground test									
Totals									

NOTE: For turboprop and turboshaft engines, the ‘‘A/B lights” and ‘‘A/B time” columns should be deleted. The TAIAA (Time at Intermediate and Above) column for turbofan and turbojet engines includes time at intermediate and above (augmented), for turboshaft and turboprop engines it includes contingency ratings.

TABLE XXXIV. Durability test runs turboprop/turboshaft engines.

The engine shall be tested with turbine temperature equal to the maximum operating or control limiting value at each power setting. The engine control shall be adjusted to permit operation at maximum allowable steady–state gas temperature at the first stage turbine rotor inlet. The relationship between measured gas temperature and turbine rotor inlet temperature shall be determined prior to test and rechecked every tenth cycle. The control shall be operated in failure mode during specified runs and cycles.

a. Maximum–idle run (60 min) –
Six periods of 10 minutes each, (5 minutes at Maximum power test temperature and shaft speed for Maximum power in table Ia, followed by 5 minutes atIidle). If engine has anti–icing, it is operated at maximum anti–icing bleed for 1 minute at the end of each period with Maximum power test temperature. On each fifth repetition of this test sequence, the control is operated in the failure mode for the last 3 minutes of each 5 minute period at Maximum power test temperature.

b. Incremental torque run (96 min) –
(1) 8 minutes at Maximum power test temperature and maximum shaft speed (torque limited)
(2) 8 minutes at Maximum power test temperature and minimum shaft speed (torque limited)
(3) 8 minutes at minimum shaft speed and torque equal to 90% of torque limit for continuous operation.
(4) 8 minutes at Maximum shaft speed and torque equal to 90% of torque limit for continuous operation for continuous operation
(5) 8 minutes at Maximum shaft speed and torque equal to 80% of torque limit for continuous operation for continuous operation
(6) 8 minutes at minimum shaft speed and torque equal to 80% of torque limit for continuous operation
(7) 4 minutes at minimum shaft speed and torque equal to 60% of torque limit for continuous operation
(8) 4 minutes at Maximum shaft speed and torque equal to 60% of torque limit for continuous operation
(9) 4 minutes at Maximum shaft speed and torque equal to 40% of torque limit for continuous operation
(10) 4 minutes at minimum shaft speed and torque equal to 40% of torque limit for continuous operation
(11) 4 minutes at minimum shaft speed and torque equal to 20% of torque limit for continuous operation
(12) 4 minutes at Maximum shaft speed and torque equal to 20% of torque limit for continuous operation
(13) 4 minutes at Maximum shaft speed and torque equal to 10% of torque limit for continuous operation
(14) 4 minutes at minimum shaft speed and torque equal to 10% of torque limit for continuous operation
(15) 4 minutes at minimum shaft speed and zero shaft torque
(16) 4 minutes at Maximum shaft speed and zero shaft torque
(17) 4 minutes at idle
(18) 4 minutes at idle and zero shaft speed

If incremental torque runs indicate critical vibrations between idle and the Maximum power test temperature, 48 minutes of the runs below can be substituted for 48 minutes of incremental torque runs on each test cycle at the option of the Using Service.

TABLE XXXIV. Durability test runs turboprop/turboshaft engines – Continued.

(19) 16 minutes at gas generator and shaft speeds producing the vibration
(20) 8 minutes at gas generator and shaft speeds 2% more than speeds producing the vibration
(21) 8 minutes at gas generator speed 2% more and shaft speed 2% less than speeds producing the vibration
(22) 8 minutes at gas generator and shaft speeds 2% less than speeds producing the vibration
(23) 8 minutes at gas generator speed 2% less and shaft speed 2% more than speeds producing the vibration

c. Power transient run (39 min) –
   (1) 4 minutes at no–load followed by 1 minute at Maximum power test temperature and shaft speed for Maximum power in table Ia, repeat for 20 minutes
   (2) 1 minute at Maximum test temperature and shaft speed for Maximum power in table Ia, followed by immediate decrease to no–load, followed by an increase to speed and temperature of previous condition, hold for 1 minute, repeat sequence for 19 minutes

d. Reverse pitch operation (9 min) –
1 minute at idle, 3 minutes at maximum reverse, 1 minute at Maximum power test temperature and shaft speed for Maximum power in table Ia, 3 minutes at maximum reverse, and 1 minute at Idle. If engine does not have reverse, substitute Intermediate power test temperature for maximum reverse.

e. Intermediate run (30 min) –
   (1) 15 minutes at Intermediate test temperature at maximum governed shaft speed
   (2) 15 minutes at Intermediate test temperature at minimum shaft speed and delivered torque as high as Intermediate power in table Ia

On every fifth repetition of this sequence, the first 26 minutes shall be with the control in the failure mode.

f. Maximum Continuous Run (20 min) –
   (1) 10 minutes at Maximum Continuous test temperature at maximum governed shaft speed
   (2) 10 minutes at Maximum Continuous test temperature at minimum shaft speed and delivered torque as high as Maximum Continuous power in table Ia

g. Intermediate Run (15 min) –
15 minutes at Intermediate test temperature and Intermediate power shaft speed

h. Maximum Continuous Run (15 min) –
15 minutes at Maximum Continuous test temperature and Maximum Continuous power shaft speed in table Ia

i. Intermediate–Maximum Run (30 min) –
5 minutes at Intermediate test temperature and Intermediate power shaft speed in table Ia followed by 5 minutes at Maximum test temperature andMaximum power shaft speed in table Ia, repeat sequence for 30 minutes

j. Maximum Continuous Run (30 min) –
16 minutes at Maximum Continuous test temperature and Maximum Continuous shaft speed in table Ia

k. Intermediate–Maximum Run (30 min) –
   (1) 15 minutes at Intermediate test temperature and Intermediate power shaft speed in table Ia

TABLE XXXIV. Durability test runs turboprop/turboshaft engines – Continued.

(2) 10 minutes at Maximum power test temperature and Maximum power shaft speed in table Ia, followed by 5 minutes at Intermediate test temperature and Intermediate power shaft speed in table Ia. At 5–minute intervals, anti–icing bleed shall be operated at maximum rate for 1 minute.

l. Contingency Power Qualification (67 min) –
5.6 minutes at Maximum Continuous followed by 5.6 minutes at contingency followed by 5.6 minutes at Maximum Continuous, repeat sequence four times.

Notes:

(1) Test turbine temperatures used in the Durability Test are based on the maximum operating or control limiting temperature for each power setting. For Army helicopter engines, this is equivalent to the initial rating temperature values plus a margin for control tolerances and deterioration. This delta is typically 30°C (54°F). The test temperature at the highest steady state power setting should be equal to maximum allowable temperature.

(2) The Contingency Qualification test is performed after completion of the Durability test.

(3) The Durability test for ISR and OCR are performed on two engines having separate fuels, oils, and torque absorber characteristics.

(4) Idle power, unless otherwise specified, is with minimum self–sustaining rotor speeds.

(5) Power setting load demand changes shall be made within 0.5 seconds, with engine control and fuel flow schedules set for maximum acceleration.

(6) Each 50 hours of testing shall be preceded by a 2–hour shutdown perio; test runs shall be run in the order above for ‘‘a–k”.

(7) If the engine does not have Maximum power, then Intermediate shall be substituted for Maximum power on those runs.

(8) Unless otherwise specified, the engine control shall be operated in the failure mode on every fifth repetition of the test sequence, each channel in a multi–channel or multi–unit electronic control shall be tested.

(9) Internal water–wash provisions shall be demonstrated on every tenth repetition of the test sequence.

(10) ‘‘Maximum shaft speed” above shall be the maximum shaft speed limit for continuous operation or the maximum governed speed limit allowed by the engine control system, whichever is lower. ‘‘Minimum shaft speed” shall be the minimum governed shaft speed allowed by the engine vcontrol system or the shaft speed corresponding to the torque limit for continuous operation, whichever is higher. If the operating speed varies between flight conditions or is different than table Ia conditions, then the most appropriate speed shall be used.

(11) The Durability Test shall be run with the inlet air temperature producing the highest operating temperature for each power setting. If this does not occur with table Ia conditions, the air temperature shall be the lowest value producing the highest operating temperatures for the test sequence. A single air temperature shall be used for test runs ‘‘a–k” with another value for ‘‘1” if needed. Alternately, the test sequence may use a table I condition specified by the Using Service.

(12) Power settings not corresponding to those above may be qualified separately like Contingency, or substituted for other power settings as appropriate. Some examples are various power settings for OEI (One Engine Inoperative). The amount of test time and the test plan shall be approved by the Using Service.

TABLE XXXV. CA&E component test.

Assembly	Part	Accelerated aging	Missionized/ operational test	Environmental tests						
				impact	vibration	humidity	fungus	sand and dust	sustained acceleration	explosion proof
load control valve	load control valve									
	LCV torquemotor									
surge control valve	surge control valve									
	SCV torquemotor									
	SCV RVDT									
starter generator	starter/generator									
ignition system	ignitor									
	ignitor lead									
	exciter									
inlet guide valve actuator	IGVA									
	operating solenoid									
bleed flow sensor assembly	delta P sensor									
	pressure sensor									
fuel control assembly	fuel control assembly									
	fuel pump									
	torquemotor									
	fuel solenoid									
	fuel filter bypass									
fuel distribution system	flow divider									
oil system	heat exchanger									
	oil filter bypass									
ECU	ECU									
miscellaneous controls and accessories	monopole									
	EGT thermocouples									
	electrical harness									
	LOP switch									
	oil level switch									
	fuel pressure switch									
plumbing	oil lines									
	fuel lines									

(T) = test, (TA) = test conducted as part of assembly, (S) = similarity, (H) = hermetically sealed, (N/A) = not applicable

Note: Environmental tests must be conducted in sequence from left to right as presented on the table. The same physical component must be used for the entire sequence of environmental tests.

TABLE XXXV. CA&E component test – Continued.

Assembly	Part	Individual tests						Reliability and maintenance test			
		generator	heat exchanger	fire	cavitation	low lubricity	ignition system	sneak circuit analysis	environ–mental stress screening	reliability growth test	built in test
load control valve	load control valve										
	LCV torquemotor										
surge control valve	surge control valve										
	SCV torquemotor										
	SCV RVDT										
starter generator	starter/generator										
ignition system	ignitor										
	ignitor lead										
	exciter										
inlet guide valve actuator	IGVA										
	operating solenoid										
bleed flow sensor assembly	delta P sensor										
	pressure sensor										
fuel control assembly	fuel control assembly										
	fuel pump										
	torquemotor										
	fuel solenoid										
	fuel filter bypass										
fuel distribution system	flow divider										
oil system	heat exchanger										
	oil filter bypass										
ECU	ECU										
miscellaneous controls and accessories	monopole										
	EGT thermocouples										
	electrical harness										
	LOP switch										
	oil level switch										
	fuel pressure switch										
plumbing	oil lines										
	fuel lines										

(T) = test, (TA) = test conducted as part of assembly, (S) = similarity, (H) = hermetically sealed, (N/A) = not applicable

Note: Environmental tests must be conducted in sequence from left to right as presented on the table. The same physical component must be used for the entire sequence of environmental tests.

TABLE XXXVI. Evaluation/Qualification cross reference.

4	EVALUATION AND QUALIFICATION (TITLE)	IFR		FFR		ISR		OCR	
4.1	Design (configuration) and construction								
4.1.1	Item and interface definition								
4.1.1.1	Item diagrams and schematics								
4.1.1.2	Item drawings								
4.1.1.3	Interface and installation diagrams and schematics								
4.1.1.4	Interface and installation drawings								
4.1.1.5	Installation changes								
4.1.1.6	Installation hardware								
4.1.1.7	Connections/Interchangeability								
4.1.1.8	Interface stresses and loads								
4.1.1.9	Mounts (title)								
4.1.1.9.1	Main mounts								
4.1.1.9.2	Ground handling mounts								
4.1.1.9.3	Engine stiffness								
4.1.1.10	Inlet connection stresses								
4.1.1.11	Control system interface and integration								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.1.1.11.1	Thrust/power demand								
4.1.1.11.1.1	Power lever angle								
4.1.1.11.1.2	Power lever torque								
4.1.1.11.2	Control signals								
4.1.1.11.3	Load demand lever								
4.1.1.11.4	Output speed lever								
4.1.1.11.5	Fuel shutoff lever								
4.1.1.12	Electrical system interface								
4.1.1.12.1	Instrumentation system interface								
4.1.1.13	Engine monitoring system (EMS) interface								
4.1.1.14	Fiber optics interface								
4.1.1.15	Bleed air interface								
4.1.1.15.1	Customer bleed air contamination								
4.1.1.16	Drains and fluid collection devices								
4.1.1.17	Power absorber interface								
4.1.1.18	Power take–off (PTO)								
4.1.1.19	Starting drivetrain								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.1.1.19.1	Starting torque and speed requirements								
4.1.1.19.2	Moment of inertia of rotating parts								
4.1.1.19.3	Torsional spring constant								
4.1.1.19.4	Starter train backlash								
4.1.1.20	Thrust reverser interface (see 3.7.12.2)								
4.1.1.21	Exhaust system interface								
4.1.1.22	Compatibility – Engine Airframe								
4.1.1.23	Control and External (C&E) component list								
4.1.1.24	Government furnished/loaned property list								
4.1.2	Physical characteristics (title)								
4.1.2.1	Dry mass (weight) of complete engine								
4.1.2.2	Mass (weight) of residual fluids								
4.1.2.3	Mass (weight) of operating fluid								
4.1.2.4	Mass (weight) of additional equipment								
4.1.2.5	Mass moment of inertia of complete engine								
4.1.3	Materials, processes and parts								
4.1.3.1	Adhesives and sealants								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.1.3.2	Electromeric materials								
4.1.3.3	‘‘O” rings seals and packings								
4.1.3.4	Corrosion protection (see 3.5.1.6)								
4.1.3.5	Nonmetallic hose								
4.1.3.6	Shot peening								
4.1.4	Fasteners								
4.1.4.1	Securing of fasteners								
4.1.4.2	Clamps								
4.1.4.3	Screw recesses								
4.1.4.4	Screw threads								
4.1.4.5	Helical coil installation								
4.1.4.6	External wrenching elements								
4.1.5	Nameplate and product marking								
4.1.5.1	Engine data plate marking								
4.1.5.2	Warranty marking								
4.1.5.3	Part number marking								
4.1.5.4	Critical parts identification and tracking								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.1.6	Transportability								
4.1.7	Interchangeability								
4.1.8	Safety								
4.1.8.1	Flammable fluid systems								
4.1.8.2	Fire shield								
4.1.8.3	Explosion proof								
4.1.8.4	Combustible fluid drains								
4.1.8.5	Air and gas leakage (3.1.2.9)								
4.1.8.6	Ground safety								
4.1.8.7	Jet wake								
4.1.9	Design control (title)								
4.1.9.1	Standardization								
4.1.9.2	Design standards								
4.1.9.3	Standard parts								
4.1.9.4	Parts list								
4.1.9.5	Assembly of components and parts								
4.1.9.6	Changes in vendors or fabrication process								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.2	Performance and operability (title)								
4.2.1	Performance characteristics								
4.2.1.1	Performance and presentation steady–state								
4.2.1.2	Performance and presentation transient								
4.2.1.3	Performance computer program								
4.2.1.4	Performance retention								
4.2.1.5	Performance evaluation								
4.2.2	Operating characteristics and limits								
4.2.2.1	Operating envelope								
4.2.2.2	Operating attitude and conditions								
4.2.2.3	Stopping								
4.2.2.4	Low power conditions (title)								
4.2.2.4.1	Idle thrust/power – (ground/flight idle)								
4.2.2.4.2	No load condition								
4.2.2.5	Stability								
4.2.2.6	Transients								
4.2.2.6.1	Overshoot and undershoot								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.2.2.6.2	Droop								
4.2.2.7	Windmilling								
4.2.2.8	Reverse thrust								
4.2.2.9	Steady–state and transient airflow								
4.2.2.10	Inlet airflow distortion								
4.2.2.10.1	Pressure and temperature rate of change								
4.2.2.11	Gas path and measurement plane temperature limits								
4.2.2.12	Surface temperature and heat rejection								
4.2.2.12.1	Component limiting temperature								
4.3	Environmental conditions (title)								
4.3.1	Atmospheric conditions (title)								
4.3.1.1	Humidity								
4.3.1.2	Fungus								
4.3.1.3	Corrosive atmosphere								
4.3.1.4	Icing conditions								
4.3.2	Ingestion capability (hazard resistance) (title)								
4.3.2.1	Bird ingestion								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.3.2.2	Foreign object damage (FOD)								
4.3.2.3	Ice ingestion								
4.3.2.4	Sand and dust ingestion								
4.3.2.5	Atmospheric liquid water ingestion								
4.3.2.6	Armament gas ingestion (AGI)								
4.3.2.7	Steam ingestion								
4.3.3	Electromagnetic environmental effects (EME)								
4.3.3.1	Electromagnetic interference (EMI)								
4.3.3.2	Intrasystem electromagnetic compatibility (EMC)								
4.3.3.3	Intersystem electromagnetic compatibility (EMC)								
4.4	Integrity								
4.4.1	Structural integrity								
4.4.1.1	Design service life								
4.4.1.1.1	Hot parts								
4.4.1.1.2	Cold parts								
4.4.1.1.3	Expendables								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.4.1.1.4	Bearings								
4.4.1.1.5	Components								
4.4.1.2	Design usage								
4.4.1.2.1	Internal environment								
4.4.1.2.2	Externally applied forces								
4.4.1.3	Material characterization								
4.4.1.4	Parts classification								
4.4.1.5	Durability								
4.4.1.5.1	High cycle fatigue (HCF) life								
4.4.1.5.2	Low cycle fatigue (LCF) life								
4.4.1.5.3	Creep								
4.4.1.6	Strength								
4.4.1.6.1	Factors of safety								
4.4.1.6.2	Blade and disk deflection								
4.4.1.6.3	Containment								
4.4.1.6.4	Blade out								
4.4.1.6.5	Overspeed/overtemperature								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.4.1.6.6	Disk burst speed								
4.4.1.6.7	Output shaft torque limits								
4.4.1.6.8	Output shaft speed limits								
4.4.1.6.9	Pressure vessel/case								
4.4.1.6.10	Pressure balance								
4.4.1.6.11	Gyroscopic moments								
4.4.1.7	Damage tolerance								
4.4.1.7.1	Residual strength								
4.4.1.7.2	Initial flaw size								
4.4.1.7.3	In–service inspection flaw size								
4.4.1.7.4	Inspection intervals								
4.4.1.7.5	Flaw growth								
4.4.1.7.6	Composites								
4.4.1.8	Vibration and dynamic response								
4.4.1.8.1	Vibration limits								
4.4.1.8.2	Critical speeds								
4.4.1.8.3	Blade, disk, and static structure vibration								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.4.1.9	Catastrophic failure								
4.4.1.10	Sustained acceleration								
4.4.1.11	Shock								
4.4.1.12	External surface foreign object damage								
4.4.2	Mechanical equipment and subsystem integrity								
4.4.3	Avionic/electronic integrity								
4.5	Reliability and maintainability								
4.5.1	Reliability								
4.5.1.1	Reliability quantitative requirements								
4.5.2	Maintainability								
4.5.2.1	Maintainability quantitative requirements								
4.5.2.1.1	Excluded maintenance functions								
4.5.2.2	Maintainability qualitative requirements								
4.5.2.3	Maintenance								
4.5.2.3.1	Modules								
4.5.2.3.2	Maintenance, inspection and repair cycle								
4.5.2.3.2.1	Maintenance inspection techniques								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.5.2.3.3	Tools								
4.5.2.4	Battle damage repair								
4.5.3	Human performance and human engineering								
4.6	Combat survivability								
4.6.1	Susceptibility								
4.6.1.1	Noise								
4.6.1.2	IR radiation								
4.6.1.2.1	IR suppression system								
4.6.1.3	UV radiation								
4.6.1.4	Smoke								
4.6.1.5	Gaseous emissions								
4.6.1.6	Fuel streaming/vapor puffing								
4.6.1.7	Water vapor contrails								
4.6.1.8	Radar cross section (RCS)								
4.6.1.9	Radar absorbent materials and coatings								
4.6.2	Vulnerability								
4.6.2.1	Ballistic weapons								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
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T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.6.2.1.1	Vulnerability – component and accessory								
4.6.2.1.2	Static structure								
4.6.2.2	Directed energy weapons effects								
4.6.2.3	Nuclear weapons effects								
4.6.2.4	Chemical/biological agent effects								
4.6.2.5	Fuel ingestion effects								
4.6.2.5.1	Fuel ingestion – steady–flow								
4.6.2.5.1.1	Fuel ingesting – cooling air contamination								
4.6.2.5.2	Fuel ingestion – transient (“Quick dump”)								
4.7	Subsystems (title)								
4.7.1	Anti–icing/de–icing system								
4.7.2	Control system								
4.7.2.1	Control system performance								
4.7.2.1.1	Backup control (BUC)								
4.7.2.2	Control system adjustments								
4.7.2.3	Engine overspeed protection system								
4.7.3	Fuel system								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

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T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.7.3.1	Fuels								
4.7.3.1.1	Primary fuel								
4.7.3.1.2	Alternate fuel								
4.7.3.1.3	Restricted fuel								
4.7.3.1.4	Emergency fuel								
4.7.3.2	Fuel system performance								
4.7.3.2.1	Fuel contamination								
4.7.3.2.2	Fuel system performance with external assistance								
4.7.3.2.3	Fuel system performance with no external assistance								
4.7.3.2.4	Fuel system performance under conditions of excessive fuel vapor								
4.7.3.2.5	Fuel pump priming								
4.7.3.2.6	Fuel lubricity								
4.7.3.2.7	Fuel system performance with water saturated fuel								
4.7.3.2.8	Fuel filter								
4.7.3.2.9	Fuel flow limits								
4.7.4	Electrical systems (title)								

P – Partial verification
F – Full verification

A	D
I	T

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I – Inspection

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TABLE XXXVI. Evaluation/Qualificiation cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.7.4.1	Electrical power								
4.7.4.1.1	Generator/alternator								
4.7.4.2	Alternate/external electrical power								
4.7.4.3	Electrical connectors and cables								
4.7.4.4	Electronic components								
4.7.4.5	Electrical grounding								
4.7.4.6	Electrical bonding								
4.7.4.7	Ground isolation								
4.7.4.8	Potting compounds								
4.7.5	Ignition systems								
4.7.5.1	Ignition system fouling (title)								
4.7.5.1.1	Carbon fouling								
4.7.5.1.2	Water fouling								
4.7.6	Engine monitoring system (EMS)								
4.7.6.1	(EMS) fault detection/isolation								
4.7.6.2	On–board engine diagnostic functions								
4.7.7	Optical systems								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.7.7.1	Fiber optic cables								
4.7.7.2	Fiber optic connectors								
4.7.7.3	Fiber optic passive couplers								
4.7.7.4	Fiber optic high altitude radiation resistance								
4.7.8	Lubrication system								
4.7.8.1	Lubrication oil								
4.7.8.1.1	Oil pressure and temperature limits								
4.7.8.1.2	Oil consumption limits								
4.7.8.1.3	Oil flow interruption/depletion								
4.7.8.2	Lubrication system components and features								
4.7.8.2.1	Oil reservoir								
4.7.8.2.1.1	Oil reservoir external features								
4.7.8.2.2	Oil drains								
4.7.8.2.3	Oil filterS								
4.7.8.2.4	Oil debris monitor								
4.7.8.2.5	Oil coolers								
4.7.9	Hydraulic system								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.7.9.1	Hydraulic ground test provisions								
4.7.9.2	Hydraulic system fire and safety hazards								
4.7.9.3	Hydraulic system air removal								
4.7.9.4	Hydraulic fluid filters								
4.7.10	Pneumatic system								
4.7.10.1	Pneumatic air contamination								
4.7.11	Starting								
4.7.11.1	Ground starts								
4.7.11.2	Air starts								
4.7.11.3	Starting limits								
4.7.11.4	Starting procedure								
4.7.11.5	Automatic relight								
4.7.12	Exhaust nozzle system								
4.7.12.1	Exhaust nozzle external asymmetrical air pressure loads								
4.7.12.2	Vectoring nozzle								
4.7.12.2.1	Vectoring nozzle angle and rate								
4.7.12.2.2	Vectoring nozzle failure accommodation								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.7.13	Augmentation system								
4.7.13.1	Afterburner (fuel injection) system								
4.7.13.2	Water injection system								
4.7.13.2.1	Water injection system fluid								
4.7.14	Wash system								
4.7.15	Brake system								
4.7.16	Negative torque limiter								
4.7.16.1	Multiple power section engines								
4.7.17	Power absorber (engine supplied)								
4.7.17.1	Manual feathering								
4.7.17.2	Reverse pitch operation								
4.7.18	Accessory gearbox								
4.7.19	Tubing/plumbing system								
4.7.19.1	Tubing/plumbing identification								
4.7.19.2	Tubing/plumbing clearances								
4.7.20	Inlet particle separator								
4.8	Software resources (title)								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.8.1	Software integrity								
4.8.2	Software performance and design								
4.8.2.1	Built–in test and inspectability								
4.8.2.2	Computer reprogramming								
4.8.3	Spare resources								
4.9	Engine system, controls and external evaluation								
4.9.1	Engine systems tests								
4.9.1.1	Endurance tests								
4.9.1.1.1	Accelerated mission test (AMT)								
4.9.1.1.2a	Durability test								
4.9.1.1.2b	Durability test								
4.9.1.2	Altitude test								
4.9.1.3	Engine systems tests pass/fail criteria								
4.9.2	Controls and externals (C&E) component tests								
4.9.2.1	C&E component operational tests								
4.9.2.1.1	C&E component calibration								
4.9.2.1.2	C&E component test procedures								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

D – Demonstration
T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.9.2.1.3	C&E component missionized tests								
4.9.2.1.4	C&E component accelerated aging								
4.9.2.1.5	C&E component recalibration, disassembly and inspection								
4.9.2.1.6	C&E component environmental test								
4.9.2.1.7	C&E component sand and dust								
4.9.2.1.8	C&E component vibration								
4.9.2.1.9	C&E component test pass/fail criteria								
4.10	Engine qualification								
4.10.1	Qualification milestone								
4.10.1.1	Initial flight release (IFR)								
4.10.1.2	Full flight release (FFR)								
4.10.1.3	Initial service release (ISR)								
4.10.1.4	Operational capability release (OCR)								
4.10.2	Evaluation responsibility								
4.10.2.1	Responsibility for compliance								
4.10.2.2	Quality conformance inspection								
4.10.2.2.1	Analysis inspection								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
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T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.10.3	Manner of testing								
4.10.3.1	Test surveillance								
4.10.3.2	Test article configuration and parts list								
4.10.3.2.1	Test article								
4.10.3.2.2	Installation interfaces								
4.10.3.3	Test equipment (title)								
4.10.3.3.1	Automatic recording equipment								
4.10.3.3.2	Vibration measuring equipment and response characteristics								
4.10.3.3.3	Test stand dynamic characteristics								
4.10.3.3.4	Starter test								
4.10.3.4	Test conditions								
4.10.3.4.1	Oil servicing and consumption								
4.10.3.4.2	Fluid properties for test (title)								
4.10.3.4.2.1	Fuel properties								
4.10.3.4.2.2	Oil properties								
4.10.3.4.2.3	Hydraulic fluid properties								
4.10.3.5	Test calibrations and procedures (title)								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
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T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.10.3.5.1	Pretest evaluation (title)								
4.10.3.5.1.1	Pretest dry mass (weight) of engine								
4.10..3.5.1.2	Power lever torque								
4.10.3.5.2	Calibration								
4.10.3.5.2.1	Control and engine monitoring system calibration								
4.10.3.5.2.2	Engine calibration								
4.10.3.5.2.2.1	Customer bleed air analysis calibration								
4.10.3.5.3	Inspections, maintenance and calibration								
4.10.3.5.4	Recalibration								
4.10.3.5.5	Engine disassembly and inspection								
4.10.4	Data and reporting								
4.10.4.1	Types of data								
4.10.4.1.1	Pretest data								
4.10.4.1.2	Preliminary data								
4.10.4.1.3	Data – steady–state								
4.10.4.1.4	Data – transient								
4.10.4.1.5	Starting data								

P – Partial verification
F – Full verification

A	D
I	T

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T – Test

TABLE XXXVI. Evaluation/Qualification cross reference – Continued.

		IFR		FFR		ISR		OCR	
4.10.4.1.6	Accreditable test time								
4.10.4.1.7	Miscellaneous data								
4.10.4.1.8	Test notes								
4.10.4.2	Accuracy of data								
4.10.4.2.1	Measurement uncertainty analysis								
4.10.4.3	Atmospheric data (title)								
4.10.4.3.1	Barometer reading								
4.10.4.3.2	Relative humidity								
4.10.4.4	Fluid data								

P – Partial verification
F – Full verification

A	D
I	T

A– Analysis
I – Inspection

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TABLE XXXVII. Data recording requirements.

	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady–state	Continuously during transients	Cal/ recal	Thrust
1. Time of day	X	X		X	X
2. Total endurance time					X
3. Power setting	X	X	X	X	X
4. Exhaust nozzle area sq. in.				X	X
5. Variable geometry position		X		X	X
6. Engine rotor(s) speed(s) rpm	X	X	X	X	X
7. Rotor speed(s) at idle rpm	X	X			
8. Rotor speed(s) at ignition rpm	X	X			
9. Rotor speed(s) at starter cutout rpm	X	X			
10. Engine fuel flow kg/hr (lbm/hr)		X	X	X	X
11. Delivered shaft power hp		X	X	X	X
12. Customer power extraction hp		X	X	X	X
13. Data for determining airflow kg/sec (lbm/sec)		X		X	
14. Engine inlet total pressure average kPa (in Hg abs)		X		X	X
15. Engine inlet total temperature average °C (°F)		X		X	X
16. Compressor discharge total pressure kPa (in Hg abs)		X	X	X	X
17. Customer bleed air total pressure kPa (psia)		X	X	X	
18. Compressor bleed air total pressure (acceleration) kPa (psia)			X		
19. Customer bleed air static pres–surekPa (psia)		X		X	X
20. Compressor bleed air static pressure (acceleration) kPa (psia)			X		

TABLE XXXVII. Data recording requirements – Continued.

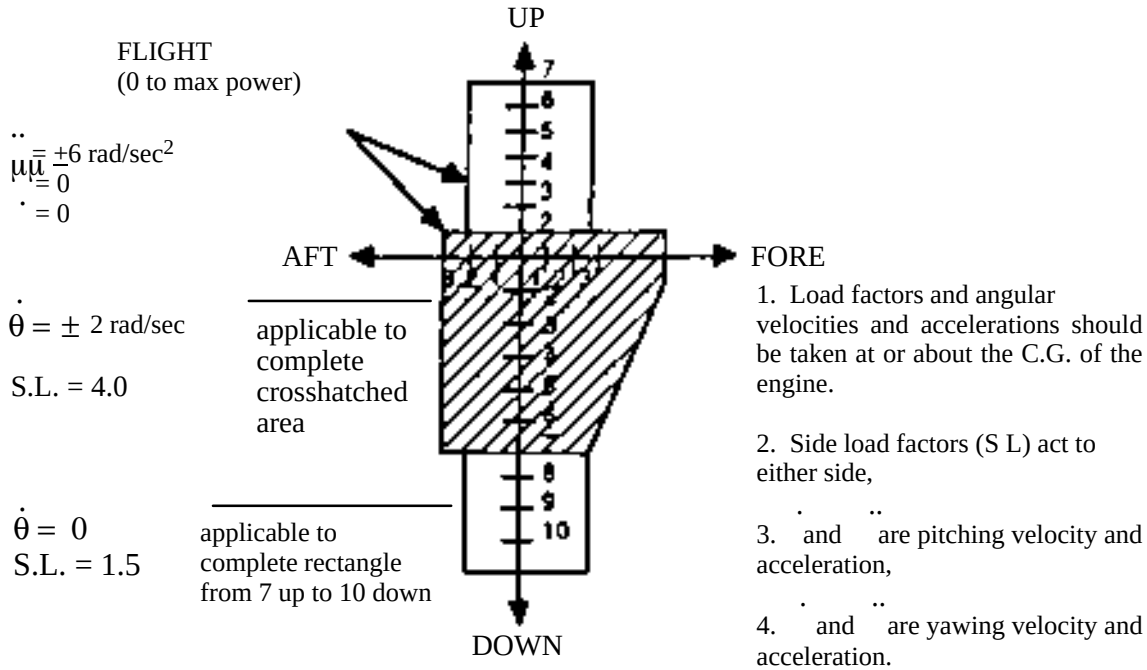
	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady–state	Continuously during transients	Cal/ recal	Thrust
21. Compressor bleed air total temperatures (all ports) °C (°F)		X		X	X
22. Compressor bleed airflow rate (all ports) kg/sec (lbm/sec)		X		X	X
23. Turbine discharge total pressure average kPa (in Hg abs)		X		X	X
24. Compressor discharge total temperature °C (°F)		X	X	X	X
25. Oil breather static pressure kPa (psig)		X		X	X
26. Gas generator turbine discharge total temperature °C (°F)	X	X	X	X	X
27. Exhaust nozzle exit total pres–sure kPa (in Hg abs)		X		X	
28. Exhaust nozzle static pressure kPa (in Hg abs)		X		X	X
29. Exhaust nozzle total temperature °C (°F)		X		X	X
30. Oil flow kg/min (lbm/min)		X		X	X
31. Oil inlet temperature at pressure pump inlet °C (°F)	X	X		X	X
32. Oil pressure at pressure pump inlet kPa (psig)	X	X		X	X
33. Oil pressure at pressure pump outlet kPa (psig)	X	X		X	X
34. Oil pressure at scavenging pump outlet kPa (psig)		X		X	X
35. Oil temperature at scavenging pump outlet °C (°F)		X		X	X
36. Oil temperature at outlet from fuel/oil cooler °C (°F)		X		X	X

TABLE XXXVII. Data recording requirements – Continued.

	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady–state	Continuously during transients	Cal/ recal	Thrust
37. Oil consumption for each cycle of the		X			
38. Fuel pressure at fuel system inlet kPA (psig)	X	X		X	X
39. Fuel pressure at point shown on engine configuration and envelope figure kPa (psig)	X	X		X	X
40. Fuel temperature at fuel system inlet °C (°F)		X		X	X
41. Fuel temperature at outlet from fuel/oil cooler °C (°F)		X		X	X
42. Measured temperature °C (°F) (specify station location)	X	X	X	X	X
43. Maximum measured temperature °C (°F) (specify station location)	X	X			
44. Liquid injection flow kg/hr (lbm/hr (e.g., water)		X		X	X
45. Engine vibration at points shown on on engine configuration and enve–lope figure velocitycm/sec (in/sec)	X	X	X	X	X
46. Accessory compartment tempera–ture(s) where applicable °C (°F)		X		X	X
47. Cooling air inlet temperature °C (°F)		X		X	X
48. Cooling air outlet temperature °C (°F)		X		X	X
49. Engine condition monitoring system data (list each parameter)		X		X	X
50. Cell static pressure kPa (in Hg abs)	X	X		X	X

TABLE XXXVII. Data recording requirements – Continued.

	Data Recording Frequency		Taken During Tests		
	Start	Once/30 min or once/cycle steady–state	Continuously during transients	Cal/ recal	Thrust
51. Ignition source voltage and current when external power is being used)	X	X			X
52. Oil leakage at accessory pads		X		X	X
53. Start number	X			X	
54. Time to ignition actuation	X				X
55. Time to light–off	X			X	X
56. Time to starter cutout	X			X	X
57. Time to stabilized idle RPM	X			X	X
58. Time to oil pressure indication (at point shown on engine configuration and envelope figure	X			X	X
59. Time to stabilize to normal oil pressure (at point shown on engine configuration and envelope figure)	X			X	X
60. Engine life counter	X	X		X	X
61. Additional data as required by the Using Service	X	X	X	X	X



1. Load factors and angular velocities and accelerations should be taken at or about the C.G. of the engine.

2. Side load factors (S L) act to either side,

3. $\dot{}$ and $\ddot{}$ are pitching velocity and acceleration,

4. $\dot{}$ and $\ddot{}$ are yawing velocity and acceleration.

5. Down loads occur during pull out.

6. Fore loads occur during arrested landing.

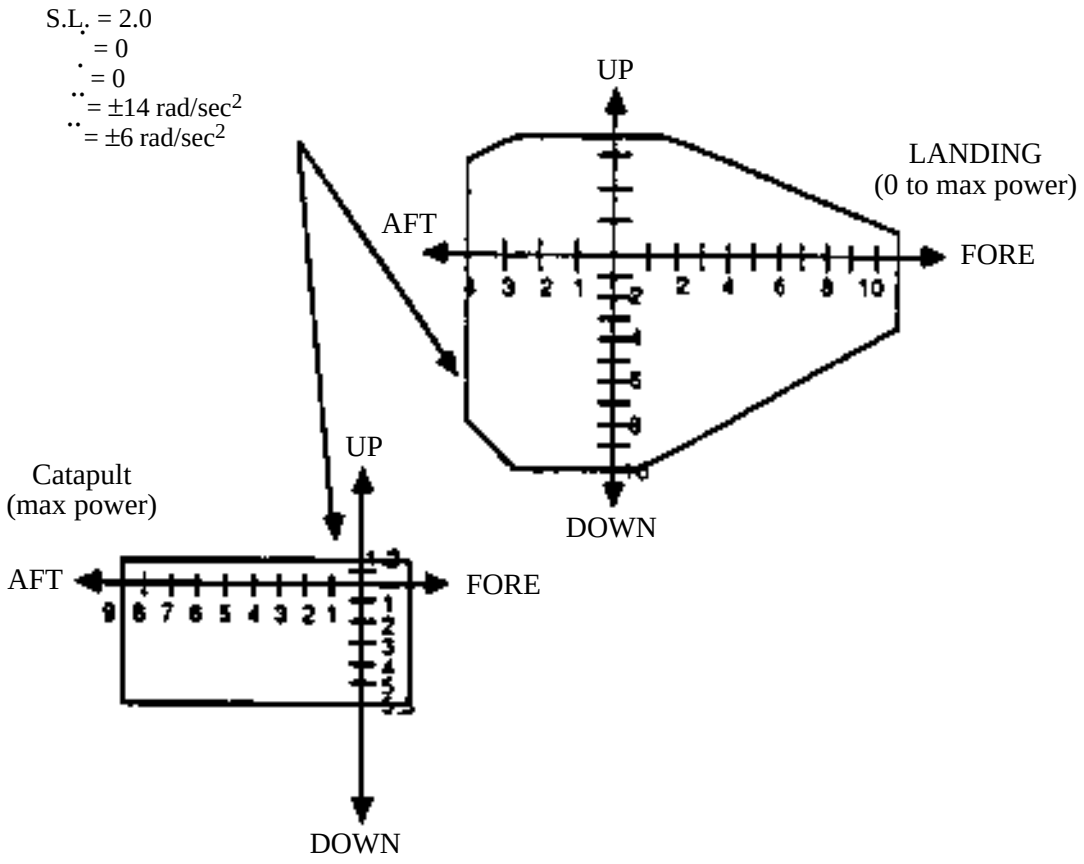


FIGURE 2. Externally applied forces (fixed wing).

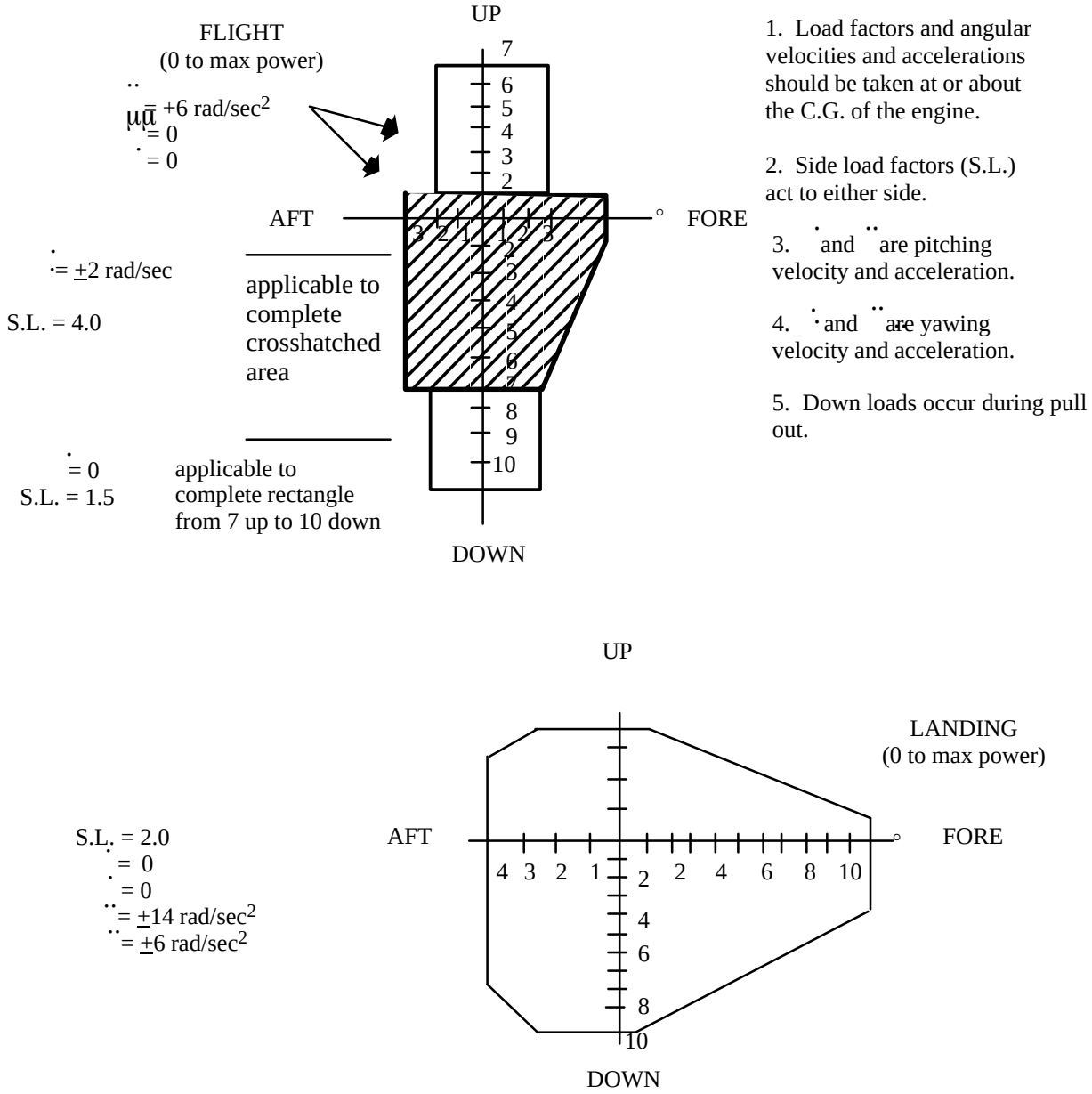
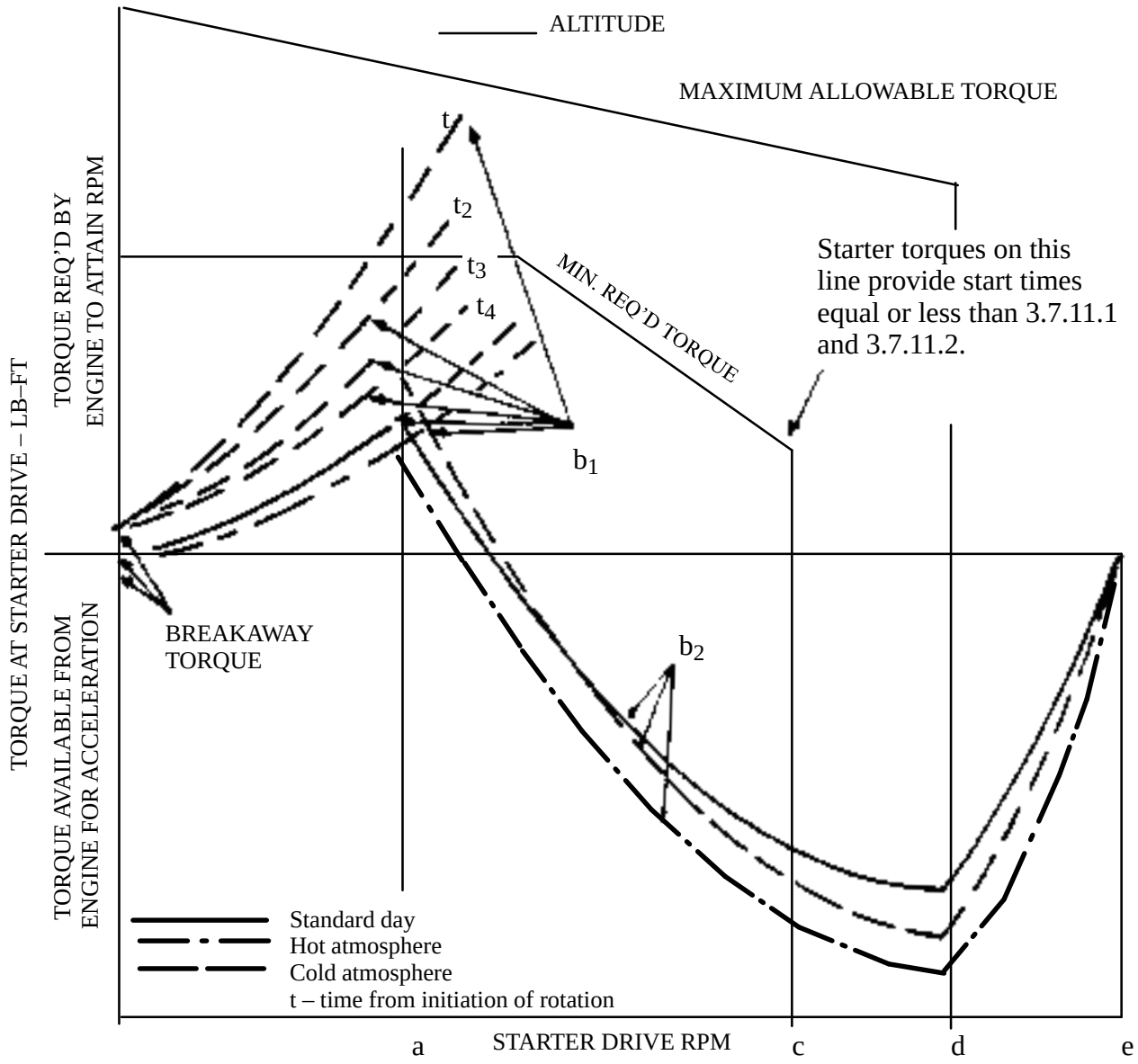


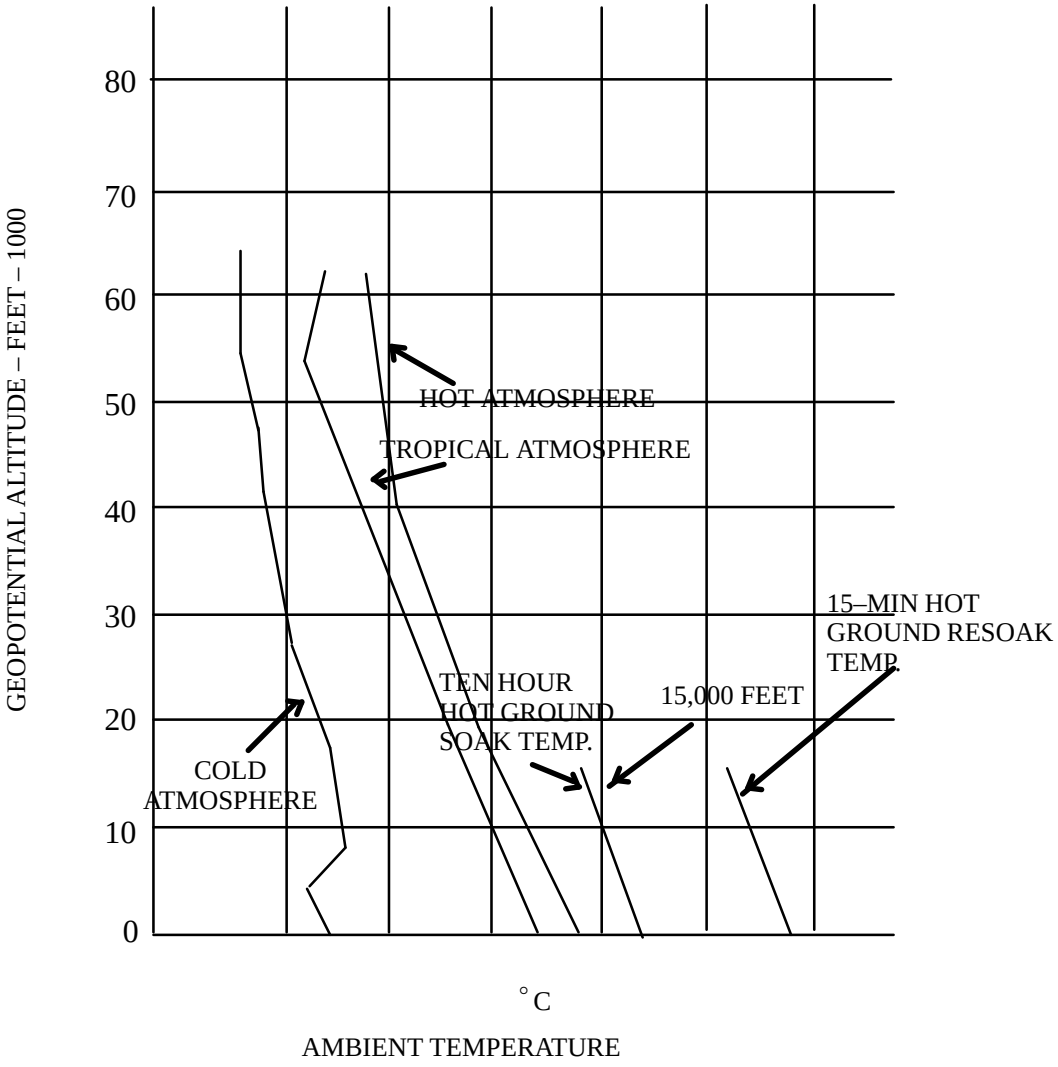
FIGURE 2. Externally applied forces (rotary wing).



NOTES:

a. Required RPM before firing. (Where applicable the engine manufacturer shall state the minimum time and the number of revolutions or any combination or conditions that must be satisfied before firing.)
   Maximum required cranking time at firing speed _____ sec.
b. Steady state torque at the starter drive.
   1. In an unfired engine. (Curves for cold atmosphere show drag torque after time from initiation of rotation.)
   2. In a fired engine.
c. Minimum starter cutout speed _____ rpm.
d. Maximum starter cutout speed _____ rpm. (Maximum cutout speed should be at least 10 percent above minimum cutout speed.)
e. Engine idle condition.

FIGURE 3. Starting torque and speed.



NOTES:

1. Hot and cold atmosphere, except at sea level, consistent with one percent risk of MIL–STD–210B, World–Wide Air Environment.
2. Sea level hot and cold atmosphere temperatures consistent with one percent risk and 20 percent risk, respectively consistent with MIL–STD–210B, Ground Environment.
3. Tropical atmosphere consistent with MIL–STD–210A.
4. The ambient static pressure variation shall be as defined in U.S. Standard Atmosphere for geopotential altitude.

FIGURE 4. Ambient temperature extremes vs. altitude.

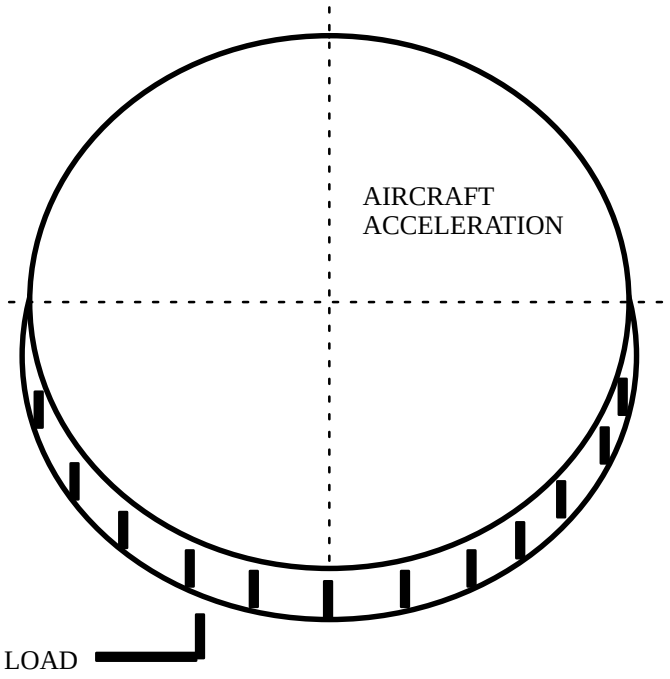


FIGURE 5. Exhaust system interface loads.

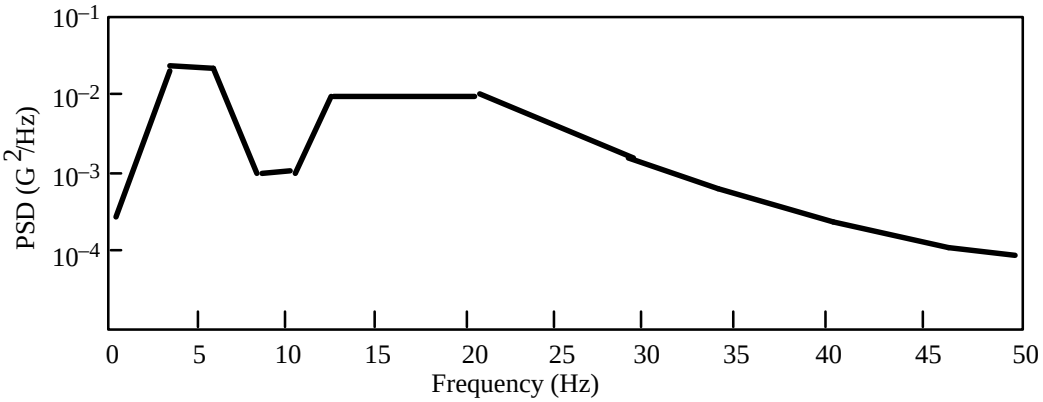


FIGURE 6. Transportability.

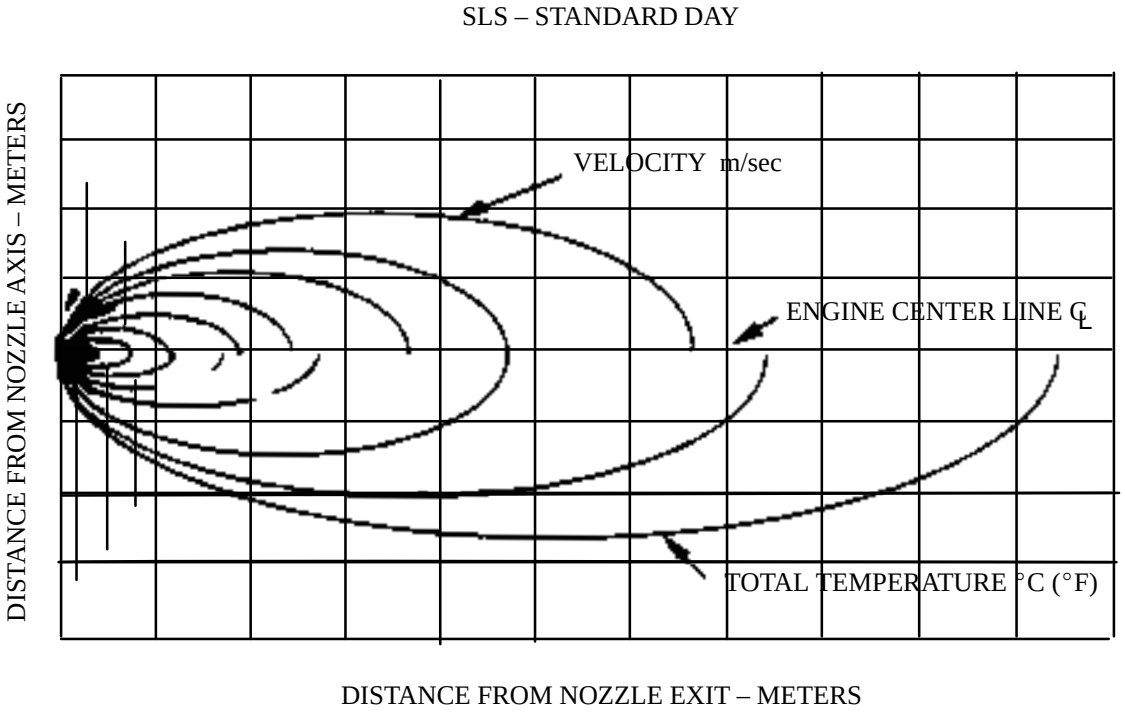
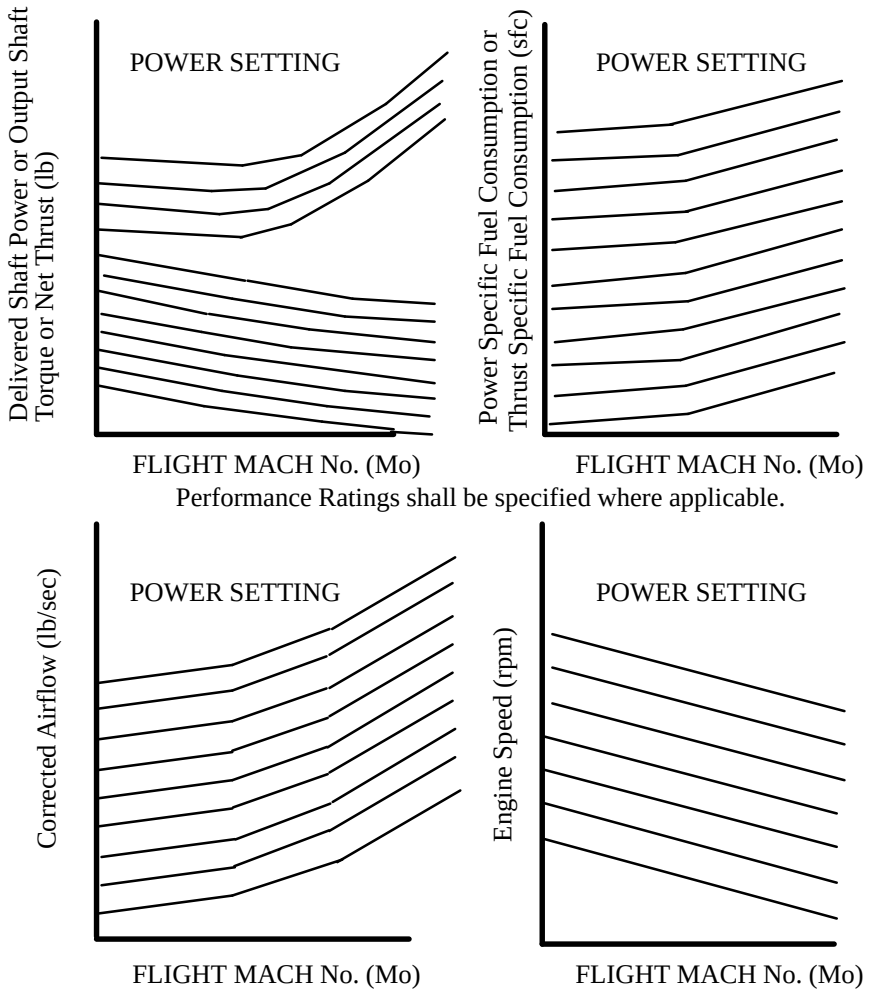


FIGURE 7. Jet wake (maximum augmented, intermediate, idle).



NOTE: For turboshaft/turboprop add residual jet thrust vs. flight Mach number.

FIGURE 8a. Performance curves.

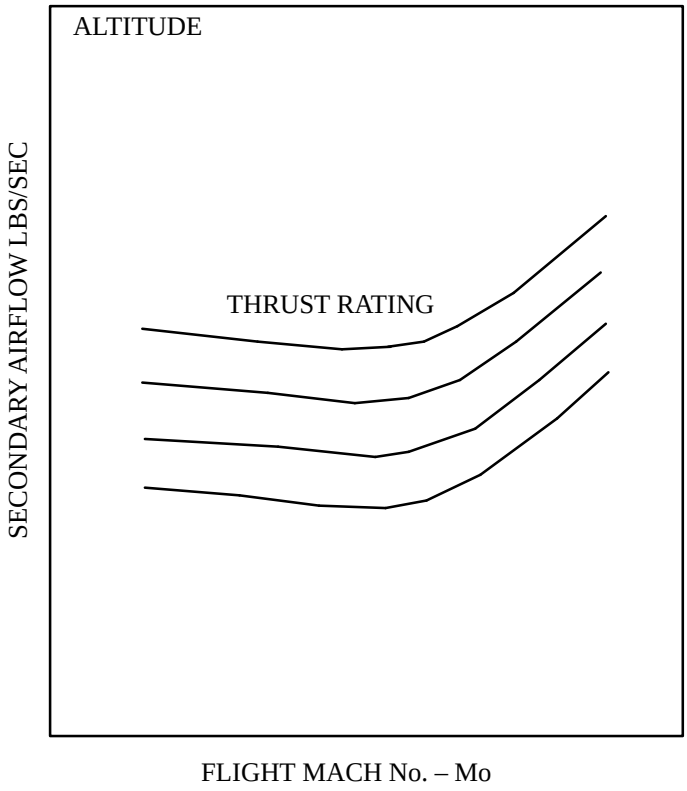


FIGURE 8b. Curves for secondary airflow.

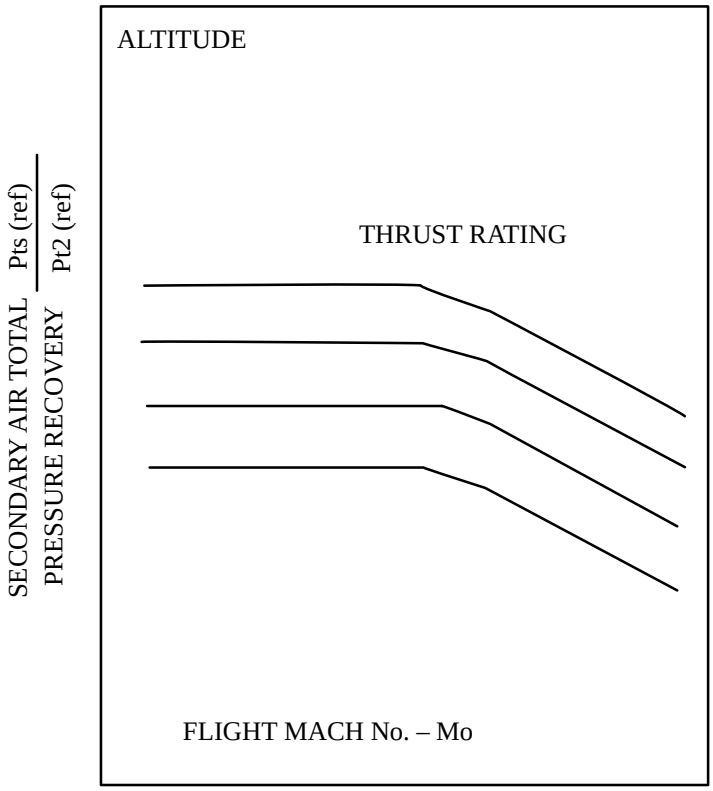


FIGURE 8c. Curves for secondary air pressure recovery.

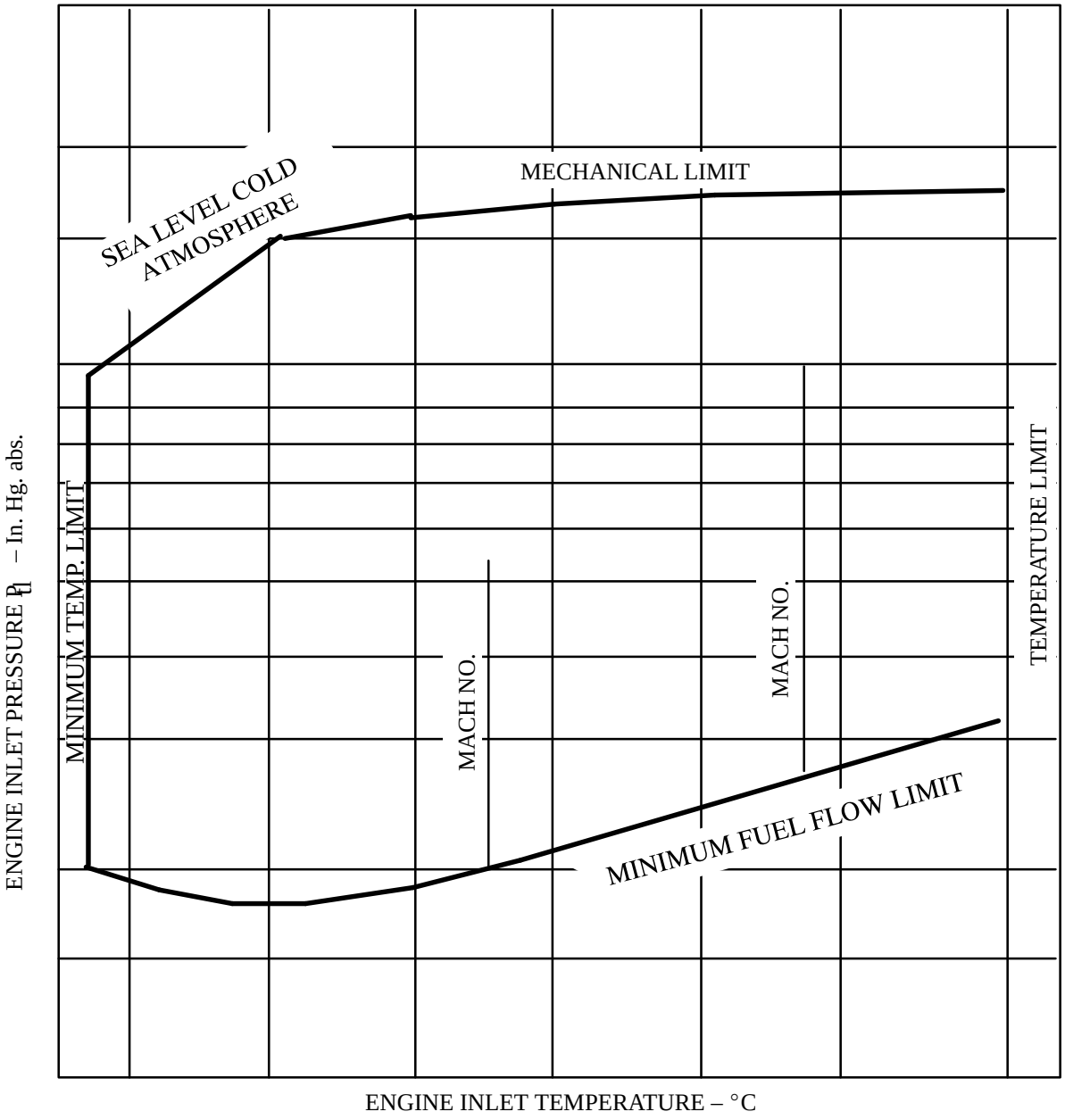
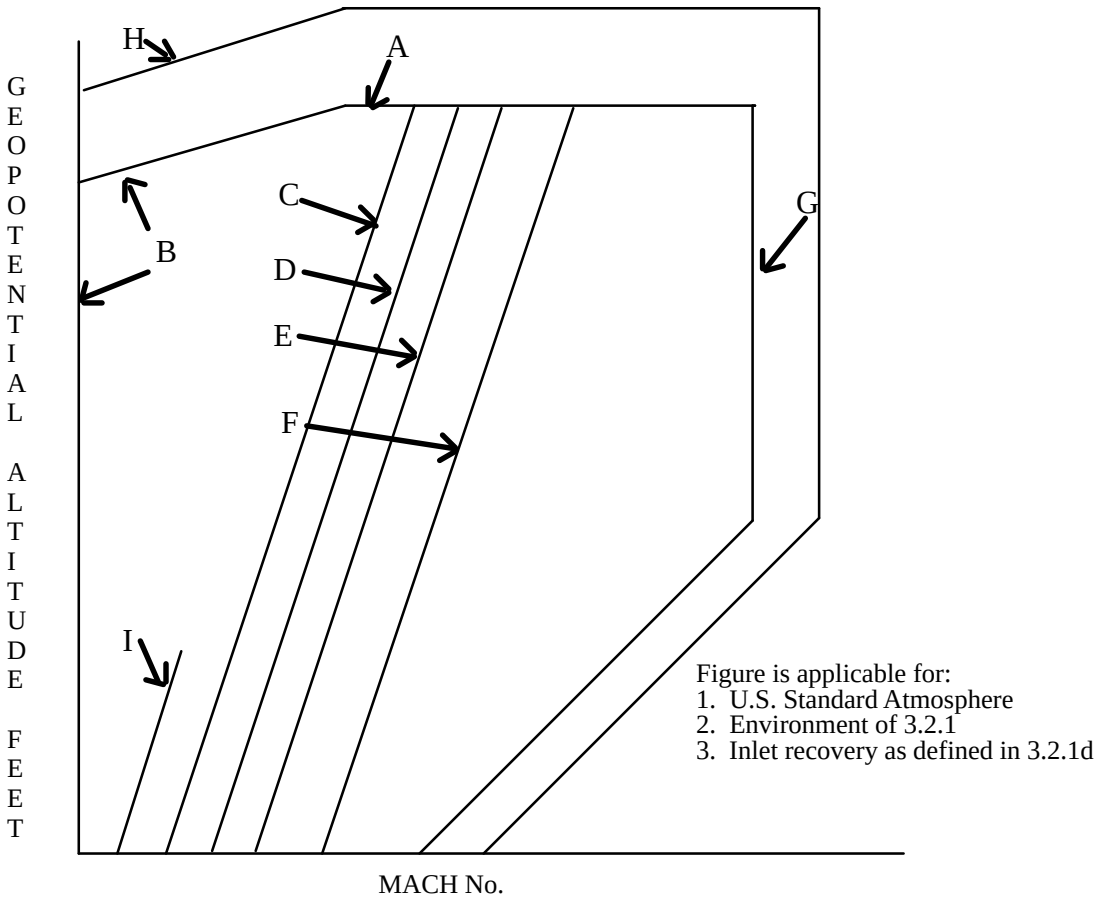


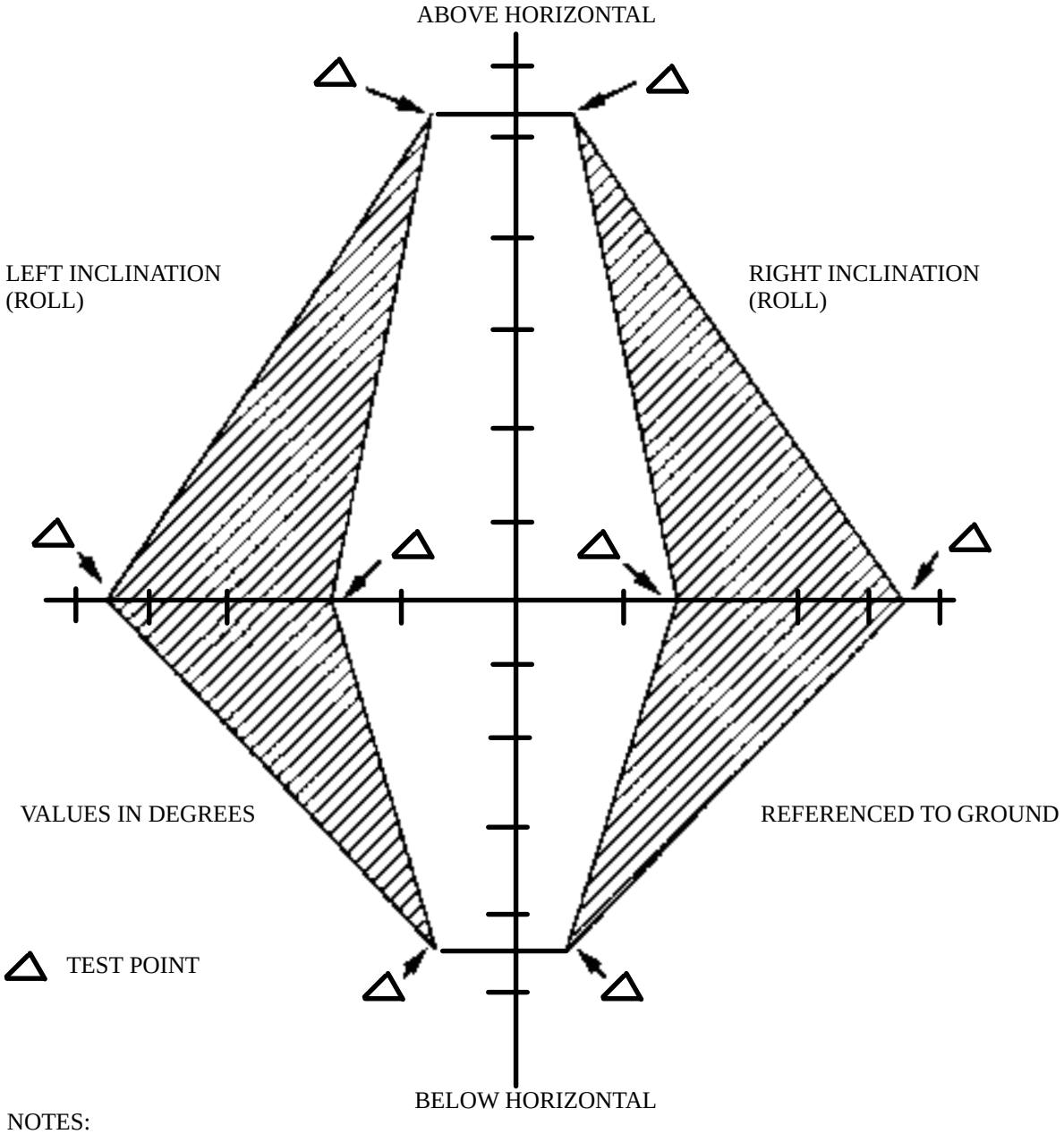
FIGURE 9. Operating limits.



NOTES:

A. Maximum starting altitude.
B. Minimum Mach No. for starter assisted starting.
C. Minimum Mach No. starting without starter assist, no customer power extraction, no customer bleed air extraction.
D. Minimum Mach No. for starting without starter assist, Maximum customer power extraction, no customer bleed air extraction.
E. Minimum Mach No. for starting without starter assist, no customer power extraction, Maximum customer bleed air extraction.
F. Minimum Mach No. for starting without starter assist, Maximum customer power extraction. Maximum customer bleed air extraction.
G. Maximum Mach No. for starting air extraction.
H. Operating envelope.
I. Post–loiter starting point.

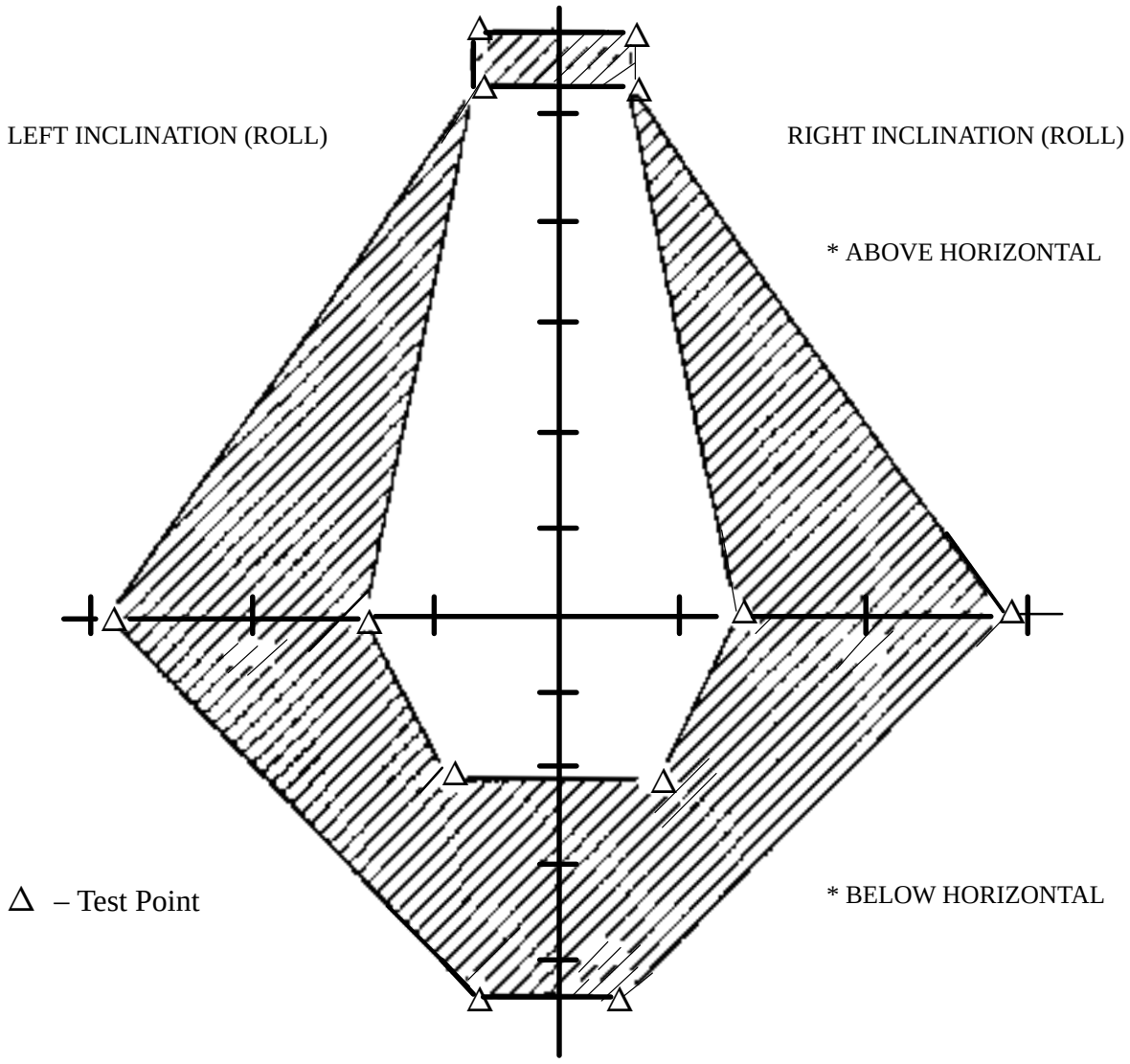
FIGURE 10. Operating envelope.



NOTES:

1. The engine shall be capable of operating at all possible acceleration conditions; however, for the purpose of defining the direction of acceleration vector from the engine CG, the figure assumes no acceleration other than gravity.
2. Engine centerline perpendicular to plane of paper.
3. Continuous operation in clear area.
4. ____ second operation in shaded area.

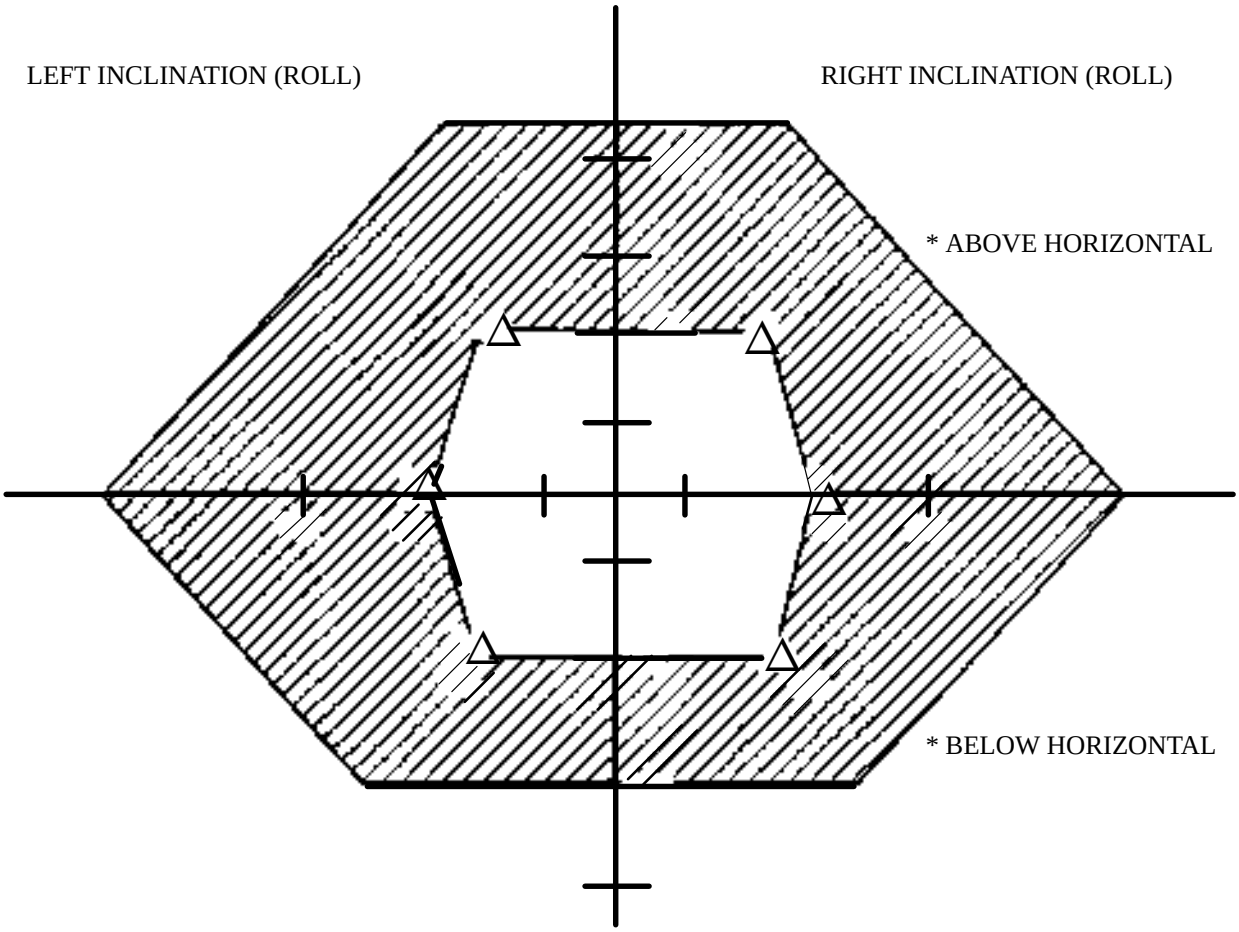
FIGURE 11. Engine attitude limits (fixed wing aircraft).



NOTES:

1. The engine shall be capable of operating at all possible acceleration conditions; however, for the purpose of defining the direction of acceleration vector from the engine CG, the figure assumes no acceleration other than gravity.
2. * —— referenced to ground.
3. Engine centerline perpendicular to plane of paper.
4. Continuous operation in clear area.
5. _____ second operation in shaded area.

FIGURE 11. Engine attitude limits (VSTOL aircraft engines).



NOTES:

1. For the purpose of defining the direction of the acceleration vector from the engine CG, the figure assumes no acceleration other than gravity; however, the engine shall be capable of operating at all possible acceleration conditions.
2. *—referenced to ground.
3. Engine centerline perpendicular to plane of paper.
4. Continuous operation in clear area.
5. ____ second operation in shaded area.
6. Symbol △ indicates points for test.

FIGURE 11. Engine attitude limits (rotary wing aircraft engines).

Output Shaft Speed: _____RPM

Output Shaft Torque: Zero

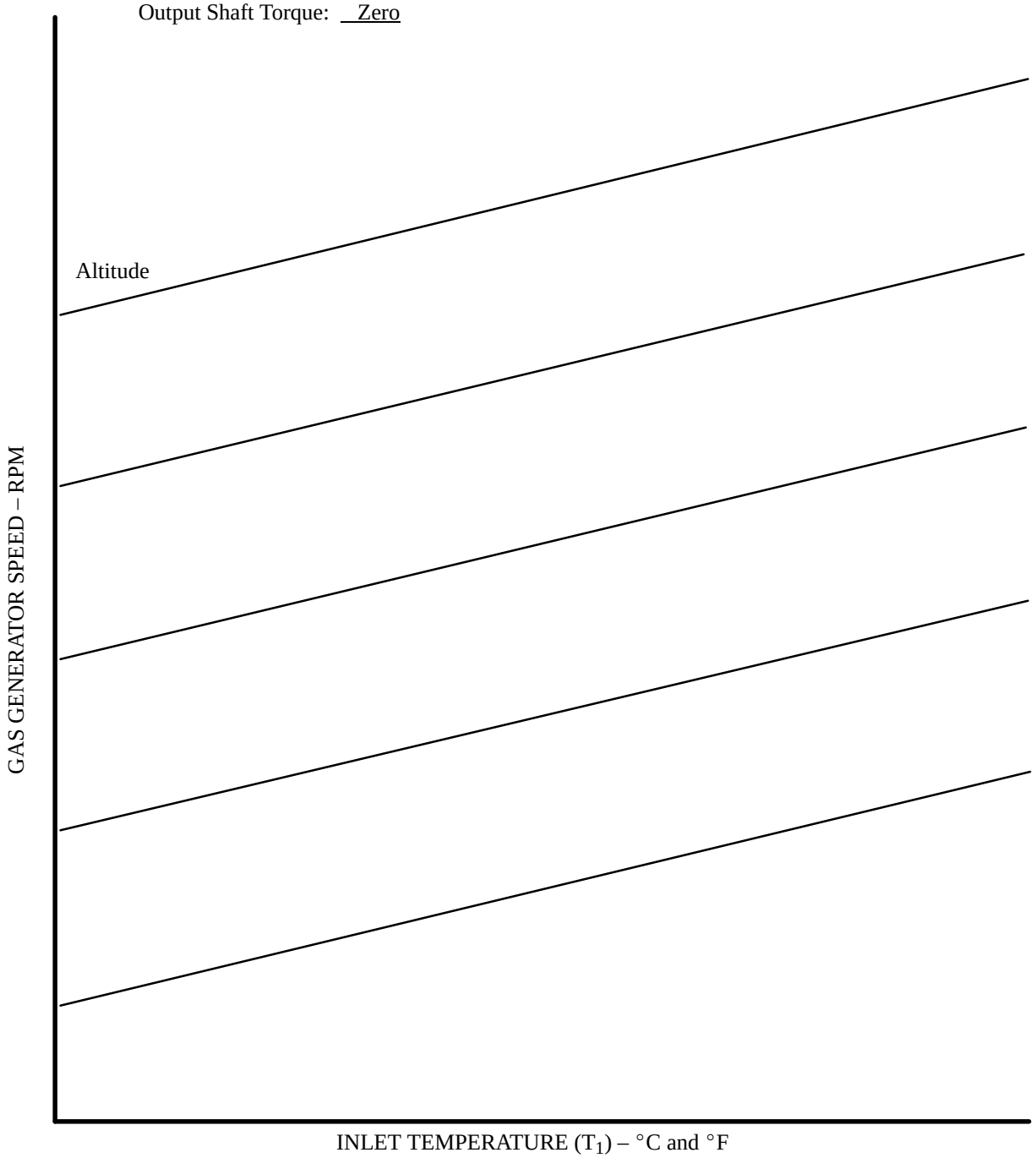
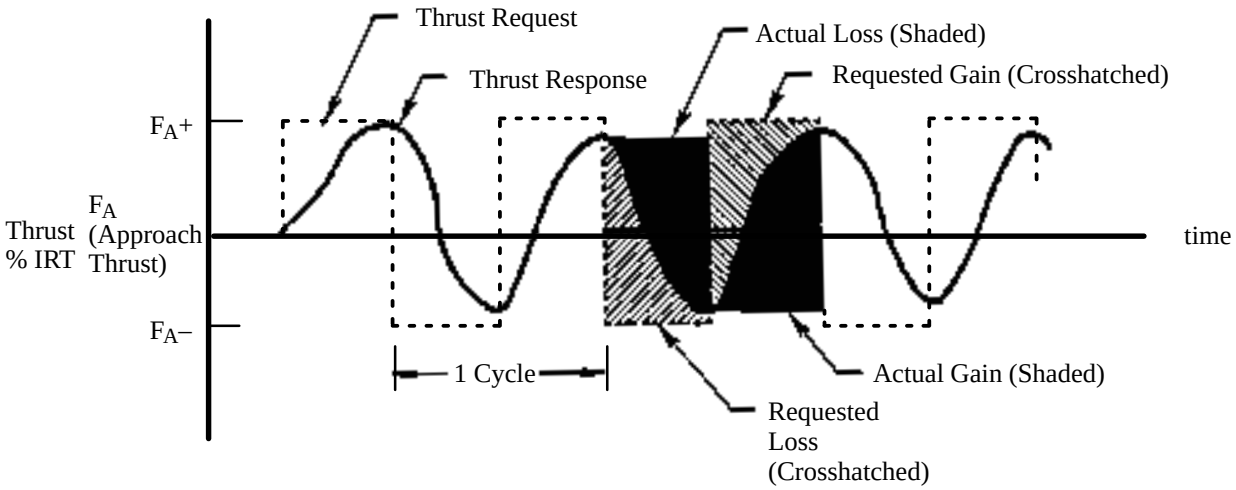


FIGURE 12. No load conditions.



$$\text{Gain/Loss Ratio} = \text{GLR} = \frac{\text{Actual Gain/Requested Gain}}{\text{Actual Loss/Requested Loss}}$$

$$\text{Average GLR} = \text{AGLR} = \frac{1}{N}\sum_{1}^{N}(\text{GLR}) \qquad N = \text{No. of Cycles Evaluated}$$

$$_____ \times \text{AGLR} \times _____ _{\text{AGLR}} \times$$

FIGURE 13. Stability.

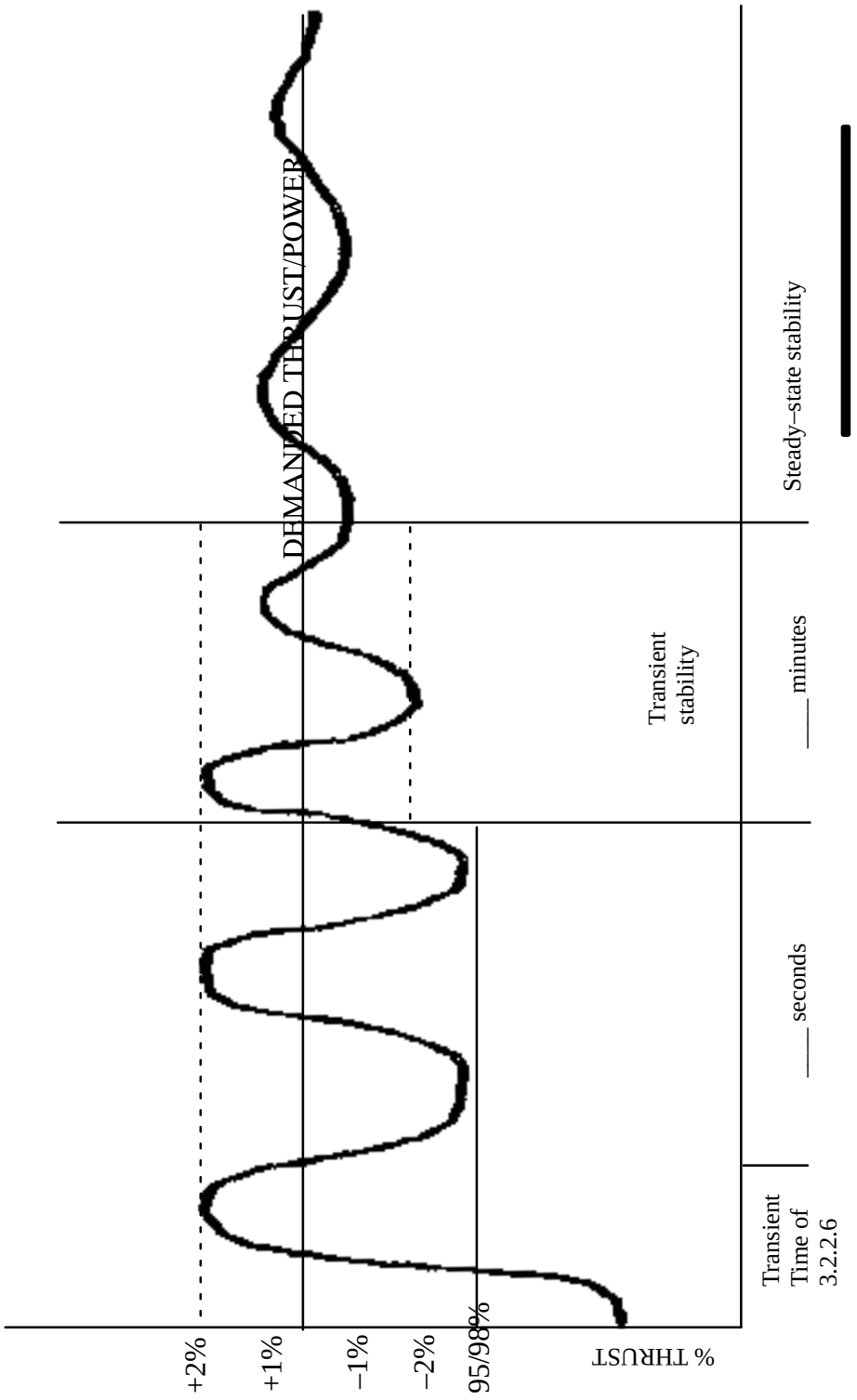


FIGURE 14. Transient.

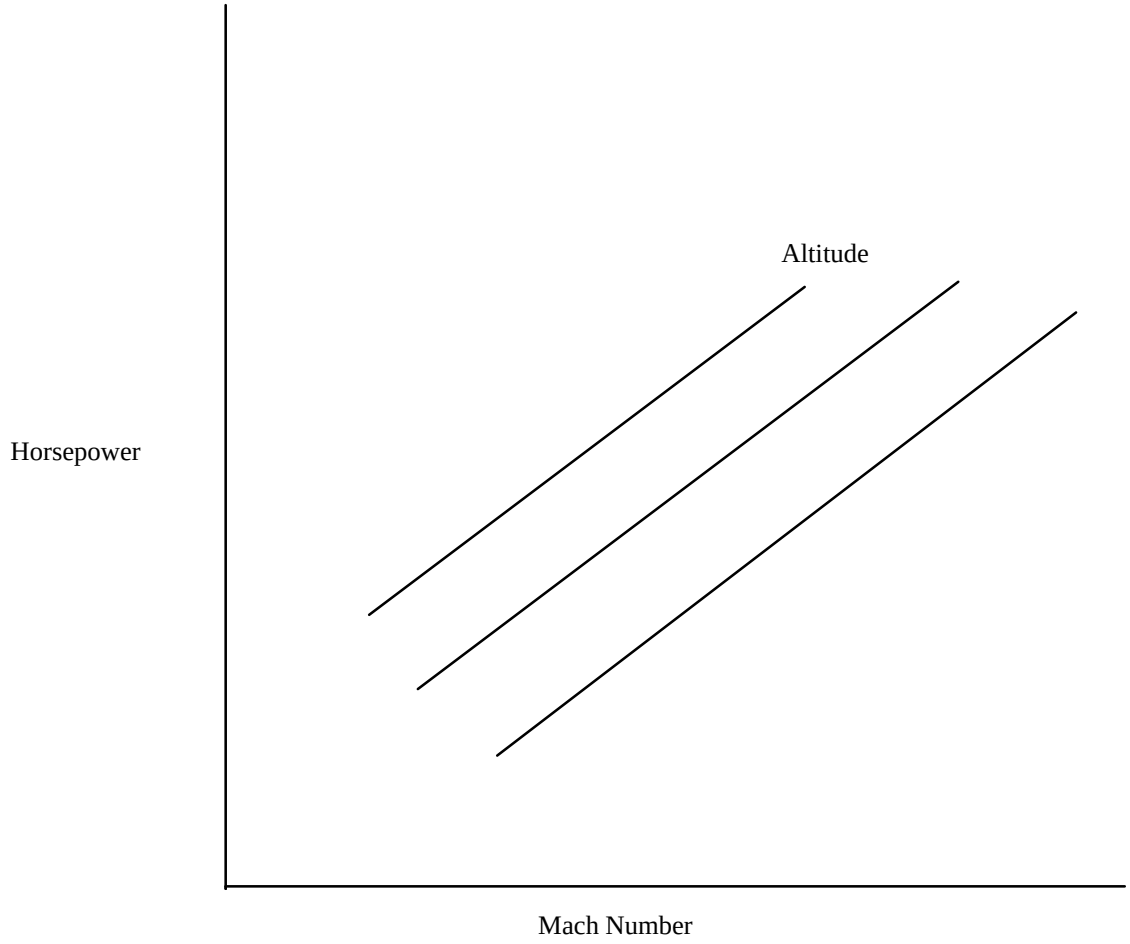


FIGURE 15a . Customer extraction during windmilling.

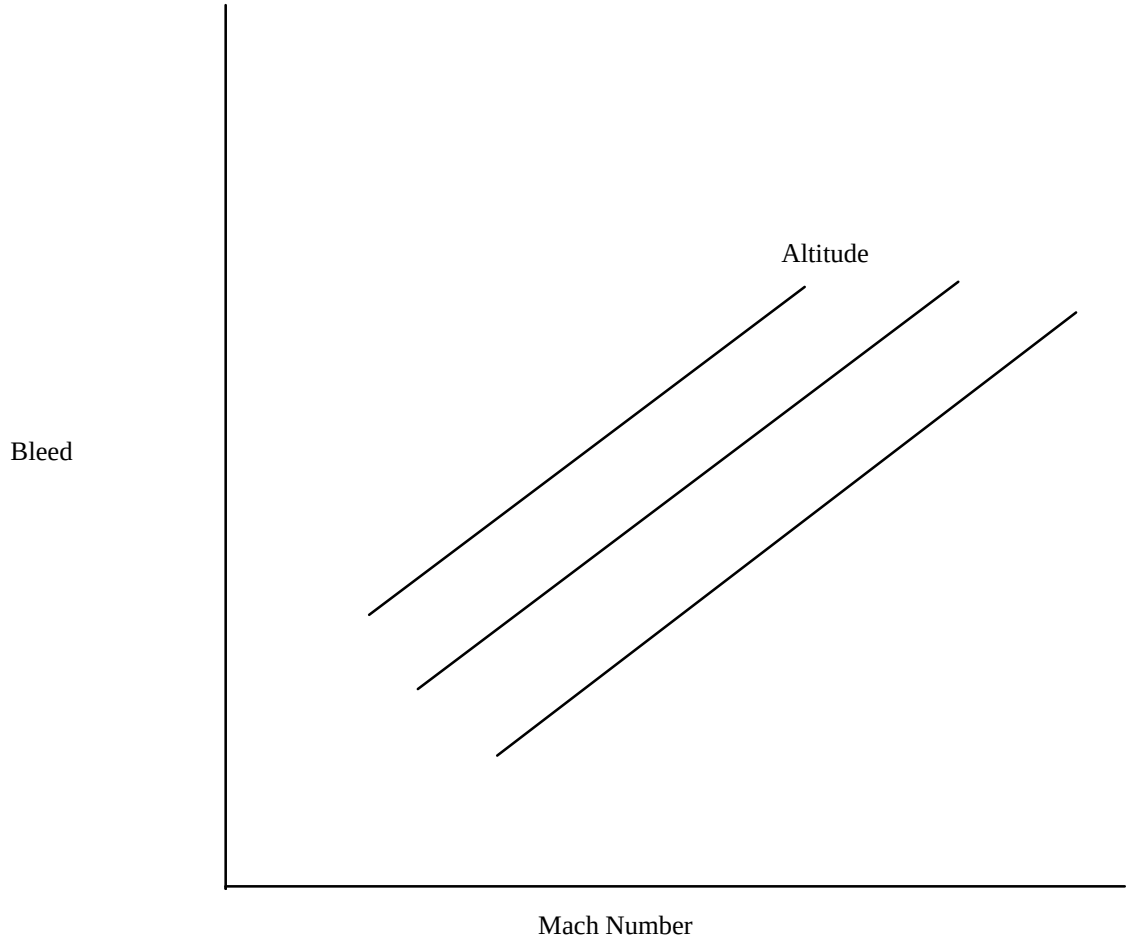
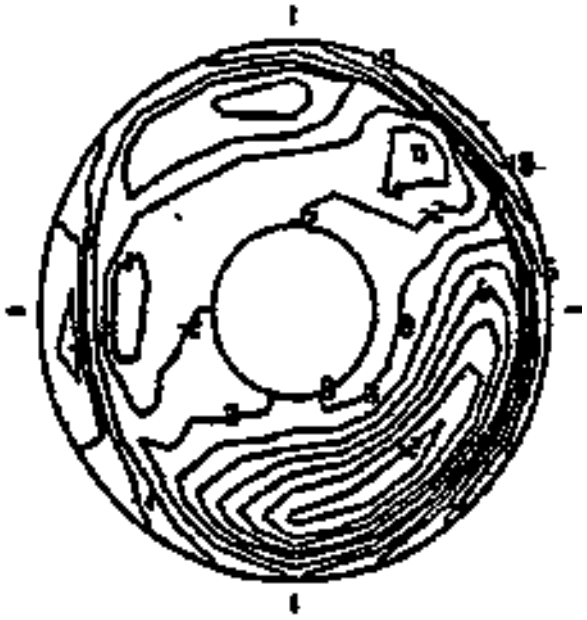


FIGURE 15b. Customer extraction during windmilling.

XM =
α =
β =
P1/PQ =
WlR =



Forward Looking Aft

Ring Radii	Rake Location (Deg.)	0	45	90	135	180	225	270	315

Inner Radius =
Outer Radius =

FIGURE 16. Inlet airflow distortion pattern.

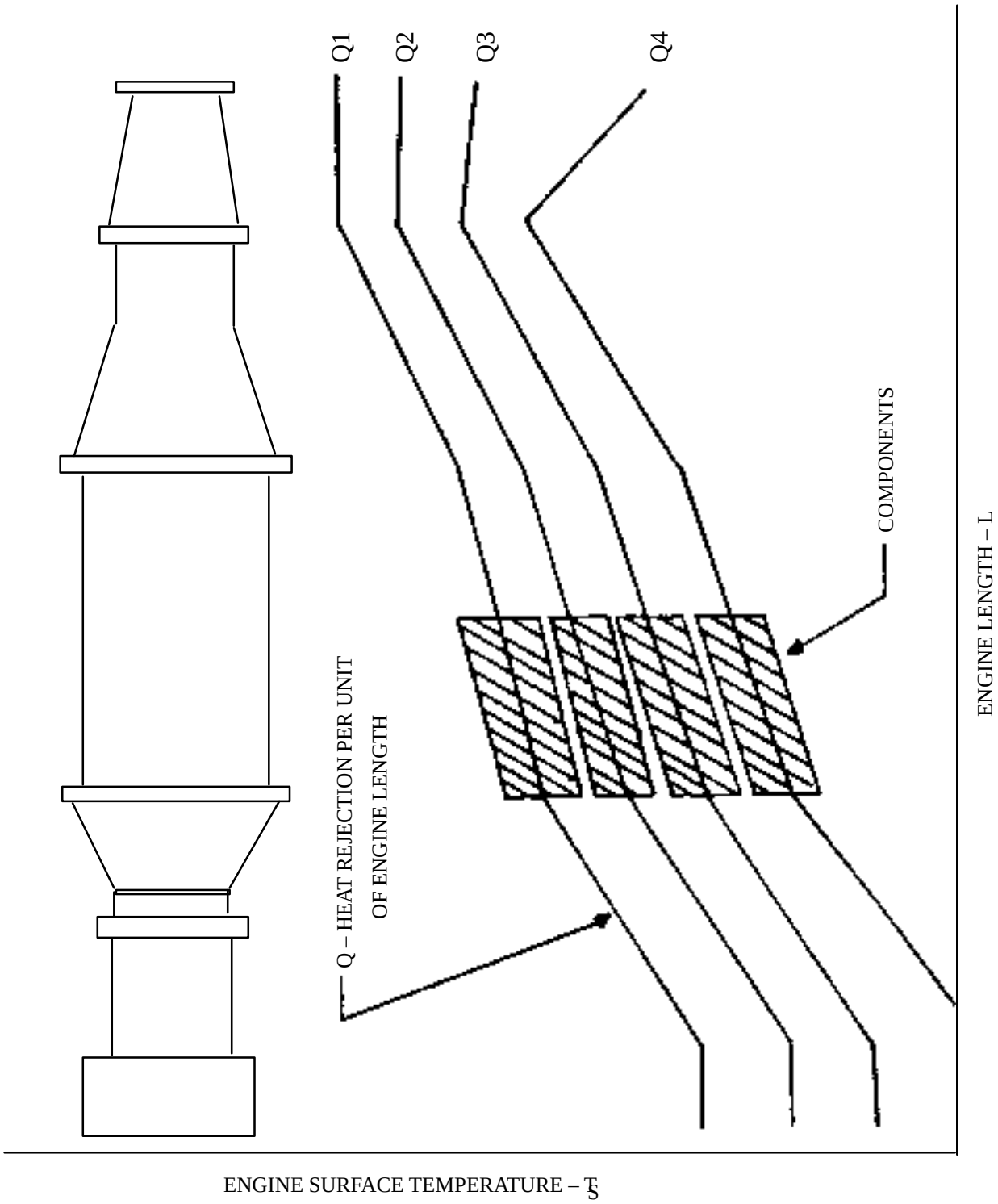
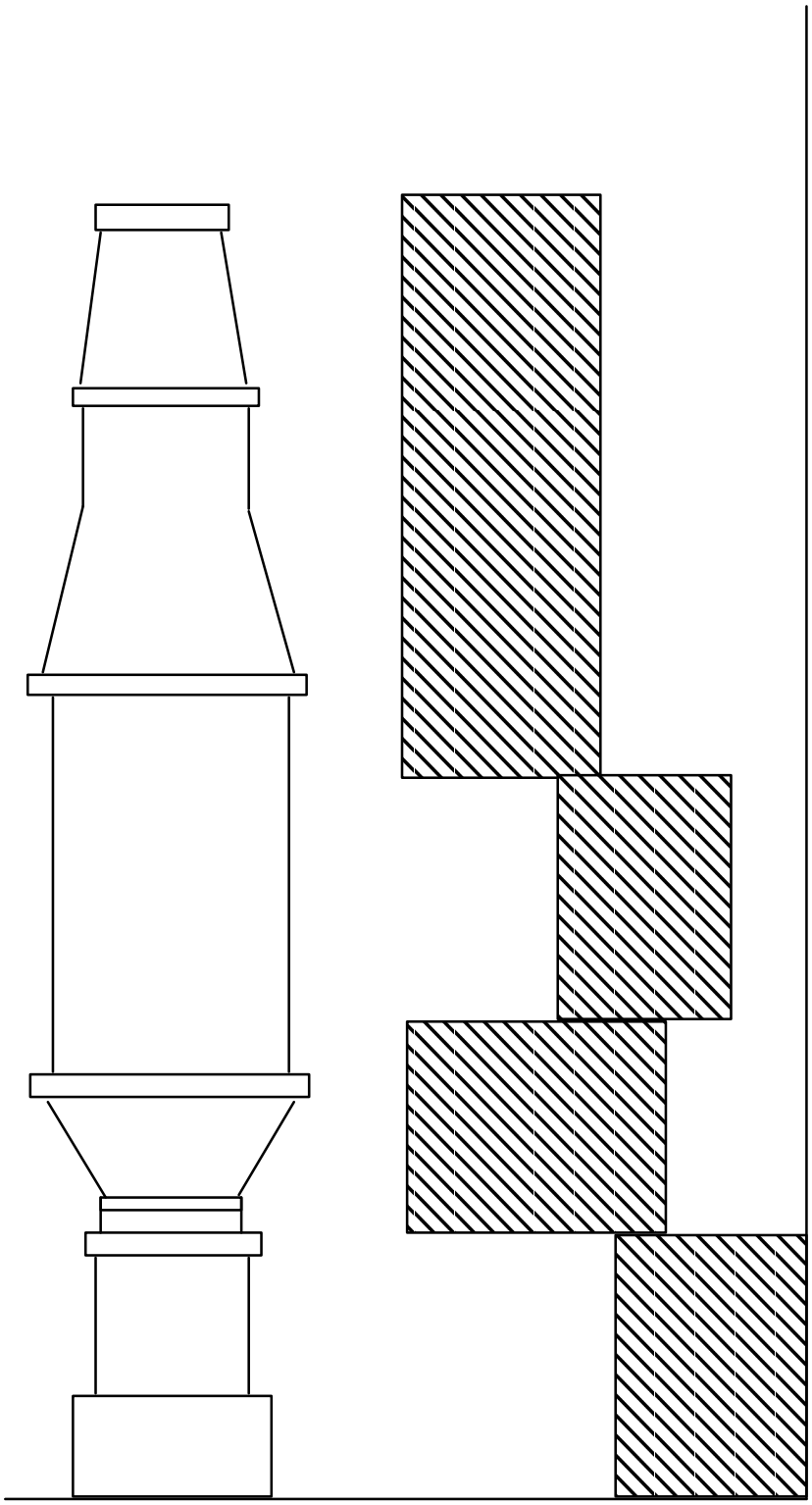


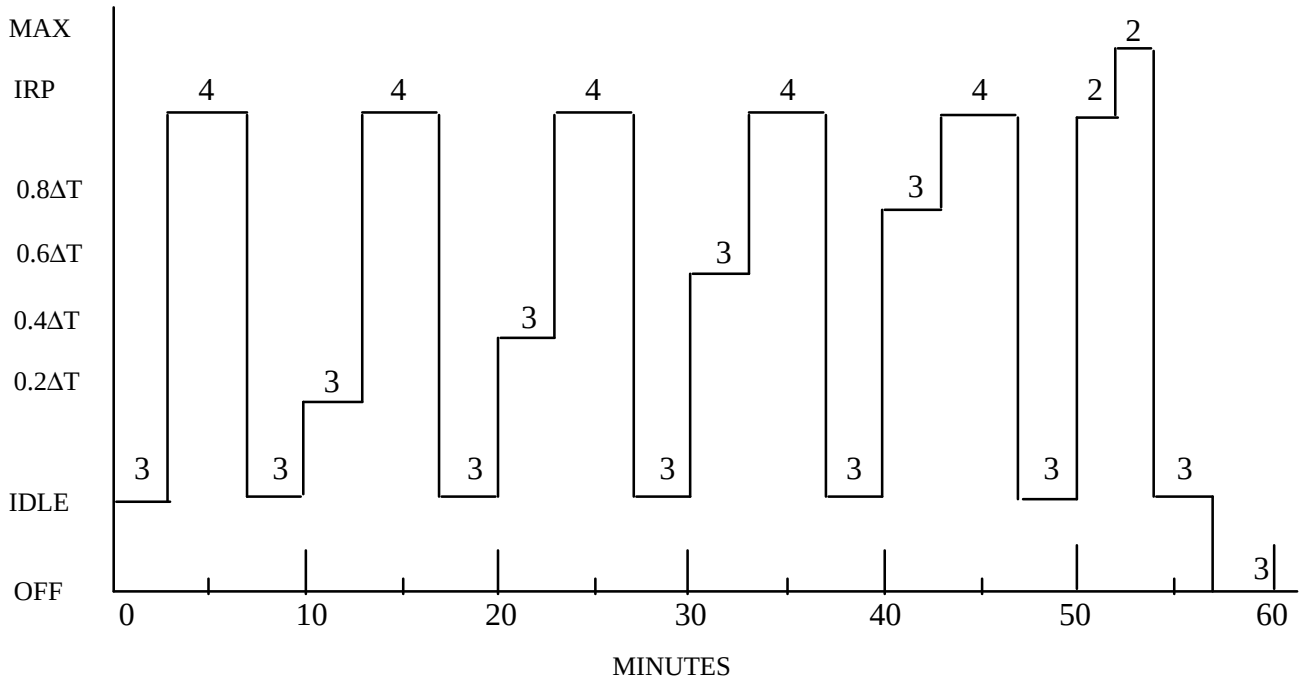
FIGURE 17a. Engine surface temperature.



1. (Restrictions or Conditions for Data)
2.
3.

SECTION	MATERIAL	EMISSIVITY
I		
II		

FIGURE 17b. Engine surface emissivity.



ΔT = Delta Temperature between IRP and Idle

FIGURE 18. Engine corrosion operating cycle.

ALTITUDE: Sea Level to 6.7 km (22,000 feet)
MAXIMUM VERTICAL EXTENT: 2.0 km (6,500 feet)

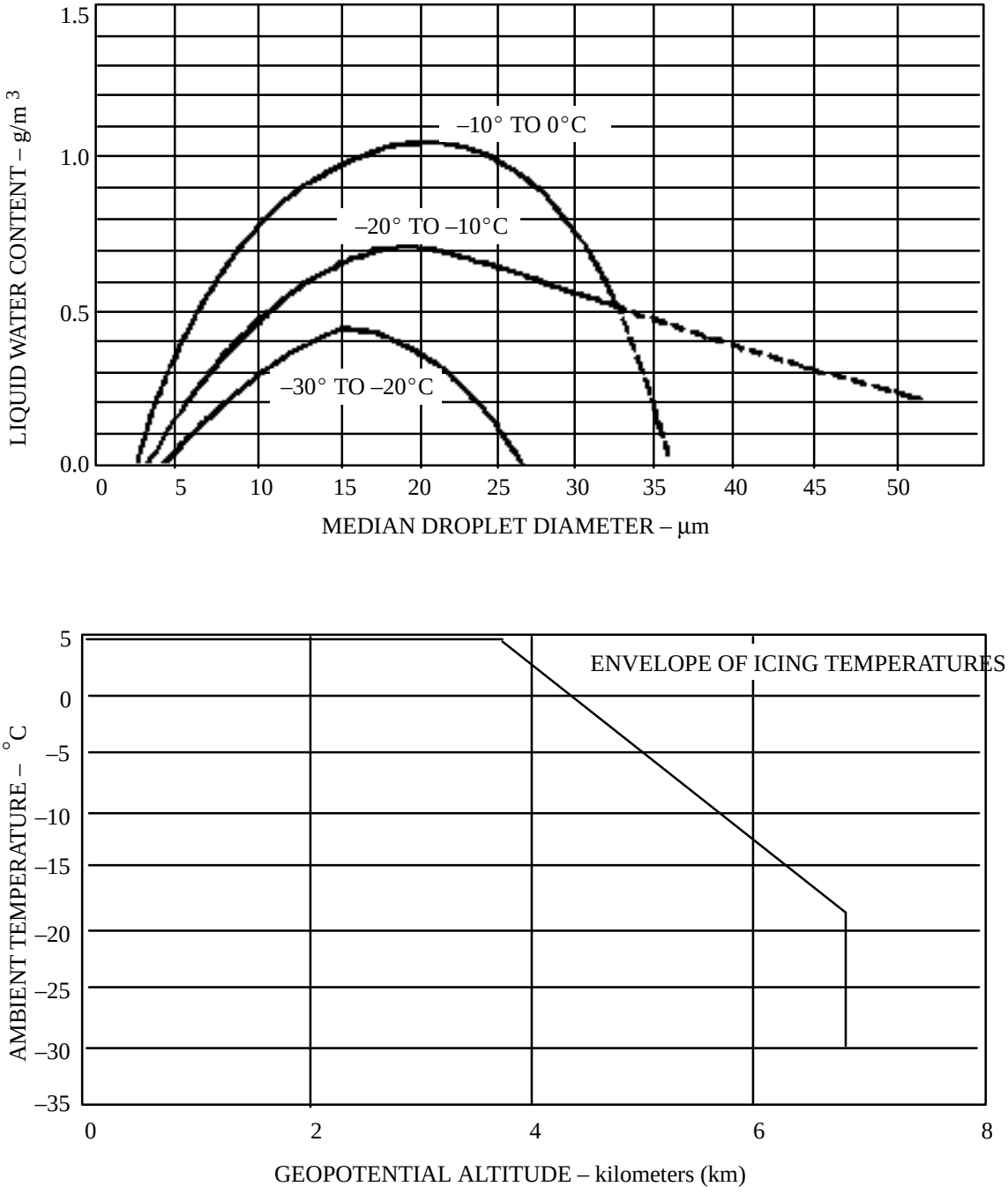


FIGURE 19a. Continuous maximum icing conditions.

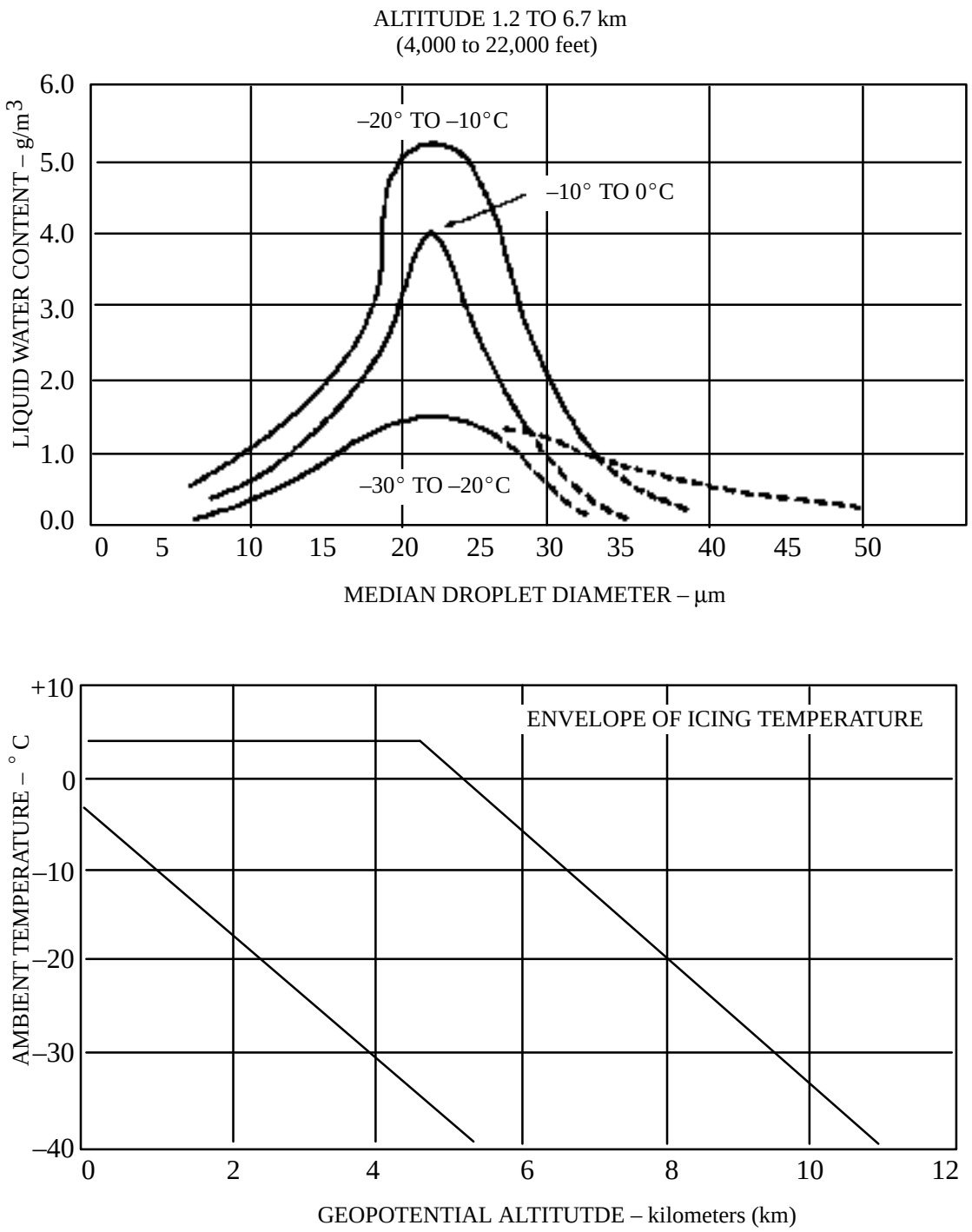
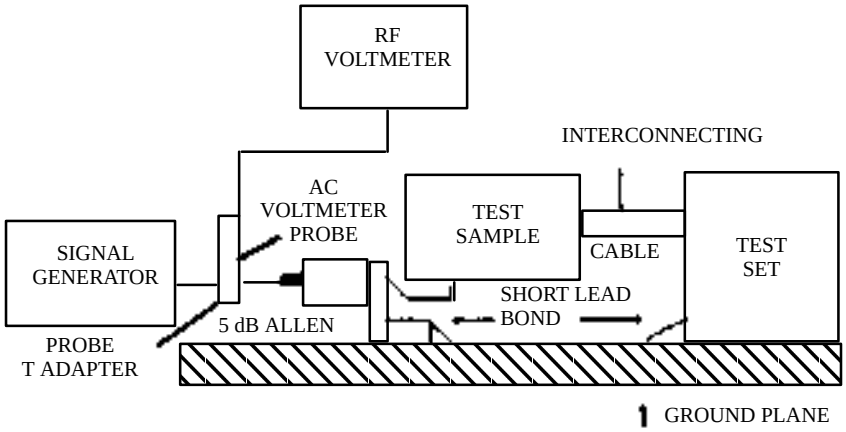
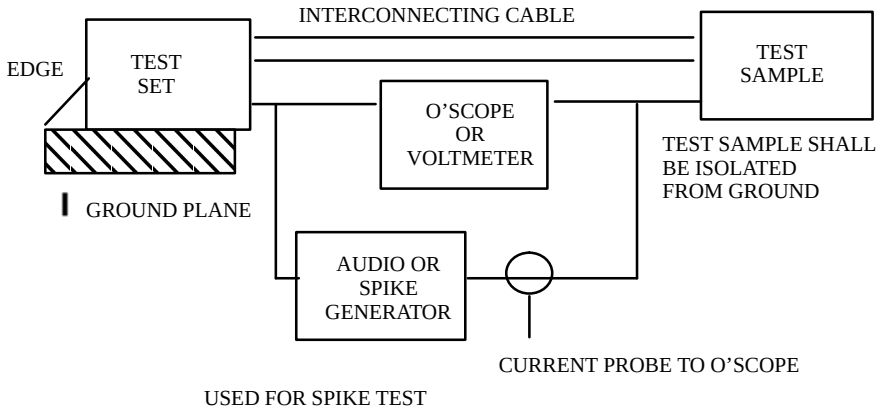


FIGURE 19b. Intermittent maximum icing conditions.

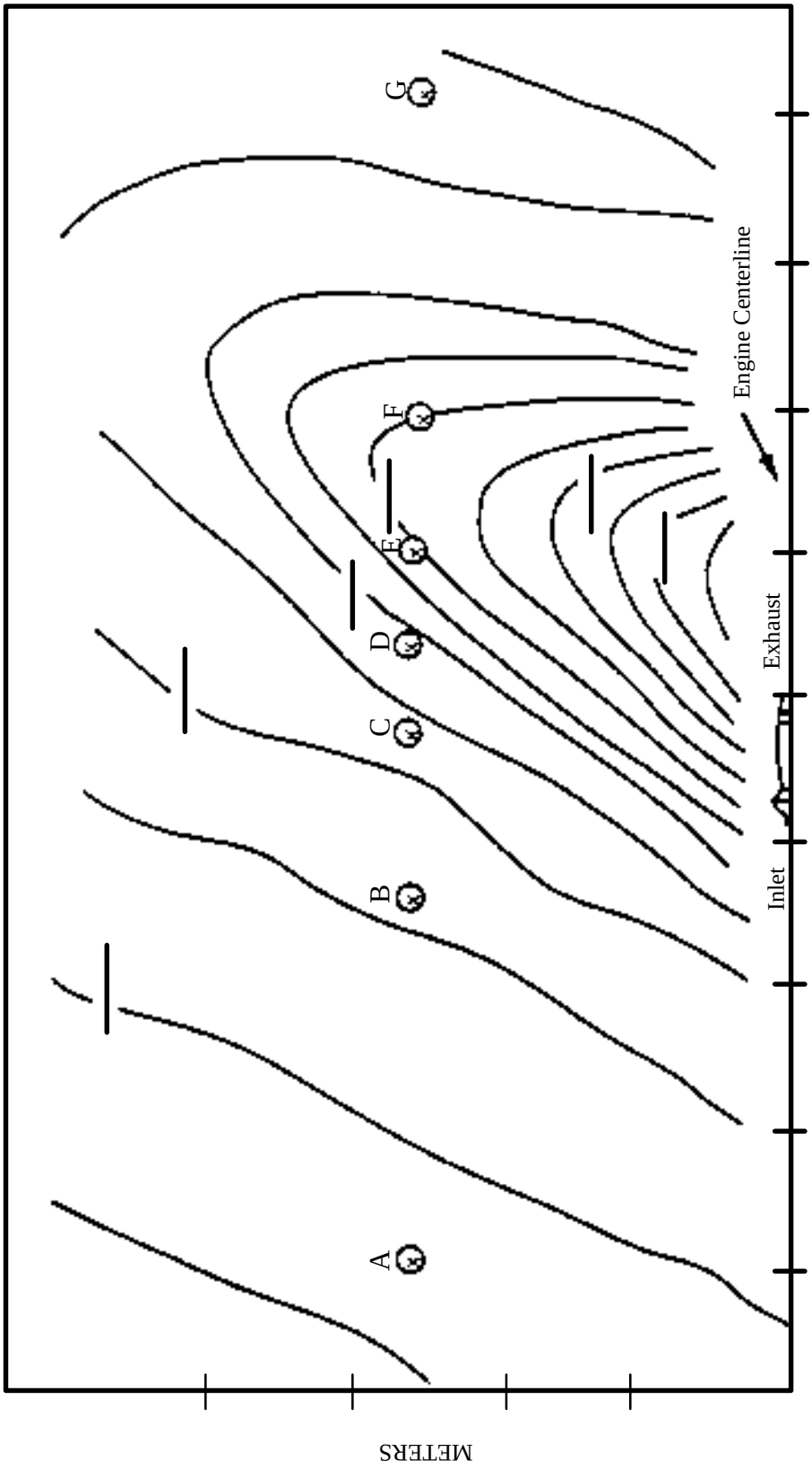


GROUND PLANE INTERFERENCE TEST SETUP (50 KHz to 100 MHz)



GROUND PLANE INTERFERENCE TEST SETUP SPIKE TEST AND
320Hz TO 50 KHz TEST

FIGURE 20. Ground plane interference test setup.



Near field octave band sound pressure level contours (dB – ref. 0.0002 μbars)

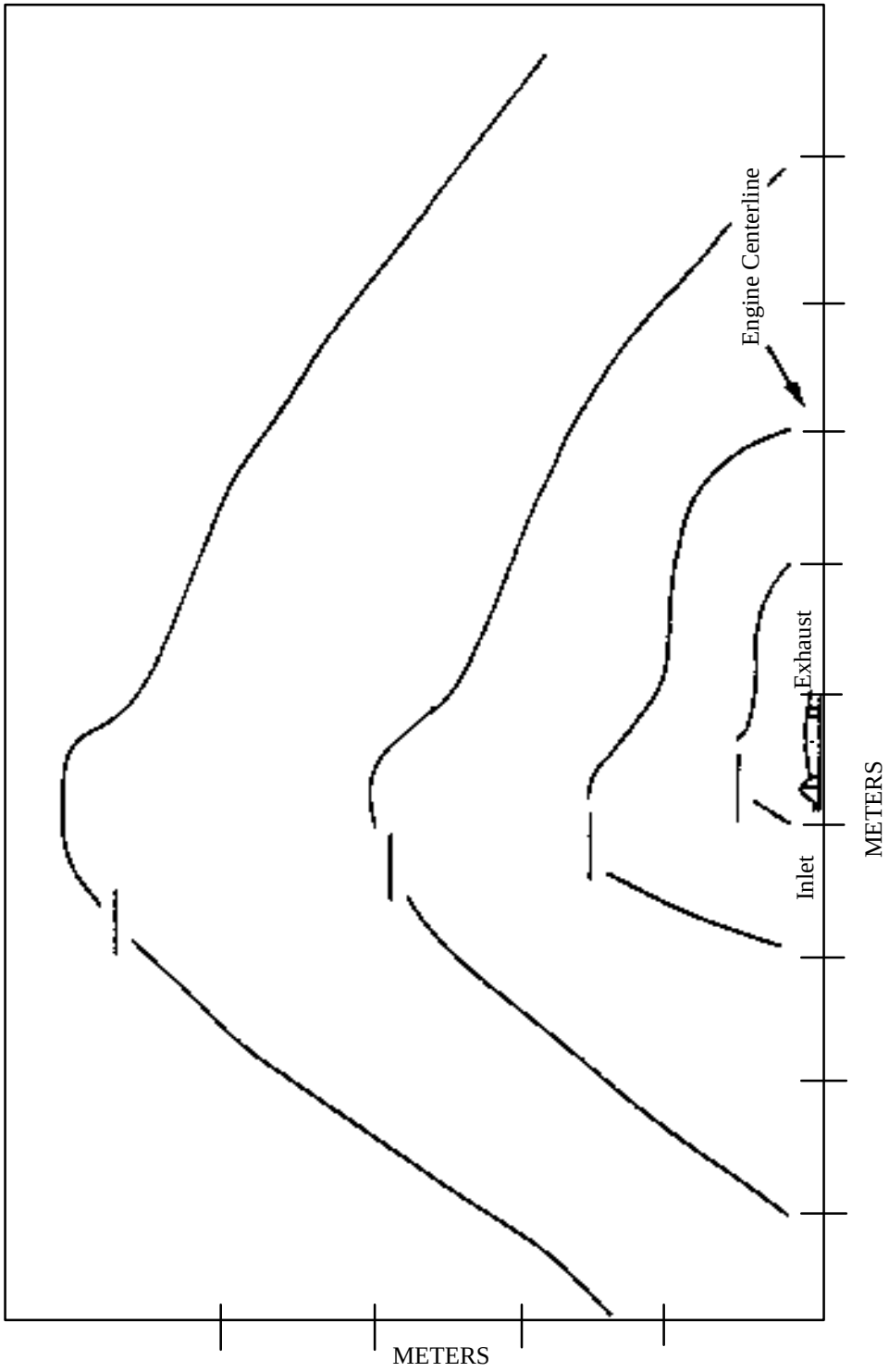


FIGURE 21b. Far field overall sound pressure level contours (dB – ref. 0.0002 μbars).
Far field overall perceived noise level contours (dB – ref. 0.0002 μ bars).

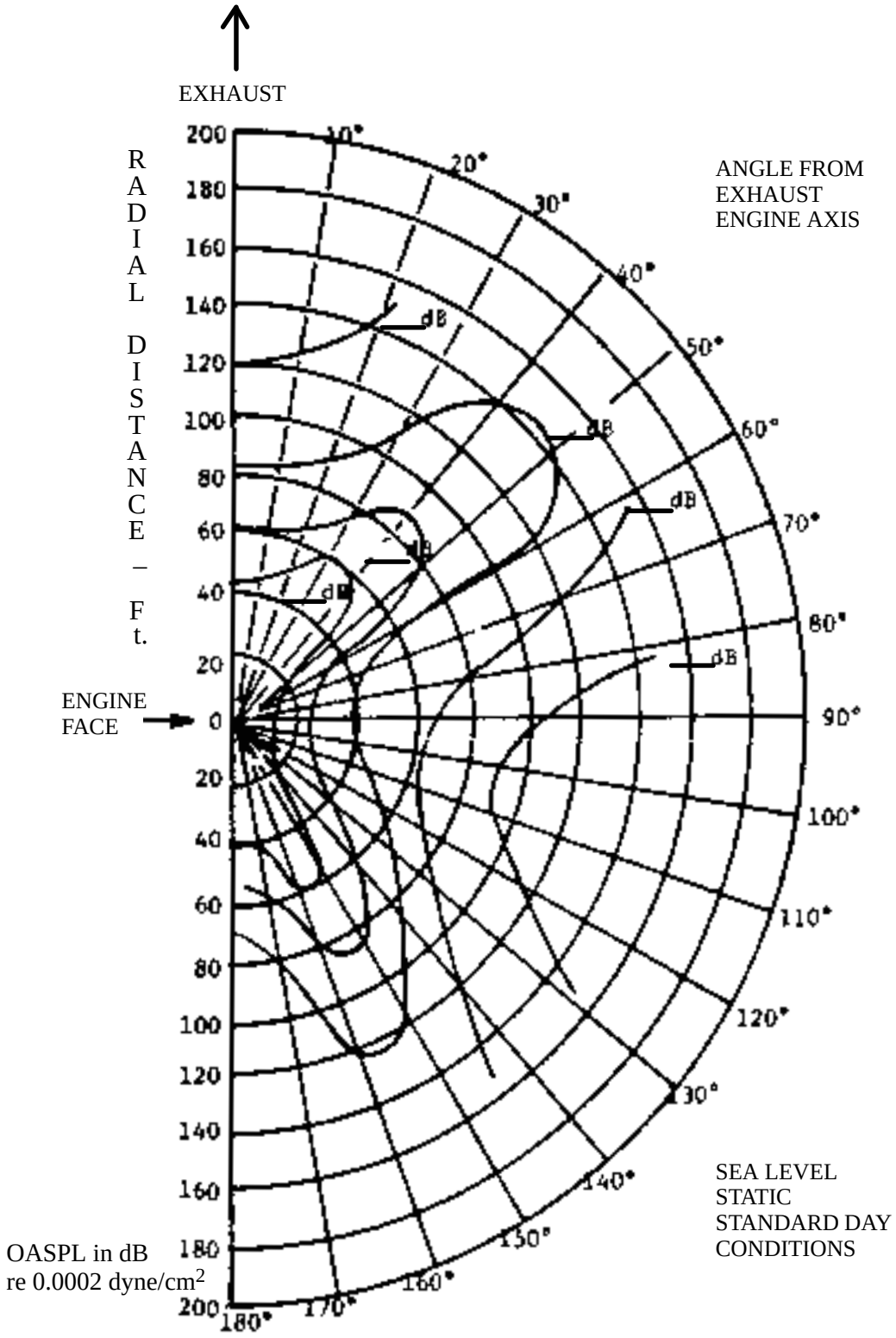
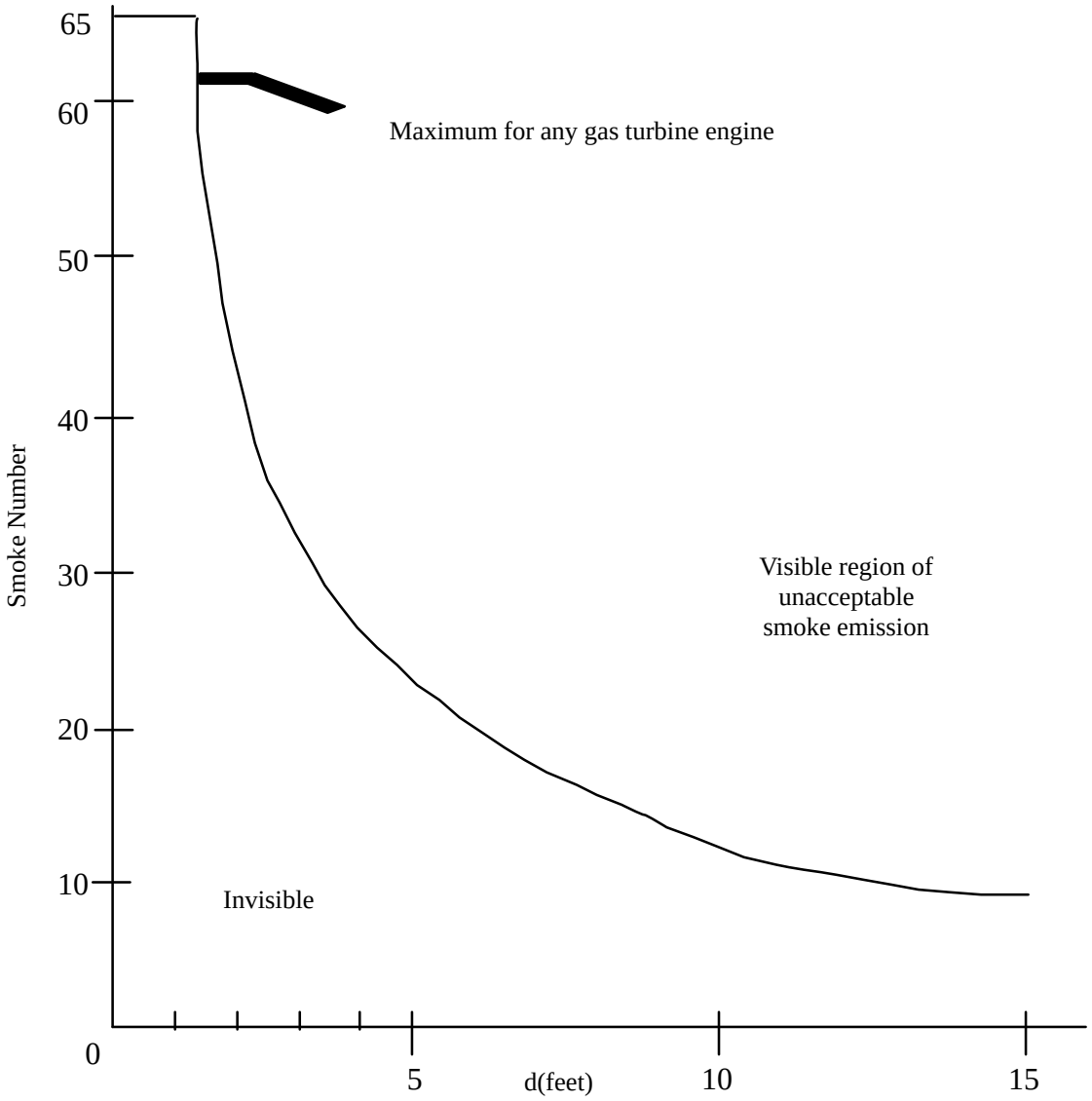


FIGURE 21c. Estimated overall sound pressure level contours at idle.



d: Diameter of the vitiated airflow exhaust nozzle at the engine exhaust exit plane (ft). The exhaust exit plane is the first downstream plane, normal to the exhaust system, that does not contain a solid surface around the stream. For engines with variable area exhaust nozzles, d shall be the mean of values attainable when not afterburning. For engines with non–circular exhaust nozzle areas, d shall be the diameter of the smallest circle circumscribing the actual nozzle. For engines with vitiated airflow leaving the engine through an annulus, d shall be the diameter of a circle having the same cross–sectional area as the annular exhaust stream.

FIGURE 22. Smoke number.

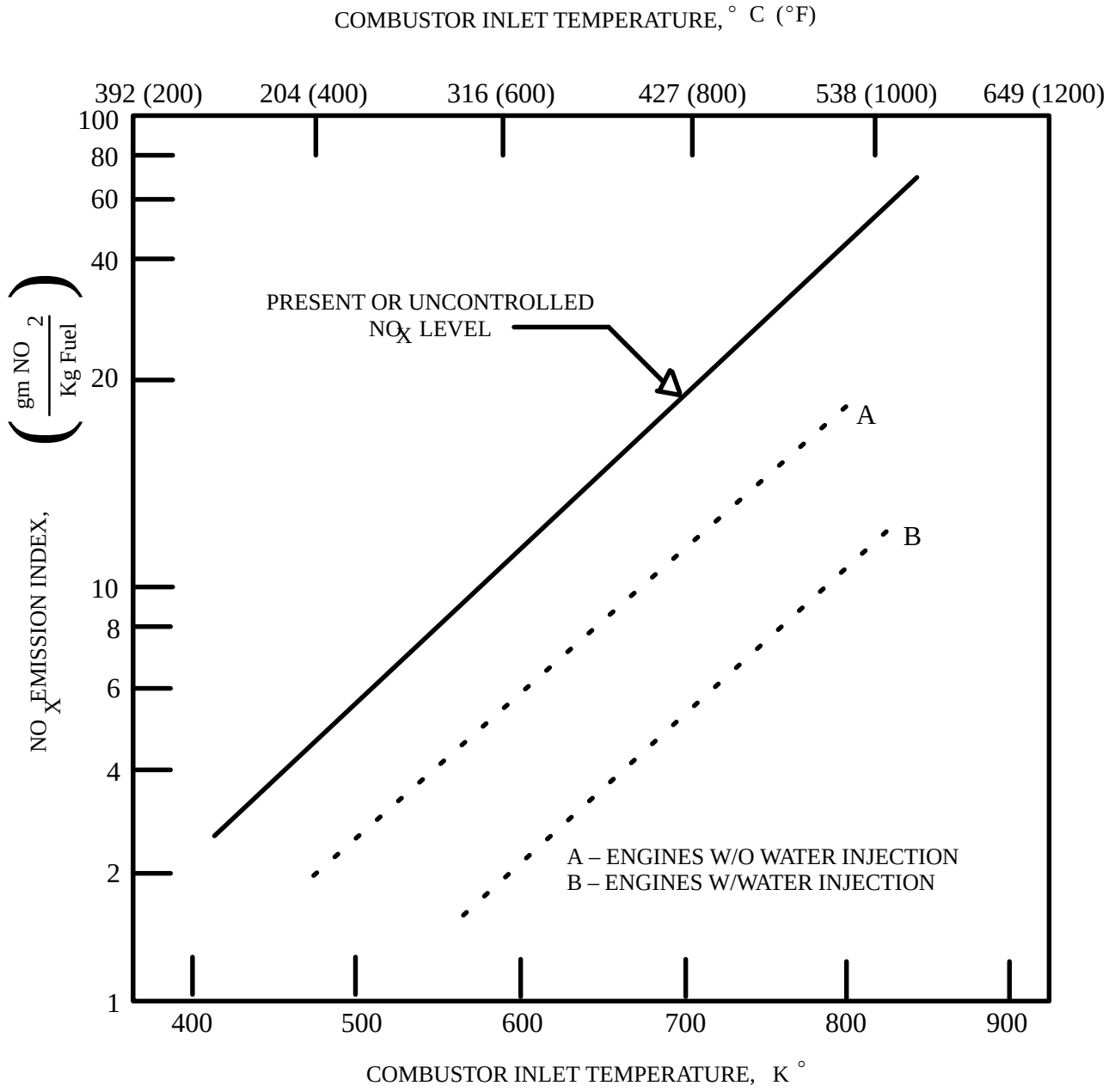


FIGURE 23a. Gaseous emissions.

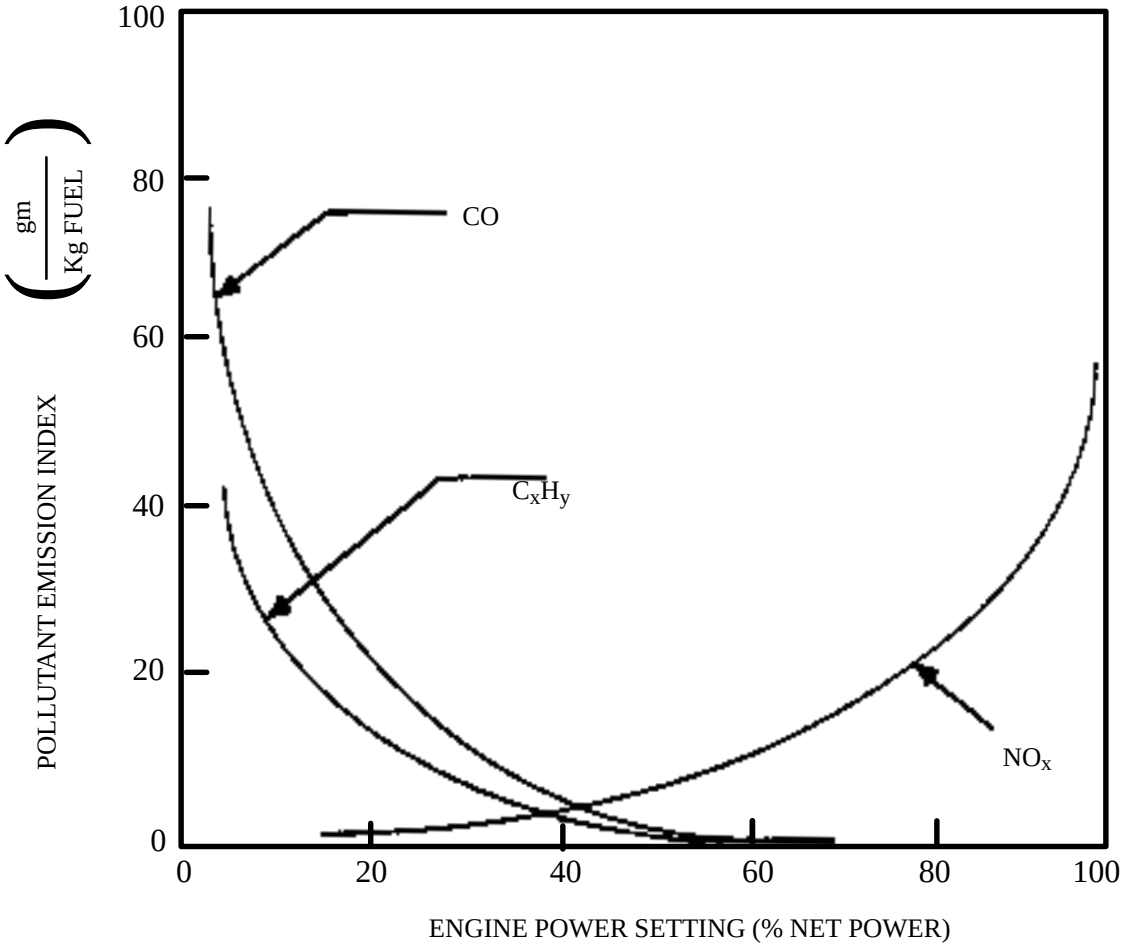


FIGURE 23b. Gaseous emissions.

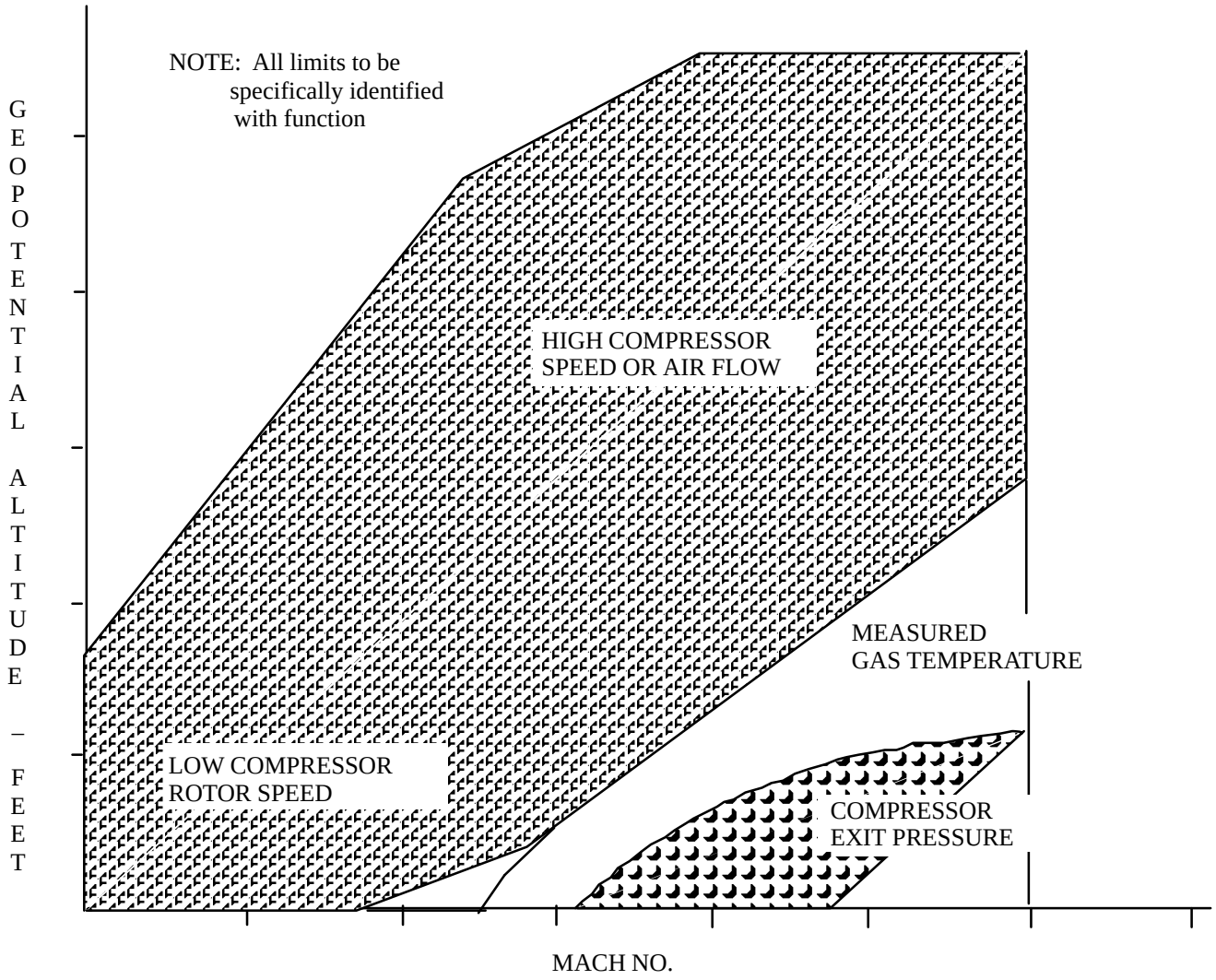


FIGURE 24. Control limiting regimes.

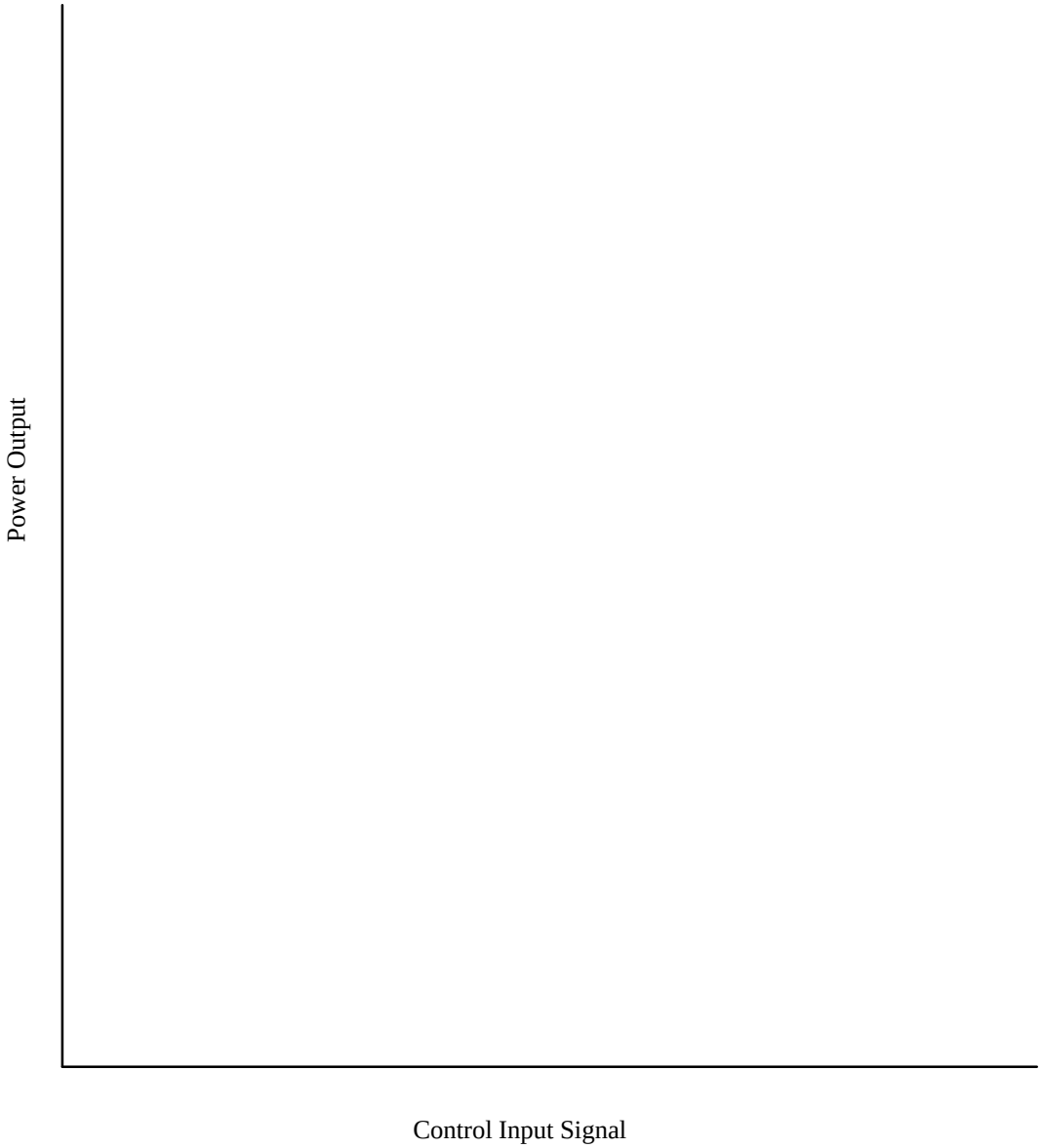


FIGURE 25. Control system performance.

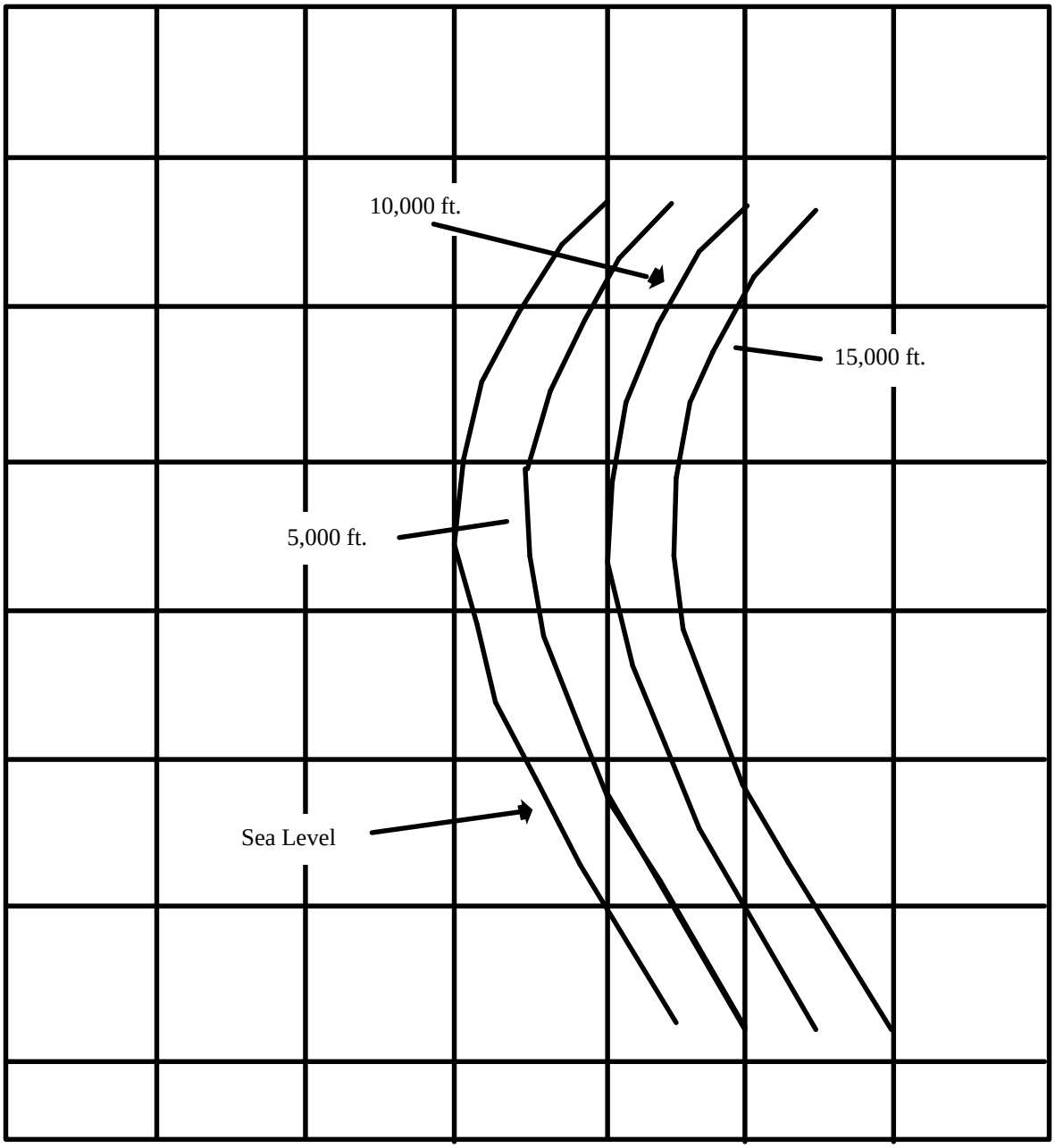
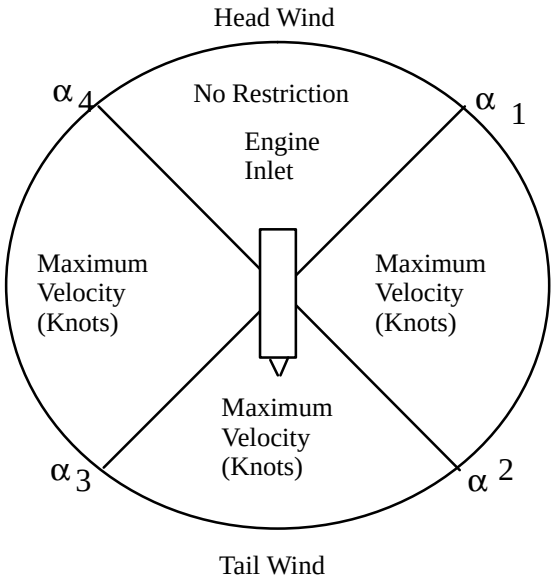


FIGURE 26a. Ground start time.

RELATIVE WIND CONDITION ENVELOPE
STANDARD STARTING PROCEDURE



NOTE:

If wind velocity equals or exceeds the value noted, DO NOT ATTEMPT A START. Refer to Corrective Measures.

a = Angular position from engine inlet centerline in degrees.

FIGURE 26b. Ground start wind envelope.

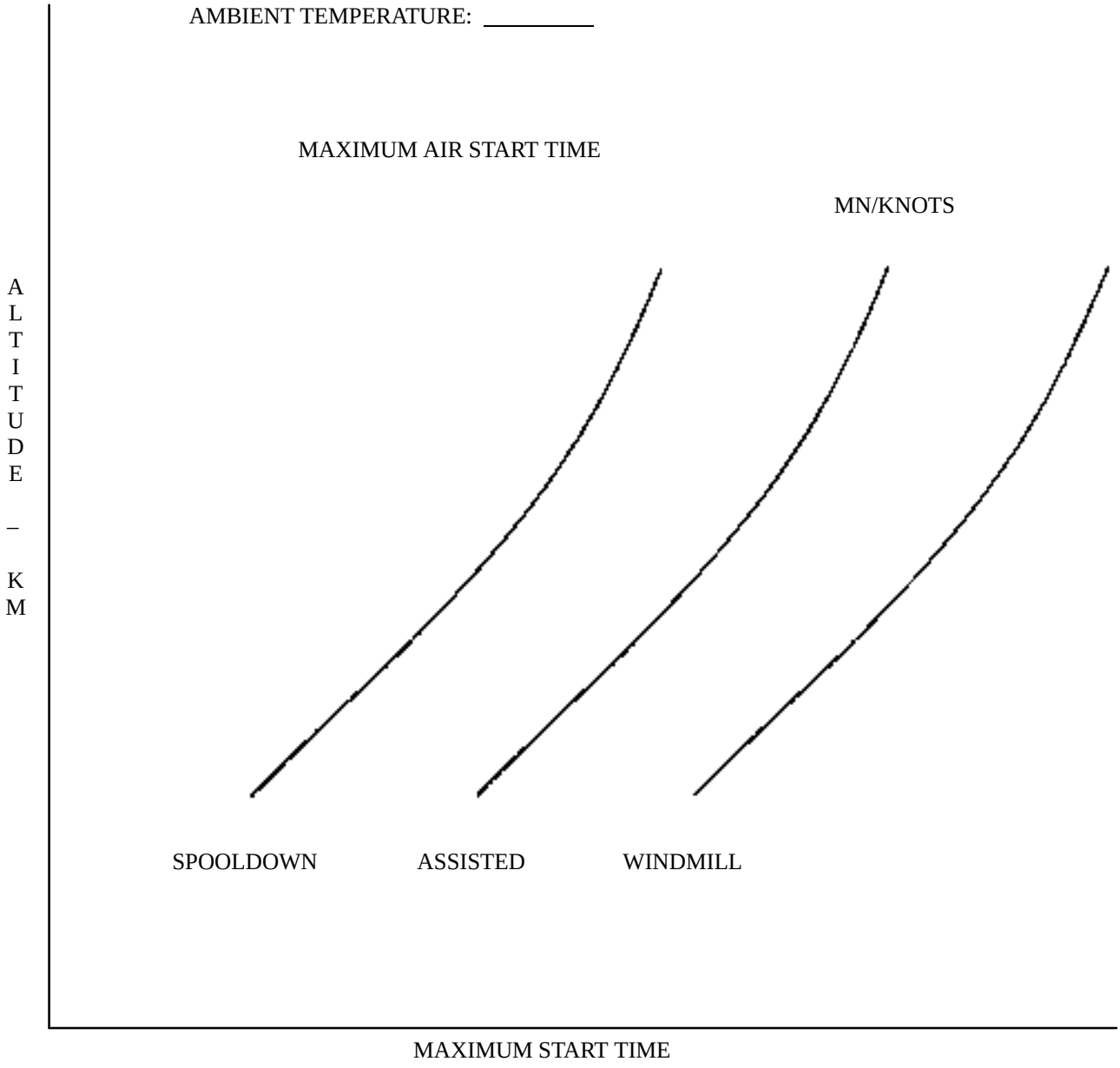


FIGURE 27. Windmill and airstart envelopes.

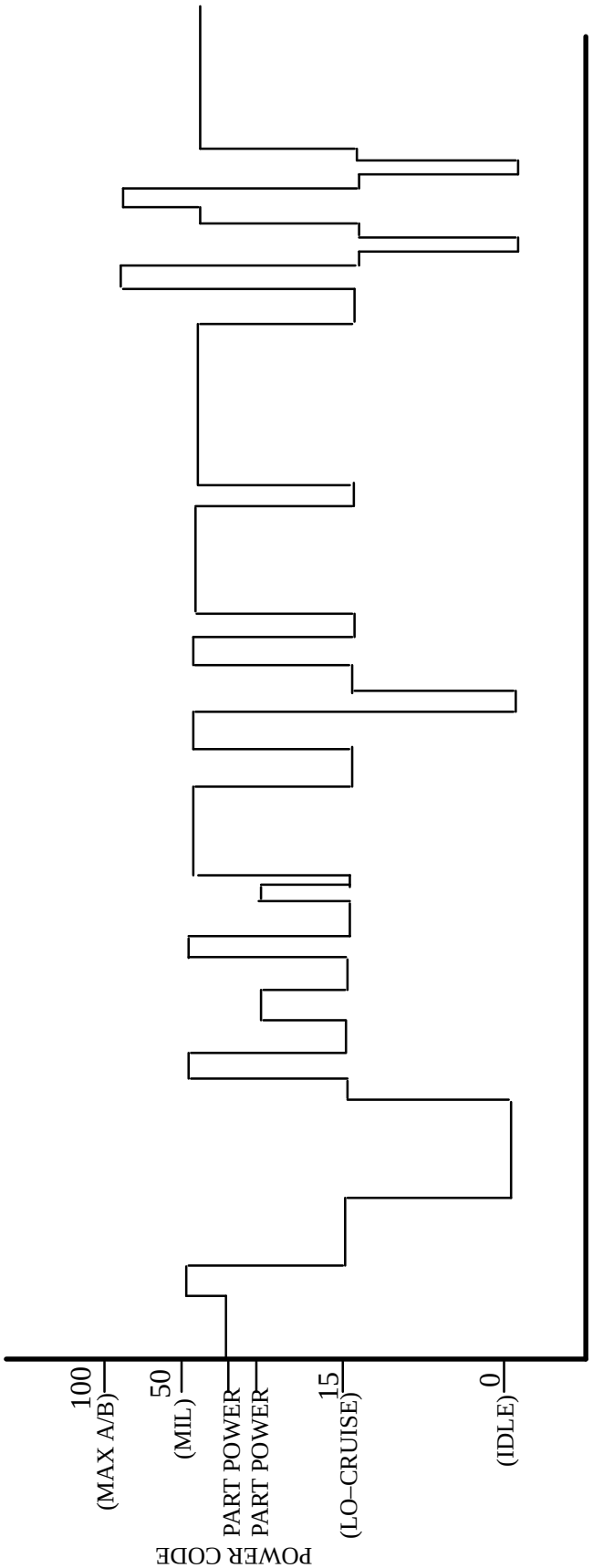


FIGURE 28a. Air to surface mission (example only).

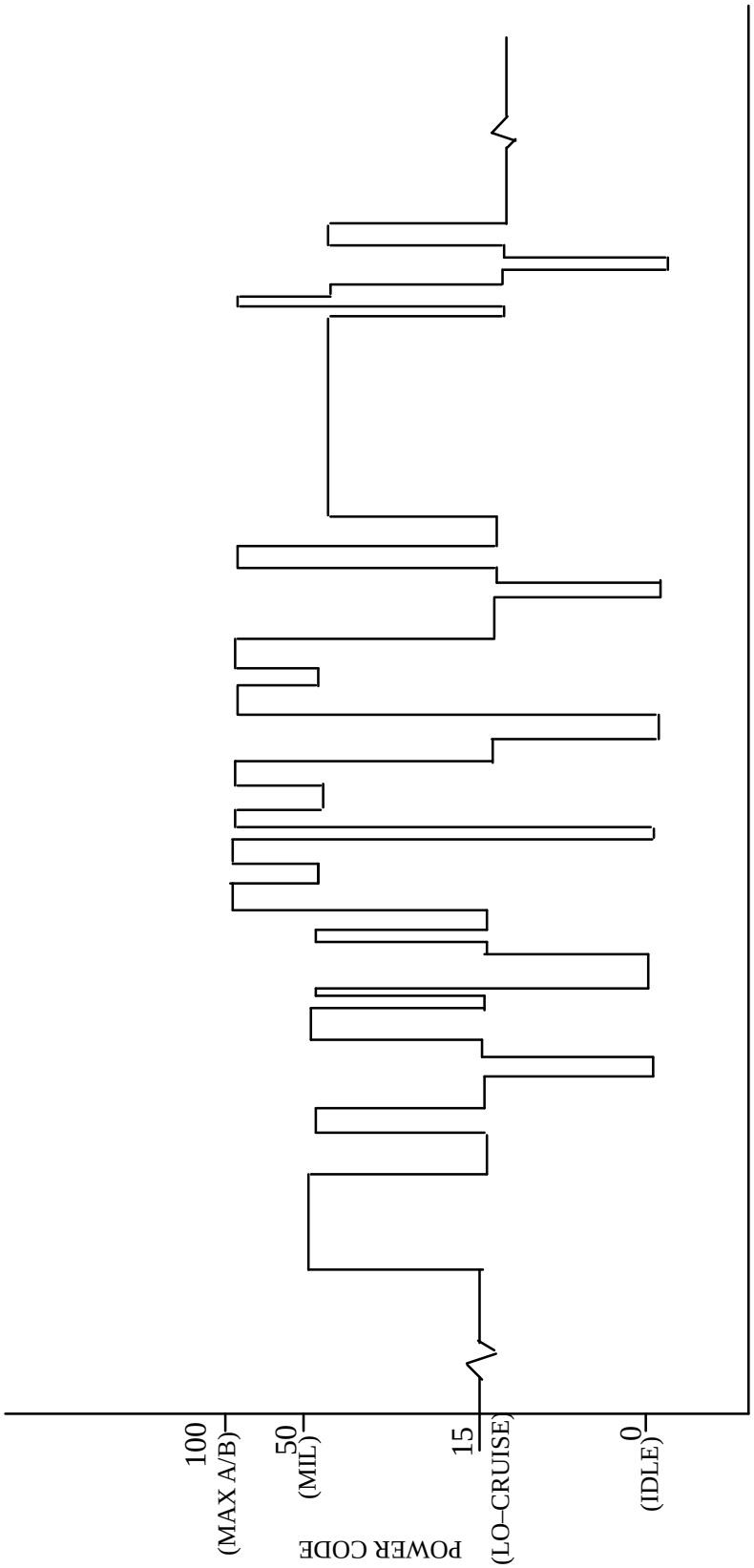


FIGURE 28b. Air combat mission 1 (example only).

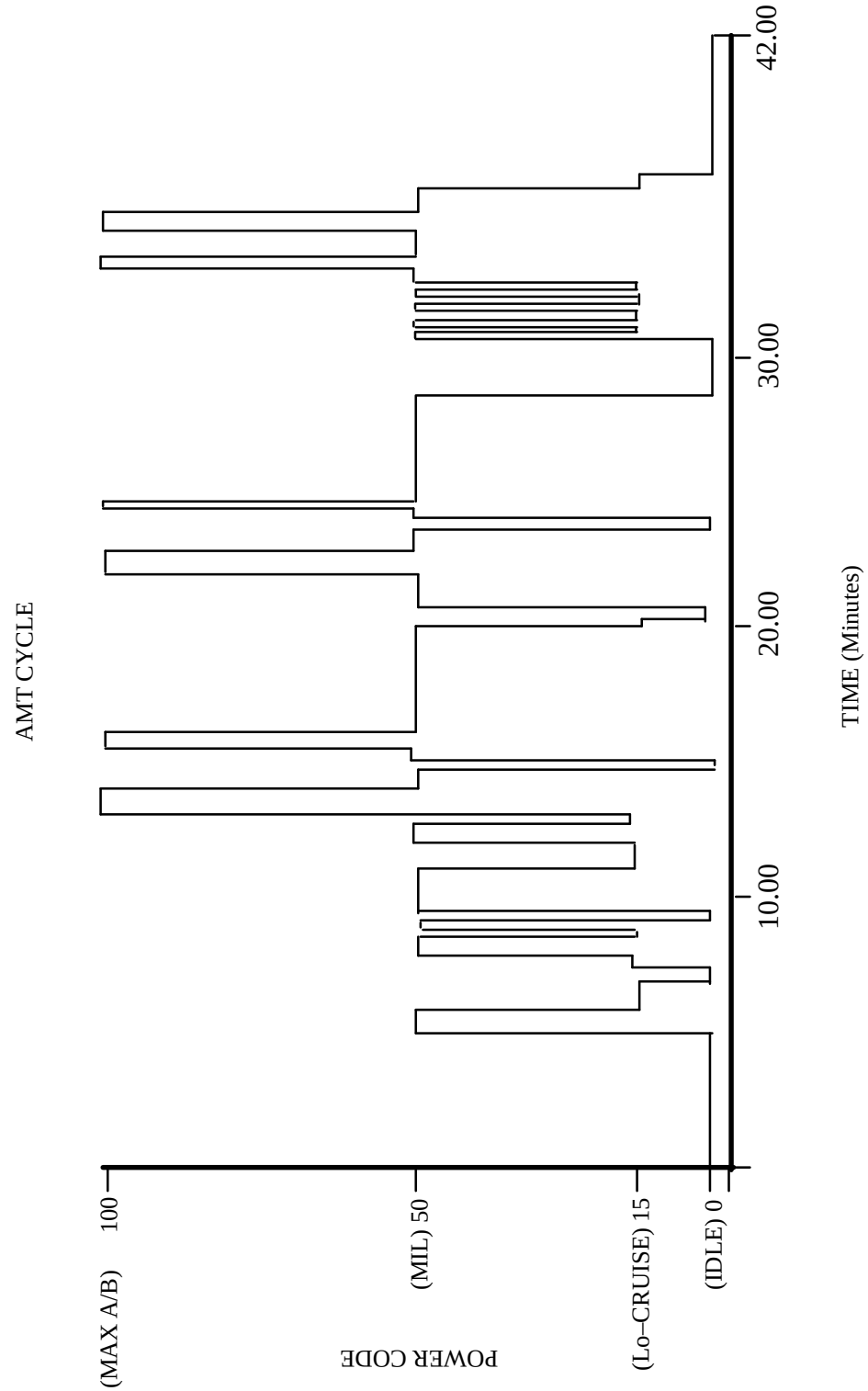


FIGURE 28c. Air combat mission 2 (example only).

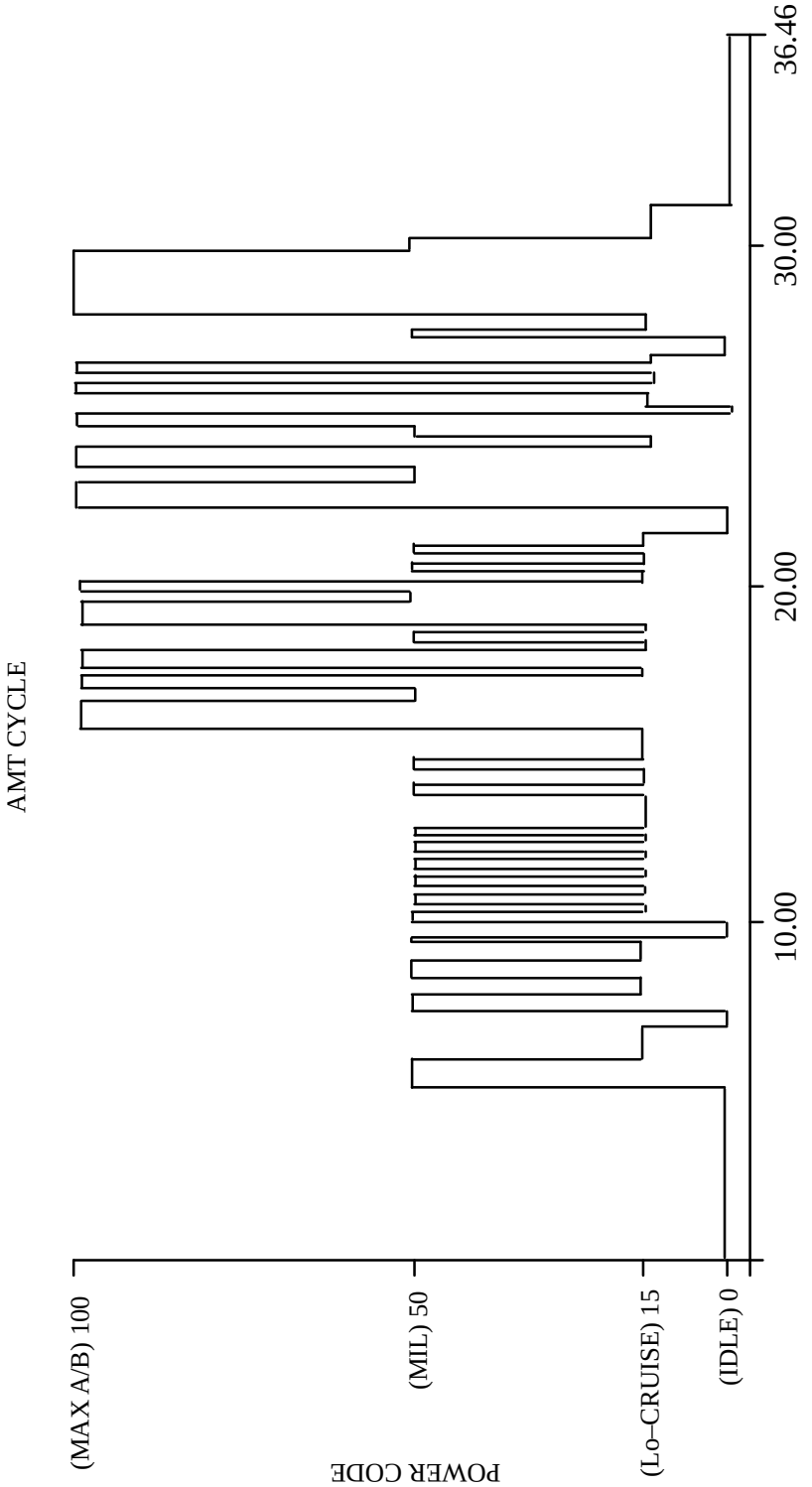


FIGURE 28d. Air combat mission 3 (example only).

Custodians:
Army – AV
Navy – AS
Air Force – 11

Preparing Activity:
Air Force – 11

(Project 2840–0655)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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3. The preparing activity must provide a reply within 30 days from receipt of the form.
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4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

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